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THE OTTAWA HOSPITAL

PARENTERAL DRUG THERAPY MANUAL

Forty-Second Edition

Editors

MIRELLA GIUDICE, B.Sc.PHARM.
NATHALIE GRÉGOIRE, B.PHARM.
ANNE MASSICOTTE, B.PHARM., M.Sc.

Pharmacy Department

Endorsed by the Pharmacy and Therapeutics Committee
and
the Medical Advisory Committee

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(613) 737-8344

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PREFACE

The Parenteral Drug Therapy Manual of The Ottawa Hospital arose out of the necessity to provide a convenient, accessible, reliable reference and guide to health professionals involved with parenteral drug administration. The initial manual was researched and developed in 1975 by Marilyn E. Brown-Morris, M.Sc., former pharmacy resident; the Manual has been maintained up-to-date by the Drug Information Service of the Pharmacy Department.

The recommendations in this Manual are policy at The Ottawa Hospital (TOH). This Manual is not a complete treatise on parenteral drugs; it is impossible to provide extensive therapeutic or compatibility information in such an abbreviated format. For such information, you are advised to contact your Drug Information Service. This Manual is designed to provide essential information on the parenteral administration of a drug once therapeutic and compatibility data have been assessed.

In this Manual, drugs that may be used parenterally are listed alphabetically by non-proprietary (generic) name at the head of the monograph. Under the heading of "Other Names", current brand names and alternative non-proprietary names are listed. In addition, some discontinued popular brand names are still listed under this section to allow easier retrieval. The "Other Names" section is not a comprehensive listing and should not be considered as an endorsement of any product nor imply that TOH stocks the particular brand listed.

This manual is based on documented statements from the literature and reflects evidence-based documented practice in a hospital setting. Therefore, the various sections of this manual are not solely based on the product monograph approved by Health Canada.

The section on "Potential Administration Hazards" does not necessarily contain all the side effects or hazards which may be encountered with parenteral therapy if these hazards are present with oral therapy. Consult other references (manufacturer's product monograph, Drug Information Service) for further information.

The doses listed in the section ''Dosage'' apply to adults, with normal renal and liver functions. Specialized dosage information (children, neonates, patients with renal and/or hepatic impairment), will be identified as such.

Statements in the "Compatibility, Stability" section assume preparation in a laminar flow hood with appropriate sterile technique. Information provided in this section refers to physical/chemical stability and not to sterility. Also consult NAPRA compounding standards.

The safe administration of drugs by Registered Nurses and Registered Practical Nurses is guided by the College of Nurses of Ontario Standards of Practice 2017 which state that knowledge, skill, judgement, and appropriate resources are necessary for the nurse to manage the patient safely. At TOH, Registered Practical Nurses require a specific skills set and must be on specific designated units for parenteral medication administration. Please refer to TOH Model of Nursing Clinical Practice Generic RPN Skills List and unit/area specific RPN Skills List for more details.

Adding Medication Below the Drip Chamber (IV Direct)

According to the Regulated Health Professionals Act (RHPA), administering intravenous medications below the drip chamber falls under the controlled act "administering a substance by injection or inhalation".

A Registered Nurse may acquire the skill of administering intravenous medication below the drip chamber by attending a general and/or unit specific lecture. Upon successfully acquiring this skill, the Registered Nurse may administer certain medications directly intravenously if all other provisions described in the Manual are satisfied.

At TOH, Registered Practical Nurses are not permitted to administer intravenous medications below the drip chamber.

Prerequisites for Medications

There may be prerequisites for the administration of certain drugs. These are defined below and the specific prerequisites are also indicated in the drug monographs. Unit-based protocols, once approved by the Pharmacy and Therapeutics Committee, will supersede these prerequisites. The drugs have been classified, according to prerequisite(s), into seven categories.

1. No prerequisite noted - Drugs which may be given by an RN in any area.

2. Respiratory Support - There must be an oral airway, manual ventilator (ambu bag), airway suctioning and oxygen equipment readily available on the unit.

3. Blood Pressure Monitoring - Drugs requiring use of non-invasive blood pressure monitoring with vital signs being monitored more frequently than q1h.

4. Continuous BP Monitoring - Drugs requiring invasive continuous arterial blood pressure monitoring and where a physician is available in-house to immediately attend STAT calls. However, if inserting an arterial line is not feasible or if the drug requiring continuous blood pressure monitoring is to be administered only for a short time period, "continuous blood pressure monitoring" can be performed by a non-invasive method such as ongoing frequent intermittent blood pressure monitoring (e.g., q10 minutes or as decided by the team).

5. Cardiac Monitoring - The patient must be connected to a cardiac monitor and the Registered Nurse giving the drug must have acquired the skill of ECG interpretation (except Neonatal ICU).

6. Ventilator Support - The patient must have an endotracheal tube or tracheostomy tube in situ and must be mechanically ventilated. The RN must have acquired the skill of ventilatory support.

7. Fetal Monitoring - The patient must be connected to an electronic fetal monitor and the Registered Nurse must have acquired the skill of interpreting the monitor strip.

Abbreviations

aPTT	Activated partial thromboplastin time	min	Minute(s)
ASA	Acetylsalicylic acid	LMWH	Low molecular weight heparin
AV	Atrioventricular	NAPRA	National Association of Pharmacy Regulatory Authorities
BP	Blood pressure	NS	Normal saline
CHF	Congestive heart failure	NSAID	Nonsteroidal anti-inflammatory drug
CNS	Central nervous system	PO	Oral
CrCl	Creatinine clearance	PRN	As needed
DEHP	Diethylhexyl phthalate	PVC	Polyvinylchloride
DMARDs	Disease-modifying antirheumatic drugs	RL	Ringer's lactate
D5W	Dextrose 5% in water	RN	Registered Nurse
D10W	Dextrose 10% in water	SC	Subcutaneous
ECG	Electrocardiogram	SNRIs	Serotonin-norepinephrine reuptake inhibitors
GI	Gastro-intestinal	SSRIs	Selective serotonin reuptake inhibitors
HIV	Human immunodeficiency virus	SWFI	Sterile water for injection
IM	Intramuscular	Temp	Temperature
INR	International normalized ratio	TOH	The Ottawa Hospital
IV	Intravenous	ULN	Upper limit of normal
LFT	Liver function tests		

Definitions

IV Direct: Also called IV Push. Administration of a relatively small volume of a concentrated solution of medication directly into the venous system via a peripheral or central venous access device. The injection is done below the drip chamber over at least 60 seconds (rapid IV direct should be given over 10-30 seconds).

Intermittent IV infusion: Administration by the intravenous route of a medication, usually mixed in an IV solution, over a short period of time and at specific times (ex: administration of IV ciprofloxacin over 60 minutes every 12 hours). Unless otherwise specified, dilute in 50-100 mL of compatible solution and administer over 15-60 minutes.

Continuous IV infusion: Administration by the intravenous route of medication usually mixed in an IV solution that is required on a continuous basis, without interruption, until the end of the treatment (ex: administration of a heparin infusion).

Physician only: Only a physician may administer this medication. This applies to all doses. The physician is expected to stay within a reasonable proximity on the patient's unit until the full effect of the medication has been observed. This may apply to a specific route of administration and/or to a specialty specific physician.

Physician for first dose only: A physician is required to administer the first dose of any new course of the medication during patient's hospitalization. The physician is expected to stay within a reasonable proximity on the patient's unit until the full effect of the medication has been observed. This may apply to a specific route of administration and/or to a specialty specific physician. A nurse may administer subsequent doses.

RN in presence of physician: A nurse may administer the medication when a physician is present. The physician is expected to be present for all doses, and stay within a reasonable proximity on the patient's unit until the full effect of the medication has been observed. This may apply to a specific route of administration and/or to a specialty specific physician.

RN in presence of physician for first dose only: A nurse may administer the medication but a physician is required to be present for the first dose of any new course during hospitalization. The physician is expected to stay within a reasonable proximity on the patient's unit until the full effect of the medication has been observed. This may apply to a specific route of administration and/or to a specialty specific physician. A nurse may administer subsequent doses.

High alert medication: Medication that has a higher risk of causing significant patient harm or fatality when used incorrectly. Heightened vigilance is required when preparing, handling, and administering this medication. TOH made its own list of high alert medications, based on that of the Institute of Safe Medication Practices (ISMP). Consult Corporate policy ADM VIII 860 (High alert medications).

Non-cytotoxic Hazardous Drug / Cytotoxic Drug: Hazardous drugs include both cytotoxic and non-cytotoxic drugs. According to The National Institute for Occupational Safety and Health (NIOSH) 2016 List of Antineoplastic and Other Hazardous Drugs in Healthcare Settings, drugs considered hazardous include those that exhibit one or more of the following six characteristics in humans or animals: carcinogenicity, teratogenicity or other developmental toxicity, reproductive toxicity, organ toxicity at low doses, genotoxicity, structure and toxicity profiles of new drugs that mimic existing drugs determined hazardous by the above criteria.

Intramuscular injection (IM):

At TOH, for IM injections to adults, consult the Corporate nursing standard operating procedure C-SOP NUR 0040 (Medication administration: Intramuscular injections for the adult) and for IM injections to newborns, consult the Department of Obstetrics, Gynecology and Newborn care policy 00482 (Intramuscular injection for the newborn).

The three recommended sites for an IM injection are: 1) ventrogluteal site (gluteal muscles); 2) vastus lateralis (middle thigh); and 3) deltoid if the patient's muscle mass is adequately developed. For children under 1 year old, only the vastus lateralis site is recommended.

The maximum volume, which may be given in any single IM injection, is 3 mL for adults and 2 mL for children. An exception is for the deltoid injection where a maximum of 2 mL should not be exceeded (approximately 1 mL or less is preferred).

Intramuscular injections should be administered using the Z-track method, except when the patient does not have sufficient muscle mass, in newborns or in an emergency situation.

Subcutaneous injection (SC):

May be given into outer aspects of upper arms, thighs and abdomen. The volume administered is usually between 0.5 and 2 mL.

POLICIES ON INTRAVENOUS SOLUTIONS AND MEDICATIONS

Objectives:

1. To minimize the risk of infection to the patients.
2. To help maintain mechanically trouble-free infusions.
3. Reduce complications related to additives.

Policies:

1. Always follow instructions given in the "Parenteral Drug Therapy" Manual.

2. Parenteral products should be examined before and after they are ready for administration, for particulate matter by viewing the contents under a bright light. Hanging intravenous solutions should be inspected periodically for the formation of precipitates or for changes in colour. Ensure that the expiry date has not lapsed.

3. The nurse who prepares and adds medication to the bag or chamber must also initiate the administration of the solution. A "Medication Added" label must be affixed to the intravenous container/burette, stating the name of the drug, the dose added, the solution used, the nurse's initials, the date and time of preparation, the date and time of start of the infusion, and the patient's name.

4. The nurse who prepares medication for injection, as ordered by a physician, must retain the ampoule or vial for verification by the physician when the drug is prepared by the nurse but administered by the physician.

5. Use one chamber (Buretrol ®, Soluset ®, minibag) per drug. Before using a chamber for a second drug or before utilizing one chamber for any two drugs either simultaneously or one following another, verify compatibility with Pharmacy.

6. Mix any two drugs in a syringe or infusion bag only if their compatibility is known and/or has been verified by Pharmacy.

7. Flush medications through tubing with IV solution after each administration, but at a rate of drug delivery no faster than recommended.

8. Leave chamber/bag and tubing connected to the Y-injection site after each administration.

9. A physician's order is required to convert a continuous infusion to a saline lock.

10. Peripheral saline locks are only recommended when the patient requires 4 or less infusion episodes per 24 hours. If more than 4 infusion episodes in 24 hours are required, the physician <u>must</u> order a continuous infusion.

For information regarding tubing and IV line changes, refer to The Ottawa Hospital Nursing Policy, Procedure + Protocol Manual.

PRIORITIES FOR USE OF PARENTERAL INFUSION PUMPS

PRIORITY 1

The following infusions "must" be administered by an infusion pump.

- A medication where cardiac monitoring, ventilator support or fetal monitoring is required.
- Alglucosidase alfa
- Alteplase (tPA, tissue plasminogen activator)
- Amino acids
- Aminophylline
- Amiodarone
- Blinatumomab
- Busulfan
- Cemiplimab
- Cidofovir
- Cisatracurium
- Clodronate (SC infusion)
- Daratumumab
- Deferoxamine (SC infusion)
- Dexmedetomidine (IV and SC infusions)
- Digoxin immune Fab
- Diltiazem
- Dipyridamole
- Dobutamine
- Dopamine
- Eculizumab
- Edrophonium
- Epidural infusions
- Epinephrine
- Epoprostenol
- Eptifibatide
- Esmolol
- Factor VIII (recombinant)
- Factor VIII/von Willebrand factor complex (human)
- Factor IX concentrate (human and recombinant)
- Fat Emulsions
- Fentanyl (IV and SC infusions)
- Filgrastim (SC infusion)
- Foscarnet
- Fosphenytoin
- Furosemide (SC infusion)
- Glycopyrrolate (SC infusion)
- Gonadorelin (SC infusion)
- Heparin
- Human Prothrombin Complex
- Hydromorphone (IV and SC infusions)
- Hyoscine butylbromide (SC infusion)
- Ibutilide
- Iloprost
- Immune globulins (IV and SC infusions)
- Insulin continuous infusions (IV and SC infusions)
- Intra-arterial infusions (infusion pump not needed when used for blood pressure monitoring)
- Intrapleural infusions
- Intravenous infusions in hemodialysis vascular accesses
- Intravenous infusions in neonates
- Isoproterenol
- Ketamine (IV and SC infusions)
- Ketorolac (SC infusion)
- Labetalol

- Levetiracetam (SC infusion)
- Lidocaine (IV and SC infusions)
- Lorazepam (SC infusion)
- Magnesium sulfate (for eclampsia, preeclampsia and fetal neuroprotection)
- Meperidine
- Methotrimeprazine (SC infusion)
- Metoprolol
- Midazolam (IV and SC infusions)
- Milrinone
- Morphine
- Nalbuphine
- Nitroglycerin
- Nitroprusside
- Norepinephrine
- Opioids
- Oxytocin (for induction of labour)
- Pancuronium
- Phenobarbital (SC infusion)
- Phentolamine
- Phenylephrine
- Potassium chloride (KCl) infusion when rate is greater than 10 mmol/hr or concentration greater than 40 mmol/L
- Procainamide
- Propofol
- Propranolol
- Quinine base
- Quinine dihydrochloride
- Remifentanil
- Rocuronium
- Salbutamol
- Scopolamine (SC infusion)
- Sodium chloride hypertonic 3%
- Subcutaneous infusions
- Succinylcholine
- Sufentanil
- Temsirolimus
- Treprostinil (IV and SC infusions)
- Vancomycin

PRIORITY 2

The preferred method of infusion is via an infusion pump for the following medications:

* Amphotericin B
* Continuous central venous infusions
* Continuous chemotherapy infusions
* Fluid restriction to less than 50 mL/hr
* High alert medication infusions
* Multiple infusions for a specific time frame
* Phenytoin (loading dose)
* Potassium chloride (KCl) infusion where rate is less than 10 mmol/hr or concentration less than 40 mmol/L

PRIORITY 3

There will always be specific situations not indicated in Priority 1 and 2 listing that will warrant the use of an infusion pump.
Buretrols ® can also be used to control an infusion rate.

PUMP SHORTAGE

In the event of infusion pump shortage, please consult the Drop Rate Calculation page located at the Pharmacy Department's internal link at myHospital, under Quick References.

USEFUL FORMULAS

Ideal or lean body weight: Male = 50 + (2.3 X height in inches above 60)
Female = 45.5 + (2.3 X height in inches above 60)

Dosing body weight: Ideal body weight + 0.4 (actual body weight - ideal body weight)

Creatinine clearance: mL/sec = $\frac{(140 - \text{age}) \times \text{weight (kg)}}{50 \times \text{serum creatinine (mcmol/L)}}$ X 0.85 (for females)

mL/min = Calculated CrCl in mL/sec X 60

NAME OF MEDICATION
ABATACEPT

OTHER NAMES
Orencia ®

CLASSIFICATION
Immunomodulator

INDICATIONS

- Treatment of moderately to severely active rheumatoid arthritis (RA) in adults.
- Treatment of moderately to severely active juvenile RA in children.
- Treatment of active psoriatic arthritis (PsA) in adults.
- May be used as monotherapy or in combination with non-biologic DMARD therapy (refer to Miscellaneous section).

ADMINISTRATION

- For IV use: reconstitute each 250 mg vial with 10 mL of SWFI, using the silicone-free disposable syringe provided with each vial and an 18-21 gauge needle, to obtain a 25 mg/mL solution. Gently swirl until contents are completely dissolved. Do NOT shake. Upon complete dissolution, the vial should be vented with a needle to dissipate any foam that may be present.
- Intermittent IV infusion: further dilute the reconstituted solution to 100 mL with NS as follows: from a 100 mL NS infusion bag, withdraw a volume equal to the volume of the reconstituted solution required for the patient's dose. The required volume of the reconstituted solution is then added to the NS bag using the silicone-free disposable syringe provided with each vial. Gently mix. Final concentration of the solution will depend upon the amount of drug added, but will be no more than 10 mg/mL. Infuse over 30 minutes, using an infusion set with a sterile non-pyrogenic low-protein-binding filter (pore size 0.2-1.2 micron).
- SC: use the 125 mg prefilled syringe or autoinjector. Allow the prefilled syringe and autoinjector to stand at room temp for 30-60 minutes prior to administration. Sites of injection are: front of the thigh (preferred for the prefilled syringe), abdomen (except for the 2-inch area around the navel), outer area of upper arms. Rotate injection sites (at least 1 inch apart). *Prefilled syringe* (adults and children): inject into pinched skin at a 45° angle. *Autoinjector* (adults only): position so that viewing window is visible. Inject into pinched skin at a 90° angle. Hold for 15 seconds and wait until blue indicator in viewing window stops moving before removing autoinjector.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; anaphylactoid reaction, anaphylaxis, hypotension, urticaria, dyspnea, erythema, rash, pruritus.
- Infusion-related reactions: reported within 60 minutes of starting the infusion and are usually mild to moderate in severity. Most common: dizziness, headache, hypertension; other symptoms include hypotension, tachycardia, bronchospasm, dyspnea, cough, myalgia, flushing, chills, throat tightness, chest discomfort.
- GI: nausea, diarrhea, abdominal pain.
- Infections: upper respiratory tract infection, nasopharyngitis, bronchitis, rhinitis, pneumonia, urinary tract infection, acute pyelonephritis, cellulitis, herpes infections, sepsis.
- Local reactions: for both IV and SC administration; hematoma, pruritus, erythema, pain, swelling.

DOSAGE

Adults:
- IV administration: if less than 60 kg: 500 mg (2 vials) IV; if 60-100 kg: 750 mg (3 vials) IV; if more than 100 kg: 1000 mg (4 vials) IV. To administer at 0, 2 and 4 weeks and then every 4 weeks.
- SC administration:
  - RA in abatacept-naïve patients: initiate with a single IV loading dose as per weight-based dosing above, followed by 125 mg SC within 24 hours and 125 mg SC once weekly thereafter. The IV loading dose may be omitted for patients who are unable to receive an infusion.
  - PsA: 125 mg SC once weekly. An IV loading dose is not needed.
- When switching from IV to SC therapy, administer the first SC injection at the time of the next scheduled IV infusion.

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DOSAGE (Cont.)

Pediatrics:
- IV administration (6 years of age and older): if less than 75 kg: 10 mg/kg/dose IV; if 75-100 kg: 750 mg IV and if greater than 100 kg: 1000 mg IV. To administer at 0, 2 and 4 weeks and then every 4 weeks.
- SC administration (children weighing 50 kg and greater): 125 mg SC once weekly. An IV loading dose is not needed.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials, prefilled syringes and autoinjectors between 2-8°C; protect from light.
- Compatible with NS.
- Reconstituted and fully diluted solutions should be clear, colourless to pale yellow; do not use if opaque particles, discolouration or foreign particles are present.
- Reconstituted solutions and solutions diluted to a maximum concentration of 10 mg/mL in NS are stable for 24 hours at room temp or in the fridge.
- Do not infuse in the same IV line as other agents.

MISCELLANEOUS

- Emergency drugs (e.g., acetaminophen, antihistamines, corticosteroids and/or epinephrine) should be immediately available for the treatment of hypersensitivity reactions.
- Live vaccines should not be given concurrently with abatacept or within 3 months of its discontinuation.
- Not recommended to be given concurrently with another biologic RA agent (i.e., adalimumab, anakinra, etanercept, infliximab) due to the increased risk of serious infection.
- Abatacept may cause false elevations in blood glucose on the day of infusion when using blood glucose monitors with glucose dehydrogenase pyrroloquinolinequinone (GDH-PQQ) based test strips; these test strips may react with maltose present in the abatacept formulation for IV use. The abatacept formulation for SC use does not contain maltose.
- Use cautiously in patients with chronic obstructive pulmonary disease (COPD) due to increased risk of COPD exacerbation, cough, rhonchi, pneumonia and dyspnea.

REFERENCES

2, 5, 74, 95, 121.

NAME OF MEDICATION
ACETAMINOPHEN

OTHER NAMES
Ofirmev, Paracetamol

CLASSIFICATION
Analgesic, Antipyretic

INDICATIONS

- Short-term management of mild to moderate pain when oral and rectal administration are not possible.
- Management of moderate to severe pain as an adjunct to opioid analgesics.
- Treatment of fever.

ADMINISTRATION:

- Intermittent IV infusion (mandatory):
  - For doses of 1000 mg: insert a vented IV set through the septum of the premixed bag of 1000 mg/100 mL and infuse over 15 minutes.
  - For doses less than 1000 mg: withdraw volume from premixed bag of 1000 mg/100 mL that corresponds to the appropriate dose and transfer to an empty sterile container (glass bottle, plastic IV container, or syringe). Infuse undiluted over 15 minutes. Discard any unused portion of the original bag.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, anaphylactic/anaphylactoid reactions, angioedema. Rarely, severe skin reactions such as acute generalized exanthematous pustulosis, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- GI: nausea, vomiting, constipation.
- CNS: headache, insomnia, agitation.
- Respiratory: atelectasis.
- Local reactions: pain.

DOSAGE

- The IV and oral doses are equivalent on a mg to mg basis.
- Do not exceed the maximum recommended single or total daily dose. All routes of acetaminophen (IV, oral and rectal) must be included in the calculation of the dose.
- Adults and adolescents weighing 50 kg or greater:
  - 650 mg IV q4h OR 1000 mg IV q6h (maximum total daily dose of 4000 mg).
  - Dosage adjustment for chronic alcoholism, malnutrition, or dehydration: 12.5 mg/kg IV q4h OR 15 mg/kg IV q6h (maximum total daily dose of 75 mg/kg, not to exceed 3750 mg).
- Adults and adolescents weighing less than 50 kg: 12.5 mg/kg IV q4h OR 15 mg/kg IV q6h (maximum single dose of 750 mg and maximum total daily dose of 75 mg/kg, not to exceed 3750 mg/day).
- Children aged 2 to 12 years: 12.5 mg/kg IV q4h OR 15 mg/kg IV q6h (maximum single dose of 750 mg for children weighing less than 50 kg and 1000 mg for children weighing 50 kg or greater; maximum total daily dose of 75 mg/kg, not to exceed 3750 mg/day for children weighing less than 50 kg and 4000 mg for children weighing 50 kg or greater).
- Children less than 2 years: 12.5 mg/kg IV q4h OR 15 mg/kg/dose IV q6h (maximum total daily dose of 60 mg/kg/day).
- Dosage in renal impairment: for CrCl of 30 mL/min and over: no dosage adjustment necessary. For CrCl between 30 to 10 mL/min: may need to extend dosing interval and/or decrease total daily dose. For CrCl less than 10 mL/min or hemodialysis: usual dose q6-8h; removed by hemodialysis.
- Dosage in hepatic impairment: use with caution in patients with mild to moderate impairment; a decrease in the total daily dose may be warranted. Do not administer to patients with severe impairment.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store premixed bags between 15-30°C. Protect from light. Do not refrigerate. Do not freeze.
- Do not remove metalized overpack until just prior to administration.
- Use within 6 hours of penetrating premixed bag or transferring to another container.
- Compatible with NS, D5W, RL.
- Stable for 84 hours at room temp undiluted in polypropylene syringes.

MISCELLANEOUS

- Contains mannitol.
- Contains povidone K12, which is eliminated unchanged by the kidneys. In children less than 2 years of age, the potential for accumulation of povidone K12 due to immature glomerular filtration is unknown. Safety data for povidone K12 is limited in this population.

REFERENCES

5, 82, 102, 135, 182, 200, 208.

New monograph 2021

NAME OF MEDICATION
AcetaZOLAMIDE

OTHER NAMES
Diamox ®

CLASSIFICATION
Diuretic, Carbonic anhydrase inhibitor

INDICATIONS

- Elevated intraocular pressure in chronic open-angle glaucoma and secondary glaucoma or in preoperative treatment of acute closed-angle glaucoma.
- Edema; diuresis.
- Metabolic alkalosis.
- Epilepsy.
- Periodic ataxia.

ADMINISTRATION

- Reconstitute each 500 mg vial with at least 5 mL of SWFI to obtain a solution of no greater than 100 mg/mL.
- IV direct (preferred): physician or RN. Administer 500 mg (or less) over at least 1 minute.
- Intermittent IV infusion: dilute dose in 50-100 mL of NS or D5W and administer over 15-30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphalaxis, urticaria, fever, rash.
- GI: nausea, vomiting, diarrhea, taste alteration, anorexia.
- CNS: drowsiness, confusion; paresthesia associated with hypokalemia.
- Electrolyte disturbances: hypokalemia.
- Endocrine and metabolic: metabolic acidosis.
- Ophthalmic: risk of developing choroidal effusion with acute myopia and/or angle-closure glaucoma, manifested as an acute onset of decreased visual acuity, blurred vision or ocular pain. May occur within hours to weeks of drug initiation. Discontinue treatment if these symptoms occur.
- Otic: hearing dysfunction, tinnitus.
- Renal: dysuria, crystalluria, polyuria, renal colic, renal failure.

DOSAGE

- Adults:
  - Chronic open-angle glaucoma: 250-1000 mg/day IV, usually given in divided doses when the total daily dose exceeds 250 mg.
  - Secondary glaucoma and preoperative treatment of acute closed-angle glaucoma: 250 mg IV q4h. Some patients may respond to 250 mg IV q12h. In certain acute cases, may give an initial dose of 500 mg IV followed by 125 or 250 mg IV q4h.
  - Edema: 5 mg/kg (usual range: 250-375 mg) IV once daily in the morning.
  - Metabolic alkalosis: 500 mg IV as a single dose; reassess need depending on acid-base status.
  - Epilepsy: 8-30 mg/kg/day IV in divided doses (usual range: 375-1000 mg/day IV); some patients may respond to lower doses. Reduce starting dose to 250 mg IV once daily in patients taking other anticonvulsants; increase dose as needed. Doses greater than 1000 mg IV per day may not offer additional benefit.
  - Periodic ataxia: 250-1000 mg/day IV.

- Pediatrics:
  - Edema: 5 mg/kg IV or 150 mg/m^2 IV once daily in the morning or every other day.
  - Glaucoma: 5-10 mg/kg IV q6h (maximum 1000 mg/day).
  - Epilepsy: 8-30 mg/kg/day IV in divided doses (usual range: 375-1000 mg/day IV); some patients may respond to lower doses. Reduce starting dose to 250 mg IV once daily in patients taking other anticonvulsants; increase dose as needed. Doses greater than 1000 mg IV per day may not offer additional benefit.

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DOSAGE (Cont.)

- Diuretic activity diminishes with continued use; if diuresis does not continue after initial response, do not increase dose but withhold drug for 1 day to allow kidney recovery. To avoid loss of diuretic effect, may be given intermittently (e.g., on alternate days or for 2 days followed by a day of rest).

- Dosage in renal impairment:

CrCl (mL/min):	greater than 50	50-10	less than 10
Dosing interval (hr):	6	12	Do not administer as drug is ineffective

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C.
- Reconstituted solution is stable for 3 days in the fridge; the manufacturer recommends use within 12 hours as it does not contain a preservative.
- Stable for 5 days at room temp or 44 days in the fridge when diluted in D5W or NS at a concentration of 0.375 mg/mL in PVC containers.

MISCELLANEOUS

- Contraindicated in patients with hyperchloremic acidosis, hypokalemia or hyponatremia, Addison's disease or other forms of adrenal failure, cirrhosis or severe renal failure.

REFERENCES

1, 2, 4, 40, 82, 95, 102, 135, 216, 543.

NAME OF MEDICATION
N-ACETYLCYSTEINE

OTHER NAMES
Mucomyst ®, Parvolex ®

CLASSIFICATION
Antidote

INDICATIONS
- Treatment of acetaminophen overdose when a potentially hepatotoxic quantity has been ingested. Decision to administer N-acetylcysteine (NAC) should be made by plotting the measured acetaminophen serum level on the Rumack-Matthew nomogram (available in the manufacturer's product monograph) or by calling a Poison Centre.

ADMINISTRATION
- Intermittent IV infusion: if using the non-standard 48-hour protocol. Dilute each dose as per a ratio of 1 mL of 20% NAC in 5 mL of D5W. Administer each dose over 60 minutes.
- Continuous IV infusion: if using the standard 21-hour IV infusion protocol or the Ontario, Manitoba and Nunavut Poison Centres protocol; refer to Dosage section.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: skin reactions: flushing, rash, pruritus, urticaria. Anaphylaxis and anaphylactoid reactions, angioedema, hypotension or hypertension, tachycardia, facial edema, respiratory symptoms (e.g., wheezing, bronchospasm). Occur soon after starting infusion; severe reactions are rare. Management depends on severity of reaction and includes interrupting infusion and/or administering diphenhydramine IV; epinephrine and corticosteroids may be required for more serious reactions. Once symptoms have resolved, restart infusion cautiously at a reduced infusion rate.
- GI: nausea, vomiting.
- Hematologic: increase or decrease in prothrombin time.
- Local reactions: injection site reactions.

DOSAGE
- NAC therapy should be started within 24 hours and preferably within 8-10 hours of acetaminophen ingestion to prevent liver damage.
- **Standard protocol: 21-hour IV infusion:**

Weight	Loading dose over 60 minutes	Followed by 2nd dose over 4 hours	Followed by 3rd dose over 16 hours
20 kg or less	150 mg/kg IV in 3 mL/kg D5W	50 mg/kg IV in 7 mL/kg D5W	100 mg/kg IV in 14 mL/kg D5W
21 to 40 kg	150 mg/kg IV in 100 mL D5W	50 mg/kg IV in 250 mL D5W	100 mg/kg IV in 500 mL D5W
Greater than 40 kg	150 mg/kg IV (maximum 15 g) in 200 mL D5W	50 mg/kg IV (maximum 5 g) in 500 mL D5W	100 mg/kg IV (maximum 10 g) in 1000 mL D5W

Note:
1) May reduce the amount of diluent for fluid-restricted patients.
2) May replace D5W with 1/2NS if more appropriate.
3) Consider administering 200 mg/kg as the 3rd dose (instead of 100 mg/kg) if massive ingestion of acetaminophen or persistently elevated acetaminophen levels.
4) Extending treatment beyond 21 hours of therapy should be considered if massive ingestion, concomitant ingestion of other substances, pre-existing hepatic disease, acetaminophen levels remain detectable, AST, ALT, or INR remain elevated or abnormalities in predictors of poor prognosis are present.

- **Ontario, Manitoba, and Nunavut Poison Centres protocol:**
  - Dilute the 20% NAC solution with D5W to a concentration of 30 mg/mL (3%) according to patient weight:

Up to 20 kg	Remove 37.5 mL from a 250 mL bag of D5W. Add 37.5 mL of 20% NAC.
21-40 kg	Remove 75 mL from a 500 mL bag of D5W. Add 75 mL of 20% NAC.
41 kg or greater	Remove 150 mL from a 1000 mL bag of D5W. Add 150 mL of 20% NAC.

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DOSAGE (Cont.)

- Loading dose for all patients: 60 mg/kg/hr (maximum of 6000 mg/hr) or 2 mL/kg/hr (maximum of 200 mL/hr) for 4 hours followed by the maintenance dose.

- Maintenance dose:
  for typical dosing scenario: 6 mg/kg/hr (maximum of 600 mg/hr) or 0.2 mL/kg/hr (maximum of 20 mL/hr) until advised to stop by the Poison Centre or patient meets the Stopping Criteria (see below).
  for high-risk patients*: 12 mg/kg/hr (maximum of 1200 mg/hr) or 0.4 mL/kg/hr (maximum of 40 mL/hr) until advised to stop by the Poison Centre or patient meets the Stopping Criteria (see below). *High risk patients are defined according to various clinical and laboratory parameters.

- Stopping Criteria: 1) when acetaminophen level is undetectable AND 2) a minimum NAC infusion of 12 hours (including the loading dose) AND 3) AST/ALT of 100 international units/L or less (or falling and less than 50% peak) AND 4) INR less than 2 AND 5) patient is well.

- **48-hour protocol (not standard: used in exceptional cases of massive overdose or if treatment with NAC is delayed until 10-24 hours after acetaminophen ingestion):**
  - 140 mg/kg IV as a loading dose, followed by 70 mg/kg IV q4h for 12 doses. Total dose = 980 mg/kg (in 13 doses) over 48 hours.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light.
- Opened vials should be stored in the fridge and used within 96 hours. The colour of the solution in the opened vial may change to a light purple under certain conditions. The colour change does not affect the quality of the product.
- Compatible with D5W and 1/2NS.
- Stable for 72 hours at room temp in D5W at a concentration of 30 mg/mL in PVC containers. However, the Ontario, Manitoba, and Nunavut Poison Centres recommend that each bag of 3% solution be changed after 24 hours.

MISCELLANEOUS

- Emergency drugs (epinephrine, antihistamines and corticosteroids) must be immediately available for the treatment of hypersentivity reactions.
- Each mL of 20% solution contains 200 mg of NAC.
- The solution diluted to 3% is slightly hyperosmolar but is still considered safe for administration via a peripheral vein.
- Serum acetaminophen concentrations should be determined between 4 to 24 hours after suspected time of acute ingestion (to ensure that peak concentrations have occurred). If an extended release preparation of acetaminophen was ingested, repeat plasma levels every 4 hours until level peaks.
- Acetaminophen levels should be repeated every 12 hours until undetectable (less than 66 mcmol/L).
- NAC has not been found to be more effective than NS infusion in the prevention of renal dysfunction associated with contrast media.

REFERENCES

1, 2, 4, 7, 95, 135, 208, 263, 339, 417, 441.

NAME OF MEDICATION
ACYCLOVIR

OTHER NAMES
Zovirax ®

CLASSIFICATION
Antiviral

INDICATIONS

- Mucosal and cutaneous herpes simplex (HSV-1 and HSV-2) infections in immunocompromised patients.
- Severe or disseminated herpes simplex infections in pregnancy.
- Severe initial genital herpes infections in immunocompetent patients.
- Herpes simplex encephalitis, varicella zoster encephalitis.
- Varicella-zoster infections in immunocompromised patients.
- Severe varicella infections in pregnant woman and immunocompromised patients.
- Herpes zoster retinitis (acute retinal necrosis).

ADMINISTRATION

- Intermittent IV infusion: dilute in D5W, NS or RL to a concentration no greater than 10 mg/mL and infuse over at least 60 minutes.
- Ensure adequate hydration, particularly during the first 2 hours following the infusion, to prevent precipitation in renal tubules. Recommended urine output is at least 500 mL/g of acyclovir infused during the 2-hour period following acyclovir infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria; rarely angioedema, anaphylaxis, Stevens-Johnson syndrome, erythema multiforme, toxic epidermal necrolysis.
- GI: nausea, vomiting.
- CNS: headache, tremors, confusion, agitation, lethargy, obtundation, hallucinations, seizures, coma.
- Hepatic: transient increases in liver enzymes.
- Renal: increased serum urea and creatinine; hematuria; possible crystallization in renal tubules if administration rate is too rapid, patient is not adequately hydrated, concomitant administration of nephrotoxic agents or pre-existing renal dysfunction.
- Local reactions: venous irritation, phlebitis, inflammation.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Adults and adolescents (12 years of age and older):
  - Herpes simplex infections (mucosal and cutaneous): 5 mg/kg IV q8h.
  - Severe initial genital herpes infections: 5-10 mg/kg IV q8h.
  - Encephalitis: herpes simplex: 10 mg/kg IV q8h; varicella zoster: 10-15 mg/kg IV q8h.
  - Varicella zoster infections: cutaneous or visceral lesions in immunocompromised patients: 10 mg/kg IV q8h.
  - Varicella infections immunocompromised patients: 10 mg/kg IV q8h.
  - Herpes zoster retinitis: 10-15 mg/kg IV q8h.

- Pregnancy:
  - Herpes simplex infections: 5-10 mg/kg IV q8h.
  - Varicella infections: 10-15 mg/kg IV or 500 mg/m^2 IV q8h.

- Pediatrics (younger than 12 years of age):
  - Herpes simplex infections (mucosal and cutaneous) and severe initial genital herpes infections: 250 mg/m^2 IV q8h OR 10 mg/kg IV q8h.
  - Herpes simplex encephalitis (3 months to 12 years of age): 10-20 mg/kg IV q8h.
  - Varicella zoster infections in immunocompromised patients (1-12 years of age): 500 mg/m^2 IV q8h OR 10-20 mg/kg IV q8h.
  - Varicella infections in immunocompromised patients (1-12 years of age): 500 mg/m^2 IV q8h OR 10 mg/kg IV q8h.

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DOSAGE (Cont.)

- Dosage in renal impairment:

CrCl (mL/min)	50-25	25-10	less than 10
Dose	100%	100%	50%
Interval (hr)	12	24	24*

* Administer after dialysis as drug is substantially removed by hemodialysis.

- Dosage in obesity: dose should be calculated using ideal body weight. Some clinicians will use the adjusted body weight in severely ill patients (e.g., encephalitis).

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C. Protect from light.
- Stable at room temp for 24 hours in Ringer's or RL.
- Stable for 37 days in the fridge or at room temp in D5W or NS at a concentration of 5 mg/mL in PVC containers.
- Stable for 63 days in the fridge (5°C +/- 3°C) in NS at a concentration of 5 mg/mL in polypropylene bags.
- Stable for 21 days at room temp or 35 days in the fridge protected from light in D5W at concentrations of 1, 7 and 10 mg/mL in PVC containers.
- Stable for 7 days at room temp or 35 days in the fridge protected from light in NS at concentrations of 1, 7 and 10 mg/mL in PVC containers.
- Precipitation frequently occurs when diluted solutions are stored in the fridge; however, precipitation may also occur at room temp. The precipitate dissolves when the solution is warmed to room temp; these solutions should be used immediately after warming to room temp due to the subsequent appearance of persistent microprecipitates. Precipitation is more common in PVC than in ethylene vinyl acetate (EVA) containers.

MISCELLANEOUS

- Use with caution in patients with underlying neurologic deficits, those with serious renal or hepatic failure, electrolyte abnormalities or significant hypoxia due to increased risk of CNS toxicity.

REFERENCES

1, 2, 4, 6, 40, 82, 95, 102, 135, 143, 200, 208, 366, 367, 462, 551.

NAME OF MEDICATION
ADALIMUMAB

OTHER NAMES
Amgevita ™, Hadlima ™, Hulio ®, Humira ®, Hyrimoz ®

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS

- Moderately to severely active rheumatoid arthritis (RA) in adults.
- Moderately to severely active polyarticular juvenile idiopathic arthritis (JIA) in children.
- Psoriatic arthritis (PsA) in adults.
- Active ankylosing spondylitis (AS) in adults.
- Moderately to severely active Crohn's disease (CD), fistulizing CD and ulcerative colitis (UC) in adults.
- Severely active CD in adolescents.
- Chronic moderate to severe plaque psoriasis in adults.
- Moderately to severely active hidradenitis suppurativa in adults and adolescents.
- Non-infectious uveitis in adults.
- Chronic non-infectious anterior uveitis in children.

ADMINISTRATION

- **Ensure that the selected brand concentration and dosage form (pen, prefilled syringe, autoinjector or vial) are adequate for administration of the required dose in an adult or a child.**
- SC: thigh or abdomen (at least 2 inches from navel); rotate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis (rare), rash, anaphylactoid reaction, angioedema, urticaria.
- Cardiovascular: exacerbation of or new onset CHF.
- CNS: headache, dizziness.
- Infections: patients receiving adalimumab are at increased risk of infections, including tuberculosis and fungal infections.
- Local reactions: erythema, itching, hemorrhage, pain, swelling.

DOSAGE

Dosage specifications for adults and pediatrics may differ slightly among the different products depending on their approved uses and available dosage forms; refer to specific manufacturer's monograph.

Adults:
- RA, PsA and AS: 40 mg SC every 2 weeks. Maintenance dose may be increased to 40 mg SC once weekly if needed.
- CD, fistulizing CD and UC: 160 mg SC at week 0 (to be given in one day: four SC injections of 40 mg OR two SC injections of 80 mg; OR split over two consecutive days: two SC injections of 40 mg OR one SC injection of 80 mg on each day) followed by 80 mg SC at week 2 (to be given in one day as two SC injections of 40 mg OR one SC injection of 80 mg), followed by a maintenance regimen of 40 mg SC every 2 weeks beginning at week 4. Maintenance dose may be increased to 40 mg SC once weekly if needed.
- Plaque psoriasis and uveitis: 80 mg SC as an initial dose (as two SC injections of 40 mg or one SC injection of 80 mg), followed by 40 mg SC every 2 weeks, starting one week after initial dose.
- Hidradenitis suppurativa: 160 mg SC at week 0 (to be given in one day: four SC injections of 40 mg OR two SC injections of 80 mg; OR split over two consecutive days: two SC injections of 40 mg OR one SC injection of 80 mg on each day), followed by 80 mg SC at week 2 (to be given in one day as two SC injections of 40 mg OR one SC injection of 80 mg), followed by a maintenance regimen of 40 mg SC once weekly beginning at week 4.

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DOSAGE (Cont.)

Pediatrics:
- Polyarticular JIA (2 years of age and older):
  - dosage based on weight: if 10 kg to less than 15 kg: 10 or 20 mg SC every 2 weeks; if 15 kg to less than 30 kg: 20 mg SC every 2 weeks; if 30 kg or greater: 40 mg SC every 2 weeks.
  - dosage based on body surface area (based on a study conducted with Humira ®): 24 mg/m^2 SC every 2 weeks, with a maximum dose of 20 mg if less than 4 years of age and 40 mg if 4-17 years of age.
- CD (13-17 years of age and weighing 40 kg or greater): 160 mg SC at week 0 (to be given in one day: four SC injections of 40 mg OR two SC injections of 80 mg; OR split over two consecutive days: two SC injections of 40 mg OR one SC injection of 80 mg on each day) followed by 80 mg SC at week 2 (to be given in one day as two SC injections of 40 mg OR one SC injection of 80 mg), followed by a maintenance regimen of 20 or 40 mg SC every 2 weeks beginning at week 4.
- Hidradenitis suppurativa (12-17 years of age and weighing at least 30 kg): 80 mg SC at week 0, followed by 40 mg SC every 2 weeks starting at week 1; dosage can be increased to 40 mg once weekly if needed.
- Uveitis (2 years of age and older):
  - if less than 30 kg: 20 mg SC every 2 weeks. A dose of 10 mg SC may be used if between 10 kg and less than 15 kg. If child is 6 years of age or older, may be preceded by a 40 mg SC loading dose one week before starting maintenance therapy.
  - if 30 kg or more: 40 mg SC every 2 weeks. If child is 6 years of age or older, may be preceded by a 80 mg SC loading dose one week before starting maintenance therapy.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Amgevita ™, Hulio ® and Humira ® formulations:
- Store between 2-8°C. Protect from light. Do not freeze.
- The pens and prefilled syringes are stable for a single period of 14 days at room temp (do not exceed 25°C) and protected from light; discard if not used with that time period.

Hadlima ™ formulation:
- Store between 2-8°C. Protect from light. Do not freeze.
- Stable for a single period of 28 days at room temp (do not exceed 25°C) and protected from light; discard if not used within that time period.

Hyrimoz ® formulation:
- Store between 2-8°C. Protect from light. Do not freeze.
- Stable for a single period of 21 days at room temp (do not exceed 25°C) and protected from light; discard if not used within that time period.

MISCELLANEOUS

- Amgevita ™, Hadlima ™, Hulio ® and Hyrimoz ® are classified as being biosimilars to Humira ®; therefore the products are not interchangeable.
- The concomitant use of adalimumab with other biologic DMARDs (e.g., anakinra and abatacept) or other TNF antagonists is not recommended due to the increased risk of infection.
- Patients should be monitored closely for infections (including tuberculosis) before, during and after treatment.
- The needle covers for Amgevita ™ prefilled syringes and autoinjectors contain latex.

REFERENCES

1, 5, 53, 95, 135, 558.

NAME OF MEDICATION
ADENOSINE

OTHER NAMES
Adenocard ®

CLASSIFICATION
Antiarrhythmic

INDICATIONS

- Conversion of paroxysmal supraventricular tachycardia to sinus rhythm.
- Diagnotic aid for broad or narrow complex supraventricular tachycardia.

ADMINISTRATION

- IV direct: physician or RN in presence of physician. ***Cardiac monitoring***. Administer undiluted and rapidly over 1-2 seconds. For pediatric patients requiring a dose of less than 0.6 mg, may dilute with NS to achieve a final concentration ranging from 0.3-1 mg/mL.
- To ensure that the drug reaches the systemic circulation, administer in a port as close as possible to the trunk and follow with a rapid NS flush (20 mL for adults and 5-10 mL for pediatrics) of the IV line OR give directly into a peripheral vein.

POTENTIAL ADMINISTRATION HAZARDS

- The following effects may appear immediately after injection and generally last less than 1 minute: facial flushing, shortness of breath/dyspnea, chest pressure, nausea, headache, lightheadedness.
- Cardiovascular: transient arrhythmias during time of conversion may appear on the ECG and these generally last only a few seconds without intervention; patients who develop high-level AV block after one dose should not receive additional doses; asystole.
- Respiratory: bronchospasm in patients with asthma; respiratory compromise in patients with chronic obstructive pulmonary disease.

DOSAGE

Adults:
- Initial dose: 6 mg IV. Reduce dose to 3 mg IV if adenosine is administered via a central line, patient is being treated with carbamazepine or dipyridamole, or has a transplanted heart.
- Additional doses: if tachycardia does not respond within 1-2 minutes, give a second dose of 12 mg (6 mg if central line) IV over 1-2 seconds. This dose may be repeated a second time, if necessary.
- Do not exceed 12 mg IV as a single bolus injection.

Pediatrics:
- For patients weighing less than 50 kg: 0.05-0.1 mg/kg IV (maximum of 6 mg IV). Patients weighing 50 kg or greater: give adult dose.
- Additional doses: if tachycardia does not respond within 1-2 minutes, may increase dose by 0.05-0.1 mg/kg increments every 1-2 minutes until sinus rhythm is established or up to a maximum dose of 0.3 mg/kg.
- Do not exceed 12 mg IV as a single bolus injection.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes at room temp between 15-30°C. Do not refrigerate as crystallization can occur. Crystals will dissolve when warmed to room temp.
- Stable for 16 days at room temp or in the fridge in D5W or NS at a concentration of 0.75 mg/mL in PVC bags or polypropylene syringes.
- Stable for 14 days at room temp or in fridge in D5W and NS at concentrations of 0.05 mg/mL, 0.1 mg/mL and 0.22 mg/mL in PVC bags.
- Stable for 7 days at room temp and 14 days in the fridge at a concentration of 3 mg/mL (undiluded) in polypropylene syringes.
- Stable for 14 days at room temp or in the fridge in D5W or NS at concentrations of 0.01 mg/mL and 0.05 mg/mL in polyolefin bags.

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MISCELLANEOUS

- Use caution in patients treated with digoxin alone or digoxin in combination with verapamil due to risk of ventricular fibrillation/flutter.
- In patients taking methylxanthines (theophylline, caffeine), adenosine may be ineffective or larger doses may be required.
- May produce a higher degree of heart block in patients taking carbamazepine.

REFERENCES

1, 4, 5, 40, 95, 102, 135, 182, 208.

NAME OF MEDICATION
AGALSIDASE BETA

OTHER NAMES
Fabrazyme ®, Recombinant Alpha-galactosidase A Enzyme

CLASSIFICATION
Enzyme, Replacement therapy

INDICATIONS
- Enzyme replacement therapy in patients with Fabry disease.

ADMINISTRATION
- **Ensure premedication has been administered as recommended; refer to Potential Administration Hazards and Dosage sections.**
- Allow vial and diluent to reach room temp prior to reconstitution (approximately 30 minutes). Reconstitute each 35 mg vial and 5 mg vial with 7.2 mL and 1.1 mL, respectively, of SWFI. Roll and tilt vial gently. Do NOT shake. Final concentration of 5 mg/mL.
- Intermittent IV infusion: further dilute with NS to a minimum total final volume of 50 mL (if patient weight is 35 kg or less), 100 mL (if weight = 35.1-70 kg), 250 mL (if weight = 70.1-100 kg) or 500 mL (if weight is over 100 kg). Gently invert infusion bag to mix the solution. Do NOT shake. Initial infusion rate of no greater than 0.25 mg/min (15 mg/hr). If well tolerated, rate may be increased in increments of 0.05 to 0.08 mg/min (3-5 mg/hr) with each subsequent infusion. For patients weighing 30 kg and over, administration time should be at least 90 minutes (based on patient tolerance). For patients weighing less than 30 kg, the maximum infusion rate is 0.25 mg/min (15 mg/hr).
- Do NOT use filter needles during preparation of the infusion.
- Administer through an in-line low-protein-binding 0.2 micron filter.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylaxis, bronchospasm, hypotension, dyspnea, angioedema, dysphagia, urticaria, pruritus, rash, flushing, nasal congestion, chest discomfort.
- Infusion-related reactions: may be severe and include chills, fever, temperature change sensation, pain in extremity, paresthesias, headache, nausea, vomiting, abdominal pain, dyspnea, chest pain/tightness, flushing, hypertension, hypotension, myalgia, edema of lip and ear, fatigue. If reaction occurs, slow infusion rate, withhold infusion and/or administer acetaminophen, antihistamines, and/or corticosteroids. **Consider premedication** with these agents.
- Cardiovascular: QTc interval prolongation, peripheral edema, hypertension, hypotension, tachycardia, bradycardia.
- CNS: dizziness, headache, paresthesias.
- Local reactions: pain, infiltration at IV site, bleeding, infection.

DOSAGE
Adults and pediatric patients (8 years of age and older):
- **Premedication is recommended prior to each infusion (e.g., acetaminophen 1000 mg PO and hydroxyzine 25-50 mg PO or diphenhydramine). A corticosteroid may be added in patients with a previous infusion reaction.**
- 1 mg/kg IV every 2 weeks.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Reconstituted and diluted solutions may be stored in the fridge for up to 24 hours.

MISCELLANEOUS
- Use cautiously in patients with compromised cardiac function due to higher risk of severe complications from infusion reactions.

REFERENCES
1, 5, 40, 95, 135.

Full revision 2020

NAME OF MEDICATION
ALBUMIN, NORMAL HUMAN SERUM

OTHER NAMES
Alburex ®, Albutein, Octalbin, Plasbumin ®

CLASSIFICATION
Plasma protein

INDICATIONS

- Hypoproteinemia with or without edema.
- Treatment of hypovolemic shock.
- Thermal (burn) injury.
- Renal disease (e.g., acute nephrotic syndrome, hypotension during hemodialysis).
- Acute respiratory distress syndrome.
- Cardiopulmonary bypass.
- Therapeutic plasma exchange.
- Ovarian hyperstimulation syndrome.
- Complications of liver diseases (e.g., hepatorenal syndrome, paracentesis in cirrhosis, spontaneous bacterial peritonitis).

ADMINISTRATION

- Available as 5% and 25% solutions.
- Continuous IV infusion (mandatory): for the 25% solution: administer undiluted or dilute in NS (preferred) or D5W (for patients who are sodium-restricted). Do NOT dilute with SWFI as this may cause hemolysis and acute renal failure. For the 5% solution: ready-to-use solution; do NOT further dilute.
- Rate of administration is variable depending on indication, blood volume, patient response and solution concentration.
  - normal blood volume: administer the 5% solution at a maximum rate of 2-4 mL/min, except in patients with hypoproteinemia where a maximum rate of 5-10 mL/min can be used. Administer the 25% solution at a maximum rate of 1-2 mL/min, except in patients with hypoproteinemia where a maximum rate of 2-3 mL/min can be used.
  - hypovolemic shock: rapid IV infusion may be initially required; as plasma volume approaches normal, slow the rate of the 5% solution to a maximum of 2-4 mL/min and the rate of the 25% solution to a maximum of 1-2 mL/min to minimize risks of circulatory overload and pulmonary edema.
- At TOH, obtain vital signs pre-infusion, at 15 minutes after the start of the infusion and after completion of the infusion.
- Consult Corporate Nursing Standard Operating procedure C-SOP NUR 0045 (Blood products and blood derivatives: Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: may be allergic in nature or due to protein overload: anaphylaxis, urticaria, rash, pruritus, chills, fever, flushing, edema, angioedema, changes in respiration, pulse and blood pressure.
- Cardiovascular: circulatory failure, cardiac failure, hypertension, tachycardia, edema.
- GI: nausea, vomiting.
- CNS: headache.
- Electrolyte disturbances.
- Hematologic: coagulation abnormalities.
- Respiratory: dyspnea, pulmonary edema.
- Risk of transmission of infectious agents, including viruses, and theoretically the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.

DOSAGE

- Dosing varies greatly depending on indication and clinical status of patient. The 5% solution should be used in hypovolemic or intravascularly depleted patients; the 25% solution should be used in patients with fluid or sodium restrictions.
- Hypoproteinemia: usual daily dose of 50-75 g IV (1000-1500 mL of 5% solution OR 200-300 mL of 25% solution).

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DOSAGE (Cont.)

- Hypovolemia: initial dose of 12.5-25 g IV (250-500 mL of 5% solution); repeat after 15 to 30 minutes as needed OR 25-50 g (100-200 mL of 25% solution).
- Acute nephrotic syndrome: 25 g IV (100 mL of 25% solution) IV once daily for 7-10 days.
- Hypotension in renal dialysis: usual dose is approximately 25 g IV (100 mL of 25% solution).
- Acute respiratory distress syndrome: 25 g IV (100 mL of 25% solution) over 30 minutes; may repeat q8h prn for 3 days.
- Therapeutic plasma exchange: 5% solution; titrate dose to plasma volume removed during procedure.
- Ovarian hyperstimulation syndrome: 50 to 100 g IV (200-400 mL of 25% solution) over 4 hours; repeat q4-12h prn.
- Hepatorenal syndrome type 1: 1 g/kg IV (4 mL/kg of 25% solution; maximum of 100 g) on day 1, followed by 100-200 mL/day of 25% solution on days 2-14.
- Paracentesis (greater than 5 L) in patients with cirrhosis: 6-8 g IV (24-32 mL of 25% solution) for every litre of fluid removed (At TOH: 100 mL of 25% solution IV over 30 minutes for patients with cirrhosis).
- Spontaneous bacterial peritonitis (non-malignant): 1.5 g/kg IV (6 mL/kg of 25% solution) on day 1, followed by 1 g/kg IV (4 mL/kg of 25% solution) on day 3.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 2-30°C. Protect from light. Do not freeze.
- Use within 4 hours after opening.
- Solution contains no preservatives; if solution shows turbidity or sediment, do not use.
- Compatible with NS, D5W and RL.

MISCELLANEOUS

- Sodium content varies between formulations and ranges between 130-160 mmol/L (5% or 25% solution).
- Contraindicated in patients with severe anemia, cardiac failure, renal insufficiency, pulmonary edema or if history of allergic reaction to albumin.
- Albumin solutions contain a low amount of aluminum (no more than 200 mcg/L); however, toxic levels may occur with high doses, prolonged use and in renal impairment.

REFERENCES

1, 4, 5, 40, 82, 95, 102, 135, 464.

NAME OF MEDICATION
ALDESLEUKIN

OTHER NAMES
Interleukin-2, Proleukin ®

CLASSIFICATION
Antineoplastic

INDICATIONS
- Metastatic renal cell carcinoma (RCC).
- Metastatic malignant melanoma.

ADMINISTRATION
- **Ensure premedication has been administered as recommended; refer to Potential Administration Hazards and Dosage sections.**
- Reconstitute each vial of 22 million international units with 1.2 mL of SWFI (do NOT use bacteriostatic water or NS) to obtain a solution of 18 million international units/mL. During reconstitution, direct diluent against side of vial; gently swirl to avoid excess foaming; do NOT shake.
- Intermittent IV infusion: dilute dose in 50 mL of D5W and administer over 15 minutes.
- Continuous IV infusion.
- Desired concentration for IV administration is 30-70 mcg/mL (0.5-1.1 million international units/mL); smaller or larger volumes of D5W may be used to maintain this concentration; refer to Compatibility, Stability section for more information.
- The manufacturer recommends that aldesleukin be administered only in a hospital setting with an intensive care unit and specialists skilled in cardiopulmonary or intensive care medicine.
- Flush line with D5W pre and post infusion.
- Do NOT use a filter for dilution or administration.
- SC.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: erythema, pruritus, hypotension (when given in combination with chemotherapy).
- Capillary leak syndrome: can occur immediately following start of infusion; loss of vascular tone and increased vascular permeability resulting in hypotension, decreased organ perfusion and function, cardiac arrhythmias/ischemia, ascites, pleural or pericardial effusions, respiratory insufficiency, GI bleeding/infarction, renal failure, edema, mental status changes; may be fatal. Management involves withholding therapy; vasopressors (e.g., dopamine or phenylephrine) may help maintain organ perfusion and diuretics may help with edema.
- Flu-like symptoms: fever, chills, rigor, arthralgia, fatigue**. Consider premedication** with acetaminophen or an NSAID (e.g., ibuprofen, indomethacin); refer to Dosage section. Meperidine may be administered to control rigors.
- Cardiovascular: hypotension, arrhythmias, cardiac arrest.
- GI: nausea, vomiting, diarrhea, GI bleeding/infarction. **Consider premedication** with an antiemetic (doses greater than 12-15 million international units/m^2 are associated with a moderate emetic potential) and a H_2 antagonist to prevent GI irritation/bleeding; refer to Dosage section.
- CNS: mental status changes (e.g., lethargy, somnolence, confusion, agitation). In patients who develop moderate to severe lethargy or somnolence, continued administration may lead to coma.
- Dermatologic: rash (may be exfoliative).
- Endocrine and metabolic: acidosis
- Hematologic: disseminated intravascular coagulopathy, bleeding.
- Renal: renal dysfunction, oliguria.
- Respiratory: adult respiratory distress syndrome, dyspnea, respiratory failure.
- Exacerbation of pre-existing or onset of new autoimmune disease or inflammatory disorders.
- Local reactions: with SC administration, development of nodules at injection site; apply cold compresses to reduce swelling and pain.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- **Consider premedication with an antipyretic (e.g., acetaminophen or ibuprofen), antiemetic, antidiarrheal and H_2 antagonist (e.g., ranitidine, famotidine); these can be discontinued 12 hours following the last dose.**
- IV:
  - RCC: 0.6 million international units/kg IV q8h for a maximum of 14 doses; repeat another 14 doses after a 9-day rest period. May also be given as: 18 million international units/m^2/day as a continuous 24-hour IV infusion for 5 days; repeat after a 5-8 day rest period.
  - Melanoma: 0.6 million international units/kg IV q8h for a maximum of 14 doses; repeat another 14 doses after a 9-day rest period. May also be given as 24 million international units/m^2 IV daily on days 12-16 and 19-23 with concurrent chemotherapy.
- SC:
  - RCC: 18 million international units daily by SC injection for 5 days, followed by a 2-day rest period. For additional cycles, 9 million international units SC on days 1 and 2, followed by 18 million international units daily SC for the next 3 days.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze.
- Reconstituted vials are stable for 48 hours between 2-25°C. Storage in the fridge is preferred.
- Incompatible with NS and bacteriostatic water for injection; will result in aggregation.
- The manufacturer only recommends dilutions between 30 and 70 mcg/mL (0.5-1.1 million international units/mL). Stable for 48 hours at room temp or in fridge in D5W at concentrations of 30-70 mcg/mL (0.5-1.1 million international units/mL) in PVC containers.
- Addition of albumin (final concentration of 0.1%) is necessary for dilutions of less than 30 mcg/mL (0.5 million international units/mL) to improve stability and decrease adsorption of aldesleukin to surfaces of PVC bags, tubing and administration sets (see chart below); albumin to be added to D5W before aldesleukin.
- PVC containers are preferred due to better drug delivery.
- Stability of diluted solutions in D5W:

Final Concentration (mcg/mL)	Final concentration (approximate) (million international units/mL)	Stability of IV infusions (intermittent & continuous) at room temp	Stability of IV continuous infusions at 32°C in CADD pump
5	0.1	---------	6 days. Add albumin
30-70	0.5-1.1	48 hours. Albumin not required	----------
71-100	1.2-1.6	Unstable	Unstable
101-500	1.7-8.2	--------	6 days. Albumin not required

MISCELLANEOUS

- Prior to therapy, patients should have normal cardiac, lung, liver, renal and CNS functions; these should be assessed regularly during treatment.
- Vital signs, weight and fluid intake/output should be monitored daily during therapy.
- 18 million international units = 1.1 mg of protein.

REFERENCES

1, 4, 5, 40, 129, 135, 165, 208, 317.

NAME OF MEDICATION
ALEMTUZUMAB

OTHER NAMES
Human CD52 antibody, Lemtrada ®, MabCampath ®

CLASSIFICATION
Antineoplastic, Immunomodulator, Monoclonal antibody

INDICATIONS
- B-cell chronic lymphocytic leukemia (B-CLL) in patients who have been treated with alkylating agents and who have failed fludarabine therapy (MabCampath ®).
- Previously untreated progressive B-CLL (MabCampath ®).
- T-cell prolymphocytic leukemia (T-PLL) (MabCampath ®).
- Relapsing remitting multiple sclerosis (RRMS) (Lemtrada ®).

ADMINISTRATION
- **Ensure premedication has been administered as recommended; refer to Potential Administration Hazards and Dosage sections.**
- Available as 30 mg/mL (MabCampath ®) and 12.2 mg/mL (Lemtrada ®).
- Do NOT shake vial prior to use.
- Intermittent IV infusion: ***blood pressure monitoring***. For both MabCampath ® and Lemtrada ®, withdraw dose of alemtuzumab from the vial into a syringe; dilute in 100 mL of NS or D5W. Gently invert the bag to mix the solution. Infuse over 2 hours (MabCampath ®) or 4 hours (Lemtrada ®). Patients should be monitored for at least 1-2 hours after completion of each IV infusion.
- SC (MabCampath ® for B-CLL only): thigh or abdomen; rotate injection site. Patients should be monitored for 1 hour post-injection until at least 3 doses of 30 mg have been administered.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis, angioedema, rash, urticaria.
- Infusion-related reactions: hypotension, chills/rigors, fever, flushing, shortness of breath, bronchospasm, urticaria, rash, nausea, vomiting, headache; occur most frequently during first week of therapy (MabCampath ®). May be severe and may occur within 24 hours or even longer post-infusion; refer to Dosage section for premedication. Infusion reactions are managed by stopping or slowing infusion rate, depending on severity of reaction, and managing symptoms. Once symptoms resolve, infusion can be restarted at a slower infusion rate, unless reaction was severe, **with consideration given to using additional premedication.** May consider switching to SC route as reactions with SC administration are less severe than those with IV administration.
- Cardiovascular: myocardial infarction, cardiac arrhythmias, cardiac arrest, hypotension, hypertension, arterial thromboembolism, stroke (ischemic or hemorrhagic; in patients with B-CLL). Administer to patients with pre-existing cardiac disease only if benefits outweigh risks.
- GI: nausea, vomiting, diarrhea.
- CNS: anxiety, paresthesia, headache, dizziness, vertigo.
- Hematologic: lymphopenia, neutropenia, thrombocytopenia, pancytopenia, autoimmune idiopathic thrombocytopenia, autoimmune hemolytic anemia. Profound lymphopenia predisposes to a variety of opportunistic infections (bacterial, viral, fungal and protozoan). Patients with B-CLL and T-PLL should receive oral anti-infective prophylaxis: trimethoprim-sulfamethoxazole equivalent to one double strength (DS) tablet twice a day 3 times a week and famciclovir 250 mg twice a day (or acyclovir 400 mg twice a day or valacyclovir 500 mg twice a day) upon initiation of therapy. Prophylaxis to be continued during treatment and for 2 months after completion of therapy or until CD4+ count is equal or greater than 200 cells/mcL, whichever occurs later. Patients with RRMS should receive oral antiviral prophylaxis with acyclovir 200 mg twice daily (or equivalent) starting on the first day of each treatment course and continuing for at least 1 month after the end of treatment.
- Renal: glomerulonephritis in patients with B-CLL.
- Respiratory: cough, dyspnea, acute respiratory distress.
- Hemophagocytic lymphohistiocytosis (a condition where a large number of immune cells attack and destroy other blood cells) in patients with B-CLL.
- Progressive multifocal leukoencephalopathy.
- Tumour lysis syndrome; reduce risk by ensuring adequate hydration and administering allopurinol for patients with high tumour load.
- Local reactions: with SC administration; injection site skin reactions (erythema, edema, pruritus, pain) of mild to moderate severity during weeks 1 and 2. Ice packs or cold compresses may alleviate symptoms.

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DOSAGE

B-CLL and T-PLL (MabCampath ®):

- **Premedication with oral/IV diphenhydramine 50 mg and acetaminophen 500-1000 mg PO 30 minutes prior to infusion is recommended prior to the first dose, when the dose is escalated and as clinically indicated to minimize the risk of infusion-related reactions. Can also consider giving corticosteroids (e.g., methylprednisolone 1000 mg IV for the first 3 doses or prednisone 10 mg PO) for the first 1-2 weeks. Add methylprednisolone 1000 mg IV or hydrocortisone 200 mg IV (or equivalent) for severe infusion-related events and meperidine 25 mg IV to lessen rigors; both are given 60 minutes prior to infusion.**
- Only the IV route should be used for T-PLL.
- Initial dose: 3 mg IV/SC daily; when this dose is tolerated (infusion-related toxicities are Grade 2 or less), increase daily dose to 10 mg IV/SC; when the 10 mg dose is tolerated, initiate maintenance dose. In most patients, escalation to maintenance dose can be accomplished in 3 to 7 days; for SC route, a longer dose escalation of 1-2 weeks may be required due to injection site reactions. Initial doses can be given daily on days 1-3 or on alternate days (days 1, 3, 5).
- Maintenance dose: 30 mg IV/SC daily administered 3 times a week on alternate days (i.e., Monday, Wednesday, Friday). Consecutive daily dosing may also be used. Administer for up to 12 weeks.
- If therapy is interrupted for 7 days or more, reinstitute therapy with gradual dose escalation.
- Single doses greater than 30 mg or cumulative doses greater than 90 mg per week should not be administered due to increased incidence of pancytopenia.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

RRMS (Lemtrada ®):

- **Premedication with methylprednisolone 1000 mg IV is recommended for the first 3 days of each treatment course; antihistamines and/or antipyretics may also be considered.**
- First treatment course: 12 mg IV once daily for 5 consecutive days.
- Second treatment course: 12 mg IV once daily for 3 consecutive days administered 12 months after the first treatment course.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from direct sunlight. Do not freeze. Do not shake.
- Single-use vial; discard unused portion.
- Stable for 8 hours at room temp or in the fridge protected from light in PVC containers.

MISCELLANEOUS

- **Do not confuse MabCampath ® with Lemtrada ®.** Both contain alemtuzumab; however, they have different concentrations, doses and uses and are therefore not interchangeable.
- Resuscitation equipment must be immediately available for the treatment of hypersensitivity and infusion-related reactions.

REFERENCES

1, 2, 40, 129, 135, 165, 472.

NAME OF MEDICATION
ALFACALCIDOL

OTHER NAMES
1-alfacalcidol, 1-Alpha-hydroxycholecalciferol, 1-Alpha-hydroxyvitamin D3, 1-Alpha-OHD_3, One-Alpha ®

CLASSIFICATION
Vitamin D analog

INDICATIONS

- Management of hypocalcemia, secondary hyperparathyroidism, and osteodystrophy in patients with chronic renal disease.

ADMINISTRATION

- Shake ampoules well prior to use.
- IV direct: physician or RN; undiluted; administer over 30 seconds.

POTENTIAL ADMINISTRATION HAZARDS

- Electrolyte disturbances: hypercalcemia, hyperphosphatemia (at excessive doses).

DOSAGE

- Initial dose of 1 mcg IV per dialysis session (2 to 3 times per week). If insufficient response after 1 week, increase dose in weekly increments of 1 mcg IV per dialysis session. Maximum of 12 mcg IV per week.
- Maintenance: usual dose is 6 mcg IV per week (range of 1.5 to 12 mcg per week divided equally between dialysis sessions).
- Titration period should not exceed 6 weeks.
- Dose should be titrated to maintain total serum calcium in the upper half of the normal range.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 2-8°C; protect from light.
- Not recommended to dilute in infusion fluids.
- May adsorb to plastic.

MISCELLANEOUS

REFERENCES

2, 95, 122, 135, 266.

Full revision 2020

NAME OF MEDICATION
ALGLUCOSIDASE ALFA

OTHER NAMES
Myozyme ®

CLASSIFICATION
Enzyme, Replacement therapy

INDICATIONS
- Replacement therapy in both infantile-onset and late-onset Pompe's disease (deficiency of lysosomal acid alpha-glucosidase [GAA]).

ADMINISTRATION
- **Consider premedication if patient experienced a hypersensitivity/infusion-related reaction; refer to Potential Administration Hazards section.**
- Allow vials to reach room temp prior to reconstitution (approximately 30 minutes).
- Reconstitute each 50 mg vial with 10.3 mL SWFI by **slowly** adding in a drop-wise manner down the inside wall of the vial. Avoid direct forceful impact of the diluent on the powder to prevent foaming. Tilt and roll the vial gently to allow dissolution. Do NOT invert, swirl, or shake the vial. Final concentration is 5 mg/mL. Do NOT use if opaque particles or discolouration are present.
- Intermittent IV infusion (mandatory): slowly withdraw required dose to avoid foaming in the syringe. Infusion volume is dependent on patient's weight; refer to table in manufacturer's monograph for correct volume of NS to be used; this volume ranges from 50-1000 mL. Remove the airspace from the NS bag to reduce the risk of particle formation due to drug sensitivity to air-liquid interface. Add the reconstituted dose slowly and directly into the NS, NOT into airspace, for a final concentration between 0.5-4 mg/mL. Gently invert bag to mix. Avoid foaming; do NOT shake the bag. Administer via an in-line low-protein-binding 0.2 micron filter to remove visible thin white strands or translucent fibres of alglucosidase alfa. Start IV infusion at 1 mg/kg/hr; if no hypersensitivity or infusion-related events occur after 30 minutes, increase rate by 2 mg/kg/hr every 30 minutes as tolerated to a maximum of 7 mg/kg/hr. The total infusion time will be approximately 4 hours. If reaction occurs, slow and/or stop the infusion.
- Do NOT use filter needles to prepare the infusion.
- Protect from light both the reconstituted and diluted solutions.
- Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylaxis, acute cardiorespiratory failure, cardiac arrest, hypotension, tachycardia, bronchospasm, decreased oxygen saturation, respiratory distress, dyspnea, urticaria, angioedema, throat tightness, chest discomfort/pain. May occur during and up to 3 hours post-infusion. Majority of reactions are moderate to severe in intensity. Management involves reducing infusion rate and/or temporarily withholding infusion as well as administering appropriate medical support, IV fluids, oxygen, antihistamines (e.g., diphenhydramine), corticosteroids (e.g., hydrocortisone IV), bronchodilators (e.g., inhaled salbutamol), and epinephrine. Permanent discontinuation of therapy may be required. Some patients have been rechallenged under close supervision. If treatment is to be continued, **consider premedication** with antihistamines, corticosteroids, and/or acetaminophen.
- Infusion-related reactions: rash, urticaria, pruritus, fever, rigors, hypotension, hypertension, tachycardia, flushing, sweating, cough, chest discomfort, tachypnea, low oxygen saturation, agitation, irritability, nausea, vomiting, headache, dizziness. May occur during and up to 2 hours post-infusion; delayed-onset reactions within 48 hours post-infusion have also occurred. Risk increases with higher infusion rates and acute underlying illness (e.g., acute infection, respiratory compromise, heart failure). Majority of reactions are mild to moderate in intensity. Management of mild to moderate reactions involves reducing infusion rate and/or temporarily withholding infusion as well as administering antipyretics (e.g., acetaminophen), antihistamines (e.g., diphenhydramine), and/or corticosteroids (e.g., dexamethasone). **Consider premedication** with these agents. Severe reactions are managed by stopping infusion, providing appropriate medical support, antihistamines, corticosteroids, IV fluids, oxygen, and/or epinephrine, as needed.
- Cardiovascular: cardiorespiratory failure, cardiac arrest, ventricular arrhythmia, bradycardia, peripheral edema.
- GI: diarrhea, constipation.
- Dermatologic: severe skin and systemic immune-mediated reactions (e.g., ulcerative and necrotizing skin lesions).
- Hematologic: anemia.
- Musculoskeletal: inflammatory arthropathy.
- Renal: nephrotic syndrome secondary to membranous glomerulonephritis, proteinuria.
- Respiratory: upper respiratory tract infection, otitis media, pneumonia, respiratory failure.

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DOSAGE

- 20 mg/kg IV every 2 weeks. Round dose up to the nearest vial size.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Do not freeze or shake.
- Reconstituted solution is stable 24 hours in the fridge protected from light. Do not freeze or shake.
- Stable for 24 hours in the fridge protected from light in NS at concentrations of 0.5 and 4 mg/mL. Do not freeze or shake.
- Thin white strands or translucent fibers composed of alglucosidase alfa particles may appear after reconstitution and dilution in NS. These increase over time, are normal, do not affect the purity or strength and are removed by the 0.2-micron low-protein-binding in-line filter during IV administration.

MISCELLANEOUS

- Use with caution in patients with cardiovascular and/or respiratory disease due to the increased risk of acute cardiorespiratory failure secondary to infusion-related reactions and/or IV fluid overload.
- Emergency drugs and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.

REFERENCES

1, 4, 5, 40, 95, 135, 208.

NAME OF MEDICATION
ALPHA$_1$-PROTEINASE INHIBITOR

OTHER NAMES
Alpha$_1$-antitrypsin, Prolastin-C ®

CLASSIFICATION
Enzyme, Replacement therapy

INDICATIONS

- Replacement therapy for congenital alpha$_1$-antitrypsin deficiency in patients with clinical evidence of emphysema.

ADMINISTRATION

- Reconstitute 1 g vial with 20 mL of SWFI (supplied). Swirl vigorously for 10-15 seconds to thoroughly break up cake and then swirl continuously until powder is completely dissolved. Some foaming will appear without affecting product quality.
- Intermittent IV infusion: Withdraw exact dose using the filter needle supplied by manufacturer. Replace filter needle with appropriate needle for injection. Transfer dose to empty sterile IV container. Administer at a rate of 0.08 mL/kg/min or greater; usual dose is administered over approximately 15 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: pruritus, rash, hives, urticaria, dyspnea, bronchospasm, hypotension, anaphylaxis.
- Cardiovascular: hypotension (rare), tachycardia.
- CNS: headache, somnolence, dizziness, lightheadedness.
- GI: nausea, diarrhea.
- Hematologic: transient leukocytosis and dilutional anemia.
- Respiratory: shortness of breath, dyspnea, wheezing.
- Fever (occurring up to 12 hours after infusion), malaise, flu-like symptoms, sweating, back pain, fatigue.
- Risk of transmission of infectious agents, including viruses, and theoretically the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.

DOSAGE

- 60 mg/kg IV once weekly.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at temperature not exceeding 25°C. Do not freeze.
- Administer within 3 hours after reconstitution.
- Do not refrigerate after reconstitution.

MISCELLANEOUS

- Patients should be immunized against hepatitis B.

REFERENCES

1, 2, 5, 40.

Full revision 2020

NAME OF MEDICATION
ALPROSTADIL

OTHER NAMES
Prostaglandin E_1, Prostin VR ®

CLASSIFICATION
Prostaglandin

INDICATIONS
- To maintain the patency of the ductus arteriosus in neonates until surgery can be performed.
- Treatment of circulatory deficiency (Raynaud phenomenon).

ADMINISTRATION
Patency of ductus arteriosus:
- Continuous IV infusion: ***respiratory support, blood pressure monitoring, cardiac monitoring***.
  Dilute 500 mcg (1 mL) in 25-250 mL of D5W or NS. (At TOH: add 200 mcg [0.4 mL] to 19.6 mL D5W for a final concentration of 10 mcg/mL).To minimize possibility of haze formation, add alprostadil directly to the IV infusion solution, avoiding contact with the walls of plastic containers; discard any solution that appears hazy as well as the plastic container. The infusion rate to deliver 0.1 mcg/kg/min can be calculated as follows:

$$\text{Infusion rate (mL/hr)} = \frac{\text{Volume (mL) containing 500 mcg X weight (kg)}}{83.3}$$

- Preferred route is IV via a large vein (peripheral or central) although alprostadil may be administered through an umbilical artery catheter positioned near ductus arteriosus.

POTENTIAL ADMINISTRATION HAZARDS
- Cardiovascular: flushing (more common with intra-arterial administration), bradycardia or tachycardia, hypotension, edema, arrhythmias, cardiac arrest.
- GI: diarrhea, gastric outlet obstruction secondary to antral hyperplasia if alprostadil is given for more than 120 hours.
- CNS: seizures.
- Endocrine and metabolic: hypokalemia, hypoglycemia.
- Hematologic: disseminated intravascular coagulation, hemorrhage, thrombocytopenia.
- Renal: anuria.
- Respiratory: respiratory depression, apnea; usually in the first hour of the infusion and in those weighing less than 2 kg at birth.
- Fever.

DOSAGE
- Maintenance of patency of ductus arteriosus: initial infusion of 0.05 to 0.1 mcg/kg/min with subsequent reduction to lowest effective dose once desired effect is achieved (this may be accomplished by reducing the dosage from 0.1 to 0.05 to 0.025 to 0.01 mcg/kg/min). Although doses up to 0.4 mcg/kg/min have been used, doses greater than 0.1 mcg/kg/min generally do not offer additional benefits.
- Infusion should ideally be initiated within 96 hours after birth.
- Infusion is usually continued until surgery is completed, usually within 24-48 hours after initation.
- Refer to specialized references for doses for other indications.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 2-8°C.
- Stable for 24 hours at room temp when diluted in D5W or NS.

MISCELLANEOUS
- Neonates receiving alprostadil for greater than 120 hours should be closely monitored for antral hyperplasia and gastric outlet obstruction.
- Use with caution in neonates with bleeding tendencies.
- Avoid use in neonates with respiratory distress syndrome.

REFERENCES
1, 4, 5, 40, 95, 102, 319, 513.

Full revision 2020

NAME OF MEDICATION	
ALTEPLASE	
OTHER NAMES	CLASSIFICATION
Activase ® rt-PA, r-TPA, Tissue plasminogen activator, tPA	Thrombolytic

Do NOT confuse Activase ® with Cathflo ®. Although both contain alteplase, only Activase ® (and not Cathflo ®) can be used for the indications stated in this monograph.

INDICATIONS

- Lysis of coronary artery thrombi associated with evolving myocardial infarction.
- Acute ischemic stroke.
- Acute massive or submassive pulmonary embolism (PE).
- Acute peripheral arterial occlusion in patients with acute limb ischemia.

ADMINISTRATION

- Reconstitute each 50 mg vial with the diluent provided (50 mL of SWFI) and according to manufacturer's instructions for a final concentration of 1 mg/mL. Gently swirl to mix; do NOT shake.
- Reconstitute each 100 mg vial with the diluent provided (100 mL of SWFI) and according to manufacturer's instructions for a final concentration of 1 mg/mL. Use the transfer device provided. Gently swirl to mix; do NOT shake. Use a vented tubing with the glass bottle.
- Slight foaming upon reconstitution is NOT unusual. Let stand undisturbed for several minutes to allow large bubbles to dissipate.
- IV direct: physician or RN. Administer bolus dose over 1-2 minutes (refer to Dosage section).
- Intermittent IV infusion: the 50 mg and 100 mg reconstituted vials may be further diluted with NS or D5W to a concentration of no less than 0.5 mg/mL; refer to Compatibility and Stability section for dilution and to Dosage section for infusion rate. Note that the 100 mg reconstituted vial may also be infused straight from the glass bottle as there is no vacuum; this is not possible with the 50 mg reconstituted vial as there is a vacuum. Must be administered by an infusion pump.
- Continuous infusion by arterial line: interventional radiologist.
- Refer to TOH stroke protocol for patient monitoring requirements.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; rash, urticaria, angioedema, laryngeal edema, anaphylactoid and anaphylactic reactions, shock. Angioedema has occurred up to 2 hours after infusion.
- Cardiovascular: arrhythmias may occur and are associated with coronary artery reperfusion.
- Hematologic: internal bleeding (intracranial, retroperitoneal, GI, genitourinary, respiratory tract), epistaxis, gingival bleeding, hematuria, hematoma.
- Local reactions: extravasation can cause ecchymosis and/or inflammation.

DOSAGE

- Myocardial infarction: 15 mg IV over 1-2 minutes followed by 0.75 mg/kg IV (maximum 50 mg) over 30 minutes followed by 0.5 mg/kg IV (maximum 35 mg) over next 60 minutes.
- Acute ischemic stroke: 0.9 mg/kg IV (maximum 90 mg) with 10% of the total dose (0.09 mg/kg) given as an IV bolus (over 1 minute) followed by the remaining 90% (0.81 mg/kg) infused over 60 minutes.
- Acute massive or submassive PE: 100 mg IV over 2 hours OR 10 mg IV bolus over 10 minutes followed by 90 mg IV infused over 2 hours.
- Acute peripheral arterial occlusion: weight-based regimen: 0.001-0.02 mg/kg/hr (maximum 2 mg/hr) intraarterially OR fixed dose regimen: 0.12-2 mg/hr intraarterially. Duration of infusion: varies depending on size and location of the occlusion; usually 6-48 hours.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-30°C. Protect from light.
- Reconstituted solution (1 mg/mL with SWFI) is stable for 8 hours between 2-30°C.
- Stable for 6 months in the freezer at -20°C in SWFI at a concentration of 1 mg/mL in polypropylene syringes.
- Stable for 22 weeks in the freezer at -30°C in SWFI at a concentration of 1 mg/mL in plastic syringes.
- Stable for 1 month in the freezer at -20°C in SWFI at a concentration of 1 mg/mL in glass vials.
- Stable for 2 weeks in the freezer at -70°C then thawed and stored at 23°C for 24 hours in SWFI at a concentration of 1 mg/mL in glass vials.
- Stable for 48 hours in the fridge and for 14 days in the freezer at -25°C and -70°C, thawed at room temp and then stored at 2°C for 48 hours in SWFI at concentrations of 0.5 mg/mL, 1 mg/mL, 2 mg/mL in polypropylene syringes.
- Compatible with D5W and NS; incompatible with RL.
- Stable for 8 hours at room temp in D5W or NS at a concentration of 0.5 mg/mL in glass or PVC containers.
- When diluted in D5W, minimum concentration is 0.5 mg/mL; dilution to 0.16 and 0.09 mg/mL in D5W resulted in precipitation immediately and in 4 hours, respectively.
- When diluted in NS to a concentration of 0.2 mg/mL, no precipitation was observed. Stable for 24 hours at room temp in NS at a concentration of 0.01 mg/mL in PVC bags.
- During the period of reconstitution and infusion, protection from light is not necessary.
- Do not use solutions containing preservatives.

MISCELLANEOUS

- Avoid IM injections and excessive handling of patient; venipunctures should be performed carefully and as infrequently as possible to avoid bleeding; if arterial puncture is required, use an upper extremity vessel that is accessible to manual compression.

REFERENCES

1, 2, 4, 5, 6, 95, 135, 208, 365.

NAME OF MEDICATION
ALTEPLASE (FOR CATHETER OCCLUSION)

OTHER NAMES
Activase ® rt-PA, Cathflo ®, r-tPA, Tissue plasminogen activator, tPA

CLASSIFICATION
Thrombolytic for catheter occlusion

INDICATIONS

- To restore function of central venous access devices (CVAD).

ADMINISTRATION

- Either the Cathflo ® product or Activase ® product may be used.
- Cathflo ®: reconstitute 2 mg vial with supplied diluent of 2.2 mL of SWFI. Do NOT use bacteriostatic water. Slight foaming is NOT unusual. Let vial stand undisturbed to allow bubbles to dissipate. Gently swirl to mix; do NOT shake. Final concentration of 1 mg/mL.
- Activase ®: reconstitute the 50 mg and 100 mg vials with the accompanying 50 mL and 100 mL, respectively, of SWFI for a final concentration of 1 mg/mL. Do NOT use bacteriostatic water. Slight foaming is NOT unusual. Gently swirl to mix; do NOT shake.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare and in cases where drug enters systemic circulation; rash, urticaria, angioedema, anaphylactoid reactions.
- Hematologic: bleeding (theoretical risk; has not been reported with Cathflo ® product).
- Systemic infection (due to potential for releasing localized infected clot or fluid into systemic circulation).

DOSAGE

- Patients weighing 30 kg or more: usual dose of 2 mg (2 mL). For patients weighing less than 30 kg: dose is 110% of the internal lumen volume of their CVAD, not to exceed 2 mL.
- Instill slowly appropriate dose into occluded catheter. After a 30 minute dwell time, inspect catheter function. If function not restored, allow to dwell for an additional 90 minutes, for a total dwell time of 120 minutes. If catheter function not restored, a second 2 mg dose may be used and the whole procedure repeated once. Longer dwell times of up to 24-72 hours have also been used.
- When catheter function restored, aspirate 4 to 5 mL of blood to remove drug and residual clot. Gently irrigate catheter with NS.
- Lower doses of 1 mg and higher doses of up to 4 mg have also been used.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials of Cathflo ® between 2-8°C. Store vials of Activase ® between 2-30°C. Protect from light.
- Reconsituted solution (1 mg/mL with SWFI) is stable for 8 hours between 2-30°C.
- Stable for 6 months in the freezer at -20°C in SWFI at a concentration of 1 mg/mL in polypropylene syringes.
- Stable for 48 hours in the fridge and for 14 days in the freezer at -25°C and -70°C in SWFI at concentrations of 0.5 mg/mL, 1 mg/mL and 2 mg/mL in polypropylene syringes.
- Compatible with D5W and NS; incompatible with RL.
- During the period of reconstitution and infusion, protection from light is not necessary.
- Do not use solutions containing preservatives.

MISCELLANEOUS

- Use caution in patients with active internal bleeding, thrombocytopenia or other hemostatic defects and in those with suspected catheter-related bloodstream infections.

REFERENCES

1, 2, 4, 5, 6, 95, 99, 208, 425.

Full revision 2020

NAME OF MEDICATION
AMIKACIN

OTHER NAMES
Amikin ®

CLASSIFICATION
Antibiotic - aminoglycoside

INDICATIONS

- Treatment of serious infections caused by gram-negative organisms (e.g., *Citrobacter, Ps. aeruginosa, Proteus, Klebsiella, E. coli, Enterobacter, Serratia sp.*) that are resistant to other agents.

ADMINISTRATION

- Intermittent IV infusion: dilute in 50-250 mL of a compatible IV solution to obtain a final concentration of 0.25-5 mg/mL. For children, volume of diluent may be less. Administer over 30-60 minutes.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, toxic epidermal necrolysis, erythema multiforme, Stevens-Johnson syndrome.
- CNS: paresthesia, muscle twitching, convulsions, neuromuscular blockade leading to respiratory paralysis (rare).
- Otic: ototoxicity: auditory toxicity (e.g., high frequency hearing loss, tinnitus), and vestibular toxicity (e.g., vertigo, nystagmus, dizziness, ataxia, nausea, vomiting); may be irreversible.
- Renal: nephrotoxicity; associated with persistently elevated trough serum levels; usually reversible if drug is discontinued at first sign of azotemia.
- Risk factors for ototoxicity and nephrotoxicity include renal impairment, advanced age, dehydration, concomitant oto- or nephrotoxic medications, high doses, and prolonged therapy (greater than 10 days).
- Local reactions: pain, irritation, burning and erythema at injection site.

DOSAGE

- Traditional dosing: 5-7.5 mg/kg/dose IV/IM with dosing interval according to CrCl. Round dose to nearest 50 mg. Maximum of 15 mg/kg/day or 1.5 g/day. Refer to Miscellaneous section for monitoring serum levels.
- Dosage in renal impairment: dosing interval must be extended, based on CrCl. Based on TOH experience:

CrCl (mL/min)	greater than 60	60-40	39-20	less than 20 or hemodialysis
Interval (hr)	8	12	24	frequency as per serum concentrations

- Extended interval dosing: 15-20 mg/kg IV/IM q24h may be used in selected patients. Round dose to nearest 250 mg. Refer to Miscellaneous section for monitoring serum levels.
- Doses should be calculated using actual body weight, except in obese patients.
- Dosage in obesity (20-30% above ideal body weight (IBW)): dose should be calculated using dosing body weight (DBW) = IBW + 0.4 X (actual body weight-IBW).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C. Protect from light.
- Faint yellowing of amikacin injection does not indicate loss of potency. Dark coloured solutions should be discarded.
- Stable for 24 hours at room temp, 60 days in the fridge and 30 days frozen in most IV solutions including D5W, NS and RL, at concentrations of 0.25 mg/mL to 5 mg/mL in PVC containers.
- Stable for 48 hours at room temp in D5W and NS at a concentration of 5 mg/mL in PVC containers.
- Stable for 24 hours at room temp and 30 days frozen in D5W at a concentration of 20 mg/mL in PVC containers.

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MISCELLANEOUS

- Safety for treatment periods exceeding 14 days has not been established.
- Monitoring serum levels:
  - Traditional dosing:
    - serum levels pre-dose (trough): taken 5 minutes before next dose; should be between 5-8 mg/L.
    - serum levels post-dose (peak): taken 30 minutes after the end of IV infusion or 60 minutes after IM injection; should be between 20-30 mg/L.
  - Extended interval dosing:
    - serum levels pre-dose (trough): taken 5 minutes before next dose; should be undetectable (i.e., less than 2.5 mg/L).
    - serum levels post-dose (peak): no need to monitor.

REFERENCES

1, 2, 4, 5, 6, 40, 95, 135, 143, 208, 303.

NAME OF MEDICATION
AMINOPHYLLINE

OTHER NAMES
Theophylline ethylenediamine

CLASSIFICATION
Bronchodilator

INDICATIONS

- Treatment of acute exacerbations of asthma, chronic obstructive pulmonary disease, chronic bronchitis and related bronchospastic disorders.
- Treatment of bronchospasm in cystic fibrosis and acute descending respiratory infections.
- Relieve the periodic apnea and increase arterial blood pH in patients with Cheyne-Stokes respiration.
- Treatment of neonatal apnea of prematurity.

ADMINISTRATION

- IV direct (NOT preferred): physician or RN. Administer the 25 mg/mL solution undiluted at a rate not exceeding 25 mg/min. NOT recommended to give IV direct to pediatric patients.
- Intermittent IV infusion: for loading dose. Dilute in 100 mL and give over 20-30 minutes; maximum rate of 25 mg/min.
- Continuous IV infusion: dilute in 500 mL to a final concentration of 1-5 mg/mL (usual dilution is 1 mg/mL for pediatric patients). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, angioedema, rash, pruritus.
- Cardiovascular: palpitation, tachycardia, extrasystoles, hypotension, flushing.
- GI: nausea, vomiting, abdominal pain, anorexia, diarrhea.
- CNS: headache, nervousness, insomnia, irritability, tremor.
- Rapid IV injection may cause flushing, dizziness, faintness, lightheadedness, palpitation, bradycardia, hypotension, precordial pain, syncope and cardiac arrest.
- Signs suggestive of toxicity in adults and children include nausea, vomiting, abdominal pain, arrhythmias, hyperreflexia, seizures and tachypnea. There may be no prodromal symptoms before the occurrence of severe toxicity.
- Signs suggestive of toxicity in neonates are tachycardia, vomiting, jitteriness, hyperreflexia and seizures.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Use ideal body weight to calculate dose.
- Theophylline serum levels should be monitored and the infusion rate adjusted accordingly. Therapeutic range is 55-110 mcmol/L, except for neonatal apnea where the range is 55-70 mcmol/L.

Adults:
- Loading dose (bronchoconstriction):
  - 6 mg/kg IV in patients who have not received aminophylline (or theophylline) in the previous 24 hours.
  - A partial loading dose of 3 mg/kg IV may be given to increase serum theophylline level by approximately 25-30 mcmol/L OR increasing the dose by 1 mg/kg IV will increase the serum theophylline level by 10 mcmol/L.
- Maintenance dose (bronchoconstriction):
  - 0.5-0.7 mg/kg/hr IV. Increase to 0.9 mg/kg/hr in smokers.
  - Geriatrics (greater than 60 years): 0.375 mg/kg/hr IV.
  - Dosage in hepatic impairment, cardiac decompensation, cor pulmonale, sepsis with multi-organ failure, and shock: 0.25 mg/kg/hr IV.

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Dosage (Cont.)

Pediatrics:
- Use with caution in children less than 1 year of age due to reduced elimination and the potential for toxicity.
- Loading dose:
  - Apnea of prematurity: 5-8 mg/kg/dose IV.
  - Bronchoconstriction: 6 mg/kg IV.
  - A partial loading dose of 3 mg/kg IV may be given for patients already receiving aminophylline or theophylline.
- Maintenance dose:
  - Apnea of prematurity: initial 2-6 mg/kg/day IV divided q8-12h OR 2.5-5 mg/kg/day IV divided q12h OR 4-9 mg/kg/day IV divided q8h.
  - Bronchoconstriction:

Age	Aminophylline IV maintenance dose
Neonates post-natal age of 24 days or less	1.25 mg/kg q12h (intermittent dosing)
Neonates post-natal age greater than 24 days to 6 weeks	1.9 mg/kg q12h (intermittent dosing)
6 weeks to 6 months	0.5 mg/kg/hr
6-12 months	0.6-0.7 mg/kg/hr
1-9 years	1-1.2 mg/kg/hr
9-12 years	0.9 mg/kg/hr
12-16 years	Non smoker: 0.625-7 mg/kg/hr Smoker: 0.9 mg/kg/hr
Adolescent 16 years of age and older	Non smoker: 0.5 mg/kg/hr Smoker: 0.9 mg/kg/hr

  - Dosage in renal impairment for neonates to 3 months of age: reduce dose. For infants greater than 3 months of age, no adjustment is necessary.
  - Dosage in hepatic impairment, cardiac decompensation or cor pulmonale: 0.25 mg/kg/hr IV OR reduce maintenance dose by 50%.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Store ampoules between 20-25°C. Protect from light. Protect from freezing.
- Stable for 48 hours at room temp in D5W and NS at concentrations of 0.5 and 2 mg/mL in PVC bags.
- Dilutions in D5W may develop a yellow discoloration, but aminophylline concentrations remain within acceptable limits.
- Do not refrigerate. May result in precipitation of aminophylline crystals.

MISCELLANEOUS

- Aminophylline is a 2:1 complex of theophylline and ethylenediamine. Aminophylline 25 mg = theophylline 19.7 mg.
- To convert from aminophylline to theophylline, multiply aminophylline dose by 0.8.

REFERENCES

1, 2, 4, 6, 40, 82, 95, 102, 135, 137, 182, 208, 323, 462, 513.

NAME OF MEDICATION
AMIODARONE

OTHER NAMES
Cordarone ®

CLASSIFICATION
Antiarrhythmic

INDICATIONS

- Treatment of documented, life-threatening, frequently recurring ventricular fibrillation and hemodynamically unstable ventricular tachycardia in patients refractory to all other treatments.
- Acute control of supraventricular tachyarrhythmia, including recent-onset atrial fibrillation.
- Cardiac arrest.
- Can be used in patients for whom oral amiodarone is indicated but who are unable to take oral medication.

ADMINISTRATION

- IV direct: physician or RN; **in cardiac arrest only**. Administer undiluted IV direct through a central or peripheral line, followed by 10 mL of D5W or NS.
- Intermittent IV infusion: ***cardiac monitoring***. Dilute dose in 100 mL D5W and administer at a rate of 15 mg/min. Administer using an in-line 0.22 micron filter. Must be administered by an infusion pump.
- Continuous IV infusion: ***cardiac monitoring***. Dilute 900 mg (18 mL from the amiodarone vials or ampoules) in 500 mL of D5W to obtain a final concentration of 1.8 mg/mL; use a glass or polyolefin container; PVC tubing may be used for administration. Concentrations of 1-6 mg/mL can also be used for subsequent maintenance infusions. Administer using an in-line 0.22 micron filter. Refer to Dosage section for administration rate. Must be administered by an infusion pump.
- Due to risk of severe phlebitis, if duration of infusion is 60 minutes or longer, the concentration should not exceed 2 mg/mL unless a central line is used.
- If duration of infusion exceeds 4 hours, a glass or polyolefin container must be used.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: severe cutaneous reactions, including Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Cardiovascular: hypotension, exacerbation of existing arrhythmia or precipitation of new arrhythmia, bradycardia, tachycardia, heart block, QT interval prolongation, heart failure, cardiac arrest, cardiogenic shock.
- GI: nausea, vomiting.
- Hepatic: increased bilirubin, increased liver enzymes.
- Respiratory: adult respiratory distress syndrome.
- Local reactions: pain, phlebitis (refer to Administration section).
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Cardiac arrest: 300 mg given by rapid IV bolus. May give a second dose of 150 mg IV (given over 10 minutes).
- Ventricular arrhythmias: initial dose of 1050 mg IV over the first 24 hours as follows:
  1. 150 mg IV over 10 minutes. Mix 150 mg (3 mL) in 100 mL of D5W (1.5 mg/mL).
  2. 360 mg IV over 6 hours. Administer this dose by using the bag containing 900 mg in 500 mL of D5W (1.8 mg/mL; refer to Administration section); infuse at 1 mg/min (33 mL/hr), then
  3. 540 mg IV over 18 hours. Use the same 1.8 mg/mL solution and infuse at 0.5 mg/min (17 mL/hr). Supplemental doses of 150 mg IV over 10 min may be given for breakthrough arrhythmias. After the initial 24 hours, continue infusion at 0.5 mg/min (17 mL/hr) (720 mg over 24 hours).
- The maintenance infusion rate may be increased to control arrhythmias.
- Maximum cumulative dosage of 2200 mg per 24 hours.
- Dosage in hepatic impairment: no specific recommendations. If LFTs exceed 3 x ULN or double in a patient with elevated baseline LFTs, consider decreasing dose or discontinuing therapy.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules, vials and prefilled syringes between 15-30°C. Protect from light.
- Stable for 24 hours at room temp, protected from light, in D5W at concentrations of 0.6 and 6 mg/mL in glass, polyolefin containers and rigid PVC bottles. Do not use an evacuated glass container since the buffer in these containers may cause precipitation of amiodarone.
- Stable for 4 hours at room temp, protected from light, in D5W at a concentration of 0.6 mg/mL in PVC bags.
- Stable for 24 hours at room temp, protected from light, in D5W at a concentration of 6 mg/mL in PVC bags.
- Stable for 5 days at room temp in D5W at a concentration of 0.6 mg/mL in glass containers.
- Stable for 21 days at room temp and 38 days in the fridge in D5W at a concentration of 2 mg/mL in polyolefin containers.
- Dilution in NS is not recommended due to conflicting stability data.

MISCELLANEOUS

- For conversion from IV to oral amiodarone, the manufacturer recommends an initial oral dose of 800-1600 mg daily for patients on IV amiodarone for less than 1 week, 600-800 mg daily for patients on IV amiodarone for 1-3 weeks, and 400 mg daily when on IV amiodarone for more than 3 weeks.

REFERENCES

1, 2, 4, 5, 40, 95, 135, 143, 183, 208, 322, 323, 462.

NAME OF MEDICATION	
AMOXICILLIN/POTASSIUM CLAVULANATE	
OTHER NAMES	CLASSIFICATION Antibiotic – penicillin with beta-lactamase inhibitor

INDICATIONS

- Treatment of the following infections in adults and children when due to susceptible strains, including beta-lactamase producing bacteria: infections of the ear, nose and throat (e.g., mastoiditis, peritonsillar infections, epiglottitis, and sinusitis), acute exacerbations of chronic bronchitis, community-acquired pneumonia, urinary tract infections, skin and soft tissue infections, bone and joint infections, intra-abdominal infections and female genital infections.
- Prophylaxis of infections associated with major surgeries (e.g., those involving the GI tract, pelvic cavity, head and neck, and biliary tract) in adults.

ADMINISTRATION

- **CHECK ALLERGY STATUS OF PATIENT.**
- Refer to Miscellaneous section for expression of vial content.
- Reconstitute each vial of 500 mg/100 mg, 1000 mg/200 mg, and 2000 mg/200 mg with 10 mL, 20 mL, and 20 mL of SWFI, respectively. After dissolution, solution may turn pink; this effect is transient, and the solution will then rapidly turn clear.
- IV direct: physician or RN. Use **only** the 5:1 ratio doses (500 mg/100 mg and 1000 mg/200 mg). Administer undiluted over 3-4 minutes.
- Intermittent IV infusion (mandatory for children less than 3 months of age): use the 5:1 ratio (500 mg/100 mg and 1000 mg/200 mg) or 10:1 ratio (2000 mg/200 mg). Dilute 500 mg/100 mg in 50 mL and dilute 1000 mg/200 mg or 2000 mg/200 mg in 100 mL of NS (preferred due to longer stability) or RL. Administer over 30 to 40 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, anaphylaxis/anaphylactoid reactions, angioedema, acute generalized exanthematous pustulosis, drug reaction with eosinophilia and systemic symptoms (DRESS), Stevens-Johnson syndrome, toxic epidermal necrolysis.
- GI: diarrhea (especially with higher clavulanic acid doses), nausea, vomiting.
- CNS: seizures in patients with impaired renal function or in those receiving high doses.

DOSAGE

- Ratio of amoxicillin to clavulanic acid is 5:1 or 10:1 (refer to Miscellaneous section for more details); for doses of amoxicillin of 1000 mg or less, use the 5:1 ratio; for doses greater than 1000 mg the 10:1 ratio is preferred to minimize clavulanate exposure.

- <u>Treatment of infections:</u>
  - Adults and children weighing 40 kg or greater:
    - 1000 mg/200 mg IV q8h of the 5:1 ratio (500 mg/100 mg and 1000 mg/200 mg) OR 2000 mg/200 mg IV q8-12h of the 10:1 ratio (2000 mg/200 mg).
    - For infections caused by penicillin-resistant S. *pneumoniae*, a dose of at least 2000 mg/200 mg IV q8-12h is required (2000 mg/200 mg of the 10:1 ratio).
    - Dosage in renal impairment: use **only** the 5:1 ratio (500 mg/100 mg and 1000 mg/200 mg).

CrCl (mL/min)	30-10	less than 10*
Dose (mg/mg)	initially 1000/200 followed by 500/100	initially 1000/200 followed by 500/100
Dosing interval (hr)	12	24

*Hemodialysis: administer q24h with a supplemental dose of 500 mg/100 mg at the end of dialysis.

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DOSAGE (Cont.)

- Children aged 3 months and older and weighing between 4 to 40 kg:
  - 25 mg/5 mg/kg IV q8h (5:1 ratio) OR 50 mg/5 mg/kg IV q8h (10:1 ratio).
  - Dosage in renal impairment: use **only** the 5:1 ratio.

CrCl (mL/min)	30-10	less than 10**
Dose (mg/mg/kg)	25/5	25/5
Dosing interval (hr)	12	24

**Hemodialysis: a supplemental dose of 12.5 mg/2.5 mg/kg IV should be given at the end of dialysis.

- Children aged less than 3 months or weighing less than 4 kg: 25 mg/5 mg/kg IV q12h (5:1 ratio) OR 50 mg/5 mg IV q12h (10:1 ratio).

- Surgical prophylaxis:
  - Adults: 1000 mg/200 mg to 2000 mg/200 mg IV given at the induction of anesthesia for procedures lasting less than 1 hour. For procedures lasting longer than 1 hour, a dose of 1000 mg/200 mg IV may be repeated twice for a total of 3 doses in 24 hours, including the dose at induction.
  - Dosage in renal impairment: administer a single infusion and use **only** the 5:1 ratio (500 mg/100 mg and 1000 mg/200 mg) for patients with a CrCl of less than 30 mL/min.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C.
- Reconstituted solution is stable for 15 minutes at room temp.
- According to the manufacturer, once diluted, stable for 60 minutes at 25°C in NS or for 4 hours at 5°C in NS (for storage at 5°C, the reconstituted solution should be added to pre-refrigerated infusion bags). The infusion should be administered immediately after reaching room temp. However, longer stability data are available (see below).
- Stable for 4 hours at room temp, exposed to light, and for 8 hours in the fridge, protected from light, in NS at concentrations of 10 mg/2 mg/mL and 20 mg/2 mg/mL in polyolefin containers.
- Stable for 60 minutes at 25°C in RL.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Vials contain:
  - Amoxicillin 500 mg and clavulanic acid 100 mg (5:1 ratio), usually expressed as 500 mg/100 mg. Also contains 31.4 mg of sodium and 19.6 mg of potassium.
  - Amoxicillin 1000 mg and clavulanic acid 200 mg (5:1 ratio), usually expressed as 1000 mg/200 mg. Also contains 62.9 mg of sodium and 39.3 mg of potassium.
  - Amoxicillin 2000 mg and clavulanic acid 200 mg (10:1 ratio), usually expressed as 2000 mg/200 mg. Also contains 125.9 mg of sodium and 39.3 mg of potassium.
- Potassium clavulanate is included as a beta-lactamase inhibitor, which extends amoxicillin's spectrum of coverage.

REFERENCES

5, 82,135, 208, 579.

NAME OF MEDICATION
AMPHOTERICIN B

OTHER NAMES
Fungizone ®

CLASSIFICATION
Antifungal

Do NOT confuse amphotericin B with liposomal amphotericin B or with amphotericin B lipid complex. This monograph is specific to AMPHOTERICIN B.

INDICATIONS
- Progressive and potentially systemic fungal infections, including aspergillosis, blastomycosis, candidiasis, coccidioidomycosis, cryptococcosis, histoplasmosis, leishmaniasis, mucormycosis and sporotrichosis.

ADMINISTRATION
- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards section.**
- **Consider prehydration with NS and sodium repletion to reduce nephrotoxicity; refer to Potential Administration Hazards section.**
- Reconstitute 50 mg vial with 10 mL of SWFI without preservatives for a concentration of 5 mg/mL. Shake vial until complete dissolution.
- Intermittent IV infusion: dilute in D5W (with a pH of greater than 4.2) to a maximum concentration of 0.1 mg/mL (i.e., 50 mg or less in 500 mL) when infused into a peripheral vein or to a maximum concentration of 0.25 mg/mL when infused via a central line. Flush the line with D5W before and after administration. Recommended infusion time is 2-6 hours; however, infusion times of 1 to 2 hours have been used in adults and older children with normal renal function. The preferred method of infusion is with an infusion pump.
- Use of an in-line membrane filter during infusion is optional; however, the pore size should be 1 micron or greater.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylactic and anaphylactoid reactions, rash, pruritus, skin exfoliation, toxic epidermal necrolysis, Stevens-Johnson syndrome.
- Infusion-related reactions: occur within 1-3 hours of starting the infusion, especially with the first few doses; fever, shaking chills, headache, anorexia, nausea, vomiting, hypotension, dyspnea, tachypnea. Rapid IV infusion (over less than 60 minutes) is associated with a more severe reaction consisting of hypotension, bronchospasm, hyperkalemia or hypokalemia, arrhythmias and shock, particularly in patients with renal impairment. In case of reaction, reduce rate of infusion and administer antihistamines and/or corticosteroids. Meperidine may be given for chills. **Consider premedication** with an antipyretic (ibuprofen, acetaminophen or ASA), antiemetic, diphenhydramine, hydrocortisone or meperidine 25-50 mg IV given 30 to 60 minutes prior to infusion.
- Cardiovascular: flushing, hypotension, hypertension.
- GI: anorexia, diarrhea, dyspepsia, epigastric pain, nausea, vomiting, stomach cramping.
- Electrolyte disturbances: hypokalemia, hypomagnesemia, hypocalcemia.
- Musculoskeletal: myalgia, arthralgia.
- Renal: tubular necrosis, decreased glomerular filtration rate, azotemia, nephrocalcinosis. Permanent renal damage possible when cumulative dose exceeds 5 g. **Consider prehydration** with 500-1000 mL of NS and sodium repletion prior to administration of amphotericin B to reduce risk of nephrotoxicity.
- Local reactions: venous pain at injection site with or without phlebitis and thrombophlebitis (use pediatric scalp-vein needles, decrease infusion concentration, change injection site, switch to alternate day therapy, add heparin 500-1000 units or 1 unit/mL to the infusion); extravasation can cause chemical irritation.

DOSAGE
Adults:
- A test dose of 1 mg IV (1 mg in 20 mL of D5W) over 20-30 minutes is sometimes recommended followed by a 2-4 hour observation period.
- Initial dose: 0.25-0.3 mg/kg IV q24h OR 5-10 mg/day in patients with impaired cardiac or renal function or in those who reacted to test dose. Gradually increase in increments of 5-10 mg/day as tolerance permits, up to a maximum of 1.5 mg/kg IV q24h. Alternate day therapy of up to 1.5 mg/kg IV every other day is also possible.
- If therapy is interrupted for longer than 7 days, restart at 0.25 mg/kg/day and increase dose gradually.

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DOSAGE (Cont.)

Pediatrics:
- A test dose of 0.1 mg/kg (maximum 1 mg) IV (in 20 mL of D5W) over 20-60 minutes is sometimes recommended.
- Initial dose: 0.25-0.5 mg/kg IV q24h, increase gradually in increments of 0.25-0.5 mg/kg/day IV until the desired dose is achieved; maximum of 1.5 mg/kg/day IV.
- Maintenance dose: 0.25-1 mg/kg IV q24h OR 1-1.5 mg/kg/day IV every other day. Rapidly progressing disease may require short-term doses of up to 1.5 mg/kg/day IV.

- Dosage in renal impairment: If CrCl is less than 10 mL/min, give full dose q24-36h; if renal impairment is due to amphotericin B, may decrease dose by 50% or give every other day.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Also stable for 2-4 weeks at room temp, protected from light.
- Reconstituted solution is stable for 24 hours at room temp (protected from light) and for 7 days in the fridge.
- Incompatible with NS, RL or SWFI that contains preservatives as diluents other than D5W may cause precipitation.
- Stable for 24 hours at room temp in D5W at a concentration of 0.1 mg/mL in PVC containers.
- Stable for 35 days in the fridge (protected from light) in D5W at concentrations of 0.1 mg/mL and 0.25 mg/mL in PVC containers.
- Stable for 5 days at room temp (exposed to light) or in the fridge (protected from light) in D5W at concentrations of 0.2 mg/mL, 0.5 mg/mL and 1 mg/mL in PVC containers.
- Although the manufacturer recommends light protection for aqueous solutions of amphotericin B, several reports indicate that short-term exposure to light for 8 to 24 hours results in little difference in potency compared to light protection.

MISCELLANEOUS

- Monitor renal function, liver function, electrolytes, blood counts and hemoglobin.

REFERENCES

1, 4, 5, 40, 95, 102, 135, 143, 182, 208, 366, 462.

NAME OF MEDICATION
AMPHOTERICIN B LIPID COMPLEX

OTHER NAMES
Abelcet ®

CLASSIFICATION
Antifungal

Do NOT confuse amphotericin B lipid complex with amphotericin B or with liposomal amphotericin B. This monograph is specific to amphotericin B LIPID COMPLEX.

INDICATIONS

- Treatment of invasive fungal infections in patients who are refractory to or intolerant of conventional amphotericin B therapy.

ADMINISTRATION

- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards section.**
- Shake vial gently until no evidence of yellow sediment at the bottom.
- Intermittent IV infusion: transfer dose using 5-micron filter needle supplied with each vial. Each filter needle may be used to filter up to 4 vials. Dilute in D5W to a concentration of 1 to 2 mg/mL. Flush the line with D5W before and after administration. Administer at a rate of 2.5 mg/kg/hr; invert infusion container several times before administration and every 2 hours if infusion time exceeds 2 hours.
- Do NOT use an in-line filter.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, anaphylactoid reactions, rash.
- Infusion-related reactions: less common than with conventional amphotericin B; fever, shaking chills/rigors, nausea, vomiting, anorexia, headache, dyspnea, tachypnea, vasodilation, hypotension, hypertension, tachycardia, rarely bronchospasm, arrhythmias, shock; occur 1-3 hours after starting infusion; more common with first few doses. For patients experiencing infusion-related reactions, **consider premedication** with a NSAID +/- diphenhydramine or acetaminophen and diphenhydramine or hydrocortisone given 30 to 60 minutes prior to infusion. Meperidine may be given for rigors.
- Cardiovascular: hypotension, hypertension, chest pain, cardiac arrest.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: headache.
- Electrolyte disturbances: hypokalemia, hypomagnesemia, hypocalcemia.
- Hematologic: anemia, leukopenia, thrombocytopenia.
- Hepatic: hyperbilirubinemia.
- Renal: nephrotoxicity: less than with conventional amphotericin B; increase in serum creatinine and renal failure; dose limiting renal toxicity may be observed.
- Respiratory: dyspnea, respiratory failure.
- Infection.
- Multiorgan failure.
- Local reactions: erythema, pain, inflammation at injection site.

DOSAGE

Adults:
- 3-5 mg/kg IV once daily.

Pediatrics:
- 2.5-5 mg/kg IV once daily.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2 to 8°C. Protect from light. Do not freeze.
- Solution is yellow and opaque in appearance.
- Stable for 48 hours in the fridge and for an additional 6 hours at room temp in D5W at concentrations of 1-2 mg/mL.
- Stable for 10 days at room temp in D5W at a concentration of 1 mg/mL.
- Incompatible with NS.

MISCELLANEOUS

- Monitor renal function, liver function, electrolytes, blood counts and hemoglobin.

REFERENCES

1, 5, 40, 102, 135, 208.

NAME OF MEDICATION
AMPHOTERICIN B (LIPOSOMAL)

OTHER NAMES
Ambisome ®

CLASSIFICATION
Antifungal

Do NOT confuse liposomal amphotericin B with amphotericin B or with amphotericin B lipid complex. This monograph is specific to LIPOSOMAL amphotericin B.

INDICATIONS

- Treatment of systemic or disseminated infections due to *Candida*, *Aspergillus*, or *Cryptococcus* in patients refractory to or intolerant of conventional amphotericin B, or in those with renal impairment.
- Empiric therapy for fungal infections in febrile neutropenic patients.
- Treatment of cryptococcal meningitis in HIV-infected patients.

ADMINISTRATION

- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards section.**
- Reconstitute 50 mg vial with 12 mL of SWFI (without a bacteriostatic agent) to obtain a concentration of 4 mg/mL; shake vial vigorously for 30 seconds to completely disperse the drug.
- Intermittent IV infusion: transfer dose using the 5 micron filter supplied with each vial and dilute in D5W to a concentration of 0.5 to 2 mg/mL. A lower concentration of 0.2 to 0.5 mg/mL may be used in infants and small children. Flush the line with D5W before and after administration. Infuse over 1 or 2 hours, depending on patient's tolerance.
- Use of an in-line filter during infusion is optional; however, pore size should be at least 1 micron or greater.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, anaphylactoid reactions, angioedema, urticaria, rash, pruritus.
- Infusion-related reactions: less than with conventional amphotericin B; fever, shaking chills/rigors, nausea, vomiting, hypertension, hypotension, tachycardia, dyspnea, tachypnea, hypoxia; occur 1 to 3 hours after starting infusion; more common with first few doses. For patients experiencing infusion-related reactions, **consider premedication** with a NSAID (e.g., ibuprofen) +/- diphenhydramine or acetaminophen with diphenhydramine or hydrocortisone given 30 to 60 minutes prior to infusion. Meperidine may be given for rigors.
- Cardiovascular: hypotension, hypertension, tachycardia, chest pain, edema.
- GI: nausea, vomiting, diarrhea, abdominal pain, constipation, anorexia, GI hemorrhage.
- CNS: headache, dizziness, confusion.
- Electrolyte disturbances: hypokalemia, hypomagnesemia, hypocalcemia, hyponatremia.
- Endocrine and metabolic: hyperglycemia, hypervolemia.
- Hematologic: anemia, leukopenia, thrombocytopenia.
- Hepatic: hyperbilirubinemia, increased LFTs.
- Musculoskeletal: back pain.
- Renal: nephrotoxicity: less than with conventional amphotericin B; increase in serum creatinine, hematuria.
- Respiratory: dyspnea, cough, epistaxis, rhinitis, pleural effusion.
- Infection.
- Local reactions: erythema, pain, inflammation, phlebitis at injection site.

DOSAGE

Adults:
- Usual range: 3 to 6 mg/kg IV daily.
- Doses up to 7.5 mg/kg/day IV have been used for severe CNS infections.

Pediatrics:
- Usual range: 3 to 5 mg/kg IV daily.

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-25°C.
- Incompatible with NS.
- Reconstituted solution (4 mg/mL) is stable for 14 days in the fridge (protected from light) in the original vials or in polypropylene syringes.
- Stable for 14 days in the fridge (protected from light) and 24 hours at room temp (exposed to light) in D5W at a concentration of 2 mg/mL in polyolefin containers.
- Stable for 11 days in the fridge (protected from light) and 24 hours at room temp (exposed to light) in D5W at a concentration of 0.2 mg/mL in polyolefin containers.
- Stable for 14 days in the fridge (protected from light) in D5W at concentrations of 0.5 mg/mL and 2 mg/mL in PVC containers.
- Stable for 11 days in the fridge (protected from light) in D5W at a concentration of 0.2 mg/mL in PVC containers.

MISCELLANEOUS

- Each 50 mg vial contains approximately 900 mg of sucrose.
- Monitor renal function, liver function, electrolytes, blood counts, and hemoglobin.

REFERENCES

1, 5, 40, 95, 102, 135, 143, 182, 465.

NAME OF MEDICATION
AMPICILLIN

OTHER NAMES
Aminobenzylpenicillin, Ampicin ®

CLASSIFICATION
Antibiotic - penicillin

INDICATIONS

- For moderate or severe infections due to organisms susceptible to ampicillin.
- Prevention of bacterial endocarditis associated with dental or respiratory tract procedures/surgery, where oral administration is not possible.
- Prevention of neonatal Group B streptococcal (GBS) disease, as an alternative to penicillin G.

ADMINISTRATION

- **CHECK ALLERGY STATUS OF PATIENT**.
- For IV use: reconstitute each vial with SWFI as per manufacturer's recommendation.
- IV direct: physician or RN. Concentration should not exceed 100 mg/mL. Administer doses up to 500 mg slowly over 3-5 minutes, and doses greater than 500 mg over 10-15 minutes, at a maximum rate of 100 mg/min.
- Intermittent IV infusion: dilute in 50-100 mL of NS to obtain a final concentration not exceeding 30 mg/mL; administer over 15-30 minutes, not exceeding 100 mg/min.
- Continuous IV infusion: the total daily dose may be administered as a 24-hour continuous infusion.
- IM: reconstitute each vial with SWFI as per manufacturer's recommendation. Inject deep into a large muscle mass.
- Consult Medical Staff Rules and Regulations MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis) for specific dosing regimens and infusion time.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: immediate (anaphylaxis) or delayed (rash, urticaria), erythema multiforme, exfoliative dermatitis.
- CNS: seizures if IV administration of large doses is too rapid.
- Dermatologic: rash frequently occurs a few days after starting ampicillin therapy in patients with a viral infection, *Salmonella* infection, lymphocytic leukemia or those who have hyperuricemia or taking allopurinol; in most cases, rash appears to be nonimmunologic.
- Hematologic: rare; anemia, leukopenia, eosinophilia, thrombocytopenia, agranulocytosis; thought to be hypersensitivity reactions.
- Hepatic: elevated AST with repeated IM injections of large doses; not indicative of liver impairment.
- Local reactions: pain (IM injection); phlebitis (IV injection).

DOSAGE

- Adults:
  - Treatment: Moderate infections: 250-2000 mg IV/IM q6h.
    Severe infections: 8-14 g/day IV/IM OR 150-200 mg/kg/day IV/IM, in divided doses q3-4h.
  - Prevention of bacterial endocarditis: 2 g IV/IM 30-60 minutes before skin incision or procedure.
  - Prevention of neonatal GBS disease: 2 g IV at onset of labour, then 1 g IV q4h intrapartum.

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DOSAGE (Cont.)

- Pediatrics:
  - Treatment: Mild to moderate infections: 25-200 mg/kg/day IV/IM, in divided doses q6h. Maximum: 8 g/day.
    Severe infections: 200-400 mg/kg/day IV/IM, in divided doses q4-6h. Maximum: 12 g/day.
  - Prevention of bacterial endocarditis: 50 mg/kg IV/IM 30-60 minutes before skin incision or procedure. Maximum: 2 g.

- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	Usual	Usual	Usual
Interval (hr):	6	6-12	12-24*

* Drug is removed by hemodialysis.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution is stable for 60 minutes at room temp and for 4 hours in the fridge.
- Stable for 24 hours at room temp or for 4 days in the fridge in NS at a concentration of 5 mg/mL in polyolefin containers.
- Stable for 24 hours in the fridge in NS at a concentration of 5 mg/mL in PVC containers.
- Stable for 24 hours at room temp or 3 days in the fridge in NS at a concentration of 10 mg/mL in ethylene vinyl acetate (EVA) containers.
- Stable for 48 hours in the fridge in NS at a concentration of 10 mg/mL in PVC containers.
- Stable for 24 hours at room temp in NS at a concentration of 12 mg/mL.
- Stable for 4 days in the fridge in NS at a concentration of 20 mg/mL in PVC containers.
- Stable for 8 hours at room temp in NS at concentration of 30 mg/mL.

MISCELLANEOUS

- There are approximately 3 mmol of sodium per gram of injectable ampicillin.
- Monitor renal, hepatic and hematologic functions during prolonged therapy.

REFERENCES

1, 2, 4, 5, 30, 40, 82, 91, 95, 135, 208, 216, 338.

NAME OF MEDICATION
ANIDULAFUNGIN

OTHER NAMES
Eraxis ®

CLASSIFICATION
Antifungal

INDICATIONS

- Treatment of invasive candidiasis/candidemia.

ADMINISTRATION

- Reconstitute each 100 mg vial with 30 mL of SWFI to obtain a concentration of 3.33 mg/mL. It may take up to 5 minutes to dissolve vial contents.
- Intermittent IV infusion: transfer contents of 1 or 2 reconstituted vials into 100 mL or 200 mL, respectively, of D5W or NS to obtain a final concentration of 0.77 mg/mL. Administer slowly at a rate not exceeding 1.1 mg/min (i.e., 1.4 mL/min).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis.
- Infusion-related reactions: histamine-mediated reactions (e.g., dyspnea, bronchospasm, flushing, hypotension, pruritus, rash, urticaria). Infrequent when infusion rate does not exceed 1.1 mg/min.
- Cardiovascular: hypotension, hypertension, peripheral edema, deep vein thrombosis.
- GI: nausea, vomitting, diarrhea.
- CNS: insomnia.
- Hepatic: increases in ALT, AST and alkaline phosphatase; hepatitis, hepatic failure.
- Electrolyte disturbances: hypokalemia, hypomagnesemia.
- Uninary tract infection.

DOSAGE

- Usual dose: loading dose: 200 mg IV on day 1.
  maintenance dose: 100 mg IV once daily, starting on day 2.
- Intravascular candidiasis (i.e., endocarditis, implantable cardiac device infection and suppurative thrombophlebitis): 200 mg IV once daily.
- Osteoarticular candidiasis (i.e., osteomyelitis and septic arthritis): 100 mg IV once daily.
- Esophageal candidiasis: 200 mg IV once daily OR 100 mg IV loading dose on day 1 followed by 50 mg IV once daily (note: 50 mg vials are not available in Canada).
- Dosage in renal impairment: no dosage adjustment required. Anidulafungin is not removed by hemodialysis.
- Dosage in hepatic impairment: no dosage adjustment required.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Excursions permitted for 96 hours up 25°C. Do not freeze.
- Reconstituted solution is stable for 24 hours between 15-30°C.
- Stable for 48 hours between 15-30°C or for 72 hours when frozen in NS or D5W at a concentration of 0.77 mg/mL.

MISCELLANEOUS

- Safety and efficacy have not been established for the treatment of meningitis caused by *Candida*.
- Contains fructose.

REFERENCES

1, 5, 95, 116, 135.

Full revision 2020

NAME OF MEDICATION
ANTI-INHIBITOR COAGULANT COMPLEX

OTHER NAMES
Activated prothrombin complex concentrate, Feiba NF ®

CLASSIFICATION
Coagulation factor

INDICATIONS

- Prevention and control of hemorrhagic episodes in patients with hemophilia A or B who have developed inhibitor antibodies to factor VIII or IX.
- Control of life-threatening bleeding in patients who have acquired inhibitor antibodies to factors VIII, XI or XII.
- Reversal of apixaban, dabigatran (if idarucizumab is not available), edoxaban or rivaroxaban in case of urgent surgery/procedure or life-threatening bleeding.
- Reversal of dabigatran activity in case of emergency surgery/procedure if idarucizumab is not available.

ADMINISTRATION

- For IV use: warm diluent and lyophilized powder to room temp. Refer to manufacturer's instructions for reconstitution and preparation of the infusion.
- Intermittent IV infusion: infuse at a rate not exceeding 2 units/kg/min. Flush IV line with NS before and after administration.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0048 (Blood products and blood derivatives: Coagulation factor concentrate – Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, angioedema, GI manifestations, bronchospasm, hypotension, anaphylaxis.
- Infusion-related reactions: fever, chills, dizziness, nausea, hypertension. With rapid infusion, stabbing pain and numbness in the face and extremities with hypotension.
- Hematologic: thromboembolic complications, including disseminated intravascular coagulation (rare), especially following high doses or in patients with thrombotic risk factors. If any sign or symptom of thromboembolic complication occurs (blood pressure and pulse rate changes, respiratory distress, cough, chest pain), discontinue the infusion and investigate patient. Thrombotic microangiopathy has been reported in patients concomitantly treated with emicizumab.
- Remote risk of transmission of infectious agents, including viruses, as this product is prepared from large pools of human plasma.
- Local reactions: pain at injection site.

DOSAGE

- Hemophilia A or B: 50-100 units/kg/dose IV, **not exceeding 100 units/kg/dose and 200 units/kg/day for all bleeding conditions listed below**. Refer below for more specific dosing; to continue treatment until bleeding is controlled.
  - Joint hemorrhage: 50-75 units/kg IV q12h; may increase to 100 units/kg IV q12h if needed.
  - Mucous membrane bleeding: 50 units/kg IV q6h; if dose is increased, prolong the interval to ensure that the maximum daily dose of 200 units/kg/day is not exceeded.
  - Soft tissue hemorrhage: 100 units/kg IV q12h.
  - Other severe hemorrhage: 100 units/kg IV q12h; may be given q6h prn but at a lower dose to ensure that the maximum daily dose of 200 units/kg/day is not exceeded.
  - Surgery: 50-100 units/kg IV q6-12h, the first dose administered immediately prior to surgery (maximum: 200 units/kg/day).
  - Routine prophylaxis: 85 units/kg (range: 70-100 units/kg) IV every other day or 3-4 times per week.
- Hemorrhage in patients with acquired inhibitor antibodies to factors VIII, XI or XII: 50-100 units/kg IV q8-12h (maximum: 200 units/kg/day).
- Reversal of apixaban, dabigatran, edoxaban or rivaroxaban activity: single dose of 50 units/kg IV (maximum of 2000 units).

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and diluent between 2-25°C. Protect from light. Do not freeze.
- Reconstituted solution is stable at room temp for 3 hours. Do not refrigerate.

MISCELLANEOUS

- Emergency drugs (epinephrine, antihistamines and corticosteroids) should be immediately available for the treatment of hypersensitivity reactions.
- Systemic antifibrinolytics, such as aminocaproic acid or tranexamic acid, should not be administered within 6-12 hours after the anti-inhibitor coagulant complex due to the risk of thromboembolic events.
- Feiba NF ® contains factors II, IX and X, mainly non-activated, factor VII mainly activated, factor VIII inhibitor bypassing activity, factor VIII coagulant antigen and Prothrombin Complex Factors.
- Febia NF ® contains approximately 3.5 mmol (80 mg) of sodium per 1000 units.
- Feiba NF ® does not contain heparin.

REFERENCES

1, 5, 132, 135, 141.

NAME OF MEDICATION
ANTITHROMBIN III HUMAN

OTHER NAMES
Antithrombin III NF ®

CLASSIFICATION
Anticoagulant

INDICATIONS

- Prophylaxis and treatment of thrombotic and thromboembolic disorders in patients with hereditary antithrombin III (ATIII) deficiency (antithrombin III activity below 70% of normal); particularly valuable during surgery or pregnancy and delivery.

ADMINISTRATION

- Allow diluent to reach room temp before reconstitution.
- Reconstitute the 450-550 and the 900-1100 international unit vials with 10 mL and 20 mL, respectively, of the provided diluent (SWFI) as per manufacturer's instructions. Use the provided double-ended needle. Do NOT administer if the solution is cloudy or contains deposits.
- IV direct: physician or RN. Use the provided filter needle to withdraw vial solution then replace it with a needle appropriate for the IV injection; administer at a maximum rate of 5 mL/min.
- Intermittent IV infusion: administer undiluted at a maximum rate of 5 mL/min with an administration set that contains a filter (range of 5-149 microns).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactic/anaphylactoid reactions, shock, wheezing, urticaria, chills, fever.
- Cardiovascular: flushing.
- GI: nausea, retching; may be signs of hypersensitivity reactions.
- CNS: dizziness, tremor.
- Hematologic: hemorrhage, especially in combination with heparin.
- Risk of transmission of infectious agents, including viruses, and theoretically, the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.

DOSAGE

- Disseminated intravascular coagulation (DIC):
  - Initial dose (international units) = [desired ATIII activity (%) - baseline ATIII activity (%)] x body weight (kg) divided by 1%.
  - Maintenance dose: when ATIII activity is less than 70%. It is calculated using the formula stated above, except that the 1% is substituted instead, with the actual increase in ATIII activity (in %) produced by 1 international unit per kg of body weight, as determined by the measurement of ATIII activity following the administration of the initial dose.
- Other antithrombin III defects:
  - Method 1:
    - Initial dose: 1500 international units;
    - Maintenance dose: half the initial dose every 8-24 hours.
  - Method 2:
    - Initial dose (international units) = [desired ATIII activity (%) - baseline ATIII activity (%)] x body weight (kg) divided by 2%.
    - Maintenance dose is also calculated using the formula stated above, except that the 2% is substituted instead, with the actual increase in ATIII activity (in %) produced by 1 international unit per kg of body weight, as determined by the measurement of ATIII activity following the administration of the initial dose.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Do not freeze.
- Use the reconstituted solutions immediately; do not refrigerate once reconstituted.

MISCELLANEOUS

- This monograph is specific to the Antithrombin III NF ® brand from Shire Canada.
- Each international unit of Antithrombin III NF ® can contain up to 0.1 international unit of heparin.
- Biological half-life is approximately 2.5 days in patients with congenital antithrombin III deficiency; however, in DIC, the half-life can be reduced to a few hours. It is also reduced when product is administered with heparin due to accelerated antithrombin turnover.

REFERENCES

1, 5, 95, 135.

NAME OF MEDICATION
ANTITHYMOCYTE GLOBULIN, EQUINE

OTHER NAMES
Antithymocyte immunoglobulin (equine), ATG, Atgam ®, Equine IgG, Horse ATG

CLASSIFICATION
Immunosuppressant

Do NOT confuse equine antithymocyte globulin with rabbit antithymocyte globulin.
This monograph is specific to EQUINE antithymocyte globulin.

INDICATIONS

- Prevention or treatment of acute allograft rejection (kidney, skin, cardiac, bone marrow).
- Treatment of aplastic anemia, lymphoma, agranulocytosis.
- Myeloablative autologous stem-cell transplantation (ASCT) for severe scleroderma.

ADMINISTRATION

- **Consider premedication to prevent febrile reactions; refer to Potential Administration Hazards section.**
- Intradermal sensitivity testing is strongly recommended prior to administration of the initial dose. Refer to manufacturer's monograph for the full procedure.
- Intermittent IV infusion: dilute in 250-1000 mL of a compatible diluent, not exceeding a concentration of 4 mg/mL. Contact of undiluted antithymocyte globulin with air may denature protein; therefore IV solution container should be inverted when antithymocyte globulin is added to prevent contact with air inside the container. Gently mix. Do NOT shake solution since excessive foaming and/or denaturation of the protein may occur. Infuse through a 0.2 to 1 micron in-line filter over at least 4 hours into a large, high-flow central vein.
- Observe patient during infusion for hypersensitivity reactions.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, rash, erythema, urticaria, wheal and flare, pruritus, toxic epidermal necrolysis, serum sickness-like reactions.
- Infusion-related reactions: febrile reaction, chills; febrile reaction tends to decrease in severity after the first few doses of the drug. To prevent or minimize febrile reactions, **premedicate** with an antipyretic (e.g., acetaminophen 650 mg PO), antihistamine (e.g., diphenhydramine 25-50 mg PO) and/or corticosteroid (e.g., hydrocortisone 25-50 mg IV).
- Hematologic: leukopenia and/or thrombocytopenia, hemolysis (rare).
- Infections.
- Risk of transmission of infectious agents, including viruses, and theoratically the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.
- Local reactions: pain, swelling, redness at infusion site, thrombophlebitis, clotted fistulas or shunts.

DOSAGE

- Expressed in terms of <u>equine</u> antithymocyte globulin.
- Renal allograft: 10-30 mg/kg/day IV for 14 days; may be followed by alternate day therapy for a total of 21 doses in 28 days. The first dose may be given within 24 hours before or after transplantation. Alternatively, the first dose may be delayed until the diagnosis of rejection.
- Aplastic anemia: 10-20 mg/kg/day IV for 8-14 days; maybe followed by alternate day therapy for a total of 21 doses in 28 days.
- Bone marrow allograft: 7-20 mg/kg IV for 3-14 doses.
- Heart transplant: 5-11 mg/kg/day IV administered intermittently.
- Skin transplant: 5-40 mg/kg IV administered ever other day or daily.
- Myeloablative ASCT for severe scleroderma: 15 mg/kg IV for 6 doses administered on days -5, -3, -1, +1, +3,+5 (day 0 being the day of the stem cell transplant).

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 2-8ºC. Protect from light. Do not freeze. Do not shake. The solution is transparent to slightly opalescent, colourless to faintly pink or brown and may contain granular or flaky deposits during storage.
- Stable for 24 hours in the fridge in NS, D5-1/2NS, D5-1/4NS at concentrations of up to 4 mg/mL.
- Stable for 24 hours in the fridge in 1/2NS or NS at a concentration of 1 mg/mL.
- Not stable in D5W since low salt concentration may lead to precipitation.
- Manufacturer recommends that solution should not be used after a period of 24 hours from time of dilution (including administration time).

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.

REFERENCES

1, 5, 95, 135, 508, 509.

NAME OF MEDICATION
ANTITHYMOCYTE GLOBULIN, RABBIT

OTHER NAMES
Antithymocyte immunoglobulin (rabbit), Rabbit ATG, Rabbit IgG, rATG, Thymoglobulin ®

CLASSIFICATION
Immunosuppressant

Do NOT confuse rabbit antithymocyte globulin with equine antithymocyte globulin.
This monograph is specific to RABBIT antithymocyte globulin.

INDICATIONS

- Prevention or treatment of acute allograft rejection (kidney, heart, bone marrow).

ADMINISTRATION

- **Consider premedication to prevent febrile reactions; refer to Potential Administration Hazards section.**
- Allow vials to reach room temperature before reconstitution. Reconstitute each 25 mg vial with 5 mL of SWFI. Rotate vial gently for complete dissolution. Do NOT shake.
- Intermittent IV infusion:
  - Central administration: dilute in a volume of NS or D5W corresponding to 50 mL per 25 mg of product (total volume usually 50-500 mL at a final concentration of approximately 0.5 mg/mL). Mix solution by inverting the bag gently once or twice. Infuse in a high-flow vein (central venous cathether) through a 0.22 micron in-line filter over a minimum of 6 hours for the first dose and at least 4 hours for subsequent doses.
  - Peripheral administration: if a high-flow vein is not available and dose needs to be administered, dilute product into 500 mL of NS (not D5W) and add 1000 units of heparin and 20 mg of hydrocortisone to the bag in order to decrease risk of thrombophlebitis and deep vein thrombosis. Mix solution by inverting the bag gently once or twice. Infuse through a 0.22 micron in-line filter over a minimum of 6 hours for the first and second doses and at least 4 hours for subsequent doses.
- Observe patient closely during infusion.
- Refer to TOH Renal Transplant Protocol if needed.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: cytokine release syndrome (hypotension, tachycardia, myocardial infarction, respiratory distress, pulmonary edema); anaphylaxis; serum sickness (fever, rash, urticaria, arthralgia and/or myalgia).
- Infusion-related reactions: fever, chills; can be minimized with slow infusion rates and **premedication** with acetaminophen (500-650 mg PO), antihistamines (e.g., diphenhydramine 25 mg IV or PO) and/or corticosteroids (e.g., methylprednisolone 2-5 mg/kg IV) 60 minutes prior to each infusion.
- Cardiovascular: hypotension, hypertension, tachycardia, peripheral edema.
- GI: nausea, diarrhea, abdominal pain.
- CNS: headache, pain.
- Electrolyte disturbances: hyperkalemia.
- Hematologic: leukopenia, thrombocytopenia.
- Infections.
- Risk of transmission of infectious agents, including viruses, and theoretically the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from human blood components.
- Local reactions: pain, swelling, erythema at injection site.

DOSAGE

- Expressed in terms of rabbit antithymocyte globulin.
- Renal allograft:
  - Prevention of rejection: 1.5 mg/kg/day IV for 4-7 days OR 1 mg/kg/day IV for 3-6 days OR 2 mg/kg/day IV for 3 days; the first dose to be administered intraoperatively before reperfusion of donor kidney.
  - Treatment of rejection: 1.5 mg/kg/day IV for 7-14 days.
  - Dosage adjustment for leukopenia and/or thrombocytopenia:

White Blood Cell (WBC) count (cells/mm^3)	Platelet count (cells/mm^3)	Dose reduction
2000-3000	50 000-75 000	50%
less than 2000	less than 50 000	avoid

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DOSAGE (Cont.)

- Bone marrow transplant (prevention of graft-versus-host disease): 0.5 mg/kg IV administered 2 days prior to transplant, then 2 mg/kg IV 1 day prior and 1 day after transplant OR 2.5 mg/kg/day IV for 3 days before transplant.
- Heart transplant:
  - Prevention of rejection: 1-1.5 mg/kg/day IV for up to 7 days.
  - Treatment of rejection: 0.75-1.5 mg/kg/day IV for 5-14 days.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Temperature excursion to up to 37°C for 10 days is permitted.
- Reconstituted solution is stable for up to 24 hours at room temp, but best to use it immediately as it does not contain a preservative.
- Stable for 24 hours at room temp in NS or D5W at a concentration of 0.5 mg/mL.
- Stable for 24 hours at room temp exposed to light in NS at concentrations of 0.2 mg/mL and 0.3 mg/mL mixed with heparin at 2 units/mL and hydrocortisone at 0.05 mg/mL in PVC bags; the same mixture precipitates in D5W.

MISCELLANEOUS

- Do not use in patients allergic to rabbit protein.
- Monitor WBC count (including lymphocytes), and platelets.
- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.

REFERENCES

1, 5, 40, 95, 135, 208, 288, 289.

NAME OF MEDICATION
APOMORPHINE

OTHER NAMES
Apokyn ®, Movapo ™

CLASSIFICATION
Antiparkinsonian

Do NOT confuse with morphine. Apomorphine lacks opioid analgesic properties. This monograph is specific to APOMORPHINE.

INDICATIONS

- Acute, intermittent treatment of hypomobility ("off" episodes) in patients with advanced Parkinson's disease as an adjunct to oral antiparkinsonian agents.

ADMINISTRATION

- SC: ***blood pressure monitoring*** only on first time dose or dose increase; upper arms, thighs, or abdomen. Rotate injection site.
- **FOR SC USE ONLY.** Crystallization with thrombus formation and pulmonary embolism if administered IV.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, angioedema, bronchoconstriction (due to presence of sulfites).
- Cardiovascular: orthostatic hypotension, syncope, edema, flushing, angina, myocardial infarction, cardiac arrest.
- GI: nausea, vomiting (initiate domperidone as an antiemetic at least 2 days prior to starting apomorphine; avoid $5HT_3$ antagonists [e.g., ondansetron, granisetron, palonosetron] due to additive hypotension).
- CNS: sudden onset of sleep, drowsiness, dizziness, dyskinesia, hallucinations, confusion, headache.
- Dermatologic: sweating, pallor.
- Respiratory: rhinorrhea.
- Yawning, priapism.
- Local reactions: bruising, granuloma, pruritus, induration, erythema, pain.

DOSAGE

- Initial dose (also called test dose): 2 mg (0.2 mL) SC; may titrate up to 6 mg (0.6 mL) prn up to 5 times per day, with at least 2 hours between doses. Maximum total daily dose of 20 mg (2 mL).
- If therapy is interrupted for more than 1 week, resume treatment with a dose of 2 mg (0.2 mL) SC and re-titrate dose according to efficacy and tolerability.
- Dosage in renal impairment: reduce initial dose to 1 mg (0.1 mL) SC in mild to moderate impairment; contraindicated in severe renal impairment.
- Dosage in hepatic impairment: no dosage adjustment is recommended in mild to moderate impairment but use with caution; contraindicated in severe hepatic impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled multi-dose pens between 15-30°C. Protect unopened pens from light.
- Opened prefilled pens can be stored at room temp and be used for a maximum of 48 hours.

MISCELLANEOUS

- Contains sulfites.
- Onset of action is 10-20 minutes. Duration of action is about 60-90 minutes.

REFERENCES

1, 5, 135, 328.

Full revision 2020

NAME OF MEDICATION	
ARGATROBAN	
OTHER NAMES	CLASSIFICATION
	Anticoagulant

INDICATIONS

- Prophylaxis or treatment of thrombosis in patients with heparin-induced thrombocytopenia (HIT).
- Anticoagulation of patients with or at risk of HIT who are undergoing percutaneous coronary intervention (PCI).

ADMINISTRATION

- For IV use: dilute 250 mg (2.5 mL) or 500 mg (5 mL) from vials in 250 mL or 500 mL, respectively, of NS, D5W or RL to obtain a final concentration of 1 mg/mL. Invert the diluent bag repeatedly for one minute to mix solution. The slight haziness that may appear upon the preparation should disappear rapidly upon mixing.
- IV direct: physician or RN; using the 1 mg/mL solution, administer loading and bolus doses over 3 to 5 minutes in a large bore IV line.
- Continuous IV infusion: using the 1 mg/mL solution, administer as per Dosage section for rate.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: mostly in patients receiving thrombolytic agents and/or contrast material; airway reactions, skin eruptions; anaphylaxis.
- Cardiovascular: hypotension, cardiac arrest, ventricular tachycardia, atrial fibrillation, bradycaria.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- Hematologic: bleeding at any site (minor or major).
- Respiratory: dyspnea, cough.
- Fever.

DOSAGE

- Prophylaxis or treatment of thrombosis:
  - Initial rate of 2 mcg/kg/min IV, not to exceed 10 mcg/kg/min; adjust rate to target steady state aPTT of 1.5-3 times the initial baseline value (not to exceed 100 seconds).
  - Critically ill patients with multiple organ dysfunction: lower rates (e.g., 0.2-1.2 mcg/kg/min IV) may be required.
  - Dosage in hepatic impairment:
    - mild impairment: infusion rate adjustments may be required; use with caution.
    - moderate impairment: initial infusion rate of 0.5 mcg/kg/min IV and adjust rate carefully as aPTT level may take longer to achieve steady-state.
    - severe impairment: further reduce infusion rate; maximum initial rate of 0.05 mcg/kg/min IV; use with caution.
- Anticoagulation of patients undergoing PCI:
  - Loading dose of 350 mcg/kg IV direct, followed by a continuous IV infusion of 25 mcg/kg/min. Five to 10 minutes after completion of loading dose, check the activated clotting time (ACT); if ACT is above 300 seconds, proceed with PCI. If ACT is less than 300 seconds, give a bolus dose of 150 mcg/kg and increase the infusion rate to 30 mcg/kg/min and recheck the ACT in 5-10 minutes. If ACT is greater than 450 seconds, decrease the infusion rate to 15 mcg/kg/min and recheck the ACT in 5-10 minutes. When a therapeutic ACT is achieved (between 300 and 450 seconds), the infusion rate in effect at that time should be continued for the duration of the procedure.
  - If dissection, impending abrupt closure, thrombus formation during PCI or inability to achieve ACT greater than 300 seconds, give a bolus dose of 150 mcg/kg followed by an increase in the infusion rate to 40 mcg/kg/min.
  - Dosage in hepatic impairment: avoid use if significant disease OR if AST/ALT levels are 3 x ULN or greater.

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DOSAGE (Cont.)

- Dosage in renal impairment: no dosage adjustement is necessary.
- Dosage in obesity: for patients with a body mass index (BMI) of up to 51 kg/m^2, initial infusion rate should be calculated by using the actual body weight.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25ºC. Do not refrigerate. Protect from light.
- Stable for 24 hours at room temp in ambient indoor light in NS, D5W or RL at a concentration of 1 mg/mL.
- Stable for 96 hours at room temp or in the fridge protected from light in NS, D5W or RL at a concentration of 1 mg/mL.
- Prepared solutions should not be exposed to direct sunlight.

MISCELLANEOUS

- No specific antidote is available.
- aPTT returns to baseline within 2-4 hours after discontinuing infusion.
- Concomitant use of argatroban and warfarin results in prolongation of the INR beyond that produced by warfarin alone. Consult product monograph for monitoring.
- Contraindicated in patients with hereditary fructose intolerance since the product contains sorbitol.

REFERENCES

1, 5, 40, 95, 135, 208.

NAME OF MEDICATION
ARGININE

OTHER NAMES
L-arginine

CLASSIFICATION
Diagnostic agent, Acidifying agent

INDICATIONS

- To evaluate the pituitary for the release of human growth hormone.
- Management of severe metabolic alkalosis.

ADMINISTRATION

- Intermittent IV infusion: dilute to a 10% solution (100 mg/mL) with SWFI, D5W or D10W. Do NOT use the solution if not clear.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis.
- Cardiovascular: flushing, especially with rapid infusion.
- GI: nausea, vomiting, especially with rapid infusion.
- CNS: headache, numbness, especially with rapid infusion.
- Electrolyte disturbances: hyperkalemia in patients with renal or hepatic impairment or with diabetes (can be life-threatening). May interfere with acid-base balance (due to chloride content).
- Endocrine and metabolic: hyperglycemia.
- Local reactions: venous irritation, especially with rapid infusion.
- Extravasation hazard: vesicant. Consult the document "Extravasation of noncytotoxic intravenous agents associated with tissue necrosis" in TOH Electronic Drug Formulary.

DOSAGE

- Adults:
  - Growth hormone reserve test: 30 g (as 300 mL of a 10% solution) IV over 30 minutes. Schedule test in morning after overnight fast and a normal night rest. Bed rest is essential for at least 30 minutes before infusion begins. Blood samples should be drawn at -30, 0 (immediately before start of infusion), 30, 60, 90, 120, and 150 minutes.
  - Severe metabolic alkalosis: administer the following dose at a rate of 10 g/hr IV:
  - Dose of arginine (g) = $\frac{\text{Desired decrease in plasma bicarbonate (mmol/L) X weight (kg)}}{9.6}$
- Pediatrics:
  - Growth hormone reserve test: 0.5 g/kg IV over 30 minutes; maximum dose of 30 g. For more instructions, refer to the adult growth hormone reserve test in the Dosage section.

- Dosage in renal impairment: use with caution, due to high nitrogen content of arginine and risk of severe hyperkalemia.
- Dosage in hepatic impairment: use with caution, due to risk of hyperkalemia.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30ºC.

MISCELLANEOUS

- Exercise caution in patients with hyperchloremic acidosis due to high chloride content (0.475 mmol/mL) of arginine.
- Contraindicated in patients with highly allergic tendencies.

REFERENCES

1, 3, 5, 95, 167, 366, 367.

Full revision 2020

NAME OF MEDICATION	
ARIPiprazole	
OTHER NAMES	CLASSIFICATION
Abilify Maintena ®	Antipsychotic

INDICATIONS

- Treatment of acute and maintenance phases of schizophrenia in adults.
- Maintenance monotherapy treatment for bipolar I disorder in adults.

ADMINISTRATION

- Vials: following manufacturer's instructions and using the provided material, reconstitute the 300 mg and 400 mg vials with 1.5 and 1.9 mL of SWFI, respectively, to obtain a final concentration of 200 mg/mL. Vigorously shake the vials for 30 seconds to obtain a uniform milky-white homogenous suspension. When the suspension is not administered immediately after reconstitution, the vial must be shaken vigorously for at least 60 seconds to re-suspend the product before withdrawing the dose. Using the provided syringes and vial adapter, withdraw the exact volume to be injected.
- Prefilled dual chamber syringes: follow the manufacturer's instructions for reconstitution; vertically and vigorously shake the syringe for 20 seconds to ensure a uniform milky-white homogenous suspension.
- IM: before injection, select the appropriate needles: *for injection into the gluteus*: 22 gauge 1.5 inch for a non-obese patient and 21 gauge 2 inches for an obese patient; *for injection into the deltoid*: 23 gauge 1 inch for a non-obese patient and 22 gauge 1.5 inch for an obese patient. Slowly inject deeply into the gluteus or deltoid; do NOT massage the injection site. Rotate injection sites between the 2 gluteal or deltoid muscles. If initiating with the two injection start, inject in two different muscles; do NOT inject into the same deltoid or gluteal muscle.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: orthostatic hypotension, tachycardia.
- GI: nausea.
- CNS: insomnia, headache, drowsiness, cognitive impairment, extrapyramidal reactions (akathisia), anxiety.
- Endocrine and metabolic: weight gain, hyperglycemia, increased serum cholesterol, increased LDL cholesterol, increased serum triglycerides, decreased HDL cholesterol.
- Local reactions: pain, redness, swelling, hematoma, pruritus and induration at injection site.

DOSAGE

- One injection start: on the day of treatment initiation, administer one injection of 400 mg IM and maintain patient on oral aripiprazole 10-20 mg per day or on current other oral antipsychotics for 14 days to maintain therapeutic concentrations.
- Two injections start: on the day of treatment initation, administer two separate injections of 400 mg IM at two separate injection sites along with a single 20 mg dose of oral aripiprazole. Ensure patients discontinue their current oral antipsychotic upon initiation of aripiprazole IM. Do not use this two injection start if patient is a known CYP2D6 poor metabolizer and concomitantly uses a strong CYP3A4 inhibitor.
- For both starting regimens, administer the next monthly 400 mg IM injection no sooner that 26 days after the previous injection. Decrease to 300 mg IM once a month if patient experienced adverse drug reactions. If an IM dose is missed, refer to manufacturer's instructions.
- Starting and/or maintenance dosage to be adjusted in patients taking strong CYP3A4 inhibitors and/or strong CYP2D6 inhibitors for more than 14 days and/or in patients known as poor CYP2D6 metabolizers; refer to manufacturer's instructions.
- Dosage in renal impairment: no dosage adjustment.
- Dosage in hepatic impairment: no dosage adjustment in mild to moderate liver impairment; use with caution in patients with severe liver impairment since data is insufficient.

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Vials:
- Store between 15-30°C.
- Reconstituted suspension in vials is stable for 4 hours at 25°C; do not store the reconstituted suspension in the syringe.

Prefilled dual chamber syringes:
- Store between 15-30°C. Protect from light. Do not freeze.

MISCELLANEOUS

- Do not use in patients taking strong CYP3A4 inducers for more than 14 days as they may render aripiprazole ineffective.

REFERENCES

1, 5, 95, 135.

NAME OF MEDICATION
ARSENIC TRIOXIDE

OTHER NAMES
Trisenox ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Induction of remission and consolidation in patients with acute promyelocytic leukemia (APL).

ADMINISTRATION

- Intermittent IV infusion: dilute dose in 100 to 250 mL of D5W or NS. Infuse over 1-2 hours or over up to 4 hours if acute vasomotor reactions occur (e.g., flushing, changes in BP, pallor).

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: pruritus, rash.
- Cardiovascular: arrhythmias, AV block, atrial fibrillation/flutter, chest pain, pallor, tachycardia, hypotension, hypertension, flushing, edema, QTc interval prolongation.
- GI: nausea, vomiting, dyspepsia, diarrhea, abdominal pain, constipation.
- CNS: headache, insomnia, paresthesia, encephalopathy (including Wernicke's; can be fatal), anxiety, depression, dizziness, tremor.
- Dermatologic: dermatitis.
- Electrolyte disturbances: hypokalemia, hyperkalemia, hypocalcemia, hypomagnesemia.
- Endocrine and metabolic: hyperglycemia.
- Hematologic: hyperleukocytosis, thrombocytopenia, anemia, epistaxis.
- Hepatic: elevated AST, ALT, alkaline phosphatase, bilirubin.
- Respiratory: cough, dyspnea, upper respiratory tract infections, oropharyngeal pain.
- APL differentiation syndrome: symptoms may include unexplained fever, dyspnea, acute respiratory distress, hypoxia, pulmonary infiltrates, pleural or pericardial effusions, weight gain, peripheral edema, hypotension and renal, hepatic or multiorgan dysfunction, with or without leukocytosis; can be fatal. If occurs, withhold therapy and immediately administer dexamethasone 10 mg IV BID, regardless of leukocyte count, for at least 3 days, until symptom resolution.
- Tumour lysis syndrome.
- Fatigue, chills, pyrexia, arthralgia, myalgia.
- Local reactions: pain, erythema, edema, bleeding, inflammation, induration at injection site.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Induction: 0.15 mg/kg/day IV until bone marrow remission; do not exceed 60 doses. Other regimens are also used.
- Consolidation: 0.15 mg/kg/day IV for 25 doses over a period up to 5 weeks, starting 3-6 weeks after completion of induction therapy. Other regimens are also used.
- Dosage in renal impairment: use with caution; consider a reduced dose if CrCl is less than 30 mL/min.
- Dosage in hepatic impairment: use with caution; consider a reduced dose if severe impairment (Child-Pugh class C).
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and vials between 15-30ºC. Do not freeze.
- According to the manufacturers, the diluted solution is stable for 24 hours at room temp and for 48 or 72 hours (depending on the brand) in the fridge in D5W or NS. However, longer stability data are available (see below).
- Stable for 14 days at room temp or in the fridge in D5W or NS at concentrations of 0.01 mg/mL and 0.08 mg/mL in PVC containers.
- Stable for 72 hours at room temp or in the fridge in D5W or NS at a concentration of 0.15 mg/mL in PVC containers.

MISCELLANEOUS

- ECG monitoring at baseline and twice a week (or more frequently if clinically indicated) while on treatment (induction and consolidation); consider continuous ECG monitoring in patients at risk of QT interval prolongation. Do not administer if QTc interval is greater than 500 msec, unless corrected and reassessed with serial ECGs. Product may be administered when QTc interval is less than 430 msec for males and less than 450 msec for females.
- Monitor electrolytes, blood glucose, hematologic, hepatic, renal and coagulation parameter tests at baseline and twice a week (or more frequently if clinically indicated) during induction and at least once weekly during consolidation.
- Avoid concomitant drugs that can prolong QT interval or induce electrolyte disorders.

REFERENCES

1, 5, 40, 95, 129, 135, 165, 208.

NAME OF MEDICATION
ARTESUNATE *

OTHER NAMES
Artesun ®

CLASSIFICATION
Antimalarial

INDICATIONS

- Treatment of severe or complicated malaria due to *Plasmodium falciparum.*

ADMINISTRATION

Product from China (Artesun ®):

- Reconstitute each vial of 120 mg of artesunate base with the contents (2 mL) of the provided ampoule of sodium bicarbonate solvent. Shake the vial (not too vigourously) for several minutes to mix well until the powder is completely dissolved and the solution is clear. The solution should clear in a few minutes. Discard if the solution does not clear or a precipitate is present.
- IV direct (preferred): add the contents (10 mL) of the supplied ampoule of NS diluent to the vial containing the reconstituted artesunate solution. Shake to mix well. Solution should be clear. Discard if the solution appears cloudy or a precipitate is present. This will yield a concentration of artesunate of **10 mg/mL.** Administer over 1-2 minutes into the tubing of a freely running IV solution of D5W or NS.
- IM (if IV access cannot be obtained): add 4 mL of the supplied NS diluent to the vial containing the reconstituted artesunate solution. Shake to mix well. Solution should be clear. Discard if the solution appears cloudy or a precipitate is present. This will yield a concentration of artesunate of **20 mg/mL.** Preferred site of injection: anterior thighs; depending on volume to administer, may need to divide dose and inject in several sites.
- Administer dose within 60 minutes of reconstitution.
- Observe patient for 30 minutes following administration for signs of a hypersensitivity reaction.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity (rare): rash, urticaria, itching, swelling, watery eyes, dyspnea, hypotension, angioedema, anaphylaxis.
- Cardiovascular: bradycardia.
- GI: anorexia, nausea, vomiting, cramps, diarrhea, taste alteration.
- CNS: dizziness, lightheadedness, headache, insomnia.
- Hematologic: reversible decrease in reticulocyte count, delayed hemolytic anemia.
- Hepatic: increased liver enzymes.
- Otic: tinnitus.
- Respiratory: cough, nasal symptoms.
- Arthralgias, muscle disorders, fatigue, fever, malaise.
- Local reactions: pain at injection site.

DOSAGE

- **For adults and children weighing 20 kg and over**: 2.4 mg/kg IV/IM at time 0, 12, 24 and 48 hours (total of 4 doses or 9.6 mg/kg).
- **For children weighing less than 20 kg**: 3 mg/kg IV/IM at time 0, 12, 24 and 48 hours (total of 4 doses or 12 mg/kg).
- First dose should be administered STAT. May change to oral therapy (refer to Miscellaneous section) **after the 3rd parenteral dose** if patient can tolerate oral therapy and parasitemia is 1% or less. Pregnant women should receive all 4 IV doses of artesunate. In the rare event that the patient cannot tolerate oral therapy after 4 doses of IV artesunate or parasitemia is still above 1%, artesunate may be continued at the same dose once a day for up to a total of 7 days (total of 9 doses).
- Dosage in renal impairment: no dosage adjustment is necessary.
- Dosage in hepatic impairment: no dosage adjustment is necessary.
- Dosage in obesity: dose should be calculated using actual body weight (i.e., no maximum dose).

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Product from China (Artesun ®):

- Store manufacturer's package containing artesunate, sodium bicarbonate and NS at room temp below 30°C. Protect from light.
- Reconstituted solution should be stored below 30°C and used within 60 minutes. Discard any unused solution.

MISCELLANEOUS

- It is essential that appropriate follow-on oral therapy be started after the last dose of artesunate (administered at least 4 hours after the artesunate dose).
- Weekly complete blood counts are required for 4 weeks after treatment with artesunate due to the risk of delayed hemolytic anemia.

REFERENCES

97, 98, 135, 163, 208, 327, 346, 553, 555.

* Available from the Canadian Malaria Network via Health Canada's Special Access Programme

NAME OF MEDICATION
ASPARAGINASE (*E. coli* source)

OTHER NAMES
Kidrolase ®, L-asparaginase

CLASSIFICATION
Antineoplastic

Do NOT confuse asparaginase of E. coli source with that of Erwinia source.
This monograph is specific to asparaginase of E. COLI source.

INDICATIONS

- Treatment of acute lymphoblastic leukemia, acute myeloid leukemia, chronic lymphocytic leukemia, Hodgkin's lymphoma, non-Hodgkin's lymphoma, melanosarcoma, reticulosarcoma.

ADMINISTRATION

- For IV use:
  - Reconstitute each 10 000 international unit vial by slowly adding 4 mL of SWFI against the vial's inner wall to obtain a concentration of 2500 international units/mL. Rotate vial gently until dissolution; do NOT shake.
  - Intermittent IV infusion: dilute in 50-250 mL of NS or D5W. Gently invert the container to mix; do NOT shake. Infuse over a minimum of 30 minutes into the tubing of a freely running IV solution of NS or D5W.
  - Filtering through a 5-micron filter during administration will remove gelatinous fibres that may develop in solution upon standing with no loss of potency.

- For IM use (preferred for intermittent treatment):
  - Reconstitute each 10 000 international unit vial by slowly adding 2 mL of NS against the vial's inner wall to obtain a concentration of 5000 international units/mL OR by slowly adding 1 mL of NS to obtain a concentration of 10 000 international units/mL. Rotate vial gently until dissolution; do NOT shake.
  - IM: inject deep into a large muscle. Volumes greater than 2 mL should be divided and administered at 2 separate sites.
  - Filtering through a 5-micron filter during administration will remove gelatinous fibres that may develop in solution upon standing with no loss of potency.

- Observe patient for 60 minutes after administration.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: <u>common</u>: rash, urticaria, chills, facial edema, fever, bronchospasm, hypotension, flushing, anaphylaxis.
- Cardiovascular: hypertension, thromboembolism (including in brain).
- GI: nausea, vomiting, pancreatitis (can be fatal).
- CNS: headache, lethargy, coma, depression, confusion, drowsiness, hallucinations, agitation, seizures, posterior reversible encephalopathy syndrome (rare).
- Endocrine and metabolic: hyperglycemia, hyperlipidemia.
- Hematologic: hemorrhage.
- Hepatic: increased transaminases, hyperbilirubinemia, hyperammoniemia, hepatic failure, hepatic steatosis.
- Tumour lysis syndrome, hyperuricemia.
- Local reactions: injection site reaction.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- A test dose of 2 international units intradermally is usually recommended before administration of the first dose or if restarting therapy after several days of interruption. Observe patient for 60 minutes.
- Daily treatment: 200-1000 international units/kg/day IV/IM for 28 days; may continue for an additional 14 days if remission not achieved.
- Intermittent treatment: to give as 3 injections per week for 4 weeks as follows: 400 international units/kg on Mondays and Wednesdays, and 600 international units/kg on Fridays; may continue for an additional 2 weeks if remission not achieved. For this regimen, the IM route is preferred over the IV route due to a lower risk of anaphylactic reactions.
- Dosage in renal impairment: no dosage adjustment necessary.
- Dosage in hepatic impairment: administer with caution in patients with mild to moderate impairment; do not administer in patients with severe impairment.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Reconstituted solution is stable for 3 hours at room temp and 72 hours in the fridge.
- The diluted solution is clear light yellow to brown.
- Stable for 72 hours in the fridge in NS.
- Stable for 24 hours in the fridge in D5W.
- Stable for 7 days in the fridge in NS and RL in polyethylene or polyolefin containers.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- When Kidrolase ® (asparaginase from *E. coli*) causes a hypersensitivity reaction (but not anaphylaxis), substitute with asparaginase of Erwina source.
- Contraindicated in patients currently with or have a history of pancreatitis.

REFERENCES

5, 95, 129, 135, 165, 208.

NAME OF MEDICATION
ASPARAGINASE (Erwinia)

OTHER NAMES
Erwinase ®, L-asparaginase

CLASSIFICATION
Antineoplastic

Do NOT confuse asparaginase of Erwinia source with that of E. coli source.
This monograph is specific to asparaginase of ERWINIA source.

INDICATIONS

- Treatment of acute lymphoblastic leukemia.
- Treatment of patients who developed hypersensitivity (but not anaphylaxis) with asparaginase derived from *E. coli*.

ADMINISTRATION

- Reconstitute each 10 000 unit vial by slowly adding 1 mL (for IV direct) or 2 mL (for IM or SC) of NS against the vial's inner wall to obtain a concentration of 10 000 units/mL or 5000 units/mL, respectively. Do NOT use SWFI for reconstitution. Swirl vial gently until dissolution; do NOT invert vial to ensure minimal contact with stopper; do NOT shake.
- IV direct (NOT preferred due to increased risk of hypersensitivity).
- Intermittent IV infusion (NOT preferred due to increased risk of hypersensitivity): dilute in 100 mL of NS; do NOT shake or squeeze bag; infuse over 1-2 hours.
- IM (preferred): maximum of 2 mL per injection site.
- SC: maximum of 2 mL per injection site.
- Filtering through a 5-micron filter during withdrawal from vial or administration will remove gelatinous fibres that may develop in solution upon standing with no loss of potency.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, urticaria, arthralgia, respiratory distress, anaphylaxis. Up to 33% of patients who developed a hypersensitivity reaction with asparaginase of *E. coli* source will also react to asparaginase of Erwinia source.
- Cardiovascular: thromboembolism.
- GI: nausea, vomiting, pancreatitis (can be fatal).
- CNS: depression, lethargy, drowsiness, coma, confusion, agitation, seizures, hallucinations, headache, posterior reversible enceophalopathy syndrome (rare).
- Endocrine and metabolic: hyperglycemia.
- Hematologic: hemorrhage.
- Hepatic: increased transaminases, hyperbilirubinemia.
- Fever, fatigue.
- Tumour lysis syndrome, hyperuricemia; can be minimized with allopurinol and alkalinization of urine.
- Local reactions: injection site reaction.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Acute lymphoblastic leukemia: various regimens exist.
- Substitute for asparaginase from *E. coli* source: usually 20 000-25 000 units/m^2 IM/IV for each scheduled dose of asparginase from *E. coli* source.
- Dosage in renal impairment: no dosage adjustment necessary.
- Dosage in hepatic impairment: administer with caution in patients with mild to moderate impairment; do not administer in patients with severe impairment.
- Consult specific protocol.

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- The reconstituted solution should be clear and colourless.
- Reconstituted solution is stable for 4 hours at room temp in glass containers or polypropylene syringes; however, best to administer within 15 minutes.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- Contraindicated in patients currently with or have a history of pancreatitis.

REFERENCES

1, 4, 5, 95, 129, 135, 165.

NAME OF MEDICATION
ATEZOLIZUMAB

OTHER NAMES
Tecentriq ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS
- Treatment of locally advanced or metastatic urothelial carcinoma.
- Treatment of locally advanced or metastatic non-small cell lung cancer.
- Treatment of extensive-stage small cell lung cancer.
- Treatment of unresectable, locally advanced or metastatic triple-negative breast cancer.
- Treatment of unresectable or metastatic hepatocellular carcinoma.

ADMINISTRATION
- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards section.**
- Intermittent IV infusion (mandatory): withdraw the required volume of solution from the vial and dilute into a 250 mL infusion bag of NS made of PVC, polyethylene, polypropylene or polyolefin to obtain a final concentration between 3.2-16.8 mg/mL. Gently invert bag to mix; do NOT shake. Administer initial dose over 60 minutes; if well tolerated, subsequent infusions may be administered over 30 minutes.
- May administer (optional) with a 0.2-0.22-micron sterile, non-pyrogenic, low-protein-binding in-line filter.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication**: consult Corporate policy ADM VIII 860 (High alert medications)
- Hypersensitivity: immune-mediated reactions, can be fatal; can involve any organ: anaphylaxis, dermatologic (pruritus, rash, bullous dermatitis, pemphigoid, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis), cardiovascular (myocarditis), GI (diarrhea, colitis), pancreatitis, hepatitis, CNS (meningoencephalitis, neuropathies), endocrine (thyroid disorders, adrenal insufficiency, type 1 diabetes mellitus including diabetic ketoacidosis, hypophysitis), hematologic (autoimmune hemolytic anemia), myositis (including rhabdomyolysis), nephritis, ophthalmic (iritis, uveitis, optic neuritis, keratitis, retinopathy), respiratory (pneumonitis, interstitial lung disease). Use with caution in patients with co-morbidities such as colitis, hepatic impairment, and respiratory or endocrine disorders. Consult manufacturer's recommendations for management of these reactions and when to withhold or discontinue treatment.
- Infusion-related reactions: dyspnea, fever, chills, hypotension. For mild or moderate infusion reactions, interrupt treatment or reduce infusion rate and closely monitor the patient; **consider premedication** with antipyretic and antihistamine. If reaction is severe, discontinue treatment.
- Cardiovascular: hypotension, peripheral edema.
- GI: nausea, vomiting, diarrhea, constipation, decreased appetite, abdominal pain.
- CNS: headache.
- Electrolyte disturbances: hypokalemia, hyponatremia, hypophosphatemia, hypomagnesemia.
- Endocrine and metabolic: hyperglycemia, hypoalbuminemia.
- Hematological: anemia, lymphopenia.
- Hepatic: elevated LFTs.
- Musculoskeletal: arthralgia, myalgia, back/neck pain, pain in extremities.
- Renal: increased creatinine, hematuria.
- Respiratory: cough, nasal congestion, dyspnea.
- Fever, chills, fatigue, asthenia, infection.

DOSAGE
- Various dosages used, depending on the type of cancer treated: 840 mg IV every 2 weeks OR 1200 mg IV every 3 weeks OR 1680 mg IV every 4 weeks.
- Dosage in renal impairment: no dosage adjustment for creatinine clearance of 30 mL/min or greater; no data when creatinine clearance is less than 30 mL/min.
- Dosage in hepatic impairment: no dosage adjustment required for mild or moderate impairment; no data in severe hepatic impairment.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8º C. Protect from light. Do not freeze. Do not shake.
- The solution in the vial is colourless to slightly yellow.
- Stable for 8 hours at room temp and 24 hours in the fridge in NS in PVC, polyethylene, polypropylene and polyolefin containers.
- Compatible only with NS.

MISCELLANEOUS

- AST, ALT, bilirubin, thyroid function, renal function, electrolytes, and glucose should be monitored prior to and periodically during treatment.

REFERENCES

1, 5, 129, 135, 165.

NAME OF MEDICATION	
ATROPINE	
OTHER NAMES	CLASSIFICATION
	Anticholinergic

INDICATIONS
- Spasmolytic effect on smooth muscle.
- Decrease secretions (salivary and bronchial secretions).
- Antidote for anticholinesterase agents (e.g., physostigmine, neostigmine), certain species of *Amanita* (mushroom), and in cases of anticholinesterase pesticide poisoning.
- Antidote for nerve gas exposure (in the context of chemical warfare).
- Treatment of symptomatic sinus bradycardia and atrioventricular block.

ADMINISTRATION
- Rapid IV direct: physician or RN. ***Cardiac monitoring***. Administer undiluted.
- IM, SC.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, skin exfoliation, urticaria, anaphylaxis.
- Cardiovascular: flushing, tachycardia, palpitations.
- GI: dry mouth and throat, dysphagia, thirst, constipation.
- CNS: dizziness, restlessness, ataxia, headache, delirium, insomnia.
- Ophthalmic: blurred vision, mydriasis, photophobia.
- Renal: urinary retention.
- Respiratory: formation of mucus plugs in the bronchi of patients with chronic lung disease.
- Fever.
- Local reactions: injection site pain.

Antidote: physostigmine (controversial; only if severe or life-threatening symptoms of anticholinergic toxicity), diazepam to control delirium, pilocarpine to counteract mydriasis.

DOSAGE
Adults:
- Smooth muscle relaxation and suppression of secretions: 0.4-1 mg IV/IM/SC 30-60 minutes preoperatively, then q4-6h prn.
- Bradycardia: 0.5 mg IV, repeated every 3-5 minutes prn up to total dose of 3 mg. Single doses lower than 0.5 mg are not recommended as they can result in paradoxical bradycardia.
- Anticholinesterase poisoning (from pesticides, mushrooms): 1-2 mg IV/IM/SC repeated every 5-60 minutes prn. In severe cases: 2-6 mg, repeated every 5-60 minutes prn.
- Nerve gas exposure: 2-6 mg IV/IM; to repeat every 5-10 minutes until secretions are minimal, the skin is dry and ventilation (assisted or spontaneous) is adequate.

Pediatrics:
- Smooth muscle relaxation and suppression of secretions: 0.01-0.02 mg/kg IV/IM/SC (minimum dose of 0.1 mg; maximum dose of 1 mg) 30-60 minutes preoperatively, then q4-6h prn.
- Bradycardia: 0.02 mg/kg IV (minimum single dose of 0.1 mg; maximum single dose of 0.5 mg in children and 1 mg in adolescents); dose may be repeated once to a maximum total dose of 1 mg in children or 2 mg in adolescents.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules, vials and prefilled syringes at room temp. Protect from light and heat. Do not freeze.

MISCELLANEOUS

REFERENCES
1, 2, 5, 82, 135, 534.

Full revision 2020

NAME OF MEDICATION
AVELUMAB

OTHER NAMES
Bavencio ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of metastatic Merkel cell carcinoma.
- Maintenance treatment of patients with unresectable locally advanced or metastatic urothelial carcinoma.

ADMINISTRATION

- **Ensure premedication has been administered; refer to Potential Administration Hazards and Dosage sections.**
- Intermittent IV infusion (mandatory): withdraw the required volume from the vial and dilute in 250 mL of NS or 1/2NS. Gently invert bag to mix; do NOT shake. Administer over 60 minutes using a PVC infusion set and a sterile, non-pyrogenic, low-protein-binding polyethersulfone 0.2 micron in-line or add-on filter. Flush the line with NS or 1/2NS after administration.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication**: consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: immune-mediated reactions; most occur during treatment but may occur after treatment discontinuation; can be severe or life-threatening; can involve any organ: hepatitis, pneumonitis, colitis, pancreatitis, endocrine (thyroid disorders, adrenal insufficiency, type 1 diabetes mellitus, hypopituitarism), renal (nephritis, renal dysfunction), myocarditis, myositis, dermatologic (rash, pruritus, psoriasis, exfoliative dermatitis, erythema multiforme, pemphigoid), uveitis, arthritis, Guillain-Barré syndrome, sepsis. Use with caution in patients with co-morbidities such as colitis, hepatic impairment, respiratory or endocrine disorders. Consult manufacturer's recommendations for management of these reactions and when to withhold or discontinue treatment.
- Infusion-related reactions: fever, chills, flushing, hypotension, dyspnea, wheezing, back pain, abdominal pain, urticaria; may be severe or life-threatening and can occur despite premedication. For mild infusion reactions, reduce the infusion rate by 50%. For moderate infusion reactions, interrupt treatment until symptoms have resolved or are mild and then restart at 50% of the infusion rate; **consider premedication** with acetaminophen and an antihistamine. For severe reactions, discontinue treatment.
- Cardiovascular: hypertension, peripheral edema.
- GI: nausea, vomiting, constipation, diarrhea, abdominal pain, decreased appetite.
- CNS: dizziness, headache.
- Electrolyte disturbances: hypokalemia, hyperkalemia, hyponatremia, hypophosphatemia, hypomagnesemia.
- Endocrine and metabolic: hyperglycemia, elevated lipase and amylase.
- Hematologic: anemia, lymphopenia, neutropenia, thrombocytopenia.
- Hepatic: elevated LFTs, bilirubin and gamma glutamyl transferase.
- Musculoskeletal: pain, arthralgia, weakness, elevated creatine phosphokinase.
- Respiratory: cough, dyspnea.
- Fever, fatigue.
- Urinary tract infection.

DOSAGE

- **Premedication with acetaminophen (500-650 mg) and an antihistamine (e.g., diphenhydramine 25-50 mg) 30-60 minutes prior to the infusion is mandatory for the first 4 infusions**. For subsequent cycles, premedication is administered based upon clinical judgment and presence/severity of prior infusion reactions.
- 10 mg/kg IV every 2 weeks.
- Dosage in renal impairment: no dosage adjustment required for CrCl greater than or equal to 15 mL/min. No data for CrCl less than 15 mL/min.
- Dosage in hepatic impairment: no dosage adjustment required for mild impairment (normal bilirubin and AST greater than ULN or bilirubin 1-1.5 x ULN) or moderate impairment (bilirubin 1.5-3 x ULN). No data available for severe impairment (bilirubin greater than 3 x ULN).

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DOSAGE (Cont.)

- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake.
- The vial solution should be clear and colourless to pale yellow.
- Stable for 8 hours at room temp or 24 hours in the fridge in NS and 1/2NS at concentrations between 0.016-8 mg/mL in polyethylene, polypropylene, and ethylene vinyl acetate infusion bags and glass bottles.
- Compatible with dextrose solutions.

MISCELLANEOUS

- Complete blood count, liver function, renal function, and thyroid function should be monitored prior to and periodically during treatment.

REFERENCES

1, 5, 129, 165, 208, 405.

NAME OF MEDICATION
AzaCITIDine

OTHER NAMES
Ladakamycin, Vidaza ®

CLASSIFICATION
Antineoplastic

INDICATIONS
- Treatment of intermediate-2 and high risk myelodysplastic syndrome in patients who are not eligible for stem cell transplant.
- Treatment of acute myeloid leukemia with 20-30% blasts and multi-lineage dysplasia in patients who are not eligible for stem cell transplant.

ADMINISTRATION
- For SC use: reconstitute each 100 mg vial by slowly injecting 4 mL SWFI to obtain a 25 mg/mL suspension. Shake vigorously until a uniform suspension is obtained; the suspension will remain cloudy. Do NOT use if the suspension contains large particles or agglomerates.
- SC: ensure suspension is at room temp before injecting. Using a 25-gauge needle, inject at a 45-90° angle in thigh, abdomen or upper arm. Rotate injection sites; administer new injection at least 2.5 cm from the previous site; avoid a red, bruised, tender or hardened area. For doses greater than 100 mg (4 mL), divide dose in 2 separate injection sites. (Note: at TOH, divide dose to a maximum volume of 2 mL per SC injection site). Just before injection, the contents of the syringe must be re-suspended by vigorously rolling it between the palms until a uniform cloudy suspension is obtained.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, pruritus, urticaria.
- GI: nausea, vomiting, diarrhea, constipation, anorexia, abodominal pain.
- CNS: headache, dizziness, insomnia, anxiety.
- Electrolyte disturbances: hypokalemia, decreased bicarbonate.
- Hematologic: anemia, neutropenia, thrombocytopenia, leukopenia.
- Renal: increased urea and creatinine, renal tubular acidosis, renal failure.
- Respiratory: dyspnea, upper respiratory tract infections.
- Tumour lysis syndrome, hyperuricemia; can be minimized with allopurinol, hydration and sodium bicarbonate.
- Fatigue, fever, arthralgia, myalgia, malaise.
- Local reactions: erythema, induration, pain, skin discolouration, pruritus, inflammation, hematoma, rash, bruising, necrosis (rare).

DOSAGE
- Usual dosage: 75 mg/m^2 SC once daily for 7 consecutive days to be repeated every 4 weeks.
- Dosage in renal impairment: refer to manufacturer's recommendations.
- Dosage in hepatic impairment: use with caution. Contraindicated in patients with advanced malignant hepatic tumours.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials between 15-30°C.
- Reconstituted suspension (25 mg/mL) is stable for 45 minutes at room temp or for 8 hours in the fridge.
- Incompatible with D5W or sodium bicarbonate.

MISCELLANEOUS
- Contains mannitol.

REFERENCES
1, 5, 129, 165.

Full revision 2020

NAME OF MEDICATION
AzaTHIOprine

OTHER NAMES
Imuran ®

CLASSIFICATION
Immunosuppressant

INDICATIONS
- For short-term administration when oral administration is not possible.
- Adjunct for the prevention of rejection in renal ransplantation.
- Adjunct for treatment of auto-immune diseases (e.g., rheumatoid arthritis).

ADMINISTRATION
- Reconstitute each 50 mg vial with 5-15 mL of SWFI; to obtain an exact concentration of 10 mg/mL, reconstitute with 5 mL of SWFI. Swirl until the solution is clear.
- IV direct: physician only. Administer over 5 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of NS, 1/2NS or D5W and infuse over 30-60 minutes (range: 5 minutes to 8 hours).

POTENTIAL ADMINISTRATION HAZARDS
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, cough, dyspnea, hypotension, Stevens-Johnson syndrome, toxic epidermal necrolysis; other symptoms of hypersensitivity include fever, rigors, arthralgia, myalgia, malaise.
- GI: nausea, vomiting, diarrhea, anorexia.
- Hematologic: anemia, thrombocytopenia, bleeding, macrophage activation syndrome. Leukopenia: temporary withdrawal or reduction in dosage may be necessary at the first sign of abnormally large fall in leukocyte count.
- Hepatic: increased alkaline phosphatase, bilirubin and transaminases.
- Infections.

DOSAGE
- Renal transplantation: initial dose of 3-5 mg/kg/day IV, usually beginning at time of transplant. Dose reduction to maintenance levels of 1-3 mg/kg/day IV (usually 50-150 mg) is generally possible. Switch to oral administration, at the same dose, when possible.
- Rheumatoid arthritis: initial dose of approximatively 1 mg/kg IV OR 50-100 mg IV as a single daily dose or in 2 divided doses. Dose can be increased by 0.5 mg/kg every 4-8 weeks; maximum of 2.5 mg/kg/day.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	75%	50% *

* A supplemental dose of 0.25 mg/kg is recommended following hemodialysis.

- Dosage in hepatic impairment: no dosage adjustment provided; use with caution.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials between 15-25°C. Protect from light.
- Reconsituted solution is stable for 16 days at room temp under fluorescent light at a concentration of 10 mg/mL in original vial or plastic syringes.
- Stable for 8 days at room temp exposed to fluorescent light or for 16 days in the fridge in D5W at a concentration of 2 mg/mL in PVC containers.
- Stable for 16 days at room temp exposed to fluorescent light or in the fridge in NS or 1/2NS at a concentration of 2 mg/mL in PVC containers.
- Incompatible with methylparaben, propylparaben and phenol preservatives.

MISCELLANEOUS
- Patients receiving allopurinol concomitantly require **a reduction in dose to 1/3-1/4 of the usual dose of azathioprine**.

REFERENCES
1, 4, 5, 40, 135, 208.

Full revision 2020

NAME OF MEDICATION
AZITHROMYCIN

OTHER NAMES
Zithromax ®

CLASSIFICATION
Antibiotic - macrolide

INDICATIONS

- Treatment of lower respiratory tract infections (community-acquired pneumonia) and genitourinary tract infections (pelvic inflammatory disease) due to susceptible organisms.

ADMINISTRATION

- Reconstitute each 500 mg vial with 4.8 mL of SWFI to obtain a concentration of 100 mg/mL. Shake vial until dissolution.
- Intermittent IV infusion (mandatory): dilute required dose in 250 mL or 500 mL of NS, D5W, RL, 1/2NS, D5-1/2NS, D5-1/3NS, D5-RL, to obtain a final concentration of 1 mg/mL or 2 mg/mL. Administer the 1 mg/mL solution over 3 hours and the 2 mg/mL solution over 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, vasculitis, angioedema, anaphylaxis, erythema multiforme, acute generalized exanthematous pustulosis, Stevens-Johnson syndrome, toxic epidermal necrolysis, drug reaction with eosinophilia and systemic symptoms (DRESS), serum sickness.
- Cardiovascular: rare; cardiac arrhythmias such as QT interval prolongation and torsades de pointes.
- GI: diarrhea, nausea, abdominal pain, vomiting.
- Hepatic: increased ALT and AST.
- Renal: increased creatinine.
- Myasthenia gravis.
- Local reactions: erythema, pain, swelling, tenderness.

DOSAGE

- Community-aquired pneumonia: 500 mg IV once daily for at least 2 days before switching to oral formulation. At TOH: for moderately ill patients, administer 500 mg IV for 1 dose then, the following day, 250 mg IV daily (switch to oral formulation when possible).
- Pelvic inflammatory disease: 500 mg IV once daily for at least 1 day before switching to oral formulation.
- Dosage in renal impairment: no dosage adjustment required.
- Dosage in hepatic impairment: no dosage adjustement required in mild to moderate impairment; use with caution in patients with severe impairment due to lack of data.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution stable for 24 hours when stored at or below 30ºC.
- Stable for 24 hours at room temp or 7 days in the fridge in NS, 1/2NS, D5W, RL, D5-RL, dextrose-saline at a concentration of 1-2 mg/mL.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 95, 135, 208, 422.

Full revision 2020

NAME OF MEDICATION
AZTREONAM *

OTHER NAMES
Azactam ®, Azthreonam

CLASSIFICATION
Antibiotic - monobactam

INDICATIONS

- Treatment of serious infections including complicated and uncomplicated urinary tract infections, lower respiratory tract infections (including those in cystic fibrosis patients), skin and skin structure infections, gynecologic and intra-abdominal infections, gonorrhea, and septicemia caused by susceptible gram-negative organisms.

ADMINISTRATION

- For IV use (preferred for doses above 1 g or for severe infections):
  - For IV direct injection: reconstitute the 1 g and the 2 g vials with 6-10 mL of SWFI. Shake vial immediately and vigorously.
  - For intermittent IV infusion: reconstitute each gram of aztreonam with at least 3 mL of SWFI. Shake vial immediately and vigorously.
- For IM use (preferred for doses up to 1 g): reconstitute each gram of aztreonam with at least 3 mL of SWFI or NS. Shake vial immediately and vigorously.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion: dilute each gram of aztreonam in at least 50 mL of NS, D5W, RL, Ringer's, dextrose-saline solution, D5-RL or sodium lactate 1/6 M, to obtain a final concentration not exceeding 20 mg/mL. Administer over 20-60 minutes. Flush line with a compatible fluid before and after administration.
- IM: deep into a large muscle.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, toxic epidermal necrolysis, erythema multiforme, exfoliative dermatitis, bronchospasm, angioedema, anaphylaxis.
- GI: diarrhea, nausea, vomiting.
- Hepatic: transient increases in AST, ALT and phosphatase alkaline.
- Local reactions: phlebitis/thrombophlebitis, induration, erythema with IV administration; discomfort, swelling, pain with IM administration.

DOSAGE

- Adults: maximum of 8 g per 24-hour period.
  - Urinary tract infections: 500 mg or 1 g IM/IV q8-12h.
  - Moderately severe systemic infections: 1 g IM/IV or 2 g IV q8-12 h.
  - Severe systemic infections, life-threatening infections, cystic fibrosis: 2 g IV q6-8h.
  - Gonorrhea/cystitis: single dose of 1 g IM.
- Pediatrics:
  - Older than 1 week of age: usual dose 90-120 mg/kg/day IV/IM divided q6-8h; maximum daily pediatric dose should not exceed adult dose.
  - Severe infections (2 years of age or older): 150-200 mg/kg/day IV/IM divided q6-8 h; maximum daily pediatric dose should not exceed adult dose.
  - For cystic fibrosis patients (*Pseudomonas aeruginosa*; older than 1 week of age): 150-200 mg/kg/day IV/IM divided q6-8h; maximum daily dose of 12 g.
- Dosage in renal impairment:

CrCl (mL/min)	30-10	less than10
Dose	initial dose: 100% of dose, then 50% of dose for subsequent doses at usual intervals	initial dose: 100% of dose, then 25% of dose for subsequent doses at usual intervals**

** Removed by hemodialysis; in serious or life-threatening infections, a supplemental dose equivalent to 12.5% (1/8) of initial dose should be given after each dialysis session.

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NAME OF MEDICATION
AZTREONAM *

OTHER NAMES
Azactam ®, Azthreonam

CLASSIFICATION
Antibiotic - monobactam

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from heat.
- Reconstituted solution is colourless to light straw yellow which may develop a slight pink tint over time (no loss of potency).
- Reconstituted solution is stable for 24 hours in the fridge.
- Stable for 48 hours at room temp and 7 days in the fridge in D5W and NS at concentrations of 10 mg/mL and 20 mg/mL in PVC containers.
- Stable for 48 hours at room temp and 7 days in the fridge in RL, Ringer's, D5-RL, dextrose-saline solutions and sodium lactate 1/6 M at concentrations up to 20 mg/mL.

MISCELLANEOUS

- Potential for cross-allergenicity with beta-lactam antibiotics.

REFERENCES

1, 4, 40, 82, 95, 135, 208, 363, 426.

* Available via Health Canada's Special Access Programme

NAME OF MEDICATION
BASILIXIMAB

OTHER NAMES
Simulect ®

CLASSIFICATION
Immunosuppresant, Monoclonal antibody

INDICATIONS

- Prophylaxis of acute organ rejection in *de novo* renal transplantation; to be used concurrently with cyclosporine (Neoral ®) and corticosteroid-based immunosuppression.

ADMINISTRATION

- Reconstitute each 20 mg vial with 5 mL of SWFI to obtain a concentration of 4 mg/mL. Shake the vial gently to dissolve the powder.
- IV direct: physician or RN. Administer undiluted over 30-60 seconds.
- Intermittent IV infusion: dilute with NS or D5W to a volume of at least 50 mL for the 20 mg dose and at least 25 mL for the 10 mg dose. Gently invert bag to avoid foaming; do NOT shake. Infuse over 20 to 30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactic/anaphylactoid reactions, including rash, urticaria, pruritus, hypotension, tachycardia, cardiac failure, respiratory difficulties and failure, capillary leak syndrome; may occur within 24 hours of initial or subsequent exposures.
- GI: nausea and vomiting with IV direct administration.
- Infections.
- Local reactions: pain at IV direct injection site.

DOSAGE

- Adults and children weighing at least 35 kg: first 20 mg IV dose within the 2 hours prior to transplantation surgery, and second 20 mg IV dose 4 days following transplantation.
- Children weighing less than 35 kg: first 10 mg IV dose within the 2 hours prior to transplantation surgery, and second 10 mg IV dose 4 days following transplantation.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Reconstituted solution should be colourless, and clear to opalescent.
- Reconstituted solution is stable for 4 hours at room temp and 24 hours in the fridge.
- Once diluted, solution should be infused immediately.

MISCELLANEOUS

- Contains mannitol.

REFERENCES

1, 5, 40, 95, 135.

Full revision 2021

NAME OF MEDICATION
BENDAMUSTINE

OTHER NAMES
Cytostasan, Treanda ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of relapsed indolent B-cell non-Hodgkin lymphoma (NHL) in adults who did not respond to or progressed during or shorlty following treatment with a rituximab-containing regimen.
- Treatment of symptomatic chronic lymphocytic leukemia (CLL) in treatment-naïve adults.

ADMINISTRATION

- **Ensure premedication has been administered if the patient experienced an infusion reaction with a prior bendamustine infusion; refer to Potential Administration Hazards section.**
- Reconstitute each 25 mg and 100 mg vial with 5 mL and 20 mL, respectively, of SWFI to obtain a concentration of 5 mg/mL. Shake well until the powder is completely dissolved (within 5 minutes). Do NOT use the reconstituted solution if particulate matter is still observed after 5 minutes.
- Intermittent IV infusion: dilute in 500 mL of NS or D2.5-1/2NS to obtain a final concentration within 0.2-0.6 mg/mL. Mix well. The solution should be clear, colourless to slightly yellow. Administer over 60 minutes when used for NHL and over 30 minutes for CLL.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, bullous exanthema, Stevens-Johnson syndrome, toxic epidermal necrolysis and drug reaction with eosinophilia and systemic symptoms (DRESS); risk of severe reaction when administered with allopurinol.
- Infusion-related reactions: fever, chills, pruritus and rash. Rarely, anaphylaxis and anaphylactoïd reactions in second and subsequent cycles. To prevent severe reactions, antihistamines (e.g., diphenhydramine), antipyretics (e.g., acetaminophen) and corticosteroids can be administered as **premedications** in subsequent cycles in patients who have previously experienced mild to moderate infusion reactions. Discontinue treatment in patients with more severe reactions.
- Cardiovascular: hypertension, hypertensive crisis, cardiac failure, myocardial infarction, palpitations, angina, arrhythmias, pericardial effusion, tachycardia, peripheral edema..
- GI: nausea, vomiting, diarrhea, constipation, stomatitis, abdominal pain, dyspepsia, anorexia, dysgeusia, dehydration.
- CNS: headache, dizziness, insomnia.
- Electrolyte disturbances: hypokalemia.
- Hematologic: neutropenia (including febrile neutropenia), thrombocytopenia, anemia, leukopenia.
- Hepatic: increased liver transaminases (ALT, AST), hyperbilirubinemia, liver injury.
- Respiratory: pneumonia, pulmonary fibrosis, dyspnea, cough.
- Tumour lysis syndrome as early as first treatment cycle (refer to Hypersensitivity section above for the risk of using allopurinol), hyperuricemia.
- Fatigue, back pain.
- Sepsis, septic shock, reactivation of hepatitis B infection, fever.
- Local reactions: pain, phlebitis, pruritus, swelling, irritation.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- NHL: 120 mg/m^2 IV as monotherapy on days 1 and 2 of a 21-day cycle, up to 8 cycles.
- CLL: 100 mg/m^2 IV as monotherapy on days 1 and 2 of a 28-day cycle, up to 6 cycles.

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DOSAGE (Cont.)

- Dosage in renal impairment: use with caution in patients with CrCl between 40-80 mL/min. Should not be used in patients with CrCl below 40 mL/min.
- Dosage in hepatic impairment: use with caution in patients with mild hepatic impairment (total bilirubin greater than ULN up to 1.5 x ULN or AST/ALT/alkaline phosphatase greater than ULN up to 2.5 x ULN). Should not be used in patients with moderate or severe hepatic impairment (bilirubin 1.5 x ULN or greater or AST/ALT/alkaline phosphatase 2.5 x ULN or greater).
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-25°C, with excursions permitted up to 30°C. Protect from light.
- The reconstituted solution is stable for 30 minutes; it should be clear, colourless to pale yellow.
- Stable for 3 hours at room temp exposed to room light or for 24 hours in the fridge in NS or D2.5-1/2NS at concentrations between 0.2-0.6 mg/mL in PVC or polyethylene containers.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment should be immediately available for the treatment of hypersensitivity and infusion-related reactions.
- Do not administer live attenuated vaccines during bendamustine treatment.
- Contains mannitol.

REFERENCES

1, 5, 40, 129, 135, 165, 208.

NAME OF MEDICATION
BENZTROPINE

OTHER NAMES
Benzatropine, Benztropine mesylate, Cogentin ®

CLASSIFICATION
Antiparkinsonian

INDICATIONS

- Treatment of drug-induced extrapyramidal symptoms (EPS) and acute dystonic reactions.
- Symptomatic treatment of parkinsonism when a more rapid response than with oral administration is desired.

ADMINISTRATION

- IV direct: physician only. Administer undiluted over 1 minute.
- IM (preferred).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash.
- Cardiovascular: tachycardia, palpitations, postural hypotension.
- GI: nausea, vomiting, dry mouth, constipation; paralytic ileus (rare).
- CNS: nervousness, dizziness, drowsiness, anxiety, headache, visual hallucinations, depression, paresthesia at extremities; confusion and excitement at high doses.
- Musculoskeletal: weakness with high doses.
- Ophthalmic: blurred vision, mydriasis.
- Renal: urinary retention, difficulty in urinating.
- Anhidrosis, hyperthermia.

DOSAGE

Adults:
- Drug-induced EPS: 1-2 mg IM/IV 1 to 3 times daily. Dose can be increased by 0.5 mg at intervals of at least 5 days; maximum of 6 mg/day.
- Acute drug-induced dystonic reactions: 1-2 mg IM/IV followed by oral doses of 1-2 mg/day.
- Parkinsonism: usual dose of 1-2 mg IM/IV per day (range of 0.5-6 mg/day). Dose can be increased gradually by 0.5 mg every 5-6 days; maximum of 6 mg/day.

Pediatrics (3 years of age and older)
- Drug-induced EPS and acute dystonic reactions: 0.02-0.05 mg/kg IM/IV once or twice daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Do not freeze.

MISCELLANEOUS

- Onset of action after IV/IM administration: a few minutes.
- Contraindicated in patients with angle-closure glaucoma.

REFERENCES

1, 2, 5, 40, 82, 95, 208.

Full revision 2021

NAME OF MEDICATION
BETAMETHASONE

OTHER NAMES
Celestone Soluspan ®

CLASSIFICATION
Corticosteroid

INDICATIONS

- For local or systemic anti-inflammatory, antirheumatic or antiallergic effects for the treatment of corticosteroid-responsive disorders.
- Antenal treatment in pregnancies (24-34 weeks gestation) at risk of preterm delivery within 7 days, to hasten fetal maturation (e.g., lungs, cerebral blood vessels).

ADMINISTRATION

- Shake well to obtain a uniform suspension.
- IM: deep into a large muscle.
- Intraarticular, intrasynovial, intralesional and at soft tissue sites. If mixing with preservative-free 1% or 2% lidocaine, first withdraw betamethasone into a syringe then withdraw lidocaine; refer to Compatibility, Stability section.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactic/anaphylactoid reactions.
- Cardiovascular: edema, hypertension.
- Electrolyte disturbances: hypernatremia, hypokalemia.
- Endocrine and metabolic: Cushing syndrome, hyperglycemia.
- Musculoskeletal: muscle weakness, myopathy; may exacerbate myasthenia gravis.

DOSAGE

- **Doses are expressed as the total amount of betamethasone per dose, 1 mg being equivalent to 0.5 mg of betamethasone sodium phosphate plus 0.5 mg of betamethasone acetate (each mL contains 3 mg of betamethasone sodium phosphate and 3 mg of betamethasone acetate)**.
- For systemic effect: 6-12 mg (1-2 mL) IM at intervals of 3 days to a week, as needed.
- For local effect: 1.2-12 mg (0.2-2 mL) depending on size of tissue/joint injected. Dose may be repeated if needed at intervals of 3 days to 6 weeks. Refer to manufacturer's recommendations.
- Antenatal fetal maturation: 12 mg (2 mL) IM every 24 hours for 2 doses.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-30ºC. Protect from light.
- Compatible with an equal volume of preservative-free 1% or 2% lidocaine; preservatives (such as parabens or phenol) may cause flocculation of the suspension.

MISCELLANEOUS

REFERENCES

1, 2, 5, 8, 135.

Full revision 2021

NAME OF MEDICATION
BEVACIZUMAB

OTHER NAMES
Anti-VEGF rhuMAb, Avastin ®, Mvasi ®, Zirabev ™

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of metastatic carcinoma of the colon or rectum.
- Treatment of unresectable, locally advanced, metastatic or recurrent non-squamous non-small cell lung cancer (NSCLC).
- Treatment of glioblastoma after relapse or disease progression, following prior therapy.
- Treatment of platinum-resistant or platinum-sensitive recurrent or high-risk epithelial ovarian, fallopian tube and primary peritoneal cancer.

ADMINISTRATION

- Intermittent IV infusion: dilute dose in 100 mL of NS (final concentration should be within 1.4-16.5 mg/mL). Infuse the first dose over 90 minutes; if well tolerated, infuse the second dose over 60 minutes; if well tolerated, may infuse subsequent doses over 30 minutes. Rapid infusions of the 5 mg/kg and 7.5 mg/kg doses over 10 minutes have also been used safely.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity and infusion-related reactions: anaphylaxis and anaphylactoid reactions; rigors, dyspnea, wheezing, oxygen desaturation, diaphoresis, hypertension, hypotension, flushing/redness, rash, chest pain. If reaction occurs, stop infusion and start appropriate medical treatment.
- Cardiovascular: hypertension, hypertensive crisis, CHF, peripheral edema.
- GI: nausea, vomiting, diarrhea, constipation, anorexia, stomatitis; abdominal pain, fistulae and GI perforation (including gallbladder): if occurs, bevacizumab should be discontinued permanently.
- CNS: headache; posterior reversible encepalopathy syndrome (rare).
- Hematologic: leukopenia, neutropenia (including febrile neutropenia), thrombocytopenia; thromboembolism, including stroke, transient ischemic attack, myocardial infarction, pulmonary embolism; tumour-associated hemorrhage and minor mucocutaneous bleeding, hemoptysis; pulmonary, GI and CNS hemorrhage.
- Renal: proteinuria, nephrotic syndrome.
- Impairment of wound healing; if wound dehiscence requires medical intervention, bevacizumab should be discontinued permanently.
- Development of non-GI fistulae, necrotizing fasciitis (rare).
- Fatigue.

DOSAGE

- Colorectal cancer: 5 mg/kg IV or 10 mg/kg IV every 2 weeks OR 7.5 mg/kg IV every 3 weeks.
- NSCLC: 15 mg/kg IV every 3 weeks.
- Glioblastoma: 10 mg/kg IV every 2 weeks.
- Ovarian, fallopian tube or peritoneal cancer:
  - Platinum-sensitive: 15 mg/kg IV every 3 weeks.
  - Platinum-resistant: 10 mg/kg IV every 2 weeks OR 15 mg/kg IV every 3 weeks.
  - High-risk cancer: 7.5 mg/kg IV every 3 weeks.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

…/Cont.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze or shake.
- Stable for 48 hours between 2-30°C in NS at concentrations between 1.4-16.5 mg/mL.
- Stable for 24 hours between 2-30°C in NS at a concentration of 1 mg/mL in PVC and polyolefin containers.
- Compatible with PVC (Avastin ®, Mvasi ®, Zirabev ™), polyolefin (Avastin ®, Mvasi ®, Zirabev ™) and ethylene vinyl acetate (EVA) (Zirabev ™) containers.
- Do not mix with dextrose solutions as a concentration-dependent degradation of bevacizumab has been observed.

MISCELLANEOUS

- Mvasi ® and Zirabev ™ are classified as being biosimilar to Avastin ®; therefore, these products are not interchangeable.
- Hold bevacizumab for at least 28 days before or after major surgery and until the surgical incision is fully healed.

REFERENCES

1, 5, 40, 129, 135, 165, 208.

NAME OF MEDICATION
BIVALIRUDIN

OTHER NAMES
Angiomax ®

CLASSIFICATION
Anticoagulant

INDICATIONS

- Anticoagulant in patients undergoing percutaneous coronary intervention (PCI) including patients with acute coronary syndromes due to ST-segment elevation myocardial infarction (STEMI).
- Anticoagulant in patients with acute coronary syndromes (unstable angina or non ST-segment elevation myocardial infarction [NSTEMI]) in whom early PCI is planned.
- Anticoagulant in patients with or at risk of heparin-induced thrombocytopenia or heparin-induced thrombocytopenia thrombosis syndrome undergoing PCI or cardiac surgery.

ADMINISTRATION

- Reconstitute each 250 mg vial with 5 mL of SWFI to obtain a concentration of 50 mg/mL. Gently swirl to dissolve. Must be further diluted as per instructions below before administration.
- IV direct: physician or RN. Dilute each reconstituted 250 mg vial in 50 mL of NS or D5W to obtain a final concentration of 5 mg/mL. Administer calculated dose as IV bolus.
- Continuous IV infusion:
  - 5 mg/mL solution: dilute each reconstituted 250 mg vial in 50 mL of NS or D5W (e.g., 250 mg in 50 mL, 500 mg in 100 mL, 1250 mg in 250 mL).
  - 0.5 mg/mL solution (for low-rate administration): dilute the reconstituted 250 mg vial in 500 mL of NS of D5W.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, anaphylaxis.
- Cardiovascular: hypotension, hypertension.
- GI: nausea.
- CNS: headache.
- Hematologic: bleeding, ranging from minor to major.
- Back pain.
- Local reactions: pain at injection site.

DOSAGE

Patients undergoing PCI:
- 0.75 mg/kg IV bolus followed by an IV infusion of 1.75 mg/kg/hr for the duration of the PCI procedure. Five minutes after bolus has been given, an Activated Clotting Time (ACT) should be performed and an additional IV bolus of 0.3 mg/kg should be given if needed. Continuation of the infusion for up to 4 hours post-procedure is optional. After 4 hours, may continue IV infusion if needed at a reduced rate of 0.2 mg/kg/hr for up to 20 hours, or at 0.25 mg/kg/hr for up to 12 hours.
- Dosage in renal impairment: bolus dose remains the same; continuous infusion rate is adjusted; the ACT should be monitored. Refer to manufacturer's monograph for more details.

CrCl (mL/min)	59-30	Less than 30	Hemodialysis
4-hour Infusion (mg/kg/hr)	1.75	1	0.25

- Dosage in obesity: dose should be calculated using actual body weight.

Acute coronary syndromes (unstable angina or NSTEMI):
- 0.1 mg/kg IV bolus followed by an IV infusion of 0.25 mg/kg/hr prior to angiography and continued through angiography and as long as needed. If patient is medically treated, the IV infusion of 0.25 mg/kg/hr may be continued for up to 72 hours. If patient proceeds to a PCI or a cardiac surgery, refer to manufacturer's monograph for more details.

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DOSAGE (Cont.)

Patients going to cardiac surgery:
- On-pump cardiac surgery: 1 mg/kg IV bolus followed by an IV infusion of 2.5 mg/kg/hr; terminate the infusion approximately 15 minutes prior to the anticipated end of cardiopulmonary bypass. Refer to manufaturer's monograph for more details.
- Off-pump cardiac surgery: 0.75 mg/kg IV bolus followed by an IV infusion of 1.75 mg/kg/hr for the duration of the procedure. Refer to manufacturer's monograph for more details.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Do not freeze.
- Reconstituted solution (50 mg/mL) is stable for 24 hours between 2-8°C. The solution should be clear to slightly opalescent, colourless to slightly yellow.
- Stable for 24 hours at room temp in NS or D5W at concentrations between 0.5-5 mg/mL in glass or PVC containers.

MISCELLANEOUS

- For patients undergoing PCI, bivalirudin is intended for use with ASA (300-325 mg daily). Clopidogrel can also be administered.
- Bivalirudin can be started 30 minutes after discontinuation of IV heparin or 8 hours after discontinuation of SC LMWH.
- May interfere with INR coagulation test.
- Contains mannitol.

REFERENCES

1, 4, 5, 40, 95, 135, 208, 436.

NAME OF MEDICATION
BLEOMYCIN

OTHER NAMES
Blenoxane ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of squamous cell carcinomas, lymphomas, testicular carcinoma, malignant pleural effusion, renal cell cancer, soft tissue sarcomas, and non small cell lung cancer.

ADMINISTRATION

- **Consider premedication to decrease severity of some side effects; refer to Potential Administration Hazards section.**
- For IV use: reconstitute each 15 unit vial with 5-10 mL of NS or SWFI to obtain a concentration of 1.5-3 units/mL.
- IV direct: physician or RN; inject reconstituted solution slowly over 10 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of NS; infuse over 10-15 minutes.
- Continuous IV infusion.
- IM, SC: reconstitute each 15 unit vial with 1-5 mL of NS or SWFI to obtain a concentration of 3-15 units/mL.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactic/anaphylactoid reactions: wheezing, hypotension, confusion, fever, chills: can occur immediately or be delayed for several hours; usually with first or 2nd dose in patients treated for lymphoma. **Premedication** with hydrocortisone may prevent or decrease the severity of fever/chills. Acetaminophen can be used to control the fever.
- Cardiovascular: chest pain, pleuropericarditis (rare; reduce IV infusion rate).
- GI: nausea, vomiting, anorexia, stomatitis, mucositis.
- Dermatologic: radiation recall reactions (rare); rash on pressure areas and abdominal skin creases.
- Respiratory: pneumonitis, dyspnea, rales.
- Local reactions: phlebitis; pain with IM/SC administration.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Treatment of lymphoma: a test dose of 2 units or less may be administered prior to the first 2 treatment doses in order to rule out hypersensitivity to bleomycin. If no acute reaction occurs after 2-4 hours, administer the regular dose.
- Usual dose: 0.25-0.5 units/kg or 10-20 units/m^2 IV/IM/SC, from twice weekly to every 4 weeks OR fixed dose of 30 units IV once weekly or every 3 weeks.
- Cumulative dosage exceeding 400 units should be given with great caution because of the increased incidence of pulmonary toxicity associated with large cumulative dosages.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	75%	50%

- Consult specific protocol.

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…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Temperature excursion at 25°C for up to 1 month is permitted.
- Solution reconstituted with NS is stable for 28 days at room temp or in the fridge protected from light.
- Solution reconstituted with SWFI is stable for 24 hours at room temp and for 48 hours in the fridge.
- Stable for 24 hours at room temp in NS at concentrations of 0.3 units/mL and 3 units/mL in glass and PVC containers.
- Stable for 24 hours at 25°C in NS at a concentration between 0.6-1.2 units/mL in PVC containers.
- Stable for 48 hours at room temp in NS at a concentration of 0.015 units/mL in glass, PVC and high-density polyethylene containers.
- Stable for 28 days at room temp protected from light in NS at a concentration of 0.15 units/mL in PVC containers.
- Do not dilute in D5W, due to lower stability.

MISCELLANEOUS

- Bleomycin is a mixture of peptides and should be described in units rather than in mg.

REFERENCES

1, 4, 5, 6, 40, 42, 95, 129, 135, 165, 200, 208.

NAME OF MEDICATION
BLINATUMOMAB

OTHER NAMES
Blincyto ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS
- Treatment of adults with relapsed or refractory B-cell precursor acute lymphoblastic leukemia (ALL).
- Treatment of children with Philadelphia chromosome-negative relapsed or refractory B-cell precursor ALL.
- Treatment of patients with Philadelphia chromosome-negative CD19 positive B-cell precursor ALL in first or second hematologic complete remission with minimal residual disease greater than or equal to 0.1%.

ADMINISTRATION
- **Ensure premedication has been administered as recommended; refer to Dosage section.**
- An IV solution stabilizer is provided with the blinatumomab package and is used to coat the IV bag prior to addition of reconstituted drug in order to prevent its adhesion to IV bags and IV lines. Do NOT use the solution stabilizer for reconstitution of blinatumomab.
- Reconstitute each 38.5 mcg blinatumomab vial with 3 mL of SWFI directed toward the side of the vial to obtain a concentration of 12.5 mcg/mL. Gently swirl contents to avoid excess foaming. Do NOT shake. Do NOT use if the solution is cloudy or contains a precipitate.
- Continuous IV infusion:

For blinatumomab 24, 48, 72 or 96-hour infusion:
- Add 270 mL of NS into an empty bag made of polyolefin, PVC non-DEHP or ethylene vinyl acetate (EVA). Transfer 5.5 mL of IV solution stabilizer to the IV bag containing NS. Gently mix the contents of the bag to avoid foaming. Discard remaining solution stabilizer vial.
- Depending on the admixture to prepare (9 mcg/day, 28 mcg/day, 5 mcg/m^2/day or 15 mcg/m^2/day, each infused over 24, 48, 72 or 96 hours), transfer the required amount of reconstituted blinatumomab to the NS bag; refer to manufacturer's specific instructions. Gently mix the contents of the bag to avoid foaming.
- Attach the IV tubing (polyolefin, PVC non-DEHP or EVA) to the IV bag with a sterile, non-pyrogenic, low-protein-binding 0.2 or 0.22 micron in-line filter. Remove air from the IV bag. Prime the tubing only with the prepared solution; do NOT prime with NS.
- The entire volume of the admixed blinatumomab will be more than the actual volume administered to the patient (240 mL) to account for the priming of the IV line and to ensure that the patient will receive the full dose. Administer dose by infusing 240 mL at 10 mL/hr for a duration of 24 hours, 5 mL/hr for a duration of 48 hours, 3.3 mL/hr for a duration of 72 hours or 2.5 mL/hr for a duration of 96 hours. Discard any remaining solution.

For blinatumomab 7-day infusion (for patients weighing 22 kg or more only):
- Add 90 mL of bacteriostatic NS to an empty IV bag made of polyolefin, PVC non-DEHP or EVA. Transfer 2.2 mL of IV solution stabilizer to the NS bag. Gently mix the contents of the bag to avoid foaming. Discard remaining solution stabilizer vial.
- Transfer reconstituted blinatumomab into the NS bag; the volume to transfer will differ depending on the patient's weight; refer to manufacturer's instructions. Gently mix the contents of the bag to avoid foaming.
- Add NS to the IV bag to a final volume of 110 mL resulting in 0.74% benzyl alcohol; refer to manufacturer's instructions for the exact volume of NS to add. Gently mix the contents of the bag to avoid foaming.
- Attach IV tubing (polyofefin, PVC non-DEHP or EVA) to the IV bag; an in-line filter is not needed for the 7-day infusion. Remove air from the IV bag. Prime the tubing only with the prepared solution; do NOT prime with NS.
- The entire volume of the admixed blinatumomab will be more than the actual volume administered to the patient (100 mL) to account for the priming of the IV line and to ensure that the patient will receive the full dose. Administer dose by infusing 100 mL at 0.6 mL/hr for a duration of 7 days. Discard any remaining solution.

- When changing the infusion bag or when at infusion completion, do NOT flush the infusion line as this will cause an inadvertent administration of a bolus dose.
- Must be administered by an infusion pump.

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POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity.
- Infusion-related reactions: symptoms are similar to those of cytokine release syndrome; most common during first infusions of the first and second cycles as well as with dose escalation.
- Cytokine release syndrome: pyrexia, asthenia, headache, hypotension, total bilirubin increased, elevation of liver enzymes (AST and ALT), nausea; may also be associated with disseminated intravascular coagulation, capillary leak syndrome, hemophagocytic histiocytosis/macrophage activation syndrome. Observe patient closely, especially during the first infusion of the first 2 cycles; median time of onset is 2 days. May need to interrupt the infusion or definitively discontinue the treatment.
- Cardiovascular: hypotension, hypertension, peripheral edema, arrhythmia, tachycardia.
- GI: abdominal pain, diarrhea, constipation, nausea, vomiting.
- CNS: dizziness, headache, insomnia, tremor, confusion, seizures, encephalopathy; more frequent in the elderly.
- Dermatologic: rash.
- Electrolyte disturbances: decreased potassium, magnesium, phosphate and calcium.
- Endocrine and metabolic: hyperglycemia.
- Hematologic: anemia, neutropenia (including febrile neutropenia), lymphopenia, thrombocytopenia, bleeding.
- Hepatic: increase in LFTs.
- Infections: can be severe; more frequent in the elderly.
- Respiratory: cough, dyspnea.
- Tumour lysis syndrome: hydrate patient and administer allopurinol or rasburicase, especially in patients with higher leukocytosis or a high tumour burden.
- Pancreatitis (rare, may be fatal).
- Fatigue, fever, musculoskeletal pain, chills.

DOSAGE

- **Intrathecal chemotherapy CNS prophylaxis is recommended before and during blinatumomab therapy to prevent CNS ALL relapse**.
- Treatment of relapsed or refractory B-cell precursor ALL (adults and pediatrics):
  - Hospitalization is recommended for at least the first 9 days of the first treatment cycle and the first 2 days of the second treatment cycle.
  - **Pre-phase treatment for patients with high tumour burden**: for patients (adults and pediatrics) with 50% or more leukemic blasts in bone marrow or if peripheral blood leukemic blast counts are greater than 15×10^9/L, treat with dexamethasone (maximum of 24 mg/day) for up to 4 days prior to the first dose of blinatumomab.
  - **Premedication: adults: dexamethasone 20 mg IV 60 minutes before first dose of blinatumomab of each cycle. Pediatrics: dexamethasone 10 mg/m^2 (maximum of 20 mg) PO or IV 6-12 hours prior to the start of blinatumomab (cycle 1 day 1) followed by dexamethasone 5 mg/m^2 PO or IV within 30 minutes of the start of blinatumomab (cycle 1 day 1).**
  - If patient weight is 45 kg or more (fixed dose): cycle 1: 9 mcg/day IV on days 1 to 7, then 28 mcg/day IV on days 8 to 28; after a 14-day treatment-free period (days 29-42), continue with cycles 2-5: 28 mcg/day IV on days 1 to 28; separate each cycle by a 14-day treatment-free period (days 29-42). Cycles 6-9 (maintenance): 28 mgc/day IV on days 1 to 28; separate each cycle by a 56-day treatment-free period (days 29-84).
  - If patient weight is less than 45 kg (body surface area-based dose): cycle 1: 5 mcg/**m^2**/day (maximum of 9 mcg/day) IV on days 1 to 7, then 15 mcg/**m^2**/day (maximum of 28 mcg/day) IV on days 8 to 28; after a 14-day treatment-free period (days 29-42), continue with cycles 2-5: 15 mcg/**m^2**/day IV (maximum of 28 mcg/day) on days 1 to 28; separate each cycle by a 14-day treatment-free period (days 29-42). Cycles 6-9 (maintenance): 15 mcg/**m^2**/day (maximum of 28 mcg/day) IV on days 1 to 28; separate each cycle by a 56-day treatment-free period (days 29-84).

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DOSAGE (Cont.)

- Treatment of minimal residual disease-positive B-cell precursor ALL (adults and pediatrics):
  - Hospitalization is recommended for at least the first 3 days of the first cycle and the first 2 days of the second cycle. For all subsequent cycles and reinititation (when treatment is interrupted for at least 4 hours), supervision by a health care professional or hospitalization is recommended.
  - **Premedication: adults: dexamethasone 16 mg IV 60 minutes before first dose of blinatumomab of each cycle. Pediatrics: dexamethasone 5 mg/m^2 (maximum of 20 mg) PO or IV within 30 minutes prior to the first dose of blinatumomab in the first cycle and when restarting an infusion after an interruption of 4 or more hours in the first cycle.**
  - If patient weight is 45 kg or more (fixed dose): cycle 1 (induction): 28 mcg/day IV on days 1 to 28; after a 14-day treatment-free period (days 29-42), continue with cycles 2-4 (consolidation): 28 mcg/day IV on days 1 to 28; separate each cycle by a 14-day treatment-free period (days 29-42).
  - If patient weight is less than 45 kg (body surface area-based dose): cycle 1 (induction): 15 mcg/**m^2**/day (maximum of 28 mcg/day) IV on days 1 to 28; after a 14-day treatment-free period (days 29-42), continue with cycles 2-4 (consolidation): 15 mcg/**m^2**/day (maximum of 28 mcg/day) IV on days 1 to 28; separate each cycle by a 14-day treatment-free period (days 29-42).
- Dosage in renal impairment: no formal studies but accumulation may occur if CrCl is between 30-59 mL/min. No information if CrCl is less than 30 mL/min or patient is on hemodialysis.
- Dosage in hepatic impairment: no formal studies but may not affect drug clearance; refer to manufacturer's recommendations.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials of blinatumomab and the solution stabilizer between 2-8°C. Protect from light. Do not freeze.
- IV solution stabilizer in vial should be clear, colourless to slightly yellow.
- Blinatumomab vial and solution stabilizer vial are stable for 8 hours between 23-27°C, protected from light.
- Reconstituted solution should be clear to slightly opalescent, colourless to slightly yellow.
- Reconstituted solution is stable for 4 hours between 23-27°C or for 24 hours in the fridge, protected from light.
- Stable for 96 hours between 23-27°C or for 10 days in the fridge in preservative-free NS.
- Stable for 7 days between 23-27°C or for 14 days in the fridge in NS with preservative (benzyl alcohol).

MISCELLANEOUS

- Do not administer live vaccines for at least 2 weeks before treatment and during treatment with blinatumomab, and until B cell recovery after the last cycle.

REFERENCES

1, 5, 129, 135, 165, 473.

NAME OF MEDICATION	
BORTEZOMIB	
OTHER NAMES Velcade ®	CLASSIFICATION Antineoplastic

INDICATIONS

- Treatment of patients with previously untreated multiple myeloma who are suitable or unsuitable for stem cell transplantation (SCT).
- Treatment of progressive/relapsed multiple myeloma.
- Treatment of previously untreated or relapsed/refractory mantle cell lymphoma.

ADMINISTRATION

- For IV use: reconstitute the 3.5 mg vial and the 2.5 mg vial with 3.5 mL and with 2.5 mL, respectively, of NS to obtain a final concentration of 1 mg/mL.
- For SC use: reconstitute the 3.5 mg vial and the 2.5 mg vial with 1.4 mL and with 1 mL, respectively, of NS to obtain a final concentration of 2.5 mg/mL. If local reaction at site of injection occurs, may use a lower concentration (i.e., 1 mg/mL) by following the reconstitution instructions for IV use.
- IV direct: physician or RN; undiluted; inject as a bolus over 3-5 seconds.
- SC: thighs or abdomen; rotate injection site. New injections should be given at least 2.5 cm from an old site. Avoid site that is tender, bruised, erythematous or indurated.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis, angioedema, rash, urticaria, pruritus.
- Cardiovascular: orthostatic/postural hypotension: may need pharmacological treatment, including hydration and/or adjustment of antihypertensive medications; CHF, peripheral edema.
- GI: nausea, vomiting, diarrhea, abdominal pain, anorexia, constipation.
- CNS: dizziness, headache, insomnia, paresthesia, seizures, posterior reversible encephalopathy syndrome (rare), peripheral neuropathy; motor neuropathy (can be fatal).
- Hematologic: thrombocytopenia, anemia, leukopenia, neutropenia, hemorrhage.
- Respiratory: dyspnea, cough, acute diffuse infiltrative pulmonary disease (e.g, pneumonitis) (rare).
- Progressive multifocal leukoencephalopathy (rare).
- Fatigue, fever, myalgia.
- Virus reactivation, including herpes zoster.
- Local reactions: erythema, pruritus with SC injection; if occurs, administer SC with a less concentrated solution (1 mg/mL instead of 2.5 mg/mL) or consider switching to IV route. Irritation with IV administration.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Previously untreated multiple myeloma (patients <u>unsuitable</u> for SCT): administer for nine 6-week treatment cycles: cycles 1-4: 1.3 mg/m^2 IV/SC twice weekly (days 1, 4, 8, 11, 22, 25, 29, 32) followed by a 10-day rest period (days 33-42); cycles 5-9: 1.3 mg/m^2 IV/SC once weekly (days 1, 8, 22, 29) followed by a 13-day rest period (days 30-42).
- Previously untreated multiple myeloma (patients <u>suitable</u> for SCT): administer for 3 to 6 treatment cycles: 1.3 mg/m^2 IV twice weekly (days 1, 4, 8, 11) followed by a rest period of up to 20 days (up to day 31) OR 1.5 mg/m^2 IV once weekly (days 1, 8, 15, 22) (4 week cycles).
- Relapsed multiple myeloma and relapsed/refractory mantle cell lymphoma: administer up to eight 3-week cycles: 1.3 mg/m^2 IV/SC twice weekly for 2 weeks (days 1, 4, 8, 11) followed by a 10-day rest period (days 12-21). For extended treatment beyond 8 cycles, maintenance schedule is once weekly for 4 weeks (days 1, 8, 15, 22) followed by a 13-day rest period (days 23-35).
- Previously untreated mantle cell lymphoma: administer up to eight 3-week cycles: 1.3 mg/m^2 IV/SC twice weekly for 2 weeks (days 1, 4, 8, 11) followed by a 10-day rest period (days 12-21).

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DOSAGE (Cont.)

- At least 72 hours should elapse between consecutive doses to minimize drug accumulation.
- Dosage in renal impairment: no dosage adjustment necessary; monitor patient closely for toxicities, especially if CrCl is 30 mL/min or less. Hemodialysis: administer dose post-dialysis.
- Dosage in hepatic impairment: no dosage adjustment necessary for mild impairment. For moderate to severe impairment (bilirubin greater than 1.5 x ULN), initial dose should be reduced to 0.7 mg/m^2; for subsequent doses, consider increasing dose to 1 mg/m^2 or decreasing dose to 0.5 mg/m^2 based on patient tolerability.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light.
- The 1 mg/mL and 2.5 mg/mL reconstituted solutions are clear and colourless.
- Stable for 42 days at 23°C or 4°C in NS at a concentration of 1 mg/mL in glass vials.
- Stable for 5 days at 22°C exposed to normal room light and for 7 days at 5°C protected from light in NS at a concentration of 1 mg/mL in polypropylene syringes.
- Stable for 21 days at 23°C exposed to fluorescent light or at 4°C protected from light in NS at a concentration of 2.5 mg/mL in glass vials and polypropylene syringes.

MISCELLANEOUS

- Contains mannitol.

REFERENCES

1, 4, 5, 129, 135, 165, 208.

NAME OF MEDICATION
BOTULINUM ANTITOXIN (EQUINE) *

OTHER NAMES
BAT ®, Botulism Antitoxin Heptavalent

CLASSIFICATION
Antitoxin

Botulinum antitoxin is available in different preparations that are not interchangeable. This monograph is specific to BAT ® from Emergent BioSolutions Canada.

INDICATIONS

- Treatment of symptomatic botulism following documented or suspected exposure to botulinum neurotoxin serotypes A, B, C, D, E, F, or G.

ADMINISTRATION

- Intermittent IV infusion (mandatory):
  - Bring vial to room temperature prior to use. If frozen, thaw contents by placing vial in the fridge at 2-8°C for approximately 14 hours. If rapid thawing is required, place vial at room temp for 60 minutes followed by a water bath at 37°C until thawed. Do NOT thaw in a microwave oven.
  - Do NOT shake vial during preparation to avoid foaming.
  - Withdraw from vial the required volume for the dose; it is important to note that the volume contained in vials will differ by lot number, ranging from 10-26 mL. If the dose is a partial vial, withdraw the entire contents from the vial to determine the total volume in the vial, then keep only the percentage partial dose required for the dose; refer to Dosage section.
  - Add the required volume to an IV bag of NS to obtain a final volume 10 times greater than that withdrawn from the vial (e.g., if the volume from vial to be diluted is 10 mL, add it to 90 mL of NS, for a total volume of 100 mL).
  - Do NOT use if the solution is turbid, cloudy or contains particulates other than translucent-to-white proteinaceous particulates.
  - Infuse via a non-pyrogenic, low-protein-binding in-line filter of 15-micron or less; smaller in-line filters (e.g., 0.2 microns or 1.2 microns) may be used but may result in slower infusion. Refer to Dosage section for infusion rates.
  - Must be administered by an infusion pump.
- Monitor vital signs throughout the infusion. Also monitor patient for signs and symptoms of hypersensitivity and infusion-related reaction during and after the infusion; refer to Potential Administration Hazards section.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis/anaphylactoid reactions, rash, urticaria, pruritus, erythema, wheezing, bronchospasm, cough, stridor, hypotension, tachycardia, laryngeal edema, angioedema, shock. Increased risk of severe reaction in patients with a history of exposure to equine-derived blood products (antivenom/antitoxin), allergy to horses, asthma or seasonal allergies; refer to Dosage section for initial infusion rate in patients at risk. Discontinue infusion should patient develop a hypersensitivity reaction.
- Infusion-related reactions: fever, chills, headache, nausea, vomiting, arthralgia, myalgia, fatigue, vasovagal reaction. Reduce infusion rate and administer symptomatic therapy; discontinue infusion if symptoms worsen.
- Risk of transmission of infectious agents, including viruses, cannot be completely eliminated as product is made from pooled equine plasma.

DOSAGE

- Adults (17 years of age and older): the dose is the total volume contained in one vial, diluted as per the Administration section. Infuse at an initial rate of 0.5 mL/min IV for 30 minutes; if tolerated, the rate can be doubled every 30 minutes, up to a maximum rate of 2 mL/min.

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DOSAGE (Cont.)

- Pediatrics:
  - 1 to 16 years old: refer to manufacturer's recommendations for weight-based pediatric dose as it ranges from 20-100% of the volume in one vial. Dilute the dose as per the Administration section. Infuse at an initial rate of 0.01 mL/kg/min (maximum of 0.5 mL/min) IV for 30 minutes; if tolerated, the rate can be increased by 0.01 mL/kg/min every 30 minutes, up to a maximum infusion rate of 0.03 mL/kg/min, not exceeding 2 mL/min.
  - Less than 1 year old: the dose is 10% of the volume of one vial, regardless of patient's weight. Dilute the dose as per the Administration section. Infuse at an initial rate of 0.01 mL/kg/min IV for 30 minutes; if tolerated, the rate can be increased by 0.01 mL/kg/min every 30 minutes, up to a maximum infusion rate of 0.03 mL/kg/min.
- For patients at risk of hypersensitivity reaction, initial infusion rate should be the lowest possible (less than 0.01 mL/min); refer to Potential Administration Hazards section.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the freezer at -15°C or below.
- Thawed solution is stable at 2-8°C for a maximum of 36 months or until 48 months from the date of manufacture, whichever comes first. Do not refreeze.
- Thawed solution should be clear or slightly opalescent.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment should be immediately available for the treatment of hypersensitivity reactions.
- BAT ® contains immune globulin fragments from horses immunized with serotypes A, B, C, D, E, F and G of botulinum toxoid and toxin.
- Contains maltose; may give falsely-elevated glucose readings with glucometers using glucose dehydrogenase pyrroloquinoline-quinone (GDH-PQQ) methods for testing.

REFERENCES

135, 212, 238, 240, 265, 271, 290.

* Available via Population and Public Health Division of Ontario Ministry of Health and Long-Term Care

NAME OF MEDICATION
BRENTUXIMAB VEDOTIN

OTHER NAMES
Adcetris ®, Monomethyl auristatin E

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of previously untreated stage IV Hodgkin lymphoma.
- Post-autologous stem cell transplant (ASCT) consolidation treatment of patients with Hodgkin lymphoma at increased risk of relapse or progression.
- Treatment of Hodgkin lymphoma after failure of ASCT or after failure of at least two multi-agent chemotherapy regimens in patients who are not ASCT candidates.
- Treatment of previously untreated systemic anaplastic large cell lymphoma, CD30-expressing peripheral T-cell lymphoma-not otherwise specified or CD30-expressing angioimmunoblastic T-cell lymphoma.
- Treatment of systemic anaplastic large cell lymphoma after failure of at least one multi-agent chemotherapy regimen.
- Treatment of primary cutaneous anaplastic large cell lymphoma or CD30-expressing mycosis fungoides who have had prior systemic therapy.

ADMINISTRATION

- **Ensure premedication has been administered if the patient experienced an infusion reaction with a prior brentuximab vedotin infusion; refer to Potential Administration Hazards section.**
- Reconstitute each 50 mg vial with 10.5 mL of SWFI to obtain a concentration of 5 mg/mL. Direct the stream toward the wall of the vial. Gently swirl the vial to aid dissolution. Do NOT shake.
- Intermittent IV infusion: dilute dose in at least 100 mL of NS, D5W or RL to obtain a final concentration between 0.4 and 1.8 mg/mL. Gently invert the bag to mix the solution. Administer over 30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, anaphylaxis, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Infusion-related reactions: immediate or up to 2 days after administration; more common in patients who develop antibodies to brentuximab vedotin. Symptoms include chills, nausea, swelling, throat tightness, urticaria, pruritus, rash, erythema, dyspnea, cough, wheezing, flushing, hypotension, dizziness and fever. In cases of mild to moderate reactions, **premedicate** with an antipyretic (e.g., acetaminophen), an antihistamine (e.g., diphenhydramine) and a corticosteroid for subsequent infusions; discontinue treatment in severe cases.
- Cardiovascular: supraventricular arrhythmia, hypotension, peripheral edema, thromboembolism.
- GI: nausea, vomiting, diarrhea, constipation, oral candidiasis, abdominal pain (can be severe), anorexia, stomatitis, dyspepsia, ileus, enterocolitis, GI perforation (rare), acute pancreatitis (can be fatal).
- CNS: anxiety, headache, insomnia, peripheral neuropathy (sensory or motor), dizziness.
- Endocrine and metabolic: hyperglycemia.
- Hematologic: thrombocytopenia, anemia, lymphopenia. Neutropenia: if grade 3 or 4, growth factor administration (e.g., filgrastim) should be considered.
- Hepatic: increased transaminases and/or bilirubin.
- Musculoskeletal: arthralgia, back pain, myalgia.
- Respiratory: pneumonia, upper respiratory tract infection, cough, dyspnea, nasal congestion, pneumonitis, pneumothorax; interstitial lung disease, acute respiratory distress syndrome (rare).
- Progressive multifocal leukoencephalopathy.
- Tumour lysis syndrome.
- Sepsis, septic shock, fatigue, fever, pain in extremity (can be severe),.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- Previously untreated stage IV Hodgkin lymphoma: 1.2 mg/kg (maximum of 120 mg) IV every 2 weeks for a maximum of 12 doses.
- All other listed indications: 1.8 mg/kg (maximum of 180 mg) IV every 3 weeks for 6 to 16 cycles.
- Dosage in renal impairment: no adjustment required if CrCl is 30 mL/min or greater. Avoid use if CrCl is less than 30 mL/min.
- Dosage in hepatic impairment:
  - Mild impairment (Child-Pugh class A): dose reduction is required:
    - Previously untreated stage IV Hodgkin lymphoma: 0.9 mg/kg/dose (maximum of 90 mg IV).
    - All other listed indications: 1.2 mg/kg/dose (maximum of 120 mg) IV.
  - Moderate to severe impairment (Child-Pugh class B or C): avoid use.
- Dosage adjustment due to toxicity: refer to the manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Vials are stable for no more than 5 days (120 hours) cumulative exposure time at 25°C.
- Reconstituted solution is stable for 24 hours at room temp or between 2-8°C. The solution should be clear to slightly opalescent, colourless and free of visible particles.
- Stable for 24 hours between 2-8°C in NS, D5W and LR at concentrations of 0.4-1.8 mg/mL in PVC, ethylene vinyl acetate (EVA), polyolefin, polypropylene or polyethylene containers.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment should be immediately available for the treatment of hypersensitivity and infusion-related reactions.
- As a minimum, a complete blood count should be performed before each dose.
- Combination with bleomycin is contraindicated due to increased pulmonary toxicity.

REFERENCES

1, 5, 129, 135, 165, 208, 337.

NAME OF MEDICATION
BUPIVACAINE

OTHER NAMES
Marcaine ®, Sensorcaine ®, Vivacaine ™

CLASSIFICATION
Local anesthetic

INDICATIONS

- Local or regional anesthesia and analgesia.

ADMINISTRATION

- Local infiltration (including for dental use).
- Peripheral nerve block including nerve plexus block.
- Retrobulbar block.
- Central nerve block: spinal (intrathecal) or epidural (including caudal block).
- Nurses may not administer epidural boluses unless using an infusion pump and following Unit policy.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity (rare): urticaria, pruritus, erythema, edema.
- Cardiovascular: cardiovascular toxicity is a risk with large bolus techniques and can result in depression of the myocardium, hypotension or hypertension, decreased cardiac output, heart block, bradycardia, ventricular arrhythmias, and cardiac arrest that is notoriously difficult to treat; usually dose-related and due to inadvertently high plasma levels.
- CNS: excitation or depression: anxiety, restlessness, dizziness, tremors, drowsiness, convulsions, loss of consciousness; usually dose-related and due to inadvertently high plasma levels.
- Ophthalmic: blurred vision.
- Otic: tinnitus.
- Respiratory: respiratory arrest.
- Local reactions: edema at insertion site.

DOSAGE

- Available as 3 concentrations: 0.25% and 0.5% solutions (with or without epinephrine 1:200 000), and 0.75% solution without epinephrine. All three concentrations may produce complete sensory block, but the effect on motor function increases as concentration increases.
- The dose is influenced by many factors (e.g., area to be anesthetized, degree of anesthesia required, individual tolerance).
- The highest concentration (0.75% solution) is not recommended for obstetrical anesthesia and analgesia.
- Local inflitration: up to 60 mL of the 0.25% solution OR up to 30 mL of the 0.5% solution.
- Peripheral nerve blocks: 5-40 mL of the 0.25% solution OR 5-30 mL of the 0.5% solution.
- Retrobulbar block: 2-4 mL of the 0.75% solution.
- Caudal block: 15-30 mL of the 0.25% or 0.5% solution.
- Epidural (other than caudal block): 5-20 mL of the 0.25% solution, 5-30 mL of the 0.5% solution OR 10-20 mL of the 0.75% solution (where the 0.75% solution is used only for single dose anesthesia). Continuous infusions may also be administered (e.g., 10 mL/hr of the 0.25% solution). Concentrations of 0.125% to 0.5% have been used for anesthesia during labour.
- Spinal: 0.8-2 mL using the 0.75% hyperbaric solution.
- A single dose is usually sufficient. Usual maximum of 175 mg per single dose (70 mL of the 0.25% solution, 35 mL of the 0.5% solution or 23 mL of the 0.75% solution) without epinephrine or 225 mg (90 mL of the 0.25% solution, 45 mL of the 0.5% solution or 30 mL of the 0.75% solution) with epinephrine. However, certain patients and procedures may require smaller or larger doses. Doses should be repeated no more often than q3h. Maximum of 400 mg (160 mL of the 0.25% solution, 80 mL of the 0.5% solution, or 53 mL of the 0.75% solution) per 24 hours.
- Dosage reductions should be considered for elderly, actutely ill, or debilitated patients, and for those with impaired cardiovascular function or severe hepatic impairment.
- Refer to manufacturers' monographs for more specific dosing information.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and ampoules at room temp. Protect solutions containing epinephrine from light. Do not freeze.
- Stable for 32 days at room temp (exposed to light) or in the fridge protected from light in NS at a concentration of 1.25 mg/mL (0.125%) in polypropylene syringes.
- Stable for 28 days at 5°C protected from light in NS at concentrations of 0.01-37.5 mg/mL when combined with either hydromorphone 0.01-43 mg/mL or morphine 0.01-43 mg/mL or fentanyl 0.15-43 mcg/mL in polypropylene syringes or non-DEHP bags.
- Stable for 60 days at room temp (exposed to light) and in the fridge protected from light in NS at a concentration of 2.5 mg/mL (0.25%) when combined with morphine 5 mg/mL in polypropylene syringes.
- Stable for 91 days at room temp (exposed to light) or in the fridge protected from light in NS at a concentration of 2.5 mg/mL (0.25%) when combined with hydromorphone 0.02 mg/mL or 0.04 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Emergency drugs, including oxygen, and resuscitation equipment should be immediately available.
- Contraindicated for IV regional anesthesia (Bier block).
- Do not use in obstetric paracervical block anesthesia.
- Local anesthetics that contain preservatives should not be used for epidural or spinal anesthesia.
- 0.25% = 2.5 mg/mL; 0.5% = 5 mg/mL; 0.75% = 7.5 mg/mL.

REFERENCES

1, 4, 5, 14, 15, 95, 135, 208.

NAME OF MEDICATION
BUSERELIN

OTHER NAMES
Suprefact ®, Suprefact ® Depot

CLASSIFICATION
Gonadotropin-releasing hormone analog

INDICATIONS

- Palliative treatment of hormone-dependent advanced prostate cancer.

ADMINISTRATION

Regular-release formulation:
- SC: rotate injection site.

Depot formulation:
- SC: in lateral abdominal wall. Implant rods should be kept horizontal before the injection. A local anesthetic may be used before the injection.

POTENTIAL ADMINISTRATION HAZARDS
- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: allergic asthma, dyspnea, anaphylactic/anaphylactoid shock, rash, pruritus.
- Cardiovascular: hot flushes.
- GI: nausea, vomiting.
- Transient disease exacerbation due to testosterone flare reaction: bone pain, neuropathy, urinary retention, urethral obstruction, hematuria, thrombosis, lymphedema, spinal cold compression.
- Local reactions: transient; pain, irritation, swelling, urticaria.

DOSAGE

Regular-release formulation:
- Induction: 500 mcg (0.5 mL) SC q8h for 7 days.
- Maintenance: 200 mcg (0.2 mL) SC daily.

Depot formulation:
- Inject SC the contents of one applicator, either 6.3 mg of buserelin every 2 months or 9.45 mg of buserelin every 3 months.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Regular-release formulation:
- Store vials between 15-25°C. Protect from light and from heat. Do not freeze.
- Vials are stable for 14 days at room temp after first opening.

Depot formulation:
- Store applicators between 15-30°C. Protect from excessive heat.

MISCELLANEOUS

- Note that the depot formulation is available as two strengths: 6.3 mg (2-month formulation) and 9.45 mg (3-month formulation).
- Adjunctive therapy with an antiandrogen (e.g., flutamide, bicalutamide, nilutamide, cyproterone) can be started about 7 days prior to the first buserelin injection and continued for 4 weeks following the first injection to prevent the initial surge in testosterone levels.

REFERENCES

5, 129, 165.

Full revision 2021

NAME OF MEDICATION
BUSULFAN

OTHER NAMES
Busulfex ™, Myleran ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Conditioning regimen (in combination with other antineoplastics and/or radiotherapy) prior to bone marrow transplantation for various malignancies.

ADMINISTRATION

- **Ensure premedication with an anticonvulsant (e.g., phenytoin, levetiracetam, a benzodiazepine or valproic acid) to prevent seizures has been administered; refer to Dosage section.**
- Intermittent IV infusion: dilute with a volume of NS or D5W that is 10 times the volume of busulfan to give a final concentration of approximately 0.5 mg/mL. Administer through a central line over 2 hours for the dose of 0.8 mg/kg and over 3-4 hours for the dose of 3.2 mg/kg. Must be administered by an infusion pump.
- Flush the central venous catheter with 5 mL of NS or D5W before and after the infusion.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid reaction, serum sickness, rash, pruritus.
- Cardiovascular: chest pain, tachycardia, hypertension, vasodilation, thromboembolism, edema.
- GI: nausea, vomiting, diarrhea, constipation, dry mouth, stomatitis, anorexia, abdominal pain.
- CNS: headache, dizziness, insomnia, anxiety, depression, seizures.
- Electrolyte disturbances: hypokalemia, hypomagnesemia, hypocalcemia, hypophosphatemia.
- Endocrine and metabolic: hyperglycemia.
- Renal: dysuria, hematuria, oliguria.
- Respiratory: dyspnea, rhinitis, cough.
- Tumour lysis syndrome.
- Fatigue, fever, chills, pain, asthenia.
- Contains the solvent dimethylacetamide, which may be associated with hepatotoxicity and neurotoxicity. The degree to which this solvent contributes to the toxicity of busulfan is unknown.
- Local reactions: inflammation and pain at injection site.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- **Premedicate all patients with an anticonvulsant (e.g., phenytoin [loading dose required], levetiracetam, a benzodiazepine or valprioc acid) beginning 12-24 hours prior to the first dose of busulfan and continue until 24-48 hours after the last dose of busulfan.**
- 0.8 mg/kg IV q6h for 4 consecutive days, for a total of 16 doses OR 3.2 mg/kg IV once daily for 4 days; the dose should be based either on the ideal body weight or actual body weight, whichever is lower, except in obese patients.
- Dosage in obesity: dose should be calculated using adjusted body weight (in kg) as follows:
  Adjusted body weight = ideal body weight + 0.25 x (actual body weight - ideal body weight)
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Do not freeze.
- Incompatible with polycarbonate. Do not use syringes, filter needles, or IV tubing that contain polycarbonate.
- Stable for 8 hours (including infusion time) at room temp in D5W or NS at a concentration of 0.5 mg/mL.
- Stable for 12 hours (including infusion time) in the fridge in NS at a concentration of 0.5 mg/mL.

MISCELLANEOUS

- Use caution when administering busulfan to patients with a history of a seizure disorder, head trauma, or receiving medications that may lower the seizure threshold.

REFERENCES

1, 4, 5, 40, 95, 129, 135, 165.

NAME OF MEDICATION
C1 ESTERASE INHIBITOR, HUMAN

OTHER NAMES
Berinert ®, C1 inhibitor, Cinryze ®, Haegarda ™

CLASSIFICATION
Blood product

INDICATIONS

- Treatment of acute abdominal, facial, or laryngeal attacks of hereditary angioedema (HAE) of moderate to severe intensity in adults and children (Berinert ®).
- Routine prevention of angioedema attacks in adults and adolescents with HAE (Cinryze ® and Haegarda ™).

ADMINISTRATION

- Ensure that Berinert ®, Cinryze ® and Haegarda ™ vials and their diluents have reached room temp.
- **Berinert ®**: reconstitute the 500 and 1500 international unit vials with 10 mL and 3 mL, respectively, of SWFI (diluent provided) using the transfer set provided (or any double-ended needle with a vented filter spike) to obtain a final concentration of 50 international units/mL or 500 international units/mL, respectively. Gently swirl until completely dissolved; do NOT shake. Transfer dose to the provided syringe. Refer to manufacturer's instructions for a more detailed procedure.
- **Cinryze** ®: reconstitute each 500 international unit vial with 5 mL of SWFI (diluent provided) using a filter transfer device (or any double-ended needle) to obtain a final concentration of 100 international units/mL. Gently swirl until completely dissolved; do NOT shake. Transfer dose to a silicone-free syringe. Refer to manufacturer's instructions for a more detailed procedure.
- **Haegarda** ™: reconstitute the 2000 and 3000 international unit vial with 4 mL and 6 mL, respectively, of SWFI (diluent provided) using the filter transfer set provided to obtain a final concentration of 500 international units/mL. Gently swirl until completely dissolved; do NOT shake. Transfer dose to the SC infusion set or the hypodermic needle. Refer to manufacturer's instructions for a more detailed procedure.
- IV direct (Berinert ® and Cinryze ®): physician or RN. Administer in a separate line. Administer Berinert ® 1500 slowly, Berinert ® 500 at a rate of 4 mL/min (200 international units/min) and Cinryze ® over 10 minutes at a rate of 1 mL/min (100 international units/min).
- SC (Haegarda ™): abdomen or other SC injection site; rotate injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, chest tightness, wheezing, hypotension, anaphylaxis; hypersensitivity symptoms (apart from pruritus and urticaria) may mimic HAE attacks.
- Cardiovascular: thrombotic events (arterial thrombosis, myocardial infarction, stroke, deep vein thrombosis, pulmonary embolism) (Berinert ® and Cinryze ®).
- GI: dysgeusia (Berinert ®), abdominal pain, nausea, vomiting, diarrhea.
- CNS: headache, dizziness.
- Musculoskeletal: muscle spasms, back pain, facial pain (Berinert ®).
- Respiratory: nasopharyngitis (Berinert ® and Haegarda ™).
- Risk of transmission of infectious agents, including viruses, and theoretically the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.
- Local reactions: pain, erythema, rash, inflammation, hematoma at injection site.

DOSAGE

- Berinert ®: 20 international units/kg IV. A single dose is usually sufficient, however, some patients may require additional doses.
- Cinryze ®: 1000 international units IV every 3-4 days; frequency may vary depending on the patient. In case of inadequate response, doses up to 2500 international units (up to 100 international units/kg) every 3-4 days may be considered on an individual patient basis.
- Haegarda ™: 40 or 60 international units/kg SC every 3-4 days.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Berinert **®**:
- Store vials between 2-30°C. Protect from light. Do not freeze.
- The reconstituted solution of the 500 international unit vials should be colourless and clear and the solution from the 1500 international unit vials should be colourless, clear to slightly opalescent.
- The reconstituted solution is stable for 8 hours at room temp; however, the Canadian manufacturer recommends infusing immediately after reconstitution. Do not refrigerate or freeze reconstituted solution.

Cinryze ®:
- Store vials between 2-25°C. Protect from light. Do not freeze.
- The reconstituted solution should be colorless to slightly blue and clear.
- The reconstituted solution is stable for 24 hours between 15-25°C; however, the Canadian manufacturer recommends infusing within 3 hours of reconstitution.

Haegarda ™:
- Store vials between 2-30°C. Protect from light. Do not freeze.
- The reconstituted solution should be colourless and clear.
- The reconstituted solution is stable for 48 hours at room temp (up to 30°C); however, the Canadian manufacturer recommends infusing withing 8 hours of reconstitution.

MISCELLANEOUS

- Emergency drugs (epinephrine, antihistamines and corticosteroids) should be immediately available for the treatment of hypersensitivity reactions.

REFERENCES

1, 5, 40, 95, 135.

NAME OF MEDICATION
CABAZITAXEL

OTHER NAMES
Jevtana ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of patients with castration-resistant (hormone refractory) metastatic prostate cancer.

ADMINISTRATION

- **Ensure premedication has been administered; refer to Potential Administration Hazards and Dosage sections.**
- Intermittent IV infusion: for 10 mg/mL vial: omit Step 1 and proceed directly to Step 2. For 40 mg/mL vial: follow both Step 1 and Step 2.
  - Step 1: (40 mg/mL vial only) withdraw the **entire** content of the diluent (5.67 mL or 5.7 mL depending on brand) and transfer to the 60 mg/1.5 mL of the cabazitaxel concentrate; inject slowly and direct the needle towards the inside wall of the vial to minimize foaming. Mix gently by repeated inversions for at least 45 seconds to obtain a clear solution; do NOT shake. Let stand for about 5 minutes to allow some foam to dissipate. The resulting concentrate-diluent solution contains 10 mg/mL of cabazitaxel, with at least 6 mL of deliverable volume. Proceed immediately (within 60 minutes) to step 2.
  - Step 2: (10 mg/mL and 40 mg/mL vials) withdraw the required amount of the 10 mg/mL solution and dilute in 250 mL in a non-PVC container of D5W or NS to obtain a final concentration between 0.1-0.26 mg/mL. Mix by gently inverting the bag. Infuse over 60 minutes.
- Do NOT use PVC bags (with or without DEHP) for dilution and polyurethane infusion sets for administration; PVC (with or without DEHP) or polyethylene infusion sets can be used for infusing the solution.
- Administer with a 0.22 micron in-line filter.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: bronchospasm, hypotension, rash, erythema; may occur within a few minutes of initiating cabazitaxel infusion, especially during first and second infusions. A **premedication** is recommended before each treatment (refer to Dosage section). Discontinue treatment if severe.
- Cardiovascular: hypotension, arrhythmias.
- GI: nausea, vomiting, abdominal pain, mucositis, dysgeusia, anorexia, weight loss, constipation, diarrhea (may be severe and lead to dehydration, electrolyte imbalance and renal failure if left untreated); GI hemorrhage and perforation, ileus, colitis.
- CNS: headache, dizziness, peripheral neuropathy.
- Hematologic: anemia, leukopenia, neutropenia (including febrile neutropenia), thrombocytopenia.
- Musculoskeletal: pain, arthralgia, muscle spasms.
- Renal: hematuria.
- Respiratory: cough, dyspnea.
- Asthenia, fatigue, edema, fever.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- **Premedicate at least 30 minutes before each infusion with an antihistamine (e.g., diphenhydramine 25 mg IV/PO or equivalent), a corticosteroid (dexamethasone 8 mg IV/PO or equivalent) and a H_2 antagonist (ranitidine 50 mg PO or famotidine 20 mg IV).**
- Cabazitaxel 20 mg/m^2 IV on day 1 every 3 weeks, with prednisone (or prednisolone) 10 mg daily on days 1-21. Cabazitaxel at a dose of 25 mg/m^2 IV may be used for certain patients, at the clinician's discretion.
- Dosage in renal impairment: use with caution if CrCl is less than 15 mL/min as information is limited.
- Dosage in hepatic impairment:
  - Mild impairment (total bilirubin greater than ULN but no greater than 1.5 x ULN or AST greater than 1.5 x ULN): reduce dose to 20 mg/m^2 IV, use with caution and monitor patient closely.
  - Moderate impairment (total bilirubin greater than 1.5 but no greater than 3 x ULN and any AST): maximum dose of 15 mg/m^2 IV; use with caution and monitor patient closely.
  - Severe impairment: contraindicated.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Do not refrigerate.
- 10 mg/mL vials only: opened vials are stable for 28 days at room temp or in the fridge if pierced by a needle or for 10 days at room temp or in the fridge if a spike is used for solution withdrawal.
- 40 mg/mL vials only: once the diluent is added to the vial, the concentrate-diluent solution (10 mg/mL) is stable for 60 minutes between 15-30°C.
- Stable for 8 hours at room temp and for 48 hours in the fridge in D5W or NS at concentrations between 0.1-0.26 mg/mL. The stability includes the 60 minutes for infusion.

MISCELLANEOUS

- Emergency drugs and resuscitation equipement must be immediately available for the treatment of hypersensitivity reactions.
- Avoid vaccination with a live or live-attenuated vaccine while on cabazitaxel.

REFERENCES

1, 5, 46, 129, 135, 165, 294.

NAME OF MEDICATION
CAFFEINE CITRATE

OTHER NAMES
Peyona ™

CLASSIFICATION
Stimulant

INDICATIONS

- Apnea of prematurity.

ADMINISTRATION

- Intermittent IV infusion: administer undiluted or diluted with D5W; infuse loading dose over 30 minutes and maintenance dose over 3-15 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: tachycardia, palpitations.
- GI: GI distention, feeding intolerance, nausea, vomiting, gastritis, increased gastric aspirate, necrotizing enterocolitis.
- CNS: irritability, restlessness, jitteriness, insomnia, seizures.
- Electrolyte disturbances.
- Endocrine and metabolic: hyperglycemia, hypoglycemia.
- Renal: increased urine output.
- Local reactions: phlebitis and inflammation at infusion site.

DOSAGE

- Loading dose: 10 mg/kg/dose (range 5-20 mg/kg/dose)(caffeine base) IV. A second loading dose of 5-10 mg/kg IV (caffeine base) may be repeated after 24 hours in cases of insufficient response. Do not exceed total cumulative dose of 40 mg/kg.
- Maintenance dose (to be started 24 hours after loading dose): 2.5-5 mg/kg/dose (caffeine base) IV once daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules of between 15-30°C.
- Stable for 24 hours at room temp in D5W, D5-1/4NS.

MISCELLANEOUS

- IV solution of caffeine citrate was used PO to treat apnea of prematurity prior to availability of oral solution.
- Therapeutic drug levels for apnea of prematurity: 40-100 micromol/L (trough level, taken 0 to 4 hours before next dose).
- Caffeine base 10 mg = caffeine citrate 20 mg.

REFERENCES

1, 4, 5, 6, 40, 82, 102, 135, 319.

Full revision 2021

NAME OF MEDICATION
CALCITONIN SALMON

OTHER NAMES
Calcimar ®, Salcatonin

CLASSIFICATION
Parathyroid hormone

INDICATIONS
- Treatment of symptomatic Paget's disease of the bone.
- Treatment of hypercalcemia emergencies when a rapid decrease in serum calcium is required.
- Analgesic adjunct in osteoporotic vertebral crush fractures or intractable cancer pain from bone metastases.
- Treatment of osteoporosis.
- Treatment of phantom limb pain.

ADMINISTRATION
- Intermittent IV infusion (for phantom limb pain only): dilute dose in 50 mL of NS and infuse over 20-60 minutes. (At TOH, dilute dose in 250 mL of NS and infuse over 6 hours).
- SC, IM (IM route is preferred if injection volume is greater than 2 mL; use multiple sites): rotate injection sites.
- Consider performing a skin test prior to initiating therapy in patients with suspected sensitivity to calcitonin salmon. Consult manufacturer's monograph for more information.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: bronchospasm, swelling of the tongue or throat, rash, pruritus, urticaria, anaphylaxis.
- Cardiovascular: transient flushing of the face, ears, hands and feet; typically begins within minutes after injection and lasts up to 2 hours; may be minimized by administration at bedtime.
- GI: nausea, vomiting, anorexia, metallic or salty taste, abdominal pain.
- CNS: sedation, dizziness, headache, vertigo, paresthesias.
- Musculoskeletal: hypocalcemic tetany, pain.
- Renal: increased frequency of urination.
- Local reactions: swelling, pain, erythema, pruritus, inflammation.

DOSAGE
- Paget's disease: Initial dose of 100 international units SC/IM daily. Maintenance dose of 50-100 international units SC/IM 3 times weekly or daily.
- Hypercalcemia: 4 international units/kg SC/IM q12h, may be increased to 8 international units/kg q12h after 1 or 2 days. If response is unsatisfactory after 2 more days, may increase further to 8 international units/kg q6h.
- Analgesic adjunct in osteoporosis vertebral fractures: 50-100 international units SC/IM daily.
- Bone metastases: 50-200 international units SC/IM daily. In the palliative setting, may initiate with 50 international units SC at bedtime for 1-2 days and increase as needed in increments of 50 international units up to 100-200 international units SC q12h.
- Osteoporosis: 100 international units SC/IM every other day.
- Phantom limb pain: 200 international units IV daily for a few days.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Product in vial is stable for 2 weeks when exposed to temperatures between 15-30°C.

MISCELLANEOUS
- Emergency drugs (e.g., epinephrine) and resuscitation equipment should be immediately available for the treatment of hypersensitivity reactions.

REFERENCES
1, 2, 5, 95, 300, 304, 362, 457.

Full revision 2021

NAME OF MEDICATION
CALCITRIOL

OTHER NAMES
Calcijex ®, 1,25-Dihydroxy-cholecalciferol, 1,25-Dihydroxyvitamin D3

CLASSIFICATION
Vitamin D analog

INDICATIONS

- Management of hypocalcemia, secondary hyperparathyroidism, and osteodystrophy in adults and children with chronic renal failure undergoing dialysis.
- Management of hypocalcemic tetany in premature infants.

ADMINISTRATION

- IV direct: physician or RN. Administer IV as a bolus, undiluted or diluted to a concentration of 0.5 mcg/mL with SWFI, NS or D5W; for patients in end-stage renal disease, administer through the catheter at the end of hemodialysis.
- Intermittent IV infusion: usually for pediatric administration; administer undiluted over 15 minutes using PVC-free bags and administration sets.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, anaphylaxis.
- Cardiovascular: hypertension.
- GI: nausea, vomiting, constipation, dyspepsia, abdominal pain, anorexia, dehydration.
- CNS: headache, somnolence, paresthesias.
- Electrolyte disturbances: hypercalcemia.
- Musculoskeletal: myalgia.
- Asthenia.
- Urinary tract infection.
- Vitamin D intoxication.
- Local reactions: pain and redness at injection site.

DOSAGE

- Adults: Initial: 0.25-0.5 mcg (0.01 mcg/kg) IV three times weekly (every other day). May be increased by 0.25-1 mcg every 2-4 weeks. Usual dose: 0.5-3 mcg (0.01-0.05 mcg/kg) IV three times weekly. Doses up to 4 mcg IV three times weekly have been used.
- Pediatrics: initial dose is based on parathyroid hormone level: 0.0075-0.025 mcg/kg IV three times weekly (every other day). Dose may be increased by 0.25-1 mcg every 2-4 weeks.
- Infants: hypocalcemic tetany in premature infants: 0.05 mcg/kg IV daily for 5-12 days.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 15-30°C. Protect from light. Do not freeze.
- Compatible with D5W and NS.
- Stable for 20 days at room temp or 7 days in the fridge undiluted in polypropylene syringes.
- Stable for 8 hours at room temp, exposed to normal room light, in SWFI, D5W or NS at a concentration of 0.5 mcg/mL in polypropylene syringes.

MISCELLANEOUS

REFERENCES

1, 2, 4, 5, 6, 40, 82, 102, 135, 143, 182, 208.

Full revision 2021

NAME OF MEDICATION
CALCIUM CHLORIDE

OTHER NAMES
Calciject

CLASSIFICATION
Electrolyte

Do NOT confuse calcium chloride with other calcium salts.
This monograph is specific to calcium CHLORIDE.

INDICATIONS

- For severe cardiotoxicity or cardio-respiratory arrest due to hyperkalemia or hypermagnesemia.
- For shock associated with beta-blocker or calcium channel blocker overdose.
- Treatment of hypocalcemia.

ADMINISTRATION

- IV direct: physician or RN. ***Cardiac monitoring***.
  - For severe cardiotoxicity or cardio-respiratory arrest due to hyperkalemia or hypermagnesemia, treatment of severe hyperkalemia, treatment of life-threatening hypocalcemia (at TOH): administer undiluted over 2-5 min.
  - For shock associated with beta-blocker or calcium channel blocker overdose: administer undiluted over 5-10 min.
  - For other indications: administer undiluted, maximum rate of 100 mg/min.
- Intermittent IV infusion: dilute each 1000 mg in a minimum of 50 mL of NS or D5W (maximum concentration of 20 mg/mL). Administer dose over 60 minutes; do NOT exceed 6000 mg/hr or 90 mg/kg/hr. **Note that the *gluconate* salt is usually preferred for administration in a bag.**
- Continuous IV infusion: dilute in NS or D5W (maximum concentration of 20 mg/mL). Refer to Dosage section.
- Should be administered slowly through a small needle into a large vein, preferably central or deep vein, to avoid venous irriation, too rapid increase in serum calcium and extravasation of calcium solution into the surrounding tissue with resultant necrosis.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: rapid IV administration may cause bradycardia, vasodilation, hypotension, cardiac arrhythmias, syncope and cardiac arrest.
- GI: metallic or chalky taste.
- Electrolyte disturbances: hypercalcemia.
- Tingling sensations, a sense of oppression or "heat waves".
- Local reactions: burning, vein irritation.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- For severe cardiotoxicity or cardio-respiratory arrest due to hyperkalemia or hypermagnesemia, treatment of severe hyperkalemia, treatment of life-threatening hypocalcemia (at TOH): 500-1000 mg IV direct. Dose may be repeated as needed.
- For shock associated with beta-blocker or calcium-channel blocker overdose:
  - 1000-2000 mg or 20 mg/kg (maximum of 1000-2000 mg/dose) IV direct; dose may be repeated 3-4 times every 10-20 minutes;
  - May also be administered as continuous IV infusion: initiate at 20-50 mg/kg/hr; titrate to response.
- Hypocalcemia (non-severe): 200-1000 mg IV. Dose may be repeated as required according to patient’s response and serum calcium levels.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store syringes between 20-25°C. Do not freeze.
- Store vials between 15-30°C. Do not freeze.
- Compatible with NS, dextrose solutions and RL.
- Stable for 7 days at room temp, exposed to light, in D5W or NS at concentrations of 10 and 13.3 mg/mL in PVC containers.
- Stable for 7 days in the fridge in NS at a concentration of 16 mg/mL.

MISCELLANEOUS

- Caution in digitalized patients; slow IV administration and close monitoring are recommended.
- Calcium chloride 10% solution contains 100 mg/mL of calcium chloride, equivalent to 1.36 mEq (0.68 mmol) calcium or 27.2 mg calcium per mL.
- Elemental calcium content is 27% (compared to 9% for the gluconate salt), which correspondes to 270 mg (13.5 mEq or 6.75 mmol) per g of calcium chloride.

REFERENCES

1, 2, 3, 4, 5, 40, 95, 124, 135, 146, 151, 208, 263, 328, 339, 366, 367, 462.

NAME OF MEDICATION
CALCIUM DISODIUM EDETATE *

OTHER NAMES
Calcium disodium EDTA, Edetate calcium disodium, Sodium calcium edetate, Calcium disodium Versenate ®

CLASSIFICATION
Chelating agent

Do NOT confuse calcium disodium edetate with calcium salts.
This monograph is specific to calcium DISODIUM EDETATE.

INDICATIONS

- Adjunct in the treatment of acute lead poisoning, lead encephalopathy and lead nephropathy.
- May be effective in the treatment of acute intoxication caused by other heavy metals such as cadmium, chromium, manganese, nickel, zinc, and possibly vanadium.
- May be effective in the treatment of poisoning by radioactive and nuclear fission products such as plutonium, thorium, uranium and yttrium.

ADMINISTRATION

- Intermittent IV infusion, Continuous IV infusion: dilute in 250-500 mL of NS or D5W to obtain a concentration of 2-4 mg/mL, (e.g., 1000 mg dissolved in 250-500 mL); maximum concentration of 5 mg/mL. Administer over 8-12 hours or continuously over 24 hours.
- Patients with lead encephalopathy and cerebral edema may experience a lethal increase in intracranial pressure following rapid IV infusion. IM route is preferred for these patients; if must be given IV, the drug must be infused slowly.
- IM: add a local anesthetic (1 mL of 1% lidocaine to each mL of calcium disodium edetate) before injection. Total daily dose should be equally divided into 2-6 doses and given at 4-12 hour intervals. Rotate injection sites.
- Adequate hydration and establishment of good urine output before drug administration will help to prevent nephrotoxicity. Avoid excess fluids in patients with lead encephalophathy.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: histamine-like reactions including sneezing, nasal congestion and lacrimation may occur 4-8 hours after IV infusion.
- Cardiovascular: hypotension, cardiac arrhythmias.
- GI: nausea, vomiting, anorexia, GI irritation.
- CNS: tremors, headache, numbness, paresthesias; increased intracranial pressure with rapid IV infusion.
- Dermatologic: cheilosis, dermatitis, rash.
- Musculoskeletal: arthralgia, myalgia.
- Renal: renal tubular necrosis with excessive daily dose.
- Fever, chills, fatigue, malaise.
- Local reactions: IV: thrombophlebitis if the concentration exceeds 5 mg/mL; IM: pain.

DOSAGE

Adults:
- 1000-1500 mg/m^2/24 hours IV/IM OR 25-75 mg/kg/day IV/IM OR 2000-4000 mg/24 hours IV/IM for 5 days. Dose depends on blood lead levels and presence of symptoms.
- After 5 days, therapy is interrupted for at least 2 days followed by an additional 5 day course of therapy, if indicated. Therapy must be interrupted for at least 7 days prior to third and subsequent courses of therapy.
- Dosage in renal impairment for patients with lead nephropathy: use with extreme caution.

Cr (mcmol/L)	Dose
177-265	500 mg/m^2 q24h X 5 days
265-354	500 mg/m^2 q48h X 3 doses
greater than 354	500 mg/m^2 once weekly

Pediatrics: consult specialized texts.

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NAME OF MEDICATION
CALCIUM DISODIUM EDETATE *

OTHER NAMES
Calcium disodium EDTA, Edetate calcium disodium, Sodium calcium edetate, Calcium disodium Versenate ®

CLASSIFICATION
Chelating agent

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Incompatible with Ringer's injection or RL.

MISCELLANEOUS

- May be given with dimercaprol for the treatment of severe lead poisoning.
- If given as continuous IV infusion, stop the infusion for at least 60 minutes before blood is drawn to avoid a falsely elevated lead concentration.

REFERENCES

1, 4, 5, 40, 95, 135, 146, 263, 339.

* Available via Health Canada's Special Access Programme

NAME OF MEDICATION	
CALCIUM GLUCONATE	
OTHER NAMES	CLASSIFICATION
	Electrolyte

Do NOT confuse calcium gluconate with other calcium salts.
This monograph is specific to calcium GLUCONATE.

INDICATIONS

- For severe cardiotoxicity or cardio-respiratory arrest due to hyperkalemia or hypermagnesemia.
- For shock associated with beta-blocker or calcium channel blocker overdose.
- Treatment of hypocalcemia.

ADMINISTRATION

- IV direct: physician or RN. ***Cardiac monitoring***:
  - For severe cardiotoxicity or cardio-respiratory arrest due to hyperkalemia or hypermagnesemia, treatment of severe hyperkalemia, treatment of life-threatening hypocalcemia (at TOH): administer undiluted over 2-5 min.
  - For shock associated with beta-blocker or calcium channel blocker overdose: administer undiluted over 5-10 min.
  - For other indications: administer undiluted, maximum rate of 200 mg/min.
- Intermittent IV infusion: dilute in NS or D5W to a concentration of 10-50 mg/mL and infuse each 1000 mg over 60 minutes (maximum rate of 200 mg/min).
- Continuous IV infusion: dilute in NS or D5W to a concentration of 5.8-10 mg/mL. Refer to Dosage section.
- Should be administered slowly through a small needle into a large vein to avoid venous irritation, too rapid increase in serum calcium and extravasation of calcium solution into the surrounding tissue with resultant necrosis.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: rapid IV administration may cause bradycardia, vasodilation, hypotension, cardiac arrhythmias, syncope and cardiac arrest.
- GI: metallic or chalky taste.
- Electrolyte disturbances: hypercalcemia.
- Tingling sensations, a sense of oppression or "heat waves".
- Local reactions: vein irritation.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- For severe cardiotoxicity or cardio-respiratory arrest due to hyperkalemia or hypermagnesemia, treatment of severe hyperkalemia, treatment of life-threatening hypocalcemia (at TOH): 1500-3000 mg IV direct.
- For shock associated with beta-blocker or calcium channel blocker overdose:
  - 3000-6000 mg or 60 mg/kg (maximum of 3000-6000 mg/dose) IV direct; dose may be repeated 3-4 times every 10-20 minutes.
  - May also be administered as a continuous IV infusion: initate at 60-150 mg/kg/hr; titrate to response.
- Hypocalcemia (non-severe): 500-4000 mg IV. Dose may be repeated as required according to patient’s response and serum calcium levels. May follow with a continuous IV infusion.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Do not freeze.
- Compatible with NS, dextrose solutions, RL and sodium lactate 1/6 M.
- Stable for 9 days in the fridge in NS or D5W at concentrations of 9.1, 10.7 and 16.7 mg/mL in PVC containers.

MISCELLANEOUS

- Contains aluminum.
- Caution in digitalized patients; slow IV administration and close monitoring are recommended.
- Calcium gluconate 10% solution contains 100 mg/mL of calcium gluconate, equivalent to 0.465 mEq (0.232 mmol) calcium or 9.3 mg calcium per mL.
- Elemental calcium content is 9% (compared to 27% for the chloride salt), which corresponds to 90 mg (4.5 mEq or 2.25 mmol) per g of calcium gluconate.

REFERENCES

1, 2, 4, 5, 6, 40, 95, 124, 135, 146, 263, 328, 339, 366, 367, 462.

NAME OF MEDICATION
CARBETOCIN

OTHER NAMES
Duratocin ®

CLASSIFICATION
Oxytocic

INDICATIONS

- Prevention of uterine atony and postpartum hemorrhage following elective or emergency cesarean section under epidural or spinal anesthesia.
- Prevention of uterine atony and postpartum hemorrhage following vaginal birth.

ADMINISTRATION

- IV direct: physician or RN. Administer undiluted (at TOH: dilute dose with 9 mL of NS immediately prior to administration) as a bolus over 1 minute, only after the birth of infant.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension, chest pain, tachycardia, flushing.
- GI: nausea, vomiting, abdominal pain, metallic taste.
- CNS: headache, dizziness, anxiety, tremor.
- Dermatologic: pruritus, sweating.
- Hematologic: anemia.
- Musculoskeletal: back pain.
- Respiratory: dyspnea.
- Feeling of warmth, chills.

DOSAGE

- Cesarean section: 100 mcg (1 mL) IV(preferred)/IM as a single dose. Administer following birth of infant, either before or after delivery of the placenta. Do not repeat dose.
- Vaginal delivery: 100 mcg (1 mL) IV/IM as a single dose. Administer as soon as possible after delivery of infant, preferably before delivery of the placenta. Do not repeat dose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 2-8°C. Do not freeze.
- Store prefilled syringes between 2-8°C. Protect from light.
- Store vials between 15-30°C.
- No compatibility data with IV solutions is available; however, experience at TOH of carbetocin injection through a Y-site connection with D5W, NS or RL has not lead to any visual compatibility problem.

MISCELLANEOUS

- Carbetocin should not be administered prior to delivery of the infant for any reason, including elective or medical induction of labour.
- Onset of uterine activity: approximately 1 minute for IV and 3-5 minutes for IM; duration of uterine activity: 60 minutes for IV administration and 2 hours for IM administration.

REFERENCES

5, 54, 86, 135, 208.

Full revision 2021

NAME OF MEDICATION
CARBOplatin

OTHER NAMES
Carboplatinum, Paraplatin ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of advanced ovarian carcinoma and numerous other neoplasms.

ADMINISTRATION

- **Consider premedication if patient previously experienced a hypersensitivity reaction to carboplatin; refer to Potential Administration Hazards section.**
- Intermittent IV infusion, Continuous IV infusion: may further dilute in 100-250 mL of D5W or NS (final concentration range 0.3-10 mg/mL); infuse over 15-60 minutes. Has also been administered as a continuous IV infusion over 24 hours.
- Do NOT use needles, IV sets or equipment containing aluminum for preparation or administration; a black platinum precipitate may form if carboplatin comes in contact with aluminium, which will decrease drug potency.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, pruritus, erythema, urticaria, exfoliative dermatitis, Stevens-Johnson syndrome, toxic epidermal necrolysis, erythema multiforme, fever, chills, rigors, hypotension or hypertension, tachycardia, chest pain, dyspnea, wheezing, bronchospasm, swelling, anaphylaxis; may occur during infusion or up to several hours to days after carboplatin infusion. The risk of reactions increases with repeated exposure to platinum agents, generally after 7 cycles. **Premedication** with corticosteroids and H_1 receptor antagonists with or without H_2 antagonists may be considered to decrease the recurrence of these reactions; however, premedication has not been shown to be very effective.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: neurotoxicity: peripheral neuropathy, decreased deep tendon reflexes; less severe and less common than with cisplatin.
- Electrolyte disturbances: hypomagnesemia, hypokalemia, hypocalcemia, hyponatremia.
- Hematologic: myelosupression, bleeding.
- Otic: hearing impairment, tinnitus.
- Renal: nephrotoxicity; less severe and less common than with cisplatin.
- Fatigue.
- Local reactions: injection site reactions.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

Area under the curve (AUC) based dosing:
- More commonly dosed based on the Calvert formula: IV Dose (mg) = Target AUC in mg/mL/min X (glomerular filtration rate [GFR]) in mL/min + 25); where AUC = 4-7, GFR may be approximated by using CrCl. Do not use Calvert formula if CrCl is less than 20 mL/min; cap GFR at 125 mL/min. Dose repeated every 2-4 weeks.

Body surface area (BSA) based dosing:
- 200-400 mg/m^2 IV every 3-4 weeks.
- Dosage in renal impairment: use following recommendations if using BSA-based dosing instead of Calvert formula:
  Method 1:

CrCl (mL/min)	40 or greater	39-20	19-10	9 or less
Initial dose (mg/m^2)	400	250	150	100

…./Cont.

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Dosage (Cont.)

Method 2:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	50%	25%

Method 3:

CrCl (mL/min)	59-41	40-16	15 or less
Dose (mg/m^2)	250	200	insufficient data

- Reduce initial dose by 20-25% in patients with prior myelosuppression or poor performance status.
- When used in combination with other antineoplastic agents, the carboplatin dose may require adjustment according to the treatment protocol.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C. Protect from light. Do not freeze.
- Stable for 21 days at room temp, protected from light, or in the fridge, protected from light, in D5W at concentrations of 0.5 and 4 mg/mL in PVC containers.
- Stable for 84 days in the fridge, protected from light, and 24 hours at room temp in D5W at concentrations of 0.7 and 2.15 mg/mL in polyolefin and PVC containers.
- Stable for 28 days at room temp or in fridge, protected from light, in D5W at a concentration of 1 mg/mL in PVC containers.
- Stable for 9 days at room temp, protected from light, in D5W at a concentration of 2.4 mg/mL in PVC containers.
- Stable for 3 days at room temp, exposed to light, and 7 days in the fridge, protected from light, in NS at a concentration of 0.5 mg/mL in PVC containers.
- Stable for 5 days at room temp, exposed to light, and 7 days in the fridge, protected from light, in NS at a concentration of 2 mg/mL in PVC containers.
- Stable for 7 days at room temp, exposed to light, and 7 days in the fridge, protected from light, in NS at a concentration of 4 mg/mL in PVC containers.
- Stable for 5 days in the fridge, protected from light, 8 days at room temp or 24 hours at 37°C at a concentration of 10 mg/mL in plastic syringes.

MISCELLANEOUS

- Emergency drugs (e.g., epinephrine, antihistamines, bronchodilators) should be immediately available for the treatment of hypersensitivity reactions.

REFERENCES

2, 4, 5, 118, 129, 135, 143, 165, 168, 169, 208.

NAME OF MEDICATION
CARBOPROST

OTHER NAMES
Carboprost tromethamine, Hemabate ®, Prostaglandin F_2 analog

CLASSIFICATION
Oxytocic

INDICATIONS

- Treatment of postpartum hemorrhage due to uterine atony which has not responded to conventional therapy (IV oxytocin, uterine massage, IM ergot preparations).

ADMINISTRATION

- Deep IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: angioedema, anaphylaxis, anaphylactoid reactions.
- Cardiovascular: hypertension, flushing.
- GI: nausea, vomiting, diarrhea. Usual doses of loperamide can be given to counteract the diarrhea associated with carboprost.
- CNS: headache.
- Respiratory: cough, bronchospasm, pulmonary edema.
- Fever, chills. Acetaminophen has been used to treat fever and chills.
- Local reactions: pain at injection site.

DOSAGE

- 250 mcg (1 mL) IM. May be repeated at 15-90 minute intervals until therapeutic response or to a maximum cumulative dose of 2 mg (8 injections).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 2-8°C.

MISCELLANEOUS

- Contraindicated in patients with known active cardiac, pulmonary, renal, hepatic disease or acute pelvic inflammatory disease.
- Use with caution in patients with seizure disorders and diabetes.

REFERENCES

1, 5, 54, 95,135, 396.

Full revision 2021

NAME OF MEDICATION	
CARFILZOMIB	
OTHER NAMES Kyprolis ®	CLASSIFICATION Antineoplastic

INDICATIONS
- Treatment of relapsed multiple myeloma.

ADMINISTRATION
- **Ensure premedication and hydration have been administered prior to each infusion; refer to Dosage section.**
- Reconstitute each 10 mg vial, 30 mg vial and 60 mg vial with 5 mL, 15 mL and 29 mL, respectively, of SWFI by using a 21-gauge or larger gauge hypodermic needle (0.8 mm or smaller external diameter needle) and injecting slowly toward the side of the vial to minimize foaming. Gently swirl or invert vial slowly for approximately 1 minute or until complete dissolution. Do NOT shake. If foaming occurs, allow the solution to settle in the vial for approximately 5 minutes until solution is clear and foaming subsides. The concentration of the reconstituted solution is 2 mg/mL.
- Intermittent IV infusion (mandatory): administer undiluted or diluted; for dilution, withdraw the calculated dose from the reconstituted vial(s) using a 21-gauge or larger gauge hypodermic needle (0.8 mm or smaller external diameter needle) and dilute in 50 or 100 mL of D5W (do NOT dilute in NS). Infuse over at least 10 minutes when administered in combination with lenalidomide and dexamethasone (i.e., 20/27 mg/m^2 regimen) or over 30 minutes when administered with dexamethasone alone (i.e., 20/56 mg/m^2 and 20/70 mg/m^2 regimens). Flush line with NS or D5W immediately before and after administration.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, pruritus.
- Infusion-related reactions: symptoms include fever, chills, arthralgia, myalgia, facial flushing, facial edema, laryngeal edema, vomiting, weakness, shortness of breath, hypotension, syncope, chest tightness or angina. Can occur immediately or up to 24 hours after administration. Hydration and **premedication** with dexamethasone are recommended to reduce the frequency and severity of these reactions.
- Cardiovascular: new or worsening cardiac failure, pulmonary edema, cardiac arrhythmias, chest pain, myocardial ischemia or infarction, deep vein thrombosis, pulmonary embolism, peripheral edema, hypotension, hypertension (including hypertensive emergency and crisis), pulmonary arterial hypertension (rare).
- GI: nausea, vomiting, diarrhea, constipation, abdominal pain, anorexia.
- CNS: headache, dizziness, insomnia, peripheral neuropathy, posterior reversible encephalopathy syndrome (rare).
- Dermatologic: erythema, hyperhidrosis.
- Electrolyte disturbances: hyperkalemia, hypokalemia, hypercalcemia, hypocalcemia, hypophosphatemia, hypomagnesemia, hyponatremia.
- Endocrine and metabolic: hyperglycemia, hyperuricemia.
- Hematologic: bleeding, anemia, neutropenia, lymphopenia, leukopenia, thrombocytopenia, thrombotic microangiopathy.
- Hepatic: increased LFTs and bilirubin, reactivation of hepatitis B virus (HBV).
- Musculoskeletal: muscle spasms, myalgia, arthralgia, back pain, bone pain, pain in extremity, muscular weakness.
- Renal: increased serum creatinine, acute renal failure.
- Respiratory: cough, respiratory tract infections, dyspnea, acute respiratory distress syndrome, infiltrative pulmonary disease.
- Progressive multifocal leukoencephalopathy.
- Tumour lysis syndrome.
- Fever, fatigue, asthenia.
- Local reactions: infusion site reaction.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- **Premedicate with dexamethasone PO or IV at least 30 minutes but no more than 4 hours before carfilzomib administration. Adequate hydration is required for cycle 1 except for days where daratumonab is given: 30 mL/kg/day PO for 48 hours before starting therapy and 250-500 mL of IV fluids before each dose of the cycle (recommended) and after each dose (prn); for subsequent cycles, hydrate as required.**
- Carfilzomib in combination with lenalidomide and dexamethasone (20/27 mg/m^2 twice weekly regimen, 28-day cycles):
  - Cycle 1: 20 mg/m^2 IV on days 1 and 2. If tolerated, increase to 27 mg/m^2 IV on days 8, 9, 15 and 16.
  - Cycles 2-12: 27 mg/m^2 IV on days 1, 2, 8, 9, 15 and 16.
  - Cycles 13 and beyond: 27 mg/m^2 IV on days 1, 2, 15 and 16.
  - Lenalidomide: 25 mg PO on days 1 to 21; dexamethasone: 40 mg PO/IV on days 1, 8, 15 and 22.
- Carfilzomib in combination with dexamethasone and daratumumab (20/56 mg/m^2 twice weekly regimen, 28-day cycles):
  - Cycle 1: 20 mg/m^2 IV on days 1 and 2. If tolerated, increase to 56 mg/m^2 IV on days 8, 9, 15 and 16.
  - Cycles 2 and beyond: 56 mg/m^2 IV on days 1, 2, 8, 9, 15 and 16.
  - Dexamethasone:
    - Up to 75 years of age: 20 mg PO/IV on days 1, 2, 8, 9, 15 and 16, and 40 mg PO/IV on day 22.
    - Older than 75 years of age:
      - Cycle 1: 20 mg PO/IV on days 1, 2, 8, 15 and 22.
      - Subsequent cycles: 20 mg PO/IV on days 1, 8, 15 and 22.
  - Daratumumab:
    - Cycle 1: 8 mg/kg IV on days 1 and 2, and 16 mg/kg IV on days 8, 15 and 22.
    - Cycle 2: 16 mg/kg IV on days 1, 8, 15 and 22.
    - Cycles 3-6: 16 mg/kg IV on days 1 and 15.
    - Subsequent cycles: 16 mg/kg IV on day 1.
- Carfilzomib in combination with dexamethasone alone (20/56 mg/m^2 twice weekly regimen, 28-day cycles):
  - Cycle 1: 20 mg/m^2 IV on days 1 and 2. If tolerated, increase to 56 mg/m^2 IV on days 8, 9, 15 and 16.
  - Cycles 2 and beyond: 56 mg/m^2 IV on days 1, 2, 8, 9, 15 and 16.
  - Dexamethasone: 20 mg PO/IV on days 1, 2, 8, 9, 15, 16, 22 and 23.
- Carfilzomib in combination with dexamethasone alone (20/70 mg/m^2 once weekly regimen, 28-day cycles) for patients who cannot adhere to twice weekly regimen due to burden of dosing frequency:
  - Cycle 1: 20 mg/m^2 IV on day 1. If tolerated, increase to 70 mg/m^2 IV on days 8 and 15.
  - Cycles 2 and beyond: 70 mg/m^2 IV on days 1, 8 and 15.
  - Dexamethasone: 40 mg PO/IV on days 1, 8, 15 and 22; to omit on day 22 of cycles 10 and beyond.
- No dosage adjustment is required for weight changes of 20% or less.
- Dosage in obesity: dose should be calculated using the body surface area, up to a maximum of 2.2 m^2.
- Dosage in renal impairment: no dosage adjustment is necessary for carfilzomib; use with caution if CrCl is less than 50 mL/min. For patients on dialysis, administer dose after the dialysis session.
- Dosage in hepatic impairment: decrease carfilzomib dose by 25% if mild to moderate impairment; no data in severe impairment (bilirubin greater than 3 x ULN).
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light.
- According to manufacturer, the reconstituted solution, in SWFI in the vial or in a syringe, and the diluted solution, in D5W in the IV bag, are stable for 4 hours at room temp or for 24 hours in the fridge. The total time from reconstitution to administration should not exceed 24 hours. However, longer stability data are available from other sources; see below.
- Reconstituted solution of 2 mg/mL with SWFI is stable for 14 days at room temp, protected from light, in polypropylene siliconized syringes and for 28 days in the fridge, protected from light, in glass vials.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 10 days at room temp, protected from light, and 28 days in the fridge, protected from light, in D5W at a concentration of 0.6 mg/mL in polyolefin containers and a concentration of 0.8 mg/mL in polypropylene siliconized syringes.

MISCELLANEOUS
- Thromboprophylaxis is recommended in patients treated with carfilzomib in combination with dexamethasone alone or with lenalidomide and dexamethasone; choice of antithrombotic agent should be based on patient's underlying risks and clinical status.
- Hypertension should be well-controlled prior to initiating treatment with carfilzomib.
- Each mL of the reconstituted solution of carfilzomib contains 0.3 mmol (7 mg) of sodium.
- Consider antiviral prophylaxis to decrease risk of herpes zoster virus reactivation.
- Patients should be tested for HBV infection before starting therapy.

REFERENCES
5, 129, 135, 165, 208.

NAME OF MEDICATION
CARMUSTINE

OTHER NAMES
BCNU, BiCNU ™

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of primary brain tumours, malignant lymphomas, malignant melanoma, multiple myeloma, and gastrointestinal carcinoma.

ADMINISTRATION

- Allow supplied diluent (absolute ethanol) to reach room temp before use. Reconstitute each 100 mg vial with 3 mL of the diluent, then add 27 mL of SWFI to obtain a concentration of 3.3 mg/mL in 10% ethanol.
- Intermittent IV infusion: dilute in 500 mL of NS or D5W in glass bottles, PVC-free and DEHP-free polypropylene or polyolefin containers. Infuse doses up to 200 mg/m^2 over 1-2 hours and infuse doses greater than 200 mg/m^2 over at least 2 hours or at a maximum rate of 3 mg/m^2/min. Use polyethylene tubing for infusions longer than 2 hours.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash.
- Infusion-related reactions: hypotension, agitation, skin and conjunctival flushing (occurs within 2 hours after rapid IV administration, is due to ethanol content, and may last 4 hours).
- Cardiovascular: tachycardia, chest pain, myocardial ischemia (with high doses).
- GI: nausea, vomiting, diarrhea, constipation, anorexia.
- CNS: ataxia, dizziness.
- Ophthalmic: blurred vision.
- Local reactions: burning, hyperemia, vasospasm, pain, chemical phlebitis at injection site.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 20-65 mg/m^2 (0.5-1 mg/kg) IV on day 1 every 4-6 weeks.
- 100 mg/m^2 IV on day 1 every 6 weeks.
- 75-120 mg/m^2 IV on days 1 and 2 or 40-80 mg/m^2 IV on days 1-3 every 6-8 weeks.
- 150-200 mg/m^2 IV every 6-8 weeks as a single dose on day 1 or divided as 2 once daily injections on days 1 and 2.
- Bone marrow transplant: 300 mg/m^2 or 500 mg/m^2 IV as a single dose (fatal without bone marrow/stem cell transplant).
- Dosage in renal impairment: discontinue if CrCl is less than 10 mL/min. Alternative recommendation:

CrCl (mL/min)	60-46	45-31	30 or less
Dose	80%	75%	Avoid

- Dosage in hepatic impairment: dose modification may be necessary; no specific recommendations exist.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Unopened vials:
  - Store vials between 2-8°C.
  - Stable for 7 days at room temp, but slow decomposition occurs (about 3% in 36 days).
  - Dry powder in unopened vials may range in appearance from lacy flakes to a congealed mass, without degradation of carmustine. Discard if product has liquified.
  - Temperatures of 30.5-32°C will cause deterioration; the powder will liquefy and appear as an oily film; vial should be discarded.
- Reconsitutited solution:
  - Solution should be clear, colourless to light yellow.
  - Stable for 8 hours at room temp, protected from light, or 24 hours in the fridge.
  - Refrigerated solution should be inspected for crystal formation before use. Redissolve crystals by warming vial to room temp and shaking vial.
- Diluted solution:
  - Stable for 8 hours at room temp, protected from light, or 24 hours in the fridge followed by an additional 6 hours at room temp, protected from light, in D5W or NS at a concentration of 0.2 mg/mL in glass or polypropylene containers.
  - Stable for 6 hours at room temp, exposed to light, in D5W at a concentration of 1 mg/mL in polyolefin containers.
- Incompatible with PVC and ethylene vinyl acetate (EVA) containers and tubing.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 95, 129, 135, 165, 200, 208, 225.

NAME OF MEDICATION
CASPOFUNGIN

OTHER NAMES
Cancidas ®

CLASSIFICATION
Antifungal

INDICATIONS

- Empiric therapy for presumed fungal infections in febrile neutropenic patients.
- Treatment of invasive aspergillosis in patients who are refractory or intolerant to other therapies.
- Treatment of invasive candidiasis.
- Treatment of esophageal candidiasis.

ADMINISTRATION

- Reconstitute each 50 mg or 70 mg vial with 10.5 mL of either NS, SWFI, or bacteriostatic water for injection with 0.9% benzyl alcohol. If refrigerated product is used, allow vial to reach room temp before reconstitution.
- Intermittent IV infusion: dilute in 250 mL of NS, 1/2NS, 1/4NS or RL; for fluid-restricted patients or in some pediatric patients: dilute in 100 mL or in a reduced volume of NS, 1/2NS, 1/4NS or RL, not to exceed a final concentration of 0.5 mg/mL. Administer over 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: histamine-related reactions (rash, facial swelling, angioedema, pruritus, sensation of warmth, bronchospasm), anaphylaxis, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Cardiovascular: tachycardia, hypotension, peripheral edema, flushing.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: headache.
- Fever, chills, sweating.
- Local reactions: swelling, phlebitis, pruritus at injection site.

DOSAGE

Adults:
- Empiric therapy in febrile neutropenic patients and invasive aspergillosis: 70 mg IV loading dose on day 1, followed by 50 mg IV daily thereafter. An increase to 70 mg daily may be considered in patients who are tolerating the drug well and have an inadequate response.
- Invasive candidiasis: 70 mg IV loading dose on day 1, followed by 50 mg IV daily thereafter.
- Esophageal candidiasis: 50 mg IV daily. A 70 mg IV loading dose on day 1 is often recommended.
- For patients receiving concomitant inducers of drug clearance (e.g., efavirenz, nevirapine, rifampin, dexamethasone, phenytoin, carbamazepine), a dose of 70 mg IV daily may be considered.
- Dosage in renal impairment: no dosage adjustment required; not removed by dialysis.
- Dosage in hepatic impairment: no dosage adjustment required in patients with mild hepatic impairment (Child-Pugh score 5 to 6). For patients with moderate hepatic impairment (Child-Pugh score 7 to 9), maintain loading dose of 70 mg IV where recommended and decrease daily dose to 35 mg IV daily. Pharmacokinetic data suggest that patients with severe hepatic impairment (Child-Pugh score greater than 9) can be dosed similarly to those with moderate hepatic impairment.
- Dosage in obesity: a dose of 70 mg IV daily may be considered for patients weighing greater than 80 kg.

Pediatrics (3 months to 17 years of age):
- For all indications: 70 mg/m^2 IV loading dose (maximum of 70 mg) on day 1, followed by 50 mg/m^2 IV (maximum of 70 mg) daily thereafter. If 50 mg/m^2 is well tolerated but the response is inadequate, the dose can be increased to 70 mg/m^2 IV (maximum of 70 mg) daily.
- For patients receiving concomitant inducers of drug clearance (e.g., efavirenz, nevirapine, rifampin, dexamethasone, phenytoin, carbamazepine), a dose of 70 mg/m^2 IV (maximum of 70 mg) daily may be considered.
- Dosage in hepatic impairment: no dosage adjustment required in patients with mild hepatic impairment (Child-Pugh score of 5-6). For patients with moderate hepatic impairment (Child-Pugh score 7-9), decrease daily dose by 30%.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials from Merck Canada and MDA between 2-8°C; store vials from Fresenius Kabi Canada between 15-25°C.
- Reconstituted solution is stable for 60 minutes between 15-25°C.
- Compatible with NS, RL; incompatible with dextrose.
- Stable for 24 hours between 15-25°C or 48 hours between 2-8°C in NS, 1/2NS, 1/4NS or RL at concentrations of 0.14, 0.2, 0.28, 0.34 and 0.47 mg/mL.
- Stable for 28 days in the fridge in NS at a concentration of 0.5 mg/mL in polyolefin bags.

MISCELLANEOUS

- Merck and MDA products contain 39 mg and 54 mg of sucrose per 50 mg and 70 mg vial, respectively; also contain mannitol.

REFERENCES

1, 5, 40, 82, 95, 102, 135, 180, 208, 363, 477.

NAME OF MEDICATION
CeFAZolin

OTHER NAMES
Ancef ®, Kefzol ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS

- For severe infections of the respiratory tract, skin and soft tissue, biliary tract, genitourinary tract, bone and joints, septicemia and endocarditis due to susceptible organisms.
- Perioperative prophylaxis.

ADMINISTRATION

- For IV use: reconstitute 500 mg and 1 g vials with 10 mL of SWFI. Shake well until dissolved. If the reconstituted solution remains cloudy after 3 minutes, do NOT administer as it is a sign that the product precipitated.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of compatible solution and infuse over 10-60 minutes.
- Continuous IV infusion: dilute in an appropriate volume of compatible solution and administer as a continuous infusion.
- IM: reconstitute 500 mg and 1 g vials with 3.8 and 2.5 mL, respectively, of SWFI to provide solutions containing approximately 125 mg/mL and 334 mg/mL OR reconstitute the 500 mg vial with 2 mL of NS to provide a solution containing approximatively 225 mg/mL. Do NOT reconstitute the 1 g vial with NS. Inject deep into a large muscle mass.
- Consult Medical staff rules and regulations: MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis) for specific dosing regimens and infusion time.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults:
  - 0.5-2 g q6-8h, with a maximum dose of 12 g/day for severe, life-threatening infections.
  - For perioperative prophylaxis: 1-3 g (depending on weight) within 60 minutes before skin incision or procedure.
  - At TOH, follow this dosing guide for preoperative prophylaxis, depending on patient's weight, as per Medical staff rules and regulations MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis).

Weight	Less than 60 kg	60-119 kg	120 kg and greater
Dose	1 g	2 g	3 g

- Pediatrics: 25-100 mg/kg/day in 2-4 divided doses; maximum of 6 g/day.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	100%	50%
Dosing Interval (hr)	8	12	24-48

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C. Protect from light.
- Reconstituted solutions may range in colour from pale yellow to yellow, without a change in potency.
- Reconstituted solution is stable for 24 hours at room temp not exceeding 25°C and for 72 hours in the fridge protected from light according to manufacturer's instructions.
- Stable for 30 days in the fridge in D5W at a concentration of 10 mg/mL when stored in PVC containers.
- Stable for 5 days at room temp and 24 days in the fridge in D5W at a concentration of 20 mg/mL when stored in PVC containers.
- Stable for 7 days at room temp and 15 days in the fridge in NS at a concentration of 20 mg/mL when stored in PVC containers.
- Stable for 13 days at room temp and 28 days in the fridge in SWFI at a concentration of 100 mg/mL and 200 mg/mL when stored in plastic syringes.
- Compatible with saline, dextrose and RL.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Positive Coombs' test has been reported.

REFERENCES

1, 2, 4, 5, 40, 82, 135, 143, 148, 216, 305, 338.

Full revision 2012; limited revision 2014, 2015, 2016, 2018, 2020

NAME OF MEDICATION
CEFEPIME

OTHER NAMES
Maxipime ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS

- Treatment of lower respiratory tract infections, exacerbations of chronic bronchitis, urinary tract infections, skin and skin structure infections, meningitis and other CNS infections, complicated intra-abdominal infections and septicemia due to susceptible organisms.
- Empiric therapy in febrile neutropenia.

ADMINISTRATION

- For IV use: reconstitute the 1 g and 2 g vials with 10 mL of SWFI, D5W, or NS.
- IV direct: physician or RN. Administer 1 to 2 g over 3 to 5 minutes.
- Intermittent IV infusion: dilute with 50-100 mL NS, D5W, D10W, D5-NS, D5-RL and infuse over 30 minutes.
- IM: reconstitute 1 g vial with 2.4 mL of SWFI, NS, D5W or 0.5 to 1% lidocaine to provide a final concentration of 280 mg/mL. Inject deep into a large muscle mass.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults: mild to moderate infections: 0.5-1 g q12h
  severe infections: 2 g q8-12h
- Pediatrics (aged 2 months to 12 years old): 50 mg/kg q12h. For febrile neutropenia: 50 mg/kg q8h. (Maximum: 2 g/dose).
- Dosage in renal impairment:

Method 1:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	50-100%	25-50%
Dosing interval (hr)	8-12	24	24

Method 2:

CrCl (mL/min)	greater than 50	50-30	29-11	less than 11	Hemodialysis *
Dosage range depending on severity of infection	1-2 g	1-2 g	0.5-2 g	0.25-1 g	500 mg
Dosing interval (hr)	8-12	12-24	24	24	24

* On dialysis days, cefepime should be administered following dialysis.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- The colour of reconstituted solutions may darken without affecting potency.
- Reconstituted solutions are stable for 24 hours at room temp and 72 hours in the fridge, protected from light, according to manufacturer's instructions.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 7 days in the fridge in NS or D5W at concentrations of 1-40 mg/mL in PVC containers.
- Stable for 14 days in the fridge and 24 hours at room temp in NS, D5W, SWFI at concentrations of 100 and 200 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Beta-lactamase resistant.
- Positive Coombs' test has been reported.
- Considered as a 4th generation cephalosporin.

REFERENCES

1, 2, 4, 40, 82, 143, 216.

NAME OF MEDICATION
CEFOTAXIME

OTHER NAMES
Claforan ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS

- For severe infections (lower respiratory tract infections, UTI, septicemia, skin and skin structure infections, intra-abdominal infections, gynecologic infections and CNS infections) due to susceptible organisms.
- Treatment of Lyme disease.
- Perioperative prophylaxis (not considered a drug of choice).

ADMINISTRATION

- For IV use: reconstitute each 500 mg, 1 g, or 2 g vial with 10 mL of SWFI.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion: further dilute with 50-100 mL NS or D5W and give over 20-30 minutes.
- Continuous IV infusion: dilute in 250-1000 mL NS or D5W and administer as a continuous infusion.
- IM: reconstitute the 500 mg, 1 g or 2 g vial with 2, 3 or 5 mL of SWFI, respectively; resultant solutions contain approximately 230, 300 or 330 mg/mL, respectively. Inject deep into a large muscle mass.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults:
  - Mild uncomplicated infections: 1 g q12h
  - Moderate to severe infections: 1-2 g q8h
  - Very severe/life-threatening infections: 2 g q4-8h
  - Maximum daily dose of 12 g
  - Perioperative prophylaxis: 1 g 30-90 minutes before skin incision
- Pediatrics (aged 1 month to 12 years old): 50-100 mg/kg/day in 3 divided doses. Severe infections: 150-200 mg/kg/day in 3 to 4 divided doses. Maximum dose of 12 g/day.
- Dosage in renal impairment:

  Method 1:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dosing interval (hr)	6-8	8-12	24

  Method 2:

  if CrCl less than 20 mL/min: decrease dose by 50%.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Reconstituted solutions may range in colour from light yellow to amber. Discolouration of the solution may indicate a loss of potency.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY (Cont.)

- Reconstituted solutions for IV or IM administration are stable for 12 hours at room temp and 24 hours in the fridge, according to Canadian manufacturers' instructions.
- Stable for 24 hours at room temp and 7 days in the fridge when diluted in NS or D5W at a concentration of 50 mg/mL in PVC bags.
- Stable for 24 hours at room temp and 22 days in the fridge when diluted in NS or D5W at a concentration of 10 mg/mL in PVC bags.
- Compatible with dextrose, saline and RL.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Beta-lactamase resistant.
- Positive Coombs' test has been reported.

REFERENCES

1, 2, 4, 5, 9, 40, 82, 143, 216.

NAME OF MEDICATION
CefOXitin

OTHER NAMES
Mefoxin ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS

- For severe infections (respiratory tract infections, skin and skin structure infections, bone and joint infections, genitourinary tract infections, intra-abdominal infections and septicemia) due to susceptible organisms.
- Perioperative prophylaxis.

ADMINISTRATION

- For IV use: reconstitute 1 g vial with 10 mL and 2 g vial with 10 or 20 mL of SWFI, NS, or D5W.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of compatible solution and infuse over 10-60 minutes.
- Continuous IV infusion: dilute in up to 1000 mL of compatible solution and administer as a continuous infusion.
- IM: reconstitute each 1 g or 2 g vial by adding 2 mL or 4 mL, respectively, of SWFI, or 0.5% or 1% lidocaine (without epinephrine) to obtain approximately 400 mg/mL. Inject deep into a large muscle mass.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults: 1-2 g q4-8h, with a maximum dose of 12 g/day for severe, life-threatening infections.
  Perioperative prophylaxis: 1-2 g 30-60 minutes before skin incision and 1-2 g q6-8h for 24 hours postoperatively.
- Pediatrics (aged 3 months to 12 years old): 80-160 mg/kg/daily in 4-6 equal doses, maximum of 12 g/day.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-30	29-10	less than 10
Dosing interval (hr)	6-8	8-12	12-24	24-48

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Dark brown solutions should not be used.
- Reconstituted solution is stable for 48 hours at room temp and 7 days in the fridge.
- Infusion stable for 24 hours at room temp or 13 days in the fridge when diluted in NS or D5W at a concentration of 20 mg/mL in PVC containers.
- Compatible with dextrose, saline and RL.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Beta-lactamase resistant.
- Positive Coombs' test has been reported.

REFERENCES

1, 2, 4, 40, 82, 143, 216.

Full revision 2012; limited revision 2015, 2017

NAME OF MEDICATION
CefTAZidime

OTHER NAMES
Ceptaz ®, Fortaz ®, Tazidime ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS
- For severe infections (lower respiratory tract infections, UTI, skin and skin structure infections, septicemia, bone and joint infections, intra-abdominal infections, gynecologic infections and CNS infections) due to susceptible organisms.
- Empiric therapy in febrile neutropenic patients.

ADMINISTRATION
- For IV use: reconstitute 1 g or 2 g vial with 10 mL of SWFI. Shake well until dissolved.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W and infuse over 15 to 30 minutes.
- Continuous IV infusion: dilute in an appropriate volume of NS or D5W and administer as a continuous infusion.
- IM: reconstitute 1 g vial with 3 mL of SWFI, or 0.5 or 1% lidocaine to obtain approximately 280 mg/mL. Inject deep into a large muscle mass.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE
- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults: range of 1-6 g/day in 2-3 divided doses; maximum dose: 6 g/day.
- Pediatrics (aged 1 month to 12 years old): 100-150 mg/kg/day in 3 divided doses; maximum dose: 6 g/day.
- Dosage in renal impairment:

Method 1:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dosing interval (hr)	8-12	12-24	24-48

Method 2:

CrCl (mL/min)	50-31	30-16	15-6	less than 6
Dosing moderate infections	1 g q12h	1 g q24h	500 mg q24h	500 mg q48h
Dosing severe infections	1.5 g q12h	1.5 g q24h	750 mg q24h	750 mg q48h

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Reconstituted solutions range from light yellow to amber. Colour changes do not necessarily indicate a loss of potency.
- Reconstituted solution stable for 24 hours at room temp and 7 days in the fridge.
- Stable for 24 hours at room temp or 7 days in the fridge in D5W or NS at concentrations from 1 to 40 mg/mL.
- Compatible with D5W, NS, RL.

MISCELLANEOUS
- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Beta-lactamase resistant.
- Positive Coombs' test has been reported.

REFERENCES
1, 2, 4, 5, 40, 82, 143, 216.

Full revision 2012; limited revision 2015

NAME OF MEDICATION
CEFTAZIDIME/AVIBACTAM *

OTHER NAMES
Avycaz ®

CLASSIFICATION
Antibiotic – cephalosporin with beta-lactamase inhibitor

INDICATIONS

- Treatment of complicated intra-abdominal infections in combination with metronidazole in adults and children.
- Treatment of complicated urinary tract infections, including pyelonephritis, in adults and children.
- Treatment of hospital-acquired bacterial pneumonia and ventilator-associated bacterial pneumonia in adults.

ADMINISTRATION

- Reconstitute each 2.5 g vial (2 g of ceftazidime and 0.5 g of avibactam) with 10 mL of SWFI, NS, D5W, or RL to obtain an approximate concentration of 167 mg/mL of ceftazidime and 42 mg/mL of avibactam in an approximate volume of 12 mL. Mix gently to ensure complete dissolution.
- Intermittent IV infusion (mandatory): dilute dose of 2.5 g (12 mL), 1.25 g (6 mL), or 0.94 g (4.5 mL) in 50 to 250 mL of same diluent used to reconstitute the vial (except SWFI) for a final concentration of 8 to 40 mg/mL of ceftazidime and 2 to 10 mg/mL of avibactam; mix gently and ensure complete dissolution; infuse over 2 hours.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, anaphylaxis, angioedema, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Cardiovascular: hypertension, chest pain.
- GI: nausea, vomiting, abdominal pain, constipation, diarrhea, *Clostridium difficile*-associated diarrhea and colitis.
- CNS: headache, dizziness, anxiety; seizures, myoclonus, encephalopathy, particularly in patients with renal impairment.
- Hepatic: increased LFTs.
- Respiratory: cough.
- Fever.
- Local reactions: phlebitis, inflammation, hematoma, thrombosis at infusion site.

DOSAGE

Adults:
- 2.5 g IV q8h.
- Dosage in renal impairment:

CrCl (mL/min)	50-31	30-16	15-6**	less than or equal to 5**
Dose (g)	1.25 g	0.94 g	0.94 g	0.94 g
Dosing interval (hr)	8	12	24	48

**Removed by hemodialysis; administer dose after dialysis on hemodialysis days.
- Dosage in hepatic impairment: no dosage adjustment required.

Pediatrics:
- 3 months to less than 6 months of age: 50 mg/kg (ceftazidime 40 mg/kg and avibactam 10 mg/kg) IV q8h; 6 months to less than 18 years of age: 62.5 mg/kg (ceftazidime 50 mg/kg and avibactam 12.5 mg/kg) IV q8h. Do not exceed adult dose.
- Dosage in renal impairment (children aged 2 years and older):

CrCl (mL/min)	50-31	30-16	15-6***	less than or equal to 5***
Dose (g) (Do not exceed adult dose)	31.25 mg/kg	23.75 mg/kg	23.75 mg/kg	23.75 mg/kg
Dosing interval (hr)	8	12	24	48

***Removed by hemodialysis; administer dose after dialysis on hemodialysis days.
- Dosage in hepatic impairment: no dosage adjustment required.

…/Cont.

NAME OF MEDICATION
CEFTAZIDIME/AVIBACTAM *

OTHER NAMES
Avycaz ®

CLASSIFICATION
Antibiotic – cephalosporin with beta-lactamase inhibitor

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light.
- Reconstituted solution must be further diluted within 30 minutes of its reconstitution.
- Diluted solution is stable for 12 hours at room temp.
- Stable for 24 hours in the fridge and then for 12 hours at room temp in D5W, NS, or RL at concentrations of 8 to 40 mg/mL of ceftazidime and 2 to 10 mg/mL of avibactam.
- Compatible with NS, D5W, RL and dextrose-saline solutions.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics
- Avibactam is included as a beta-lactamase inhibitor, which extends ceftazidime's spectrum of activity.
- Positive Coombs' test has been reported.
- Monitor CrCl at least daily in patients with changing renal function and adjust dosage accordingly.
- Each 2.5 g vial contains 146 mg of sodium.

REFERENCES

1, 4, 5, 40, 135.

* Available via Health Canada's Special Access Programme

New monograph 2020

NAME OF MEDICATION
CEFTOBIPROLE

OTHER NAMES
Ceftobiprole medocaril sodium, Zevtera ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS

- Treatment of adults with hospital-acquired pneumonia (excluding ventilator-associated pneumonia) and community-acquired pneumonia caused by susceptible strains.

ADMINISTRATION

- Reconstitute the 500 mg vial with 10 mL of SWFI or D5W to obtain a concentration of 50 mg/mL; shake vigorously to dissolve, which may take up to 10 minutes. Allow foam to dissipate.
- Intermittent IV infusion: withdraw 5 mL of reconstituted solution for a 250 mg dose or 10 mL for a 500 mg dose and dilute in 125 mL or 250 mL, respectively, of NS, D5W or RL to obtain a final concentration of 2 mg/mL; gently invert bag 5-10 times to mix; do NOT shake. Infuse over 2 hours in patients with a CrCl of 150 mL/min or less and over 4 hours in patients with a CrCl of greater than 150 mL/min.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, anaphylaxis.
- Cardiovascular: phlebitis.
- GI: nausea, vomiting, diarrhea, colitis.
- CNS: seizures (more common in patients with pre-existing CNS/seizure disorders), headache, agitation.
- Electrolyte disturbances: hyponatremia.
- Hematologic: agranulocytosis, thrombocytopenia.
- Hepatic: increase in LFTs.
- Renal: renal failure.
- Local reactions: redness, pain, swelling, irritation at injection site.

DOSAGE

- Hospital-acquired pneumonia: 500 mg IV q8h for 7-14 days.
- Community-acquired pneumonia: 500 mg IV q8h for 4-14 days. Consider switching to oral treatment after at least 3 days of IV ceftobiprole, if clinically appropriate.
- Dosage in renal impairment:

CrCl (mL/min)	80-50	49-30	less than 30	end-stage renal disease with or without hemodialysis
Dose	500 mg	500 mg	250 mg	250 mg*
Interval (hr)	8	12	12	24

*drug is removed by dialysis; administer after dialysis on dialysis days.

- Dosage in obesity: no dosage adjustment necessary in morbidly obese patients.

…/Cont.

OTHER NAMES
Ceftobiprole medocaril sodium, Zevtera ®

CLASSIFICATION
Antibiotic - cephalosporin

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light.
- Reconstituted solution is stable for 60 minutes at room temp (25ºC) or for 24 hours in the fridge.
- Diluted solution should be clear to slightly opalescent and yellowish in colour.
- Stable for 8 hours at room temp unprotected from light, 12 hours at room temp protected from light or 72 hours in the fridge protected from light in D5W at a concentration of 2 mg/mL in PVC, polyethylene, and glass containers.
- Stable for 8 hours at room temp unprotected from light, 24 hours at room temp protected from light, or 72 hours in the fridge protected from light in NS at a concentration of 2 mg/mL in PVC, polyethylene, and glass containers
- Stable for 8 hours at room temp unprotected from light or 24 hours at room temp protected from light in RL at a concentration of 2 mg/mL in PVC, polyethylene, and glass containers Do not refrigerate solutions diluted in RL.
- Compatible with RL; incompatible with other calcium-containing solutions.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Considered as a 5th generation cephalosporin.
- Positive Coombs test reported.
- Potential for false-positive result with alkaline picrate assay (Jaffé reaction) to measure serum creatinine and with urine glucose tests using the copper reduction technique (cupric sulfate).
- Each 500 mg dose contains approximately 1.3 mmol (29 mg) of sodium.
- 666.6 mg of ceftobiprole medocaril sodium corresponds to 500 mg of ceftobiprole.

REFERENCES

5, 135, 200, 208.

New monograph 2019

NAME OF MEDICATION
CEFTOLOZANE/TAZOBACTAM

OTHER NAMES
Zerbaxa ™

CLASSIFICATION
Antibiotic – cephalosporin with beta-lactamase inhibitor

INDICATIONS
- Treatment of infections due to susceptible strains including beta-lactamase producing bacteria in the following conditions: complicated intra-abdominal infections, complicated urinary tract infections (cUTI) including pyelonephritis, and nosocomial pneumonia including ventilator-associated pneumonia.

ADMINISTRATION
- Reconstitute each 1.5 g vial (1 g ceftolozane and 500 mg tazobactam) with 10 mL of SWFI or NS to obtain a volume of approximatively 11.4 mL; swirl to dissolve.
- Intermittent IV infusion: withdraw 22.8 mL (2 entire vials), 17.1 mL, 11.4 mL (1 entire vial), 5.7 mL, 3.5 mL, 2.9 mL or 1.2 mL for a dose of 3 g (2 g ceftolozane and 1 g tazobactam), 2.25 g (1.5 g ceftolozane and 750 mg tazobactam) 1.5 g (1 g ceftolozane and 500 mg tazobactam), 750 mg (500 mg ceftolozane and 250 mg tazobactam), 450 mg (300 mg ceftozolane and 150 mg tazobactam), 375 mg (250 mg ceftolozane and 125 mg tazobactam) or 150 mg (100 mg ceftolozane and 50 mg tazobactam) respectively, and transfer into a 100 mL bag of NS or D5W; infuse over 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, urticaria, anaphylaxis.
- GI: nausea, diarrhea.
- CNS: headache.
- Fever.

DOSAGE
- Complicated intra-abdominal infections and cUTI: 1.5 g IV q8h for 4-14 days.
- Nosocomial pneumonia: 3 g IV q8h for 8-14 days.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-30	29-15	end-stage renal disease on hemodialysis
Dose:				
Intra-abdominal/cUTI	1.5 g	750 mg	375 mg	loading dose 750 mg maintenance dose* : 150 mg
Nosocomial pneumonia	3 g	1.5 g	750 mg	loading dose: 2.25 g Maintenance dose*: 450 mg
Interval (hr)	8	8	8	8

  * on hemodialysis days, to administer at the earliest possible time after dialysis since approximately 2/3 of the dose is removed by hemodialysis.
- Dosage in hepatic impairment: no dosage adjustment is necessary.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials between 2-8°C; protect from light.
- Reconstituted solution is stable for 1 hour at room temp before its transfer and further dilution into a bag. Do not freeze.
- Diluted solution is stable for 24 hours at room temp or 7 days in the fridge in NS or D5W in PVC bags.
- Diluted solution should be clear and may vary from colourless to slightly yellow.

MISCELLANEOUS
- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Positive Coombs' test has been reported.
- Tazobactam is included as a beta-lactamase inhibitor, which extends ceftolozane spectrum of coverage to include certain beta-lactamase producing bacteria.
- Each 1.5 g vial contains 1 g of ceftolozane and 500 mg of tazobactam. Doses are usually ordered as 1.5 g (1 g ceftolozane and 500 mg tazobactam), 750 mg (500 mg ceftolozane and 250 mg tazobactam), 375 mg (250 mg ceftolozane and 125 mg tazobactam), or 150 mg (100 mg ceftolozane and 50 mg tazobactam).

REFERENCES
5, 40, 95, 135.

New monograph 2017; limited revision 2020

NAME OF MEDICATION
CefTRIAXone

OTHER NAMES
Rocephin ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS

- For severe infections (lower respiratory tract infections, septicemia, bone and joint infections, UTI, skin and skin structure infections, intra-abdominal infections, endocarditis, gonorrhea, gynecologic infections, and CNS infections) due to susceptible organisms.
- Perioperative prophylaxis (not considered a drug of choice).

ADMINISTRATION

- For IV use: reconstitute the 250 mg, 500 mg, 1 g and 2 g vials with 2.4 mL, 4.8 mL, 9.6 mL or 19.2 mL, respectively, of SWFI for a final concentration of 100 mg/mL.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion (preferred): dilute in 50-100 mL of NS, D5W or dextrose-saline solutions and infuse over 15-30 minutes.
- IM: reconstitute the 250 mg, 500 mg, 1 g and 2 g vials with 0.9 mL, 1.7 mL, 3.3 mL or 6.6 mL, respectively, of SWFI, NS, D5W, or 1% lidocaine to obtain approximately 250 mg/mL. More concentrated solutions (350 mg/mL) may be obtained by reconstituting the 500 mg, 1 g and 2 g vials with 1.1 mL, 2.2 mL or 4.4 mL, respectively, of above stated solutions. The more concentrated solution (350 mg/mL) is NOT recommended for the 250 mg vial size. Inject deep into a large muscle mass.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults: usual dose: 1-2 g once daily or in equally divided doses twice daily.
  meningitis or intracranial brain abscess: 2 g IV q12h.
  gonorrhea: 250 mg IM as a single dose.
- Pediatrics: usual dose: 50-75 mg/kg/day, given in 1 or 2 divided doses per day; maximum dose 2 g/day.
  meningitis: 100 mg/kg/day given once daily or divided q12h. A loading dose of 100 mg/kg may be given at start of therapy. Maximum of 4 g/day.
  gonorrhea: 125 mg IM as a single dose.
- Dosage in renal impairment: no dosage adjustment required.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Ceftriaxone is contra-indicated in neonates (less than or equal to 28 days of age) if they require (or are expected to require) treatment with calcium containing IV solutions (including TPN, RL) because of the risk of precipitation of ceftriaxone with calcium. A crystalline material was observed in the lungs and kidneys at autopsy of a small number of neonates who received ceftriaxone and calcium containing solutions.

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COMPATIBILITY, STABILITY (Cont.)

- Do not use diluents containing calcium (e.g., RL, Hartmann's solution) to reconstitute or dilute ceftriaxone because a precipitate can form.
- Do not administer ceftriaxone simultaneously with calcium-containing IV solutions (including TPN, RL) via a Y-site because precipitation of ceftriaxone-calcium may occur. However, in patients other than neonates, ceftriaxone may be administered in the same line as a calcium containing solution, provided they are administered sequentially of one another and the infusion lines are thoroughly flushed between infusions with a compatible fluid.
- No interactions between ceftriaxone and oral calcium-containing products or between IM ceftriaxone and calcium-containing products (IV or oral) have been reported.
- Stable for 2 days at room temp and 10 days in the fridge when solutions are reconstituted with SWFI, D5W or NS to a concentration of 100 mg/mL for IV use.
- Stable for 24 hours at room temp and 3 days in the fridge when solutions are reconstituted with SWFI, D5W, NS or lidocaine 1% to a concentration of 250 mg/mL and 350 mg/mL for IM use.
- Stable for 3 days at room temp and 14 days in the fridge when diluted in D5W at a concentration of 40 mg/mL in a PVC container.
- Stable for 3 days at room temp and 30 days in the fridge when diluted in NS at a concentration of 40 mg/mL in a PVC container.
- Stable for 48 hours at room temp and 72 hours in the fridge when diluted in NS at a concentration of 10 mg/mL.
- Stable for 7 days at room temp and 12 weeks in the fridge when diluted in D5W at a concentration of 10 mg/mL in a glass bottle.
- Stable for 180 days at -20°C, 40 days in the fridge or 3 days at room temp when diluted with SWFI at a concentration of 100 mg/mL and stored in polypropylene syringes.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Beta-lactamase resistant.
- Positive Coombs' test has been reported.

REFERENCES

1, 2, 4, 5, 82, 95, 135, 143, 216.

Full revision 2012; limited revision 2015, 2016, 2018, 2020

NAME OF MEDICATION
CEFUROXIME

OTHER NAMES
Kefurox ®, Zinacef ®

CLASSIFICATION
Antibiotic - cephalosporin

INDICATIONS

- For severe infections (respiratory tract infections, genitourinary tract infections, soft tissue infections, skin and skin structure infections, bone and joint infections, and septicemia) due to susceptible organisms.
- Perioperative prophylaxis.
- Has been used in the treatment of meningitis but is no longer considered a drug of choice; third generation cephalosporins are now preferred as they have been found to be clinically superior.

ADMINISTRATION

- For IV use: reconstitute the 750 mg and 1500 mg vials with 8 mL and 16 mL of SWFI, respectively, to get an approximate concentration of 90 mg/mL. Shake well until dissolved.
- IV direct: physician or RN. Administer slowly over 3-5 minutes.
- Intermittent IV infusion (preferred): dilute in 50-100 mL of D5W or NS and infuse over 15-60 minutes.
- Continuous IV infusion: dilute in 500-1000 mL of D5W or NS and administer as a continuous infusion.
- IM: reconstitute 750 mg vial with 3 mL of SWFI to obtain approximately 220 mg/mL. Inject deep into a large muscle mass.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, urticaria, pruritus, anaphylaxis.
- GI: diarrhea, nausea, vomiting, colitis.
- Hematologic: eosinophilia, leukopenia, neutropenia, thrombocytopenia.
- Hepatic: transient increase in LFT.
- Renal: transient increase in serum creatinine and urea.
- Local reactions: phlebitis (with IV administration); pain (with IM administration).

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults:
  - Usual dose: 750 mg-1.5 g q8h.
  - Surgical prophylaxis: 1.5 g 30-60 minutes before skin incision; repeat dose in the operating room after 4 hours when surgery is prolonged.
- Pediatrics (aged 1 month to 12 years old): 75-150 mg/kg/day in 3 divided doses. Maximum dose 6 g/day.
- Dosage in renal impairment:

<u>Method 1:</u>

CrCl (mL/min)	greater than 50	50-10	less than 10
Dosing interval (hr)	8	8-12	24

<u>Method 2:</u>

CrCl (mL/min)	greater than 20	20-10	less than 10
Dosing interval (hr)	750 mg-1.5 g q8h	750 mg q12h	750 mg q24h

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Reconstituted solutions may range in colour from light yellow to amber; solutions may darken without affecting potency.

…/Cont.

NAME OF MEDICATION
CEFUROXIME

OTHER NAMES
Kefurox ®, Zinacef ®

CLASSIFICATION
Antibiotic - cephalosporin

…/Cont.

COMPATIBILITY, STABILITY (Cont.)

- Reconstituted solution (90-220 mg/mL) is stable 24 hours at room temp or 48 hours if in the fridge.
- Stable for 24 hours at room temp and 7 days in the fridge in NS or D5W at concentrations of 7.5 and 15 mg/mL in PVC containers.
- Stable for 30 days in the fridge in D5W or NS at concentrations of 5 and 10 mg/mL in PVC containers.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.
- Beta-lactamase resistant.
- Positive Coomb's test has been reported.

REFERENCES

1, 2, 4, 5, 40, 82, 135, 143, 216, 338.

Full revision 2012; limited revision 2015

NAME OF MEDICATION
CEMIPLIMAB

OTHER NAMES
Libtayo ™

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of metastatic or locally advanced cutaneous squamous cell carcinoma.

ADMINISTRATION

- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards and Dosage sections.**
- Intermittent IV infusion (mandatory): do NOT shake vials; dilute in 50-100 mL of NS or D5W for a final concentration between 1-20 mg/mL. Gently invert bag to mix the solution. Do NOT shake. Administer over 30 minutes through an IV line containing a sterile in-line or add-on filter (pore size of 0.2-5 microns). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication**: consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: immune-mediated reactions, can be fatal; can involve any organ: anaphylaxis, dermatologic (pruritus, rash, pemphigoid, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis), cardiovascular (myocarditis, pericarditis), GI (diarrhea, colitis), hepatitis, CNS (meningitis, myasthenia gravis, myelitis, encephalomyelitis, Guillain-Barré syndrome, encephalitis, neuropathies), endocrine (thyroid disorders, adrenal insufficiency, type 1 diabetes mellitus including diabetic ketoacidosis, hypophysitis), arthritis, myositis (including rhabdomyolysis), nephritis, ophthalmic (iritis, uveitis), pneumonitis, thrombocytopenic purpura, vasculitis. Consult manufacturer's recommendations for management of these reactions and when to withhold or discontinue treatment.
- Infusion-related reactions: nausea, fever, rash, dyspnea. For mild or moderate infusion reactions, interrupt infusion until resolution of reaction or reduce infusion rate and closely monitor the patient; **consider premedication** with diphenhydramine 50 mg PO (or equivalent) and/or acetaminophen 325 to 1000 mg PO and, if necessary, hydrocortisone 25 mg IV (or equivalent) 30 minutes prior to subsequent infusions. For more severe reactions, discontinue treatment permanently.
- Cardiovascular: hypertension.
- GI: nausea, vomiting, diarrhea, constipation, anorexia.
- CNS: headache.
- Hematologic: anemia, lymphopenia.
- Musculoskeletal: arthralgia, pain.
- Respiratory: cough, pneumonia, upper respiratory tract infection.
- Fatigue.
- Graft-versus-host disease or hepatic veno-occlusive disease in patients receiving allogeneic stem cell transplantation.
- Solid organ transplant rejection.

DOSAGE

- **In patients who previously experienced an infusion-related reaction, consider premedication with diphenhydramine 50 mg PO (or equivalent) and/or acetaminophen 325 to 1000 mg PO, and, if necessary, hydrocortisone 25 mg IV (or equivalent) 30 minutes prior to cemiplimab infusion.**
- 350 mg IV every 3 weeks OR 3 mg/kg IV every 2 weeks for patients with low body weight.
- Dosage in renal impairment: no dosage adjustment required if CrCl is 30 mL/min or greater; no dosage recommendation can be made if CrCl is less than 30 mL/min as data is limited.
- Dosage in hepatic impairment: no dosage adjustment required in mild impairment (total bilirubin greater than 1 to 1.5 x ULN or AST greater than ULN) based on limited data; no data for moderate (total bilirubin greater than 1.5 to 3 x ULN and any AST) to severe (total bilirubin greater than 3 x ULN and any AST) impairment.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Vial solution is clear to slightly opalescent, colourless to pale yellow and may contain trace amounts of translucent to white particles.
- Stable for 8 hours at room temp and 24 hours between 2-8°C in NS or D5W at concentrations between 1-20 mg/mL (includes time for infusion administration). Allow solution to reach room temp prior to administration.

MISCELLANEOUS

- Monitor blood counts, blood glucose and liver, renal and thyroid function at baseline, before each dose and as clinically indicated.

REFERENCES

5, 129, 135, 165.

New monograph 2021

NAME OF MEDICATION
CERTOLIZUMAB PEGOL

OTHER NAMES
Cimzia ®

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS
- Treatment of moderate to severe rheumatoid arthritis, with or without methotrexate.
- Treatment of moderate to severe psoriatic arthritis, with or without methotrexate, in patients who have failed one or more DMARDs.
- Treatment of active ankylosing spondylitis in patients who had an inadequate response to conventional therapy.
- Treatment of severe non-radiographic axial spondyloarthritis in patients who had an inadequate response or are intolerant to NSAIDs.
- Treatment of moderate to severe plaque psoriasis in patients who are candidates to systemic therapy.

ADMINISTRATION
- SC: in the thigh or abdomen. Rotate site. Do not inject in skin that is tender, bruised, hard or red. If a 400 mg dose is given, use 2 separate injection sites.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rare; angioedema, dyspnea, hypotension, rash, serum sickness, urticaria. May occur after first administration; discontinue if occurs.
- Cardiovascular: hypertension, rate and rhythm disorders, CHF (new onset or worsening).
- GI: diarrhea, abdominal pain, gastritis, vomiting, dry mouth.
- CNS: headache, anxiety, dizziness.
- Elevation of liver enzymes.
- Infections: patients receiving certolizumab pegol may be more prone to develop infections, including tuberculosis, fungal or opportunistic infections. Discontinue if serious infection occurs.
- Fatigue, arthralgia, myalgia, back pain, fever.
- Local reactions: pain, erythema, hematoma, discolouration.

DOSAGE
- Loading dose for all indications (except plaque psoriasis): 400 mg SC (2 injections of 200 mg each) at weeks 0, 2 and 4.
- Maintenance dose for all indications (except plaque psoriasis): 200 mg SC every 2 weeks or alternatively, 400 mg SC (2 injections of 200 mg each) every 4 weeks.
- Plaque psoriasis: 400 mg SC (2 injections of 200 mg each) every 2 weeks OR, after 3 doses of 400 mg SC (at weeks 0, 2, 4), may consider 200 mg SC every 2 weeks.
- Missed dose: if the next scheduled dose is within a week, do not administer missed dose and wait until next dose is due. If the next scheduled dose is one week or longer away, administer the missed dose as soon as possible then follow with the next scheduled dose.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled syringes in the fridge; do not freeze; protect from light.
- Stable for 10 days at room temp, protected from light; to discard after this period of time.

MISCELLANEOUS
- Do not administer live or live attenuated vaccines concurrently with this drug.
- Do not use with other biologic DMARDs (e.g., abatacept, anakinra, rituximab, tocilizumab) because of possible increased risk of infections and similar toxicities.
- Patient should be tested for hepatitis B virus infection and tuberculosis before initiating therapy.
- May cause false elevation of aPTT.

REFERENCES
1, 5.

New monograph 2015; limited revision 2016, 2020

NAME OF MEDICATION
CETUXIMAB

OTHER NAMES
Erbitux ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of patients with epidermal growth factor receptors (EGFR) expressing metastatic colorectal cancer with the wild-type K-RAS gene: 1) in combination with irinotecan for patients who are refractory to other irinotecan-based chemotherapy; 2) as a single agent therapy for patients who are intolerant of irinotecan-based chemotherapy; or 3) as single agent therapy for patients who have failed both irinotecan- and oxaliplatin-based regimens and who have received a fluoropyrimidine.
- Initial treatment of locally or regionally advanced squamous cell carcinoma of the head and neck, in combination with radiation therapy.

ADMINISTRATION

- **Ensure premedication has been administered prior to each infusion; refer to Dosage section.**
- Intermittent IV infusion (mandatory): do NOT shake or dilute the solution. Transfer desired dose to an empty sterile IV bag, bottle, or syringe. Prime administration line with NS before administration. Infuse dose by piggy backing to the patient's infusion line via a low-protein-binding 0.22 micron in-line filter. Administer initial dose over 2 hours and maintenance dose over 60 minutes; do NOT administer at a rate exceeding 10 mg/min. Flush line with NS after administration.
- Monitor patient for 60 minutes following each infusion and for longer in those who experienced infusion reactions.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Infusion-related reactions: bronchospasm, stridor, hoarseness, urticaria, hypotension, shock, loss of consciousness, myocardial infarct and/or cardiac arrest. Consider **premedications** to lessen these infusion reactions, but they can still occur despite their use; refer to Dosage section. For severe infusion reactions, stop the infusion immediately and discontinue permanently treatment with cetuximab. For mild to moderate reactions, slow the infusion rate by 50% (5 mg/min maximum) and treat symptoms as appropriate.
- GI: diarrhea, nausea, vomiting, abdominal pain, constipation, anorexia, mucositis, dehydration.
- Fever, headache, asthenia, fatigue, insomnia.
- Respiratory: dyspnea, cough, interstitial lung disease, pulmonary embolus.
- Hypomagnesemia, hypokalemia, hypocalcemia.
- Dermatologic: acneiform rash; limit sun exposure during treatment and for two months after stopping therapy.

DOSAGE

- **Premedicate all patients with diphenhydramine 50 mg IV (at TOH, give PO) 30-60 minutes prior to each cetuximab dose. Adding a corticosteroid (e.g., dexamethasone 8 mg IV) at least before the first infusion (unless medically contraindicated) reduces the incidence of severe infusion reactions. Consider premedicating with a corticosteroid prior to each subsequent infusions**.
- Colorectal cancer:
  - initial dose, either as monotherapy or combined with irinotecan: 400 mg/m^2 IV. Subsequent maintenance dose: 250 mg/m^2 IV weekly until disease progression or unacceptable toxicity.
  - 500 mg/m^2 IV every 2 weeks until disease progression or unacceptable toxicity.
- Squamous cell carcinoma of the head and neck: initial dose of 400 mg/m^2 IV administered 1 week prior to initial course of radiation therapy. Subsequent maintenance dose: 250 mg/m^2 IV weekly for the duration of radiation therapy. Complete infusion 60 minutes prior to radiation therapy.
- Dosage adjustment due to toxicity: decrease dose based on severity of acneiform rash; refer to manufacturer's recommendations.
- Consult specific protocol.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store in the fridge. Protect from freezing.
- Solution should be clear and colourless and may contain a small amount of easily visible, white, amorphous cetuximab particulates.
- Do not mix or dilute with other drugs.
- Prepared dose and unused portion in the vial are stable for 12 hours in the fridge or 8 hours at room temp (according to manufacturer).

MISCELLANEOUS

- Cetuximab is contraindicated in patients with known hypersensitivity to murine protein.
- Administration of a test dose does not reliably identify patients at risk for severe allergic reactions and is not recommended.

REFERENCES

1, 2, 6, 40, 95, 129, 165, 208, 552.

Full revision 2012; limited revision 2013, 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
CHLORAMPHENICOL

OTHER NAMES
Chloromycetin ®

CLASSIFICATION
Antibiotic

INDICATIONS

- For serious infections caused by susceptible organisms, where potentially less toxic drugs are ineffective or contraindicated: acute *Salmonella typhi* infections, meningeal infections, bacteremia, Rocky Mountain spotted fever, psittacosis.
- Cystic fibrosis regimens.
- Treatment of plague and tularemia (in the context of bioterrorism).

ADMINISTRATION

- Reconstitute 1000 mg vial with 10 mL of SWFI or D5W to obtain an approximate concentration of 100 mg/mL.
- IV direct: physician only. Administer over at least 1 minute.
- Intermittent IV infusion (preferred): dilute in 50-100 mL of D5W or NS; final concentration should not exceed 20 mg/mL. Infuse over 10-30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: fever, rash, angioedema, urticaria, anaphylaxis.
- GI: nausea, vomiting, diarrhea; unpleasant taste with rapid IV administration.
- CNS: headache, confusion, depression, delirium; optic and peripheral neuritis with long-term use.
- Hematologic: dose-related, reversible bone marrow depression (common); idiosyncratic, non-dose-related, irreversible aplastic anemia with a high mortality rate (rare).
- Gray syndrome toxicity (type of circulatory collapse) in premature and full-term infants; has occurred in children up to 2 years of age.

DOSAGE

- Adults and pediatrics (full term infants from age 2 weeks and above):
  - Usual: 50 mg/kg/day IV in equally divided doses q6h.
  - Plague: 25 mg/kg IV q6h for 10 days.
  - Tularemia: 15 mg/kg IV q6h for 14-21 days.
  - Maximum dosage: 100 mg/kg/day, not to exceed 4 g/day.
- Premature infants, full term infants under 2 weeks of age, older infants with immature metabolic processes: 25 mg/kg/day IV in equally divided doses q6h.
- In patients with impaired renal and/or hepatic function, dosage must be reduced in proportion to the degree of impairment; monitor plasma chloramphenicol concentrations.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp (15-30°C).
- Reconstituted solution is stable for 30 days at room temp. Cloudy solutions should not be used.
- Stable 24 hours at room temp, exposed to light, in D5W or NS at a concentration of 10 mg/mL.

MISCELLANEOUS

- Hematologic studies (complete blood count) are recommended every 2 days during therapy to monitor for dose-related bone marrow suppression.
- Desired plasma levels: trough = 5-10 mcg/mL (5-15 mcg/mL for meningitis); peak = 10-20 mcg/mL (15-25 mcg/mL for meningitis). Take trough serum level 5 minutes before the next dose and peak serum level 30-90 minutes after the completion of IV dose.
- Sodium content: 2.25 mEq (52 mg)/g.

REFERENCES

1, 2, 4, 5, 6, 40, 82, 95, 208, 216, 536, 537.

Full revision 2012; limited revision 2015

NAME OF MEDICATION	
CIDOFOVIR	
OTHER NAMES	CLASSIFICATION
Mar-Cidofovir, Vistide ®	Antiviral

INDICATIONS

- Treatment of cytomegalovirus (CMV) retinitis in patients with acquired immunodeficiency syndrome (AIDS).
- Treatment of disease due to adenovirus.
- Treatment of BK polyomavirus-associated hemorrhagic cystitis.

ADMINISTRATION

- **Ensure protocol to reduce risk of renal impairment has been initiated; refer to specific protocol in the Dosage section**.
- **Consider premedication to decrease hypersensitivity reactions; refer to Potential Administration Hazards section.**
- Intermittent IV infusion: dilute dose in 100 mL NS; infuse over 60 minutes at a constant rate. Flush line with 10 mL NS before and after infusion. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, fever, chills; can treat with antihistamine and/or acetaminophen; **consider premedication** for subsequent doses.
- GI: nausea, vomiting, diarrhea, anorexia.
- CNS: headache, pain.
- Hematologic: neutropenia, anemia.
- Ophthalmic: iritis, uveitis, decreased intraocular pressure.
- Renal: dose-dependent nephrotoxicity, proteinuria, Fanconi's syndrome; refer to specific protocol in Dosage section to reduce risk of renal impairment.
- Respiratory: cough, dyspnea.
- Infection.
- Metabolic acidosis.
- Asthenia.

DOSAGE

- Induction dose: 5 mg/kg IV once weekly for 2 consecutive weeks.
- Maintenance dose: 5 mg/kg IV once every 2 weeks. Maintenance dose to be started 2 weeks after completion of induction treatment.
- **To reduce risk of renal impairment, the protocol below must be followed with each dose of cidofovir:**
  - Probenecid (not marketed in Canada): 2 g orally 3 hours before cidofovir dose, followed by 1 g orally at 2 and 8 hours after completion of cidofovir infusion for a total probenecid dose of 4 g for each cidofovir dose.
  - Hydration: infuse at least 1 liter of NS over 1-2 hours immediately before starting cidofovir infusion; for patients who can tolerate additional fluid, infuse a second liter of NS over 1-3 hours. This second liter can be initiated either at the start of the cidofovir infusion or immediately afterward.
- Dosage in renal impairment: decrease maintenance dose to 3 mg/kg for remainder of therapy if serum creatinine increases by 27-35 mcmol/L from baseline; discontinue therapy if serum creatinine increases by 44 mcmol/L or greater from baseline or if proteinuria of 2^+ or greater develops; do not initiate therapy if baseline creatinine clearance is 55 mL/min or lower or if baseline proteinuria is 2^+ or greater (100 mg/dL or greater).

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C.
- Stable for 24 hours in the fridge or at room temp in D5W or NS at concentrations of 0.21 mg/mL and 8.12 mg/mL in PVC or polyolefin containers; must be administered within 24 hours of preparation.
- Stable for 5 days in the fridge in NS at concentrations of 0.2 mg/mL and 8.1 mg/mL in PVC or polyolefin containers.
- Stable for 150 days in the fridge, at 25°C (exposed to or protected from light) and at 32°C in NS at a concentration of 6.25 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Contraindicated if patient is receiving concomitant nephrotoxic drugs (e.g., aminoglycosides, amphotericin B, foscarnet, NSAIDs, etc); at least 7 days should elapse between discontinuation of such drugs and administration of cidofovir.
- Serum creatinine and urine protein levels must be determined within 48 hours prior to each dose of cidofovir.
- White blood cell counts with differential should be monitored prior to each dose.

REFERENCES

1, 4, 5, 40, 95, 135, 143, 427, 479, 533.

Full revision 2012; limited revision 2014, 2015, 2018, 2019, 2020

NAME OF MEDICATION
CIPROFLOXACIN

OTHER NAMES
Cipro ®

CLASSIFICATION
Antibiotic - quinolone

INDICATIONS

- Broad spectrum antibiotic coverage (including some methicillin-resistant *Staph aureus*, Enterobacteriaceae and *Pseudomonas aeruginosa*) indicated for use in a variety of moderate to severe infections when oral administration is not possible.
- Treatment of anthrax, plague and tularemia (in the context of bioterrorism).

ADMINISTRATION

- Intermittent IV infusion: use premixed bag of 2 mg/mL or dilute the concentrate of 10 mg/mL for the required dose in NS, D5W or other compatible solution to a concentration of 1-2 mg/mL. Infuse over 60 minutes into a large vein. Flush line with D5W before and after administration of ciprofloxacin. Refer to Compatibility, Stability section.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, fever, eosinophilia, anaphylaxis (rare).
- Cardiovascular: aortic aneurism/aortic dissection (rare, but serious); avoid using in patients with known history or at risk of aortic aneurism.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: dizziness, sedation, agitation, headache, confusion, tremors, insomnia.
- Myasthenia gravis exacerbation (rare, but serious); avoid using in patients with this condition.
- Local reactions: phlebitis, burning, pain, pruritus, paresthesia, erythema, swelling; more frequent if infused rapidly (e.g., over 30 minutes) or via a small vein.

DOSAGE

- Adults:
  - Usual dose is 200-400 mg IV q12h; maximum dose 400 mg IV q8h.
  - Anthrax: 400 mg IV q12h; to ADD 1 or 2 additional antimicrobials to which *Bacillus anthracis* is likely susceptible (rifampin, vancomycin, penicillin, ampicillin, chloramphenicol, imipenem, clindamycin and clarithromycin).
  - Plague: 400 mg IV q12h for 10-14 days.
  - Tularemia: 400 mg IV q12h for at least 10 days.
- Pediatrics:
  - Usual dose: 20-30 mg/kg/day IV in divided doses q12h; maximum 800 mg/day.
  - Anthrax: 20-30 mg/kg/day IV in divided doses q12h; maximum 400 mg/dose; maximum 1000 mg/daily; initiated as part of a multiple anti-infective regimen, add 1 or 2 additional antimicrobials to which *Bacillus anthracis* is likely susceptible (rifampin, vancomycin, penicillin, ampicillin, chloramphenicol, imipenem, clindamycin and clarithromycin).
  - Plague: 30 mg/kg/day IV in divided doses q12h; maximum 1000 mg/daily; for 10-14 days.
  - Tularemia: 30 mg/kg/day IV in divided doses q12h; maximum 1000 mg/daily; for at least 10 days.
  - Cystic fibrosis and in select cases: 30 mg/kg/day IV in divided doses q8-12h; maximum 1200 mg/day.
- Dosage in renal impairment:

  Adults:

  Method 1:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	50-75%	50%

  Method 2:

  CrCl less than 30 mL/min: administer usual dose q18-24h.

  Pediatrics: adjust dosing interval in moderate to severe renal failure to q18-24h.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Available as a premixed, ready-to-use solution of 2 mg/mL in D5W and a concentrate of 10 mg/mL liquid which requires further dilution.
- Store vials and premixed bags at room temp. Protect from light and freezing.
- Premixed bag, out of overwrap, is stable for 14 days when protected from sunlight.
- Diluted concentrate is stable for 14 days in the fridge or at room temp in D5W, D10W, NS, dextrose-saline solutions, or RL at a concentration of 2 mg/mL.
- Ciprofloxacin is incompatible with many medications; flushing the line with D5W before and after its administration may reduce the risk of drug precipitation. From TOH experience, flushing with NS is not as effective to prevent some incompatibilities.

MISCELLANEOUS

- Switch to oral route when feasible. Equivalent regimens, IV to PO (regular release formulation):
  200 mg IV q12h = 250 mg PO q12h; 400 mg IV q12h = 500 mg PO q12h; 400 mg IV q8h = 750 mg PO q12h.
- Use in pediatrics is not routinely recommended but may be justified in special circumstances.

REFERENCES

1, 2, 4, 5, 9, 28, 40, 44, 82, 95, 135, 143, 195, 216, 536, 537, 538.

Full revision 2012; limited revision 2015, 2020

NAME OF MEDICATION
CISATRACURIUM

OTHER NAMES
Nimbex ®

CLASSIFICATION
Neuromuscular blocker

INDICATIONS
- Adjunct to general anesthesia.
- To facilitate non-emergency endotracheal intubation.
- To provide skeletal muscle relaxation during surgery or mechanical ventilation.

ADMINISTRATION
- IV direct: physician trained in anesthesiology; RN may administer subsequent doses; ***ventilator support; cardiac monitoring***; may administer undiluted over 5-10 seconds.
- Continuous IV infusion: ***ventilator support; cardiac monitoring***. May further dilute to 0.1 mg/mL or 0.4 mg/mL in D5W or NS. At TOH: withdraw 20 mL from a 100 mL minibag of D5W or NS and discard. Add 40 mg (20 mL from cisatracurium 2 mg/mL vials) to the minibag to obtain a final concentration of 0.4 mg/mL. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylactic or anaphylactoid reactions, bronchospasm, rash, itching.
- Cardiovascular: hypotension, bradycardia, flushing.

Antidote: anticholinesterase agents such as neostigmine or edrophonium, in conjunction with an anticholinergic agent such as atropine or glycopyrrolate.

DOSAGE
- Cisatracurium should be administered by or under the supervision of experienced clinicians familiar with the agent. Dosage must be individualized in each case.
- Adults:
  - Initial: 0.15 to 0.2 mg/kg/dose for a duration of action of 55 and 61 minutes, respectively.
    Onset of action: 1.5 to 2 minutes.
  - Maintenance: 0.03 mg/kg/dose. Duration of action of approximately 20 minutes.
  - Continuous Infusion: initial: 3 mcg/kg/min.
    maintenance: 1-2 mcg/kg/min; up to 10 mcg/kg/min in the intensive care setting.
- Pediatrics (above 2 years of age):
  - Initial: 0.1 mg/kg/dose. Onset of action within 2.8 minutes; duration of action of 28 minutes.
  - Maintenance: 0.03 mg/kg/dose. Duration of action of approximately 20 minutes.
  - Continuous Infusion: 1-4 mcg/kg/min.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store in the fridge. Protect from light and freezing. Stable 45 days at room temp in original vial.
- Stable 30 days at room temp or in the fridge undiluted at a concentration of 2 mg/mL in polypropylene plastic syringes.
- Stable 30 days in the fridge in D5W or NS at a concentration of 0.1, 2, and 5 mg/mL in PVC containers.
- Stable 14 days at room temp in NS and stable 7 days at room temp in D5W at a concentration of 0.1 mg/mL in PVC containers.
- Stable 30 days at room temp in NS and stable 14 days at room temp in D5W at a concentration of 0.2 mg/mL in PVC containers.
- Stable 30 days at room temp in D5W or NS at a concentration of 5 mg/mL in PVC containers.
- Incompatible with alkaline solutions (e.g., barbiturate solutions).
- Incompatible with RL.

MISCELLANEOUS
- Product is hypotonic.

REFERENCES
1, 2, 4, 5, 40, 82, 135, 139, 208.

Full revision 2012; limited revision 2015, 2016, 2017, 2019, 2020

NAME OF MEDICATION
CISplatin

OTHER NAMES
Cisplatinum, Platinol ®, Platinol-AQ ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Used in combination with other agents and/or therapies in the treatment of testicular, ovarian, cervical, bladder, head and neck, and lung cancers.
- Also used in a variety of other neoplasms.

ADMINISTRATION

- Intermittent IV infusion: For low doses: dilute in 50-100 mL of NS; infuse over 15-30 minutes. For higher doses: dilute in 2 L of NS; infuse over 6-8 hours; consult specific protocol.
- Continuous IV infusion: dilute in 1-2 L of NS and administer as a continuous infusion; consult specific protocol.
- A 50 mg vial is available in powder form via Health Canada Special Access Programme for certain indications.
- Do NOT use needles, IV sets or equipment containing aluminum for preparation or administration; a black platinum precipitate will form if cisplatin comes in contact with aluminum and potency will decrease.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid (facial edema, flushing, bronchoconstriction, hypotension and tachycardia); rash, urticaria.
- Cardiovascular: arterial thromboembolism, bradycardia, conduction disorder, hypertension.
- GI: nausea, vomiting, hiccups.
- Electrolyte disturbance, increased serum amylase, transient increase of liver enzymes and bilirubin.
- Hematologic: myelosuppression, anemia.
- Nephrotoxicity with hypokalemia and hypomagnesemia; pre-treatment hydration with NS (supplemented possibly with potassium chloride and magnesium sulfate) and concomitant mannitol or furosemide-induced diuresis are recommended to reduce nephrotoxicity. Consult specific protocol.
- Otic: ototoxicity; dose related, cumulative and irreversible.
- Neurotoxicity, peripheral neuropathy.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Numerous dosing protocols exist (dependent on clinical criteria and concomitant therapy).
- Single agent therapy: 50-100 mg/m^2 IV every 3-4 weeks.
- For testicular cancer in combination therapy: 20 mg/m^2 IV daily for five consecutive days, every 3 weeks for 3-4 courses.
- Doses greater than 100 mg/m^2/cycle IV every 3-4 weeks are rarely used; confirmation of higher doses is advised.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	75%	50% or discontinue

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from light. Do not put in fridge or freezer as precipitate may form; however, the precipitate should dissolve at room temp without loss of potency.

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COMPATIBILITY, STABILITY (Cont.)

- Punctured vial is stable for 28 days at room temp if protected from light and 7 days at room temp if exposed to fluorescent room light.
- Diluted solution is stable for at least 24 hours at room temp in NS.
- At concentrations greater than 0.6 mg/mL precipitation occurs if refrigerated. More dilute solutions may be refrigerated. Discard diluted solutions with precipitate; they should not be warmed for redissolution.
- Minimise exposure to light.

MISCELLANEOUS

REFERENCES

1, 2, 4, 6, 40, 95, 129, 165, 208, 216.

NAME OF MEDICATION
CLADRIBINE

OTHER NAMES
2-CdA, 2-Chlorodeoxyadenosine, Leustatin ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of hairy cell leukemia.

ADMINISTRATION

- Intermittent IV infusion (daily), Continuous IV infusion (daily): dilute in 500 mL of NS; infuse over 2 or 24 hours.
- Continuous IV infusion (7 days): use a medication cassette; dilute to 100 mL in bacteriostatic NS (benzyl alcohol preserved); both the drug and NS should be passed through a 0.22 micron filter. Cassettes prepared for patients weighing more than 85 kg may have reduced preservative effectiveness.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rare.
- GI: nausea (mild), anorexia, vomiting, diarrhea.
- Fever, chills, dizziness, headache, insomnia, fatigue.
- Hyperuricemia: can be minimized with allopurinol and hydration.
- Dermatologic: rash.
- Local reactions: redness, swelling, pain, thrombosis, phlebitis.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 0.09 mg/kg/day IV as an infusion for 5-7 days.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	75%	50%

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store in the fridge. Protect from light. Precipitate may form with low temperatures. In the presence of precipitate or if the vial freezes: allow the vial to warm naturally to room temp; do not heat or microwave. Vials with precipitate may be shaken. Potency and expiry are not affected, however, thawed vials should not be refrozen.
- Diluted solution may be stored in the fridge for up to 8 hours prior to initiation of infusion.
- Stable at room temp for at least 24 hours in NS.
- Incompatible in D5W.
- Stable 30 days at 4°C and 18°C in NS at a concentration of 0.016 mg/mL in PVC containers.
- Stable 14 days at 4°C in NS at a concentration of 0.024 mg/mL in glass containers.
- Dilutions for infusions in medication cassettes are stable for at least 7 days.

MISCELLANEOUS

REFERENCES

1, 2, 4, 40, 95, 129, 165, 216.

Full revision 2012; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
CLINDAMYCIN

OTHER NAMES
Dalacin C ®

CLASSIFICATION
Antibiotic

INDICATIONS

- For severe infections caused by susceptible strain of gram positive staphylococci, streptococci (except *E. faecalis*), pneumococci and anaerobic bacteria when oral administration is not possible.
- For *P. carinii* pneumonia in patients infected with HIV who failed conventional therapy.
- Perioperative prophylaxis, prevention of bacterial endocarditis.

ADMINISTRATION

- Intermittent IV infusion: dilute to a concentration not exceeding 18 mg/mL or use premixed bag and administer at a rate not exceeding 30 mg/min.
- Recommended concentration and minimum infusion time from the manufacturer:

Dose	Volume of diluent	Infusion time
300 mg	50 mL	10 minutes
600 mg	50 mL	20 minutes
900 mg	100 mL (or if use premixed bag: 50 mL)	30 minutes
1200 mg	100 mL	45 minutes

- Administration of more than 1200 mg in a single one hour infusion is not recommended.
- IM (deep).
- Consult Medical staff rules and regulations MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis) for specific dosing regimens and infusion time.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: maculopapular rash, urticaria, pruritus, fever, hypotension, rarely polyarthritis, anaphylactoid reactions.
- GI: unpleasant or metallic taste, nausea, vomiting, abdominal pain and tenesmus. Severe diarrhea and cramping; occasionally *C. difficile* associated diarrhea and colitis may develop, even up to several weeks following discontinuation of therapy.
- Hepatic and hematologic abnormalities.
- Local reactions: IV administration: thrombophlebitis, erythema, pain, swelling. IM administration: pain, induration, sterile abscess, reversible increase in serum creatine kinase.

DOSAGE

- Dosage and route of administration (IM/IV) should be determined by severity of infection, susceptibility of the causative organism(s), and condition of the patient. The IV route is preferable for patients with severe or life-threatening infections, or with lowered resistance resulting from debilitating conditions (e.g., malnutrition, trauma, surgery, diabetes, heart failure or malignancy, particularly if shock is present or pending).
- Adults:
  - Perioperative prophylaxis: 600-900 mg IV 30 minutes before skin incision.
  - Prevention of bacterial endocarditis: 600 mg IV/IM 30-60 minutes before procedure.
  - Treatment: usual: 600-2700 mg/day divided in 2 to 4 doses. For *P. carinii* pneumonia: 600-900 mg IV q6-8h in combination with oral primaquine. For life-threatening infections, dose may be increased up to 4800 mg/day.
- Single IM doses greater than 600 mg are not recommended.
- Pediatrics (over one month):
  - Prevention of bacterial endocarditis: 20 mg/kg IV 30-60 minutes before procedure. Maximum: 600 mg.
  - Treatment: 15-40 mg/kg/day divided in 3 to 4 doses, depending on type and severity of infection. Maximum: 3600 mg/day.

…/Cont.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Crystals may form with low temperature storage or freezing; allow vial to warm naturally to room temp for crystal dissolution.
- Stable 22 days at room temp and 54 days in the fridge in D5W or NS at a concentration of 6 and 12 mg/mL in PVC containers.
- Stable 2 days at room temp under fluorescent light undiluted at a concentration of 150 mg/mL in polypropylene syringes.
- Compatible with RL.

MISCELLANEOUS

REFERENCES

1, 2, 4, 9, 40, 82, 95, 143.

Full revision 2012; limited revision 2014, 2015, 2020

NAME OF MEDICATION
CLODRONATE

OTHER NAMES
Bonefos ®, Clasteon ®, Clodronic acid

CLASSIFICATION
Bisphosphonate

INDICATIONS

- Management of hypercalcemia of malignancy following hydration.
- Adjunct in the management of osteolysis resulting from bone metastases of malignant tumours.

ADMINISTRATION

- Intermittent IV infusion: dilute daily dose in 500 mL of NS, D5W and administer over at least 2-6 hours. Infuse 1500 mg dose over at least 4 hours.
- SC infusion (chest, abdomen, thigh): dilute in 50 to 250 mL of NS or D5W and infuse over 2 to 3 hours; can also dilute in 1 L of NS or D5W and infuse over 6-24 hours. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria; bronchospasm with acetylsalicylic acid sensitivity (rare).
- Cardiovascular: thrombophlebitis (with rapid infusion).
- GI: anorexia, vomiting, diarrhea, gastric pain.
- Hypocalcemia, hypophosphatemia, transient hyperparathyroidism.
- Nephrotoxicity: renal failure, proteinuria, increase in serum creatinine.
- Musculoskeletal pain.
- Osteonecrosis of the jaw; patients should have dental examinations prior to therapy and avoid invasive dental procedures while on treatment.
- Local reactions: SC infusion: pain, swelling, bruising, redness; can be alleviated by decreasing the rate of administration or changing the site of administration; application of heat may reduce local irritation.

DOSAGE

- Dehydration must be corrected with NS prior to treatment.
- Hypercalcemia: 300 mg IV daily for 2-5 days (maximum 7-10 days), until serum calcium returns to normal; or single 1500 mg IV dose.
- Bone metastases: 1500 mg IV as one dose every 3-4 weeks.
- Dosage in renal impairment:

<u>Method 1</u>:

CrCl (mL/min)	80-50	49-12	less than 12
Dose	75-100%	50-75%	50% or discontinue

<u>Method 2:</u>

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	25-50%	avoid

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Bonefos ®: diluted solution is stable for 24 hours at room temp in D5W or NS.
- Clasteon ®: diluted solution is stable for 12 hours at room temp in D5W or NS.
- Stable 3 days at room temp exposed to light in D5W or NS at a concentration of 0.6 mg/mL in PVC or glass containers.
- Incompatible with bivalent ions and must not be mixed with calcium containing solutions such as RL.

MISCELLANEOUS

- Hypercalcemia: onset of action = 2-3 days; usually normalized after 3-6 days of therapy.

REFERENCES

2, 95, 129, 165, 208, 216, 457, 519, 520.

Full revision 2012; limited revision 2015, 2020

NAME OF MEDICATION
CLOFARABINE

OTHER NAMES
Clolar ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of relapsed or refractory acute lymphoblastic leukemia (ALL)
- Treatment of refractory acute myeloid leukemia (AML).
- Treatment of refractory Langerhans cell histiocytosis in children 1 year of age and older.

ADMINISTRATION

- Intermittent IV infusion: ***Blood pressure monitoring***. Transfer required dose via a 0.2 micron syringe filter in a sufficient amount of D5W or NS to obtain a final concentration of 0.15-0.4 mg/mL. Administer over 2 hours; some specific protocols may allow for 1 hour infusion.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, pruritus, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Cardiovascular: tachycardia, flushing, edema; hypotension: if hypotension occurs during the 5 days of administration, stop the infusion; treatment can be reinstituted at a lower dose if hypotension resolves on its own.
- GI: nausea, vomiting, diarrhea, abdominal pain, anorexia; enterocolitis (more frequent within 30 days of treatment and in combination therapy).
- CNS: headache, anxiety.
- Hematologic: myelosuppression can be severe: anemia, leukopenia, febrile neutropenia, thrombocytopenia.
- Hepatic: elevated ALT/AST, hyperbilirubinemia, hepatotoxicity.
- Renal: elevated creatinine.
- Infection, fever, palmar-plantar erythrodysesthesia syndrome, chills, fatigue, pain in extremities, epistaxis.
- Tumour lysis syndrome, hyperuricemia: consider prophylaxis with antihyperuricemics and alkalinization of urine. Administer continuous IV fluids during the 5-day treatment period.
- Systemic inflammatory response syndrome, capillary leak syndrome (symptoms of tachypnea, tachycardia, hypotension, pulmonary edema): may be prevented by administration of corticosteroids (e.g., 100 mg/m^2 IV hydrocortisone on days 1 to 3). If occurs, discontinue the drug and institute supportive measures. Therapy may be restarted (usually at a lower dose) once patient is stable.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Adults:
  - ALL: 30-52 mg/m^2 IV per day for 5 days, every 2 to 6 weeks.
  - AML: 15-40 mg/m^2 IV per day for 3-5 days, every 3 to 6 weeks.
- Pediatrics (1 year of age and older):
  - ALL: 52 mg/m^2 IV per day for 5 days, every 2 to 6 weeks.
  - Refractory Langerhans cell histiocytosis: 25 mg/m^2/day IV for 5 days, every 28 days.
- Dosage in renal impairment: administer 50% of the dose if CrCl is 30-60 mL/min and use with caution. Contra-indicated in patients with severe renal impairment (CrCl less than 30 mL/min).
- Dosage in hepatic impairment: use with caution in patients with mild to moderate impairment. Contra-indicated in patients with severe hepatic impairment (AST and/or ALT more than 5 x ULN, and/or bilirubin more than 3 x ULN).
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at 25°C; excursions permitted between 15-30°C.
- Stable 24 hours at room temp diluted in D5W or NS at concentrations of 0.15-0.4 mg/mL.
- Stable for 28 days at room temp exposed to light or refrigerated protected from light undiluted (1 mg/mL) or in NS or D5W at concentrations of 0.2 and 0.6 mg/mL in polypropylene or polyethylene bags.

MISCELLANEOUS

- Contra-indicated in patients with a history of serious cardiac, renal, liver or pancreatic disease, and in patients with symptomatic CNS involvement.

REFERENCES

1, 5, 6, 95, 128, 135, 208.

New monograph 2016; limited revision 2017, 2020

NAME OF MEDICATION
ClomiPRAMINE *

OTHER NAMES
Anafranil ®

CLASSIFICATION
Antidepressant

INDICATIONS

- Treatment of depressive disorders, obsessive compulsive disorders, phobias, cataplexy accompanying narcolepsy and chronic painful conditions when other routes of administration have failed.

ADMINISTRATION

- Intermittent IV infusion: dilute in 250-500 mL of D5W or NS and infuse over 1.5 to 3 hours.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, anaphylaxis (rare).
- Cardiovascular: hypotension, arrhythmias, conduction defects may occur with excessive doses.
- GI: nausea, increased appetite.
- CNS: drowsiness, restlessness, dizziness, tremor, headache, myoclonus.
- Anticholinergic side effects: dry mouth, sweating, constipation, blurred vision, disturbances of micturition.
- Fatigue.
- Local reactions: thrombophlebitis and burning sensation with IV administration.

DOSAGE

- Presence of hypokalemia must be corrected before starting treatment.
- IV: initiate with 50-75 mg once daily for 3-5 days.
- IM: initiate with 25-50 mg daily and increase by 25 mg daily to maximum of 100-150 mg daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from light and excessive heat.
- Stable 3 days at room temp in D5W or NS at a concentration of 0.2 mg/mL in PVC or glass containers.

MISCELLANEOUS

- Switch to oral route when feasible; clomipramine 25 mg IV is generally equivalent to clomipramine 50 mg PO; subsequent adjustment may be required.

REFERENCES

3, 29, 45, 208.

* Available via Health Canada's Special Access Programme

Full revision 2012; limited revision 2015

NAME OF MEDICATION
CLOXACILLIN

OTHER NAMES

CLASSIFICATION
Antibiotic - penicillin

INDICATIONS

- Treatment of suspected or confirmed infections caused by susceptible penicillinase-producing staphylococci.
- Treatment of endocarditis, osteomyelitis, pneumonia, sepsis and soft-tissue infections caused by methicillin-sensitive *S.aureus* (MSSA).

ADMINISTRATION

- **CHECK ALLERGY STATUS OF PATIENT**.
- For IV use: reconstitute vials of 500, 1000, and 2000 mg with 4.8, 9.6, and 6.8 mL of SWFI, respectively, to obtain a final concentration of 100 mg/mL, 100 mg/mL and 250 mg/mL, respectively.
- IV direct: physician or RN. Administer slowly over 2-4 minutes.
- Intermittent IV infusion (preferred): dilute in 50-100 mL of solution and administer over 30-40 minutes.
- IM: reconstitute the 500 mg vial with 1.7 mL of SWFI to obtain a final concentration of 250 mg/mL; shake well.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, fever, chills, pruritus, serum-sickness, anaphylaxis.
- GI: nausea, vomiting, epigastric discomfort, flatulence, loose stools.
- Local reactions: phlebitis and thrombophlebitis with IV administration.

DOSAGE

- Adults: usual dose of 1-2 g q6h up to 2 g q4h for severe infections.
- Pediatrics:
  - Mild to moderate infections: 50-100 mg/kg/day in divided doses q6h (maximum 6 g/day).
  - Severe infections: 150-200 mg/kg/day in divided doses q4-6h; up to 300 mg/kg/day in select cases (dose limit: maximum 12 g/day; 2 g/dose).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Reconstituted solution is stable for 24 hours at room temp or 48 hours in the fridge.
- Stable 18 days in the fridge and 24 hours at room temp, in NS or D5W at a concentration of 5-50 mg/mL in PVC containers.

MISCELLANEOUS

- Cross-allergenicity with other beta-lactam antibiotics.

REFERENCES

2, 4, 5, 9, 149.

Full revision 2012; limited revision 2015, 2019, 2020

NAME OF MEDICATION
CODEINE

OTHER NAMES

CLASSIFICATION
Opioid analgesic

INDICATIONS

- For symptomatic relief of mild to moderate pain.
- Antitussive agent.

ADMINISTRATION

- IM.
- SC.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis (rare); pseudoallergic reactions due to histamine release: pruritus, urticaria.
- Cardiovascular: hypotension, syncope.
- GI: nausea, vomiting, constipation.
- CNS: sedation, dizziness, dysphoria, anxiety, somnolence.
- Respiratory depression.
- Sweating, flushing.

Antidote: naloxone (for respiratory depression).

DOSAGE

- Pain: 30 mg IM/SC q4h as needed (range: 15-60 mg).
- Antitussive dose: 10-20 mg IM/SC q4-6h, as needed (maximum 120 mg/day).
- Dosage in renal impairment:

Crcl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	75%	50%

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from light and avoid freezing. Discard if discoloured or contains a precipitate.
- Compatible in syringe with dimenhydrinate, glycopyrrolate or hydroxyzine.

MISCELLANEOUS

- The efficacy of codeine is dependent on its conversion to morphine by the body. There can be a high degree of variability in the extent of this metabolism amongst patients. Ultra-rapid metabolizers can be at increased risk of morphine toxicity and poor metabolizers may experience an absence of effect. Breastfed babies of nursing mothers taking codeine should be monitored for any signs of morphine toxicity; limit use to less than 3-4 days in nursing mothers.

REFERENCES

1, 2, 4, 95, 140, 216.

Full revision 2012; limited revision 2016, 2017, 2020

NAME OF MEDICATION
COLISTIMETHATE

OTHER NAMES
Colistin, Coly-Mycin M ®

CLASSIFICATION
Antibiotic - polymyxin

INDICATIONS

- Treatment of infections due to certain gram-negative organisms (e.g., *Acinetobacter baumannii*, *Escherichia coli*, *Klebsiella pneumoniae* and *Pseudomonas aeruginosa*) when less toxic alternatives are contraindicated or ineffective.

ADMINISTRATION

- Reconstitute each 150 mg vial with 2 mL of SWFI. Swirl gently to avoid frothing; final concentration is 75 mg/mL of colistin base activity (CBA). Refer to Miscellaneous section for equivalence between CBA and colistimethate sodium (CMS).
- IV direct: physician or RN. Administer over 3-5 minutes.
- Intermittent IV infusion: dilute dose in NS, D5W, dextrose-saline solutions or RL and administer over 30-60 minutes.
- Continuous IV infusion: further dilute with 100-1000 mL of NS, D5W, dextrose-saline solutions or RL; refer to Dosage section for more information.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, rash, pruritus.
- GI: stomach upset, colitis.
- CNS: neurotoxicity: peripheral paresthesia, tingling of the extremities or of the tongue, dizziness, vertigo, blurred vision, slurred speech, ataxia, confusion, coma, psychosis, seizures, neuromuscular blockade.
- Renal: nephrotoxicity (dose-dependent; may be reversible if drug is discontinued; avoid concomitant administration of other nephrotoxic drugs).
- Respiratory: respiratory arrest, respiratory tract paralysis.
- Local reactions: severe pain with IM administration.

DOSAGE

- **Dosages are expressed in terms of colistin base activity (CBA).**

Method 1 (manufacturer's recommendations):
- 2.5-5 mg/kg/day IM/IV in 2-4 divided doses depending on the severity of the infection.
- Although the manufacturer recommends weight-based dosing, pharmacokinetic data show that dosing based on body weight is not necessary (see Method 2).
- Continuous IV infusion: give half of the total daily dose IV direct, then 1-2 hours later give the other half by IV infusion over 22-23 hours.
- Dosage in renal impairment:

CrCl (mL/min)	79-50	49-30	29-10
Dose	2.5-3.8 mg/kg/**day**	2.5 mg/kg/**day**	1.5 mg/kg/**dose**
Dosing Interval (hr)	12	12-24	36
Maximum total daily dose (mg)	150-230	133-150	100

- Dosage in obesity: dose should be calculated using ideal body weight.

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DOSAGE (Cont.)

Method 2 (consensus guidelines; to achieve average steady-state plasma concentration [$C_{ss,avg}$] of 2 mg/L):

- Loading dose is recommended in patients where rapid attainment of therapeutic levels is required (e.g., critically ill patients). The following equation is used to calculate the loading dose:
  Loading dose (mg) = $C_{ss,avg}$ target (mg/L) x 2 x body weight (kg), where $C_{ss,avg}$ target = 2 mg/L.
  For obese patients, use ideal body weight. The loading dose should not exceed 300 mg IV.
- Maintenance dose: 300-360 mg/day IV in 2 divided doses administered q12h. The maintenance dose is started 12 hours after the loading dose.
- Dosage in renal impairment: renal function should be monitored and the daily dose adjusted:

CrCl (mL/min)	Total daily dose	CrCl (mL/min)	Total daily dose
90 or greater	360 mg	39-30	195 mg
89-80	340 mg	29-20	175 mg
79-70	300 mg	19-10	160 mg
69-60	275 mg	9-5	145 mg
59-50	245 mg	0	130 mg
49-40	220 mg		

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Reconstituted solution is stable for 7 days in the fridge or at room temp.
- Reconstituted solution is stable for 24 hours at room temp or 48 hours in the fridge in polypropylene syringes.
- Diluted solution is stable for 24 hours in NS, D5W, dextrose-saline solutions, or RL.

MISCELLANEOUS

- Colistimethate is the inactive prodrug of colistin.
- Unit equivalency: 1 mg of CBA is equivalent to 30 000 international units; 1 mg of CMS is equivalent to 12 500 international units; 1 million international units is equivalent to 33 mg of CBA and to 80 mg of CMS.

REFERENCES

1, 2, 4, 40, 95, 135, 196, 423, 424.

Full revision 2012; limited revision 2015, 2019, 2020

NAME OF MEDICATION
COPPER

OTHER NAMES
Cupric sulfate, Micro Cu

CLASSIFICATION
Trace element

INDICATIONS

- As a supplement to total parenteral nutrition (TPN) to prevent or correct copper deficiency.
- Treatment of copper deficiency resulting from malabsorption or excessive losses (e.g., bariatric bypass surgery, major burns).

ADMINISTRATION

- Intermittent IV infusion (mandatory): dilute in at least 100 mL of compatible fluid (at TOH: dilute 2 mg in 250 mL of D5W and infuse over 2 hours).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity
- Hepatic: hepatic failure with excessive doses.
- Local reactions: phlebitis.

DOSAGE

- As a supplement to TPN: 0.3 to 1.5 mg/day IV added to TPN solution.
- Copper deficiency: 2 mg IV once daily for 5 days; repeat periodically prn; doses up to 5 mg IV daily have been suggested.
- Dosage in renal impairment: no dosage adjustment is recommended.
- Dosage in hepatic impairment: use with caution; in patients with significant cholestasis or hepatic impairment, may decrease dose by 50% or discontinue if required.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp not exceeding 28°C.
- Stable for 24 hours after dilution in TPN.
- Compatible with dextrose and TPN solutions.

MISCELLANEOUS

REFERENCES

5, 135, 466, 468, 469.

New monograph 2020; limited revision 2021

NAME OF MEDICATION
COSYNTROPIN

OTHER NAMES
Cortrosyn ®, Synacthen ® Depot

CLASSIFICATION
Synthetic adrenocorticotropic hormone

INDICATIONS

- Diagnosis of adrenocortical insufficiency.
- Diagnosis of primary aldosteronism by adrenal vein sampling.
- For management of post-lumbar puncture headaches.
- For management of various collagen, dermatologic, endocrine, eye, and hemolytic diseases (depot formulation).

ADMINISTRATION

Conventional cosyntropin (Cortrosyn ®):
- Reconstitute 250 mcg vial with 1 mL of NS.
- IV direct: physician or RN. Dilute with 2-5 mL of NS and administer over 2 minutes.
- Intermittent IV infusion:
  - For a greater stimulus, when screening for adrenocortical insufficiency: dilute 250 mcg in a large volume of NS or D5W and administer over 4-8 hours (usually 40 mcg/hr over 6 hours).
  - Post-lumbar puncture headache: dilute dose in 250-1000 mL NS and infuse over 0.5-2 hours.
- Continuous IV infusion: for adrenal vein sampling; at TOH, dilute 250 mcg in 250 mL of NS or D5W.
- IM.

Depot formulation (Synacthen ® Depot):
- IM administration only: shake slightly to resuspend suspension; administer slowly and deeply into the gluteal region.

- Observe patient for 60 minutes following a dose for any serious or potentially fatal allergic reactions.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity (rare): urticaria, pruritus, rash, dyspnea, severe malaise; preparation much less allergenic than corticotropin (ACTH) but cross-reactivity exists between cosyntropin and ACTH.
- Cardiovascular: bradycardia, tachycardia, hypertension, peripheral edema, flushing.
- Local reactions: pain, induration with IM administration.

DOSAGE

Adults:
- Diagnostic test for adrenocortical insufficiency:
  - Standard dose: 0.25 mg (250 mcg) - 0.75 mg (750 mcg) IV/IM of conventional cosyntropin.
  - Low dose: 0.001 mg (1 mcg) IV of conventional cosyntropin (serial dilutions to be prepared by pharmacy to obtain a 1 mcg/dose).
  - Depot formulation: 1 mg IM.
- Adrenal vein sampling: continuous IV infusion at a rate of 50 mcg/hr, starting 30 minutes before adrenal vein catheterization and continuing for the duration of the procedure.
- Post-lumbar puncture headache: 0.75-1 mg or 0.015 mg/kg as a single infusion; may repeat, if necessary, 24 hours later. Has also been given IM.
- Other diseases: depot formulation: average dose is 0.5-1 mg IM twice weekly (range of 0.5-1 mg every 2-3 days up to 2 mg at weekly or even longer intervals). Individualize to the smallest dose with the longest interval possible to achieve disease control.

Pediatrics:
- Diagnostic test: older than 2 years of age: 0.035 mg/kg IV/IM; dose limit 0.25 mg (250 mcg) of conventional cosyntropin.
  2 years old or younger: 0.125 mg (125 mcg) of conventional cosyntropin.
- Other diseases: depot formulation: refer to manufacturer's information.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store conventional cosyntropin at room temp.
- Store depot formulation in the fridge and protect from light.
- Conventional cosyntropin: reconstituted solution is stable for 24 hours at room temp or 21 days in the fridge. Since it contains no preservatives, prolonged storage after reconstitution is not advised.
- Diluted solution is stable for 12 hours at room temp in NS and dextrose solution.
- Incompatible with blood or plasma products.

MISCELLANEOUS

- Epinephrine must be available in case of acute hypersensitivity reaction.
- Conventional cosyntropin and depot formulation of cosyntropin are not interchangeable.
- Patients taking inadvertent doses of cortisone or hydrocortisone on the day of the test could have abnormally high plasma cortisol levels at baseline and a paradoxical decrease in plasma cortisol level after cosyntropin.
- Prednisone, dexamethasone, betamethasone will not interfere with test when fluorometric analysis is used.
- Patients on estrogens may have abnormally high cortisol plasma levels before and after cosyntropin, but a normal incremental response to cosyntropin still occurs.
- Avoid pre-test doses of cortisone, hydrocortisone and spironolactone on the day of the test when fluorometric analysis is used. Spironolactone can be given on the day of the test if radioimmunoassay or competitive protein binding methods of analysis are used.
- Cosyntropin 0.25 mg is equivalent to 25 international units ACTH.

REFERENCES

1, 2, 9, 40, 82, 95, 135, 142, 144, 153, 160, 217, 416, 440.

Full revision 2012; limited revision 2015, 2016, 2019

NAME OF MEDICATION
CO-TRIMOXAZOLE

OTHER NAMES
Septra ®, TMP-SMX, Trimethoprim/sulfamethoxazole

CLASSIFICATION
Antibiotic - sulfonamide

INDICATIONS

- Treatment of septicemia and meningitis, genito-urinary tract, pulmonary, gastrointestinal and skin and soft tissue infections due to susceptible organisms.
- Treatment of *Pneumocystis jiroveci* (*carinii*, PCP) pneumonia.

ADMINISTRATION

- Intermittent IV infusion: dilute each 5 mL ampoule (equivalent to 80 mg of trimethoprim and 400 mg of sulfamethoxazole) with 100-125 mL NS, D5W, dextrose-saline or Ringer's solutions; may dilute each 5 mL ampoule with 50-75 mL NS or D5W in fluid restricted patients. Administer a single dose over 60-90 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pulmonary infiltrates, cough, shortness of breath, serum sickness, anaphylaxis.
- GI: nausea, vomiting, anorexia, diarrhea.
- Hyperkalemia.
- Hematologic: anemia, thrombocytopenia, leukopenia; with high doses or prolonged duration of treatment.
- Local reactions: irritation, pain, phlebitis, inflammation.

DOSAGE

- Adults:
  - Severe infections: 160-240 mg of trimethoprim component IV q6-12h.
  - Meningtitis: 5 mg/kg of trimethoprim component IV q6-12h.
  - PCP: 16-20 mg/kg/day of trimethoprim component IV in 3-4 divided doses.

- Pediatrics:
  - Usual: 5-12 mg/kg/day of trimethoprim component IV, divided q6-12h.
  - PCP: 16-20 mg/kg/day of trimethoprim component IV, in 3-4 divided doses.

- Dosage in renal impairment:

CrCl (mL/min)	greater than 25	25-15	less than 15
Dose	100%	50%	not routinely recommended
Interval	Usual	Usual	----

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from light.
- Use punctured vials within 48 hours.
- According to manufacturer: start the infusion within 5 hours of dilution. However, longer stability data are available from other sources; see below.
- Stable for 48 hours at room temp in D5W or NS and 24 hours at room temp in RL when diluted 1:25 (i.e., 5 mL or 80 mg of trimethoprim component in 125 mL) in glass containers.
- Stable for 24 hours at room temp in D5W or RL and 14 hours at room temp in NS when diluted 1:20 (i.e., 5 mL or 80 mg of trimethoprim component in 100 mL) in glass containers.
- Stable for 24 hours at room temp in D5W or NS when diluted 1:15 (i.e., 5 mL or 80 mg of trimethoprim component in 75 mL) in PVC containers.
- Stable for 24 hours at room temp in RL when diluted 1:15 (i.e., 5 mL or 80 mg of trimethoprim component in 75 mL) in glass containers.

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COMPATIBILITY, STABILITY (Cont.)
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Stable for 24 hours at room temp in D5W or NS when diluted 1:10 (i.e., 5 mL or 80 mg of trimethoprim component in 50 mL) in PVC containers.
- Observe for cloudiness or evidence of precipitation especially with concentrated solutions and with solutions in glass containers.

MISCELLANEOUS

- Do not use in patients allergic to sulfa.
- Maintain adequate fluid intake to prevent crystalluria and stone formation.

REFERENCES

1, 2, 4, 9, 40, 82, 95, 135, 208, 422, 484.

NAME OF MEDICATION
COVID-19 VACCINE (mRNA)

OTHER NAMES
Comirnaty, COVID-19 Vaccine Moderna, Pfizer-BioNTech COVID-19 vaccine, Tozinameran

CLASSIFICATION
Vaccine

Do NOT confuse COVID-19 vaccine (mRNA) with COVID-19 vaccines from other classes (e.g., viral vector). This monograph is specific to the mRNA class of COVID-19 vaccines.

INDICATIONS

- For active immunization against the coronavirus disease 2019 (COVID-19) caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) in individuals 12 years of age and older.

ADMINISTRATION

Pfizer-BioNTech vaccine:

- Allow vial to thaw between 2-8°C, which may take up to 3 hours, or thaw at room temp up to 25°C for 30 minutes. After thawing, each vial contains 0.45 mL.
- Allow vial to reach room temp. Vial must be further diluted prior to administration and dilution must occur within 2 hours of exposure to room temp. Prior to dilution, gently invert vial 10 times. Do NOT shake. Undiluted vaccine suspension is white to off-white in colour and may contain white to off-white opaque amorphous particles. Dilute each vial with 1.8 mL of NS (do NOT use bacteriostatic NS or any other diluent) using a transfer syringe with a 21-gauge or narrower needle. Prior to removing the needle from the vial, equalize vial pressure by withdrawing 1.8 mL of air into the empty syringe. Gently invert vial 10 times. Do NOT shake. The diluted vaccine suspension should be off-white in colour and free of particulates.
- Each vial provides 6 doses of 0.3 mL. Low dead-volume syringes and/or needles can extract 6 doses per vial. If standard syringes and needles are used, it may be possible to extract only 5 doses per vial.

Moderna vaccine:

- Allow vial to thaw between 2-8°C for 2.5 hours. Let vial stand at room temp for 15 minutes prior to use. Alternatively, thaw vial between 15-25°C for 60 minutes. Gently swirl vial after thawing and between each dose withdrawal. Do NOT shake. Do NOT further dilute. Vaccine dispersion is white to off-white in colour and may contain white or translucent product-related particulates.
- Each vial provides 10 doses of 0.5 mL.

Pfizer-BioNTech and Moderna vaccines:

- IM: inject into the deltoid (preferred) or at the anterolateral thigh.
- Observe patient for at least 15 minutes after administration; a longer interval of 30 minutes is preferred for patients if there is a concern for a potential vaccine reaction.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, angioedema, anaphylaxis. Patients experiencing a mild to moderate (i.e., non-anaphylactic) immediate allergy (within 60 minutes) following a previous dose of an mRNA COVID-19 vaccine can receive a subsequent dose following an evaluation of the risks versus benefits.
- Cardiovascular: myocarditis, pericarditis; very rare; occur within a few days following vaccination; more common after second dose, in males, and in adolescents and young adults less than 30 years of age; if occurs after first dose, defer second dose until more information is available.
- GI: nausea, vomiting, diarrhea.
- CNS: headache.
- Musculoskeletal: myalgia, arthralgia.
- Fatigue, fever, chills.
- Lymphadenopathy.
- Local reactions: injection site pain, swelling, redness, tenderness.

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DOSAGE

- Pfizer-BioNTech vaccine: 12 years of age and older: 2 doses of 0.3 mL (30 mcg) IM separated by 21 days. The interval between doses can range from a minimum of 19 days to an extended interval of up to 16 weeks.
- Moderna vaccine: 12 years of age and older: 2 doses of 0.5 mL (100 mcg) IM separated by at least 28 days. The interval between doses can range from a minimum of 21 days to an extended interval of up to 16 weeks.
- For completion of the vaccination series with a different brand or class of COVID-19 vaccine, consult the latest national and provincial public health recommendations.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Pfizer-BioNTech vaccine:
- Store vials between -80°C to -60°C. Protect from light.
- Vials can also be stored between -25°C to -15°C for 2 weeks. Protect from light. Vials stored between -25°C to -15°C for 2 weeks may be returned once to the recommended storage of -80°C to -60°C. Total cumulative storage time between -25°C to -15°C should be tracked, not to exceed 2 weeks.
- Thawed undiluted vials are stable in the fridge for 1 month or 2 hours at room temp. Do not refreeze.
- After dilution, vials are stable between 2-25°C for 6 hours. Do not refreeze.
- Consult manufacturer's information for further information on storage and transportation of frozen vials as well as transportation of thawed vials.

Moderna vaccine:
- Store vials between -25°C to -15°C. Protect from light.
- Vials can also be stored between 2-8°C for 30 days and between 8-25°C for 24 hours. Do not refreeze.
- Punctured vials are stable for 24 hours in the fridge or at room temp. Do not refreeze.
- Consult manufacturer's information for further information on transportation of thawed vials.

MISCELLANEOUS

- Emergency drugs (e.g., epinephrine) should be immediately available for the treatment of hypersensitivity reactions.
- Administration should be delayed, if possible, during an acute febrile illness or active infection of moderate to severe intensity.
- Concurrent administration of other live or inactivated vaccines should be avoided, unless required for post-exposure prophylaxis or other valid reason. At least 28 days should elapse after either dose of the mRNA COVID-19 vaccine before administering another vaccine and at least 14 days should elapse after the administration of another vaccine prior to giving the mRNA COVID-19 vaccine. Refer to the National Advisory Committee on Immunization's (NACI) recommendations for further details.
- There is a theoretical risk that mRNA vaccines may temporarily affect cell-mediated immunity, resulting in false-negative tuberculin skin test (TST) or Interferon Gamma Release Assay (IGRA) test results. If TST or an IGRA test is required, it should be administered and read prior to the mRNA COVID-19 vaccine or delayed until at least 4 weeks after vaccination.
- COVID-19 vaccines should not be given simultaneously with anti-SARS-CoV-2 monoclonal antibodies or convalescent plasma.
- Both Pfizer-BioNTech and Moderna vaccines contain sucrose and polyethylene glycol (PEG). There is the potential for cross-reactivity between PEG and polysorbates.
- The Moderna vaccine contains tromethamine (trometamol).

REFERENCES

5, 135, 576.

New monograph 2021

NAME OF MEDICATION
CYCLOPHOSPHAMIDE

OTHER NAMES
Cytoxan ®, Procytox ®

CLASSIFICATION
Antineoplastic, Immunosuppressant

INDICATIONS

- Treatment of breast cancer, lung cancer, multiple myeloma, neuroblastoma, ovarian cancer, retinoblastoma, mycosis fungoides, lymphomas, leukemias, and in various protocols for other neoplasms.
- Treatment of some autoimmune disorders.

ADMINISTRATION

- Reconstitute the 200 mg, 500 mg, 1000 mg or 2000 mg vials with 10 mL, 25 mL, 50 mL or 100 mL, respectively, with NS without preservatives to give a concentration of 20 mg/mL; do NOT use bacteriostatic NS that contains benzyl alcohol. Shake to dissolve for a clear solution.
- IV direct: physician or RN; undiluted; doses less than 500 mg; inject each 100 mg over at least 1 minute into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion: dilute in 100-1000 mL of D5W or NS; infuse over 15 minutes to 4 hours, depending on doses and protocol.
- Continuous IV infusion: as per protocol.
- Aluminium-containing IV needles, syringes, or sets should NOT be used to prepare or administer cyclophosphamide.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rare anaphylactoid reactions; rash, pruritus, hives.
- Infusion-related reactions: nasal congestion and facial discomfort with IV doses administered too rapidly (flushing, runny eyes, rhinorrhea, sinus congestion, sneezing during and immediately after infusion); slow the rate of administration or give as an infusion rather than IV direct.
- GI: nausea, vomiting.
- CNS: headache, dizziness.
- Hemorrhagic cystitis.
- Hyponatremia, syndrome of inappropriate anti-diuretic hormone secretion.
- Hyperuricemia.
- Radiation recall reaction (rare).
- Local reactions: chemical phlebitis, redness, swelling, pain.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 500 mg/m^2 to 1200 mg/m^2 IV every 3 weeks.
- 300 mg/m^2 IV once weekly.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose method 1	100%	100%	75%
Dose method 2	100%	50-75%	50%

- Consult specific protocol.

…/Cont.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp (do not exceed 25°C). Protect from direct light. Do not use vial with melted content.
- Reconstituted vial is stable for 24 hours at room temp and 72 hours in the fridge.
- Diluted solutions in NS, D5W, or D5W in NS are stable 24 hours at room temp or for 72 hours in the fridge.
- Compatible with RL.

MISCELLANEOUS

- Encourage patient to drink plenty of fluids during therapy and to void frequently to help prevent hemorrhagic cystitis.

REFERENCES

1, 2, 4, 40, 129, 147, 165, 216.

Full revision 2012; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
CycloSPORINE

OTHER NAMES
Cyclosporine A, Neoral ®, Sandimmune ®

CLASSIFICATION
Immunosuppressant

INDICATIONS

- Prophylaxis or treatment of rejection after solid organ transplantation or bone marrow transplantation.
- For patients unable to take the oral solution pre- or postoperatively.

ADMINISTRATION

- Intermittent IV infusion, Continuous IV infusion: dilute each 50 mg (1 mL) with 20 to 100 mL (each 5 mL ampoule with 100 to 500 mL) of D5W or NS and infuse over 2 to 6 hours or continuously over 24 hours. Glass or non-PVC containers are preferred over PVC bags for infusion. Use non-PVC IV administration sets.
- The initial dose should be given 4-12 hours prior to transplantation.
- Observe patient for at least 30 minutes after the start of infusion and frequently thereafter for signs of any hypersensitivity.

POTENTIAL ADMINISTRATION HAZARDS

- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: the cyclosporine injectable formulation contains Cremophor EL (polyoxyethylated castor oil) which has been reported to cause anaphylactoid reactions.
- Cardiovascular: hypertension.
- GI: nausea, vomiting, diarrhea.
- CNS: tremor, headache.
- Renal: dose-related nephrotoxicity.
- Hepatic: increase in serum bilirubin and liver enzymes.

DOSAGE

Adults/Pediatrics:
- Initiate with 3 to 5 mg/kg/day IV. Adjust dose to the desired blood or plasma concentration.
- For patients unable to take the oral formulation: treat with IV at one-third of the oral dose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp. Protect from light and freezing.
- The Cremophor EL and ethanol vehicle cause significant leaching of DEHP plasticizer from PVC containers and PVC administration sets.
- Cyclosporine is subject to significant sorption to PVC containers and PVC administration sets. Sorption is higher with dilutions in NS than with D5W.
- Stable for 48 hours at room temp or in the fridge in D5W at a concentration of 2 mg/mL in glass bottles.
- Stable 14 days at room temp in D5W or NS at a concentration of 0.2 and 2.5 mg/mL in polypropylene-polyolefin bags.
- Stable for 14 days at room temp in NS and D5W at a concentration of 2.5 mg/mL in ethylene vinyl acetate bags.
- Stable for 7 days at room temp in NS and D5W at a concentration of 0.2 mg/mL in ethylene vinyl acetate bags.
- Short term (less than 10 minutes) use of polypropylene syringes for preparation is considered safe. Cyclosporine should not be stored in plastic syringes due to leaching of rubber components from rubber tips of plungers.

MISCELLANEOUS

REFERENCES

1, 2, 4, 6, 40, 48, 95, 135, 143, 340.

Full revision 2012; limited revision 2014, 2015, 2017, 2018, 2019, 2020

NAME OF MEDICATION
CYTARABINE

OTHER NAMES
Arabinosyl cytosine, Ara-C, Cytosar ®, Cytosine arabinoside

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of acute myelocytic leukemia, chronic myelocytic leukemia, acute lymphocytic leukemia and erythroleukemia.
- Treatment of non-Hodgkin's lymphoma.
- Combined with other antineoplastic drugs in various protocols.

ADMINISTRATION

- Reconstitute vials of 100 mg, 1 g, and 2 g with 5, 10, and 20 mL with D5W, NS, SWFI or bacteriostatic water for injection, respectively, to obtain a final concentration of 20 mg/mL, 100 mg/mL, and 100 mg/mL, respectively. For high dose therapy use preservative free NS only.
- Also available as a 100 mg/mL aqueous solution.
- IV direct: physician or RN; undiluted; each 100 mg over 1-3 minutes into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion: doses less than 1 g/m^2: dilute further with 50-100 mL of D5W or NS and infuse over 15-60 minutes. Dilute larger doses in 250 mL of D5W or NS and infuse over 1-3 hours.
- Continuous IV infusion.
- IM.
- SC.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid (rare).
- Cytarabine syndrome: flu-like syndrome, fever, rash, conjunctivitis. Symptoms usually resolve with drug discontinuation; corticosteroids can be used for treatment and prophylaxis. Cytarabine can be continued if patient responds to corticosteroids.
- GI: nausea, vomiting (especially with rapid IV administration of high doses).
- CNS: cerebellar syndrome, decreased consciousness and personality change (high doses); seizures.
- Respiratory: adult respiratory distress syndrome (ARDS) (high doses).
- Hematologic: leukopenia, neutropenia, thrombocytopenia.
- Hyperuricemia.
- Local reactions: pain and inflammation (rare); thrombophlebitis, cellulitis.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Usual in combination therapy: 2-6 mg/kg/day IV or 100-200 mg/m^2/day IV for 5 to 10 days.
- High dose: 1-3 g/m^2 IV q12h for 2-6 days.
- Dosage in renal impairment: no dosage modification for standard doses; dosage modification only for high-dose therapy:

CrCl (mL/min)	greater than 60	60-45	45-30	less than 30
Dose	100%	60%	50%	avoid

- Consult specific protocol.

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NAME OF MEDICATION
CYTARABINE

OTHER NAMES
Arabinosyl cytosine, Ara-C, Cytosar ®, Cytosine arabinoside

CLASSIFICATION
Antineoplastic

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from light.
- Reconstituted vial with bacteriostatic water for injection is stable for 48 hours at room temp. Vials reconstituted with preservative-free diluents should be used immediately. Discard hazy solutions.
- Vials of 100 mg/mL aqueous solution should be discarded 24 hours after opening.
- Stable for 7 days at room temp or in the fridge in D5W or NS at concentrations of 8, 24, and 32 mg/mL in PVC or glass containers.
- Stable 7 days at room temp in D5W or NS at a concentration of 0.5 mg/mL in PVC or glass containers.

MISCELLANEOUS

REFERENCES

1, 2, 4, 5, 40, 129, 143, 165, 200.

Full revision 2012; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
DACARBAZINE

OTHER NAMES
DTIC ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Palliative therapy of metastatic malignant melanoma.
- May also be used for Hodgkin's disease, neuroblastoma and soft tissue sarcomas.

ADMINISTRATION

- Reconstitute 200 mg and 600 mg vials with 19.7 mL and 59.1 mL of SWFI, respectively, to obtain 10 mg/mL.
- IV direct: physician or RN; undiluted; inject over at least 1 minute into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion (preferred): dilute in 250-500 mL of D5W or NS; may be infused with an additional 250 mL of IV solution run at same time by piggyback to reduce vein irritation; infuse over 15-120 minutes. Longer infusion times (1-2 hours) are preferable to minimize pain.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid reactions (rare), rash.
- Cardiovascular: hypotension (high doses i.e., greater than 850 mg/m^2).
- GI: nausea, vomiting, diarrhea.
- CNS: confusion, seizures.
- Flu-like syndrome: fever, myalgia, malaise (during or after treatment).
- Dermatologic: facial flushing, photosensitivity (rare).
- Local reactions: pain at injection site, phlebitis.
- Extravasation hazard: irritant. Subcutaneous or perivascular extravasation may result in tissue damage (necrosis) and severe pain. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Various dosing regimens: 2-4.5 mg/kg/day IV for 10 days every 3 weeks or 200-250 mg/m^2/day IV for 5 days every 3-4 weeks or 375 mg/m^2 IV on days 1 and 15 every 4 weeks.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store unopened vials in the fridge, protect from light.
- Protect solutions from light while in storage.
- A colour change of the solution from its normal pale yellow colour or colourless to pink or red indicates decomposition.
- Reconstituted solution is stable for 24 hours at room temp and 96 hours in the fridge when protected from light.
- Stable for 8 hours at room temp (under normal room light conditions) or 24 hours in the fridge diluted with up to 500 mL of D5W or NS.
- Stable for 72 hours at room temp and 7 days in the fridge, protected from light, in D5W at a concentration of 1.4 mg/mL in PVC bags.

MISCELLANEOUS

REFERENCES

1, 4, 5, 6, 40, 129, 143, 165.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
DACTINomycin

OTHER NAMES
Actinomycin D, Cosmegen ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of Wilms' tumour, rhabdomyosarcoma, Ewing's sarcoma and gestational trophoblastic neoplasia.
- May also be used for testicular carcinoma, ovarian germ cell tumour and Kaposi's sarcoma.

ADMINISTRATION

- Reconstitute with 1.1 mL of SWFI without preservative to a final concentration of 0.5 mg/mL. Do NOT use SW with preservatives as it may form a precipitate.
- IV direct: physician or RN; undiluted; inject over 2-3 minutes into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion: dilute in 50 mL NS or D5W; do NOT dilute to a concentration lower than 10 mcg/mL; infuse over 10-15 minutes.
- Do NOT use an in-line cellulose ester membrane filter.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid (rare), rash.
- GI: nausea, vomiting, anorexia, abdominal pain.
- Dermatologic: flushing of face and torso, radiation recall reaction (rare).
- Hypocalcemia (rare).
- Fatigue, fever, chills.
- Local reactions: pain.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Wilms' tumour: 45 mcg/kg IV on day 1 every 3 weeks.
- Rhabdomyosarcoma: 15 mcg/kg/day or 0.4-0.6 mg/m^2/day IV for 5 days in various schedules.
- Ewing's sarcoma: 1.25 mg/m^2 IV in various schedules.
- Gestational trophoblastic neoplasia: 12 mcg/kg/day IV for 5 days as a single agent or 500 mcg IV on days 1 and 2 in combination with other antineoplastic agents.
- Dosage in obesity: dose should be calculated using body surface area.
- For edematous patients, dose should be calculated using body surface area.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp; protect from light.
- Reconstituted solution is stable for 24 hours at room temp or fridge.
- Stable for 24 hours at room temp exposed to light following dilution in D5W in glass and PVC containers.
- Compatible with NS, D5W.

MISCELLANEOUS

REFERENCES

1, 4, 5, 6, 40, 95, 129, 165.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
DALTEPARIN

OTHER NAMES
Fragmin ®

CLASSIFICATION
Anticoagulant

INDICATIONS

- Prophylaxis of thrombosis related to surgery.
- Prophylaxis of thrombosis in medical patients who are at risk of thromboembolism due to severely restricted mobility during acute illness.
- Treatment of acute deep vein thrombosis (DVT).
- Unstable coronary artery disease (unstable angina and non-Q-wave myocardial infarction).
- Extended treatment and prevention of venous thromboembolism (VTE) in patients with cancer.
- Prevention of clotting in the extracorporeal system during hemodialysis and hemofiltration in connection with acute renal failure or chronic renal insufficiency.

ADMINISTRATION

- Continuous IV infusion: dilute 10 000 international units in 500 mL NS or D5W to obtain a final concentration of 20 international units/mL. Infuse at approximately 8 international units/kg/hr (for 100 international units/kg/12 hours).
- SC: in the fat tissue of the lower abdomen u-shaped area around the navel; as an alternative, can also be injected into the side of the thigh (but not into the muscle tissue). With the thickness of skin held between the operator's thumb and forefinger, insert the entire length of the needle vertically into the skin. To minimize bruising, injection sites should not be massaged after injection. Rotate site of injection daily.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylactoid reactions, pruritus, rash.
- Hematologic: bleeding (major and minor), thrombocytopenia.
- Risk of spinal or epidural hematoma that can result in permanent paralysis when epidural or spinal anesthesia or spinal puncture is used in conjunction with dalteparin.
- Transient increases in liver transaminases (AST, ALT); reversible on discontinuation of the drug.
- Local reactions: hematoma and pain at injection site; skin necrosis (very rare).

Antidote: The anticoagulant effect of dalteparin can be partially neutralized by protamine. Refer to protamine in this manual for more details.

DOSAGE

Prophylaxis of thrombosis related to surgery:
- General surgery with associated risk of thromboembolic complications: 2500 international units SC 1-2 hours before surgery, then 2500 international units SC once daily each morning until patient is mobilized.
- General surgery associated with additional risk factors and elective hip surgery: 5000 international units SC the evening before surgery, then 5000 international units SC once daily in the evening until patient is mobilized.
  OR
  As an alternative: 2500 international units SC 1-2 hours before surgery and 2500 international units SC 4-8 hours after surgery then 5000 international units SC each morning on the following days.
- Dosage in obesity: may use a higher prophylactic dose.

Prophylaxis of thrombosis in medical patients with severely restricted mobility:
- 5000 international units SC once daily.

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DOSAGE (Cont.)

Treatment of acute DVT:
- 200 international units/kg SC once daily.
- If patient is at increased risk of bleeding, give 100 international units/kg SC twice daily or IV (as a continuous 12 hour infusion) for a total of 200 international units/kg over 24 hours.
- Dosage in obesity: dose should be calculated using actual body weight.

Unstable coronary artery disease:
- 120 international units/kg SC twice daily, for up to 6 days; maximum single dose 10 000 international units/12 hours.

Extended treatment of VTE in patients with cancer:
- Month 1: 200 international units/kg SC once daily for the first 30 days.
- Month 2-6: 150 international units/kg SC once daily; refer to manufacturer's monograph for more details.
- Dosage in obesity: dose should be calculated using actual body weight.

Anticoagulation for hemodialysis and hemofiltration:
- Chronic renal failure, patients with no other bleeding risk: hemodialysis and hemofiltration for a maximum of 4 hours: IV bolus 5000 international units; hemodialysis and hemofiltration for more than 4 hours: IV bolus 30-40 international units/kg, followed by IV infusion 10-15 international units/kg/hr.
- Acute renal failure, patients with high bleeding risk: IV bolus 5-10 international units/kg, followed by an infusion of 4-5 international units/kg/hr.

Dosage in renal impairment: use with caution when creatinine clearance is below 30 mL/min; consider dosage adjustment and monitoring of anti-Xa levels.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Stable for 24 hours at room temp in D5W or NS at a concentration of 20 international units/mL in glass or plastic containers.
- Stable for 30 days at room temp or in the fridge when drawn from multi-dose vials and stored in plastic syringes.

MISCELLANEOUS

- Dalteparin has only a moderate prolonging effect on clotting time assays, such as aPTT or thrombin time. For laboratory monitoring of effect, anti-Xa methods are recommended (although not routinely done, monitoring is recommended in special cases such as obesity, pregnancy and renal failure).
- Dalteparin should not be given to patients with a history of heparin-induced thrombocytopenia (risk of cross-reactivity), unless an in-vitro platelet aggregation test is negative.
- As the multi-dose vial contains benzyl alcohol, it is not recommended for use during pregnancy.

REFERENCES

1, 2, 4, 55, 95, 119, 120.

Full revision 2013; limited revision 2015, 2016, 2017, 2020

NAME OF MEDICATION
DANAPAROID

OTHER NAMES
Orgaran ®

CLASSIFICATION
Anticoagulant

INDICATIONS

- Prophylaxis of deep vein thrombosis (DVT) following orthopedic, major abdominal and thoracic surgery or in non-hemorrhagic stroke.
- Treatment of DVT or pulmonary embolism (PE) in patients with an acute episode of heparin-induced thrombocytopenia (HIT) and for DVT prophylaxis in patients with a history of HIT.
- Anticoagulant during cardiac procedures, peripheral vascular bypass, cardiopulmonary bypass, and hemodialysis in patients with HIT who require further anticoagulation.

ADMINISTRATION

- IV direct: physician or RN. For loading dose only. Administer undiluted or diluted in a small volume of IV solution over 5 minutes.
- Intermittent IV infusion: for loading dose only. Dilute in 50-100 mL of compatible IV solution and administer over 15-30 minutes.
- Continuous IV infusion: dilute 2250 anti-Xa units (1.8 mL of 750 anti-Xa units/0.6 mL = 3 amps) in 250 mL of D5W or NS to obtain a final concentration of 9 anti-Xa units/mL. More concentrated solutions may also be prepared.
- SC: anterolateral & posterolateral abdominal wall; rotate injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, allergic reactions.
- GI: nausea, constipation.
- Hematologic: major bleed, hematoma, bruises, hematuria, thrombocytopenia.
- Risk of spinal or epidural hematoma that can result in permanent paralysis when spinal/epidural anesthesia is used in conjunction with danaparoid.
- Local reactions: pain, hematoma.

Antidote: protamine is not a neutralizing agent for the activity of danaparoid. If critical bleeding occurs, fresh frozen plasma and plasmapheresis may be used.

DOSAGE

- Prophylaxis of DVT following surgery: 750 anti-Xa units SC q12h up to 14 days. Start preoperatively and to give the last preoperative dose 1-4 hours before surgery.
- Prophylaxis of DVT in nonhemorrhagic stroke: 1000 anti-Xa units IV bolus, then 750 anti-Xa units SC q12h for 7-14 days.
- HIT: DVT prophylaxis (current or past HIT):
  - 90 kg or less (current or past HIT): 750 anti-Xa units SC q8h or q12h for 7-10 days
  - more than 90 kg: current HIT: 1250 anti-Xa units SC q8h or q12h for 7-10 days
    past HIT: 1250 anti-Xa units SC q12h or 750 anti-Xa units SC q8h for 7-10 days
  - Note: Regardless of weight, may give an initial IV bolus of 1250 anti-Xa units for rapid attainment of prophylaxis levels if clinically necessary (i.e., current HIT)

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DOSAGE (Cont.)

- HIT: DVT and PE treatment:
  If thrombosis is less than 5 days old:
  - IV loading dose: 55 kg or less: 1500 anti-Xa units (1.2 mL = 2 amps)
    55-90 kg: 2250 anti-Xa units (1.8 mL = 3 amps)
    more than 90 kg: 3750 anti-Xa units (3 mL = 5 amps)
  - Maintenance dose (after loading dose): IV infusion: 400 anti-Xa units/hr X 4 hours, then 300 anti-Xa units/hr X 4 hours, then 150-200 anti-Xa units/hr for 5-7 days. Adjust infusion rate to maintain plasma anti-Xa level between 0.5-0.8 anti-Xa units/mL.
    OR
    SC injections for 4-7 days: 55 kg or less: 1500 anti-Xa units q12h
    55-90 kg: 2000 anti-Xa units q12h
    more than 90 kg: 1750 anti-Xa units q8h

  If thombosis is 5 days old or greater:
  - IV loading dose: 1250 anti-Xa units (regardless of weight)
  - Maintenance dose with SC injections:
    90 kg or less: 750 anti-Xa units q8h or q12h
    more than 90 kg: 750 anti-Xa units q8h or 1250 anti-Xa units q8h or q12h
- HIT: as an anticoagulant during hemodialysis: 1500-3750 anti-Xa units IV before dialysis session; dosage to vary depending on plasma anti-Xa levels, frequency of dialysis and weight of the patient. Refer to manufacturer's monograph for more detailed information.
- Consult manufacturer's monograph for more specific dosing information according to each indication.
- Dosage in renal impairment: reduce dose by 25 to 50%; monitor plasma anti-Xa levels for further dosage adjustment.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 2-30°C; protect from light.
- Stable 48 hours at room temp when diluted in NS, D5W, D5-1/2NS and RL.

MISCELLANEOUS

- Cross-reactivity between danaparoid and heparin-induced antibody is less than 10%.
- Monitor anticoagulant activity (via plasma anti-Xa levels) for patients with a weight outside the range of 55 to 90 kg or with renal failure. Consult manufacturer's monograph for more information.

REFERENCES

2, 5, 6, 95, 135, 445, 452, 545.

Full revision 2013; limited revision 2015, 2016, 2017, 2020, 2021

NAME OF MEDICATION
DANTROLENE

OTHER NAMES
Dantrium ®

CLASSIFICATION
Skeletal muscle relaxant

INDICATIONS

- Treatment of malignant hyperthermia crisis.
- For preoperative prophylaxis in individuals at risk of malignant hyperthermia.
- Treatment of neuroleptic malignant syndrome.

ADMINISTRATION

- Reconstitute each 20 mg vial with 60 mL SWFI (without preservative). Shake vial until solution is clear. Note: using warmed (up to 40°C) SWFI without preservative may decrease shaking time to obtain a clear solution from 3 minutes to 30 seconds.
- IV direct: physician or RN. For active treatment. Give as rapid IV injection.
- Intermittent IV infusion: for preoperative prophylaxis. Transfer the reconstituted dose to an empty sterile IV bag and infuse over 60 minutes. The solution should NOT be in contact with NS or D5W.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, rash, pruritus, urticaria.
- GI: nausea, diarrhea.
- CNS: drowsiness, dizziness, lightheadedness, malaise.
- Muscle weakness which rarely may result in slurred speech, drooling, difficulty in swallowing, choking, decreased grip strength, difficulty in walking, and enuresis.
- Pulmonary edema, fatigue.
- Local reactions: thrombophlebitis (extravasation may cause local tissue damage due to high pH).

DOSAGE

- Treatment of malignant hyperthermia crisis in adults and pediatrics: initial dose of 2.5 mg/kg; repeat dose until signs of malignant hyperthermia are reversed. Cumulative doses greater than 10 mg/kg (up to 30 mg/kg) may be necessary. When symptoms are controlled, give 1 mg/kg IV q4-6h for at least 24 hours (if oral administration is not possible).
- For prevention of malignant hyperthermia crisis prior to surgery: 2.5 mg/kg infused IV over 60 minutes - start 75 minutes before induction of anesthesia.
- Treatment of neuroleptic malignant syndrome: initial dose of 1-2.5 mg/kg; repeat dose, if needed, to a maximum cumulative dose of 10 mg/kg. When symptoms are controlled, 1 mg/kg q6h or 1 mg/kg/daily may be used.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution is stable for 6 hours at room temp; protect from direct light.
- Incompatible with D5W or NS or bacteriostatic water.
- Incompatible with glass bottles.
- Do not use solution if cloudy or contains a precipitate.

MISCELLANEOUS

- Avoid concurrent use with verapamil or diltiazem due to possible cardiac depression and hyperkalemia.
- Each 20 mg vial contains 3 g of mannitol.

REFERENCES

1, 2, 6, 40, 95, 135, 208.

Full revision 2013; limited revision 2015, 2016

NAME OF MEDICATION
DAPTOmycin

OTHER NAMES
Cubicin ® RF

CLASSIFICATION
Antibiotic

INDICATIONS

- Treatment of complicated skin and skin structure infections caused by susceptible strains of gram-positive microorganisms.
- Treatment of *Staphylococcus aureus* bloodstream infections including those with right-sided *Staphylococcus aureus* infective endocarditis (native valve) caused by methicillin-susceptible and methicillin-resistant strains.
- Treatment of vancomycin-resistant *Enterococcus faecium* (VREfm) infections.
- Treatment of other infections such as CNS, bone, joint and cardiac infections (e.g., left sided endocarditis), however, it should NOT be used for the treatment of pneunomia.

ADMINISTRATION

- Reconstitution for Cubicin ® RF: reconstitute the 500 mg vial by injecting 10 mL of SWFI or bacteriostatic water for injection, using a 21 gauge transfer needle (or smaller in diameter) and pointing it toward the wall of the vial. Rotate or swirl the vial for a few minutes to dissolve the contents. Slowly remove the reconstituted solution (50 mg/mL concentration) from the vial using a 21 gauge needle (or smaller in diameter).
- IV direct: physician or RN. For adult patients only: administer over 2 minutes.
- Intermittent IV infusion: Cubicin ® RF: for adults and children 7-17 years of age: dilute dose in 50-100 mL of NS to get a final concentration in the range of 1 to 14 mg/mL (typically 10 mg/mL) and infuse over 30 minutes. For children 1-6 years of age: dilute dose in 25 mL of NS and infuse over 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: pruritus, hives, shortness of breath, difficulty swallowing, truncal erythema, anaphylaxis.
- Cardiovascular: hypotension, hypertension.
- GI: diarrhea, constipation, nausea, vomiting.
- CNS: headache, dizziness, insomnia.
- Myopathy: muscular pain, weakness, creatine phosphokinase elevations.
- Abnormal LFTs.
- Local reactions.

DOSAGE

Adults:
- Complicated skin and skin structure infections: 4 mg/kg IV q24h for 7 to 14 days.
- *Staphylococcus aureus* bloodstream infections: 6 mg/kg IV q24h for 10 to 42 days, with an option for an additional 14 days.
- Endocarditis: 8-12 mg/kg IV q24h.
- VREfm infections: 8-12 mg/kg IV q24h.
- Other infections: 6-10 mg/kg IV q24h.
- Dosage in renal impairment: if CrCl is less than 30 mL/min, administer dose q48h. If possible, administer dose following completion of hemodialysis on hemodialysis days.
- Dosage in hepatic impairment: no dosage adjustment required if mild to moderate impairment (Child-Pugh class A or B); no data if severe impairment (Child-Pugh class C).

Pediatrics:
- Complicated skin and skin structure infections: treat for up to 14 days:
  - 12-17 years of age: 5 mg/kg IV q24h.
  - 7-11 years of age: 7 mg/kg IV q24h.
  - 2-6 years of age: 9 mg/kg IV q24h.
  - 1 to less than 2 years of age: 10 mg/kg IV q24h.

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DOSAGE (Cont.)

- *Staphyloccus aureus* bloodstream infections: treat for up to 42 days:
  12-17 years of age: 7 mg/kg IV q24h.
  7-11 years of age: 9 mg/kg IV q24h.
  1-6 years of age: 12 mg/kg IV q24h.
- Dosage in renal impairment: if CrCl 30 mL/min and above: usual dose. If CrCl less than 30 mL/min, dosage has not been established.
- Dosage in hepatic impairment: no dosage adjustment required in mild to moderate impairment (Child-Pugh class A or B); no data in severe impairment (Child-Pugh class C).

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Cubicin ® RF:
- Store vials between 15-30°C.
- Reconstituted solution should be in range of colour from pale yellow to light brown.
- Reconstituted solution with SWFI: stable for 1 day at room temp and 3 days in the fridge when stored in the original vial or in a polypropylene syringe.
- Reconstituted solution with bacteriostatic water for injection: stable for 2 days at room temp when stored in the original vial or in a polypropylene syringe, 3 days in the fridge when stored in the original vial and 5 days in the fridge in a polypropylene syringe.
- Diluted solution in NS: if reconstituted originally with SWFI: stable for 19 hours at room temp and 3 days in the fridge. If reconstituted originally with bacteriostatic water for injection: stable for 2 days at room temp and 5 days in the fridge.
- Compatible with NS; incompatible with dextrose solutions.

MISCELLANEOUS

- Daptomycin is inactive against gram-negative bacteria.
- Plasma creatine phosphokinase levels should be measured at baseline and at least once weekly during therapy for all patients.

REFERENCES

1, 5, 6, 40, 82, 135, 306, 422, 428, 510.

Full revision 2013; limited revision 2014, 2015, 2018, 2020, 2021

NAME OF MEDICATION
DARATUMUMAB

OTHER NAMES
Darzalex ®, Darzalex ® SC, HuMax-CD38

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of patients with newly diagnosed multiple myeloma who are eligible for autologous stem cell transplant (ASCT) (to be used in combination with bortezomib, thalidomide and dexamethasone). (IV formulation only).
- Treatment of patients with newly diagnosed multiple myeloma who are ineligible for ASCT (to be used in combination with lenalidomide and dexamethasone, or with bortezomib, melphalan and prednisone). (IV and SC formulations).
- Treatment of patients with multiple myeloma who have received at least one prior therapy (to be used in combination with lenalidomide and dexamethasone, or bortezomib and dexamethasone). (IV and SC formulations).
- Treatment of patients with multiple myeloma who have received at least three prior lines of therapy including a proteasome inhibitor (PI) and an immunomodulatory agent (IMiD), or who are refractory to both a PI and an IMiD. (IV and SC formulations).
- Treatment of patients with newly diagnosed light chain amyloidosis (to be used in combination with bortezomib, cyclophosphamide and dexamethasone). (SC formulation only).

ADMINISTRATION

- **There are 2 formulations: one for IV use (20 mg/mL vial) and one for SC use (120 mg/mL vial). Ensure using the correct one.**
- **Ensure premedication has been administered as recommended and follow with post-infusion/post-injection medication if needed; refer to Dosage section**.
- Intermittent IV infusion:
  1. Ensure using the IV formulation. From a NS IV bag (see below for bag size), remove a volume of NS equal to the required volume for the daratumumab dose. Add required dose of daratumumab into the NS bag. The IV bags can be made of PVC, polypropylene, polyethylene or polyolefin blend. Gently invert to mix. Do NOT shake; do NOT freeze. Very small translucent to white proteinaceous particles may develop; do not use if discoloured or if visibly opaque or foreign particles are detected.
     - For first infusion: use a NS bag size of 1000 mL; infuse at 50 mL/hr for the first hour; if no infusion reactions occur, increase rate by 50 mL/hr every hour to a maximum rate of 200 mL/hr.
     - For the second infusion: can use a 500 mL bag only if no infusion reactions occurred during the first 3 hours of the first infusion (otherwise, follow same instructions as per first infusion above); infuse at 50 mL/hr for the first hour; if no infusion reactions occur, increase rate by 50 mL/hr every hour to a maximum rate of 200 mL/hr.
     - For the third and subsequent infusions: use a 500 mL bag and infuse at 100 mL/hr only if no infusion reactions occurred with rates of at least 100 mL/hr with the first 2 infusions (otherwise, follow same instructions as per second infusion above); if no infusion reactions occur, increase rate by 50 mL/hr every hour to a maximum rate of 200 mL/hr. May also administer a 500 mL bag following an accelerated infusion protocol: 20% of the dose over 30 minutes (200 mL/hr) then 80% of the dose over 60 minutes for a total infusion time of 90 minutes.
  - Use an in-line sterile non-pyrogenic low-protein-binding polyethersulfone filter (0.2-0.22 microns) for administration. The following administration sets are compatible: polyurethane, polybutadiene, PVC, polypropylene and polyethylene.
  - Must be administered by an infusion pump.
- SC: ensure using the SC formulation; allow vial to reach room temp. Inject SC into abdomen, approximately 7.5 cm to the right or left of navel, over 3-5 minutes. Alternate injection site. If patient experiences pain during the injection, pause or slow down the injection rate; if pain is not alleviated, finish administering the dose at the opposite site of the abdomen. Ensure site of injection is free of scars and is not red, bruised, tender or hard.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus, urticaria, anaphylaxis.

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POTENTIAL ADMINISTRATION HAZARDS (Cont.)

- Infusion-related reactions (with IV administration): frequent during the first infusion and infrequent with subsequent infusions. Symptoms include: nasal congestion, cough, throat irritation, rhinitis, dyspnea, bronchospasm, nausea, chills, hypertension, pulmonary edema and hypoxia. Premedication and post-infusion medications decrease the incidence of reactions. Can occur during or up to 4 hours after the infusion (and up to 48 hours after the infusion without post-infusion medications). If a reaction occurs, stop the infusion and manage symptoms; once reaction has resolved, treatment can be restarted at half the last infusion rate, unless the reaction is severe (grade 4 severity or recurrence for a third time of a grade 3 reaction); in such case, discontinue the drug permanently.
- Injection-related systemic reactions (with SC administration): frequent following the first injection and infrequent with subsequent injections. Usually occur within 4 hours after the injection and the majority on the day of the injection. Symptoms include: nasal congestion, cough, throat irritation, rhinitis, wheezing, pyrexia, chest pain, pruritus, chills, vomiting, nausea, hypotension. Premedication and post-injection medications decrease the incidence of the reactions.
- Cardiovascular: hypertension, hypotension, palpitations, arrhythmia, cardiac failure/arrest, peripheral edema.
- GI: nausea, vomiting, anorexia, diarrhea, constipation, abdominal pain.
- CNS: headache, insomnia, peripheral sensory neuropathy.
- Electrolyte disturbances: hypercalcemia, hypocalcemia, hyponatremia.
- Hematologic: anemia, neutropenia, thrombocytopenia.
- Hepatic: AST increase.
- Renal: serum creatinine increase.
- Respiratory: cough, dyspnea, bronchospasm, rhinitis, upper respiratory tract infections, pneumonia.
- Infections (including cytomegalovirus infection, reactivation of hepatitis B virus and herpes zoster virus; refer to Miscellaneous section).
- Fatigue, musculoskeletal pain, fever.
- Local reactions: erythema (IV); pain, bruising, rash, hemorrhage, pruritus (SC).

DOSAGE

For IV administration:
- **Premedication for daratumumab monotherapy: to be administered 1-3 hours before darutumumab infusion: antipyretics (acetaminophen 650-1000 mg PO), antihistamines (diphenhydramine 25-50 mg IV/PO) and corticosteroids (methylprednisolone 100 mg IV or an equivalent dose of a long- or intermediate-acting corticosteroid) for the first infusion; may use a lower dose of the corticosteroid (e.g., methylprednisolone 60 mg IV) for subsequent infusions.**
- **Premedication for daratumumab combination therapy: same as above but, instead of methylprednisolone, administer dexamethasone 20 mg IV for the first infusion then consider PO for subsequent infusions**.
- **Post-infusion medication for daratumumab monotherapy: administer an oral corticosteroid (20 mg of methylprednisolone or an equivalent dose of a long- or intermediate-acting corticosteroid) on the first and second day after each infusion (beginning the day after the infusion).**
- **Post-infusion medication for daratumumab combination therapy: consider administering low-dose oral methylprednisolone (20 mg or less) or equivalent the day after the infusion. However, if dexamethasone is part of the chemotherapy regimen and is scheduled to be administered the day after daratumumab infusion, additional post-infusion medications may not be needed.**
- **In all patients with chronic obstructive pulmonary disease, respiratory infusion reactions may necessitate post-infusion administration of short- and long-acting bronchodilators with inhaled corticosteroids**.
- Newly diagnosed multiple myeloma eligible for ASCT, in combination with bortezomib, thalidomide and dexamethasone: 16 mg/kg IV as per the following frequency:
  - Induction weeks 1-8: weekly (total of 8 doses);
  - Induction weeks 9-16: every 2 weeks (total of 4 doses);
  - Stop for high-dose chemotherapy and ASCT;
  - Consolidation weeks 1-8 (post ASCT): every 2 weeks (total of 4 doses).

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DOSAGE (Cont.)

- Newly diagnosed multiple myeloma ineligible for ASCT, in combination with bortezomib, melphalan and prednisone: 16 mg/kg IV as per the following frequency:
  - Weeks 1-6: weekly (total of 6 doses);
  - Weeks 7-54: every 3 weeks (total of 16 doses);
  - Weeks 55 onwards until disease progression: every 4 weeks.
- Newly diagnosed multiple myeloma and multiple myeloma with at least one prior therapy, in combination with lenalidomide and dexamethasone, or multiple myeloma with at least 3 prior lines of therapy, as monotherapy: 16 mg/kg IV as per the following frequency:
  - Weeks 1-8: weekly (total of 8 doses);
  - Weeks 9-24: every 2 weeks (total of 8 doses);
  - Weeks 25 onwards until disease progression: every 4 weeks.
- Multiple myeloma with at least one prior therapy, in combination with bortezomib and dexamethasone: 16 mg/kg IV as per the following frequency:
  - Weeks 1-9: weekly (total of 9 doses);
  - Weeks 10-24: every 3 weeks (total of 5 doses);
  - Weeks 25 onwards until disease progression: every 4 weeks.

For SC administration:

- **Premedication for daratumumab monotherapy: to be administered 1-3 hours before each dose of daratumumab: antipyretics (acetaminophen 650-1000 mg PO), antihistamines (diphenhydramine 25-50 mg IV/PO or equivalent), corticosteroids (methylprednisolone 100 mg IV/PO or an equivalent dose of a long- or intermediate-acting corticosteroid) for the first SC injection; may use a lower dose of the corticosteroid (methylprednisolone 60 mg) for subsequent injections.**
- **Premedication for daratumumab combination therapy: same as above but instead of methylprednisolone, administer dexamethasone 20 mg IV/PO. If dexamethasone is part of the chemotherapy regimen, the dexamethasone treatment dose will serve as the premedication on daratumumab injections days. Additional regimen specific corticosteroid (e.g., prednisone) should not be taken on daratumumab injections days when patients receive dexamethasone as a premedication.**
- **Post-injection medication for daratumumab monotherapy: administer an oral corticosteroid (20 mg of methylprednisolone or an equivalent dose of a long- or intermediate-acting corticosteroid) of the first and second day after each SC injection of daratumumab (beginning the day of the injection).**
- **Post-injection medication for daratumumab combination therapy: consider administering low-dose oral methylprednisolone (20 mg or less) or equivalent the day after the SC injection of daratumumab. However, if a corticosteroid (e.g., dexamethasone or prednisone) is part of the chemotherapy regimen and is scheduled to be administered the day after daratumumab SC injection, additional post-injection medications may not be needed.**
- Post-injection medication may be discontinued if no major injection-related reaction is observed after the first 3 SC injections.
- In all patients with chronic obstructive pulmonary disease, respiratory injection reactions may necessitate post-injection medications including short- and long-acting bronchodilators and inhaled corticosteroids. These medications may be discontinued if no major injection-related reactions is observed with the first 4 SC injections.
- Dose: 1800 mg SC as per the following schedule:
  - In combination with lenalidomide and dexamethasone or as monotherapy (4-week cycle regimens):
    - Weeks 1-8: weekly (total of 8 doses);
    - Weeks 9-24: every 2 weeks (total of 8 doses);
    - Weeks 25 onwards until disease progression: every 4 weeks.
  - In combination with bortezomib, melphalan and prednisone (6-week cycle regimens):
    - Weeks 1-6: weekly (total of 6 doses);
    - Weeks 7-54: every 3 weeks (total of 16 doses);
    - Weeks 55 onwards until disease progression: every 4 weeks.

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DOSAGE (Cont.)

- In combination with bortezomib and dexamethasone (3-week cycle regimens):
  - Weeks 1-9: weekly (total of 9 doses);
  - Weeks 10-24: every 3 weeks (total of 5 doses);
  - Weeks 25 onwards until disease progression: every 4 weeks.
- In combination with bortezomib, cyclophosphamide and dexamethasone (4-week cycle regimens):
  - Weeks 1-8: weekly (total of 8 doses);
  - Weeks 9-24: every 2 weeks (total of 8 doses);
  - Weeks 25 onwards until disease progression or a maximum of 2 years: every 4 weeks.

For both IV and SC administrations:

- Dosage adjustment due to toxicity: no dosage reduction should be done; in case of myelotoxicity, delay dose until bone marrow recovers and then administer full dose as above.
- Dosage in renal impairment: no dosage adjustment is necessary.
- Dosage in hepatic impairment: no dosage adjustment in mild impairment; not studied sufficiently in moderate or severe impairment.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8ºC. Do not freeze or shake. Protect from light.
- Unopened vials for SC administration (120 mg/mL) are stable for 24 hours at room temp and ambient light.
- The undiluted solution (IV and SC) in vial should be colourless to yellow.
- The diluted IV solution (with NS) is stable for 24 hours in the fridge, protected from light, followed by 15 hours (including infusion time) at room temp (15-25ºC) and exposed to room light.
- The SC injection solution is stable for 4 hours at room temp at ambient light in polypropylene or polyethylene syringes.
- The SC injection solution is compatible with polypropylene, polyethylene and PVC SC infusion sets.

MISCELLANEOUS

- Emergency drugs and resuscitation equipment must be immediately available for the treatment of infusion/injection-related reactions.
- Hepatitis B virus (HBV) reactivation may occur with daratumumab; perform HBV screening before starting therapy; monitor patients for HBV reactivation during and for at least 6 months following end of therapy.
- Herpes zoster virus reactivation may occur with daratumumab; start an antiviral prophylaxis within one week before intial treatment and continue until 3 months after completion of treatment.
- Daratumumab may interfere with cross-matching and red blood cell antibody screening; interference with indirect Coombs test may persist for up to 6 months after the last dose. Type and screen patient's blood before intiating therapy if possible.

REFERENCES

5, 129, 165, 406, 429.

New monograph 2018; limited revision 2019, 2020, 2021

NAME OF MEDICATION
DARBEPOETIN

OTHER NAMES
Aranesp ®, Darbepoetin alpha

CLASSIFICATION
Hematopoietic

INDICATIONS

- Treatment of anemia associated with chronic kidney disease.
- Treatment of chemotherapy-induced anemia in non-myeloid malignancies.

ADMINISTRATION

- IV direct (preferred in patients on hemodialysis): physician or RN. Administer undiluted from vial or use prefilled syringe, over at least 1 minute.
- SC: in the outer area of upper arms, abdomen (at least 2 inches from navel), front of middle thighs, buttocks. Rotate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, angioedema, bronchospasm, rash, urticaria, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Cardiovascular: hypertension, hypotension, chest pain.
- GI: nausea, vomiting, diarrhea.
- CNS: headache, dizziness.
- Dyspnea.
- Peripheral edema.
- Infection.
- Myalgia, arthralgia, fatigue, fever.
- Thrombotic events.
- Pure red cell aplasia (rare).
- Local reactions: injection site pain or local redness, swelling or itching, vascular access thrombosis.

DOSAGE

Anemia associated with chronic kidney disease:
- Initial dose:
  - Patients of dialysis: 0.45 mcg/kg once weekly or 0.75 mcg/kg every 2 weeks; can be administered IV or SC but IV preferred.
  - Patients not on dialysis: 0.45 mcg/kg SC/IV every 4 weeks.
- Titrate dose in steps of 25% to target hemoglobin of 100 to 120 g/L. Do not exceed a hemoglobin level of 120 g/L. Do not exceed increase in hemoglobin of greater than 10 g/L in a 2-week period. Do not increase dose more frequently than once a month.
- Consult the manufacturer's monograph for more details on dosing, including how to switch patients from erythropoietin to darbepoetin.

Chemotherapy-induced anemia:
- Do not initiate if hemoglobin is 100 g/L or greater.
- Two dosing regimens may be used: 500 mcg SC every 3 weeks or 2.25 mcg/kg SC once weekly with a maximum of 4.5 mcg/kg once weekly. Adjust dose to maintain the lowest hemoglobin level sufficient to avoid the need for red blood cell transfusion and not to exceed 120 g/L.
- Discontinue darbepoetin following completion of a chemotherapy course or after 8 weeks of therapy with no response.
- Consult manufacturer's monograph for more details on dosage adjustment.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes betweent 2-8°C. Protect from light. Do not freeze.
- Do not shake. Do not dilute.
- Single use only.
- Do not use if discoloured or precipitate present.

MISCELLANEOUS

- When darbepoetin therapy is started or the dose adjusted, hemoglobin should be measured on a weekly basis until stabilized and then monthly thereafter.
- High hemoglobin and/or high rate of rise of hemoglobin increases the risk of cardiovascular and neurological events.
- Blood pressure should be followed closely. Darbepoetin should not be used in patients with uncontrolled hypertension.
- Iron status should be evaluated, as the majority of patients will require iron supplementation.
- Patients experiencing pure red cell aplasia with other erythropoietin products should not be switched to darbepoetin.
- Needle cover of all prefilled syringes contains latex.

REFERENCES

1, 2, 40, 95, 135, 292.

Full revision 2013; limited revision 2016, 2017, 2018

NAME OF MEDICATION
DAUNOrubicin

OTHER NAMES
Cerubidine ®, Daunomycin

CLASSIFICATION
Antineoplastic

INDICATIONS
- Treatment of myeloblastic and acute lymphoblastic leukemias.
- Induction of remission in chronic myeloid leukemia, reticulosarcoma, Ewing or Wilms' tumours, lymphosarcoma and other malignant tumours.

ADMINISTRATION
- Reconstitute with 4 mL SWFI for a final concentration of 5 mg/mL. Shake gently to dissolve.
- IV direct: physician or RN; dilute in 10-15 mL of NS; inject over 2-3 minutes into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W; infuse 50 mL over 10-15 minutes; infuse 100 mL over 30-45 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid (rare); fever and rash.
- Cardiovascular: transient arrhythmias, facial flushing with rapid injection.
- GI: nausea and vomiting.
- Dermatologic: radiation recall reaction (rare).
- Hyperuricemia.
- Red discolouration of urine for 1-2 days.
- Local reactions: pain, flare reaction (histamine release).
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- Usual adult dose: initial therapy: 30-60 mg/m^2/day IV for 3-6 days every 3-4 weeks (monotherapy) or 45 mg/m^2/day IV for 2-3 days every 3-4 weeks (combination therapy). Maintenance: 30 mg/m^2 IV once weekly.
- To avoid cardiotoxicity, total cumulative dose should not exceed 900 mg/m^2 in adults with normal cardiac function or 400-450 mg/m^2 in adults with cardiac risk factors or in combination with thoracic radiation.
- Dosage adjustment needed in hepatic or renal impairment.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials at room temp. Protect from light.
- Reconstituted solution stable for 7 days at room temp. Protect from light.
- Contact between daunorubicin 5 mg/mL and aluminium may result in darkening of solution and formation of black patches on the aluminium surface after 12-24 hours.
- Colour change of reconstituted solution from red to blue-purple indicates decomposition; discard these solutions.
- Stable for 43 days at room temp, 4°C and -20°C protected from light in NS and D5W at a concentration of 0.1 mg/mL in PVC bags.
- Stable for 28 days at room temp protected from light in D5W, NS and RL at a concentration of 0.1 mg/mL in plastic syringes.
- Stable for 43 days at 4°C, protected from light in SWFI at a concentration of 2 mg/mL in polypropylene syringes.
- Stable for 80 hours at room temp in NS and D5W at a concentration of 5 mg/mL.
- Stable for 54 hours at room temp in RL at a concentration of 5 mg/mL.

MISCELLANEOUS
- Not interchangeable with liposomal daunorubicin.

REFERENCES
1, 2, 4, 5, 6, 40, 95, 129, 143, 165.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
DECITABINE *

OTHER NAMES
5-Aza-2'-deoxycytidine, Dacogen ™

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of myelodysplastic syndrome.

ADMINISTRATION

- Reconstitute each 50 mg vial with 10 mL SWFI to obtain a 5 mg/mL solution.
- Intermittent IV infusion: immediately dilute reconstituted solution with 50 to 250 mL of cold (2-8°C) NS, D5W, or RL to a concentration of 0.1-1 mg/mL and infuse over 1 or 3 hours, depending on protocol.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- GI: nausea, diarrhea, constipation.
- CNS: headache, insomnia.
- Hematologic: neutropenia, thrombocytopenia, anemia.
- Fever, fatigue, cough.
- Hyperglycemia.
- Petechiae.
- Local injection site reactions: swelling, erythema, pain.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 15 mg/m^2 IV infused over 3 hours; repeat q8h for 3 days. Repeat this cycle every 6 weeks for a minimum of 4 cycles; treatment to continue as long as patient continues to benefit.
- 20 mg/m^2 IV infused over 60 minutes once daily for 5 days. Repeat this cycle every 4 weeks.
- Dose to be adjusted or delayed based on hematologic values, renal function, hepatic function, and in the presence of active or uncontrolled infection; refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Use reconstituted solution within 15 minutes. Drug concentration decreases by 10% after 5 hours at room temp.
- Stable for 7 hours in the fridge if prepared using cold (2-8°C) infusion fluids.

MISCELLANEOUS

- Caution when administering to patients with pre-existing hepatic or renal impairment.

REFERENCES

1, 5, 40, 60, 95, 152, 178.

* Available via Health Canada's Special Access Programme

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
DEFEROXAMINE

OTHER NAMES
Desferal ®

CLASSIFICATION
Chelating agent

INDICATIONS

- An adjunct in the treatment of acute iron intoxication.
- Treatment of chronic iron overload due to transfusion-dependent anemias.
- Diagnosis of aluminium overload (deferoxamine infusion test).
- Treatment of chronic aluminium overload in patients with end stage renal failure (ESRF) under maintenance dialysis.

ADMINISTRATION

- For IV & SC use: reconstitute the 500 mg vial and 2 g vial with 5 mL and 20 mL of SWFI respectively for a final concentration of 95 mg/mL (taking into account volume of powder). Do NOT reconstitute with NS.
- For IM use: reconstitute the 500 mg vial and 2 g vial with 2 mL and 8 mL of SWFI respectively for a final concentration of 213 mg/mL (taking into account volume of powder). Do NOT reconstitute with NS.
- Intermittent IV infusion, Continuous IV infusion (preferred for treatment of acute iron intoxication): dilute with NS, D5W or RL; maximum concentration of 250 mg/mL. Refer to Dosage section for rates.
- IM (for treatment of chronic iron toxicity).
- SC infusion (for treatment of chronic iron toxicity): must be administered by an infusion pump.
- It should NOT be administered at the same time as blood transfusions (even if in 2 separate lines) to avoid confusion of the causal agent should hypersensitivity or infusion-related reactions occur.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, angioedema, rash, urticaria.
- Infusion-related reactions: following rapid IV infusion (greater than 15 mg/kg/hr), generalized erythema, urticaria, flushing, tachycardia, hypotension and shock may occur (due to possible histamine release).
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: headache, dizziness, neuropathy, paresthesia, precipitation of aluminium-related dialysis encephalopathy.
- Discolouration of urine (reddish colour).
- Arthralgia, myalgia.
- Fever.
- Increased frequency of opportunistic infections, especially *Yersinia enterocolitica*, *Yersinia pseudotuberculosis* and possibly fungi.
- Adult respiratory distress syndrome-like condition at high doses.
- Local reactions: pain, pruritus, erythema, swelling, burning, and induration at the site of injection or infusion.

DOSAGE

Adults:
- Acute iron intoxication:
  - Continuous IV infusion: 15 mg/kg/hr IV; may titrate up to 40 mg/kg/hr in life-threatening situations. Infuse for 12 hours in moderate poisoning and for up to 24 hours in severe poisoning.
  - Intermittent IV infusion: 1 g IV initially at maximum rate of 15 mg/kg/hr, followed by 500 mg IV q4h for 2 doses at a maximum rate of 125 mg/hr. Depending on the clinical response, subsequent doses of 500 mg IV may be administered q4-12h. Maximum dose: 6 g/day. Maximum single dose should not exceed 2 g.
- Chronic iron overload: 0.5-1 g IM daily or 1-4 g/day by SC infusion or IV infusion (may be given as a 12-hour infusion, but in some patients, iron excretion is further increased by infusing the same dose over 24 hours).
- Diagnosis of aluminium overload: Adults with ESRF: 5 mg/kg IV single dose following dialysis or during the last 60 minutes of the hemodialysis session.
- Chronic aluminium overload: 5 mg/kg IV once weekly following dialysis or during the last 60 minutes of the hemodialysis session. Dosage and length of treatment should be individually determined.

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DOSAGE (Cont.)

Pediatrics:
- Acute iron intoxication:
  - Continuous IV infusion: 15 mg/kg/hr IV; may titrate up to 35 mg/kg/hr in life-threatening situations. Infuse for 12 hours in moderate poisoning and for 24 hours in severe poisoning. Maximum dose of 6 g/day.
  - Intermittent IV infusion: 20 mg/kg (maximum 1 g) IV at a maximum rate of 15 mg/kg/hr, followed by 10 mg/kg (maximum 500 mg) IV q4h for 2 doses at a rate not exceeding 125 mg/hr. Depending on the clinical response, subsequent dose of 10 mg/kg IV (maximum 500 mg) may be administered q4-12h. Maximum dose of 6 g/day.
  - IM: initial dose of 90 mg/kg IM followed by 45 mg/kg IM q4-12h. Maximum dose of 6 g/day; maximum single dose should not exceed 1 g.
- Chronic iron overload: 0.5-1 g IM daily or 20-50 mg/kg/day by SC infusion or IV infusion over 8-12 hours. Maximum dose: 2 g/day.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C. Protect from light.
- Reconstituted solution is stable for 24 hours at room temp; do not refrigerate.
- Reconstituted solution is clear and colourless to slightly yellow. Turbid solutions should not be used.
- Stable for 24 hours at room temp when diluted in NS, D5W and RL solutions.
- Stable for 14 days at 35°C in D5W or NS at a concentration of 5 mg/mL in natural rubber elastomeric reservoirs.
- Stable for 14 days at 30°C in SWFI at a concentration of 250 mg/mL when stored in polypropylene infusion pump syringes.

MISCELLANEOUS

- Deferoxamine is not a substitute for standard treatment of iron intoxication (emesis, gastric lavage, suction and maintenance of airways, control of peripheral vascular failure and correction of acidosis).
- Contraindicated in patients with severe renal disease or anuria not on dialysis.
- Theoretically, 1 g of deferoxamine is capable of sequestering 85 mg of iron (as the ferric ion) and 41 mg of aluminium.

REFERENCES

1, 2, 4, 5, 6, 40, 82, 95, 135, 143, 208, 339.

Full revision 2013; limited revision 2014, 2015, 2016, 2018, 2019, 2020

NAME OF MEDICATION
DEFIBROTIDE

OTHER NAMES
Defitelio ™

CLASSIFICATION
Thrombolytic

INDICATIONS

- Treatment of adults and pediatric patients with hepatic sinusoidal obstruction syndrome (SOS; formerly called hepatic veno-occlusive disease) following hematopoietic stem cell transplantation.

ADMINISTRATION

- Intermittent IV infusion: dilute dose in NS or D5W to obtain a concentration of 4 to 20 mg/mL. Gently mix. Solution should be clear, colourless to light yellow and without visible particles. Infuse over 2 hours using a 0.2 micron in-line filter. Flush with NS or D5W immediately before and after administration.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: pruritus, rash, wheezing, angioedema.
- Cardiovascular: hypotension.
- GI: nausea, vomiting, diarrhea.
- Hematologic: bleeding (e.g., GI tract, epistaxis, intracranial, cerebral, pulmonary); stop therapy if significant bleeding occurs.

DOSAGE

Adults and pediatrics (one month of age and older):
- 6.25 mg/kg IV q6h (25 mg/kg/day) for at least 21 days.
- Dosage in renal impairment: no dosage adjustment is necessary, including hemodialysis patients.
- Dosage in hepatic impairment: no dosage adjustment is necessary.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 20-25°C (excursions permitted between 15-30°C). Do not freeze.
- Diluted solution stable for 4 hours at room temp and 24 hours in the fridge.

MISCELLANEOUS

- Do not administer concurrent medications that increase risk of hemorrhage such as warfarin, heparin, LMWHs (except for maintenance of central IV line and dialysis machine filtration), alteplase or other systemic anticoagulants or fibrinolytic therapy.
- Monitor carefully if patient receives antiplatelet therapy.
- Discontinue infusion at least 2 hours prior to an invasive procedure.
- Use with caution in patients with pig allergies as defibrotide is derived from porcine tissue.

REFERENCES

5, 95, 135.

Full revision 2018; limited revision 2020

NAME OF MEDICATION
DENOSUMAB

OTHER NAMES
Prolia ®, Xgeva ®

CLASSIFICATION
Bone resorption inhibitor, Monoclonal antibody

INDICATIONS

- Treatment of osteoporosis in postmenopausal women at high risk for fracture or who have failed or are intolerant to other osteoporosis therapy (Prolia ®).
- Treatment of osteoporosis in men at high risk for fracture (Prolia ®).
- To increase bone mass in men with nonmetastatic prostate cancer who are receiving androgen deprivation therapy and are at high risk for fracture (Prolia ®).
- To increase bone mass in women receiving adjuvant aromatase inhibitor therapy for nonmetastatic breast cancer (Prolia ®).
- Treatment and prevention of glucocorticoid-induced osteoporosis in women and men at high risk of fracture (Prolia ®).
- For reducing the risk of developing skeletal-related events (SRE) in patients with multiple myeloma and in patients with bone metastases from breast cancer, prostate cancer, non-small cell lung cancer and other solid tumours (Xgeva ®).
- Treatment of adults and skeletally mature adolescents with giant cell tumour of bone that is unresectable or where surgical resection is likely to result in severe morbidity (Xgeva ®).
- Treatment of hypercalcemia of malignancy that is refractory to IV bisphosphonates (Xgeva ®).

ADMINISTRATION

- Prior to administration, bring to room temp in original container (allow to stand 15-30 minutes).
- SC: in upper arm, upper thigh or abdomen.

POTENTIAL ADMINISTRATION HAZARDS

- GI: nausea, diarrhea.
- CNS: headache, insomnia, depression.
- Respiratory: dyspnea, cough.
- Dermatologic: dermatitis, eczema, rash; most of the reactions are not localized to the injection site.
- Back pain, extremity pain and musculoskeletal pain.
- Hypocalcemia especially in patients with severe renal failure (CrCl less than 30 mL/min or on dialysis); hypophosphatemia (Xgeva ®).
- Fatigue, asthenia.
- May increase risk of infections.

DOSAGE

- Osteoporosis/to increase bone mass (Prolia ®): 60 mg SC once every 6 months.
- Preventing SRE in patients with multiple myeloma and in patients with bone metastases (Xgeva ®): 120 mg SC once every 4 weeks.
- Giant cell tumour of bone and hypercalcemia of malignancy (Xgeva ®): initial dose of 120 mg SC on days 1, 8 and 15 of month 1; maintenance dose of 120 mg SC every 4 weeks starting on day 1 of month 2.
- Dosage in renal impairment: no dosage adjustment required.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled syringes and vials in the fridge; protect from light; do not freeze; avoid vigorous shaking.
- Stable for 30 days at room temp (up to 25°C), protected from light.

MISCELLANEOUS

- Ensure adequate intake of calcium and vitamin D.
- Monitor serum calcium level; level should be corrected before starting therapy.

REFERENCES

1, 2, 5, 129, 135.

Full revision 2013; limited revision 2015, 2016, 2019

NAME OF MEDICATION
DESMOPRESSIN

OTHER NAMES
DDAVP ®, Octostim ®

CLASSIFICATION
Antidiuretic hormone analog

INDICATIONS
- Replacement therapy for central diabetes insipidus.
- To control bleeding in patients with hemophilia A and in Von Willebrand's Disease (Type I).
- For prolonged bleeding times and hemorrhagic tendencies in uremic patients.

ADMINISTRATION
- IV direct: physician only. For diabetes insipidus only. Administer over 1 minute.
- Intermittent IV infusion: dilute in 50 mL (10 mL for children weighing 10 kg or less) of solution and administer over 20-30 minutes. For bleeding disorders only.
- IM, SC: preferred for diabetes insipidus; SC route can also be used for bleeding disorders.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylaxis (rare).
- Cardiovascular: facial flushing, mild hypotension or hypertension, tachycardia.
- GI: nausea, abdominal cramps.
- CNS: headache.
- Vulvar pain.
- Potential for water intoxication and hyponatremia when used for its hemostatic effects. Watch for signs and symptoms associated with hyponatremia: headache, nausea, vomiting, weight gain, restlessness, fatigue, lethargy, disorientation, depressed reflexes, loss of appetite, irritability, muscle weakness and/or spasms, abnormal mental status (e.g., confusion, decreased consciousness, hallucinations). May lead to seizures and coma.
- Thrombotic events (rare).
- Local reactions: erythema, swelling, burning pain at site of injection; occasional burning or pain along the vein of injection.

DOSAGE
- Adults:
  - Diabetes insipidus: 1-4 mcg once daily IV, IM, SC as a single dose or in two divided doses.
  - Bleeding disorders: 0.3 mcg/kg or 10 mcg/m^2 IV or SC.

- Pediatrics:
  - Diabetes insipidus: 0.4 mcg once daily IV, IM, SC.
  - Bleeding disorders: 0.3 mcg/kg SC or IV.

- When used preoperatively, administer 30 minutes before procedure.
- When used for prevention/treatment of bleeding, repeated administration at intervals of less than 48 hours may increase the possibility of tachyphylaxis. However, if needed, a second dose may be administered 8-24 hours after the first dose.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules in the fridge; do not freeze.
- Stable for 24 hours at room temp in NS or D5W at concentrations of 0.3 mcg/mL and 0.4 mcg/mL in PVC bags.

MISCELLANEOUS
- Peak effect on blood clotting occurs 30 minutes after an IV dose and 60-90 minutes after a SC dose.
- Desmopressin 1 mcg = 4 international units.
- Equivalent dosage: 1 mcg IV = 10 mcg intranasal.

REFERENCES
1, 2, 5, 40, 71, 73, 75, 95, 135, 449, 450.

Full revision 2013; limited revision 2015, 2016, 2018

NAME OF MEDICATION
DEXAMETHASONE

OTHER NAMES
Decadron ®

CLASSIFICATION
Corticosteroid

INDICATIONS

- For short-term emergency corticosteroid therapy.
- Situations when a rapid and intense hormonal effect is desired, e.g., shock, hypersensitivity reactions, status asthmaticus, organ transplants, cerebral edema, hypercalcemia, bacterial meningitis.
- Antiemetic for chemotherapy-induced nausea and vomiting.
- Adrenocortical insufficiency.
- As an adjunct in the treatment of coronavirus disease 2019 (COVID-19; SARS-CoV-2 infection) in hospitalized patients requiring invasive mechanical ventilation or supplemental oxygen.

ADMINISTRATION

- IV direct: physician or RN. Only for doses of 8 mg or less; administer over 1 minute.
- Intermittent IV infusion: dilute in 50-100 mL with NS or D5W and administer over 5-15 minutes.
- Continuous IV infusion.
- IM, SC

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactoid reactions, anaphylaxis, angioedema, urticaria.
- IV related reactions: burning and tingling in the perineal area may occur with rapid IV injections; slowing down may reduce the occurrence.
- GI: peptic ulcer, perforation, hemorrhage, nausea, pancreatitis, abdominal distension, increased appetite.
- CNS: mental disturbances (euphoria, mood swings, depression, personality changes, psychoses), insomnia, headache, vertigo.
- Dermatologic: hypo- or hyper-pigmentation, bruising, scarring, induration, and sterile abscess.
- Electrolyte disturbances: hypokalemia.
- Endocrine and metabolic: hyperglycemia.
- Ophthalmic: increased intraocular pressure.
- Immunosuppression.

DOSAGE

- Doses are highly variable; average dose range is 0.5-24 mg IM or IV daily; may be divided in 2-4 doses. Total dose usually does not exceed 80 mg/24 hours.
- Shock: 1-6 mg/kg IV as a single injection or 40 mg IV repeated q2-6h if needed or 20 mg IV followed by continuous IV infusion of 3 mg/kg/24 hours. High-dose therapy is continued until patient condition stabilizes and usually should not be used longer than 48-72 hours.
- Hypersensitivity reactions: 4-8 mg IV on the first day, then follow for the next 5 days with decreasing oral doses (1.5 mg q12h on days 2-3; 0.75 mg q12h on day 4 and 0.75 mg daily on days 5-6).
- Cerebral edema: 10 mg IV followed by 4 mg IM q6h until symptoms subside. Reduce dose after 2-4 days and discontinue gradually over 5-7 days. If recurrent or inoperable brain tumours, 2 mg IM or IV 2 or 3 times daily to relieve symptoms of increased intracranial pressure.
- Bacterial meningitis: 0.15 mg/kg IV q6h X 4 days; start first dose 10-20 minutes before or concurrently with first dose of antibiotic.
- Antiemetic for chemotherapy-induced nausea and vomiting: 8-20 mg IV before chemotherapy; additional lower doses may be given over the next 24-72 hours.
- As an adjunct in the treatment of COVID-19: 6 mg IV daily for up to 10 days or hospital discharge if sooner.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Stable for 55 days at room temp at a concentration of 10 mg/mL (original undiluted solution) when repackaged in 3 mL plastic syringes.
- Stable for 14 days at room temp protected from light in D5W or NS at concentrations of 0.66 mg/mL and 0.092 mg/mL in PVC bags.

MISCELLANEOUS

- Extreme caution in patients with diabetes, active tuberculosis, active peptic ulcer, fungal, viral or bacterial infections not controlled by medications.

REFERENCES

1, 4, 5, 6, 40, 95, 135, 457, 506, 514.

NAME OF MEDICATION
DEXMEDETOMIDINE

OTHER NAMES
Precedex ™

CLASSIFICATION
Sedative

INDICATIONS

- Sedation of initially intubated and mechanically ventilated post-surgery patients in an intensive care setting for a maximum of 24 hours.
- Conscious sedation of non-intubated patients before and/or during procedures involving monitored anesthesia care (MAC) or awake fiberoptic intubation (AFI) in the operating room setting.
- Sedation for intractable pain/delirium in palliative care patients.

ADMINISTRATION

- Continuous IV infusion: ***cardiac monitoring***; ***continuous BP monitoring***. Prepare the infusion using one of the following 2 methods: 1) dilute 2 mL (200 mcg of base drug) from the concentrated solution with 48 mL of NS for a final volume of 50 mL and a concentration of 4 mcg/mL; shake gently to mix well; OR 2) use the ready-to-use solution (4 mcg/mL, 50 and 100 mL vials). To be used for loading and maintenance doses. Must be administered by an infusion pump.
- Continuous SC infusion (for palliative care only): at TOH: when using a 50 mL minibag of NS, withdraw 10 mL from the bag and discard; add 1000 mcg (10 mL from dexmedetomidine 100 mcg/mL concentrated solution) to the minibag to obtain a final concentration of 20 mcg/mL. Shake gently to mix well. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension, bradycardia, sinus arrest. (**Antidote** for severe bradycardia: atropine or glycopyrrolate). Transient hypertension can occur primarily during the administration of the loading dose. Decreasing the infusion rate may alleviate cardiovascular side effects.
- GI: dry mouth, nausea.
- Respiratory depression.
- Withdrawal symptoms: if administered for more than 24 hours and then stopped abruptly: nervousness, agitation, headache, rapid rise in blood pressure, elevated plasma catecholamine concentration.

DOSAGE

- Intensive care unit sedation (duration of infusion should not exceed 24 hours):
  Loading dose: 1 mcg/kg IV over 10-20 minutes if required.
  Maintenance dose: 0.2-0.7 mcg/kg/hr IV to adjust to desired level of sedation. Although infusion rates up to 2.5 mcg/kg/hr have been used, doses over 1.5 mcg/kg/hr do not add any further benefit.
- Conscious sedation for MAC or AFI (duration of infusion should not exceed 24 hours):
  MAC: Loading dose: 0.5 or 1 mcg/kg IV over 10 minutes.
  Maintenance dose: 0.6 mcg/kg/hr IV to start, with subsequent titration from 0.2-1 mcg/kg/hr.
  AFI: Loading dose: 0.5 or 1 mcg/kg IV over 10 minutes.
  Maintenance dose: 0.7 mcg/kg/hr IV.
- Palliative care sedation: as a continuous SC infusion, initial rate at 0.2 mcg/kg/hr with titration increments of 0.1 mcg/kg/hr q1h prn. Usual range: 0.2-0.7 mcg/kg/hr. Maximum infusion rate of 1.4 mcg/kg/hr.
- Dosage in renal impairment, hepatic impairment or in patients over 65 years of age: consider dose reduction for both the loading and maintenance infusions.
- Dose reductions may be required when co-administered with sedatives, hypnotics, anesthetics or opioids.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Compatible with D5W, NS, RL and mannitol 20%.
- Stable for 48 hours at room temp in NS at concentrations of 4, 8, 12 and 20 mcg/mL in PVC containers.
- Stable for 48 hours at room temp and for 14 days in the fridge when diluted in NS at a concentration of 4 mcg/mL in polypropylene syringes.
- May adsorb to some types of natural rubber; consider using components made with synthetic or coated natural rubber gaskets.

MISCELLANEOUS

REFERENCES

1, 2, 40, 94, 95, 135, 208, 307, 525, 526, 530.

Full revision 2013; limited revision 2014, 2015, 2017, 2021

NAME OF MEDICATION
DEXRAZOXANE

OTHER NAMES
Zinecard ®

CLASSIFICATION
Chelating agent

INDICATIONS

- Reduce the incidence and severity of cardiomyopathy associated with cumulative doses of doxorubicin exceeding 300 mg/m^2 in women with breast cancer who would benefit from continuing doxorubicin therapy above this cumulative dose.

ADMINISTRATION

- Reconstitute vials of 250 mg and 500 mg with 25 mL and 50 mL, respectively, of SWFI only to obtain a concentration of 10 mg/mL.
- Intermittent IV infusion: dilute with RL to a final concentration of 1.3-3 mg/mL; infuse over 15 minutes. After the completion of dexrazoxane infusion, administer doxorubicin within 30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Dermatologic: urticaria.
- Local reactions: pain.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 500 mg/m^2 IV in a ratio of 10:1 relative to the doxorubicin dose. Dose of 500 mg/m^2 should not be exceeded.
- Dosage in renal impairment: when CrCl is less than 40 mL/min, reduce dose by 50% (i.e., give 250 mg/m^2).
- Dosage in hepatic impairment: as doxorubicin dose is decreased, dexrazoxane will be decreased accordingly, keeping a ratio of 10:1.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution is stable for 30 minutes at room temp or for up to 3 hours in the fridge.
- Stable for 60 minutes at room temp or for 4 hours in the fridge in RL at a concentration of 1.3-3 mg/mL.

MISCELLANEOUS

- Not indicated for use when initiating doxorubicin therapy; may interfere with antitumour effects of combination regimens.
- Even though controversial, dexrazoxane is considered an antineoplastic agent.

REFERENCES

1, 2, 4, 6, 40, 95, 129, 165.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2020, 2021

NAME OF MEDICATION
DEXTRAN 40

OTHER NAMES
Gentran 40 ®, LMD 10%,
Low molecular weight dextran

CLASSIFICATION
Plasma volume expander

INDICATIONS

- Fluid replacement and plasma volume expansion in the adjunctive treatment of certain types of shock, when whole blood or blood products are not available.
- May also be used for prophylaxis during surgical procedures with a high incidence of venous thrombosis and pulmonary emboli.
- In extracorporeal circulation, as a priming fluid in pump oxygenators.

ADMINISTRATION

- Continuous IV infusion (mandatory).
- Available as a 10% solution in NS or D5W ready for administration at a rate dependant on the amount of fluid loss.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactoid reactions manifested as urticaria, chest tightness, nausea and vomiting, wheezing, hypotension; closely monitor patient during the first 5 minutes of infusion.
- Cardiovascular: vascular overload and pulmonary edema especially with large doses; increase in central venous pressure.
- Renal failure in extremely dehydrated patients.
- Hematologic: prolongation of bleeding time, decrease in plasma protein concentrations.

DOSAGE

- Shock: 10 to 20 mL/kg infused IV in the first 24 hours. Thereafter, 10 mL/kg daily for maximum of 5 days. The first 500 mL may be infused rapidly while monitoring central venous pressure closely.
- Prophylaxis of venous thrombosis and pulmonary embolism: 10 mL/kg IV initiated during surgery. Can give 500 mL/day for an additional 2 to 3 days. Thereafter, may give 500 mL every 2-3 days for up to 2 weeks.
- Extracorporeal circulation: dose varies with volume of pump oxygenators; generally 10-20 mL/kg is added to the perfusion circuit. Do not exceed 20 mL/kg.
- Dosage in renal impairment: adjust dose as excessive dosing may precipitate renal failure and anuria.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Use only clear solution.
- Dextran flakes may form in the solution; to redissolve, heat solution in water bath at 100°C until it becomes clear or autoclave at 110º C for 15 minutes.

MISCELLANEOUS

- State of hydration should be assessed. Monitor urine output.
- Caution in patients in whom sodium intake is restricted: 500 mL of 10% Dextran-40 in NS contains 77 mmol of Na and Cl. May wish to use dextran in dextrose.
- Depending on assay methods used, dextran may interfere with tests for blood typing and cross-matching, blood glucose, total proteins and bilirubin. To avoid misleading results, draw blood samples before initiation of dextran.

REFERENCES

1, 4, 40, 95, 135.

Full revision 2013; limited revision 2015

NAME OF MEDICATION
DEXTROSE

OTHER NAMES
Glucose

CLASSIFICATION
Carbohydrate

INDICATIONS
- For fluid and calorie replacement (5% and 10% solutions).
- Severe hypoglycemia (25% and 50% solutions).
- As an adjunct to insulin for the treatment of hyperkalemia.

ADMINISTRATION
- IV direct: physician or RN.
- Intermittent IV infusion.
- Continuous IV infusion.
- Rapid rates of administration of hypertonic solutions predispose to pain and phlebitis if a peripheral vein is used. Do NOT exceed a concentration of 10-12.5% for continuous infusion into a peripheral vein.
- Rate of administration: 0.5 g/kg/hr. Maximum: 0.8 g/kg/hr. Caution with higher rates of administration because of the risk of glucosuria. However, in certain circumstances, may be administered at a faster rate, i.e., up to 200 mg/kg/min; refer to Dosage section for more details.

POTENTIAL ADMINISTRATION HAZARDS
- CNS: hyperosmolar syndrome (mental confusion, loss of consciousness).
- Electrolyte disturbances: hypokalemia.
- Endocrine and metabolic: hyperglycemia and glycosuria; reactive hypoglycemia after abrupt discontinuation of concentrated dextrose infusion; fluid disturbances.
- Local reactions: pain, vein irritation, phlebitis.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE
Adults:
- For treatment of severe hypoglycemia: 10-25 g (20-50 mL of 50% solution) IV over 1-3 minutes.
- For treatment of hyperkalemia: 5-10 units of regular insulin IV direct **followed immediately** by 25-50 g of dextrose (50-100 mL of 50% solution) administered IV over 5 minutes.

Pediatrics:
- For treatment of hypoglycemia: infants 6 months of age and younger: 0.25-0.5 g/kg/dose IV; maximum: 25 g/dose.
  Infants older than 6 months of age and children: 0.5-1 g/kg/dose IV; maximum: 25 g/dose.
- For treatment of hyperkalemia: infants and children: 0.5-1 g/kg IV (using 25% or 50% solution) combined with 1 unit of regular insulin for every 4-5 g dextrose given; infuse over 30 minutes to 2 hours.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Dextrose has an acidic pH (4-4.2) and therefore specific compatibility information should be consulted when injected into an IV line containing another drug.

MISCELLANEOUS
- If dextrose 50% is ordered in a non-emergency situation (i.e., patient is stable), verify if patient can take an oral form of glucose.
- Dextrose solutions are an excellent media for microbial growth.
- Each g of IV dextrose provides 3.4 kcal; osmolarity of D10W: 505 mOsm/L and osmolarity of D25W: 1330 mOsm/L.

REFERENCES
1, 40, 82, 95, 135, 236, 328, 366, 367, 462.

Full revision 2013; limited revision 2015, 2017, 2021

NAME OF MEDICATION
DiazePAM

OTHER NAMES
Diazemuls ®, Valium ®

CLASSIFICATION
Benzodiazepine

INDICATIONS

- Where very rapid response desired and oral administration is not possible, e.g., in acute anxiety and agitated states; oral absorption more reliable than IM.
- To alleviate symptoms of acute alcoholic withdrawal.
- To alleviate anxiety and provide sedation, light anesthesia and anterograde amnesia prior to endoscopic and surgical procedures or cardioversion.
- Adjunctive in status epilepticus.
- Adjunctive in skeletal muscle spasm.
- To prevent/treat seizures associated with nerve gas exposure (in the context of chemical warfare).

ADMINISTRATION

- IV direct (preferred): physician; RN for doses up to 10 mg/dose. ***Respiratory support***. Administer slowly at a rate of 5 mg/min for adults and over a 3-minute period for infants and children but do not exceed a rate of 0.25 mg/kg over 3 minutes. Administer directly into a large vein to avoid thrombosis. If not possible, give into the tubing of a flowing IV solution as close as possible to the vein insertion.
- Intermittent IV infusion: refer to Compatibility and Stability section.
- IM: rarely justified because absorption is slow and erratic.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension (**Antidote**: norepinephrine or dopamine), bradycardia.
- CNS: amnesia, ataxia, confusion, drowsiness, dizziness.
- Respiratory depression (especially when dose exceeds 6 mg or when given too quickly).
- Laryngospasm: appropriate apparatus must be readily available to counteract.
- Local reactions: pain with injection, phlebitis.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

Antidote: flumazenil will reverse all sedative effects.

DOSAGE

Adults:
- For anxiety: 2-10 mg IV; may repeat in 3-4 hours.
- For alcohol withdrawal: depending on severity of symptoms, 5-20 mg IV; to be repeated as needed to keep patient comfortable (intervals from 5 minutes to 4 hours have been used).
- For sedation, endoscopic procedure: 10 mg or less IV immediately prior to procedure; maximum dose up to 20 mg or 5-10 mg IM 30 minutes prior to procedure.
- For sedation, preoperative: 10 mg IV before surgery; maximum dose up to 20 mg.
- For sedation, cardioversion: 5 to 15 mg IV 5-10 minutes prior to procedure.
- For status epilepticus: 5-10 mg IV; may repeat every 10-15 minutes, up to a total dose of 30 mg. May repeat initial dose in 2-4 hours. Maximum dose in 24 hours is 100 mg.
- For muscle spasm: 5-10 mg IV; may repeat in 3-4 hours.
- For seizures associated with nerve gas exposure: 5-10 mg IV; repeat as indicated by respiratory and seizure status.

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DOSAGE (Cont.)

Pediatrics:
- For sedation, muscle relaxation or anxiety: 0.04-0.3 mg/kg/dose IV q2-4h as needed to a maximum of 0.6 mg/kg in an 8-hour period. Maximum single dose: 10 mg.
- For status epilepticus:
  - If child is younger than 5 years old: 0.2-0.5 mg/dose IV every 2-5 minutes to a maximum total dose of 5 mg; repeat in 2-4 hours as needed.
  - If child is 5 years old or older: 1 mg/dose IV every 2-5 minutes to a maximum total dose of 10 mg; repeat in 2-4 hours as needed or 0.2-0.5 mg/kg/dose IV every 15-30 minutes to a maximum total dose of 10 mg.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Protect from light.
- Do not mix with any other injectable drug.
- Dilution for IV infusion is not recommended. Precipitation can occur.
- Considerable amount of diazepam is adsorbed by plastic, PVC bags, PVC tubing, catheters, and infusion sets with a burette chamber. Patients may receive variable concentrations of the drug, depending on flow rate, type of administration set used and length of tubing.
- If dilution is imperative, add dilution solution to diazepam, not diazepam to solution. Polyethylene or glass containers and polyethylene/polypropylene plastic syringes are recommended for use.
- Diazepam is temporarily soluble diluted at least 1 mg/10 mL in NS, D5W, Ringer's or RL in glass containers. Dilutions of 1:20 in NS, D5W, Ringer's or RL should be used within 4 hours. Dilutions of 1:40 in NS, D5W, Ringer's or RL should be used within 6 hours. Dilutions of 1:100 in NS, D5W or RL should be used within 24 hours. Do not use if cloudy or if precipitation present.
- Diazemuls ® may be diluted <u>only</u> in its own emulsion base (Intralipid or Nutralipid). Mixture must be used within 6 hours.

MISCELLANEOUS

- Oral route preferred when possible.
- Caution required for patients with low albumin, hepatic or renal impairment or respiratory insufficiency.
- The injectable preparation can be given rectally; it is given undiluted through a rectal cannula at a dose of 0.5 mg/kg (maximum: 20 mg).
- Diazemuls ® should not be used in the management of status epilepticus because of its delayed action.

REFERENCES

1, 4, 5, 6, 40, 82, 95, 135, 263, 367, 407, 462, 527, 534.

Full revision 2013; limited revision 2015, 2017, 2020, 2021

NAME OF MEDICATION
DIGOXIN

OTHER NAMES
Digoxin injection C.S.D., Lanoxin ®

CLASSIFICATION
Cardiac glycoside

INDICATIONS
- Treatment of congestive heart failure emergencies which cannot be controlled by oral medication, or when a rapid effect is desired.
- Control of ventricular response in supraventricular dysrhythmias.

ADMINISTRATION
- IV direct: physician; RN for doses up to 0.25 mg. Administer (undiluted or diluted at least fourfold with SWFI, NS, D5W or RL) over at least 5 minutes.
- Intermittent IV infusion: dilute dose in 50 mL of NS or D5W and infuse immediately over 5-10 minutes.
- IM: not advised due to erratic absorption, severe local irritation, and pain.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: angioedema, rash, pruritus, urticaria.
- Cardiovascular: atrial tachycardia with or without block, asystole, junctional tachycardia, AV dissociation, AV block, PR prolongation, premature ventricular contractions, ST segment depression, ventricular tachycardia, ventricular fibrillation.
- GI: nausea, vomiting, diarrhea, abdominal pain (clinical signs of digoxin toxicity).
- CNS: headache, dizziness, drowsiness, confusion, hallucinations (clinical signs of digoxin toxicity).
- Muscle weakness, fatigue.
- Blurred vision, halos, yellow or green vision (clinical signs of digoxin toxicity).
- Local reactions: extravasation can cause local irritation and sloughing.

Antidote: digoxin immune fab if severe toxicity.

DOSAGE
- IV loading dose: 50% of calculated dose given as first dose, then 25% given at 4 to 8 hours intervals for 2 doses.
  Adults: (10 years and older): 8-12 mcg/kg of lean body weight IV (usually a total dose of 0.5-1 mg).
  Pediatrics: Full term: 20-30 mcg/kg IV — 2-5 years: 25-35 mcg/kg IV
  1-24 months: 30-50 mcg/kg IV — 5-10 years: 15-30 mcg/kg IV
- IV maintenance dose: 25-35% of IV loading dose (usually 0.1-0.4 mg) given once daily. Infants and children less than 10 years of age should receive the daily maintenance dose divided q12h.
- Dosage in renal impairment:

CrCl (mL/min)	50-10	less than 10
Loading dose:	100%	50%
Maintenance dose:	25-75%	10-25%

- When switching from oral or IM to IV therapy, decrease dosage by 20-25%.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp; protect from light.
- Stable for 48 hours at room temp or refrigerated in NS or D5W at a concentration of 2.5 mcg/mL.

MISCELLANEOUS
- Hypokalemia, hypercalcemia, hypomagnesemia and hypothyroidism predispose patient to toxicity.
- Following a single IV dose, effects are noticeable in 5-30 minutes and develop fully in 1-4 hours. (Half-life of IV digoxin is 33-44 hours if renal function is normal).
- For digoxin levels, draw blood sample just prior to a dose or at least 6-8 hours after the last dose.
- To convert serum digoxin concentration from nmol/L to ng/mL, multiply by 0.78.

REFERENCES
1, 4, 6, 40, 82, 95, 135, 462.

Full revision 2013; limited revision 2015, 2021

NAME OF MEDICATION
DIGOXIN IMMUNE FAB

OTHER NAMES
DigiFab ®

CLASSIFICATION
Antidote

INDICATIONS

- For the treatment of life-threatening digoxin intoxication.

ADMINISTRATION

- Reconstitute each 40 mg vial with 4 mL SWFI to obtain a solution of approximately 10 mg/mL. Gently mix; do NOT shake. Solution should be clear and colourless.
- IV direct: physician only. ***Cardiac monitoring***. Reserve in situations where cardiac arrest is imminent.
- Intermittent IV infusion (preferred): ***cardiac monitoring***. Dilute total dose in any desired amount of NS (at TOH, dilute in 50-100 mL) and infuse over at least 30 minutes. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, anaphylaxis.
- Cardiovascular: worsening of underlying condition e.g., atrial fibrillation, congestive heart failure.
- Hypokalemia.

DOSAGE

- A 40 mg vial will neutralize approximately 0.5 mg of digoxin.
- For dosing in patients in whom amount of acute ingestion of digoxin or serum digoxin concentration is <u>unknown</u>:
  - Adults: Chronic toxicity: 6 vials are usually adequate.
    Acute toxicity: 10-20 vials are usually adequate.
  - Pediatrics: Chronic toxicity (up to 20 kg): one vial is usually sufficient.
    Acute toxicity: may consider administering 10 vials, and observe patient. If needed, another 10 vials may be administered. Watch for volume overload.

- For dosing in patients in whom amount of acute ingestion of digoxin or serum digoxin concentration is <u>known</u>, refer to calculations below:
  - Calculations when ingestion of known amount (to round up to the nearest whole vial):
    Formula 1:

$$\text{Dose (in \# vials)} = \frac{\text{number of tablets x strength of tablet x 0.8 (bioavailability)}}{\text{0.5 mg of digoxin bound/vial}}$$

  - Calculations based on steady-state serum digoxin concentrations (to round up to the nearest vial):
    Formula 2: Calculation with digoxin in **nmol/L (S.I. units):**

$$\text{Dose (in \# vials)} = \frac{\text{(serum digoxin concentration in } \mathbf{nmol/L} \text{ x 0.781) (weight in kg)}}{100}$$

  - For infants and small children weighing 20 kg or less: dose calculated above as number of vials can be translated in mg for a more precise dose to administer:
    Formula 3: Dose (in mg) = (Dose in # of vials) x (40 mg/vial)

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store unopened vials in the fridge.
- Reconstituted solutions stable for 4 hours in the fridge.
- Compatible with NS.

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MISCELLANEOUS

- Since digoxin immune Fab will interfere with digitalis assay, digoxin concentration measurement may be misleading until Fab fragments are eliminated from the body (may take several days in patients with normal renal function or a week or longer in patients with renal impairment).
- Digoxin immune Fab is derived from sheep Fab immunoglobulins fragments; patients with allergies to sheep proteins may be at higher risk for hypersensitivity reactions.
- Skin testing has not proved useful in predicting allergic response.
- To convert serum digoxin concentration from nmol/L to ng/mL multiply by 0.78.

REFERENCES

1, 2, 40, 95.

Full revision 2013; limited revision 2015, 2020

NAME OF MEDICATION
DIHYDROERGOTAMINE

OTHER NAMES

CLASSIFICATION
Ergot alkaloid

INDICATIONS

- Symptomatic relief of migraine or other types of vascular headaches.

ADMINISTRATION

- IV direct: physician only for first dose; RN can administer repeat doses. Administer at a rate of 1 mg/min.
- IM, SC (both preferred).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, facial edema, dyspnea.
- Cardiovascular: hypertension, myocardial ischemia, transient tachycardia or bradycardia.
- GI: nausea, vomiting, abdominal pain, diarrhea.
- CNS: dizziness.
- Weakness in the legs, muscle pain in the extremities, numbness and tingling of fingers and toes.
- Local reactions: edema and itching.

DOSAGE

- Migraine attack, cluster headache: 1 mg initially IV/IM/SC. May repeat q1h to a maximum of 3 mg in a 24-hour period (SC or IM route) or 2 mg in a 24-hour period (IV route); total weekly dose should not exceed 6 mg (by any route of injection).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp below 25°C. Protect ampoules from light and heat. Do not refrigerate.
- Do not use if solution becomes discoloured.

MISCELLANEOUS

- Use of antiemetics (e.g., metoclopramide 10 mg) prior to each dose is recommended due to the high incidence of nausea/vomiting.
- Contraindicated in patients with ischemic or vasospastic heart disease, coronary artery disease, peripheral vascular disease, uncontrolled hypertension, severely impaired hepatic or renal functions, pregnancy, breastfeeding, sepsis, basilar or hemiplegic migraine.
- Onset of action occurs in 15-30 minutes following IM administration and within a few minutes (usually less than 5 minutes) after IV administration. Duration of action is 3-4 hours following IM administration.

REFERENCES

1, 3, 5, 6, 40, 95, 135.

Full revision 2013

NAME OF MEDICATION
DiltiaZEM

OTHER NAMES
Cardizem ®

CLASSIFICATION
Antiarrhythmic

INDICATIONS

- Rapid conversion to sinus rhythm of paroxysmal supraventricular tachycardia (PSVT).
- Temporary control of rapid ventricular rate in atrial fibrillation or flutter.

ADMINISTRATION

- IV direct: first dose physician only. ***Cardiac monitoring; blood pressure monitoring***. Administer undiluted over 2 minutes.
- Continuous IV infusion: ***cardiac monitoring; blood pressure monitoring***. At TOH: dilute 125 mg (25 mL from a diltiazem 5 mg/mL vial) in 100 mL of D5W, NS or D5-1/2NS to obtain a final concentration of 1 mg/mL (125 mg/125 mL). For larger volumes of solution, may also dilute 250 mg (50 mL) in 250 mL of D5W, NS or D5-1/2NS (final concentration of 0.83 mg/mL) or 250 mg (50 mL) in 500 mL of D5W, NS or D5-1/2NS (final concentration of 0.45 mg/mL). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension, flushing, vasodilation, cardiac conduction blocks, heart failure, arrhythmia.
- CNS: headache, dizziness.
- Local reactions: burning and itching at injection site.

DOSAGE

- Initial bolus of 0.25 mg/kg IV over 2 minutes.
- If response inadequate after 15 minutes, may give a second bolus of 0.35 mg/kg IV over 2 minutes.
- For continued heart rate control, an infusion may be started following the bolus dose(s) at a rate of 5 mg/hr IV and titrated to effect in increments of 5 mg/hr, but not to exceed 15 mg/hr for up to 24 hours.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store unopened vials in the fridge. Do not freeze. Stable for 1 month at room temp, then should be discarded.
- Stable for 24 hours at room temp or in the fridge when diluted with D5W, NS or D5-1/2NS at concentrations up to 1 mg/mL in glass or PVC containers.

MISCELLANEOUS

- Concomitant IV diltiazem and IV beta-blockers may cause significant cardiac depression. Therefore, co-administration (within a few hours) is contraindicated.
- Maximal heart rate reduction occurs within 2-7 minutes of bolus and persists for 1-3 hours.
- Contraindicated in patients with sick sinus syndrome (except if pacemaker in place), in patients with second- or third-degree AV block (except if pacemaker in place), in patients with severe hypotension or cardiogenic shock, in patients with ventricular tachycardia, in patients with atrial fibrillation or flutter when associated with an accessory bypass tract.

REFERENCES

1, 4, 6, 40, 95.

Full revision 2013; limited revision 2014, 2015, 2017, 2019, 2020

NAME OF MEDICATION	
DimenhyDRINATE	
OTHER NAMES	CLASSIFICATION
Gravol ®	Antiemetic

INDICATIONS

- To alleviate nausea and vomiting due to surgery, pregnancy, motion sickness.
- Treatment and prevention of radiation sickness.
- Symptomatic treatment of nausea and vertigo due to Menière's disease and other vestibular disturbances.

ADMINISTRATION

- IV direct: physician or RN. If using the IM formulation (50 mg/mL), dilute each 50 mg (1 mL) with 10 mL of NS or D5W and administer slowly over 2 minutes. If using the IV formulation (10 mg/mL), may administer undiluted over 2 minutes.
- Intermittent IV infusion.
- IM, SC: use the 50 mg/mL concentration.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension, palpitations, tachycardia.
- GI: dry mouth.
- CNS: drowsiness, dizziness, headache, paradoxical CNS stimulation especially in children.
- Local reactions: pain at the injection site; undiluted solution of IM formulation given IV is irritating to the veins and may produce sclerosis.

DOSAGE

- For the treatment of nausea, vomiting, dizziness and vertigo associated with motion sickness: 50-100 mg q4-6h as needed. Maximum dose: 400 mg/24 hours.
- For nausea and vomiting due to surgery: 50 mg preoperatively; to be repeated postoperatively as needed to a maximum of 400 mg/24 hours.
- For radiation sickness: 50-100 mg 30-60 minutes before treatment; repeat as necessary after treatment. Maximum dose: 400 mg/24 hours.
- Acute attacks of vertigo due to Menière's disease: 50 mg q6h.
- Pediatrics less than 12 years of age: 5 mg/kg/day or 150 mg/m^2/day given in divided doses q6h. Maximum dose: 300 mg/day.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Stable for 7 days at room temp with exposure to light or 91 days in the fridge protected from light diluted in NS or D5W at concentrations of 0.5 mg/mL, 1 mg/mL and 2 mg/mL in PVC minibags.
- Stable for 60 days at room temp with exposure to light or in the fridge protected from light when diluted in NS at concentrations of 2.5 mg/mL, 5 mg/mL and 10 mg/mL in polypropylene syringes.
- Compatible with dextrose, saline and RL solutions.

MISCELLANEOUS

- Anti-emetic effects occur almost immediately after IV dose, 20-30 minutes after IM dose; duration of action is 3-6 hours.

REFERENCES

1, 2, 4, 6, 9, 56, 82, 95, 457.

Full revision 2013; limited revision 2015

NAME OF MEDICATION
DiphenhydrAMINE

OTHER NAMES
Benadryl ®

CLASSIFICATION
Antihistamine

INDICATIONS

- Treatment of severe allergic reactions.
- To reduce nausea and vomiting due to surgery, malignancy or antineoplastic agents.
- Prevention of allergic reactions during transfusion.
- Relief of pruritus.
- As a sedative for preoperative medication.
- Management of tremor early in parkinsonian syndrome and of drug-induced extrapyramidal reactions.

ADMINISTRATION

- IV direct (preferred): physician or RN. Administer undiluted or diluted in 10 mL with NS. Give at a maximum rate of 25 mg/min.
- Intermittent IV infusion.
- IM.
- SC (palliative care): do NOT exceed 50 mg/dose. Inject in upper arms, thighs or abdomen.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactoid reactions, rash, urticaria.
- Cardiovascular: hypotension, rapid pulse.
- CNS: sedation, dizziness, headache, paradoxical excitement.
- Anticholinergic effects (blurring of vision, constipation, difficulty in urination, dry mouth).
- Hemolytic anemia (very rare).
- Local reactions: irritation following subcutaneous or perivascular administration.

DOSAGE

- Adults: 25-50 mg q4h, with a maximum single dose of 100 mg. Maximum daily dosage: 400 mg.
- Pediatrics: 5 mg/kg/day or 150 mg/m^2/day, given in 4 divided doses. Maximum daily dosage: 300 mg.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from light as darkens on exposure to light. Do not freeze.
- Stable for 91 days at room temp or in the fridge, protected from light, diluted in NS or D5W, at concentrations of 0.25 mg/mL, 0.5 mg/mL and 1 mg/mL in PVC minibags.
- Stable for 14 days in the fridge in NS or D5W at concentrations of 0.2 mg/mL and 1 mg/mL in PVC minibags.
- Stable for 28 days at room temp or in the fridge, protected from light, diluted in NS at concentrations of 1.25 mg/mL, 2.5 mg/mL and 5 mg/mL in polypropylene syringes.
- Compatible with NS, D5W, RL.

MISCELLANEOUS

REFERENCES

1, 4, 6, 40, 82, 95, 143, 188, 432, 457, 539.

Full revision 2013; limited revision 2015, 2020, 2021

NAME OF MEDICATION
DIPHTHERIA ANTITOXIN *

OTHER NAMES

CLASSIFICATION
Antitoxin (equine)

INDICATIONS

- Treatment of diphtheria.

ADMINISTRATION

- **In patients who have previously received horse proteins, refer to Dosage section for test dose and desensitization**.
- Intermittent IV infusion (restricted to very severe cases): dilute half the dose in 250-500 mL NS and infuse slowly over 2-4 hours; the other half should be given IM.
- IM (preferred).
- SC: for test dose only.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis: urticaria, respiratory distress, vascular collapse, hypotension; serum sickness.
- GI: nausea, vomiting.
- Dry cough, hoarseness.
- Local reactions: pain, erythema and urticaria at injection site.

DOSAGE

Test dose:
- For patients who have previously received horse proteins without allergic reaction: administer 0.2 mL SC; if no allergic reaction occurs after at least 30 minutes, the remainder of the treatment dose can be given IM.
- For patients who have previously received horse proteins with allergic reactions (local or systemic): desensitize with 0.2 mL of a 1:10 dilution given SC; after 30 minutes, continue desensitization with 0.2 mL SC of undiluted solution. If no reaction after 30 minutes, the remainder of the treatment dose can be given IM.

Treatment dose:
- Pharyngeal or laryngeal disease of 48 hours duration: 250 international units/kg IM for one dose.
- Nasopharyngeal lesions: 500 international units/kg IM for one dose.
- Pretoxic diphtheria: 750-1000 international units/kg IM for one dose.
- Toxic diphtheria: 1000-2000 international units/kg IM for one dose.
- Diphtheria croup: 100 000 international units IM for one dose.
- If antitoxin is administered 3 days after onset of disease, double the dose.
- Refer to Administration section for IV administration in very severe cases.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store in the fridge.
- Once vial is opened, the solution must be used immediately.

MISCELLANEOUS

- Emergency drugs (such as epinephrine) must be immediately available for the treatement of hypersensitivity reactions.
- Appropriate antibiotic therapy (i.e., erythromycin, penicillin G) should be given simultaneously.

REFERENCES

31, 95, 176, 255.

* Available through Ontario Public Health

Full revision 2013; limited revision 2015, 2020

NAME OF MEDICATION
DIPYRIDAMOLE

OTHER NAMES
Persantine ®

CLASSIFICATION
Diagnostic agent, Platelet aggregation inhibitor

INDICATIONS

- Stress testing with Thallium-201; to induce vasodilation for myocardial perfusion imaging in the diagnosis of coronary artery disease.

ADMINISTRATION

- Intermittent IV infusion: ***cardiac monitoring***. Dilute with at least twice the injection volume with NS, 1/2NS, or D5W (i.e., each 1 mL of dipyridamole to be diluted with a minimum of 2 mL of fluid = 3 mL) for total volume of approximately 20-50 mL and infuse over 4 minutes for stress test. Parenteral aminophylline should be readily available. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity (rare): rash, urticaria, angioedema, laryngospasm, bronchospasm, anaphylactoid reactions.
- Cardiovascular: hypotension, ECG abnormalities, chest pain in patients with coronary artery disease; rare cases of myocardial ischemia, ventricular fibrillation, ventricular tachycardia.
- GI: nausea.
- CNS: headache, dizziness, seizures.
- Cerebrovascular events (rare): stroke, transient ischemic attack.
- Flushing.

Antidote: aminophylline 50-250 mg as a slow IV injection (maximum rate of 50 mg over 30 seconds). If symptoms (bronchospasm, chest pain) are not relieved, sublingual nitroglycerin may be given.

DOSAGE

- Dipyridamole Stress Testing: usual dose is 0.57 mg/kg (maximum 60 mg), infused at a rate of 0.142 mg/kg/min for 4 minutes. Thallium-201 should be injected within 5 minutes following completion of dipyridamole administration.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Do not freeze. Protect from light.

MISCELLANEOUS

- The use of oral theophylline, aminophylline or caffeine may cause a false-negative thallium imaging. Stop 36 hours before test.
- Oral dipyridamole treatment should be stopped 24 hours prior to testing, as the sensitivity of the test may be impaired.

REFERENCES

1, 2, 40, 95.

Full revision 2013; limited revision 2015, 2020

NAME OF MEDICATION
DOBUTamine

OTHER NAMES
Dobutrex ®

CLASSIFICATION
Sympathomimetic

INDICATIONS
- Used as an inotropic agent to increase cardiac output.

ADMINISTRATION
- Continuous IV infusion (mandatory): ***cardiac monitoring, continuous BP monitoring***. (Note: refer to the new definition of continuous BP monitoring on page 2 of the manual's preface). Must dilute before use. Concentration of solution will depend on dosage and fluid requirement of each patient.
- Dilute in a compatible solution to obtain a final concentration of 1 mg/mL (250 mg/250 mL or 500 mg/500 mL), 0.5 mg/mL (250 mg/500 mL or 500 mg/1000 mL) or 0.25 mg/mL (250 mg/1000 mL). At TOH, add 500 mg (40 mL from 12.5 mg/mL dobutamine vials) to 500 mL of D5W or NS to obtain a final concentration of 1 mg/mL. Maximum concentration of 5 mg/mL (250 mg/50 mL or 500 mg/100 mL). Infusion rates are titrated to response; refer to Dosage section.
- Administer into a large vein.
- Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: increased heart rate, ectopic beats, premature ventricular beats, palpitations, chest pain, angina; rarely ventricular tachycardia. Elevation of blood pressure, especially systolic pressure.
- GI: nausea, vomiting.
- CNS: headache, fever.
- Dyspnea.
- Hypokalemia.
- Tingling sensation, paresthesia, mild leg cramps.
- Local reactions: phlebitis at the infusion site.
- Extrasavation hazard: vesicant. **Antidote**: phentolamine 5-10 mg diluted with 10-15 mL of NS. Consult the document "Extravasation of noncytotoxic intravenous agents associated with tissue necrosis" in TOH Electronic Drug Formulary.

DOSAGE
- 2-20 mcg/kg/min. Doses as low as 0.5 mcg/kg/min and up to 40 mcg/kg/min have been used occasionally.
- Rate of infusion adjusted according to patient's response, as determined by heart rate, presence of ectopic activity, blood pressure, urine flow, and, where possible, measurement of central venous or pulmonary wedge pressure, and cardiac output.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Stable for 48 hours at room temp and 7 days under refrigeration in NS and D5W at a concentration of 0.25 mg/mL in PVC bags.
- Stable for 48 hours at room temp in NS and D5W at a concentration of 1 mg/mL in PVC and glass containers.
- Pink colour change may occur, due to slight oxidation of drug. No significant loss of potency over 48 hours at room temp.
- Incompatible with calcium gluconate, potassium phosphate or alkaline solutions (e.g., sodium bicarbonate 5%).

MISCELLANEOUS
- Hypovolemia should be corrected prior to use of dobutamine.
- Use with extreme caution following myocardial infarction.
- May be ineffective if beta-blockers have been given.
- Unlike dopamine, dobutamine has no selective effect on renal blood flow.
- Onset of action occurs within 2 minutes; peak plasma levels of the drug and peak effects occur within 10 minutes after starting IV infusion. The effects of the drug cease shortly after discontinuing the infusion.

REFERENCES
1, 4, 6, 40, 82, 95, 135, 143, 366, 367, 462.

Full revision 2013; limited revision 2015, 2016, 2017, 2019, 2020, 2021

NAME OF MEDICATION
DOCEtaxel

OTHER NAMES
Taxotere ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of operable node-positive, locally advanced or metastatic breast carcinoma, locally advanced or metastatic non-small cell lung cancer, metastatic ovarian cancer, recurrent and/or metastatic squamous cell carcinoma of the head and neck, and hormone-refractory metastatic prostate cancer.

ADMINISTRATION

- **Ensure premedication with oral dexamethasone has been administered; refer to Dosage section for more information**.
- Taxotere ® concentrated solution requires reconstitution as follows: allow the vials to stand at room temp for about 5 minutes if stored under refrigeration. Reconstitute vial with provided diluent (ethanol 13%) for a final concentration of 10 mg/mL. Gently rotate for 45 seconds to allow full mixture of the concentrate and diluent. Do NOT shake. Allow the solution to stand for 5 minutes to allow most of the foam to dissipate; this premix solution should be clear.
- Accord, Pfizer and Sandoz brands of docetaxel solution are ready for dilution and do not need a prior reconstitution with a solvent.
- Intermittent IV infusion: dilute with 250 mL of NS or D5W for a final concentration of 0.3-0.74 mg/mL. Do NOT exceed the 0.74 mg/mL concentration; if a dose larger than 200 mg is required, increase the volume accordingly. Rotate the bag gently and thoroughly. Infuse over 60 minutes. Small doses have been given over less time.
- To prevent leaching of DEHP plasticizer from containers, prepare solutions in non-DEHP containers and administer using non-DEHP administration sets.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis (rare); rash, pruritus, flushing, chest tightness, back pain, dyspnea, drug fever, chills; most likely to occur during the first two cycles of docetaxel treatment, within the first few minutes after the infusion is started. Hypotension, bronchospasm or general rash/erythema are more severe reactions and may request stopping of the infusion for further assessment.
- Cardiovascular: arrythmia, cardiotoxicity, hypertension.
- GI: nausea, vomiting, diarrhea, stomatitis, taste perversion.
- Dermatologic: desquamation, hand and foot syndrome.
- Hematologic: febrile neutropenia, leukopenia, anemia, thrombocytopenia.
- Neurologic: paresthesia, pain, burning sensation.
- Fluid retention (may need diuretic therapy): peripheral edema, pleural and pericardial effusion, ascites.
- Local reactions: hyperpigmentation, swelling, erythema, dry skin, phlebitis.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- **Premedicate all patients with oral dexamethasone in order to reduce incidence/severity of fluid retention and also to reduce the severity of hypersensitivity reactions. For docetaxel regimens given every 3 weeks: dexamethasone 8 mg BID for 3 days, starting one day prior to treatment (except for prostate cancer where premedication should be dexamethasone 8 mg to take 12 hours, 3 hours and 60 minutes before docetaxel infusion). For weekly doxetacel regimens: dexamethasone 4-8 mg BID for 3 doses, starting 12 hours before docetaxel infusion (except for prostate cancer where premedication should be dexamethasone 8 mg 60 minutes before docetaxel infusion). May also replace the multiple dose oral regimen of dexamethasone by a single 20 mg IV of dexamethasone administered just before the docetaxel infusion.**

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DOSAGE (Cont.)

- 75-100 mg/m^2 IV every 3 weeks.
- Other possible dosage for breast cancer: 35-40 mg/m^2 IV every week for 3 weeks (4-week cycles) or every week for 6 weeks (8-week cycles).
- Dosage in hepatic impairment: consult manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge or at room temp and protect from bright light.
- Taxotere ® reconstituted solution (10 mg/mL) is stable for 21 days in the fridge or at room temp.
- Pfizer brand multi-dose vials can be punctured up to 3 times in a 28 day period when kept in the fridge.
- Stable for 28 days at room temp or in the fridge and protected from light in D5W or NS at 0.3 to 0.9 mg/mL in glass or polypropylene containers.
- Stable for 56 days at room temp or in the fridge and protected from light in D5W or NS at 0.3 mg/mL and 0.7 mg/mL in polyolefin containers.
- Stable for 35 days at room temp in NS at 0.4 mg/mL and 0.8 mg/mL in polypropylene-polyethylene copolymer bags.
- Manufacturers recommend to complete infusion within 4 hours of preparation.

MISCELLANEOUS

REFERENCES

1, 5, 6, 40, 95, 129, 143, 165, 208, 308, 454, 495.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION	
DOPamine	
OTHER NAMES	CLASSIFICATION
	Sympathomimetic

INDICATIONS
- To correct hemodynamic imbalances in shock-like status (e.g., myocardial infarction, trauma, endotoxic septicemia, open-heart surgery).
- To increase renal blood flow and urine output.
- To improve cardiac output and stroke volume in severe congestive heart failure (CHF).

ADMINISTRATION
- Continuous IV infusion (mandatory): ***cardiac monitoring; continuous BP monitoring***. Available only as premixed bags of 800 mcg/mL, 1600 mcg/mL and 3200 mcg/mL in D5W. Infusion rates are titrated to response; refer to Dosage section.
- Preferable to administer via a central line; however dopamine may be administered through a large peripheral vein (i.e., antecubital vein).
- Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: ectopic beats, tachycardia, anginal pain, palpitation, vasoconstriction, hypotension, cardiac conduction abnormalities, widened QRS complex, bradycardia, hypertension. Ventricular arrhythmias may occur with very high doses.
- GI: nausea, vomiting.
- CNS: headache, anxiety.
- High doses and infusion rates that are too rapid can result in hypertension and reduced urine output.
- Endocrine and metabolic: hypoprolactinemia, hyperglycemia (since premixed with D5W).
- Respiratory: bronchospasm, dyspnea.
- Renal: azotemia.
- Piloerection.
- Local reactions: ischemia, can lead to gangrene.
- Extravasation hazard: vesicant. **Antidote**: phentolamine 5-10 mg diluted with 10-15 mL of NS. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE
- 1-5 mcg/kg/min IV initially with gradual increases of 1-4 mcg/kg/min IV at 10-30 minute intervals up to 20-50 mcg/kg/min IV if required. An initial dose of 5-10 mcg/kg/min IV may be needed to correct hypotension in seriously ill patients. Infusion rate should be adjusted according to patient’s response.
- Infusion rates greater than 50 mcg/kg/min IV have been used in patients with advanced states of circulatory decompensation.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store bags at room temp. Protect from light. Do not freeze.
- Inactivated in alkaline media (e.g., sodium bicarbonate solutions), by iron salts and oxidizing agents.
- Solutions darker than slightly yellow or discoloured should not be used.
- Stable for 7 days at room temp after removal of manufacturer’s protective overwrap.

MISCELLANEOUS
- Contains sulfites.
- Contraindicated in patients with pheochromocytoma.

REFERENCES
1, 4, 5, 6, 40, 95, 135, 143, 366, 367, 462.

Full revision 2013; limited revision 2015, 2016, 2017, 2020, 2021

NAME OF MEDICATION	
DOXOrubicin	
OTHER NAMES	CLASSIFICATION
Adriamycin ®	Antineoplastic

Do NOT confuse doxorubicin with liposomal doxorubicin or with pegylated liposomal doxorubicin. This monograph is specific to DOXORUBICIN.

INDICATIONS

- Treatment of acute lymphocytic leukemia, acute myeloblastic leukemia, bladder cancer, breast cancer, bronchogenic cancer, endometrial cancer, gastric cancer, head and neck cancer, lung cancer, lymphoma, neuroblastoma, osteogenic sarcoma, ovarian and gynaecologic cancer, soft tissue sarcoma, squamous cell carcinoma of the head and neck, testicular cancer, thyroid cancer, and Wilms' tumour.
- Other uses include adrenocortical cancer, carcinoid syndrome (small bowel), chronic lymphocytic leukemia, Ewing's sarcoma, gynecological sarcoma, hepatic cancer, islet cell cancer, mesothelioma, multiple myeloma, pancreatic cancer, prostate cancer, retinoblastoma, rhabdomyosarcoma, thymoma and urothelial carcinoma.

ADMINISTRATION

- Reconstitute 10 mg vial with 5 mL, 50 mg vial with 25 mL and 150 mg vial with 75 mL of SWFI, D5W or NS to obtain approximately 2 mg/mL. Do not reconstitute with diluents containing preservatives. Shake the vial and allow to dissolve. The solution is clear, red to red-orange.
- Also available as a premixed solution (2 mg/mL).
- IV direct: physician or RN; undiluted; inject over 3-10 minutes into the tubing of a freely running IV solution of D5W or NS, preferably via a butterfly needle into a large vein.
- Continuous IV infusion: dilute large doses in 50-1000 mL of D5W or NS and infuse via a central line as per protocol.
- Aluminium-containing apparatus should not be used in the preparation or administration of doxorubicin as the solution could darken and precipitate.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis (rare), urticaria.
- Infusion-related reactions: rash, fever, chills.
- Cardiovascular: transient arrhythmias (can be fatal), abnormal ECG findings, bradycardia, AV and bundle-branch block. Delayed cumulative dose-dependent cardiomyopathy can be reduced with the administration of dexrazoxane.
- GI: acute nausea, vomiting, anorexia, diarrhea, mucositis, stomatitis, esophagitis, typhlitis, hemorrhage, necrotizing colitis, hyperpigmentation of the oral mucosa.
- Dermatologic: facial flushing with too rapid IV injection; photosensitivity; radiation recall reaction (rare).
- Hematologic: myelosuppression, leukopenia, neutropenia, thrombocytopenia, anemia.
- Hepatic: liver dysfunction, increased LFT.
- Hyperuricemia secondary to tumour lysis syndrome; can be minimized with allopurinol and hydration.
- Renal: red discolouration of urine for 1-2 days.
- Local reactions: urticaria, phlebitis, pain at injection site, cellulitis (can be severe), streaking along the vein proximal to injection site with too rapid IV injection, phlebosclerosis if small vein is used or with repeated injections in same vein.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- Common monotherapy dose: 60-75 mg/m^2 IV every 3 weeks. The lower dose should be given to patients with inadequate marrow reserves due to old age, prior therapy or neoplastic marrow infiltration, as well as in the obese OR 20-30 mg/m^2 IV on each of 3 successive days repeated every 4 weeks (causes more stomatitis) OR 10-20 mg/m^2 IV weekly (causes less cardiotoxicity).
- Common dose in combination chemotherapy: 40-60 mg/m^2 IV every 3 to 4 weeks.
- Due to risk of cardiotoxicity, maximum cumulative dose is 550 mg/m^2 for 3 week cycle and 700 mg/m^2 for 1 week cycle in patients with no cardiac risk factor. Maximum cumulative dose is 400 mg/m^2 if previous or concomitant therapy, cardiotoxic agents or other risk factors.
- Dosage in hepatic impairment:

Bilirubin (mcmol/L)	% usual dose
21-51	50%
52-85	25%
greater than 85	Do not administer

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Premixed solution should be stored in the fridge and protected from light. Refrigeration can lead to the formation of a gel which will return to a slightly viscous to a mobile solution after keeping 2 to 4 hours at room temp. Stable at room temp for 7 days.
- Store vials that require reconstitution at room temp. Protect from light.
- Reconstituted solution in NS is stable for 7 days at room temp, and 15 days in the fridge.
- Reconstituted solution in SWFI is stable for 6 months when refrigerated, and for at least 30 days frozen at -20°C.
- Reconstituted solution in D5W is stable for 4 days at room temp.
- Stable for 43 days at -20°C, 4°C and room temp in NS or D5W at a concentration of 0.1 mg/mL when stored in PVC bags in the dark.
- Stable for 28 days in the fridge and at room temp and for 7 days at 35°C, protected from light in D5W or NS at a concentration of 1.25 mg/mL in ethylene vinyl acetate pump reservoir.
- Stable for 124 days at room temp and in the fridge in NS at 1 mg/mL and 2 mg/mL in plastic syringes.
- Compatible with RL.
- Incompatible with fluorouracil, heparin and alkaline solutions.

MISCELLANEOUS

REFERENCES

1, 4, 5, 6, 40, 95, 129, 135, 143, 165, 202, 208.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021

NAME OF MEDICATION
DOXOrubicin (liposomal)

OTHER NAMES
Myocet ™

CLASSIFICATION
Antineoplastic

Do NOT confuse liposomal doxorubicin with doxorubicin or pegylated liposomal doxorubicin. This monograph is specific to LIPOSOMAL doxorubicin.

INDICATIONS

- Metastatic breast cancer (in combination with cyclophosphamide).

ADMINISTRATION

- For reconstitution details, consult package insert.
- Intermittent IV infusion (mandatory): dilute in NS or D5W to a concentration equal or greater than 0.04 mg/mL. Infuse over 60 minutes piggyback into a running IV line of NS or D5W.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Infusion-related reactions: flushing, dyspnea, fever, facial swelling, headache, back pain, chills, tightness in chest and throat, hypotension. May be avoided by slowing infusion rate.
- Cardiovascular: arrhythmia, chest pain, pericardial effusion. Delayed cumulative dose-dependent cardiomyopathy, although incidence may be lower than with conventional doxorubicin.
- GI: nausea, vomiting, diarrhea, mucositis, stomatitis, anorexia.
- CNS: dizziness, malaise.
- Dermatologic: rash, radiation recall reaction.
- Hematologic: myelosuppression, neutropenia, thrombocytopenia.
- Fatigue, asthenia.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 60 to 75 mg/m^2 IV given every 3 weeks.
- Due to risk of cardiotoxicity, maximum cumulative dose is 750 mg/m^2 (including previous conventional doxorubicin).
- Dosage in hepatic impairment: if bilirubin is normal though ALT/AST are elevated: 75% of the usual dose.

Bilirubin (mcmol/L)	% usual dose
21-51	50%
greater than 51	25% or avoid

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge (2-8°C). Do not freeze.
- Diluted product should be a red-orange, opaque, homogenous dispersion.
- Diluted solutions are stable for 8 hours at room temp or 72 hours in the fridge.

MISCELLANEOUS

- Should not be administered to a patient allergic to eggs or egg products.

REFERENCES

5, 165.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2019, 2020

NAME OF MEDICATION
DOXOrubicin (pegylated liposomal)

OTHER NAMES
Caelyx ®

CLASSIFICATION
Antineoplastic

Do NOT confuse pegylated liposomal doxorubicin with doxorubicin or liposomal doxorubicin. This monograph is specific to PEGYLATED LIPOSOMAL doxorubicin.

INDICATIONS
- AIDS-related Kaposi's sarcoma.
- Metastatic breast cancer.
- Advanced ovarian cancer.

ADMINISTRATION
- Intermittent IV infusion (mandatory): for doses less than 90 mg, dilute in 250 mL of D5W. For doses of 90 mg or above, dilute in 500 mL of D5W. The infusion should be connected through a side port of a D5W IV infusion. For Kaposi's sarcoma, administer over 30 minutes. For ovarian and breast cancer, initial maximum infusion rate of 1 mg/min recommended to minimize the risk of infusion-related reactions. In case of such reactions, administer the dose over 90 minutes as 5% of the dose over 15 minutes; if tolerated, then double the rate for the next 15 minutes; if still tolerated, complete the infusion over the next 60 minutes. If no infusion-related reactions, administer over 60 minutes for subsequent doses. Avoid rapid flushing of the infusion line.
- **Do NOT use with any filter.**

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid reactions.
- Infusion-related reactions: may occur within minutes of start of infusion; asthma, uriticarial rash, chest pain, fever, hypertension, tachycardia, pruritus, sweating, flushing, shortness of breath, facial swelling, headache, chills, back pain, tightness in chest and throat, hypotension, vasodilation, dyspepsia, nausea, dizziness, pharyngitis, convulsions. Usually associated with first infusion and may resolve by temporarily stopping the infusion. In rare cases, may need to administer antihistamines, corticosteroids, epinephrine and anticonvulsants.
- Cardiovascular: hypotension, tachycardia, ventricular arrhythmia. Delayed cumulative dose-dependent cardiomyopathy, although incidence is lower than with conventional doxorubicin.
- GI: nausea, vomiting, anorexia, stomatitis, constipation, dysgeusia.
- CNS: confusion, dizziness, seizure, vertigo.
- Dermatologic: rash, radiation recall reaction; palmar-plantar erythrodysesthesia (pyridoxine and corticosteroids can be used prophylactically for subsequent doses).
- Hematologic: myelosuppression, leukopenia, anemia, thrombocytopenia, neutropenia.
- Respiratory: dyspnea.
- Asthenia, fatigue.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- Kaposi's sarcoma: 20 mg/m^2 IV every 2-3 weeks.
- Ovarian and breast cancer: 50 mg/m^2 IV every 4 weeks.
- Dosage in hepatic impairment:

Bilirubin (mcmol/L)	% of usual dose for Kaposi's sarcoma	* % of initial usual dose for breast or ovarian cancer
21-51	50%	75%
greater than 51	25%	50%

* Subsequent doses may be increased depending on liver function tests.

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…/Cont.

DOSAGE (Cont.)

- Subsequent doses may be modified according to toxicities.
- Due to the risk of cardiotoxicity, maximum cumulative dose of doxorubicin is 550 mg/m^2 in patients with no risk factors or 400 mg/m^2 if previous or concomitant therapy or cardiotoxic agents.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge.
- Diluted solutions are stable for 24 hours in the fridge.
- The red liposomal dispersed product is visible in the solution.
- Do not use with any diluent other than D5W. Do not use any bacteriostatic agent (e.g., benzyl alcohol).

MISCELLANEOUS

REFERENCES

1, 4, 5, 6, 40, 129, 165.

Full revision 2013; limited revision 2014, 2015, 2016, 2017, 2018, 2019, 2020

NAME OF MEDICATION
DOXYCYCLINE *

OTHER NAMES
Vibramycin ®

CLASSIFICATION
Antibiotic – tetracycline

INDICATIONS

- For infections due to sensitive organisms when oral administration is not possible.
- Acute Pelvic Inflammatory Disease (chlamydia, gonorrhea), in combination with cefoxitin.
- Treatment of syphilis.
- Treatment of anthrax, plague or tularemia in the context of bioterrorism.
- Treatment of severe malaria caused by *P. falciparum*, in combination with quinine or quinidine.

ADMINISTRATION

- Reconstitute the 100 mg vial with 10 mL of SWFI, NS, D5W, Ringer's or RL.
- Intermittent IV infusion: further dilute each 100 mg with 100 mL to 1000 mL of NS, D5W, Ringer's or RL to obtain a concentration of 0.1-1 mg/mL. Infuse over at least 60 minutes if final concentration is 0.5 mg/mL and over 2 hours if final concentration is 1 mg/mL. May be given over 4 hours. Protect from direct sunlight during infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: maculopapular and erythematous rash, anaphylaxis, urticaria, angioneurotic edema, anapylactoid purpura, pericarditis.
- Cardiovasular: intracranial hypertension.
- GI: nausea, vomiting, diarrhea, anorexia, dysphagia, enterocolitis. Anogenital lesions.
- CNS: dizziness, lightheadedness.
- Dermatologic: photosensitivity.
- Hematologic: hemolytic anemia, thrombocytopenia, neutropenia, eosinophilia.
- Renal: dose-related increase in urea.
- Local reactions: thrombophlebitis with prolonged use.

DOSAGE

- Adults and pediatrics weighing over 45 kg:
  - 200 mg IV on day one given in one or in two divided infusions followed by 100 mg IV daily from day 2 onward. In severe infections, the daily dose may be increased to 200 mg IV (given in one or two infusions).
  - Anthrax: 100 mg IV q12h for at least 60 days; ADD 1 or 2 additional antimicrobials to which *B. anthracis* is likely susceptible (rifampin, vancomycin, penicillin, ampicillin, chloramphenicol, imipenem, clindamycin and clarithromycin).
  - Plague: 100 mg IV q12h or 200 mg IV once daily for 10-14 days.
  - Tularemia: 100 mg IV q12h for 14-21 days.
  - Syphilis: 100-150 mg IV q12h for at least 10 days.
  - Pelvic inflammatory disease: 100 mg IV q12h for 14 days.
  - Malaria: 100 mg IV q12h for 7 days.

- Pediatrics (45 kg or less):
  - The usual dose is 4.4 mg/kg IV on day one given in one in or two divided infusions followed by 2.2-4.4 mg/kg/day IV given in one or two infusions from day 2 onward.
  - Anthrax, plague and tularemia: 2.2 mg/kg IV q12h.
  - Malaria: 2.2 mg/kg IV q12h for 7 days.

…/Cont.

NAME OF MEDICATION
DOXYCYCLINE *

OTHER NAMES
Vibramycin ®

CLASSIFICATION
Antibiotic – tetracycline

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Stable for 72 hours in the fridge protected from light in NS, D5W or in Ringer's at a concentration of 0.1-1 mg/mL. Once removed from the fridge, completion of infusion must be done within 12 hours.
- Stable for 48 hours at room temp protected from direct sunlight in NS or D5W at a concentration of 0.1-1 mg/mL.
- Stable for 6 hours at room temp protected from direct sunlight in RL at a concentration of 0.1-1 mg/mL.
- Stable for 12 hours at room temp protected from direct sunlight in Ringer's at a concentration of 0.1-1 mg/mL.

MISCELLANEOUS

- Change to oral formulation when feasible.
- Can cause permanent discolouration of teeth if used in children under 8 years of age.
- Check expiry date on vials as outdated product can cause nephrotoxicity.

REFERENCES

1, 4, 5, 40, 82, 143.

* Available via Health Canada's Special Access Programme

Full revision 2013; limited revision 2015, 2020

NAME OF MEDICATION
DROPERIDOL

OTHER NAMES

CLASSIFICATION
Antiemetic

INDICATIONS

- Prevention and treatment of postoperative nausea and vomiting in patients for whom other treatments are ineffective or inappropriate.

ADMINISTRATION

- IV direct: physician or RN; ***cardiac monitoring****, ***blood pressure monitoring***. Administer slowly over 1-2 minutes. Pediatrics: administer over 2-5 minutes.
- Intermittent IV infusion: ***cardiac monitoring*****.** Further dilute and administer slowly at a rate no faster than those as IV direct.

* All patients should undergo a 12-lead ECG prior to administration of droperidol to determine if a prolonged QTc interval (i.e., QTc longer than 440 msec for males and 450 msec for females) is present; do NOT administer if there is a prolonged QT interval. Cardiac monitoring should start with treatment and continued for 2 to 3 hours after completing treatment.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis.
- Cardiovascular: hypotension and tachycardia with rapid administration; QT interval prolongation, torsades de pointes, ventricular arrhythmias, cardiac arrest, hypertension; severe hypotension in postoperatively period: do not use epinephrine as it may further lower BP, use norepinephrine or phenylephrine instead.
- CNS: anxiety, somnolence, delirium, hallucinations, nightmares, dizziness, depression, restlessness.
- Respiratory: respiratory depression or arrest, apnea, bronchospasm, laryngospasm.
- Shivering, chills, muscle rigidity, extrapyramidal reactions, neuroleptic malignant syndrome.

DOSAGE

- Adults: 0.625 to 1.25 mg IV; elderly: 0.625 mg IV.
- Pediatrics: older than 2 years of age and adolescents: 20-50 mcg/kg IV, up to a maximum of 1.25 mg IV.
- Can be repeated at least 6 hours later as needed; administration of subsequent doses should be done with caution.
- Dosage in renal or hepatic impairment (adults): 0.625 mg IV.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Stable for 7 days at room temp in NS or D5W at a concentration of 20 mg/L in PVC and glass containers.
- Stable for 7 days at room temp in RL at a concentration of 20 mg/L in glass containers.
- Stable for 24 hours at room temp in RL at a concentration of 20 mg/L in PVC containers.
- Stable for 28 days at room temp protected from light undiluted (2.5 mg/mL) in polypropylene syringes.
- Stable for 180 days at room temp protected from light in NS at a concentration of 0.625 mg/mL in polypropylene syringes.

MISCELLANEOUS

- May reduce pulmonary arterial pressure and interferes with hemodynamic monitoring.
- Potentiates the effects of opioids, barbiturates, tranquillizers, benzodiazepines, general anaesthetics, and alcohol.

REFERENCES

1, 4, 5, 40, 82, 95, 135, 203, 214.

Full revision 2013; limited revision 2015, 2020

NAME OF MEDICATION
DURVALUMAB

OTHER NAMES
Imfinzi ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS
- Treatment of locally advanced or metastatic urothelial carcinoma.
- Treatment of locally advanced, unresectable, stage III non-small cell lung cancer (NSCLC).
- Treatment of extensive-stage small cell lung cancer (ES-SCLC).

ADMINISTRATION
- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards section.**
- Intermittent IV infusion: dilute in NS or D5W to obtain a final concentration of 1 to 15 mg/mL. Mix by gently inverting bag; do NOT shake. Administer through a sterile, low-protein-binding 0.2 or 0.22 micron in-line filter over 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication**: Consult policy ADM VIII 860 (High alert medications).
- Hypersensitivity: immune-mediated reactions, can be fatal; can involve any organ. Most common reactions include pneumonitis, hepatitis, colitis, endocrinopathies (hyper/hypothyroidism, adrenal insufficiency, hypophysitis/hypopituitarism, type 1 diabetes mellitus), nephritis, rash, pruritus, dermatitis; less common reactions include meningitis, myocarditis, myositis, thrombocytopenia, hemolytic anemia. Consult manufacturer's recommendations for management of these reactions and when to withhold or discontinue treatment.
- Infusion-related reactions: fever, chills, flushing, rash, pruritus, dizziness, angioedema, dyspnea, back or neck pain. For mild reactions (grade 1 or 2), interrupt infusion until resolution of reaction or slow infusion rate by half; **consider premedication** for subsequent infusions (e.g., diphenhydramine 50 mg PO, acetaminophen 325 mg to 975 mg PO, hydrocortisone 25 mg IV given 30 minutes prior to infusion). For more severe reactions (grade 3 or 4), permanently discontinue durvalumab.
- Cardiovascular: peripheral edema, chest pain, hypertension.
- GI: nausea, vomiting, diarrhea, constipation, abdominal pain, anorexia.
- CNS: dizziness, headache, paresthesia.
- Electrolyte disturbances: hypo/hyperkalemia, hyponatremia, hypermagnesemia, hypo/hypercalcemia.
- Endocrine and metabolic: hyperglycemia, hypoalbuminemia.
- Hematologic: anemia, neutropenia, lymphopenia.
- Hepatic: increased LFTs, hyperbilirubinemia.
- Musculoskeletal: back pain, arthralgia, myalgia, pain in extremities.
- Renal: increased creatinine.
- Respiratory: cough, dyspnea, dysphonia.
- Fever, fatigue, asthenia, night sweats.
- Infections.

DOSAGE
- Urothelial carcinoma: 10 mg/kg IV every 2 weeks.
- NSCLC: 10 mg/kg IV every 2 weeks OR 20 mg/kg IV every 4 weeks.
- ES-SCLC:
  - Patients weighing more than 30 kg: 1500 mg IV every 3 weeks for 4 cycles in combination with chemotherapy (etoposide and carboplatin or cisplatin) then 1500 mg IV every 4 weeks as monotherapy until disease progression or unacceptable toxicity.
  - Patients weighing 30 kg or less: as per above regimen, but use a dose of 20 mg/kg IV (instead of 1500 mg).
- Dosage in renal impairment: no dose adjustment needed for patients with CrCl of 30 mL/min or greater; no data available if CrCl is less than 30 mL/min.
- Dosage in hepatic impairment: no dose adjustment needed for patients with mild hepatic impairment (bilirubin less than or equal to ULN and AST greater than ULN OR bilirubin greater than 1-1.5 x ULN and any AST); no data available for moderate to severe hepatic impairment.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not shake. Do not freeze.
- Stable for 24 hours in fridge and 12 hours at room temp in NS or D5W. This stability includes the time of vial puncture to the start of the infusion.

MISCELLANEOUS

REFERENCES

1, 5, 129, 135, 165.

New monograph 2020; limited revision 2021

NAME OF MEDICATION	
ECULIZUMAB	
OTHER NAMES Soliris ®	CLASSIFICATION Antihemolytic, Immunomodulator, Monoclonal antibody

INDICATIONS

- Treatment of paroxysmal nocturnal hemoglobinuria (PNH) to reduce hemolysis.
- Treatment of atypical hemolytic uremic syndrome (HUS) to reduce/inhibit complement-mediated thrombotic microangiopathy.
- Treatment of generalized myasthenia gravis (gMG) in adults.
- Treatment of neuromyelitis optica spectrum disorder (NMOSD).

ADMINISTRATION

- Intermittent IV infusion (mandatory): withdraw the required volume from vial(s) for the dose (i.e., 30 mL for a 300 mg dose, 60 mL for a 600 mg dose, 90 mL for a 900 mg dose, or 120 mL for a 1200 mg dose); transfer the dose to an empty infusion bag and further dilute with D5W, NS, 1/2NS or Ringer's injection to obtain a final concentration of 5 mg/mL (since the concentration in the vials is 10 mg/mL, the volume of diluent to add in the bag is equal to the total volume of eculizumab for the dose as indicated above). Gently invert the bag to mix. Do NOT shake. Allow the diluted solution to reach room temp before administration. Infuse over 35 minutes in adults (do NOT exceed 2 hours if rate is slowed due to infusion reaction) and over 1-4 hours in pediatrics. Must be administered by an infusion pump.
- Monitor patient for at least 60 minutes following completion of infusion for signs and symptoms of hypersensitivity and infusion reaction.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, anaphylaxis.
- Infusion-related reactions: cardiovascular instability, respiratory compromise; stop the infusion in such case.
- Cardiovascular: hypertension, hypotension, tachycardia, peripheral edema.
- GI: nausea, vomiting, diarrhea, abdominal pain, gastroenteritis.
- CNS: headache, insomnia.
- Hematologic: anemia, leukopenia.
- Infections: nasopharyngitis, sinusitis, upper respiratory tract infections, urinary tract infections, herpes simplex infections, meningococcal infections (even if patient has been vaccinated).
- Respiratory: cough, nasal congestion.
- Back pain, arthralgia, myalgia, weakness.
- Fatigue, fever.

DOSAGE

Adults:
- PNH: induction: 600 mg IV weekly for 4 doses; maintenance: 900 mg IV at week 5, then 900 mg IV every 2 weeks.
- Atypical HUS, gMG and NMOSD: induction: 900 mg IV weekly for 4 doses; maintenance: 1200 mg IV at week 5, then 1200 mg IV every 2 weeks.

Pediatrics (2 months to less than 18 years old):
- Atypical HUS:
  - Weighing 5 kg to less than 10 kg: induction: 300 mg IV once; maintenance: 300 mg IV at week 2, then 300 mg IV every 3 weeks.
  - Weighing 10 kg to less than 20 kg: induction: 600 mg IV once; maintenance: 300 mg IV at week 2, then 300 mg IV every 2 weeks.
  - Weighing 20 kg to less than 30 kg: induction: 600 mg IV weekly for 2 doses; maintenance: 600 mg IV at week 3, then 600 mg IV every 2 weeks.
  - Weighing 30 kg to less than 40 kg: induction: 600 mg IV weekly for 2 doses; maintenance: 900 mg IV at week 3, then 900 mg IV every 2 weeks.
  - Weighing 40 kg and above: same regimen as adults.

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DOSAGE (Cont.)

For adults and pediatrics:
- Administer doses at these time points or within 2 days of these time points.
- Supplemental doses are required when there is concomitant support with plasmapheresis/plasma exchange or fresh frozen plasma infusion in order to maintain therapeutic plasma concentrations:
  - Plasmapheresis/plasma exchange: if most recent dose is 300 mg: administer 300 mg IV within 60 min after each session. If most recent dose is 600 mg or more: administer 600 mg IV within 60 min after each session.
  - Fresh frozen plasma infusion: if most recent dose is 300 mg or more: administer 300 mg IV 60 minutes prior to each fresh frozen plasma infusion.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Vials kept in their original carton are stable for a maximum of 3 days during a single excursion at room temp, not exceeding 25°C.
- Diluted solution is stable for 24 hours at room temp and in the fridge.

MISCELLANEOUS

- Administer vaccinations for the prevention of infection due to *Neisseria meningitidis* (serotypes A, C, Y, W-135 and B), *Streptococcus pneumoniae* and *Haemophilus influenzae* type b (Hib), according to the National Advisory Committee on Immunization (NACI) guidelines. If eculizumab is started in a patient less than 2 years of age or less than 2 weeks after the administration of meningococcal vaccine, prophylactic antibiotic therapy should be administered until 2 weeks after vaccination.
- Continue established anticoagulant therapy during eculizumab therapy.
- Monitor patients with PNH for at least 8 weeks after discontinuation of eculizumab to detect serious hemolysis.
- Monitor patients with atypical HUS for at least 12 weeks after discontinuation of eculizumab for signs and symptoms of thrombotic microangiopathy complications.
- In patients with PNH, serum LDH levels may be used to monitor response to therapy.
- In patient with atypical HUS, platelet counts, serum LDH and creatinine levels may be used to monitor response to therapy.

REFERENCES

1, 5, 40, 135.

New monograph 2018; limited revision 2019, 2020

NAME OF MEDICATION
EDARAVONE

OTHER NAMES
Radicava ™

CLASSIFICATION
Neuroprotector

INDICATIONS

- To slow the loss of function in adult patients with amyotrophic lateral sclerosis (ALS).

ADMINISTRATION

- Check the oxygen indicator on the overwrapped package prior to use: it should be pink. Do NOT use if the oxygen indicator has turned blue or purple before opening the package.
- Intermittent IV infusion (mandatory): administer each 60 mg dose as two consecutive 30 mg/100 mL premixed IV infusion bags; infuse total dose over 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: redness, wheals, erythema multiforme, anaphylaxis (urticaria, hypotension, dyspnea).
- CNS: gait disturbance, headache.
- Dermatologic: dermatitis, eczema, tinea infection.
- Endocrine and metabolic: glycosuria.
- Hematologic: hematoma.
- Musculoskeletal: back pain, myalgia.
- Respiratory: dyspnea, hypoxia, respiratory failure.

DOSAGE

- Initial treatment cycle: 60 mg IV once daily for 14 days, followed by a 14-day drug-free period.
- Subsequent treatment cycles: 60 mg IV once daily for 10 out of 14 days, followed by a 14-day drug-free period.
- Dosage in renal impairment: no dosage adjustment required.
- Dosage in hepatic impairment: no dosage adjustment required for mild to moderate hepatic impairment; no data available for severe hepatic impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 15-30º C in overwrapped package. Protect from light.
- Use within 24 hours once the overwrap package is opened. The overwrap package contains an oxygen indicator and an oxygen absorber to minimize oxidation.

MISCELLANEOUS

- Contains sulfites.

REFERENCES

1, 5, 135.

New monograph 2019; limited revision 2020

NAME OF MEDICATION
EDROPHONIUM

OTHER NAMES
Enlon ®, Tensilon ®

CLASSIFICATION
Cholinesterase inhibitor

INDICATIONS

- Diagnostic agent in suspected cases of myasthenia gravis (MG).
- Assessment of dosage adequacy of oral anticholinesterase therapy in MG.
- Differentiation of cholinergic and myasthenic crisis.
- Reversal of the effects of nondepolarizing neuromuscular blocking agents (e.g., pancuronium, vecuronium).
- Differential diagnosis or slowing of supraventricular tachyarrhythmias.
- Treatment of paroxysmal atrial tachycardia.
- Evaluation of the function of demand pacemakers.

ADMINISTRATION

- IV direct: physician only. May be given undiluted.
  - For indications related to MG: refer to Dosage section.
  - For reversal of the effects of nondepolarizing neuromuscular blocking agents: single dose over 30-45 seconds.
  - For cardiac uses: ***cardiac monitoring***. Administer as a bolus injection.
- Continuous IV infusion: ***cardiac monitoring***. Refer to Dosage section. Must be administered by an infusion pump.
- IM, SC (when IV administration is not possible).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis.
- Cardiovascular: bradycardia, arrhythmia, hypotension, cardiac arrest.
- GI: hypersalivation, nausea, vomiting, diarrhea, abdominal cramps, dysphagia, increased peristalsis.
- CNS: seizures, dysarthria, dysphonia.
- Ophthalmic: lacrimation, miosis, spasm of accommodation, diplopia, conjunctival hyperemia.
- Renal: urinary frequency and incontinence.
- Respiratory: bronchospasm, increased tracheobronchial secretions, laryngospasm, respiratory depression, respiratory arrest.
- Muscle weakness, cramps and fasciculations, diaphoresis.
- Local reactions: thrombophlebitis with IV administration.

Antidote: atropine.

DOSAGE

- Diagnosis of MG:
  IV:
  - Adults: draw 10 mg in a syringe; give 2 mg IV over 15-30 seconds; if no reaction in 45 seconds, give the remaining 8 mg. After 30 minutes, the test may be repeated if necessary.
  - Children weighing more than 34 kg: give 2 mg IV over 15-30 seconds; if no reaction in 45 seconds, give 1 mg IV every 30-45 seconds OR 0.04 mg/kg or 1.2 mg/m^2 IV in 60 seconds, and followed by 0.16 mg/kg or 4.8 mg/m^2 IV if no response occurs within 45 seconds. Maximum total dose of 10 mg IV.
  - Children weighing 34 kg or less: give 1 mg IV over 15-30 seconds; if no reaction in 45 seconds, give 1 mg IV every 30-45 seconds OR 0.04 mg/kg or 1.2 mg/m^2 IV in 60 seconds, and followed by 0.16 mg/kg or 4.8 mg/m^2 IV if no response occurs within 45 seconds. Maximum total dose of 5 mg IV.
  - Infants: 0.1 mg IV, followed by 0.4 mg IV if no response. Maximum total dose of 0.5 mg IV.

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DOSAGE (Cont.)

- Diagnosis of MG (Cont.):
  IM/SC:
  - Adults: 10 mg IM as a single dose; if there is a reaction, 2 mg IM should be given 30 minutes later to rule out a false-negative reaction.
  - Children weighing more than 34 kg: 5 mg IM.
  - Children weighing 34 kg or less: 2 mg IM.
  - Infants: 0.5-1 mg IM or SC.
- Assessment of oral anticholinesterase therapy in MG:
  - Adults: 1-2 mg IV administered 1 hour after oral intake of the other agent; to be repeated every 1-3 days until establishment of the proper dosage for the anticholinesterase agent.
  - Children: 0.04 mg/kg IV administered 1 hour after oral intake of the other agent.
- Differentiation of cholinergic and myasthenic crisis: 1 mg IV. If no further impairment within 1 minute, may repeat the dose.
- Reversal of nondepolarizing neuromuscular blocking agents: 10 mg IV. May repeat every 5-10 minutes as needed. If doses of 15 mg or above are required, they should be preceded by atropine sulfate IV. Maximum total dose: 40 mg.
- Diagnosis of supraventricular tachyarrhythmias, treatment of paroxysmal atrial tachycardia or evaluation of demand pacemakers: single dose of 5-10 mg IV.
- Slowing of supraventricular tachyarrhythmias: Test dose of 2 mg IV. Repeat 2 mg IV every minute until arrhythmia is controlled; total dose of 10 mg IV. If heart rate decreases during bolus administration, may administer as an IV infusion at a rate of 0.25-2 mg/min.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Compatible with NS and D5W.

MISCELLANEOUS

- Use with caution in patients with asthma or taking a cardiac glycoside (e.g., digoxin).
- Does not antagonize, and in fact may prolong the paralysis caused by succinylcholine.
- Contains sulfites.

REFERENCES

1, 5, 40, 82, 135.

NAME OF MEDICATION
EMICIZUMAB

OTHER NAMES
Hemlibra ®

CLASSIFICATION
Hemostatic, Monoclonal antibody

INDICATIONS

- Prevention of bleeding or reduction of bleeding frequency in adults, adolescents, and children with hemophilia A (congenital factor VIII deficiency) with or without factor VIII inhibitors.

ADMINISTRATION

- SC: do NOT shake vial. Abdomen, upper outer arm or thigh. Rotate injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- GI: diarrhea.
- CNS: headache.
- Hematologic: thrombotic microangiopathy and thromboembolism in patients having received an average cumulative amount of greater than 100 units/kg/24 hours of activated prothrombin complex concentrate (i.e., anti-inhibitor coagulant complex, Feiba ® NF) administered for 24 hours or more. Ideally, activated prothrombin complex concentrate should be discontinued at least 24 hours before emicizumab is initiated. If not possible, the initial dose of activated prothrombin complex concentrate should not exceed 50 units/kg.
- Musculoskeletal: arthralgia, myalgia.
- Fever.
- Local reactions: erythema, pain, pruritus at the injection site.

DOSAGE

- Loading dose in adults and pediatrics: 3 mg/kg SC once weekly for the first 4 weeks.
- Maintenance dose in adults and pediatrics (to start 1 week after the last loading dose): 1.5 mg/kg SC once weekly OR 3 mg/kg SC every 2 weeks OR 6 mg/kg SC every 4 weeks.
- Dosage in renal impairment: based on limited clinical data, no dosage adjustment is necessary when CrCl is 30 mL/min or greater; not studied in patients with a CrCl lower than 30 mL/min.
- Dosage in hepatic impairment: based on limited clinical data, no dosage adjustment is necessary in mild to moderate hepatic impairment; not studied in severe hepatic impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Unopened vials can be stored at less than 30°C for up to 7 days; after storage at room temp, unopened vials can be returned to the fridge if cumulative time has not exceeded 7 days.

MISCELLANEOUS

REFERENCES

5, 95, 135.

New monograph 2021

NAME OF MEDICATION
ENALAPRILAT

OTHER NAMES
Vasotec ® IV

CLASSIFICATION
ACE Inhibitor, Antihypertensive

INDICATIONS

- Treatment of hypertension when oral therapy is not practical.
- Treatment of hypertensive crises.

ADMINISTRATION

- IV direct: physician only. Inject slowly over at least 5 minutes.
- Intermittent IV infusion: dilute in 50 mL of D5W, NS, D5-NS or D5-RL. Infuse over 60 minutes in patients at risk for severe hypotension.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, anaphylactoid reactions, angioedema, rash.
- Cardiovascular: hypotension, syncope.
- GI: nausea, constipation, diarrhea.
- CNS: headache, dizziness.
- Renal: oliguria, acute renal failure, progressive azotemia, increase in serum creatinine.
- Respiratory: cough.
- Hyperkalemia.
- Fatigue.

DOSAGE

- Hypertension: 1.25 mg IV q6h. In patients on a diuretic: 0.625 mg IV q6h; may be repeated 60 minutes later if clinical response is not adequate, with subsequent doses of 1.25 mg IV q6h. Doses up to 5 mg IV q6h have been used for up to 36 hours.
- Hypertensive crises: 1.25-5 mg IV q6h prn.
- Dosage in renal impairment: If CrCl is less than 30 mL/min, 0.625 mg IV q6h; may be repeated 60 minutes later if clinical response is not adequate, with subsequent doses of 1.25 mg IV q6h. For dialysis patients, initial dose of 0.625 mg IV q6h.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Stable for 24 hours at room temp in D5W, NS, D5-NS, D5-RL.

MISCELLANEOUS

- Contraindicated in patients with a history of angioedema to any ACE inhibitor.
- Contraindicated in combination with aliskiren in patients with diabetes mellitus or moderate to severe renal impairment.
- Initial blood pressure response may take 15 minutes to 60 minutes; peak effect of the first dose may not occur until 4 hours after administration; peak effect of subsequent doses may be greater than that of the first dose.
- Switching to oral enalapril: from enalaprilat 1.25 mg IV q6h to oral enalapril at 5 mg once daily, and from enalaprilat 0.625 mg IV q6h to oral enalapril 2.5 mg once daily; titrate dose as needed.

REFERENCES

1, 5, 40, 135, 208.

Full revision 2014; limited revision 2015, 2016

NAME OF MEDICATION
ENFORTUMAB VEDOTIN *

OTHER NAMES
Padcev ™

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS
- Treatment of locally advanced or metastatic urothelial cancer.

ADMINISTRATION
- Reconstitute each vial of 20 mg and 30 mg with 2.3 mL and 3.3 mL of SWFI, respectively, to obtain a concentration of 10 mg/mL. Direct the stream of SWFI along the wall of the vial. Gently swirl the vial until complete dissolution. Do NOT shake. Do NOT expose to direct sunlight. Allow vial to settle for at least 1 minute until bubbles are gone.
- Intermittent IV infusion (mandatory): dilute required dose in NS, D5W or RL to obtain a final concentration between 0.3 and 4 mg/mL. Gently invert the bag to mix the solution. Do NOT shake. Do NOT expose to direct sunlight. Administer over 30 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, pruritus, symmetrical intertriginous and flexural exanthema, bullous dermatitis, exfoliative dermatitis, palmar-plantar erythrodysesthesia, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- GI: nausea, vomiting, diarrhea, dysgeusia, anorexia.
- CNS: peripheral neuropathy.
- Dermatologic: dry skin.
- Electrolyte disturbances: hypokalemia, hypophosphatemia.
- Endocrine and metabolic: hyperglycemia, diabetic ketoacidosis, increased lipase.
- Hematologic: anemia, leukopenia, lymphopenia, neutropenia.
- Ophthalmic: ocular disorders, dry eye (consider artificial tears as prevention).
- Renal: increased serum creatinine.
- Fatigue.
- Extravasation hazard: irritant; reactions may be delayed (e.g., 24 hours). Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- 1.25 mg/kg IV (maximum of 125 mg) on days 1, 8, and 15 of each 28-day cycle.
- Dosage in renal impairment: no adjustment required for mild, moderate, or severe impairment.
- Dosage in hepatic impairment: no adjustment required for mild impairment; avoid in moderate to severe impairment (i.e., total bilirubin over 1.5 x ULN and any AST) due to lack of data.
- Dosage adjustment due to toxicity: refer to the manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials between 2-8°C in original carton. Do not freeze. Do not shake.
- Reconstituted solution should be clear to slightly opalescent and colourless to light yellow.
- Reconstituted vials are stable for 4 hours between 2-8°C protected from light.
- Stable for 8 hours between 2-8°C, protected from light, in NS, D5W or RL at concentrations of 0.3 to 4 mg/mL.

MISCELLANEOUS
- Complete blood count with differential, LFTs and blood glucose should be obtained at baseline and at regular intervals during treatment.
- Closely monitor blood glucose levels in patients with or at risk of diabetes or hyperglycemia. Interrupt treatment if blood glucose is greater than 13.9 mmol/L.

REFERENCES
129, 135, 574.

* Available via Health Canada's Special Access Programme

New monograph 2021

NAME OF MEDICATION
ENFUVIRTIDE

OTHER NAMES
Fuzeon ®

CLASSIFICATION
Antiretroviral

INDICATIONS

- In combination with other antiretroviral agents for the treatment of HIV-1 infection in treatment-experienced patients.

ADMINISTRATION

- Reconstitute each vial with 1.1 mL of SWFI provided by the manufacturer to obtain a concentration of 90 mg/mL. Tap vial gently for 10 seconds and then allow to stand until powder goes completely into solution (which can take up to 45 minutes). The resulting solution should be clear, colourless and without bubbles or particulate matter. If the content is foamy or has gelled, allow more time for complete dissolution.
- SC: upper arm, anterior thigh, or abdomen. Rotate injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, fever, nausea, vomiting, chills, rigors, hypotension, and elevated serum transaminases.
- GI: nausea, diarrhea.
- CNS: insomnia.
- Hematology: eosinophilia.
- Bacterial pneumonia, sinusitis, cough, weight loss.
- Fatigue.
- Local reactions: frequent, may last over a week; discomfort, pain, induration, erythema, nodules, cysts, pruritus, ecchymosis, infection.

DOSAGE

- Adults: 90 mg SC BID.
- Pediatrics (6 years and older): 2 mg/kg SC BID, up to a maximum of 90 mg SC BID.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution is stable for 24 hours in the fridge.

MISCELLANEOUS

REFERENCES

1, 5, 135, 208.

Full revision 2014; limited revision 2015

NAME OF MEDICATION
ENHERTU ™ TRASTUZUMAB DERUXTECAN

OTHER NAMES
Enhertu ™, Fam-Trastuzumab deruxtecan, Trastuzumab deruxtecan, Trastuzumab DXd

CLASSIFICATION
Antineoplastic, Monoclonal antibody

Do NOT confuse Enhertu ™ trastuzumab deruxtecan with Kadcyla ® trastuzumab emtansine or trastuzumab (Herceptin ®, Herzuma ®, Kanjinti ™, Ogivri ™ and Trazimera ®). Enhertu ™ trastuzumab deruxtecan and Kadcyla ® trastuzumab emtansine both contain a cytotoxic component, unlike trastuzumab. These products have different uses, toxicities, administration and dosages. This monograph is specific to ENHERTU ™ TRASTUZUMAB DERUXTECAN.

INDICATIONS
- Treatment of unresectable or metastatic HER2-positive breast cancer in patients who have received prior treatment with trastuzumab emtansine (Kadcyla ® trastuzumab emtansine).

ADMINISTRATION
- Reconstitute each 100 mg vial by slowly injecting 5 mL of SWFI to obtain a concentration of 20 mg/mL. Gently swirl vial until completely dissolved. Do NOT shake.
- Intermittent IV infusion (mandatory): dilute dose in a 100 mL bag of D5W made of PVC or polyolefin. Gently invert the bag to mix the solution. Do NOT shake. Protect from light. If diluted product is stored in the fridge, it should be brought to room temp prior to administering to the patient. Administer initial dose over 90 minutes. If well tolerated, may administer subsequent doses over 30 minutes. Administer using an in-line 0.2 or 0.22 micron filter made of polyethersulfone or polysulfone. Monitor patient closely during and following the infusion for infusion-related reactions.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication**: consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug**: consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, pruritus, urticaria.
- Infusion-related reactions: fever, chills, dyspnea, wheezing, flushing, dizziness, tachycardia; occur within 24 hours of the infusion. For mild to moderate reactions, reduce the infusion rate or temporarily interrupt the infusion. For severe reactions, discontinue treatment.
- Cardiovascular: reduced left ventricular ejection fraction.
- GI: nausea, vomiting, diarrhea, constipation, anorexia, abdominal pain.
- CNS: headache, dizziness.
- Electrolyte disturbances: hypokalemia.
- Hematologic: neutropenia, leukopenia, thrombocytopenia, anemia.
- Hepatic: increased LFTs and bilirubin.
- Respiratory: cough, dyspnea, epistaxis, upper respiratory tract infection, interstitial lung disease, pneumonitis.
- Fatigue.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- 5.4 mg/kg IV on day 1 of each 21-day cycle.
- Dosage in renal impairment: no dosage adjustment required for CrCl of 60 mL/min to less than 90 mL/min; no dosage adjustment required for CrCl of 30 mL/min to less than 60 mL/min, but monitor patient closely; no data for CrCl less than 30 mL/min.
- Dosage in hepatic impairment: no dosage adjustment required for mild impairment (total bilirubin less than or equal to ULN and AST greater than ULN or total bilirubin greater than 1 to 1.5 x ULN and any AST); insufficient data for moderate impairment (total bilirubin greater than 1.5 to 3 x ULN and AST greater than ULN), but monitor closely for increased toxicities related to the cytotoxic component (deruxtecan); no data for severe impairment (total bilirubin greater than 3 to 10 x ULN and any AST).
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Reconstituted solution is stable for 24 hours in the fridge protected from light. It should be clear and colourless to light yellow.
- Diluted solution is stable for 4 hours at room temp, protected from light, and 24 hours in the fridge, protected from light (includes the time needed for preparation and infusion), in D5W in PVC and polyolefin containers.
- Incompatible with NS.

MISCELLANEOUS

- Baseline cardiac assessment should be done for all patients and repeated at regular intervals during treatment as clinically indicated.
- Complete blood counts should be monitored at baseline, prior to each dose and as clinically indicated.

REFERENCES

5, 135, 575.

New monograph 2021

NAME OF MEDICATION
ENOXAPARIN

OTHER NAMES
Lovenox ®

CLASSIFICATION
Anticoagulant

INDICATIONS
- Treatment of unstable angina and non-Q-wave myocardial infarction.
- Treatment of acute ST-segment elevation myocardial infarction (STEMI), including patients to be managed medically or with subsequent percutaneous coronary intervention (PCI).
- Prophylaxis of thromboembolic disorders.
- Treatment of deep vein thrombosis with or without pulmonary embolism.
- Bridging anticoagulation in patients who require temporary interruption of warfarin therapy.
- Prevention of thrombus formation in the extracorporeal circulation during hemodialysis.

ADMINISTRATION
- IV direct: physician or RN. Can be administered using the multidose vial or prefilled syringe. Flush IV line with sufficient amount of NS or D5W prior to and following the administration of enoxaparin. If a partial dose is to be administered from a prefilled syringe, ensure to expel the air bubble prior to administration. For doses less than 30 mg (i.e., for the additional bolus of 0.3 mg/kg for PCI): in order to ensure accuracy of small volume to be injected, may further dilute drug with D5W or NS to get a 3 mg/mL solution. Preparation: from a 50 mL minibag, withdraw 30 mL of solution; inject 60 mg of enoxaparin into the remaining 20 mL to obtain a 3 mg/mL solution; gently mix the bag. Volume to inject of diluted solution (mL) = patient weight (kg) x 0.1.
- SC: inject in the anterolateral or posterolateral abdominal girdle, alternating the left and right sides. With the thickness of skin held between the operator's thumb and finger, introduce the entire length of the needle vertically into the skin. To minimize bruising, injection sites should be not massaged after injection.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus, urticaria, angioedema, cutaneous vasculitis, anaphylaxis, anaphylactoid reaction.
- GI: nausea, diarrhea.
- CNS: confusion.
- Risk of spinal or epidural hematoma that can result in permanent paralysis when epidural or spinal anesthesia or spinal puncture is used in conjunction with enoxaparin. No spinal invasion should be performed for at least 12-24 hours after the last dose of enoxaparin and the next dose should be held until at least 2-4 hours after the procedure.
- Hematologic: bleeding, anemia, thrombocytopenia, thrombocytosis.
- Dermatologic: ecchymosis.
- Hepatic: transient asymptomatic elevations of liver transaminases (AST, ALT); reversible in 3-7 days after discontinuing the drug.
- Renal: hematuria.
- Respiratory: dyspnea.
- Fever, edema.
- Local reactions: hematoma, pain, irritation, skin necrosis (rare).

Antidote: The anticoagulant effect of enoxaparin can be partially neutralized by protamine. Refer to protamine in this manual for more details.

DOSAGE

- Treatment of unstable angina and non-Q-wave myocardial infarction: 1 mg/kg (up to 100 mg) SC q12h for 2-8 days.

- Treatment of acute STEMI: 30 mg IV single bolus dose plus 1 mg/kg SC dose, followed by 1 mg/kg SC q12h (maximum 100 mg/dose for the first 2 SC doses only). In patients of 75 year old and greater: no IV bolus; 0.75 mg/kg SC q12h (maximum 75 mg/dose for the first 2 SC doses only). Enoxaparin injection should be given between 15 minutes before and 30 minutes after the start of fibrinolytic therapy. Duration: 8 days or until hospital discharge.

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DOSAGE (Cont.)

- For patients managed with PCI:
  - In patients who have not received prior anticoagulant: 0.5-0.75 mg/kg IV bolus.
  - If last SC dose of enoxaparin was administered less than 8 hours before balloon inflation: no additional dose.
  - If last SC dose of enoxaparin was administered more than 8 hours before balloon inflation:0.3 mg/kg IV bolus.

- Prophylaxis of thromboembolic disorders:
  - Orthopedic surgery: 30 mg SC q12h, starting 12-24 hours after surgery; alternatively for hip-replacement surgery, 40 mg SC q24h, starting 12 hours before or after surgery. Duration: 10-35 days.
  - Abdominal or colorectal surgery: 40 mg SC q24h for 7-10 days, starting 2 hours before surgery.
  - Bedridden medical patients at risk of deep vein thrombosis: 40 mg SC q24h until mobilization or hospital discharge.
  - Dosage in obesity: if body mass index (BMI) of 40 kg/m^2 or higher OR if weight greater than 100 kg: 40 mg SC q12h.

- Treatment of deep vein thrombosis with or without pulmonary embolism:
  - 1 mg/kg SC q12h or 1.5 mg/kg SC q24h.
  - Dosage in obesity: dose should be calculated using actual body weight (favor q12h dosing). If BMI of 40 kg/m^2 or higher: 0.7-0.8 mg/kg (may cap to 150 mg) SC q12h.

- Bridging anticoagulation: 1 mg/kg SC q12h or 1.5 mg/kg SC q24h. Do not administer enoxaparin within the 24 hour period before surgery. For procedures associated with a high risk of bleeding, resume enoxaparin 48-72 hours after surgery when adequate hemostasis has been achieved.

- Prevention of thrombus formation in the extracorporeal circulation during hemodialysis:
  - Initial dose: 0.5-1 mg/kg into the arterial line of the circuit at the beginning of the dialysis session.
  - For subsequent dialysis sessions: adjust dose if necessary.

- Dosage in renal impairment: use with caution; monitoring of anti-Xa levels; dosage adjustments recommended in cases of severe renal impairment (CrCl under 30 mL/min). Refer to manufacturer's recommendation.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes at room temp.
- Enoxaparin, when drawn from multi-dose vials and stored in plastic syringes, is stable for 10 days in the fridge. Change needle when ready to inject drug after storage.
- Compatible by y-site with NS and D5W.

MISCELLANEOUS

- Enoxaparin should not be given to patients with a history of heparin-induced thrombocytopenia (risk of cross-reactivity), unless an *in vitro* platelet aggregation test is negative.
- For laboratory monitoring of effect, anti-Xa methods are recommended (although not routinely done, monitoring is recommended in special cases such as extremes of weights, pregnancy and renal failure). The recommended time for monitoring anti-Xa activity is 4 hours after a SC injection (peak activity).
- Enoxaparin potency in international units: 1 mg = 100 international units.
- As the multi-dose vial contains benzyl alcohol, it is not recommended for use during pregnancy or in neonates.
- Derived from porcine tissue.

REFERENCES

1, 5, 87, 95, 119, 208, 408, 409.

Full revision 2014; limited revision 2015, 2016, 2017, 2019, 2020

NAME OF MEDICATION
EPHEDrine

OTHER NAMES

CLASSIFICATION
Sympathomimetic

INDICATIONS

- Treatment of hypotension secondary to shock or anesthesia.
- Treatment of bronchospasm.

ADMINISTRATION

- IV direct: physician; RN may administer 10 mg or less. ***Continuous BP monitoring, cardiac monitoring***. Best to dilute the dose to facilitate administration: dilute 50 mg (1 mL) with 9 mL of NS to get a concentration of 5 mg/mL. Administer at a rate of 10 mg/min.
- IM, SC.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: palpitations, tachycardia, angina, hypertension (may lead to intracranial hemorrhage); hypotension after several doses (due to direct cardiac depression and vasodilation).
- CNS: nervousness, restlessness, excitation, anxiety, insomnia, tremors, irritability. Dizziness, lightheadedness, vertigo, confusion, delirium, euphoria and hallucinations may occur with higher doses.
- Renal: painful urination, urinary retention.
- Weakness, mydriasis.

DOSAGE

- Hypotension: IV: 5-25 mg with additional doses every 5-10 minutes if needed.
  SC, IM: 25-50 mg (range 10-50 mg).
- Bronchospasm: 12.5-25 mg IM or SC.
- Maximum parenteral dosage: 150 mg/24 hours.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp; protect from light.
- Physically compatible with D5W, NS, dextrose-saline solutions, RL.
- Stable for 4 days at room temp undiluted at a concentration of 50 mg/mL in polypropylene and glass syringes.
- Stable for 60 days at room temp exposed to light or in the fridge in NS at a concentration of 5 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Should not be used or use with caution in patients with diabetes, hypertension, cardiovascular disease, hyperthyroidism, and narrow-angle glaucoma.
- Blood volume depletion should be corrected as fully as possible before ephedrine is instituted.
- Absorption and onset of action of ephedrine are more rapid following IM administration (within 10-20 minutes) than following SC administration.
- Ephedrine is less potent but longer acting than epinephrine and norepinephrine.

REFERENCES

1, 4, 5, 40, 208.

Full revision 2014; limited revision 2017, 2020

NAME OF MEDICATION	
EPINEPHrine	
OTHER NAMES	CLASSIFICATION
Adrenalin, Adrenaline	Sympathomimetic

INDICATIONS

- Treatment of severe allergic (anaphylactic or anaphylactoid) reactions.
- Bronchospasm.
- In advanced cardiovascular life support (ACLS) during cardiopulmonary resuscitation (CPR): pulseless ventricular tachycardia, ventricular fibrillation, pulseless electrical activity, and asystole.
- Bradycardia and hypotension in post-cardiac arrest care.

ADMINISTRATION

- **Check label carefully to ensure concentration of the solution used**.
- IV direct: physician or RN in presence of physician; ***cardiac monitoring***. The 0.1 mg/mL solution prepacked in syringe does not require further dilution before giving it IV direct. If the 1 mg/mL solution is used, dilute dose 1 in 10 with NS to make a 0.1 mg/mL solution. Follow with 20 mL IV flush to ensure drug delivery to systemic circulation if injection done through a peripheral vein.
- Continuous IV infusion: ***cardiac monitoring***. Dilute 1 mg in 250 or 500 mL of solution to obtain a final concentration of 4 or 2 mcg/mL, respectively. At TOH, dilute 1 mg in 250 mL or 4 mg in 1000 mL of solution to obtain a final concentration of 4 mcg/mL. More concentrated solutions such as 16 mcg/mL (4 mg/250 mL) can also be used. Administer preferably via central venous access to reduce the risk of extravasation and ensure good bioavailability. Must be administered by an infusion pump.
- IM (preferred route over SC): use the 1 mg/mL solution; inject into the anterolateral aspect of the thigh; avoid IM injection into buttocks.
- SC: use the 1 mg/mL solution; delayed absorption vs. IM injection.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: palpitation, tachycardia, hypertension, anginal pain, arrhythmias, cerebral hemorrhage.
- GI: nausea, vomiting.
- CNS: restlessness, nervousness, lightheadedness, fear, anxiety, tension, headache, dizziness, tremor.
- Renal: anuria, renal dysfunction.
- Respiratory: pulmonary edema, dyspnea.
- Weakness, pallor, mydriasis.
- Local reactions: ischemia, ulceration, necrosis.
- Extravasation hazard: vesicant. **Antidote**: phentolamine 5-10 mg diluted with 10-15 mL of NS. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

Adults:
- Hypersensitivity Reaction:
  - Usual initial dose of 0.1-0.5 mg IM (preferred) or SC. May repeat dose every 5-15 minutes if anaphylactic shock, or every 20 minutes to 4 hours if asthma.
  - Severe anaphylactic shock: 0.1 to 0.25 mg IV over 5 to 10 minutes; dose may be repeated every 5-15 minutes or followed by a continuous IV infusion at an initial rate of 1 mcg/min and increasing to 4 mcg/min if necessary.
- ACLS during CPR: 1 mg IV over 1 minute or faster; may repeat every 3 to 5 minutes.
- Post-cardiac arrest care: initial rate of 2-10 mcg/min IV; range 0.1-0.5 mcg/kg/min IV.

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DOSAGE

Pediatrics (over 1 month of age):
- Hypersensitivity Reaction:
  - 0.01 mg/kg or 0.3 mg/m^2 IM (preferred) or SC (maximum 0.5 mg). May repeat dose every 5-15 minutes if anaphylactic shock.
  - Severe anaphylactic shock: initial dose of up to 0.1 mg IV over 5-10 minutes followed by a continuous infusion at an initial rate of 0.1 mcg/kg/min (to a maximum of 1.5 mcg/kg/min).
- ACLS during CPR: 0.01 mg/kg IV (maximum 1 mg); may repeat every 3 to 5 minutes.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and prefilled syringes at room temp and protect from light.
- Do not use solution if coloured or contains a precipitate.
- Compatible in D5W, NS, Ringer's, RL

MISCELLANEOUS

- Should not be used or use with caution in patients with diabetes, cerebral arteriosclerosis, hypertension, cardiovascular disease, hyperthyroidism, shock (other than anaphylactic shock), narrow-angle glaucoma.
- Contains sulfites.
- Refer to TOH anaphylaxis algorithm.

REFERENCES

1, 2, 4, 5, 40, 95, 126, 135, 186, 230, 235, 266, 267, 364, 366, 367, 462.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2019, 2020, 2021

NAME OF MEDICATION
EpiRUBicin

OTHER NAMES
Pharmorubicin ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Alone and in combination with other chemotherapeutic agents in a variety of tumour types such as lymphoma, cancer of the lung, breast, ovary and stomach.

ADMINISTRATION

- IV direct: physician or RN; administer undiluted; inject over at least 3-5 minutes into the tubing of a freely running IV solution of NS or D5W into a large vein. Flush with 20 mL of NS or D5W to ensure drug delivery to systemic circulation.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W; infuse over 10-20 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity (rare): anaphylaxis, rash, pruritus, urticaria, chills, fever, shock.
- Cardiovascular: sinus tachycardia, ECG abnormalities, arrhythmias (acute and transient), delayed cumulative dose-dependent cardiomyopathy and heart failure.
- GI: nausea, vomiting, diarrhea, mucositis.
- Dermatologic: radiation recall reaction (rare), flushing (indicates of too rapid administration), photosensitivity(rare), skin hyperpigmentation (rare).
- Hematologic: leukopenia, neutropenia, febrile neutropenia, anemia, thrombocytopenia, thromboembolism (including fatal pulmonary embolism).
- Ophthalmic: conjunctivitis, keratitis.
- Renal: red colouration of urine (for 1-2 days after administration).
- Hyperuricemia, tumour lysis syndrome, fatigue.
- Local reactions: phlebitis, erythematous streaking (histamine release) and/or transient urticaria along the vein proximal to site of injection indicates a too rapid administration; usually subsides within 30 minutes.
- Extravasation hazard: vesicant. Can cause severe tissue necrosis without symptoms. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Weekly: 12.5-25 mg/m^2 IV.
  Every 2 weeks: 35 mg/m^2 IV.
  Every 3 to 4 weeks: 50-150 mg/m^2 IV.
  Every 4 weeks: 60 mg/m^2 IV days 1 and 8, or days 1 and 15.
- Due to risk of cardiotoxicity, maximum lifetime cumulative dose is 650 mg/m^2 for patients with cardiac risk factors and 900-1000 mg/m^2 for patients without cardiac risk factors.
- Dosage in renal impairment: lower starting doses in patients with severe renal impairment.
- Dosage in hepatic impairment:

Serum Bilirubin		AST	Dose
21-51 mcmol/L	OR	2-4 x ULN	50%
above 51 mcmol/L	OR	above 4 x ULN	25%

  Severe hepatic impairment: contraindicated
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge, protected from light.
- Stable for 43 days at 4°C and room temp protected from light in NS and D5W at a concentration of 0.1 mg/mL in PVC bags.
- Stable for 84 days at 8°C in NS at a concentration of 1 mg/mL in polypropylene syringes.
- Stable for 28 days in the fridge or at room temp in NS at a concentration of 0.5 mg/mL in polypropylene syringes.
- Avoid contact with any alkaline solution as it will result in hydrolysis of the drug.

MISCELLANEOUS

- Live and live attenuated vaccines should not be given concurrently with epirubicin.
- Cardiac monitoring in patients who have received mediastinal radiotherapy, therapy other anthracyclines (e.g., daunorubicin, doxorubicin, idarubicin) or anthracenediones (e.g., mitoxatrone), with pre-existing cardiac disease or already received epirubicin at cumulative doses above 650 mg/m^2.

REFERENCES

1, 5, 129, 143, 165.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION	
EPOPROSTENOL	
OTHER NAMES	CLASSIFICATION
Caripul ®, Flolan ®	Vasodilator

INDICATIONS

- Long-term treatment of primary pulmonary hypertension and secondary pulmonary hypertension due to scleroderma spectrum of diseases in New York Heart Association Class III and Class IV patients who did not respond adequately to conventional therapy.

ADMINISTRATION

- For Flolan ®: reconstitute each 0.5 mg or 1.5 mg vial with 5 mL **only** with diluent provided. Withdraw the required amount and further dilute in the reservoir cassette to 100 mL with the diluent provided. Consult manufacturer's monograph for details. Avoid contact with material containing polyethylene terephthalate (PET) or polyethylene terephthalate glycol (PETG) as this can lead to cracking or damage. Materials (infusion pump, cassette, external tubing) provided to patients through the Horizons Patient Support Program do not contain PET or PETG.
- For Caripul ®: reconstitute each 0.5 mg or 1.5 mg vial with 5 mL of SWFI or NS only. Withdraw the required amount and further dilute in the reservoir cassette to 100 mL with the identical diluent as used for the reconstitution. Consult manufacturer's monograph for details.
- Continuous IV infusion (mandatory): ***continuous BP monitoring, cardiac monitoring while patient is in critical care areas***. Refer to full monograph for infusion rate and concentration tables. During initiation of treatment, may be given peripherally. For chronic continuous administration, infusion via central venous catheter is necessary. Must be administered by an infusion pump (refer to manufacturer's monograph for pump specifications); infuse through a 0.22 micron in-line filter.
- **Interruptions in drug delivery or sudden large reductions in administration rate can lead to rapid return of pulmonary hypertension and can be fatal**.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: chest pain, bradycardia, hypotension, tachycardia.
- GI: nausea, vomiting, diarrhea, abdominal pain, anorexia.
- CNS: anxiety, nervousness, dizziness, headache, tremor.
- Dermatologic: flushing, eczema, rash, urticaria.
- Jaw pain, flu-like symptoms, myalgia, chills, fever, sepsis, hemorrhage, hyperglycemia; epoprostenol should be discontinued in case of pulmonary edema during dosage initiation.

DOSAGE

Initiation of treatment:
- Initiate at 2 ng/kg/min IV and increase infusion rate in increments of 2 ng/kg/min at intervals of at least 15 minutes until dose-limiting pharmacological effects occur.

Dosage adjustments during continuous chronic infusion:
- If symptoms of pulmonary hypertension persist, recur or worsen, increase infusion rate by 1-2 ng/kg/min increments at intervals of at least 15 minutes until clinical response obtained.
- If side effects are not tolerated, decrease infusion rate gradually by 2 ng/kg/min decrements at intervals of at least 15 minutes until tolerable.
- In patients receiving lung transplants, dose was tapered after the initiation of cardiopulmonary bypass.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

For Flolan ®:
- Store epoprostenol and diluent vials between 15-25°C and protect from light.
- Flolan® is compatible only with specific sterile diluent accompanying the product.
- Before being transferred to an infusion pump: stable for 24 hours in the fridge and protected from light.
- Once placed in an infusion pump: stable for 24 hours between 2-8°C (change frozen gel packs q12h, or q8h if ambient temperature is near 30°C) and protected from direct sunlight.

For Caripul ®:
- Store vials between 15-30°C.
- If the diluted solution is used immediately:
  - stable for 48 hours at 25°C and protected from direct sunlight
  - stable for 24 hours at temperatures above 25°C up to 30°C and protected from direct sunlight.
- If not used immediately: stable for 8 days in the fridge protected from light, then;
  - stable for 24 hours at 25°C protected from direct sunlight at concentrations from 3000 ng/mL to less than 15 000 ng/mL.
  - stable for 48 hours at 25°C protected from direct sunlight at concentrations of 15 000 ng/mL and above.
  - stable for 24 hours at temperatures above 25°C up to 30°C and protected from direct sunlight (all concentrations).
- Short excursions to higher temperatures are permitted. Consult manufacturer's monograph for details.

MISCELLANEOUS

- Contraindicated in patients with congestive heart failure secondary to severe left ventricular systolic dysfunction.
- Unless contraindicated, anticoagulant therapy should be administered in patients on epoprostenol to reduce the risk of pulmonary thromboembolism or systemic embolism.
- For the Flolan ® Information and Patient Care Program, contact Pharmaprix/Shoppers Drug Mart Specialty Health Network at 1-800-361-7090.
- For the Caripul ® Patient Support Program, contact Pharmaprix/Shoppers Drug Mart Specialty Health Network at 1-877-227-4785.

REFERENCES

1, 5, 40, 264, 268, 385.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
EPTIFIBATIDE

OTHER NAMES
Integrilin ®

CLASSIFICATION
Platelet aggregation inhibitor

INDICATIONS

- Treatment of patients with unstable angina, non-ST-segment elevation myocardial infarction and those subsequently undergoing percutaneous coronary interventions (PCI).
- Treatment of patients undergoing PCI.

ADMINISTRATION

- IV direct: physician or RN. For bolus dose only. Do not shake vials. Administer IV direct undiluted over 1-2 minutes, using the 2 mg/mL solution (from 10 mL vial).
- Continuous IV infusion: initiate immediately after bolus. Administer undiluted, using the 0.75 mg/mL solution (from 100 mL vial); spike vial with a vented infusion set within the circle of the stopper top. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis (rare).
- Cardiovascular: hypotension.
- Hematologic: thrombocytopenia (may be severe); major bleed, including intracranial hemorrhage/stroke, a hemoglobin drop greater than 5 g/dL, bleed from arterial access site, GI or genito-urinary tracts, lungs and retriperitoneal area; minor bleed, including gross hematuria, hematemesis, or a hemoglobin drop greater than 3-4 g/dL. Risk is increased in patients weighing less than 70 kg.

DOSAGE

- Unstable angina/non-ST-segment elevation myocardial infarction: bolus dose of 180 mcg/kg IV, (maximum 22.6 mg) followed by a continuous IV infusion at a rate of 2 mcg/kg/min (maximum 15 mg/hr) until hospital discharge or between 2-4 hours before CABG surgery, up to 72 hours. For patients subsequently undergoing PCI during the initial 72 hours, continue infusion until hospital discharge or for up to 18-24 hours after the procedure, whichever comes first (total infusion time up to 96 hours).
- In PCI: bolus dose of 180 mcg/kg IV (maximum 22.6 mg) immediately before the initiation of PCI, followed by a continuous IV infusion at a rate of 2 mcg/kg/min (maximum 15 mg/hr); administer a second bolus 10 minutes after the first bolus. Continue IV infusion until hospital discharge or for up to 18-24 hours, whichever comes first (minimum of 12 hours).
- Dosage in renal impairment: in patients with a CrCl between 30-50 mL/min, no dosage adjustment of the bolus doses but reduce infusion rate to 1 mcg/kg/min (maximum 7.5 mg/hr). Eptifibatide treatment is not recommended if CrCl is below 30 mL/min.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge protected from light. Unopened vials are stable for 2 months at room temp and protected from light.
- Compatible with NS, D5-NS.

MISCELLANEOUS

- Eptifibatide is intended for use with ASA and heparin. Refer to product monograph for dosing.
- Use care in handling patients (bleeding precautions). When obtaining IV access, use compressible sites only.
- Restoration of platelet function (greater than 50%) towards baseline 4 hours after infusion discontinuation.
- Bleeding time returns to baseline within 4-6 hours after infusion is discontinued.

REFERENCES

1, 5, 40, 135.

Full revision 2014; limited revision 2015, 2016, 2017, 2020

NAME OF MEDICATION
ERGONOVINE

OTHER NAMES
Ergometrine

CLASSIFICATION
Oxytocic

INDICATIONS

- Prevention and treatment of postpartum and postabortal hemorrhage due to uterine atony.
- As an adjunct in the diagnosis of coronary artery spasm in patients with variant angina.

ADMINISTRATION

- IV direct: physician only. ***Blood pressure monitoring***. Dilute dose with 5 mL NS and administer at a rate of 0.2 mg/min.
- IM (preferred).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity.
- Cardiovascular: hypertension (if injected undiluted or too rapidly), palpitation, bradycardia, transient chest pain, thrombophlebitis.
- GI: nausea and vomiting (with too rapid injection), abdominal pain, diarrhea.
- CNS: dizziness, headache, hallucination, vertigo.
- Otic: tinnitus.
- Respiratory: dyspnea.
- Ergotism (intense vasoconstriction which may lead to gangrene) on prolonged therapy.

DOSAGE

- For postpartum/postabortal hemorrhage: 0.2 mg IV or IM; may be repeated q2-4h to a maximum of 1 mg (total of 5 doses).
- For diagnosis of coronary artery spasm: 0.05 mg IV during coronary angiography. Repeat every 5 minutes until onset of chest pain or a total dose of 0.4 mg. Dose range: 0.1-0.4 mg.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules in the fridge although stable for 60 days at room temp; protect from light.
- Do not use if discolouration occurs.
- Compatible with NS, D5W and RL. Should not be mixed with fluids for infusions.

MISCELLANEOUS

- Should not be used prior to delivery of the placenta.
- Onset of action: IM: 2-3 minutes; IV: 1 minute.
- Duration of action (uterine effect): IM: 3 hours; IV: 45 minutes.

REFERENCES

1, 2, 3, 6, 95, 135.

Full revision 2014; limited revision 2016

NAME OF MEDICATION
EriBULin

OTHER NAMES
Halaven ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of metastatic breast cancer in patients who have previously received at least two chemotherapeutic regimens (including an anthracycline and a taxane) for the treatment of metastatic disease.
- Treatment of adult patients with unresectable advanced or metastatic liposarcoma subtype of soft tissue sarcoma. Prior therapy should have included an anthracycline-containing regimen if appropriate.

ADMINISTRATION

- IV direct: administer undiluted IV over 2 to 5 minutes.
- Intermittent IV infusion: dilute in up to 100 mL of NS; administer over 2-5 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity
- Cardiovascular: QT interval prolongation, hypertension, edema, venous thromboembolism (rare).
- GI: nausea, vomiting, diarrhea, constipation, anorexia, abdominal pain, mucositis, dysgeusia, pancreatitis (rare).
- CNS: peripheral neuropathy (can be severe and last more than a year), headache, vertigo, dizziness, insomnia.
- Hematologic: neutropenia (can be severe), thrombocytopenia, anemia, leukopenia, bleeding (can be severe).
- Hepatic: abnormal LFTs.
- Dermatologic: rash, palmar-plantar erythrodysesthesia syndrome.
- Electrolyte disturbances: hypokalemia, hyperkalemia, hypocalcemia, hypercalcemia, hypomagnesemia, hypermagnesemia, hypophosphatemia.
- Renal: increased creatinine.
- Respiratory: dyspnea, cough, pharyngolaryngeal pain.
- Hyperglycemia, fatigue, fever, arthralgia, myalgia, back pain, bone pain, pain in extremity, conjunctivitis.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 1.4 mg/m^2 IV on days 1 and 8 of a 21-day cycle. Refer to manufacturer's instructions for dosage adjustment in case of toxicity.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-15	less than 15
Dose	1.4 mg/m^2	1.1 mg/m^2	Not recommended

- Dosage in hepatic impairment: 1.1 mg/m^2 if mild hepatic impairment (Child-Pugh A), 0.7 mg/m^2 if moderate hepatic impairment (Child-Pugh B). Not recommended in patients with severe hepatic impairment (Child-Pugh C).
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Excursions to 30°C are permitted. Do not freeze.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 14 days between 18-22°C or between 2-6°C exposed to light or protected from light, undiluted in polypropylene syringes.
- Stable for 24 hours at room temp exposed to light or 48 hours refrigerated in NS at concentrations of 0.005 mg/mL to 0.2 mg/mL in IV bags.
- Stable for 14 days between 18-22°C or between 2-6°C exposed to light or protected from light in NS at concentrations of 0.0175 mg/mL and 0.049 mg/mL in multilayer polyolefin containers.
- Incompatible with dextrose solutions.

MISCELLANEOUS

- Use with extreme caution in patients with significant cardiovascular impairment or pre-existing neuropathy.
- Correct electrolytes prior to starting treatment; monitor ECG in patients with cardiac conditions and electrolyte disturbances.

REFERENCES

5, 95, 135, 165, 208.

New monograph 2014; limited revision 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
ERTAPENEM

OTHER NAMES
Invanz ™

CLASSIFICATION
Antibiotic - carbapenem

INDICATIONS

- For the following moderate to severe infections caused by susceptible microorganisms: complicated intra-abdominal infections, complicated skin and skin structure infections, community acquired pneumonia, complicated urinary tract infections, and acute pelvic infections.

ADMINISTRATION

- For IV use: reconstitute the 1 g vial with 10 mL of SWFI, NS or bacteriostatic water for injection to get approximately 100 mg/mL of solution. Shake well to dissolve.
- Intermittent IV infusion: further dilute in 50 mL of NS (or if using a lower volume for pediatrics, do not exceed 20 mg/mL as a final concentration); infuse over 30 minutes.
- For IM use: reconstitute the 1 g vial with 3.2 mL of 1% lidocaine injection (without epinephrine) to get approximately 280 mg/mL of solution. Shake well.
- IM (deep IM injection).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, angioedema, anaphylaxis.
- GI: nausea, vomiting, diarrhea.
- CNS: headache.
- Local reactions: injection site reactions (thrombophlebitis).

DOSAGE

Adults and pediatrics 13 years of age and older:
- 1 g IV/IM once daily.
- Dosage in renal impairment: in patients with CrCl less than 30 mL/min including those on hemodialysis, the dose is reduced to 500 mg once daily. If ertapenem is given within 6 hours prior to hemodialysis, a supplementary dose of 150 mg is recommended.

Pediatrics:
- 3 months-12 years of age: 15 mg/kg/dose IV/IM q12h, but not to exceed 1 g/day.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials at room temp.
- Reconstituted solutions for IM use must be used within 60 minutes of preparation.
- Reconstituted solutions for IV use must be used within 6 hours of preparation.
- Reconstituted solutions for IV use may be stored in the fridge for up to 24 hours and then used within 4 hours after removal from the fridge.
- Stable for 20 hours at room temp and 6 days in the fridge in NS at a concentration of 10 mg/mL in PVC containers.
- Stable for 6 hours at room temp and 5 days in the fridge in NS at a concentration of 20 mg/mL in PVC containers.
- Stable for 30 minutes at room temp or 4 hours at room temp if refrigerated for 24 hours before, reconstituted to a concentration of 100 mg/mL with NS in a polypropylene syringe.
- Solutions may range from colourless to pale yellow without affecting potency.
- Do not mix drug in dextrose solutions but can be piggybacked in the same line as dextrose.

MISCELLANEOUS

- Potential for cross-allergenicity with other beta-lactam antibiotics.

REFERENCES

2, 40, 82, 135, 143, 222, 342.

Full revision 2014; limited revision 2015

NAME OF MEDICATION
ERYTHROMYCIN

OTHER NAMES
Erythrocin ®

CLASSIFICATION
Antibiotic - macrolide

INDICATIONS

- Severe infections due to penicillin-sensitive organisms in patients hypersensitive to penicillin.
- *Mycoplasma pneumoniae* and *Legionella pneumophila* infections.
- Treatment of diabetic gastroparesis.

ADMINISTRATION

- Reconstitute 500 mg or 1 g vial with 10 or 20 mL, respectively, of SWFI without preservatives to provide a solution of 50 mg/mL. If NS or solutions with preservatives are used to reconstitute, precipitation may occur (this restriction does not apply to the drug in piggyback containers).
- Intermittent IV infusion: dilute 1/4 of the daily dose in 100-1000 mL of NS or RL to obtain a concentration between 1 to 5 mg/mL. Infuse over 20-60 minutes.
- Continuous IV infusion: dilute entire daily dose in NS or RL to get a concentration of 1 mg/mL and infuse over 24 hours.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis.
- Cardiovascular: prolongation of the QT interval and development of ventricular arrythmias (rare).
- GI: abdominal cramping and discomfort, nausea, vomiting, diarrhea.
- Local reactions: local pain and burning on injection; venous irritation and thrombophlebitis; minimize by further diluting drug and infusing slowly.

DOSAGE

- Adults:
  - Infections: usual dose 250 mg-1 g IV q6h.
  - Diabetic gastroparesis: 200 mg IV before each meal; switch to oral therapy (250 mg TID, 30 minutes before meals) when practical.

- Pediatrics:
  - Infections: 15-20 mg/kg IV daily in divided doses or 300-600 mg/m^2/day. For severe infections, up to 50 mg/kg daily but not to exceed 4 g daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solutions stable for 24 hours at room temp and 2 weeks if refrigerated.
- Infusions stable for 24 hours at room temp or in the fridge in NS.
- Solutions are most stable when the pH is in the range of 6-8 and become unstable and rapidly loose their potency at pH of 5.5 or below and at pH of 10 or above.
- Compatible with NS and RL. If a dextrose containing solution is used, add 1 mL of $NaHCO_3$ 4% per 100 mL of solution for enhanced stability.

MISCELLANEOUS

REFERENCES

1, 2, 4, 5, 40, 82, 95.

Full revision 2014; limited revision 2015, 2016

NAME OF MEDICATION
ERYTHROPOIETIN

OTHER NAMES
Epoetin alfa, Eprex ®,
Recombinant human erythropoietin

CLASSIFICATION
Hematopoietic

INDICATIONS

- Treatment of anemia associated with chronic renal failure.
- Treatment of anemia associated with zidovudine-treated/HIV-infected patients.
- Treatment of anemia in patients with non-myeloid malignancies, where anemia is due to the effect of concomitantly administered chemotherapy.
- To facilitate autologous blood collection and/or reduce allogeneic blood exposure in patients undergoing major elective surgery.

ADMINISTRATION

- IV direct: physician or RN. May be given undiluted over 1 to 5 minutes.
- SC: anterior abdominal wall and limbs. Maximum volume per site is 1 mL; if a larger volume needs to be administered, use more than one site. Alternate injection site. SC administration may increase the risk of immunogenicity.
- Selection of route of administration (IV or SC) may vary depending on the indication. Refer to Dosage section for more information.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, angioedema, anaphylaxis.
- Cardiovascular: dose-dependent increase in blood pressure (most often in chronic renal failure patients).
- CNS: headache, seizures.
- Dermatologic: erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Hematologic: thromboembolic events such as shunt thrombosis, deep vein thrombosis, pulmonary embolism, etc.
- Flu-like symptoms (arthralgia, myalgia, fever, headache, malaise, chills and diaphoresis) may appear 90-120 minutes after an IV bolus dose and disappear within 10-12 hours with no further treatment.
- Hyperkalemia, hypokalemia.
- Pure red cell aplasia (rare): mainly observed with SC route in patients with chronic renal failure.
- Local reactions: burning, pain, erythema.

DOSAGE

- Chronic renal failure: initiate therapy with 50-100 international units/kg IV (preferred if patient on hemodialysis) or SC three times weekly. Titrate in increments or decrements of 25 international units/kg to achieve target. Recommended hemoglobin range: 100-115 g/L, not to exceed 120 g/L. Rate of hemoglobin increase should not exceed 10 g/L in any 2-week period.
- HIV-infected patients: if endogenous serum erythropoetin levels are 500 mU/mL or below, initiate therapy with 100 international units/kg IV or SC three times weekly for 8 weeks. Titrate in increments of 50-100 international units/kg three times weekly to achieve target. Maximum dose of 300 international units/kg three times weekly. Hemoglobin levels should not exceed 120 g/L.
- Cancer patients: do not initiate if hemoglobin is 100 g/L or greater. Starting doses: 150 international units/kg SC three times weekly or 40 000 international units SC once weekly. Use the lowest dose sufficient to avoid blood transfusion. Hemoglobin levels should not exceed 120 g/L. Rate of hemoglobin increase should not exceed 10 g/L in any 2-week period. Discontinue when chemotherapy course is completed.
- To reduce allogeneic blood exposure in surgery patients: 600 international units/kg SC once weekly for 3 weeks prior to surgery (days -21, -14, -7) and on the day of surgery. If period prior to surgery is less than 3 weeks, give 300 international units/kg/day SC for 10 consecutive days prior to surgery, on the day of surgery and for 4 consecutive days immediately thereafter.
- To facilitate autologous blood collection: 600 international units/kg IV twice weekly for 3 weeks prior to surgery.
- Consult the manufacturer's monograph for more details on dosing and titration.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store in fridge; do not freeze. Protect from light.
- Do not shake as it may denature the glycoprotein.
- Prefilled syringes are stable for 7 days at room temp (not exceeding 25°C), exposed at normal room light (not direct sunlight).

MISCELLANEOUS

- Iron status should be evaluated for all patients; supplemental iron is recommended to support erythropoietin-stimulated erythropoiesis.
- Patients receiving erythropoietin prior to elective surgery, for the purposes of reducing the requirements for allogeneic blood transfusion, should receive adequate antithrombotic prophylaxis in order to reduce the incidence of deep venous thrombosis.

REFERENCES

1, 5, 95, 135, 522.

NAME OF MEDICATION
ESMOLOL

OTHER NAMES
Brevibloc ®

CLASSIFICATION
Beta-blocker

INDICATIONS

- Perioperative management of tachycardia and hypertension in patients at risk of myocardial ischemia.
- Rapid control of ventricular response in atrial fibrillation or flutter in situations requiring a short-acting agent.

ADMINISTRATION

- IV direct: physician or RN; ***cardiac monitoring, continuous BP monitoring***. Using the 10 mg/mL-10 mL vial, give dose undiluted over 30 seconds.
- Continuous IV infusion: ***cardiac monitoring, blood pressure monitoring***. Use 10 mg/mL-250 mL premixed bag. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: symptomatic hypotension (with diaphoresis, dizziness and headache), asymptomatic hypotension, bradycardia, heart block, flushing.
- GI: nausea.
- Bronchospasm.
- Local reactions: inflammation and induration at the site of injection.

DOSAGE

- For perioperative tachycardia and hypertension: initial dose of 1.5 mg/kg IV to a maximum of 100 mg over 30 seconds. For persistent tachycardia and hypertension, may follow with an infusion of 0.15-0.3 mg/kg/min.
- For atrial fibrillation or flutter: initial loading dose of 0.5 mg/kg IV over 1 minute followed by a maintenance infusion rate of 0.05 mg/kg/min for 4 minutes. If no response, repeat the 0.5 mg/kg loading dose and increase infusion rate to 0.1 mg/kg/min. Continue this titration process of 0.5 mg/kg loading doses and incremental increases of 0.05 mg/kg/min infusion rates. As the desired endpoint is approached, omit the loading dose and reduce the incremental infusion rate to 0.025 mg/kg/min. Maximum recommended infusion rate is 0.3 mg/kg/min.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and premixed bags at room temp.

MISCELLANEOUS

- Contraindicated in patients with second or third-degree AV block, sinus bradycardia, cardiogenic shock or overt cardiac failure.
- Concomitant use of IV esmolol with IV diltiazem or IV verapamil may cause significant cardiac depression. Therefore co-administration is contraindicated. Note that esmolol beta-blockade effects are dissipated 20-30 minutes after discontinuation of an infusion.
- Caution in diabetic patients as esmolol may mask tachycardia occurring during hypoglycemia (but dizziness and sweating may not be masked).
- Duration of action is very short (half-life = 9 minutes).
- Effect is relatively selective for beta-1 receptors.

REFERENCES

1, 2, 4, 5, 40.

Full revision 2014; limited revision 2015, 2016, 2017, 2020

NAME OF MEDICATION
ESTROGENS, CONJUGATED

OTHER NAMES
Premarin ®

CLASSIFICATION
Estrogen

INDICATIONS

- Treatment of abnormal uterine bleeding due to hormonal imbalance.
- Treatment of bleeding associated with uremia.

ADMINISTRATION

- Reconstitute each 25 mg vial with 5 mL of SWFI to obtain a straw-coloured solution at a concentration of 5 mg/mL. During reconstitution, add diluent slowly and direct it against the side of the vial. Mix solution gently; do NOT shake.
- IV direct: physician only. Administer slowly at a rate of 5 mg/min.
- Intermittent IV infusion: for controlling bleeding associated with uremia: dilute the dose in 50 mL of NS and infuse over 30-40 minutes.
- IM (NOT preferred because of delayed absorption).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, anaphylactoid reactions, angioedema, rash, urticaria.
- Cardiovascular: hypotension, flushing (with too rapid administration).
- GI: nausea, vomiting, ischemic colitis (rare).
- CNS: dizziness, headache, migraine, nervousness; increased risk of ischemic stroke in patients affected by migraines with aura.
- Hematologic: thrombotic disorder, including DVT, thrombophlebitis, retinal vascular thrombosis, cerebral embolism/stroke, pulmonary embolism.
- Edema.
- Local reactions: phlebitis, pain and edema at injection site.

DOSAGE

- For abnormal uterine bleeding: 25 mg IM or IV; repeat in 6-12 hours if necessary OR 25 mg IV q4-6 for 24 hours.
- For bleeding associated with uremia: 0.6 mg/kg/day IV (maximum 60 mg/day) for 5 days.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge.
- Reconstituted solution is stable for one month in the fridge.
- Compatible with NS, D5W, D10W.
- Incompatible with acid solutions.

MISCELLANEOUS

- Contains lactose.

REFERENCES

1, 5, 40, 81, 95, 208, 276, 277, 278.

Full revision 2014; limited revision 2015, 2016

NAME OF MEDICATION
ETANERCEPT

OTHER NAMES
Brenzys ®, Enbrel ®, Erelzi ®

CLASSIFICATION
Immunomodulator

INDICATIONS

- Moderately to severely active rheumatoid arthritis (RA) in adults.
- Moderately to severely active polyarticular juvenile idiopathic arthritis (JIA) in children aged 4 to 17 years.
- Active psoriatic arthritis in adults.
- Active ankylosing spondylitis in adults.
- Moderate to severe chronic plaque psoriasis in adults and severe chronic plaque psoriasis in children aged 4 to 17 years.
- Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) in adults to accelerate skin healing.

ADMINISTRATION

- **For pediatrics: ensure using a dosage form that allows a pediatric dose**.
- For multiple-use vial: reconstitute vial by adding very slowly 1 mL of supplied sterile bacteriostatic water for injection (contains benzyl alcohol) to obtain a clear and colourless solution of 25 mg/mL. Some foaming will occur and this is normal. Do NOT shake; swirl gently until dissolution (less than 10 minutes). Do NOT use vial adapter if multiple doses are to be withdrawn from the vial; use instead a 25-gauge needle for mixing and withdrawing the product and a 27-gauge needle for the injection.
- Do NOT filter reconstituted solution during preparation or administration.
- The single-use prefilled syringe and the single-use prefilled autoinjector contain 50 mg of etanercept and are ready to use; allow time to reach room temp (15-30 minutes) prior to injection without removing the needle cap.
- SC: sites of injection include the thigh, abdomen or upper arm, at least 1 inch from previous sites. Rotate injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, angioedema, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Cardiovascular: chest pain, CHF, hypertension.
- CNS: headache, dizziness, seizures, paresthesia.
- GI: nausea, vomiting, diarrhea, dyspepsia, abdominal pain.
- Dermatologic: vasculitis, SC nodules, lupus erythematosus, erythema multiforme.
- Hematologic: rare though can be severe: neutropenia, leukopenia, anemia, pancytopenia, aplastic anemia.
- Infections: reactivation or new onset of infections, including tuberculosis and hepatitis B.
- Asthenia.
- Local reactions: generally occur during 1st month of therapy, last 3-5 days and do not require discontinuation of therapy: erythema, pruritus, pain, swelling, bleeding, bruising. Redness at previous injection sites.

DOSAGE

Adults:
- RA, psoriatic arthritis, ankylosing spondylitis: 50 mg SC per week.
- Plaque psoriasis: 50 mg SC twice a week (administered 3 or 4 days apart) for 3 months then 50 mg SC per week, although some patients may still need 50 mg SC twice a week as their maintenance dose.
- SJS and TEN: if patient weight is 65 kg or less: 25 mg SC twice a week. If patient weight is greater than 65 kg: 50 mg SC twice a week. Continue until skin lesions are healed.

Pediatrics:
- JIA: 0.8 mg/kg (maximum 50 mg per dose) SC once weekly OR 0.4 mg/kg (maximum 25 mg per dose) SC twice weekly given 72-96 hours apart.
- Plaque psoriasis: 0.8 mg/kg (maximum of 50 mg per dose) SC once weekly.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials, prefilled syringes and prefilled autoinjectors in the fridge protected from light; do not freeze. Do not shake.
- Enbrel ® and Brenzys ®: vials, prefilled syringes and prefilled autoinjectors are stable for 60 days at up to 27°C and protected from direct sunlight and humidity.
- Erelzi ®: prefilled syringes and prefilled autoinjectors are stable for 28 days at room temp (20-25°C) and protected from light and heat.
- Enbrel ®: reconstituted solution prepared with the provided diluent may be stored in the original vial for up to 14 days in the fridge and 12 hours at room temp.

MISCELLANEOUS

- Brenzys ® and Erelzi ® are classified as being biosimilar to Enbrel ®; therefore, these products are not interchangeable.
- Etanercept should not be administered in patients with active infection.
- Do not administer live vaccines while on etanercept.
- Prefilled syringes and prefilled autoinjectors contain latex (Enbrel ® only).

REFERENCES

1, 5, 82, 135, 556.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2020, 2021

NAME OF MEDICATION
ETHACRYNIC ACID

OTHER NAMES
Sodium Edecrin ®, Sodium ethacrynate

CLASSIFICATION
Diuretic

INDICATIONS

- Adjunct in treatment of acute pulmonary edema.
- For edema due to congestive heart failure, nephrotic syndrome or for hepatic cirrhosis with ascites when oral administration is not possible.
- When edema has become refractory to other agents.

ADMINISTRATION

- Reconstitute each 50 mg vial with 50 mL of NS to obtain a solution of 1 mg/mL. Reconstitution with D5W is permitted if pH is above 5; otherwise, may result in a hazy or opalescent solution which should not be used.
- IV direct: physician or RN. Administer at a rate of 10 mg/min into the tubing of a freely running compatible IV solution.
- Intermittent IV infusion: infuse over 20-30 minutes.
- In case dose is repeated, use alternate administration site to avoid thrombophlebitis.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, Henoch-Schönlein purpura (vasculitis).
- Cardiovascular: hypotension (from excessive diuresis), tachycardia.
- GI: anorexia, malaise, abdominal discomfort or pain, nausea, vomiting, diarrhea (discontinue if severe), dysphagia, dry mouth, thirst, GI bleeding (especially if patient is heparinized), acute necrotizing pancreatitis.
- CNS: apprehension, headache, paresthesia, chills, vertigo, confusion.
- Hematology: rare: thrombocytopenia, neutropenia and agranulocytosis.
- Hepatic: increased LFT, jaundice.
- Otic: tinnitus, sense of fullness in ear; temporary or permanent deafness (especially if given with ototoxic drugs (such as aminoglycosides) or in patients with renal impairment).
- Hyperglycemia, hyponatremia, hypokalemia, hypocalcemia, hypomagnesemia, hypochloremic alkalosis; uricosuria, hyperuricemia, fever, hematuria, blurred vision, weakness, muscle cramps, fatigue.
- Local reactions: irritation, pain, thrombophlebitis.

DOSAGE

- Adults: 0.5-1 mg/kg/dose IV; usual dose: 50 mg; maximum dose: 100 mg.
- Pediatrics (children only, not for use in infants): 1 mg/kg/dose IV; maximum dose: 100 mg.
- Repeat doses not routinely recommended; however, if needed, repeat q8-12h.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution is stable for 24 hours from preparation.
- Compatible with D5W, NS, D5-NS, Ringer's and RL.
- Stable for 14 days at room temp or 22 days in fridge in NS at a concentration of 1 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Potassium supplements should be used concomitantly in cirrhotic, nephrotic, or digitalized patients.

REFERENCES

1, 5, 6, 40, 82, 208, 279.

Full revision 2014; limited revision 2015, 2020

NAME OF MEDICATION
ETOMIDATE

OTHER NAMES
Tomvi ™

CLASSIFICATION
General anesthetic, Sedative

INDICATIONS

- For induction of general anesthesia.
- As a supplement of subpotent anesthetic agents during anesthesia for short operative procedures.
- For induction during rapid sequence intubation.

ADMINISTRATION

- **Ensure premedication has been administered prior to etomidate administration; refer to Dosage section.**
- IV direct: physician trained in anesthesiology; ***respiratory support***. Administer undiluted over 30-60 seconds; avoid administration into small veins.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis and anaphylactoid reactions, Stevens-Johnson syndrome.
- Cardiovascular: hypertension, hypotension, arrhythmias, tachycardia, bradycardia.
- GI: nausea, vomiting.
- CNS: dyskinesia/transient skeletal muscle movements, such as myoclonus; these may be minimized with **premedication** of fentanyl with diazepam; refer to Dosage section.
- Respiratory: apnea and respiratory depression with high doses; hypoventilation, hyperventilation, cough, stridor, laryngospasm, hiccups, snoring.
- Endocrine and metabolic: decreased plasma cortisol and aldosterone levels persisting for 6-8 hours following administration and unresponsive to ACTH.
- Local reactions: transient venous pain at injection site.

DOSAGE

- **Premedication: small IV doses of fentanyl and diazepam 1 to 2 minutes before induction.**
- Induction of general anesthesia: 0.3 mg/kg (range: 0.2-0.6 mg/kg) IV, with a maximum of 60 mg as a total dose.
- As a supplement of subpotent anesthetic agents: indivudialize dosage which should be lower than the original induction dose.
- Induction during rapid sequence intubation: 0.3 mg/kg IV; if patient unstable: 0.15-0.2 mg/kg IV.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C.
- Compatible with D5W and NS.

MISCELLANEOUS

- Following induction, onset of hypnosis occurs within 1 minute; duration of action is 7 minutes.
- Should be used with caution in critically ill patients, including septic patients.
- Etomidate has minimal cardiovascular and respiratory effects and provides a greater hemodynamic stability compared to propofol and barbiturates.
- Decreases cerebral blood flow and lowers intracranial pressure.
- Does not cause histamine release.
- Etomidate exhibits no analgesic activity.
- Tomvi ™ contains propylene glycol.

REFERENCES

1, 5, 40, 95, 135, 280, 282.

Full revision 2021

NAME OF MEDICATION
ETOPOSIDE

OTHER NAMES
VePesid ®, VP-16

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of lung cancer (small cell, non-small cell), malignant lymphoma and testicular cancer.
- Can also be used in combination with other antineoplastics for a variety of tumour types.

ADMINISTRATION

- Intermittent IV infusion:
  - Dilute (preferred) in NS or D5W to obtain a final concentration of 0.2-0.4 mg/mL; do not exceed 0.4 mg/mL as more concentrated solutions in NS or D5W will lead to crystal formation. Use non-DEHP containers and non-DEHP tubing with an in-line 0.22 micron filter. Infuse over at least 30-60 minutes.
  - High doses for conditioning prior to bone marrow transplantation: At TOH, may be administered undiluted from a non-DEHP container, via a silicone central line, without a filter. Consult protocol for administration time.
- Continuous IV infusion: dilute in NS or D5W to obtain a final concentration of 0.2-0.4 mg/mL; do not exceed 0.4 mg/mL as more contentrated solutions in NS or D5W will lead to crystal formation. Use containers without DEHP and non-DEHP tubing with an in-line 0.22 micron filter.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid reaction (rare), rash, urticaria, pruritus.
- Cardiovascular: hypotension (associated with rapid IV administration of less than 30 minutes and with high doses).
- GI: nausea, vomiting, mucositis, esophagitis, anorexia, stomatitis, constipation, abdominal pain, diarrhea.
- CNS: headache, vertigo, seizures, confusion, somnolence, peripheral neuropathy.
- Dermatologic: palmar erythema (with high doses), radiation recall reaction (rare).
- Hematologic: anemia, thrombocytopenia, neutropenia, granulocytopenia, leukopenia.
- Weakness, fatigue, fever; metabolic acidosis and increased LFT at high doses.
- Local reactions: phlebitis at site of injection.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 35-150 mg/m^2/day IV for 1-5 days every 3-7 weeks.
- 1800-2400 mg/m^2/cycle as continuous IV infusion.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 50	50-10	less than 10
Dose	100%	75%	50% or discontinue

- Dosage in hepatic impairment:

Serum bilirubin (mcmol/L)	less than 25	25-50	50-85	greater than 85
Dose	100%	50%	25%	do not administer

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp; protect from light.
- Stable for 96 hours at room temp in NS or D5W at 0.2 mg/mL in polyethylene containers.
- Stable for 22 days at room temp or 14 days in the fridge in NS at 0.2 mg/mL in glass containers.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY (Cont.)

- Stable for 2 days at room temp or 7 days in the fridge in NS at 0.3 mg/mL in glass containers.
- Stable for 6 hours at room temp in NS or D5W at 0.4 mg/mL in polyethylene containers.
- Stable for 2 days in the fridge in NS at 0.4 mg/mL in glass containers.
- Stable for 8 hours at room temp in RL at 0.2 or 0.4 mg/mL in glass containers.
- Polysorbate 80 from the etoposide formulation can cause leaching of DEHP from PVC containers and tubing; therefore, non-DEHP containers and tubing are recommended for preparation and administration of etoposide solutions.

MISCELLANEOUS

REFERENCES

1, 5, 129, 165, 208, 287.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
EXENATIDE

OTHER NAMES
Bydureon ®, Byetta ®

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of type 2 diabetes mellitus; Byetta ® is used in combination with insulin glargine, metformin and/or a sulfonylurea, and Bydureon ® is used as monotherapy, in combination with metformin and/or a sulfonylurea, or in combination with basal insulin with or without metformin.

ADMINISTRATION

Bydureon ®:
- Let the pen at room temp for 15 minutes. Reconstitute the 2 mg dose with the diluent provided within the pen. Mix well by tapping until a uniform cloudy suspension is obtained. Refer to manufacturer's instructions for more details.
- SC: in the upper arm, thigh, or abdomen; press the injection button with the thumb until hearing a click; hold in place for 10 seconds. Rotate injection sites.

Byetta ®:
- Ready-to-use solution within the pen.
- SC: in the upper arm, thigh or abdomen.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: generalized pruritus and/or urticaria, rash, angioedema, anaphylaxis, antibody formation.
- GI: nausea, vomiting, dyspepsia, gastroesophageal reflux disease, diarrhea, constipation, pancreatitis.
- CNS: dizziness, nervousness, headache.
- Hypoglycemia (risk increased if used in combination with insulin glargine or a sulfonylurea; refer to Miscellaneous section); rapid weight loss.
- Renal dysfunction.
- Asthenia.
- Local reactions: pain, erythema, rash, itching, bruising, bleeding; in addition, for Bydureon ®: induration, nodules, abscess, cellulitis, necrosis.

DOSAGE

- Bydureon ®: 2 **mg** SC once weekly at any time of the day, with our without meals.
- Byetta ®: 5 **mcg** SC BID. If required, after one month, dose can be increased to 10 **mcg** SC BID. Administer dose within 60 minutes before the morning and evening meal (or before the two main meals of the day, at least 6 hours apart). Do not administer after a meal.
- Dosage in renal impairment: use cautiously in patients with moderate renal impairment (CrCl 30-50 mL/min) and in those with renal transplantation. Not to be used in patients with end-stage renal disease, severe renal impairment (CrCl less than 30 mL/min), or when receiving dialysis.
- Dosage in hepatic impairment: no dosage adjustment expected as drug is mainly excreted by the kidney.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled pens between 2-8°C. Protect from light. Do not freeze.
- Bydureon ®: before reconstitution, the pen may be kept at room temp (not to exceed 30°C) for 28 days. Once reconstituted, solution should be injected immediately and the pen discarded.
- Byetta ®: when in use, the pen may be kept between 2-25°C for 30 days; do not store the pen with the needle attached.
- Do not mix with insulin.

…/Cont.

NAME OF MEDICATION
EXENATIDE

OTHER NAMES
Bydureon ®, Byetta ®

CLASSIFICATION
Antidiabetic

…/Cont.

MISCELLANEOUS

- Exenatide is a glucagon-like peptide-1 (GLP-1) receptor agonist, an incretin mimetic agent.
- Exenatide should not be used in the treatment of type 1 diabetes mellitus or diabetic ketoacidosis.
- When exenatide is added to insulin glargine or a sulfonylurea, consider decreasing the dose of the latter agents to reduce the risk of hypoglycemia.
- When switching from Byetta ® to Bydureon ®, increased glycemia can occur and should improve within 2 weeks.

REFERENCES

1, 2, 5, 95, 135.

Full revision 2014; limited revision 2015, 2016, 2017, 2019

NAME OF MEDICATION
FACTOR VIIA (RECOMBINANT)

OTHER NAMES
Eptacog alfa (activated), NiaStase RT ®

CLASSIFICATION
Coagulation factor

Do NOT confuse factor VIIA with other coagulation factors.
This monograph is specific to factor VIIA.

INDICATIONS

- Treatment of bleeding episodes (including treatment and prevention of those occurring during and after surgery) in patients with hemophilia A or B with inhibitors to factors VIII or IX.

ADMINISTRATION

- For IV use: powder vial and diluent (histidine solvent) should be at room temp before reconstitution. If not at room temp, hold vials to bring them to room temp. Reconstitute powder vial by adding 1.1 mL of diluent for the 1 mg vial, 2.1 mL of diluent for the 2 mg vial, 5.2 mL of diluent for the 5 mg vial and 8.1 mL of diluent for the 8 mg vial. Inject diluent slowly along wall of vial; Do NOT inject diluent directly onto the powder. Gently swirl until dissolved.
- IV direct: physician or RN. Administer over 2-5 minutes.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0048 (Blood products and blood derivatives: Coagulation factor concentrate – Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, anaphylaxis (rare).
- Cardiovascular: hypertension, hypotension, edema.
- GI: nausea, vomiting.
- CNS: headache.
- Hematologic: thrombotic events (arterial and venous), hemorrhage.
- Fever, pain.
- Local reactions: redness and pain at injection site.

DOSAGE

- Dosage will depend on the nature and severity of the coagulation defect, the site of bleeding and the type of intervention.
- 35-90 mcg/kg IV q2h until achievement of hemostasis; dosage interval may then be extended from 2 to 6 hours.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store powder vial and diluent at room temp or in the fridge (2-25°C). Protect powder and diluent from light. Do not freeze.
- Reconstituted vials are stable for 6 hours at room temp or 24 hours in the fridge.
- The reconstituted solution is stable for 24 hours at 25°C in polypropylene syringes.
- Do not mix with infusion solutions.

MISCELLANEOUS

- Known hypersensitivity to mouse, hamster or bovine proteins is a contraindication to the use of the product.

REFERENCES

2, 5, 95, 135, 571.

Full revision 2014; limited revision 2017, 2019, 2020, 2021

NAME OF MEDICATION
FACTOR VIII (RECOMBINANT)

OTHER NAMES
Advate ®, Adynovate, Antihemophilic factor, Eloctate ®, Jivi ®, Kovaltry ®, Obizur ®

CLASSIFICATION
Coagulation factor

Do NOT confuse factor VIII with other coagulation factors.
This monograph is specific to factor VIII.

INDICATIONS
- In patients with hemophilia A for the prevention and treatment of hemorrhagic episodes.
- Acquired hemophilia with low levels of factor VIII inhibitors.

ADMINISTRATION
- Prior to reconstitution, bring diluent and concentrate to room temp. Reconstitute the concentrate with diluent following manufacturer's instructions.
- IV direct: physician or RN. Rate as per manufacturer's recommandation.
- Intermittent IV infusion, Continuous IV infusion: transfer in a PVC reservoir. Infusion rate as per manufacturer's recommendation. Must be administered by an infusion pump.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0048 (Blood products and blood derivatives: Coagulation factor concentrate – Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: fever, chills, facial swelling, flushing, rash, hives, hypotension, wheezing, anaphylaxis (rare).
- CNS: headache.
- Local reactions: burning, pruritus and erythema at injection site.

DOSAGE
- Should be carefully individualized based on coagulation studies performed prior to and during therapy.
- Dosage depends upon degree of deficiency, the antihemophilic factor (AHF) level desired, the weight of the patient, severity of bleeding and presence of factor VIII inhibitors. Refer to manufacturer's recommendation.
- Approximate dose required to achieve a particular increase in AHF level:
  international units required = body weight (kg) X 0.5 X desired AHF increase (in % of normal).
- Dose of 1 international unit of factor VIII/kg body weight administered raises the plasma VIII activity by approximately 2%.
- Consult specialized references for more detailed dosing information.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store between 2-8°C. Protect from light. Do not freeze.
- Adynovate: stable for 3 months at 30°C.
- Advate ® and Eloctate ®: stable for 6 months at 30°C.
- Jivi ®: stable for 6 months at 25°C and for 3 months at 30°C.
- Kovaltry ®: stable for 12 months at 25°C.
- Reconstituted solutions for IV direct injection should be kept at room temp and administered within 3 hours of reconstitution, except for Eloctate ® which can be administered within 6 hours of reconstitution.
- Reconstituted solutions of Advate ® for IV infusion are stable for 12 hours at room temp in a PVC container.

MISCELLANEOUS
- Contraindicated in patients with known hypersensitivity to mouse or hamster proteins, except for Eloctate ® which does not contain any animal proteins.
- One international unit is defined as the AHF activity present in 1 mL of normal, fresh-pooled plasma.
- Adynovate, Eloctate ® and Jivi ® have a prolonged circulating half-life compared to Advate ®, Kovaltry ® and Obizur ®.

REFERENCES
1, 5, 23, 40, 95, 135.

Full revision 2014; limited revision 2015, 2017, 2019, 2020, 2021

NAME OF MEDICATION **FACTOR VIII/ VON WILLEBRAND FACTOR COMPLEX (HUMAN)**

OTHER NAMES
Humate-P ®, Wilate ®

CLASSIFICATION
Coagulation factor

Do NOT confuse factor VIII/von Willebrand factor complex with other coagulation factors. This monograph is specific to FACTOR VIII/ VON WILLEBRAND FACTOR COMPLEX.

INDICATIONS
- Hemophilia A: treatment and prophylaxis of bleeding in adult patients with congenital or acquired Factor VIII (FVIII) deficiency.
- Von Willebrand disease (VWD): for adults and pediatrics: 1) treatment and prevention of spontaneous and traumatic bleeding episodes unresponsive to desmopressin therapy or if contraindicated; 2) for prevention or treatment of excessive bleeding during or after surgery.
- Refer to manufacturer's monograph for specific indications, depending on brand used.

ADMINISTRATION
- **Humate-P** ®: ensure that Humate-P ® vial and its diluent have reached room temp. Using the transfer set provided, reconstitute each vial of 250 international (int.) units FVIII/600 int. units von Willebrand factor (VWF) and 500 int. units FVIII/1200 int. units VWF with 5 and 10 mL respectively of diluent provided, to obtain a final concentration of 50 int. units FVIII/120 int. units VWF per mL. Reconstitute the 1000 int. units FVIII/2400 int. units VWF vial with 15 mL of the diluent provided to obtain a final concentration of 67 int. units FVIII/160 int. units VWF per mL. Swirl vial to dissolve; do NOT shake. The solution should be clear or slightly opalescent. Transfer dose into a syringe. Refer to manufacturer's instructions for a more detailed procedure.
- **Wilate ®:** ensure that Wilate ® vial and its diluent have reached room temp. Using the transfer set provided, reconstitute the 500 international (int.) units FVIII/500 int. units von Willebrand factor (VWF) vial and the 1000 int. units FVIII/1000 int. units VWF vial with 5 mL and 10 mL respectively of the provided diluent, to obtain a final concentration of 100 int. units FVIII/100 int. units VWF per mL. Swirl vial to dissolve. The reconstituted solution should be clear or slightly opalescent, colourless or slightly yellow. Transfer dose into a syringe. Refer to manufacturer's instructions for a more detailed procedure.
- Intermittent IV infusion: slowly administer the syringe content at a rate not exceeding 4 mL/min for Humate-P ® and at a rate not exceeding 2-3 mL/min for Wilate ®. Must be administered by an infusion pump.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0048 (Blood products and blood derivatives: Coagulation factor concentrate – Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, urticaria, pruritus, angioedema, wheezing, chest tightness, hypotension, anaphylaxis.
- Cardiovascular: edema (facial and peripheral), thrombotic events.
- GI: nausea, vomiting.
- CNS: dizziness, headache.
- Paresthesia, chills.
- Risk of transmission of infectious agents, including viruses, and theoretically, the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.
- Local reactions: pruritus and pain at injection site.

DOSAGE
Hemophilia:
- Dosage and duration of the substitution therapy depend on the severity of the FVIII deficiency, the location and extent of the bleeding, the patient's clinical condition, and the product selected; refer to manufacturer's monograph for specific dosage recommendations.
- As a general rule, 1 int. unit of FVIII coagulant activity per kg body weight will increase the circulating FVIII level by approximately 2 int. units/dL.

VWD:
- Dosage should be adjusted according to the extent and location of bleeding, the type of surgery, the patient's clinical condition, and the product selected; refer to manufacturer's monograph for specific dosage recommendations.

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Humate-P ®:
- Store Humate-P ® vials and diluent in the fridge or at room temp (2-25°C). Do not freeze.
- Humate-P ® solution should be used within 3 hours after its reconstitution; do not refrigerate once reconstituted.

Wilate ®:
- Store Wilate ® vials and diluent between 2-8°C. Protect from light. Do not freeze.
- Wilate ® and its diluent can be stored on a single occasion for up to 6 months at room temp and must not return to the fridge.
- Wilate ® reconstituted solution should be used immediately.

MISCELLANEOUS

REFERENCES

5, 40, 135.

New monograph 2017; limited revision 2019, 2020

NAME OF MEDICATION	FACTOR IX CONCENTRATE (HUMAN & RECOMBINANT)
OTHER NAMES Alprolix ®, Benefix ®, Immunine ® VH, Nonacog alfa, Rebinyn ®	CLASSIFICATION Coagulation factor

Do NOT confuse factor IX concentrate with other coagulation factors.
This monograph is specific to factor IX concentrate.

INDICATIONS
- Prevention and control of bleeding episodes caused by factor IX deficiency, also known as Hemophilia B or Christmas disease.

ADMINISTRATION
- Prior to reconstitution, bring diluent and concentrate to room temp. Do not exceed 37°C. Reconstitute according to manufacturer's instructions.
- IV direct: physician or RN. Rate as per manufacturer's recommendation.
- Intermittent IV infusion, Continuous IV infusion: rate as per manufacturer's recommendation. Must be administered by an infusion pump.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0048 (Blood products and blood derivatives: Coagulation factor concentrate – Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, urticaria, chills (rigors), flushing, angioedema, dyspnea, wheezing, tachycardia, hypotension, anaphylaxis.
- GI: nausea, abnormal taste.
- CNS: headache, dizziness.
- Risk of transmission of infectious agents, including viruses, and theoretically, the Creutzfeldt-Jakob disease agent, cannot be completely eliminated (human-derived product only: Immunine ® VH).
- Local reactions: pain or burning along injection site, phlebitis.

DOSAGE
- Individualized depending upon degree of deficiency, the desired level of deficient factor, weight of patient and severity of bleeding.
- Should be based on coagulation studies performed prior to surgery and at regular intervals during treatment.
- Units required: body weight (kg) X 1.2 unit/kg X desired factor IX increase (in % of normal).
- For specific dosing recommendations, consult manufacturer's information.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store Immunine ® VH between 2-8°C although stable at room temp for 3 months.
- Store Benefix ® between 2-30°C. Do no freeze.
- Store Alprolix ® and Rebinyn ® between 2-8°C although stable between 15-30°C for 6 months. Protect from light. Do not freeze.
- Following reconstitution:
  - Immunine ® VH: use immediately; do not refrigerate after reconstitution.
  - Benefix ® and Alprolix ®: use within 3 hours to avoid bacterial contamination. Do not refrigerate reconstituted solution.
  - Rebinyn ®: use within 4 hours when stored at room temp (up to 30°C) and within 24 hours when stored in the fridge.

MISCELLANEOUS
- Benefix ® and Rebinyn ® are contraindicated in patients with known hypersensitivity to hamster proteins.
- Immunine ® VH is contraindicated in patients with known hypersensitivity to heparin, disseminated intravascular coagulation or hyperfibrinolysis.
- One international unit of Factor IX corresponds to the activity of Factor IX present in 1 mL of fresh normal plasma.
- Alprolix ® and Rebinyn ® have a prolonged circulating half-life compared to Benefix ® and Immunine ® VH.

REFERENCES
1, 2, 5, 40, 95, 135.

Full revision 2014; limited revision 2015, 2016, 2017, 2019, 2020, 2021

NAME OF MEDICATION
FAMOTIDINE

OTHER NAMES
Pepcid ®

CLASSIFICATION
Histamine (H_2) antagonist

INDICATIONS

- Treatment of intractable gastric or duodenal ulcers.
- Treatment of hypersecretory conditions (e.g., Zollinger-Ellison Syndrome).
- For short-term use in patients who are unable to take oral medications.
- Stress ulcer prophylaxis in critical care patients.

ADMINISTRATION

- IV direct: physician or RN. Dilute 20 mg (2 mL) in 3 mL or 8 mL of SWFI, NS, D5W or other compatible solution to obtain a final volume of 5 mL or 10 mL, respectively. Administer at a maximum rate of 10 mg/min.
- Intermittent IV infusion: further dilute each 20 mg dose in 100 mL and infuse over 15-30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, angioedema, bronchospasm, anaphylaxis.
- GI: constipation, diarrhea.
- CNS: headache, dizziness.
- Local reactions: transient irritation at the injection site.

DOSAGE

- Adults: 20 mg IV q12h. For hypersecretory conditions: 20 mg IV q6h; higher doses may be needed; adjust to clinical needs.
- Pediatrics (1 to 16 years): 0.25 mg/kg IV q12h, up to 40 mg/day.
- Dosage in renal impairment: when CrCl is 50 mL/min or less, reduce dose by half or increase dosing interval to 36 to 48 hours. Not removed by hemodialysis.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge and protect from light. Do not freeze.
- Stable for 15 days at room temp or 63 days in the fridge in NS or D5W at a concentration of 0.2 mg/mL in PVC bags.
- Stable for 15 days at room temp in D5W or NS at a concentration of 0.2 mg/mL when stored in polypropylene syringes.
- Stable for 8 weeks in the freezer in D5W, NS or SWFI at a concentration of 2 mg/mL when stored in polypropylene syringes.
- Stable for 14 days in the fridge in D5W, NS or SWFI at a concentration of 2 mg/mL when stored in plastic syringes.
- Compatible with NS, D5W, D10W, RL.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 82, 95, 135.

Full revision 2014; limited revision 2015, 2016

NAME OF MEDICATION
FAT EMULSIONS

OTHER NAMES
ClinOleic ®, Fatty acids, Intralipid ®, Lipid emulsions, Liposyn ®, SMOFlipid ®

CLASSIFICATION
Nutritional supplement

INDICATIONS

- As a source of calories and essential fatty acids for patients requiring total parenteral nutrition (TPN) over extended periods (over 5 days usually).
- Prevention or treatment of essential fatty acid deficiency (Intralipid ®).
- Treatment of local anesthetic-induced cardiac arrest unresponsive to conventional resuscitation.

ADMINISTRATION

- IV direct: physician only. For treatment of local anesthetic toxicity; administer bolus dose over 1 minute (20% solution).
- Intermittent IV infusion, Continuous IV infusion: in a peripheral or central vein as solution is isotonic.
  Rate:
  a) as a source of calories and essential fatty acids for patients requiring TPN and for prevention of fatty acid deficiency (Intralipid ®):
     - Adults, 10% solution: 1 mL/min for the first 15-30 minutes. If no untoward effects, increase rate to 2 mL/min.
     - Adults, 20% solution: 0.5 mL/min for the first 15-30 minutes. If no untoward effects, increase rate to 1 mL/min.
     - Pediatrics, 10% solution: 0.1 mL/min for the first 10-15 minutes. If no untoward effects, increase rate to 1 mL/kg/hr. Do not exceed 100 mL/hr.
     - Pediatrics, 20% solution: 0.05 mL/min for the first 10-15 minutes. If no untoward effects, increase rate to 0.5 mL/kg/hr. Do not exceed 50 mL/hr.
  b) as a source of calories and essential fatty acids for patients requiring TPN (ClinOleic ®, SMOFlipid ®):
     - ClinOleic ® (adults only): 0.5 mL/min for the first 10 minutes. If no untoward effects, increase rate gradually to desire rate after 30 minutes. Do NOT exceed 0.75 mL/kg/hr. Usual infusion time is between 12-24 hours per day.
     - SMOFlipid ® (adults only): increase gradually until a maximum rate of 0.75 mL/kg/hr. Usual infusion time is between 12-24 hours per day.
  c) for treatment of local anesthetic toxicity: refer to Dosage section.
- Must be administered by an infusion pump.
- A filter is not necessary when fat emulsions are administered as a single agent. A filter of 1.2 micron size may be required if fat emulsions are part of a 3-in-1 admixture. Consult manufacturer's monograph for specific requirements.
- Use a non-DEHP administration set for all brands of fat emulsions.

POTENTIAL ADMINISTRATION HAZARDS

- GI: nausea, vomiting.
- Fever, chills, headache.
- Back or chest pain with dyspnea or cyanosis.
- Hyperlipidemia.

DOSAGE

Adults:
- TPN (Intralipid ®): initial dose: 1 g/kg/day IV or 500 mL of 10% or 20% IV on the first day. May increase dose gradually by 1 g/kg/day if needed.
- TPN (ClinOleic ®, SMOFlipid ®): usual dose of 1-2 g/kg/day (5-10 mL/kg/day) IV.
- Do not exceed 60% of patient's total caloric intake or 2.5-3 g/kg/day.
- Essential fatty acid deficiency (Intralipid ®): prevention: 2-4% of total caloric intake IV (e.g., 500 mL of a 10% solution twice a week). Treatment: 8-10% of total caloric intake IV; may infuse up to once daily.

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DOSAGE (Cont.)

- Treatment of local anesthetic toxicity: using a 20% solution, initial dose of 1.5 mL/kg IV over 1 minute, followed by a continuous infusion of 0.25 mL/kg/min IV. May repeat dose as a bolus once or twice (leave 5 minutes between boluses) and/or increase rate of the continuous infusion to 0.5 mL/kg/min if re-emergence or persistence of hemodynamic instability. Do not exceed a maximum cumulative dose of 12 mL/kg.

Pediatrics (Intralipid ®):
- TPN: 0.5-1 g/kg/day IV. May increase dose gradually by 0.5 g/kg/day to a maximum of 3-4 g/kg/day.
- Essential fatty acid deficiency: prevention: 2-4% of total caloric intake IV. Treatment: 8-10% of total caloric intake IV.

Premature infants (Intralipid ®):
- 0.5 g/kg/day IV using the 20% solution (avoid using the 10% solution in premature infants as its greater phospholipid load per g of triglyceride has been associated with a greater accumulation of plasma lipids). May increase dose gradually if needed to a maximum of 3-3.5 g/kg/day.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Intralipid ®, SMOFlipid ®: store up to 25°C. Do not freeze.
- ClinOleic ®: store between 15-30°C. Do not freeze.
- Exposure to light, particularly phototherapy light, has been associated with increased lipid oxidation; the clinical significance remains to be determined.
- Do not use if emulsion appears to be layering out.
- Compatible through a Y-site connection with amino acids and dextrose.

MISCELLANEOUS

- Use with caution in pulmonary disease, severe liver disease, blood coagulation disorders, anemia or when there is a danger of fat embolism.
- Use with caution in jaundiced or premature infants as fatty acids may displace bilirubin bound to albumin.
- Contra-indicated in pancreatitis with hyperlipemia, pathologic hyperlipidemia, lipoid nephrosis, and severe egg or soybean allergies.
- Monitor serum triglycerides and free fatty acids.
- A 10% solution provides 1.1 kcal/mL and a 20% solution 2 kcal/mL.
- Fat emulsions have also been used in the management of overdose with highly lipophilic cardiotoxic medications other than local anesthetics.
- When fat emulsions are used in the context of drug overdoses, collect blood samples for lab values <u>before</u> their administration. Indeed, fat emulsions may produce falsely elevated readings of albumin, magnesium and glucose (colorimetric method only for glucose); also amylase, lipase, phosphate, creatinine, total protein, alanine aminotransferase, creatine kinase and bilirubin may not be measurable. A brief centrifugation of the blood sample may minimize the interference.

REFERENCES

1, 5, 40, 78, 82, 95, 135, 181, 185, 191, 268, 344, 345, 347, 390, 391, 513.

Full revision 2014; limited revision 2015, 2016, 2018, 2021

NAME OF MEDICATION	
FentaNYL	
OTHER NAMES	CLASSIFICATION
	Opioid analgesic

INDICATIONS

- An adjunct in the induction and maintenance of general and regional anaesthesia.
- As an anesthetic agent with oxygen and a skeletal muscle relaxant to provide anesthesia without the use of additional anesthetic agents in selected patients.
- For short-term analgesia during perioperative period.
- For short-duration minor surgery and in diagnostic procedures or treatments that require the patient to be awake or very lightly anesthetized (e.g., burn dressings, bronchoscopy, catheterization).
- For control of pain in palliative care and cancer patients.

ADMINISTRATION

- IV direct: physician or RN; ***respiratory support***. Administer undiluted or dilute dose to 10 mL with NS or D5W and give slowly over 1-5 minutes.
- Continuous IV infusion: at TOH, transfer 5 mg (100 mL from fentanyl 50 mcg/mL vials) to an empty sterile minibag (final concentration: 50 mcg/mL). Must be administered by an infusion pump.
- IM, SC injection, SC infusion (must be administered by an infusion pump).

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis (extremely rare). Pseudoallergic reactions due to histamine release (rare): sweating, flushing, warmness of the face, neck and upper thorax, pruritus, urticaria.
- Cardiovascular: bradycardia (**Antidote**: atropine), hypotension, hypertension.
- GI: nausea, vomiting, constipation, dry mouth.
- CNS: dizziness, mental clouding, sedation, confusion, drowsiness, headache.
- Respiratory depression.
- Skeletal and thoracic muscle rigidity associated with hypoventilation especially with rapid IV administration.
- Fatigue.

Antidote: naloxone (for respiratory depression).

DOSAGE

- Anesthetic Adjunct: Regional anesthesia: 50-100 mcg IM or IV when analgesia required.
  General anesthesia:
  - Minor surgery: 2 mcg/kg IV.
  - Moderate-major surgery: 2-20 mcg/kg IV followed by 25-100 mcg IV as necessary for maintenance.
  - Open heart/complicated surgery: 20-50 mcg/kg IV followed by dose of 25 mcg to one half the initial dose IV as necessary for maintenance.
- As a general anesthetic: 50-100 mcg/kg (up to 150 mcg/kg may be required in certain cases) IV with oxygen and a skeletal muscle relaxant.
- Analgesia in perioperative period: intermittent dosing: 50-100 mcg IM or IV q1-2h as needed. As a continuous IV infusion: 25-200 mcg/hr or 1-2 mcg/kg/hr. Titrate to effect.
- Analgesia in palliative care and cancer patients: usual dosage range can be exceeded and is titrated to effect.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from light. Protect from freezing.
- Stable for 48 hours at room temp in D5W and NS at concentration of 5 mcg/mL in glass or PVC containers.
- Stable for 30 days at room temp or in the fridge in NS at a concentration of 20 mcg/mL in PVC containers.

…/Cont.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 28 days at room temp (and exposed to light) or in the fridge (and protected from light), undiluted (50 mcg/mL) in polypropylene syringes or PVC bags.
- Stable for 93 days at 5°C (protected from light) or 22°C (exposed to light) in NS at a concentration of 10 mcg/mL or 50 mcg/mL (undiluted) in polyolefin bags.

MISCELLANEOUS

- Onset of action after IV administration is almost immediate; duration of 30-60 minutes.
- Caution in patients on monoamine oxidase inhibitors (risk of serotonin syndrome) or taking strong CYP3A4 inhibitors (risk of fentanyl toxicity).

REFERENCES

1, 4, 5, 33, 40, 95, 135, 241, 252, 457.

NAME OF MEDICATION
FERRIC GLUCONATE COMPLEX

OTHER NAMES
Ferrlecit ®, Sodium Ferric Gluconate complex

CLASSIFICATION
Hematinic

INDICATIONS

- Treatment of iron deficiency anemia in patients undergoing chronic hemodialysis who are receiving supplemental erythropoietin therapy.

ADMINISTRATION

- IV direct: physician or RN; undiluted solutions may be administered slowly at a rate up to 1 mL/min (12.5 mg/min).
- Intermittent IV infusion (at TOH, mandatory): dilute each 62.5 mg of elemental iron (5 mL) in at least 50 mL of NS. Infuse at a rate of 125 mg/hr. At TOH: always infuse at a maximum rate of 62.5 mg/hr.
- Observe patient for signs and symptoms of hypersensitivity during and for at least 30 minutes after administration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; rash, pruritus, wheezing, hypotension, anaphylaxis.
- Infusion-related reactions: hypotension associated with fatigue, malaise, lightheadedness, weakness, flushing or severe pain in the chest, back, flanks or groin with too rapid administration; not associated with drug hypersensitivity. Usually resolve within 1-2 hours; if symptomatic, administer volume expanders. Restart infusion when resolved.
- GI: nausea, vomiting, diarrhea.
- CNS: seizures.
- Local reactions at injection site.

DOSAGE

- Initial: 125 mg of elemental iron (10 mL) IV during the dialysis session, up to a cumulative dose of 1000 mg of elemental iron administered over 8 sequential dialysis sessions (At TOH, first dose should not exceed 62.5 mg).
- Maintenance: lowest dose necessary to maintain the target levels of hemoglobin, hematocrit and iron.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp. Protect from light. Do not freeze.
- Must only be mixed with NS; use immediately after dilution in NS.
- Do not mix with other medications or add to parenteral nutrition solutions.
- Stable for 24 hours at room temp in NS and protected from light at a concentration of 0.25 mg/mL (62.5 mg/250 mL NS) in a plastic bag.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- Not dialyzable.
- Ampoule contains 62.5 mg/5 mL of elemental iron.

REFERENCES

1, 5, 40, 135, 348.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
FIBRINOGEN CONCENTRATE (HUMAN)

OTHER NAMES
Fibryga ®, RiaSTAP ®

CLASSIFICATION
Blood product, Hemostatic

INDICATIONS

- To control acute bleeding episodes and perioperative prophylaxis in patients with congenital fibrinogen deficiency, including afibrinogenemia and hypofibrinogenemia.

ADMINISTRATION

- Ensure vials of fibrinogen and diluent (SWFI) have reached room temp. Following manufacturer's instructions and using an appropriate transfer device or syringe, reconstitute fibrinogen vial with 50 mL of SWFI to obtain an approximate concentration of 20 mg/mL. Gently swirl the vial to ensure the product is fully dissolved (within 5 to 10 minutes). Do NOT shake the vial to avoid formation of foam. After reconstitution, the solution should be colourless and clear to slightly opalescent. Use a syringe and the provided filter to withdraw reconstituted solution from the vial.
- IV intermittent infusion: infuse directly from the reconstituted vial (RiaSTAP ®) or syringe (Fibryga ®) at a rate not exceeding 100 mg/min (5 mL/min) through a separate IV line. (At TOH: rinse infusion line with NS after administration).
- Consult Corporate nursing standard operating procedure C-SOP NUR 0048 (Blood products and blood derivatives: Coagulation factor concentrate – Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, chest tightness, wheezing, dyspnea, hypotension, tachycardia, shock, anaphylaxis.
- Cardiovascular: thrombotic events (myocardial infarction, pulmonary embolism, deep vein thrombosis, arterial thrombosis).
- GI: nausea, vomiting.
- CNS: headache.
- Fever, chills.
- Risk of transmission of infectious agents, including viruses, and theoretically, the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.

DOSAGE

- Dose and frequency of administration are based on extent of bleeding, laboratory values and clinical condition. Determine fibrinogen level before and after the dose. Usual target is 1 g/L (100 mg/dL) for minor bleeding (e.g., epistaxis, IM bleeding, or menorrhagia) or minor surgery and 1.5 g/L (150 mg/dL) for major bleeding (e.g., head trauma, or intracranial hemorrhage) or major surgery.
- If fibrinogen level is unknown:
  - RiaSTAP ®: 70 mg/kg IV until target level is obtained.
  - Fibryga ®: 60 mg/kg IV until target level is obtained.
- If level of fibrinogen is known:
  - RiaSTAP ™: calculate dose by using this equation:

    $$\text{Dose of fibrinogen (mg/kg)} = \frac{\text{Target level (g/L)} - \text{measured level (g/L)}}{0.017}$$

  - Fibryga ®: calculate dose by using this equation:

    $$\text{Dose of fibrinogen (mg/kg)} = \frac{\text{Target level (g/L)} - \text{measured level (g/L)}}{0.018}$$

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store RiaSTAP ® vials between 2-8°C and Fibryga ® vials between 2-25°C. Protect from light. Do not freeze.
- RiaSTAP ®: reconstituted solution is stable for 8 hours at room temp (20-25°C). Do not freeze reconstituted solution.
- Fibryga ®: reconstituted solution is stable for 24 hours at 25°C.

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MISCELLANEOUS

- Emergency drugs (epinephrine, antihistamines, corticosteroids) should be immediately available for the treatment of hypersensitivity reactions.
- Each single-use RiaSTAP ® vial contains between 900-1300 mg of fibrinogen and each single-use Fibryga ® vial contains 1000 mg of fibrinogen.

REFERENCES

1, 5, 40, 135, 480, 481.

New monograph 2017; limited revision 2020, 2021

NAME OF MEDICATION
FILGRASTIM

OTHER NAMES
G-CSF, Grastofil ®, Neupogen ®, Nivestym ™

CLASSIFICATION
Hematopoietic

Do NOT confuse filgrastim with pegfilgrastim.
This monograph is specific to FILGRASTIM.

INDICATIONS

- To decrease the incidence of infection (febrile neutropenia) in patients with non-myeloid malignancies receiving myelosuppressive antineoplastic agents.
- To decrease duration of neutropenia, fever, antibiotic use and hospitalization following induction and consolidation treatment for acute myeloid leukemia.
- To decrease duration of neutropenia and related clinical problems in post bone marrow transplant, peripheral stem cell mobilization, HIV-associated neutropenia and drug-induced neutropenia.
- To increase neutrophil counts and to reduce incidence and duration of infection in patients with congenital, cyclic or idiopathic chronic neutropenia.

ADMINISTRATION

- Do NOT shake vigorously.
- Intermittent IV infusion: dilute in 50-100 mL of D5W to obtain a final concentration between 5 and 15 mcg/mL; administer over 15-30 minutes or 4 hours (refer to Compatibility, Stability section).
- Continuous IV infusion: dilute in 50-100 mL of D5W and infuse over 24 hours (refer to Compatibility, Stability section).
- SC infusion: dilute in 10-50 mL of D5W and infuse over 24 hours (refer to Compatibility, Stability section). Must be administered by an infusion pump.
- SC: outer area of upper arms, front of middle thighs, abdomen (except 2 inches around the navel), buttocks. Rotate injection site.
- Do NOT administer in the 24-hour period prior to or after chemotherapy administration, bone marrow infusion or stem cell infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; rash, urticaria, facial edema, wheezing, dyspnea, hypotension, tachycardia, anaphylaxis.
- Cardiovascular: capillary leak syndrome manifested as hypotension, edema, hypoalbuminemia, hemoconcentration and circulatory shock.
- Hematologic: marked leukocytosis and thrombocytopenia (rare).
- Bone pain (most common side effect of filgrastim), back pain.
- Severe sickle cell crisis in susceptible patients.
- Spleen rupture (rare).
- Local reactions: erythema, swelling or pruritus at site of injection; change site of administration frequently to minimize.

DOSAGE

Adults:
- Myelosuppressive chemotherapy: initial dose of 5 mcg/kg/day as a SC injection, short IV infusion (over 15-30 minutes), continuous SC or continuous IV infusion. Doses may be increased in increments of 5 mcg/kg/day for each chemotherapy cycle according to the previous cycle's neutrophil response. Therapy should be discontinued when the neutrophil count reaches 10×10^9/L after the nadir.
- Post bone marrow transplant: initial dose of 10 mcg/kg/day as an IV infusion over 4-24 hours or as a continuous 24-hour SC infusion.
- Peripheral stem cell mobilization: 10 mcg/kg/day as a SC injection (in the morning if treatment includes plerixafor) or continuous 24-hour infusion, to start at least 4 days before the first leukapheresis procedure and continue up to the last leukapheresis procedure.

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DOSAGE (Cont.)

- HIV-associated neutropenia: initial dose of 300 mcg SC 3 times a week.
- Idiopathic or cyclic neutropenia: initial dose of 5 mcg/kg/day SC (single or divided dose).
- Severe chronic neutropenia: initial dose of 12 mcg/kg/day SC (single or divided dose).
- Refer to manufacturer's product monograph for more dosage information.

Pediatrics:
- Oncology patients: 5 mcg/kg/day SC.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Neupogen ®, Grastofil ® and Nivestym ™:
- Store in the fridge; do not shake since bubbles or foam can form. Protect from light.
- Vials and prefilled syringes are stable for 14 days (Neupogen ®) or 15 days (Grastofil ®, Nivestym ™) at room temp (maximum 25°C); discard product if not used during this time period.
- At concentrations of 5-15 mcg/mL in D5W, human albumin must first be added to the D5W at a final concentration of 2 mg/mL to minimize filgrastim adsorption to infusion containers and equipment.
- Not stable in D5W at concentrations less than 5 mcg/mL (even in presence of albumin) or at any concentration in NS.
- Neupogen ® and Nivestym ™: when diluted in D5W or in D5W plus albumin, compatible with glass bottles, PVC and polyolefin bags, and polypropylene syringes.
- Grastofil ®: when diluted in D5W, compatible with glass bottles; when diluted with D5W plus albumin, compatible with PVC and polyolefin bags.

Neupogen ® only:
- Accidental exposure of vials or prefilled syringes to room temp (up to 30°C) or freezing temperatures does not affect the stability.
- Stable for 24 hours at room temp and 7 days in the fridge undiluted when stored in tuberculin syringes.
- Stable for 24 hours at room temp and 7 days in the fridge in D5W at concentrations between 5-15 mcg/mL (with albumin) and greater than 15 mcg/mL in plastic syringes and PVC containers.

Grastofil ® only:
- Accidental one-time exposure to temperatures up to 30°C or exposure to freezing temperatures (less than 0°C) does not affect the stability. Discard product if exposure has been greater than 24 hours or product has been frozen more than once.

MISCELLANEOUS

- Grastofil ® and Nivestym ™ are classified as being biosimilar to Neupogen ®; therefore, these products are not interchangeable.
- The needle covers for the Grastofil ® and Neupogen ® prefilled syringes contain latex, but not those from Nivestym ™.

REFERENCES

1, 4, 5, 40, 135.

Full revision 2014; limited revision 2015, 2017, 2018, 2019, 2020, 2021

NAME OF MEDICATION
FLUCONAZOLE

OTHER NAMES
Diflucan ®

CLASSIFICATION
Antifungal

INDICATIONS

- For systemic fungal infections (serious candidal infections, cryptococcal meningitis) and/or in patients who cannot tolerate oral fluconazole.

ADMINISTRATION

- Ready-to-use sterile bottles of 200 mg/100 mL.
- Intermittent IV infusion: administer at maximum rate of 200 mg/hr.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, angioedema, facial edema, anaphylaxis.
- GI: abdominal pain, nausea, vomiting, diarrhea.
- CNS: headache and dizziness.
- Transient elevation of LFTs.

DOSAGE

- Adults:
  - Loading dose: in patients with acute infections, may administer twice the recommended maintenance dose on the first day, not to exceed a maximum loading dose of 800 mg IV.
  - Maintenance dose: 100-400 mg IV once daily depending on infection. Higher doses (up to 800 mg daily) have been used in life-threatening infections.
- Pediatrics:
  - Loading dose: may administer twice the recommended maintenance dose on the first day, not to exceed 12 mg/kg IV.
  - Maintenance dose: 3-12 mg/kg IV once daily.
- Dosage in renal impairment: the manufacturer recommends to give initial loading dose of 50-400 mg for adults and to adjust maintenance dose as follows:

CrCl (mL/min)	greater than 50	50-21	20-11	Hemodialysis
% of Maintenance Dose	100%	50%	25%	100% after each dialysis

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Do not freeze.
- The diluent in the ready-to-use bottle is NS.
- Stable for 24 hours at room temp in D5W and RL at a concentration of 1 mg/mL in PVC containers.

MISCELLANEOUS

REFERENCES

1, 4, 5, 95, 135, 221.

Full revision 2014; limited revision 2015

NAME OF MEDICATION
FLUDARABINE

OTHER NAMES
Fludara ®, 2-Fluoro-Ara-A Monophosphate

CLASSIFICATION
Antineoplastic

INDICATIONS

- For the treatment of refractory chronic lymphocytic leukemia.
- For the treatment of refractory low-grade non-Hodgkin's lymphoma.

ADMINISTRATION

- Intermittent IV infusion: dilute with 50-100 mL of NS or D5W to a maximum concentration of 1 mg/mL; infuse over 30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash (rare).
- GI: nausea, vomiting.
- Fever, chills.
- Hyperuricemia, tumour lysis syndrome.
- Extravasation hazard: non-vesicant/non-irriant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 25-30 mg/m^2/day IV for 5 consecutive days, every 28 days.
- Dosage in renal impairment: reduce dose to 50% (e.g., 20 mg/m^2/day IV for **3** consecutive days) if CrCl is between 30-70 mL/min; do not administer if CrCl is less than 30 mL/min.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge; do not freeze.
- Compatible with NS and D5W.
- Stable for at least 16 days at room temp and under normal light conditions in NS or D5W at a concentration of 1 mg/mL in glass vials. However, manufacturer recommends that the solution should be used within 8 hours of its preparation, as it does not contain any preservatives.

MISCELLANEOUS

REFERENCES

1, 2, 4, 129, 143, 165.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
FLUMAZENIL

OTHER NAMES
Anexate ®

CLASSIFICATION
Benzodiazepine antagonist

INDICATIONS

- For reversal of sedation in patients who received a benzodiazepine for general anesthesia or for conscious sedation.
- For the diagnosis or management of deliberate or accidental benzodiazepine overdosage.

ADMINISTRATION

- IV direct: physician or RN. Administer over 15 or 30 seconds as per Dosage section below.
- Continuous IV infusion: further dilute 1 or 2 mg in 50 mL of NS or D5W to obtain a final concentration of 0.02 mg/mL or 0.04 mg/mL, respectively. At TOH, withdraw 10 mL from a 50 mL minibag of D5W or NS and discard, then add 1 mg (10 mL from flumazenil 0.5 mg/5 mL vials) to the minibag to obtain a final concentration of 0.02 mg/mL. Rate as per Dosage section below.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: transient increases in blood pressure and heart rate, flushing.
- GI: nausea, vomiting.
- May precipitate benzodiazepine withdrawal reactions (dizziness, confusion, agitation, anxiety, tachycardia, diaphoresis, seizures) in patients on long-term benzodiazepine therapy.
- Local reactions: pain and redness at the site of injection. Avoid extravasation into perivascular tissues as irritant.

DOSAGE

- To reverse benzodiazepine-induced general anesthesia and conscious sedation:
  - Initial dose of 0.2 mg IV over 15 seconds. If no response at 60 seconds, may give additional 0.1 mg over 15 seconds and repeat at 60 second intervals to a maximum total dose of 1 mg (usual total dose is 0.3-0.6 mg).

- Known or suspected benzodiazepine overdosage:
  - Initial dose of 0.3 mg IV over 30 seconds followed by additional 0.3 mg injections, each administered over 30 seconds, at 60 second intervals to a maximum total dose of 2 mg. If sedation recurs, may use an infusion of 0.1-0.4 mg/hr.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Stable for 24 hours at room temp when diluted in NS, D5W or RL.

MISCELLANEOUS

- As flumazenil has a short duration of action (approximate half-life = 60 minutes) patients should be monitored (for 2 or more hours) for recurrence of sedation after flumazenil injection.
- Although flumazenil can reverse benzodiazepine-induced sedation, it has variable effects on benzodiazepine-induced respiratory depression.

REFERENCES

1, 4, 5, 40, 135.

Full revision 2014; limited revision 2015, 2019

NAME OF MEDICATION
FLUORESCEIN

OTHER NAMES
AK Fluor ®, Diofluor ®, Fluorescite ®

CLASSIFICATION
Diagnostic agent

INDICATIONS

- For ophthalmic fluorescence imaging of the fundus and of the iris vasculature.
- For scanning or imaging vascular flow in detached digits/flaps.
- For evaluation of ureteral patency during cystoscopy, as an alternative to indigotindisulfonate sodium.

ADMINISTRATION

- IV direct: physician or RN; administer rapidly for ophthalmic fluorescence imaging and evaluation of ureteral patency or over 5 minutes for scanning/imaging vascular flow in detached digits/flaps.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, bronchospasm, anaphylaxis. If allergy is suspected, refer to Dosage section for test dose.
- Cardiovascular: syncope.
- GI: nausea, vomiting, abdominal discomfort.
- CNS: dizziness, headache.
- Dermatological: pruritus; skin attains a temporary yellowish discolouration which fades in 6-12 hours.
- Urine attains a bright yellow colour which fades in 24-36 hours.
- Local reactions: thrombophlebitis.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- For ophthalmic fluorescence imaging: 500 mg (5 mL of 10% solution) IV.
- For scanning or imaging vascular flow in detached digits/flaps:
  - For gross visual inspection: 15-20 mg/kg IV (black patients will require the higher dose of 20 mg/kg due to darker skin pigmentation).
  - For monitoring arterial insufficiency in revascularized digits for 24-72 hours: 1-1.5 mg/kg IV, repeated q2h.
- For evaluation of ureteral patency: 25-50 mg IV when starting cystoscopic evaluation.
- Test dose in case of suspected allergy: 0.05 mL intradermally; observe for 30-60 minutes.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp or fridge.
- Do not mix or dilute with other solutions.

MISCELLANEOUS

- Emergency drugs and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- When used for evaluation of ureteral patency, fluorescein allows visualization approximately 5 minutes after IV administration.

REFERENCES

5, 95, 386, 387, 389.

Full revision 2014; limited revision 2015, 2016, 2018, 2020

NAME OF MEDICATION
FLUOROURACIL

OTHER NAMES
Adrucil ®, 5-Fluorouracil, 5-FU

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of breast, bladder, colorectal, gastric, head and neck, ovarian, pancreatic and prostate cancer.

ADMINISTRATION

- IV direct: physician or RN; undiluted; inject over 1-3 minutes into the tubing of a freely running IV solution of NS or D5W.
- Intermittent IV infusion: dilute in 50 mL of D5W or NS; infuse over 15 minutes.
- Continuous IV infusion: dilute in 500-1000 mL of D5W or NS; infuse as per protocol.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis (rare).
- Cardiovascular: asymptomatic ECG changes, arrhythmia, cardiac ischemia.
- GI: nausea and rarely vomiting.
- Radiation recall reaction.
- Excessive lacrimation, conjunctivitis.
- Local reactions: chemical phlebitis.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Adults: various dosing regimens used:
  - 250-500 mg/m^2 IV daily X 5 consecutive days every 4 weeks.
  - 500-600 mg/m^2 IV on days 1 and 8 every 4 weeks.
  - 1000 mg/m^2 IV over 24 hours X 2 consecutive days every week or X 3 consecutive days every 3 weeks or X 4 consecutive days every 4-6 weeks.
- Pediatrics:
  - 500 mg/m^2/day IV once or daily X 5 days;
  - 800-1200 mg/m^2 as a continuous infusion over 24-120 hours.
- Dosage in renal impairment: no adjustment required but consider reduction in severe renal impairment.
- Dosage in hepatic impairment: omit dose if bilirubin is above 4 x ULN.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Slight discolouration does not adversely affect potency; if dark yellow, discard as indicates greater decomposition. A precipitate may form if exposed to cold; redissolve by gently heating to 60°C and vigorous shaking.
- Compatible with NS, D5W, dextrose-saline and D5-RL.
- Stable for 14 days at room temp and in the fridge in D5W or NS at concentrations of 1 mg/mL and 10 mg/mL in PVC containers.
- Stable for 16 weeks in the fridge in D5W at a concentration of 10 mg/mL in PVC containers.
- Stable for 91 days in the fridge in NS at a concentration of 5 mg/mL in both glass and PVC containers.
- Stable for 72 hours at room temp (protected from light) in NS or D5W at concentrations of 12 mg/mL and 40 mg/mL in polypropylene syringes.
- Stable for 21 days at 30°C at a concentration of 50 mg/mL (undiluted) in polypropylene syringes.

MISCELLANEOUS

REFERENCES

1, 4, 5, 82, 129, 165.

Full revision 2014; limited revision 2015, 2016, 2017, 2018, 2020

NAME OF MEDICATION
FLUPENTIXOL DECANOATE

OTHER NAMES
Fluanxol depot, Flupenthixol decanoate

CLASSIFICATION
Antipsychotic

INDICATIONS

- Long-acting depot medication used for treatment of chronic psychosis.

ADMINISTRATION

- IM: use at least a 21 gauge dry needle, give deep IM into large muscle, use Z track method, rotate sites.
- Aspirate before injection to avoid intravascular injection.
- Do NOT massage injection site.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypertension, hypotension, ECG changes.
- CNS: frequent extrapyramidal effects (akathisia, dystonia, dyskinesia, rigidity, tremor, tardive dyskinesia, etc); both alerting and sedating effects reported; can cause excitation.
- Neuroleptic malignant syndrome.
- Local reactions: pain at injection site and indurations seen rarely (at high doses).

DOSAGE

- For patients not previously treated with long-acting depot antipsychotic: initial dose: 5 mg-20 mg IM.
- For patients previously treated with long-acting depot antipsychotics and had good tolerance: initial dose: 20-40 mg IM.
- Maintenance dose: 20-40 mg IM may be given 4-10 days after initial dose; adjust according to patient's response; usual range from 20-80 mg every 2-4 weeks.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp; protect from light.
- Do not let drug stand in syringe for longer than 15 minutes.

MISCELLANEOUS

- Peak plasma levels are obtained 4-7 days after dose administration.

REFERENCES

2, 51, 135.

Full revision 2014; limited revision 2015

NAME OF MEDICATION
FOLIC ACID

OTHER NAMES
Folate

CLASSIFICATION
Vitamin

INDICATIONS

- Treatment of macrocytic and megaloblastic anemias resulting from folate deficiency when GI absorption is impaired or oral therapy not possible (e.g., nutritional macrocytic anemia, megaloblastic anemias of pregnancy (adjuvant), infancy and childhood, megaloblastic anemia associated with primary liver disease, alcoholism, and cirrhosis, intestinal strictures, anastomoses, or sprue).
- Adjuvant in the treatment of methanol poisoning, in replacement for leucovorin calcium.
- Folic acid alone not indicated in pernicious anemia.

ADMINISTRATION

- IV direct: physician or RN. Administer 5 mg or less undiluted at a rate of 5 mg/min.
- Intermittent IV infusion: may dilute up to 5 mg with 50 mL NS or D5W to get a concentration of up to 0.1 mg/mL. Infuse over 30 minutes (over 15-30 minutes for methanol poisoning).
- Continuous IV infusion: may be added to maintenance IV solutions.
- IM, SC.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; rash, urticaria, bronchospasm, anaphylactoid reactions.
- GI: rare, occur at high doses e.g., 15 mg/day: anorexia, nausea, abdominal distention, flatulence and a bitter/bad taste.
- CNS: rare, occur at high doses e.g., 15 mg/day: altered sleep patterns, difficulty in concentrating, irritability, agitation, excitement, depression, confusion, impaired judgement.
- Slight flushing or feeling of warmth.

DOSAGE

Adults:

- Anemias:
  - Therapeutic dose: 0.25-1 mg IV/IM/SC daily.
  - Maintenance dose: 0.4 mg IV/IM/SC daily.
  - Tropical sprue dose: 3-15 mg IV/IM/SC daily.
- Methanol ingestion: in absence of metabolic acidosis: 1 mg/kg IV q6h; in presence of metabolic acidosis: 1-2 mg/kg IV q4-6 h. Adjust dose and interval based on severity of acidosis and clinical condition.

Pediatrics:

- Anemias:
  - Therapeutic dose: 0.25-1 mg IV/IM/SC daily.
  - Maintenance dose: 0.1 mg IV/IM/SC daily for infants, 0.3 mg IV/IM/SC daily for children 1-4 years, and 0.4 mg IV/IM/SC daily for older children.
- Methanol ingestion: in absence of metabolic acidosis: 1 mg/kg IV q6h; in presence of metabolic acidosis: 1-2 mg/kg IV q4-6h. Adjust dose and interval based on severity of acidosis and clinical condition.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp; protect from light.
- Stable for 24 hours at room temp and 28 days in the fridge in NS and D5W.

MISCELLANEOUS

- IV and oral doses are equivalent on a mg per mg basis.
- Administer with extreme caution to patients with undiagnosed anemia since folic acid may mask diagnosis of pernicious anemia.

REFERENCES

1, 2, 4, 6, 40, 82, 95, 135, 208, 433.

Full revision 2014; limited revision 2015, 2020

NAME OF MEDICATION
FOMEPIZOLE

OTHER NAMES
Antizol ®

CLASSIFICATION
Antidote

INDICATIONS

- Antidote for ethylene glycol (such as antifreeze) or methanol (such as windshield washer fluid) poisoning.

ADMINISTRATION

- Intermittent IV infusion: dilute in a minimum of 100 mL of NS or D5W and infuse over 30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: mild reactions (rash, eosinophilia).
- GI: nausea and bad taste/metallic taste.
- CNS: headache, dizziness and drowsiness.
- Local reactions: venous irritation and phlebosclerosis (especially if not diluted properly or administered too fast).

DOSAGE

- Initial dose of 15 mg/kg IV, followed by doses of 10 mg/kg q12h for 4 doses.
- If therapy is still needed beyond this 48 hour period, continue infusion with 15 mg/kg IV q12h thereafter until ethylene glycol or methanol levels are undetectable or below 3.2 mmol/L and 6 mmol/L, respectively, and the patient is asymptomatic with normal pH.
- If dialysis is performed, administer fomepizole q4h and consult Product Monograph for further dosing information.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Undiluted fomepizole solidifies at temperatures below 25°C. Liquefy by warming (holding in the hand or running under warm water). Solidification does not affect the efficacy, safety, or stability of fomepizole.
- Stable for at least 48 hours at room temp or in the fridge when diluted in D5W or NS; however, manufacturer recommends that the solution not be used beyond 24 hours due to absence of preservatives.

MISCELLANEOUS

REFERENCES

1, 2, 5, 40, 95, 263.

Full revision 2014; limited revision 2015, 2016

NAME OF MEDICATION	
FONDAPARINUX	
OTHER NAMES	CLASSIFICATION
Arixtra ®	Anticoagulant

INDICATIONS

- Prophylaxis of thrombosis following orthopedic surgery of the lower limbs.
- Prophylaxis of thrombosis in patients undergoing abdominal surgery who are at high risk of thromboembolic complications such as cancer patients.
- Treatment of acute deep vein thrombosis (DVT) and pulmonary embolism (PE).
- Management of unstable angina (UA) or non-ST-segment elevation myocardial infarction (NSTEMI) for the prevention of death and subsequent myocardial infarction.
- Management of ST-segment elevation myocardial infarction (STEMI) for the prevention of death and myocardial infarction in patients managed with thrombolytics or who are to receive no reperfusion therapy initially.

ADMINISTRATION

- IV direct: physician or RN. Administer undiluted. Flush with NS after to ensure drug has been all administered.
- Intermittent IV infusion: dilute in 25-50 mL of NS and administer over 1-2 minutes.
- SC: in the fat tissue of the lower abdomen. With the thickness of skin held between the operator's thumb and finger, introduce the entire length of the needle vertically into the skin. Alternate the left and right side of the abdomen at each injection.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: skin rash; rarely angioedema and anaphylatic/anaphylactoid reactions.
- Hematologic: bleeding, thrombocytopenia.
- Risk of spinal or epidural hematomas that can result in permanent paralysis when epidural or spinal anesthesia or spinal puncture is used in conjunction with fondaparinux.
- Transient increase of liver transaminases (AST, ALT).
- Local reactions: mild irritation at injection site.

DOSAGE

- Prophylaxis of thrombosis related to orthopedic surgery: 2.5 mg SC once daily; do not start initial dose earlier than 6 hours after surgical closure. Treatment to be continued for as long as the risk of thrombosis persists (usually around 7 days, although has been continued for up to 32 days in some cases).
- Prophylaxis of thrombosis related to abdominal surgery: 2.5 mg SC once daily; initial dose to start 6 to 8 hours after surgery. Treatment to be continued for 5 to 9 days (up to 10 days in some cases).
- Treatment of DVT and PE: if body weight is below 50 kg: 5 mg SC once daily; if body weight 50-100 kg: 7.5 mg SC once daily; if body weight is over 100 kg: 10 mg SC once daily. Concomitant anticoagulation with warfarin should be started within 72 hours. Fondaparinux should be continued for at least 5 days, or until warfarin effect is within target range (INR 2 to 3).
- Management of UA/NSTEMI: 2.5 mg SC once daily to continue for up to 8 days or until hospital discharge.
- Management of STEMI: 2.5 mg once daily; give first dose IV and subsequent doses SC. To continue for up to 8 days or until hospital discharge.
- Dosage in renal impairment: use with caution in patients with moderate renal impairment (CrCl: 30-50 mL/min). Do not use if CrCl is below 30 mL/min.
- Dosage in hepatic impairment: no dosage adjustment in mild to moderate impairment; no data in severe liver impairment.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Stable for 24 hours at room temp, diluted in 25-50 mL of NS.

MISCELLANEOUS

- Fondaparinux does not usually modify in a significant way the routine clotting tests (i.e., INR, aPTT). Although monitoring activity of fondaparinux is generally not required, the anti-Xa assay is the preferred test to measure its anticoagulant activity.
- Does not cross-react with serum from patients with heparin-induced thrombocytopenia (HIT); however, it should be used with caution in patients with a history of HIT.
- There is no specific antidote for fondaparinux overdosage.
- Fondaparinux should not be used as the sole anticoagulant during percutaneous coronary intervention (PCI) because of an increased risk of guiding catheter thrombosis. An effective anti-thrombin regimen such as unfractionated heparin should be administered as an adjunct to PCI, according to standard practice.
- In patients who are to undergo coronary artery bypass graft surgery, fondaparinux should not be given during the 24 hours before surgery and may be restarted 48 hours postoperatively.

REFERENCES

1, 2.

Full revision 2014; limited revision 2015, 2017, 2020

NAME OF MEDICATION
FOSAPREPITANT

OTHER NAMES
Emend IV ®

CLASSIFICATION
Antiemetic

INDICATIONS

- Prevention of nausea and vomiting induced by emetogenic cancer chemotherapy; to be used in combination with a 5-HT_3 antagonist (e.g., ondansetron) and dexamethasone.

ADMINISTRATION

- Reconstitute the 150 mg vial of fosaprepitant (equivalent to 245.3 mg of fosaprepitant dimeglumine) with 5 mL of NS. To prevent foaming, add the NS slowly to the side of the vial and gently swirl the vial to aid dissolution. Avoid shaking and jetting saline into the vial.
- Intermittent IV infusion: transfer the contents of the vial to a sterile IV bag containing 145 mL NS to give a total volume of 150 mL and a final concentration of 1 mg/mL OR dilute in a 250 mL bag of NS to obtain a final concentration of 0.6 mg/mL. Gently invert the bag 2-3 times to mix. Infuse over 20-30 minutes for the 150 mL bag and over 30 minutes for the 250 mL bag.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; flushing, erythema, dyspnea.
- Cardiovascular: hypertension.
- GI: nausea, hiccups, anorexia, diarrhea, abdominal pain, dyspepsia.
- CNS: headache, dizziness.
- Fatigue.
- Proteinuria, increased LFTs.
- Local reactions: pain, induration, erythema, pruritus, thrombophlebitis. A more diluted solution, i.e., 0.6 mg/mL instead of 1 mg/mL, is associated with a lower incidence of local reactions.

DOSAGE

- 150 mg IV 30 minutes prior to chemotherapy on Day 1 only.
- Dosage in renal impairment: no dosage adjustment is necessary.
- Dosage in hepatic impairment: no dosage adjustment is necessary in mild to moderate insufficiency (Child-Pugh score 5-9); no data in severe insufficiency (Child-Pugh score greater than 9).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. No significant loss of potency when vials are stored at room temp (at or below 25°C) over a period of up to 10 days.
- Reconstituted solution is stable for 12 days in the fridge exposed to light.
- Stable for 48 hours at room temp and in the fridge in NS or D5W at a concentration of 1 mg/mL.
- Stable for 7 days at room temp and in the fridge (both protected from light) and for 14 days at room temp and in the fridge (both exposed to light) in NS at a concentration of 0.6 mg/mL.
- Incompatible with any solutions containing divalent cations (e.g., Ca, Mg, RL, Hartmann's solution).

MISCELLANEOUS

- Fosaprepitant is the pro-drug of aprepitant.

REFERENCES

2, 95, 135, 165, 197, 559, 560.

Full revision 2014; limited revision 2015, 2021

NAME OF MEDICATION
FOSCARNET

OTHER NAMES
Foscavir ®, Vocarvi ™

CLASSIFICATION
Antiviral

INDICATIONS

- Treatment of cytomegalovirus (CMV) retinitis infection or CMV GI tract infections (e.g., colitis, esophagitis) in HIV-infected patients.
- Treatment of mucocutaneous herpes simplex virus (HSV) infections unresponsive to acyclovir in immunocompromised patients.

ADMINISTRATION

- Intermittent IV infusion: if using a central line, can administer the undiluted solution of 24 mg/mL; for peripheral vein administration, must dilute with D5W or NS to a concentration of 12 mg/mL. Infuse the 40-60 mg/kg dose over at least 60 minutes, the 90-120 mg/kg dose over 2 hours. Administration rate not to exceed 1 mg/kg/min.
- Must be administered by an infusion pump.
- Refer to Miscellaneous section for concurrent hydration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, anaphylaxis, angioedema.
- Cardiovascular: QT interval prolongation.
- GI: nausea, vomiting, diarrhea.
- CNS: seizures, headache.
- Electrolyte disturbances: hypocalcemia, hypo- or hyperphosphatemia, hypomagnesemia, hypokalemia.
- Hematologic: anemia, granulocytopenia.
- Renal: nephrotoxicity; may occur at any time but typically during week 2 of induction therapy.
- Fever.
- Local reactions: irritation if administered undiluted in peripheral veins.

DOSAGE

- CMV retinitis: induction: 60 mg/kg IV q8h or 90 mg/kg IV q12h for 14-21 days. Maintenance: 90-120 mg/kg IV once daily.
- CMV infections of GI tract: 60 mg/kg IV q8h OR 90 mg/kg IV q12h for 21-42 days or until resolution of symptoms.
- HSV infections: 40 mg/kg IV q8h or q12h for 14-21 days or until healed.
- Dosage in renal impairment: refer to manufacturer's information.
- Dosage in hepatic impairment: no dosage adjustment is necessary.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store bottles between 15-30°C. Do not refrigerate; protect from freezing.
- Stable for 35 days at room temp or in the fridge in NS or D5W at a concentration of 12 mg/mL in PVC containers. However, the manufacturer recommends that the diluted solution be stored at room temp and used within 24 hours.

MISCELLANEOUS

- Adequate hydration is recommended during treatment with foscarnet to minimize renal toxicity: 750-1000 mL of NS or D5W prior to first dose to establish diuresis. For subsequent infusions, hydration can be given concurrently with each dose: 750-1000 mL of fluid with doses of 90-120 mg/kg and 500 mL of fluid with doses of 40-60 mg/kg. Volume of fluid may be decreased if clinically appropriate; oral hydration may also be possible in some patients.

REFERENCES

1, 4, 5, 40, 95, 135, 143, 462.

Full revision 2014; limited revision 2015, 2018, 2019, 2020, 2021

NAME OF MEDICATION
FOSPHENYTOIN

OTHER NAMES
Cerebyx ®

CLASSIFICATION
Anticonvulsant

Do NOT confuse fosphenytoin with phenytoin.
This monograph is specific to FOSPHENYTOIN.

INDICATIONS
- For control of generalized convulsive status epilepticus.
- For treatment or prophylaxis of seizures in neurosurgical patients.
- As a short-term parenteral replacement for oral phenytoin.

ADMINISTRATION
- Intermittent IV infusion: ***cardiac monitoring, blood pressure monitoring, respiratory support;*** continue the monitoring 10 to 20 minutes after completion of the injection. Monitoring required only for the loading dose. Dilute dose in NS or D5W for a concentration of 1.5-25 mg PE/mL (2.5-40 mg/mL) and infuse at a maximum rate of 150 mg PE/min (refer to Miscellaneous section for PE explanation). Must be administered by an infusion pump.
- IM (not for status epilepticus).

POTENTIAL ADMINISTRATION HAZARDS
- Cardiovascular: hypotension, shock, cardiovascular collapse, arrhythmias. Minimize by administering at a rate not exceeding the rates above. Stop infusion if these occur.
- CNS: dizziness, headache, ataxia, tremor, agitation, somnolence.
- Dermatologic: rash, pruritus.
- Ophthalmic: diplopia, nystagmus.
- Otic: tinnitus.
- Burning, itching and/or paresthesia. Minimize by slowing or temporarily stopping the infusion.

DOSAGE
- Loading dose for status epilepticus: 15-20 mg PE/kg IV.
- Non-emergent loading dose: 10-20 mg PE/kg IV or IM.
- Initial maintenance therapy: 4-7 mg PE/kg/day IV or IM in 2 to 4 divided doses. Adjustments may be needed as determined by serum levels i.e., 40-80 mcmol/L (10-20 mcg/mL).
- Dosage in obesity:

  Method 1: Loading dose (mg): [14 mg PE/kg x IBW] + [19 mg PE/kg x (TBW- IBW)]
  Maintenance dose: use IBM

  Method 2: Loading dose (mg): 15 mg PE/kg x TWB for males
  20 mg PE/kg x TWB for females
  Maintenance dose: use IBM

  IBW: ideal body weight; TBW: total body weight

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials in the fridge; do not store at room temp for more than 48 hours.
- Stable for 30 days at room temp or in the fridge in D5W or NS at concentrations of 1-20 mg PE/mL in PVC bags.
- Stable for 7 days at room temp in D10W, D5-1/2NS, D5-RL and RL at concentrations of 1-20 mg PE/mL in PVC bags.
- Stable for 30 days at room temp and under refrigeration undiluted when packaged in polypropylene syringes.

MISCELLANEOUS
- The dosage of fosphenytoin is expressed in terms of phenytoin equivalents (PE) to avoid the need to do molecular weight-based adjustments when converting between the 2 agents (as each 75 mg of fosphenytoin is equivalent to 50 mg of phenytoin).
- Do not monitor phenytoin serum concentrations until conversion from fosphenytoin to phenytoin is complete, i.e., 2 hours after the end of IV infusion or 4 hours after IM injection.

REFERENCES
4, 5, 40, 135, 189, 435, 436.

Full revision 2014; limited revision 2015, 2019, 2020

NAME OF MEDICATION
FULVESTRANT

OTHER NAMES
Faslodex ®

CLASSIFICATION
Estrogen antagonist

INDICATIONS

- Hormonal treatment of locally advanced or metastatic breast cancer in postmenopausal women, regardless of age, who have disease progression following prior anti-estrogen therapy.
- Treatment of estrogen receptor-positive, human epidermal growth receptor 2 (HER2)-negative locally advanced or metastatic breast cancer in postmenopausal women not previously treated with endocrine therapy.

ADMINISTRATION

- IM: inject slowly (1 to 2 minutes per injection) into the buttock. Refer to manufacturer's monograph for detailed instructions on assembly and proper use of the safety needle (SafetyGlide systems).

POTENTIAL ADMINISTRATION HAZARDS

- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, urticaria and angioedema. May occur shortly after injection or several days after injection.
- GI: nausea, vomiting, constipation, diarrhea, anorexia.
- Hepatic: increase in transaminases, bilirubin and alkaline phosphatase.
- Hot flushes (vasodilation).
- Arthralgia, myalgia, back pain, pain in extremities, fatigue.
- Local reactions: injection site reactions with mild transient pain and inflammation; local pruritus and urticaria may occur even after uneventful injections and may develop with time into a systemic allergic response (e.g., widespread urticaria).

DOSAGE

- 500 mg (2 X 5 mL; 5 mL into each buttock) IM on days 0, 14, 28 and then q28 days thereafter.
- Dosage in renal impairment: no dosage adjustment if CrCl greater than 30 mL/min; not evaluated if CrCl is less than 30 mL/min.
- Dosage in hepatic impairment: no dosage adjustment in mild to moderate disease; not recommended in severe insufficiency.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store in the original container between 2-8°C.
- To ensure patient comfort, fulvestrant injection may be removed from the fridge and kept at room temp for up to 60 minutes or rolled gently between the hands before administration.

MISCELLANEOUS

- Due to structural similarity of fulvestrant and estradiol, fulvestrant can interfere with estradiol measurement by immunoassays, resulting in falsely elevated levels of estradiol.
- LFTs should be performed on a regular basis or when clinically indicated.

REFERENCES

1, 5, 129, 165.

Full revision 2014; limited revision 2016, 2017, 2018, 2019, 2020

NAME OF MEDICATION
FUROSEMIDE

OTHER NAMES
Lasix ®

CLASSIFICATION
Diuretic

INDICATIONS

- Treatment of edema (diuretic effect) associated with congestive heart failure, renal disease (including the nephrotic syndrome) and hepatic cirrhosis when oral administration is not possible.
- Adjunct in treatment of acute pulmonary edema.
- To increase renal excretion of calcium in the setting of hypercalcemia.

ADMINISTRATION

- IV direct: physician or RN; maximum dose to be given direct is 100 mg. Administer over 2 minutes.
- Intermittent IV infusion, Continuous IV infusion: dilute in 50-250 mL of D5W, NS or RL. Doses greater than 100 mg must be given at a rate not exceeding 4 mg/min.
- IM.
- SC, SC infusion (palliative care). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, dermatitis, angioedema, anaphylaxis, interstitial nephritis and cutaneous vasculitis.
- Cardiovascular: hypovolemia; may cause postural hypotension. Elderly and sodium-restricted patients are more susceptible.
- GI: nausea, vomiting, anorexia, diarrhea.
- CNS: blurred vision, dizziness.
- Electrolyte disturbances: hypokalemia, hypochloremia, hypomagnesemia, hypocalcemia, hyponatremia.
- Endocrine and metabolic: hyperglycemia in diabetics, hyperuricemia.
- Otic: deafness (reversible), hearing impairment (reversible or permanent) or tinnitus with large parenteral doses administered rapidly.
- Local reactions: transient pain (IM injection) and thrombophlebitis (IV administration).

DOSAGE

Adults:
- Usual diuretic dose for edema: 20-40 mg IV or IM as a single injection. May repeat q2h. If no response, may increase dose in 20 mg increments.
- For oliguria of acute renal failure, where conventional doses produce inadequate diuresis: 250 mg IV. If inadequate response, a 500 mg IV dose may be infused.
- For acute pulmonary edema: 40 mg IV. If not satisfactory within 60 minutes, may increase to 80 mg IV.
- Continuous infusion: 10-40 mg/hr IV.
- Hypercalcemia: 40-80 mg IV q1-4h after volume expansion.

Pediatrics:
- 1-2 mg/kg/dose IM or IV q6-12h.
- Continuous infusion: 0.05 mg/kg/hr IV; titrate to effect.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Exposure to light may cause discolouration. Do not use if solution is yellow.
- Stable for 24 hours at room temp in D5W at concentrations of 0.2-0.6 mg/mL.
- Stable for 24 hours at room temp in NS at concentrations of 0.2-1 mg/mL in PVC containers.
- Stable for 24 hours at room temp and 26 days in the fridge in NS at a concentration of 1 mg/mL in PVC containers.
- Stable for 24 hours at room temp in RL at concentrations of 0.6-1 mg/mL.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 24 hours at room temp at a concentration of 10 mg/mL (undiluted) in polypropylene syringes.
- Stable for 84 days at room temp (protected from light) and in the fridge (protected from light) in NS at concentrations of 1.2-3.4 mg/mL stored in PVC bags and at concentrations of 1-8 mg/mL packaged in polypropylene syringes. All samples stable for another 7 days at room temp when exposed to fluorescent light.
- Incompatible via Y-site administration with Baxter premixed nitroglycerin solution in bottle.

MISCELLANEOUS

- Onset of action after IV administration: within 5 minutes; peak response within 20-60 minutes; duration of action: 2 hours.
- Average equivalent doses: 20 mg IV = 40 mg PO; caution as there can be great variability.

REFERENCES

1, 2, 4, 6, 40, 76, 81, 82, 95, 135, 143, 457.

NAME OF MEDICATION
GANCICLOVIR

OTHER NAMES
Cytovene ®

CLASSIFICATION
Antiviral

INDICATIONS

- Treatment of cytomegalovirus (CMV) infections in immunocompromised individuals.
- Prevention of CMV disease in transplant recipients at risk for CMV disease.

ADMINISTRATION

- Reconstitute the 500 mg vial with 10 mL of SWFI to obtain a 50 mg/mL solution. Do not use SWFI that contains parabens as a preservative as this may cause precipitation. Shake well until complete dissolution.
- Intermittent IV infusion: further dilute in 50-250 mL (usually 100 mL) of NS, D5W, Ringer's or RL to a maximum concentration of 10 mg/mL. Administer over at least 60 minutes in a large peripheral vein or via a central line.

POTENTIAL ADMINISTRATION HAZARDS

- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash.
- GI: nausea and vomiting.
- CNS: confusion, headache.
- Hematologic: frequent and can be severe: neutropenia, thrombocytopenia, anemia.
- Increase in serum creatinine, abnormal LFT.
- Local reactions: pain, phlebitis and inflammation at injection site (to minimize, use a large vein with adequate blood flow).
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

For adults and pediatrics older than 3 months of age:
- Treatment of CMV infections:
  - Induction: 5 mg/kg IV q12h for 14-21 days.
  - Maintenance: 5 mg/kg IV once daily 7 days per week OR 6 mg/kg IV once daily 5 days per week.
- Prevention of CMV disease:
  - Initial: 5 mg/kg IV q12h for 7-14 days.
  - Maintenance: 5 mg/kg IV once daily 7 days per week OR 6 mg/kg IV once daily 5 days per week.

- Dosage in renal impairment:

CrCl (mL/min)	greater than 70	69-50	49-25	24-10	less than 10
Induction/initial dose (mg/kg)	5	2.5	2.5	1.25	1.25
Interval (hr)	12	12	24	24	3 times/week*
Maintenance dose (mg/kg)	5	2.5	1.25	0.625	0.625
Interval (hr)	24	24	24	24	3 times/week*

* Administer following hemodialysis

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Reconstituted solutions are stable for 12 hours at room temp and should not be refrigerated as precipitate may form.
- Incompatible with parabens-preserved solutions.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 35 days refrigerated and at room temp when diluted in NS or D5W at concentration of 1-5 mg/mL and stored in PVC containers.
- Stable for 35 days refrigerated and protected from light when diluted in D5W at 10 mg/mL and stored in PVC containers.
- Stable for 7 days at room temp and 80 days under refrigeration diluted in NS at concentrations of 1.4, 4 and 7 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Ensure adequate hydration during treatment with ganciclovir.
- Monitor neutrophil, platelet counts and serum creatinine frequently.

REFERENCES

1, 2, 4, 5, 82, 135, 143.

Full revision 2015; limited revision 2018, 2019, 2020

NAME OF MEDICATION
GANIRELIX

OTHER NAMES
Orgalutran ®

CLASSIFICATION
Gonadotropin-releasing hormone antagonist

INDICATIONS

- As a component of infertility regimens to inhibit premature luteinizing hormone surges in women undergoing controlled ovarian hyperstimulation.

ADMINISTRATION

- SC: abdomen, upper thigh. Rotate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rare; rash, facial swelling, dyspnea.
- GI: nausea, abdominal pain.
- CNS: headache, dizziness.
- Hot flushes, vaginal bleeding.
- Local reactions: redness, bruising.

DOSAGE

- 250 mcg SC once daily during the early to mid follicular phase of the menstrual cycle, usually for around 5 days (range 1 to 19 days).
- Follicle-stimulating hormone (FSH) treatment is usually started on the morning of day 2 or 3 of the menstrual cycle and ganirelix is started on the morning of day 7 or 8 of the cycle (or day 6 of FSH therapy).
- Treatment is continued until an adequate follicular response is achieved, at which time human chorionic gonadotropin (hCG) administration begins.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 15 and 30°C. Protect from light.

MISCELLANEOUS

- The needle cover of the prefilled syringe contains latex.

REFERENCES

1, 5, 95, 135.

Full revision 2015; limited revision 2019, 2020

NAME OF MEDICATION	
GemCITABINE	
OTHER NAMES	CLASSIFICATION
Gemzar ®	Antineoplastic

INDICATIONS

- Locally advanced or metastatic non-small cell lung cancer (alone or in combination with cisplatin).
- Locally advanced or metastatic adenocarcinoma of the pancreas.
- Locally advanced or metastatic transitional cell carcinoma of the bladder in combination with cisplatin.
- Unresectable locally recurrent or metastatic breast cancer in patients who have good performance status and have relapsed following adjuvant anthracycline-based chemotherapy (to give in combination with paclitaxel).

ADMINISTRATION

- There are 2 types of formulation available: a lyophilized powder that requires reconstitution and a ready-to use solution (38 mg/mL, 40 mg/mL and 100 mg/mL).
- Reconstitution of the lyophilized powder: add 5 mL, 25 mL or 50 mL of NS without preservatives to the 200 mg, 1 g and 2 g vials respectively. Invert vial to dissolve. Final concentration is 38 mg/mL.
- Intermittent IV infusion: for the reconstituted solution: administer undiluted or further dilute in NS to concentrations as low as 0.1 mg/mL. For the 38 mg/mL and 40 mg/mL ready-to use solutions: administer undiluted or further dilute in NS or D5W to concentrations as low as 0.1 mg/mL. For the 100 mg/mL ready-to use solution: further dilute in NS to obtain a final concentration between 0.1 and 10 mg/mL. All types of preparations are often diluted in 250 mL of NS. To infuse all types of preparations over 30 minutes (extending infusion time beyond 60 minutes will increase toxicity).

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, anaphylaxis (rare).
- GI: nausea, vomiting.
- Respiratory: acute shortness of breath, bronchospasm.
- Flu-like symptoms (fever, chills, fatigue, myalgia).
- Local reactions: injection site reactions.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Non-small cell lung carcinoma: 1000 mg/m^2 IV on days 1, 8, 15 of a 28 day cycle OR 1250 mg/m^2 IV on days 1, 8 of a 21 day cycle. For both regimens, cisplatin to be given at 100 mg/m^2 on day 1 of each cycle (to administer after gemcitabine).
- Pancreatic carcinoma:
  - Initial cycle: 1000 mg/m^2 IV once weekly for up to 7 weeks, followed by a one week rest.
  - Subsequent cycles: 1000 mg/m^2 IV weekly for 3 weeks followed by a one week rest.
- Bladder carcinoma: 1000 mg/m^2 IV on days 1, 8, 15 of each 28 day cycle; cisplatin to be given at 70 mg/m^2 on day 1 of each 28 day cycle (to administer after gemcitabine).
- Breast cancer: 1250 mg/m^2 IV on days 1, 8 of each 21 day cycle; paclitaxel to be given at 175 mg/m^2 over 3 hours on day 1 of each cycle (to administer before gemcitabine).
- Dosage in renal or hepatic impairment: insufficient information; use with caution.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Lyophilized powder vial:
- Store lyophilized powder vials between 15-30°C.
- Reconstituted solution (38 mg/mL in NS) is stable for 35 days at room temp and for 7 days in the fridge (crystals may develop when solution refrigerated for longer).
- Reconstituted solution (38 mg/mL in NS) is stable for 35 days at room temp and under refrigeration in polypropylene syringes. Caution as refrigeration of reconstituted solution may cause crystal to develop.
- Stable for 35 days at room temp and in the fridge in D5W or NS at concentrations of 0.1 mg/mL and 10 mg/mL in PVC containers.

Ready-to use solution (38 and 40 mg/mL):
- Store vials between 2-8°C. Do not freeze.
- Stable for 24 hours between 15-30°C when added to an empty container.
- Stable for 24 hours between 15-30°C in NS or D5W in concentrations as low as 0.1 mg/mL.
- Stable for 35 days at room temp (exposed to light) and for 116 days in the fridge (protected from light) in NS at concentrations of 0.1-26 mg/mL in PVC containers.
- Stable for 35 days at room temp (exposed to light) in D5W at concentrations of 0.1-26 mg/mL in PVC containers.

Ready-to use solution (100 mg/mL):
- Store vials at room temp (15-25°C).
- Stable for 60 days at room temp and in the fridge in NS at concentrations between 0.1 to 10 mg/mL.

MISCELLANEOUS

REFERENCES

1, 4, 5, 26, 40, 129, 165.

Full revision 2015; limited revision 2016, 2017, 2018, 2020

NAME OF MEDICATION
GEMTUZUMAB OZOGAMICIN

OTHER NAMES
Mylotarg ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of adult patients with previously untreated, *de novo* CD33-positive acute myeloid leukemia (AML), except acute promyelocytic leukemia (in combination with daunorubicin and cytarabine).

ADMINISTRATION

- **Ensure premedication has been administered prior to each infusion; refer to Potential Administration Hazards and Dosage sections.**
- This product is light sensitive and should be protected from light at all times by using an ultraviolet protector (amber bag or foil) to cover the vial and IV bag or IV syringe during its preparation and administration. The IV line does not need to be protected from light.
- Allow vials to reach room temp before reconstitution (approximately 5 minutes). Reconstitute each 4.5 mg vial with 5 mL of SWFI to obtain a concentration of 1 mg/mL. Gently mix; do NOT shake. Protect the reconstituted solution from light if not used immediately. The reconstituted solution may contain small white to off-white, opaque to translucent, and amorphous to fiber-like particles.
- Intermittent IV infusion (mandatory):
    - Doses less than 3.9 mg: add the dose to a syringe with NS.
    - Doses of 3.9 mg or greater: add the dose to a syringe with NS or dilute in a bag containing NS.
    - All doses: the final concentration should be between 0.075 mg/mL and 0.234 mg/mL. Containers made of PVC with DEHP or polyolefin (polypropylene and/or polyethylene) should be used. Gently mix; do NOT shake. Protect from light. Administer over 2 hours. If dose is given in an IV bag, use an infusion set with a low-protein-binding 0.2 micron polyethersulfone (PES) filter. If dose is given by IV syringe, use a small bore infusion line (microbore) with an in-line, low-protein-binding 0.2 micron PES filter. Protect the syringe or bag from light during the infusion; the IV line does not need light protection if infusion time is 2 hours or less. IV lines made of PVC (DEHP- or non-DEHP-containing) or polyethylene should be used. Flush the IV line with NS after each dose.
- Observe the patient during and for at least 60 minutes post-infusion for infusion-related reactions.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, anaphylaxis.
- Infusion-related reactions: fever, chills, hypotension, tachycardia, dyspnea, bronchospasm, hypoxia, respiratory failure; may occur during or within 24 hours following infusion. For mild to moderate reactions, interrupt infusion; may **repeat premedication** doses of a corticosteroid (e.g., dexamethasone 20 mg IV OR methylprednisolone 1 mg/kg IV), acetaminophen 650 mg PO, and diphenhydramine 50 mg PO/IV. Once symptoms resolve, consider restarting the infusion at no greater than half the previous rate. For severe reactions, discontinue treatment.
- Cardiovascular: QT interval prolongation.
- GI: nausea, vomiting, stomatitis, diarrhea.
- CNS: headache.
- Hematologic: neutropenia, febrile neutropenia, thrombocytopenia, anemia, leukopenia, lymphopenia, pancytopenia, bleeding/hemorrhagic events.
- Hepatic: elevation of liver enzymes and bilirubin, venoocclusive disease/sinusoidal obstruction syndrome, hepatic failure.
- Fatigue.
- Infection.
- Tumour lysis syndrome; reduce risk by ensuring adequate hydration, administering antihyperuricemic or other agents for treatment of hyperuricemia and consider leukoreduction.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- **Premedicate before each infusion with a corticosteroid (e.g., dexamethasone 20 mg IV 1 hour prior to the infusion or methylprednisolone 1 mg/kg IV within 30 minutes prior to the infusion), an antihistamine (e.g., diphenhydramine 50 mg PO/IV) and acetaminophen (e.g., 650 mg PO), both given 1 hour prior to the infusion.**
- Induction: 3 mg/m^2 IV (maximum of 4.5 mg) on days 1, 4, and 7 in combination with daunorubicin on days 1 to 3 and cytarabine as a continuous IV infusion on days 1 to 7. If a second induction cycle is required, omit gemtuzumab ozogamicin and administer only daunorubicin and cytarabine. Consult specific protocol for dosing.
- Consolidation (up to 2 cycles): 3 mg/m^2 IV (maximum of 4.5 mg) on day 1 (first and second consolidation cycles) in combination with daunorubicin IV on day 1 (first consolidation cycle) or on days 1 and 2 (second consolidation cycle) and cytarabine on days 1 to 4 (first and second consolidation cycles).
- Leukoreduction for patients with hyperleukocytosis: refer to manufacturer's recommendations for regimen adjustment.
- Dosage in renal impairment: no adjustment required for CrCl of 30 mL/min or greater. No data for CrCl less than 30 mL/min.
- Dosage in hepatic impairment: no adjustment is required if total bilirubin is no greater than 2 x ULN and AST and/or ALT is no greater than 2.5 x ULN. No data are available for patients with total bilirubin greater than 2 x ULN or AST/ALT greater than 2.5 x ULN.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze.
- The reconstituted solution is stable for 6 hours between 2-8°C and 3 hours at room temp below 25°C. Protect from light. Do not freeze.
- Stable for 12 hours between 2-8°C, protected from light, and for an additional 6 hours at room temp below 25°C (includes the 2-hour infusion time and, if needed, the 1 hour to allow for the refrigerated solution to reach room temp), protected from light, in NS at concentrations between 0.075 mg/mL and 0.234 mg/mL in PVC with DEHP or polyolefin (polypropylene and/or polyethylene) containers.
- Compatible with NS only.

MISCELLANEOUS

- Resuscitation equipment should be immediately available for the treatment of hypersensitivity and infusion-related reactions.
- Monitor blood counts, liver and renal function, sodium, potassium, magnesium, calcium, and phosphate at baseline, before each dose and as clinically indicated. Monitor ECG at baseline and as clinically indicated.

REFERENCES

5, 129, 135, 165, 208.

New monograph 2021

NAME OF MEDICATION
GenTAMICIN

OTHER NAMES
Garamycin ®

CLASSIFICATION
Antibiotic - aminoglycoside

INDICATIONS

- Treatment of serious infections caused by gram-negative organisms such as *Citrobacter, P. aeruginosa, Proteus, Klebsiella, E. coli, Enterobacter,* and *Serratia sp*. and for synergy in infections caused by gram-positive organisms such as *Enterococcus*.
- Treatment of plague (in the context of bioterrorism).
- Treatment of tularemia (in the context of bioterrorism).
- Perioperative prophylaxis.

ADMINISTRATION

- Intermittent IV infusion: dilute dose in 50-200 mL of D5W or NS (for children, volume of diluent should be consistent with the patient's needs). Pre-mixed bags also available. Administer over 30 minutes to 2 hours.
- IM.
- Consult Medical staff rules and regulations MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis) for specific dosing regimens and infusion time.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; urticaria, rash, pruritus.
- Otic: vestibular and auditory toxicity (risk factor: peak serum levels greater than 12 mg/L with traditional dosing).
- Renal: nephrotoxicity (risk factor: trough serum levels greater than 2 mg/L).
- Potential for neuromuscular blockade (rare).
- Local reactions: pain at injection site; phlebitis.

DOSAGE

- Traditional dosing: 1-2 mg/kg/dose IV or IM with dosing interval according to creatinine clearance:

CrCl (mL/min)	greater than 60	60-40	39-20	less than 20 or hemodialysis
Interval (hr)	8	12	24	frequency as per serum concentrations

  Refer to Miscellaneous section for monitoring serum levels.
- Extended interval dosing: 4-7 mg/kg/dose IV or IM once daily in selected patients. Refer to Miscellaneous section for monitoring serum levels.
- Synergy (with beta-lactams) for infections with gram-positive organisms (e.g., endocarditis): 3 mg/kg/day IV or IM in 1 to 3 divided doses.
- Treatment of plague: 5 mg/kg IV or IM once daily OR 2 mg/kg loading dose followed by 1.7 mg/kg IV or IM q8h for 10 days.
- Treatment of tularemia: 5 mg/kg IV or IM once daily for 10 days.
- Perioperative prophylaxis: 5 mg/kg (3 mg/kg at TOH) IV within 60 minutes before skin incision.
- Doses should be calculated using actual body weight, except in obese patients.
- Dosage in obesity (20-30% above ideal body weight (IBW)): dose should be calculated using dosing body weight: (DBW) = IBW (kg) + 0.4 X (actual body weight-IBW)

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Compatible with D5W, NS and RL.
- Stable for 48 hours at room temp and in the fridge in D5W and NS at a concentration of 1.2 mg/mL in PVC bags.
- Stable for 30 days at -20°C in NS or D5W at concentrations of 1 mg/mL in PVC bags.
- Stable for 30 days at room temp or in the fridge at a concentration of 40 mg/mL (undiluted) in polypropylene syringes.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 12 weeks in the fridge in NS at a concentration of 10 mg/mL in glass syringes.
- Incompatible with heparin.

MISCELLANEOUS

- Monitoring serum levels:
  - Traditional dosing: serum levels pre-dose: taken 5 minutes before next dose; should be less than 2 mg/L.
    serum levels post-dose: taken 30 minutes after the end of IV infusion or 60 minutes after the IM injection; should be less than 12 mg/L.
  - Extended interval dosing: serum levels pre-dose: taken 5 minutes before next dose; should be undetectable (i.e., less than 0.2 mg/L).
    serum levels post-dose: no need to monitor.

REFERENCES

1, 4, 5, 6, 95, 135, 143, 208, 303, 338.

NAME OF MEDICATION
GLATIRAMER ACETATE

OTHER NAMES
Copaxone ®, Glatect ™

CLASSIFICATION
Immunomodulator

INDICATIONS

- Treatment of patients with relapsing-remitting multiple sclerosis.

ADMINISTRATION

- Allow prefilled syringe to reach room temp before the injection by removing it from refrigeration for at least 20 minutes.
- Do NOT expel small air bubbles in the syringe.
- SC: arms (upper back portion), thighs (front and outer part, 5 cm above the knee and 5 cm below the groin), hips (upper outer rear quadrant) and abdomen (at least 5 cm away from the navel). Rotate site of injection; do NOT inject in the same area more than once a week.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; anaphylactoid reactions, rash.
- Immediate post-injection reaction: transient self-limited systemic reaction occurring (sporadically and unpredictably) in 14% of patients within a few seconds to minutes of injection: facial flushing, chest tightness, dyspnea, palpitations, urticaria and anxiety. Lasts up to 30 minutes; usually resolves spontaneously without sequelae. Generally occurs several months after initiating glatiramer.
- GI: nausea.
- Local reactions: erythema, pruritus, pain, induration, edema, lipoatrophy, skin necrosis (rare).

DOSAGE

- Copaxone ®: 20 mg SC once daily (avoid giving 2 injections in the same 12-hour period) OR 40 mg SC 3 times weekly (at least 48 hours between doses).
- Glatect ™: 20 mg SC once daily (avoid giving 2 injections in the same 12-hour period).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled glass syringes in the fridge; protect from light. Do not freeze.
- Stable for one month at room temp protected from light.

MISCELLANEOUS

- Copaxone ® 20 mg/1 mL and 40 mg/1 mL products are not interchangeable.

REFERENCES

1, 5, 38, 95.

Full revision 2015; limited revision 2017, 2018

NAME OF MEDICATION
GLUCAGON (RECOMBINANT)

OTHER NAMES
GlucaGen ®, GlucaGen Hypokit ®

CLASSIFICATION
Hyperglycemic agent

INDICATIONS

- In the emergency treatment of hypoglycemic reactions in diabetic patients receiving insulin. The drug is useful only if liver glycogen stores are normal.
- Diagnostic aid for radiologic examination of stomach, duodenum, small bowel and colon.
- Refractory cardiac failure (possesses inotropic and chronotropic actions).
- Beta-blocker overdose and calcium channel blocker overdose (possesses inotropic and chronotropic actions).
- To decrease motility and aid in removing foreign bodies in the lower esophagus, including food boluses.

ADMINISTRATION

- Reconstitute powder with manufacturer's diluent. Swirl gently until complete dissolution.
- IV direct: physician; RN may administer 1 mg or less in presence of physician. Administer at a rate of 1 mg/min.
- Continuous IV infusion: further dilute in D5W (at TOH: dilute 5 mg in 50 mL of D5W to obtain a concentration of 0.1 mg/mL).
- IM, SC.
- Do not use at concentrations greater than 1 mg/mL.
- Although GlucaGen ® (Novo Nordisk) is not approved by Health Canada for IV or SC administration, there is strong documentation in the literature supporting its efficacy and safety by these routes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, respiratory distress, rarely hypotension and anaphylactic shock.
- Cardiovascular: transient increase in BP and heart rate. May administer phentolamine 5-10 mg IV if sudden increases in BP.
- GI: nausea and vomiting.
- Hypokalemia.

DOSAGE

Adults:
- Hypoglycemic reactions: 1 mg SC, IM or IV. If no response within 15 minutes, may repeat 1-2 times; consider also use of parenteral glucose.
- Diagnostic aid: 0.25-2 mg IV OR 1-2 mg IM before start of procedure.
- Beta-blocker or calcium channel blocker overdose: 3-10 mg IV over 3-5 minutes OR 0.05-0.15 mg/kg IV over 1-2 minutes. Follow with a continuous IV infusion of 1-5 mg/hr (maximum: 10 mg/hr). A patient may require up to 50 mg over a 24-hour period.
- Cardiac failure: 0.01-0.05 mg/kg IV bolus OR 1-3 mg/hr as continuous IV infusion.
- Foreign body obstruction in the esophagus: 0.5-2 mg IV. Repeat in 10 minutes if necessary.

Pediatrics:
- Hypoglycemic reactions: if weight more than 20 kg: 1 mg SC, IM or IV.
  if weight less than 20 kg: 0.5 mg OR 0.02-0.03 mg/kg (maximum: 0.5 mg) SC, IM or IV.
- If no response within 15 minutes, may repeat 1-2 times; consider also use of parenteral glucose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store Glucagon ® (Lilly) between 15-30°C. Store GlucaGen ® (NovoNordisk) between 2-8°C, protected from light; do not freeze.
- The reconstituted product is clear and has water-like consistency. Discard any unused portion.

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COMPATIBILITY, STABILITY (Cont.)

- Soluble at a pH of less than 3 or greater than 9.5; be cautious of the pH of any IV solution to which it is to be added.
- Stable in dextrose solutions; stability data in NS is conflicting.

MISCELLANEOUS

- When used for hypoglycemia, failure to respond necessitates immediate administration of dextrose IV.
- Carbohydrates should be given as soon as patient awakens. Use glucagon with dextrose in water for patients in hypoglycemic coma.
- 1 mg = 1 unit.
- Contains lactose.

REFERENCES

1, 5, 40, 82, 95, 205, 368.

NAME OF MEDICATION
GLYCOPYRROLATE

OTHER NAMES
Glycopyrronium Bromide, Robinul ®

CLASSIFICATION
Anticholinergic

INDICATIONS
- Adjunctive management of peptic ulcer disease when oral medication is not tolerated or when a rapid effect is desired.
- Preoperatively to reduce secretions.
- Intraoperatively to prevent cholinergic effects.
- Postoperatively during reversal of neuromuscular blockade to protect against the muscarinic effects (e.g., bradycardia, excessive secretions) of cholinergic agents such as neostigmine or pyridostigmine.
- To reduce upper airways secretions in palliative care patients.

ADMINISTRATION
- IV direct: physician or RN. May be given undiluted or may dilute with compatible fluids. Administer at a maximum rate of 0.2 mg/min.
- IM; SC (palliative care).
- SC infusion (palliative care); must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, urticaria, pruritus, anaphylaxis.
- Cardiovascular: tachycardia, flushing.
- GI: vomiting, constipation, dry mouth, gastroesophageal reflux.
- CNS: headache.
- Ophthalmic: blurred vision, increased intraocular pressure (especially in patients with angle-closure glaucoma).
- Local reactions: pruritus, edema, erythema, pain at injection site.

DOSAGE
Adults:
- Adjunctive treatment of peptic ulcer disease: 0.1-0.2 mg IM or IV q4h, 3 or 4 times daily.
- Preoperative: 0.004 mg/kg IM 30-60 minutes before anesthesia induction.
- Intraoperative: 0.1 mg IV; may repeat every 2-3 minutes.
- Postoperative: 0.2 mg IV for each 1 mg of neostigmine or 5 mg of pyridostigmine, given either in same syringe as or before the anticholinesterase agent.
- In palliative care patients, to reduce upper airways secretions: 0.2-0.6 mg SC q2-4h prn (often given q4-8h prn) OR 1.2-2 mg/24 hours as a continuous SC infusion.

Pediatrics (children between 2 and 12 years of age):
- Preoperative: 0.004-0.01 mg/kg IM 30-60 minutes before anesthesia induction.
- Intraoperative: 0.004 mg/kg IV up to a maximum of 0.1 mg. May repeat every 2-3 minutes.
- Postoperative: 0.2 mg IV for each 1 mg of neostigmine or 5 mg of pyridostigmine, given either in same syringe as OR before the anticholinesterase agent.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials between 15-30°C.
- Stable for 48 hours at room temp in D5W, NS, Ringer's, or D5-1/2NS at a concentration of 0.8 mg/L.
- Stable for 24 hours at room temp or in the fridge in NS at a concentration of 0.1 mg/mL in polypropylene syringes.
- Stable for 90 days at room temp or in the fridge at a concentration of 0.2 mg/mL (undiluted) in polypropylene syringes.
- Incompatible with drugs or solutions with an alkaline pH.

MISCELLANEOUS
- Retards gastric emptying; if more than 2 successive parenteral doses are given, nasogastric intubation is often necessary.
- Onset: 1 minute following IV injection and 15-30 minutes following IM or SC injection.

REFERENCES
1, 4, 5, 6, 40, 82, 95, 135, 208, 457, 505.

Full revision 2015; limited revision 2016, 2020

NAME OF MEDICATION
GOLIMUMAB

OTHER NAMES
Simponi ®, Simponi ® IV

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS

- Treatment of moderate to severe rheumatoid arthritis, in combination with methotrexate (SC and IV formulations).
- Treatment of moderate to severe psoriatic arthritis (PsA), as monotherapy or in combination with methotrexate (SC and IV formulations).
- Treatment of active ankylosing spondylitis (AS) in patients with an inadequate response or intolerance to conventional therapies (SC and IV formulations).
- Treatment of severe active non-radiographic axial spondyloarthritis in patients with an inadequate response or an intolerance to NSAIDs (SC formulation).
- Treatment of moderate to severe ulcerative colitis in patients who did not respond adequately or have contraindications to conventional therapy (including 5-ASA, corticosteroids, azathioprine, 6-mercaptopurine) (SC formulation).

ADMINISTRATION

- **There are two formulations: one for IV use (vial) and one for SC use (prefilled syringe and autoinjector). Ensure using the correct one.**
- Intermittent IV infusion: ensure to use the IV solution which has a concentration of 50 mg/4 mL (12.5 mg/mL); dilute dose by slowly adding it to a minibag to obtain a final volume of 100 mL NS (withdraw a volume of NS solution from a 100 mL minibag equal to the volume of golimumab solution that will be added and discard the withdrawn solution). Mix gently. Infuse over 30 minutes (+/- 10 minutes) with an in-line low-protein-binding filter of 0.22 microns or less.
- SC: use the autoinjector (50 mg/0.5 mL or 100 mg/1 mL) or prefilled syringe (50 mg/0.5 mL or 100 mg/1 mL). Warm at room temp for 30 minutes before administration. Do NOT shake. Inject at a 90° angle for the autoinjector and at a 45° angle for the prefilled syringe (with the latter, pinch skin) into the thighs, abdomen (at leach 2 inches below the navel) or outer area of the upper arms. If multiple injections are required, use a different site for each injection. Do NOT inject into areas where skin is tender, bruised, red, scaly or hard, has scars or stretch marks. Rotate injection sites. Refer to manufacturer instructions for more details.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria. Rarely anaphylaxis and serum sickness. May occur after first administration.
- Cardiovascular: hypertension, CHF (new onset or worsening).
- GI: constipation.
- CNS: dizziness.
- Elevation of liver enzymes.
- Infections: patients receiving golimumab may be more prone to develop infections, including tuberculosis, fungal or opportunistic infections. Discontinue if serious infection occurs.
- Anemia, leukopenia.
- Pyrexia.
- Local reactions: erythema, urticaria, induration, pain, bruising, paresthesia, irritation, pruritus.

DOSAGE

- Rheumatoid arthritis, PsA and AS: 2 mg/kg IV at weeks 0, 4 and every 8 weeks thereafter OR 50 mg SC once a month.
- Non-radiographic axial spondyloarthritis and active ankylosing spondylitis: 50 mg SC once a month.
- Ulcerative colitis: 200 mg SC at week 0, 100 mg SC at week 2 and then 50 mg SC every 4 weeks thereafter. A maintenance dose of 100 mg SC every 4 weeks can also be considered.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials, autoinjectors and prefilled syringes between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Vials, autoinjectors and prefilled syringes may be stored at temperature not exceeding 25°C for a single period of 30 days, protected from light; do not refrigerate afterwards.
- The diluted IV solution is stable for 6 hours at RT, protected from light, in NS.

…/Cont.

NAME OF MEDICATION	
GOLIMUMAB	
OTHER NAMES	CLASSIFICATION
Simponi ®, Simponi ® IV	Immunomodulator, Monoclonal antibody

…/Cont.

MISCELLANEOUS

- The needle cover of both the autoinjector and prefilled syringe contains latex.
- Do not administer live or live attenuated vaccines concurrently with this drug.
- Do not use with other biologic DMARDs (e.g., abatacept, anakinra, etanercept) because of possible increased risk of infections and similar toxicities.
- Patient should be tested for hepatitis B virus infection and tuberculosis before initiating therapy.

REFERENCES

5.

New monograph 2015; limited revision 2016, 2017, 2019

NAME OF MEDICATION
GONADORELIN

OTHER NAMES
GnRH, Lutrepulse ®, Synthetic gonadotropin-releasing hormone

CLASSIFICATION
Ovulatory agent

INDICATIONS

- Induction of ovulation in women with primary hypothalamic amenorrhea.

ADMINISTRATION

- Reconstitute each vial of 0.8 mg with 8 mL of diluent provided (NS), immediately prior to use. Shake vial a few seconds to obtain a clear and colourless solution. Transfer to a polypropylene plastic reservoir.
- Intermittent IV infusion, Intermittent SC infusion: must use a suitable pulsatile pump and infusion catheter. Pump should be set to deliver the doses over a pulse period of 1 minute, repeated every 90 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis.
- GI: nausea, vomiting, diarrhea, abdominal distension/discomfort; may be signs of ovarian hyperstimulation syndrome.
- Ovarian hyperstimulation syndrome (rare): sudden ovarian enlargement, ascites, pain, pleural effusion.
- Local reactions: thrombophlebitis, hematoma, swelling, infection, redness, induration at site of injection.

DOSAGE

- Recommended starting dose is 5 mcg IV/SC every 90 minutes. Dose range: 1 to 20 mcg.
- Response usually occurs within 2 to 3 weeks following initiation of therapy.
- When ovulation occurs, therapy should be continued for another 2 weeks to maintain the corpus luteum.
- Recommended treatment interval before dosage adjustment is 21 days.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C.
- Reconstituted solution is stable for 45 days between 24-37°C when stored in vials and reservoir bags.
- Stable for up to 16 hours in catheter tubing.

MISCELLANEOUS

REFERENCES

2, 5, 95, 135.

Full revision 2015; limited revision 2016, 2020

NAME OF MEDICATION
GOSERELIN

OTHER NAMES
Zoladex ®, Zoladex ® LA

CLASSIFICATION
Gonadotropin - releasing hormone analog

INDICATIONS
- Treatment of prostate cancer and breast cancer.
- Treatment of endometriosis.
- As an endometrial thinning agent prior to endometrial ablation.
- Treatment of uterine leiomyoma.

ADMINISTRATION
- SC (anterior abdominal wall below the navel line).

POTENTIAL ADMINISTRATION HAZARDS
- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis (rare), rash, erythema, urticaria.
- Cardiovascular: hot flashes, hypotension, hypertension, heart failure, myocardial infarction, QT interval prolongation.
- GI: nausea, vomiting, diarrhea.
- CNS: mood changes, depression, headache, dizziness.
- Tumour flare reaction (e.g., increased bone pain, spinal cord depression, urinary tract obstruction).
- Pitituary apoplexia (symptoms of sudden headache, vomiting, visual changes, altered mental status with sometimes cardiovascular collapse); may appear within the hour following injection and can be severe.
- Anemia, hyperglycemia, fatigue, sexual dysfunction, arthralgia.
- Local reactions: pain, erythema, bruising at injection site.

DOSAGE
- Prostate cancer:
  - 3.6 mg SC every 28 days.
  - 10.8 mg SC every 13 weeks using the “LA” preparation.
- Breast cancer and uterine leiomyoma:
  - 3.6 mg SC every 28 days.
- Endometriosis:
  - 3.6 mg SC every 28 days.
  - 10.8 mg SC every 12 weeks, using the “LA” preparation.
- Endometrial thinning:
  - 2 doses of 3.6 mg SC given 4 weeks apart, with surgery planned for between 0 and 2 weeks after the second injection.
- Dosage in renal impairment: no dosage adjustment required.
- Dosage in hepatic impairment: no dosage adjustment required.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store between 2 and 25°C. Protect from light and moisture.

MISCELLANEOUS
- Note that two strengths are available: 3.6 mg (1 month formulation) and 10.8 mg (3 month formulation).
- To prevent the initial disease flare in patients with prostate cancer, begin antiandrogen therapy (e.g., bicalutamide, cyproterone, flutamide, nilutamide) 1 week prior to commencing goserelin therapy and continue for 2 weeks following the first injection.

REFERENCES
1, 5, 95, 129, 165.

Full revision 2015; limited revision 2016, 2019, 2020

NAME OF MEDICATION
GRANISETRON

OTHER NAMES
Kytril ®

CLASSIFICATION
Antiemetic

INDICATIONS

- Prevention of nausea and vomiting associated with chemotherapy.
- Prevention and treatment of postoperative nausea and vomiting.

ADMINISTRATION

- IV direct: physician or RN. Administer undiluted over 30 seconds.
- Intermittent IV infusion: dilute dose in NS, D5W, or dextrose-saline to a total volume of 20-50 mL prior to administration. Infuse dose over 5 minutes, beginning within 30 minutes before initiation of chemotherapy.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, dyspnea, hypotension, urticaria.
- Cardiovascular: QT interval prolongation (rare).
- GI: constipation, diarrhea, abdominal pain.
- CNS: headache, somnolence.
- Serotonin syndrome (e.g., agitation, confusion, tachycardia, muscle twitching or stiffness, fever, loss of consciousness or coma) especially when administered with other serotonergic drugs.
- Asthenia.

DOSAGE

- Adults:
  - Chemotherapy-induced nausea and vomiting: 10-40 mcg/kg IV OR 1 mg IV.
  - Postoperative nausea and vomiting: 1 mg IV.
- Pediatrics:
  - Chemotherapy-induced nausea and vomiting: 10-40 mcg/kg IV.
  - Prevention of postoperative nausea and vomiting: 40 mcg/kg IV (maximum 0.6 mg).
- Dosage adjustment is not required in patients with renal or hepatic impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light. Do not freeze.
- Compatible with NS, D5W, dextrose-saline, mannitol 20%.
- Stable for 35 days at room temp or in the fridge in NS or D5W at concentrations of 0.008-0.53 mg/mL in PVC bags with or without dexamethasone 0.05-0.35 mg/mL.
- Stable for 7 days in NS and 14 days in D5W in the fridge at a concentration of 0.02 mg/mL in a disposable elastomeric infusion system.
- Stable for 14 days at room temp or in the fridge in NS or D5W at a concentration of 0.05, 0.07 or 0.1 mg/mL in polypropylene syringes.
- Stable for 24 hours at room temp exposed to light in NS or D5W at a concentration of 0.2 mg/mL in polypropylene syringes.
- Stable for 15 days at room temp or in fridge protected from light undiluted in polypropylene syringes.

MISCELLANEOUS

REFERENCES

1, 4, 5, 6, 65, 69, 82, 95, 143.

Full revision 2015; limited revision 2016

NAME OF MEDICATION
HAEMOPHILUS B CONJUGATE VACCINE

OTHER NAMES
Act-HIB ®, Hiberix ®

CLASSIFICATION
Vaccine

INDICATIONS

- For routine immunization of all children between 2 and 59 months of age.
- For older children and adults with chronic conditions associated with increased risk of invasive *Haemophilus influenzae* type b disease such as persons with splenic dysfunction (e.g., sickle cell disease, asplenia), antibody deficiency, HIV infection or certain malignancies, solid organ transplant, as well as candidates and recipients of cochlear implants.

ADMINISTRATION

- For Act-Hib ®: reconstitute vaccine with 0.5 mL of provided diluent; shake the vial gently until obtaining a clear and colourless solution. It can also be reconstituted with the vaccine Quadracel ® for additional immunization coverage; in this case, shake well until obtaining a cloudy uniform suspension.
- For Hiberix ®: reconstitute vaccine with the entire content of the diluent vial. Shake well until obtaining a clear and colourless solution. It can also be reconstituted with the vaccine Infanrix ®-IPV for additional immunization coverage; in this case, shake well until obtaining a cloudy uniform suspension.
- IM: into the deltoid muscle; for infants younger than 1 year of age into the anterolateral aspect of the mid-thigh.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity (rare): urticaria, pruritus, rash, facial and laryngeal edema, angioedema.
- GI: vomiting.
- CNS: drowsiness, irritability.
- Fever.
- Local reactions: pain, redness, swelling, induration at injection site.

DOSAGE

- 0.5 mL IM.
- Refer to specific manufacturer's monograph for complete dosage schedule.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Do not freeze.
- Use the vaccine immediately after its reconstitution.
- Consult the manufacturer's monograph for information on additional vaccines that can be administered concomitantly (providing separate syringes and different sites are used).

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The *Haemophilus* b conjugate vaccine is an inactivated vaccine.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.
- The rubber stopper of the diluent vial to reconstitute the Act-HIB ® vaccine contains latex.

REFERENCES

1, 5, 31.

Full revision 2015; limited revision 2016

NAME OF MEDICATION
HALOPERIDOL

OTHER NAMES
Haldol ®

CLASSIFICATION
Antipsychotic

INDICATIONS
- To achieve rapid symptom control in acute and chronic psychosis (including schizophrenia and mania), acute delirium tremens, or severe malignant catatonic excitement.
- To control agitation in critically ill patients.
- For postoperative nausea and vomiting.

ADMINISTRATION
- IV direct: physician or RN; ***blood pressure monitoring*** for doses greater than 5 mg. Give at a rate no faster than 5 mg/min.
- Intermittent IV infusion: dilute dose in D5W.
- Continuous IV infusion: dilute in D5W to a concentration of 3 mg/mL or less (refer to Compatibility, Stability section).
- Although haloperidol is not approved by Health Canada for IV administration, the IM product may be given IV.
- IM, SC (palliative care).

POTENTIAL ADMINISTRATION HAZARDS
- Cardiovascular: venous thromboembolism, hypotension, tachycardia, angina, QT interval prolongation, rare reports of torsades de pointes. The risk of torsades de pointes is dose-dependant and increases as total daily dosages of 35-50 mg or more are exceeded and when the IV route is used.
- CNS: sedation, seizures, insomnia, restlessness, anxiety, euphoria, agitation, hallucinations, depression, confusion, headache, lethargy, vertigo, extrapyramidal reactions, such as akathisia or Parkinson-like condition.
- Neuroleptic malignant syndrome.
- Anticholinergic effects such as dryness of mouth, constipation, urinary retention, blurred vision, diaphoresis.

Antidote: severe hypotension can be alleviated by administration of norepinephrine or phenylephrine. Epinephrine is contraindicated.

DOSAGE
- Prompt control of agitated patients: 2-5 mg IM (in the elderly 0.5-1 mg IM) q4-8h prn; may repeat q1h prn. Dose must be individualized.
- Delirium: initial dose of 1-2 mg IV (in the elderly 0.25-0.5 mg IV) q2-4h.
- Continuous IV infusion: 10 mg initial IV bolus followed by initial IV infusion of 5-10 mg/hr. If no response after 30 minutes, repeat 10 mg IV bolus and increase infusion rate by 5 mg/hr.
- Postoperative nausea and vomiting: 0.5-2 mg IV/IM.
- Dosage in renal impairment: no adjustment required.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store between 15-30°C. Protect from light. Do not freeze.
- Stable for 7 days at room temp in NS when diluted between 0.1 and 0.75 mg/mL in glass vials.
- Stable for 7 days at room temp in D5W when diluted between 0.1 and 3 mg/mL in glass vials.
- Stable for 7 days at room temp in 1/2NS and RL at concentrations of 0.1-1 mg/mL in glass vials.
- Stable for 38 days at room temp in D5W at a concentration of 0.1 mg/mL in PVC and amber glass bottles.
- Incompatible with heparin.

MISCELLANEOUS
- Use in extreme caution, if at all, in patients with Parkinsonism and in the elderly.
- Consider a baseline ECG to assess for QTc interval prolongation.

REFERENCES
1, 4, 5, 6, 49, 51, 95, 135, 143, 369, 457.

Full revision 2015; limited revision 2016

NAME OF MEDICATION
HALOPERIDOL DECANOATE (LONG ACTING)

OTHER NAMES
Haldol L.A. ®

CLASSIFICATION
Antipsychotic

INDICATIONS

- Long-acting depot medication used for treatment of chronic schizophrenia.

ADMINISTRATION

- Deep IM: use at least 21 gauge dry needle, give deep IM into large muscle, use Z track method, rotate sites.
- Aspirate before injection to avoid intravascular injection. Do NOT massage injection site.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: venous thromboembolism, hypotension, angina, tachycardia, QT interval prolongation, torsades de pointes, ventricular arrhythmias.
- CNS: extrapyramidal effects such as akathisia or Parkinson-like condition, seizures, drowsiness/insomnia, tardive dyskinesia.
- Neuroleptic malignant syndrome.
- Anticholinergic side effects: dry mouth, constipation, urinary retention, blurred vision, diaphoresis.
- Local reactions: inflammation and nodules at injection site (more common in gluteus).

DOSAGE

- Usual initial IM dose: 10 to 20 times the daily oral dose. If the IM dose is greater than 100 mg IM, administer in 2 injections 3 to 7 days apart.
- Usual maintenance dose: 50-200 mg IM every 4 weeks, adjusted according to patient's response. Maximum of 450 mg/month.
- Dosage in renal and hepatic impairment: use with caution if severe impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15 and 30°C. Protect from light. Do not freeze.
- Do not let drug stand in syringe for longer than 15 minutes.

MISCELLANEOUS

- Peak plasma levels can occur 1-9 days after injection; however it may take up to three months to reach steady-state levels with monthly dosing.

REFERENCES

1, 5, 51, 135.

Full revision 2015

NAME OF MEDICATION
HEMIN

OTHER NAMES
Hematin, Panhematin ®

CLASSIFICATION
Blood product

INDICATIONS

- Management of recurrent attacks of acute intermittent porphyria temporally related to the menstrual cycle in susceptible women.

ADMINISTRATION

- Reconstitute each 268 mg vial with 48 mL of SWFI to obtain a final concentration of 5.4 mg/mL. Shake well for 2 to 3 minutes.
- Intermittent IV infusion: infuse undiluted directly from reconstituted vial over at least 30 minutes into a large arm vein or central venous catheter; use a sterile filter (pore size 0.45 microns or less). Flush vein with 100 mL of NS after the infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, anaphylaxis.
- CNS: headache.
- Endocrine and metabolic: risk of iron overload with multiple infusions.
- Hematologic: mild, transient anticoagulation (avoid concurrent anticoagulation therapy).
- Fever.
- Risk of transmission of infectious agents, including viruses, and theoretically, the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.
- Local reactions: infusion site reactions, phlebitis.

DOSAGE

- Standard dose in clinical practice: 2.3 to 3.1 mg/kg/day IV (range of 0.8 to 3.1 mg/kg/day IV) for 3 to 14 days based on clinical signs. For more severe cases, the dose may be repeated no earlier than every 12 hours. Maximum of 4.6 mg/kg/day.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility).

- Store vials between 20-25°C.

MISCELLANEOUS

- Monitor iron and serum ferritin in patients receiving multiple infusions of hemin due to the risk of iron overload.
- Vials contain sorbitol.

REFERENCES

5, 95, 135.

New monograph 2021

NAME OF MEDICATION
HEPARIN

OTHER NAMES
Unfractionated heparin

CLASSIFICATION
Anticoagulant

INDICATIONS
- Treatment and prophylaxis of thromboembolic disorders.
- ST elevation myocardial infarction.
- Non-ST elevation acute coronary syndrome/unstable angina.
- Anticoagulation during percutaneous coronary intervention (PCI).
- As a heparin lock flush solution to maintain patency of indwelling catheters.
- As an adjunct in the treatment of coronavirus disease 2019 (COVID-19; SARS-CoV-2 infection) in non-critically ill patients.

ADMINISTRATION
- IV direct: physician or RN. Administer undiluted over 1 minute.
- Intermittent IV infusion, Continuous IV infusion (preferred): may dilute with NS or D5W and administer as intermittent or continuous infusion. Except for fluid restricted patients, use a 100 unit/mL infusion solution of heparin by either: 1) diluting 25 000 units in 250 mL of D5W or NS (invert bag at least six times for complete mixing), OR 2) using a 250 mL premixed bag of 100 units/mL in D5W. Must be administered by an infusion pump.
- SC: pinch the skin for deep SC administration. Use a 25- or 26-gauge 1/2-inch needle. Administer on abdomen, above iliac crest. Rotate sites. Do NOT massage injection site.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: fever, chills, urticaria, rhinitis, lacrimation, asthma, anaphylactoid reactions; allergic vasospastic reactions.
- Hemorrhage. Increased risk of hemorrhage in patients with subacute bacterial endocarditis, renal or hepatic disease, severe hypertension. Continuous IV infusion has a lower incidence of bleeding complications than intermittent IV infusion.
- Heparin-induced thrombocytopenia (HIT).
- Hepatic: increased LFTs.
- Local reactions: irritation, erythema, pain, hematoma, ulceration at deep SC administration site.

Antidote: The anticoagulant effect of heparin can be neutralized by protamine. Refer to protamine in this manual for more details.

DOSAGE
- Treatment of thromboembolism: 5000 units (or 80 units/kg) IV bolus followed by a continuous IV infusion of 1300 units/hr (or 18 units/kg/hr) OR 5000 units IV bolus (bolus may also be given as 333 units/kg SC) followed by intermittent SC injections of 250 units/kg q12h OR 10 000 units IV bolus followed by intermittent IV infusions of 5000-10 000 units q4-6h.
- Prophylaxis of thromboembolism: 5000 units SC q8-12h. If administered prior to surgery: give first dose of 5000 units SC 2 hours prior to surgery.
- ST elevation myocardial infarction (alteplase or tenecteplase are also given): bolus of 60 units/kg IV (maximum 4000 units), followed by a continuous IV infusion of 12 units/kg/hr (maximum 1000 units/hr).
- Non-ST elevation acute coronary syndrome/unstable angina: initial bolus of 60 units/kg IV (maximum 4000 units), followed by an initial infusion of 12 units/kg/hr (maximum 1000 units/hr).
- PCI anticoagulation: if patient was not on anticoagulation therapy, 50 to 100 units/kg IV bolus; if patient has received prior anticoagulation, 2000 to 5000 units IV bolus.
- Infusion rates/injections to be adjusted to maintain therapeutic aPTT or heparin assay.
- To maintain patency of indwelling catheters: inject a solution of 10 units/mL or 100 units/mL in a sufficient quantity to fill the device.
- As an adjunt in the treatment of COVID-19: use a therapeutic continuous IV heparin infusion such as the one for the treatment of thrombosis.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Do not freeze.
- Compatible in NS, D5W, dextrose-saline solutions, Ringer's and RL.
- Solutions are colourless to slightly yellow.
- Stable for 48 hours at room temp in NS or D5W at concentrations of 20 units/mL and 40 units/mL in PVC bags.
- Stable for 8 hours at room temp and in the fridge in D5W or SWFI at a concentration of 300 units/mL in polypropylene syringes.
- Stable for 21 days at room temp and in the fridge in NS at a concentration of 500 units/mL in polypropylene syringes.

MISCELLANEOUS

- Derived from porcine tissues.
- At TOH, 2 order sets are available for use: 1) in acute coronary syndromes or atrial fibrillation; 2) venous thrombosis.

REFERENCES

1, 4, 5, 40, 70, 135, 143, 208, 540.

NAME OF MEDICATION
HEPATITIS A VACCINE

OTHER NAMES Avaxim ®, Avaxim-Pediatric ®, Havrix 1440 ®, Havrix 720 Junior ®, Vaqta ®, Vaqta ® Pediatric/Adolescent

CLASSIFICATION
Vaccine

INDICATIONS

- For active immunization against infection caused by hepatitis A virus.

ADMINISTRATION

- Shake well to obtain a uniform suspension.
- IM: into the deltoid muscle for older children and adults and into the anterolateral area of the thigh for infants.
- SC: only for patients with bleeding disorders at high risk of hemorrhage with IM administration (e.g., thrombocytopenia, hemophilia).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis/anaphylactoid reactions (rare), serum sickness-like reaction, angioedema, erythema multiforme, urticaria, rash, pruritus.
- GI: nausea, vomiting, diarrhea, abdominal pain, loss of appetite.
- CNS: headache, irritability, drowsiness.
- Fever, myalgia, arthralgia, malaise, fatigue, asthenia.
- Local reactions: pain, redness, warmth and swelling at injection site. Nodules with SC administration.

DOSAGE

- Adults:
  - Havrix 1440 ® and Vaqta ®: 1 mL IM.
  - Avaxim ®: 0.5 mL IM.
- Pediatrics: Avaxim-Pediatric ®, Havrix 720 Junior ® and Vaqta ® Pediatric/Adolescent: 0.5 mL IM.
- Refer to specific manufacturer's monograph and the Canadian Immunization Guide for complete dosage schedule and ages for which the individual vaccines are to be used.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes between 2-8°C. Do not freeze.
- Vaqta ® and Vaqta ® Pediatric/Adolescent are stable for 72 hours at up to 25°C.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- Contains trace amounts of neomycin.
- The hepatitis A vaccine is an inactivated vaccine.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.
- Seroconversion following SC administration is slower than with IM administration.
- Human immune globulin does not appear to interfere with the response to the hepatitis A vaccine. When given at the same time, the two products should be administered at separate sites.

REFERENCES

1, 5, 31, 135.

Full revision 2015; limited revision 2016, 2020

NAME OF MEDICATION
HEPATITIS B VACCINE

OTHER NAMES
Engerix ®-B, Recombivax HB ®,
Recombivax HB Adult Dialysis ®

CLASSIFICATION
Vaccine

INDICATIONS

- Primary immunization against hepatitis B infection and to promote active immunity to hepatitis B infection in individuals considered at high risk of potential exposure to hepatitis B virus or hepatitis B surface antigen positive material (e.g., blood, blood products, body fluids).

ADMINISTRATION

- Shake well to obtain a uniform suspension.
- IM: into the deltoid for older children and adults, and into the anterolateral area of the thigh for infants.
- SC: only for patients with bleeding disorders at risk of hemorrhage with IM administration (e.g., thrombocytopenia, hemophilia).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis/anaphylactoid reactions (rare), serum sickness-like reaction, angioedema, erythema multiforme, urticaria, rash, palpitations, dyspnea, arthritis, hypotension.
- GI: nausea, diarrhea, abdominal pain.
- CNS: headache, irritability, drowsiness.
- Respiratory: pharyngitis, symptoms of upper respiratory infection.
- Fever, fatigue, weakness, malaise.
- Local reactions: pain, pruritus, erythema, swelling, warmth, ecchymoses, induration; nodules with SC administration.

DOSAGE

- Usual adult dose: 20 mcg (Engerix ®-B) or 10 mcg (Recombivax HB ®) IM.
- Usual pediatric dose: 10 to 20 mcg (Engerix ®-B) or 2.5 to 10 mcg (Recombivax HB ®) IM depending on immunization schedule and age.
- Patients undergoing hemodialysis and immunocompromised patients often require doubling the dose and monitoring of antibody titers.
- Refer to specific manufacturer's monograph and the Canadian Immunization Guide for complete dosage schedule and ages for which the individual vaccines are to be used.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Do not freeze.
- Recombivax HB ® is stable for 72 hours at room temp and between 0-2°C.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The hepatitis B vaccine is an inactivated vaccine.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.
- Contraindications: sensitivity to yeast (Engerix ®-B and Recombivax HB ®) or to latex (Recombivax HB ®).
- Hepatitis B immune globulin does not appear to interfere with the response to hepatitis B vaccine. When given at the same time, the two products should be administered at separate sites.

REFERENCES

1, 5, 31.

Full revision 2015; limited revision 2020

NAME OF MEDICATION
HUMAN PAPILLOMAVIRUS VACCINE

OTHER NAMES
Cervarix ®, Gardasil ® 9

CLASSIFICATION
Vaccine

INDICATIONS

- Cervarix ®: for girls and women 9-45 years of age for the prevention of infection caused by Human Papillomavirus (HPV) types 16 and 18 and diseases associated with these HPV types (such as cervical cancer, intraepithelial neoplasia and adenocarcinoma *in situ*).
- Gardasil ® 9: for patients 9-45 years of age for the prevention of infection caused by HPV types 6, 11, 16,18, 31, 33, 45, 52, 58 and diseases associated with these HPV types (such as cervical, vulvar, vaginal, anal, penile cancers and related intraepithelial neoplasia; genital warts; cervical adenocarcima *in situ)*.

ADMINISTRATION

- Shake well before use.
- IM: into the deltoid region of the upper arm (Gardasil ® 9 and Cervarix ®) or in the higher anterolateral area of the thigh (Gardasil ® 9).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, bronchospasm, anaphylaxis/anaphylactoid reactions, angioedema, erythema multiforme.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: dizziness, syncope with or without tonic-clonic movements, headache.
- Fever, fatigue, myalgia, arthralgia.
- Local reactions: pain, swelling, erythema, pruritus, hematoma at injection site.

DOSAGE

- 0.5 mL IM; refer to specific manufacturer's monograph and The Canadian Immunization Guide for complete dosage schedule.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes between 2-8°C. Protect from light. Do not freeze.
- Gardasil ® 9 is stable for 72 hours between 8-25°C and between 0-2°C. Do not use if the content has frozen.
- Cervarix ® is stable for 72 hours between 8-25°C and for 24 hours between 25-37°C.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- Gardasil ® 9 is contraindicated in patients allergic to yeast protein.
- Inactivated vaccine made from virus-like particles.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.

REFERENCES

1, 5, 31.

Full revision 2015; limited revision 2016, 2020, 2021

NAME OF MEDICATION
HUMAN PROTHROMBIN COMPLEX

OTHER NAMES
Beriplex ® P/N, Octaplex ®

CLASSIFICATION
Coagulation factor

INDICATIONS

- Reversal of warfarin therapy or vitamin K deficiency in patients exhibiting major bleeding manifestations or requiring urgent (less than 6 hours) surgical procedure.
- Reversal of rivaroxaban, apixaban and edoxaban activity in case of life-threatening bleeding.
- Reversal of dabigatran, if idarucizumab unavailable, in case of life-threatening bleeding.

ADMINISTRATION

- For IV use: warm diluent and concentrate to room temp. Do not exceed 37º C. Reconstitute the powder with the provided diluent and transfer filter set according to manufacturer's instructions.
- Intermittent IV infusion: transfer dose to an empty sterile plastic bag (note: at TOH, this step is done by Transfusion Medicine). Before administration, flush IV line with NS or D5W; during administration, do NOT run in the same line with any other products including fluids with no medications (e.g., saline or dextrose solutions); after administration is completed, reflush with NS or D5W. Must be administered by an infusion pump.
  Manufacturers' recommendation: For Octaplex ®: administer at an initial rate of 1 mL/min; after 10 minutes, if patient is tolerating well the infusion, may increase rate up to 3 mL/min. For Beriplex ® P/N: administer at the maximum rate of 8 mL/min.
- At TOH, administer both Octaplex ® and Beriplex ® P/N at a rate of 6 mL/min.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0045 (Blood products and blood derivatives: Administration of).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, urticaria, flushing, chills, wheezing, angioedema, hypotension, chest tightness, tachycardia.
- GI: nausea.
- CNS: headache.
- Pyrexia.
- Thromboembolic complications (rare).
- Remote risk of transmission of infectious agents, including viruses, as this product is prepared from large pools of human plasma.
- Local reactions: burning at injection site.

DOSAGE

Reversal of warfarin therapy/vitamin K deficiency:
- Dose is expressed in mL and is based on the INR; a physician experienced in the treatment of coagulation disorders may be consulted. The following dosing regimen is suggested (this may differ from recommendations of the manufacturers):

Initial INR	Less than 3	3 - 5	Greater than 5
Dose in mL	40 mL IV	80 mL IV	120 mL IV

- If the INR is unknown and major bleeding is present: 80 mL IV.
- Single dose should not exceed 120 mL.
- Dose may be repeated in patients who continue to bleed or require urgent surgery if INR 10-30 minutes post administration is not corrected and there is insufficient time to wait for vitamin K to take effect.

Reversal of rivaroxaban, apixaban, edoxaban and reversal of dabigatran activity if idarucizumab unavailable:
- 50 units/kg (maximum of 3000 units) IV. At TOH: no maximum amount IV.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store powder and diluent between 2 to 25°C; do not freeze; protect from light.
- Reconstituted solution should be clear or slightly opalescent.
- According to manufacturers' respective monographs, the solution should be used immediately; however, if not administered immediately, storage should not exceed 3 hours at room temp for Beriplex ® P/N or 8 hours at room temp or in the fridge for Octaplex ®. Of note, the manufacturer of Octaplex ® has additional data to support that the reconstituted product is stable for 24 hours at room temp.

MISCELLANEOUS

- Contains heparin; do not give to patients with a history of heparin-induced thrombocytopenia.
- Human prothrombin complex concentrate contains the following: factors II, VII, IX, X, Protein C and Protein S.
- Administration of vitamin K_1 (10 mg IV) is strongly recommended if INR reversal is required for longer than 6 hours (which is the half-life of human prothrombin complex); onset of action of vitamin K_1 is 4-6 hours when given IV.
- Onset of action is within 10-30 minutes and effect persists for 6-8 hours.

REFERENCES

5, 10, 132, 135, 174, 231, 242, 244, 273, 329, 437.

Full revision 2015; limited revision 2017, 2019, 2020

NAME OF MEDICATION
HydrALAZINE

OTHER NAMES
Apresoline ®

CLASSIFICATION
Antihypertensive

INDICATIONS
- In essential and early malignant hypertension, as well as in hypertensive emergencies associated with pregnancy.
- IV use indicated only for conditions in which drugs cannot be taken orally or when there is an urgent need to lower blood pressure.

ADMINISTRATION
- IV direct: physician or RN in presence of physician for first dose only; ***blood pressure monitoring***. May be given undiluted or diluted in 10-20 mL of NS. Inject over 1 minute in adults and over 3 to 5 minutes in children. To avoid hypotension, injection should be stopped frequently when blood pressure is falling.
- Intermittent IV infusion (preferred): RN in presence of physician for first dose only; ***blood pressure monitoring***. Dilute dose in 50 mL of NS and infuse over 15 minutes.
- Continuous IV infusion (preferred): RN in presence of physician for start of the infusion only; ***blood pressure monitoring***. Dilute 100 mg in 500 mL to 1000 mL of NS. Give slowly and adjust rate according to the blood pressure response.
- IM.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, pruritus, urticaria, vasculitis, vascular collapse.
- Cardiovascular: postural hypotension, palpitations, flushing, tachycardia, angina.
- GI: nausea, vomiting, diarrhea, anorexia.
- CNS: depression, anxiety, headache, neuropathy, dizziness.
- Increased cerebral ischemia in predisposed patients.

DOSAGE
Adults:
- Hypertensive emergency in a non-pregnant patient: usual dose: 10-20 mg IV or IM (up to 40 mg has been used). Repeat as needed. Maximum is 300-400 mg/24 hours.
- Infusion: 0.02-0.3 mg/min until response; usual maintenance dose of 0.05-0.15 mg/min.
- Preeclampsia, eclampsia: initial dose of 5-10 mg IV or IM. Then follow by 5-20 mg IV or 5-10 mg IM every 20-40 minutes as required OR infuse at rate of 0.5-10 mg/hr IV (maximum total cumulative dose of 20 mg IV or 30 mg IM).

Pediatrics:
- 0.1-0.2 mg/kg (maximum 20 mg) IV/IM repeated q4-6h up to 1.7-3.5 mg/kg/day OR 50-100 mg/m^2/day IV/IM in 4 to 6 divided doses; OR continuous IV infusion: 1.5 **mcg**/kg/min.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light. Do not refrigerate as it could precipitate or crystallize.
- Compatible with NS, Ringer's, RL, sodium lactate 1/6 M solutions.
- Incompatible with D5W.
- Colour change over a period of 8-12 hours does not affect potency if stored below 30°C.
- Stable for 24 hours at room temp in NS at a concentration of 0.2 mg/mL in PVC bags.
- Stable for 52 days at room temp in NS at a concentration of 0.35 mg/mL in PVC bags.
- Reacts with various metals to give yellow or pink solutions. Caution with metal filters. Minimize contact time with needles.

MISCELLANEOUS
- 25-50 mg IV is approximately equal to 75-100 mg oral hydralazine.
- After IV administration, the effect begins within 5-20 minutes. Maximum effect seen in 10-80 minutes and lasts for 2-6 hours. Effect begins within 10-30 minutes after IM administration.

REFERENCES
1, 2, 4, 5, 6, 40, 82, 95, 135, 143, 170.

Full revision 2015

NAME OF MEDICATION
HYDROCORTISONE

OTHER NAMES
Solu-Cortef ®

CLASSIFICATION
Corticosteroid

INDICATIONS

- Treatment of status asthmaticus, severe allergic reactions and a variety of other inflammatory conditions.
- Treatment of existing or potential states of acute adrenocortical insufficiency when oral administration is not possible or a rapid and intense hormonal effect is required.
- Treatment of septic shock in adults unresponsive to volume resuscitation and vasopressors.

ADMINISTRATION

- For IV use: reconstitute powder of the Act-O-Vial with supplied diluent. Reconstitute powder of the NOVO/TEVA product with SWFI: 1.8 mL for the 100 and 250 mg vials, 3.8 mL for the 500 mg vial and 7.3 mL for the 1 g vial.
- IV direct: physician or RN. Administer undiluted over 30 seconds to several minutes.
- Intermittent IV infusion, Continuous IV infusion: may dilute with D5W or NS to a concentration of 0.1-1 mg/mL. Dose of 100-3000 mg may be added to 50 mL of D5W or NS in fluid restricted patients. Intermittent IV infusions can be given over 20-30 minutes.
- It is recommended that doses of 500 mg or greater be given over at least 10 minutes.
- IM (NOT preferred over IV for initial emergency use due to slower absorption): reconstitute as directed. Administer in a large muscle, not the deltoid as may cause subcutaneous atrophy.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis/anaphylactoid reactions, angioedema, urticaria, bronchospasm, laryngeal edema.
- Cardiovascular: congestive heart failure, hypertension.
- GI: peptic ulcer, GI perforations with high doses. Prophylactic antacid may be considered.
- CNS: headache, seizures, euphoria, increased intracranial pressure, insomnia, mood swings, psychotic reactions.
- Ophthalmic: increased intraocular pressure.
- Hypernatremia, hypokalemia, fluid retention, hyperglycemia, hypothalamic-pituitary-adrenal axis suppression.
- Increased risk of severe infection.
- Burning or tingling of the perineal area. Acute myopathy with high doses.
- Local reactions: dermal and subdermal atrophy.

DOSAGE

Adults:
- Acute adrenal insufficiency: 100 mg IV bolus. Then 50-75 mg IV or IM q6h OR 10 mg/hr as a continuous IV infusion for 24 hours, then taper the dose over the next 72 hours.
- Stress dosing (surgery) in adrenally-suppressed patients: dosage range of 25 to 150 mg IV per day. Administer in 2 or 3 divided doses for daily doses greater than 25 mg.
- Anti-inflammatory or immunosuppression: 15-240 mg IV or IM q12h.
- Status asthmaticus: 1-2 mg/kg IV q6h for 24 hours then 0.5-1 mg/kg IV q6h.
- Septic shock: 50 mg IV q6h OR 200 mg/24 hours IV as a continuous infusion for at least 3 days.

Pediatrics:
- Acute adrenal insufficiency: 1-2 mg/kg IV bolus, then 25-250 mg/day IV or IM in divided doses q6-8h.
- Anti-inflammatory or immunosuppression: Children: 1-5 mg/kg/day or 30-150 mg/m^2/day IV or IM divided q12-24h. Adolescents: 15-240 mg q12h IM or IV.
- Status asthmaticus: 1-2 mg/kg IV q6h for 24 hours, then 0.5-1 mg/kg IV q6h.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Reconstituted solution in the standard vials is stable for 3 days at room temp or in the fridge, if protected from light. Any unused portion must be discarded.
- Solu-Cortef ®, Act-O-Vial: stable for 24 hours at room temp in NS or D5W, at a concentration of 1 mg/mL.
- Novo/Teva product: stable for 4 hours at room temp in NS or D5W, at a concentration of less than 1 mg/mL.
- Solu-Cortef ®, Act-O-Vial and Novo/Teva product: 100 mg to 3 g stable for at least 4 hours in 50 mL (2 to 60 mg/mL) of D5W, NS or D5-NS.
- Compatible with D5W, D5-NS, NS, Ringer's and RL.

MISCELLANEOUS

REFERENCES

4, 5, 6, 40, 82, 95, 135.

NAME OF MEDICATION
HYDROmorphone

OTHER NAMES
Dilaudid ®

CLASSIFICATION
Opioid analgesic

INDICATIONS

- Relief of moderate to severe pain.

ADMINISTRATION

- IV direct: physician or RN; ***respiratory support***. Administer undiluted or dilute dose to 10 mL of NS or D5W and give over at least 2-3 minutes. At TOH: always use the 2 mg/mL solution concentration to administer any dose by IV direct route.
- Intermittent IV infusion. Must be administered by an infusion pump.
- Continuous IV Infusion: add 20 mg (2 mL from hydromorphone 10 mg/mL solution concentration) to 100 mL of NS or D5W to obtain a final concentration of 0.2 mg/mL (200 mcg/mL) OR add 20 mg (10 mL from hydromorphone 2 mg/mL solution concentration) to 90 mL NS or D5W to obtain a final concentration of 0.2 mg/mL (200 mcg/mL). Must be administered by an infusion pump.
- IM (NOT preferred due to variable absorption).
- SC (preferred): using a 30 gauge needle will cause less discomfort. At TOH: use the 2 mg/mL solution concentration for SC administration of doses of 4 mg or less; use the 10 mg/mL solution concentration only for doses greater that 4 mg.
- SC infusion: can be diluted with SWFI, NS or D5W. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis (extremely rare); pseudoallergic reactions due to histamine release: sweating, flushing, pruritus, urticaria, rash.
- Cardiovascular: bradycardia, palpitations, tachycardia; circulatory depression, including hypotension (increased incidence with rapid IV injection).
- GI: dry mouth, constipation, nausea, vomiting.
- CNS: dizziness, sedation, lightheadedness, seizures (at high doses).
- Respiratory depression with rapid IV administration.
- Local reactions: pain at injection site (SC and IM); wheal/flare over vein (IV).

Antidote: naloxone (for respiratory depression).

DOSAGE

- Opioid-naïve patients: SC/IM: 0.5-1 mg q4h prn; IV: 0.25-0.5 mg q1h prn. Titrate as required.
- These initial doses are for the average, healthy patient. Elderly patients and those with reduced renal or hepatic function may require lower doses while opioid experienced patients may need higher doses.
- For IV infusion initiate with 200 mcg/hr (0.2 mg/hr) and titrate as required. Usual dosage range: 500-3000 mcg/hr (0.5-3 mg/hr).
- Analgesia in palliative care and cancer patients: usual dosage range can be exceeded and is titrated to effect.
- Pediatrics: infants older than 6 months weighing more than 10 kg and children weighing less than 50 kg: initial dose of 0.01 to 0.015 mg/kg/dose IV q3-6h prn OR 0.003 to 0.005 mg/kg/hr as continuous IV infusion (maximum rate of 0.2 mg/hr).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and ampoules at room temp, not in the fridge due to possible precipitation or crystallization. Protect from light.
- Stable for 42 days at room temp or in the fridge in NS or D5W at concentrations of 1 and 5 mg/mL in PVC containers.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 60 days at 23°C exposed to light and refrigerated protected from light in NS at concentrations of 1.5 mg and 80 mg/mL in polypropylene syringes.
- Stable for 30 days at 30°C either undiluted at 10 mg/mL or diluted in NS to 0.1 mg/mL in polypropylene syringes.
- A slight yellowish discolouration has not been associated with loss of potency.
- Compatible with NS, D5W, D5-NS, D5-1/2NS, 1/2NS, Ringer's, RL.

MISCELLANEOUS

- Hydromorphone is 5-7 times more potent than morphine on a mg to mg basis.
- Use with caution in patients with respiratory insufficiency.

REFERENCES

1, 2, 4, 5, 40, 82, 95, 135, 143, 170, 253.

Full revision 2015; limited revision 2016, 2017, 2020

NAME OF MEDICATION
HYDROXOCOBALAMIN

OTHER NAMES
Cyanokit ®

CLASSIFICATION
Antidote

Do NOT confuse hydroxocobalamin with cyanocobalamin.
This monograph is specific to HYDROXOCOBALAMIN.

INDICATIONS

- For the treatment of known or suspected cyanide poisoning.

ADMINISTRATION

- Reconstitute each 5 g vial with 200 mL NS (preferred; may use D5W or RL if NS not available), using the supplied transfer device. Rock or invert the vial for at least 60 seconds to dissolve. Do NOT shake. Reconstituted solution is dark red, at a concentration of 25 mg/mL. Use the supplied IV tubing with filter for infusion. (Note: the supplied IV tubing is NOT compatible with current TOH smart pumps; in this situation, the product should be administered by gravity.) Refer to manufacturer's guide provided in the kit.
- Intermittent IV infusion: infuse first dose over 15 minutes; if a second dose is needed, infuse it over 15 minutes (for extremely unstable patients) to 2 hours based on patient's condition.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, chest tightness, dyspnea, edema, pruritus, urticaria and rash.
- Cardiovascular: transient hypertension, chest discomfort.
- GI: nausea, abdominal discomfort.
- CNS: headache, dizziness.
- Dermatologic: reversible red colouration of the skin and mucous membranes that may last up to 15 days; acne-like rash.
- Hematologic: decreased lymphocyte count.
- Dark red colouration of urine that may last up to 35 days. Eye redness.
- Local reactions at injection site.

DOSAGE

- Adults: 5 g IV. Repeat the dose if required. Maximum total dose is 10 g.
- Pediatrics (infants and children): 70 mg/kg IV (maximum dose: 5 g). Repeat dose if required. Maximum total dose is 140 mg/kg IV, not to exceed 10 g.
- No dosage adjustment is required in renal or hepatic impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store kit at room temp.
- Reconstituted solution is stable 6 hours in the fridge and at room temp, not to exceed 40°C. Do not freeze.
- Incompatible through Y-site with sodium thiosulfate, sodium nitrite and vitamin C.

MISCELLANEOUS

- The deep red colour of hydroxocobalamin interferes with many clinical laboratory tests and possibly with the function of hemodialysis machines; refer to manufacturer's literature.
- Advise patients to avoid direct sun while their skin remains discoloured.
- 5 g of hydroxocobalamin neutralizes approximately 40 mcmol/L (1.04 mg/L) of cyanide in the blood.

REFERENCES

1, 5, 40, 270, 542.

Full revision 2015; limited revision 2019, 2021

NAME OF MEDICATION
HYDROXYETHYL STARCH 130/0.4

OTHER NAMES
Volulyte ®, Voluven ®

CLASSIFICATION
Plasma volume expander

INDICATIONS

- Treatment of hypovolemia when plasma volume expansion is required.
- NOT a substitute for red blood cells or coagulation factors in plasma.

ADMINISTRATION

- Continuous IV infusion: rate of infusion depends on the clinical situation and the individual patient. The initial 10 to 20 mL is to be infused slowly, keeping the patient under close observation (due to possible anaphylactic/anaphylactoid reactions).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactic/anaphylactoid reactions (mild influenza-like symptoms, bradycardia, tachycardia, bronchospasm, non-cardiac pulmonary edema). If these reactions occur, discontinue administration immediately and initiate proper treatment and supportive measures.
- Increase in serum amylase (can interfere with the diagnosis of pancreatitis).
- Pruritus: more common with prolonged use of high doses; may be delayed in onset (typically 1 to 6 weeks after exposure). Generally unresponsive to treatment.
- Hematologic: at high doses, dilution of blood components (coagulation factors, plasma proteins, hematocrit), rare coagulation disorders.
- Overloading of the circulatory system (e.g., pulmonary edema).
- Renal toxicity, liver failure and increased mortality (in septic patients) in critically ill patients.

DOSAGE

- Total volume to give depends on the clinical situation and the individual patient, with a maximum of 50 mL/kg/day IV.
- Can be administered repetitively over several days, depending on clinical status of the patient.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Do not freeze.
- Use immediately after removing the overwrap.
- Compatible with NS, D5W and RL.

MISCELLANEOUS

- Monitor liver function during treatment.
- Available as a 6% solution in NS (Voluven ®: Na: 154 mmol/L, Cl: 154 mmol/L. Osmolarity: 308 mosmol/L) or in an isotonic electrolyte solution (Volulyte ®: Na: 137 mmol/L, K: 4 mmol/L, Mg: 1.5 mmol/L, Cl: 110 mmol/L, acetate: 34 mmol/L. Osmolarity: 286.5 mosmol/L).
- Note that in critically ill patients, use of crystalloid solution is favored over hydroxyethyl starch.
- Contraindicated in the following situations: patients with fluid overload (e.g., pulmonary edema, CHF), renal failure with oliguria or anuria not related to hypovolemia, sepsis, severe liver disease, patients on dialysis, severe hypernatrenia or hyperchloremia, severe hyperkalemia (for Volulyte ® only), intracranial bleeding, pre-existing coagulation or bleeding disorder, known hypersensitivity to hydroxyethyl starch.
- Has a volume effect of approximately 100% (of the infused volume) with at least 6 hour duration of effect.

REFERENCES

5, 95, 206, 269, 309.

Full revision 2015

NAME OF MEDICATION
HydrOXYzine

OTHER NAMES
Atarax ®

CLASSIFICATION
Anxiolytic, Antihistamine

INDICATIONS

- Treatment of anxiety in the acutely disturbed or hysterical patient.
- Treatment of the acute or chronic alcoholic with anxiety, withdrawal symptoms or delirium tremens.
- As pre- and postoperative, pre- and post-partum adjunctive medication to allay anxiety, to permit reduction in opioid dosage and to control emesis.
- Treatment of nausea and vomiting in the palliative patient.

ADMINISTRATION

- IM: deep into large muscle, preferably into the gluteus muscle in adults, or the midlateral muscles of the thigh in adults and children. Use in the deltoid area only if well developed.
- SC (palliative care).

POTENTIAL ADMINISTRATION HAZARDS

- GI: dryness of the mouth.
- CNS: drowsiness, confusion.
- Local reactions: marked discomfort, sterile abscess, erythema, irritation, tissue necrosis at IM injection site. Marked induration, necrosis, sloughing, swelling, petechial hemorrhage and abcess with SC administration.

DOSAGE

Adults:
- Psychiatric and emotional emergencies including acute alcohol withdrawal: 50-100 mg IM initially, repeated q4-6h prn.
- Nausea, vomiting, pre- and post-partum and pre- and postoperative adjunctive medication: 25-100 mg IM.
- Antiemetic in palliative patients: 10-50 mg IM or SC q6-8h.

Pediatrics:
- Anxiety: 0.5-1 mg/kg IM q4-6h prn. Maximum 100 mg/dose.
- Nausea, vomiting, pre- and postoperative adjunctive medication: 1.1 mg/kg IM. Maximum 100 mg/dose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp and protect from light. Do not freeze.

MISCELLANEOUS

REFERENCES

1, 2, 4, 5, 82, 95, 135, 457.

Full revision 2015; limited revision 2016

NAME OF MEDICATION
HYOSCINE BUTYLBROMIDE

OTHER NAMES
Buscopan ®, Butylscopolamine, Scopolamine butylbromide

CLASSIFICATION
Anticholinergic

Do NOT confuse hyoscine butylbromide with hyoscine hydrobromide (scopolamine hydrobromide). This monograph is specific to hyoscine BUTYLBROMIDE.

INDICATIONS

- For the relief of acute smooth muscle spasm and pain in gastrointestinal and genitourinary disorders in acutely ill patients.
- To prevent spasm with certain diagnostic procedures.
- To reduce respiratory and gastrointestinal secretions in palliative care patients.

ADMINISTRATION

- IV direct: physician or RN; administer undiluted at a rate of 20 mg/min.
- IM.
- SC, SC infusion (palliative care). For SC infusion, can dilute with SWFI, NS or D5W; must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis/anaphylactoid reaction (rare), urticaria, rash, pruritus, angioedema.
- Cardiovascular: tachycardia, hypotension, flushing.
- GI: constipation, nausea, dry mouth.
- CNS: dizziness.
- Ophthalmic: increased intraocular pressure, mydriasis.
- Renal: urinary retention.
- Respiratory: shortness of breath.

DOSAGE

- Treatment of acute spasm: 10-20 mg IM, SC or IV per dose. Maximum of 100 mg/day.
- Prevention of spasm: 10-20 mg IM, 10 to 15 minutes before the diagnostic procedure.
- To reduce respiratory secretions: 20 mg SC every hour prn OR 20 mg SC stat then 20-120 mg/24 hours as a continuous SC infusion.
- To reduce GI secretions in inoperable intestinal obstruction with colic: 20 mg SC every hour prn OR 20 mg SC stat, then 60-300 mg/24 hours as a continuous SC infusion.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and ampoules between 15 and 30°C. Protect from light and heat. Do not freeze.
- Stable for 8 hours in NS, D5W, D10W, RL, Ringer's.

MISCELLANEOUS

- Contraindicated in patient with untreated narrow angle glaucoma, benign prostatic hypertrophy with urinary retention, megacolon, GI stenotic lesions, myasthenia gravis, tachycardia, angina and cardiac failure.

REFERENCES

5, 6, 135, 256, 457.

Full revision 2015; limited revision 2019, 2020

NAME OF MEDICATION
IBUPROFEN (WITH ARGININE)

OTHER NAMES
Caldolor ®

CLASSIFICATION
Nonsteroidal anti-inflammatory

Do NOT confuse ibuprofen (with arginine) with ibuprofen lysine or ibuprofen (with tromethamine). This monograph is specific to ibuprofen with ARGININE.

INDICATIONS

- Reduction of fever in adults.
- Management of moderate to severe pain in adults, as an adjunct to intravenous opioids.

ADMINISTRATION

- Intermittent IV infusion: dilute dose in 50 to 250 mL of D5W, NS or RL to obtain a final concentration of 4 mg/mL or less (i.e., an 800 mg dose requires at least 200 mL; a 400 mg dose, 100 mL; a 200 mg dose, 50 mL). Administer over 30 minutes.
- Ensure adequate hydration before administration to reduce risk of adverse effect on renal function.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactic/anaphylactoic reactions; angioedema.
- Cardiovascular: fluid retention leading to hypertension and CHF; hypotension.
- GI: nausea, vomiting, flatulence, diarrhea, abdominal discomfort or pain, dyspepsia, GI ulceration and bleeding.
- CNS: dizziness, headache.
- Hematologic: prolonged bleeding time, anemia, neutropenia; hemolysis if administered undiluted.
- Urinary retention, renal failure.
- Hypernatremia, hyperkalemia, hypoproteinemia, increased urea.
- Dermatologic: serious skin reactions such as exfoliative dermatitis, Stevens-Johnson syndrome, toxic epidermal necrolysis.

DOSAGE

- Fever: 200-400 mg IV q4-6h prn.
- Pain: 400-800 mg IV q6h prn.
- Maximum 3200 mg/24 hours.
- Dosage in renal or hepatic impairment: not to use if moderate to severe impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Stable for 24 hours at room temp, exposed to light, at a concentration of 4 mg/mL or less.
- Compatible with D5W, NS and RL.

MISCELLANEOUS

- Do not administer to patients with ASA intolerance (i.e., nasal polyps/rhinosinusitis, urticaria/angioedema or asthma after taking ASA or other NSAIDs) as cross-hypersensitivity has been reported.
- Do not administer for perioperative pain in patients undergoing coronary artery bypass graft surgery.

REFERENCES

1, 2, 5, 6, 40, 208.

Full revision 2015; limited revision 2019

NAME OF MEDICATION
IBUPROFEN (WITH TROMETHAMINE *)

OTHER NAMES
Pedea ®

CLASSIFICATION
Nonsteroidal anti-inflammatory

Do NOT confuse ibuprofen (with tromethamine) with ibuprofen lysine or ibuprofen (with arginine). This monograph is specific to ibuprofen with TROMETHAMINE.

INDICATIONS

- Treatment of hemodynamically significant patent ductus arteriosus in preterm infants less than 34 weeks gestational age when usual treatments are ineffective.

ADMINISTRATION

- Intermittent IV infusion: undiluted (preferred) or diluted in NS or D5W to a convenient volume; infuse over 15 minutes.
- Before and after administration, flush infusion line over 15 minutes with 1.5 to 2 mL of NS or D5W to avoid contact with any acidic solution.

POTENTIAL ADMINISTRATION HAZARDS

- GI: necrotizing enterocolitis, intestinal perforation.
- CNS: intraventricular hemorrhage, periventricular leukomalacia.
- Hematologic: thrombocytopenia, neutropenia.
- Respiratory: bronchopulmonary dysplasia, pulmonary hemorrhage.
- Renal: oliguria, fluid retention, hematuria, serum creatinine increase.
- May displace bilirubin from its binding site to albumin and cause bilirubin encephalopathy.

DOSAGE

- A course of therapy is defined as 3 IV doses given at 24 hour intervals: 1st dose: 10 mg/kg; 2nd and 3rd doses: 5 mg/kg.
- If the ductus arteriosus does not close 48 hours after the last injection or if it reopens, a second course of 3 doses may be given.
- Dose to be held until urine output returns to normal levels if anuria or manifest oliguria occurs after administration of 1st or 2nd dose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Compatible with NS and D5W.

MISCELLANEOUS

- As ibuprofen may inhibit platelet aggregation, monitor premature neonates for signs of bleeding.

REFERENCES

88.

* Available via Health Canada's Special Access Programme

Full revision 2015; limited revision 2019

NAME OF MEDICATION
IBUPROFEN LYSINE *

OTHER NAMES
NeoProfen ®

CLASSIFICATION
Nonsteroidal anti-inflammatory

Do NOT confuse ibuprofen lysine with ibuprofen (with arginine) or ibuprofen (with tromethamine). This monograph is specific to ibuprofen LYSINE.

INDICATIONS

- To close a clinically significant patent ductus arteriosus in preterm infants weighing between 500 and 1500 g who are no more than 32 weeks of gestational age when usual treatments are ineffective.

ADMINISTRATION

- Intermittent IV infusion: dilute in D5W or NS to a convenient volume; infuse over 15 minutes via the closest port to insertion site.
- Administration via an umbilical arterial line has not been studied.
- Should not be administered simultaneously in the same IV line with total parenteral nutrition (TPN); if needed, stop TPN for 15 minutes **b**efore and after ibuprofen lysine administration, keeping line open with dextrose or saline.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: edema.
- GI: enterocolitis (with/without necrosis), GI perforation.
- CNS: intraventricular hemorrhage.
- Endocrine and metabolic: adrenal insufficiency, hypoglycemia, hypocalcemia, hypernatremia.
- Hematologic: anemia.
- Renal: oliguria, uremia, serum creatinine increase, hematuria, renal failure, urinary tract infection.
- Respiratory: apnea, respiratory infection, respiratory failure, atelectasis.
- Sepsis.
- Local reactions: irritating to tissue; avoid extravasation.

DOSAGE

- A course of therapy is defined as 3 IV doses given each at 24 hour intervals: 1st dose: 10 mg/kg; 2nd and 3rd doses: 5 mg/kg. Use birth weight for dosing.
- If the ductus arteriosus does not close or reopens, a second course may be given.
- Dosage in renal impairment: dose to be held at any time if urine output is less than 0.6 mL/kg/hr; resume course when renal function is back to normal.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Use diluted solution within 30 minutes of its preparation.
- Compatible with D5W and NS.
- Incompatible with TPN.

MISCELLANEOUS

- As ibuprofen may inhibit platelet aggregation, monitor premature neonates for signs of bleeding.
- Use with caution in neonates with elevated total bilirubin as ibuprofen lysine displaces bilirubin from albumin binding sites.

REFERENCES

1, 5, 40, 82, 135.

* Available via Health Canada's Special Access Programme

New monograph 2016; limited revision 2019

NAME OF MEDICATION
IBUTILIDE

OTHER NAMES
Corvert ®

CLASSIFICATION
Antiarrhythmic

INDICATIONS

- Rapid conversion of recent onset atrial fibrillation or atrial flutter to sinus rhythm, as an alternative to electric cardioversion.

ADMINISTRATION

- IV direct: physician only; ***cardiac monitoring***; undiluted; administer over 10 minutes.
- Intermittent IV infusion: physician only; ***cardiac monitoring***; dilute dose in 50 mL of NS or D5W and infuse over 10 minutes.
- ***Cardiac monitoring*** required for at least 4 hours following administration or until QTc has returned to baseline. Longer period of monitoring required if arrhythmia develops or in patients with liver dysfunction.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: arrhythmias can occur up to 3 hours post infusion. Infusion should be stopped immediately if new arrhythmia develops, if original arrhythmia worsens or if sustained or nonsustained ventricular tachycardia or marked prolongation of QT or QTc occurs.
- GI: nausea.
- CNS: headache.

DOSAGE

- Adults: greater or equal to 60 kg: 1 mg
  less than 60 kg: 0.01 mg/kg
- Post-cardiac surgery patients: greater or equal to 60 kg: 0.5 mg
  less than 60 kg: 0.005 mg/kg.
- Infusion should be stopped as soon as arrhythmia is terminated.
- If the arrhythmia does not terminate within 10 minutes after the end of the initial infusion, dose may be repeated.
- It is not recommended to give more than 2 doses due to risk of adverse events associated with QT interval prolongation.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Stable for 24 hours at room temp and 48 hours in the fridge when mixed in NS or D5W.

MISCELLANEOUS

- Not recommended for patients with QTc longer than 440 msec.
- Hypokalemia and hypomagnesemia should be corrected before treatment to reduce the potential for proarrhythmia.
- Patients with bradycardia, CHF or low ejection fraction have increased risk of developing arrhythmias with ibutilide.
- Patients with atrial arrhythmias of longer duration are less likely to respond to ibutilide.
- Patients with atrial fibrillation of more than 2-3 days must be anticoagulated for at least 2 weeks before ibutilide is given.

REFERENCES

1, 2, 40, 95, 208.

Full revision 2015

NAME OF MEDICATION
ICATIBANT

OTHER NAMES
Firazyr ®

CLASSIFICATION
Anti-angioedema agent

INDICATIONS

- Treatment of acute attacks of hereditary angioedema in adults and children 2 years of age and older with C-1 esterase inhibitor deficiency.

ADMINISTRATION

- Preparation: for adults and children weighing over 65 kg: attach the provided 25 gauge needle to the syringe hub and screw on securely. Do NOT use a different needle. Children weighing 65 kg or less: transfer the exact dose from the prefilled syringe to the empty graduated syringe using the provided connector; refer to the manufacturer's instructions for more details.
- SC: inject SC over at least 30 seconds in the abdominal area, 2 to 4 inches below navel, and at least 2 inches away from any scar or area that is bruised, swollen or painful.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus.
- GI: nausea.
- CNS: headache, dizziness.
- Hepatic: increase of transaminases (AST and ALT).
- Fever, fatigue.
- Local reactions: hematoma, irritation, hypoesthesia, erythema, swelling, warm sensation, burning, itching, pain.

DOSAGE

- Adults: 30 mg (3 mL) SC; repeat dose in no less than 6 hours if response not complete or symptoms recur. Maximum of 90 mg (3 injections) per 24-hour period. Safety of more than 8 injections per month has not been studied.
- Pediatrics (2-17 years old): weight of 12-25 kg: 10 mg (1 mL); 26-40 kg: 15 mg (1.5 mL); 41-50 kg: 20 mg (2 mL); 51-65 kg: 25 mg (2.5 mL); over 65 kg: 30 mg (3 mL). No more than 1 injection per attack has been studied.
- Dosage in renal impairment: no dosage adjustment necessary
- Dosage in hepatic impairment: no dosage adjustment necessary.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled syringes between 2-25°C; do not freeze.

MISCELLANEOUS

- Caution when administering to patients on an angiotensin-converting enzyme (ACE) inhibitor as icatibant may decrease its hypotensive activity.
- Do not use if patient has only pre-attack symptoms (paresthesia, or erythema).

REFERENCES

1, 5, 95, 135.

New monograph 2016; limited revision 2020

NAME OF MEDICATION
IdaruBICIN

OTHER NAMES
Idamycin PFS ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of acute non-lymphocytic leukemia (ANLL) in adults (front line or refractory/relapsed disease).
- Treatment of acute lymphocytic leukemia (ALL) in adults and children (second line).

ADMINISTRATION

- IV direct: physician or RN; undiluted; inject over 5-10 minutes into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W; infuse over 10-15 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis.
- Cardiovascular: transient arrhythmias (rare); facial flushing with rapid injection.
- GI: nausea and vomiting.
- Radiation recall reaction (rare).
- Hyperuricemia.
- Red discolouration of urine for 1-2 days.
- Local reactions: chemical phlebitis at injection site; local erythematous streaking along the vein with rapid injection.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Adults: ANLL: used in combination: 12 mg/m^2 IV daily for 3 days.
  used alone: 8 mg/m^2 IV daily for 5 days.
  ALL: used alone: 12 mg/m^2 IV daily for 3 days.
- Pediatrics: ALL: used alone: 8-10 mg/m^2 IV daily for 3 days.
- Dosage in renal impairment: CrCl 10-50 mL/min: 75% of the dose
  CrCl less than 10 mL/min: 50% of the dose
- Dosage in hepatic impairment: if bilirubin of 40-85 mcmol/L, give 50% of the dose. If bilirubin greater than 85 mcmol/L, omit dose.
- Dose may be reduced in the presence of other toxicities.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C and protect from light.
- Compatible with NS, dextrose, dextrose-saline, and RL.
- Stable for 4 weeks at room temp if kept in the dark in NS or D5W at a concentration of 0.1 mg/mL.
- Stable for 72 hours at room temp if protected from light and for 6 hours at room temp when exposed to light in NS or D5W at a concentration of 0.01 mg/mL.
- Prolonged contact with alkaline solutions will result in degradation of the drug.

MISCELLANEOUS

REFERENCES

1, 2, 4, 40, 129, 135, 165.

Full revision 2015; limited revision 2016, 2017, 2018, 2020

NAME OF MEDICATION
IdaruCIZUmab

OTHER NAMES
Praxbind ®

CLASSIFICATION
Antidote, Monoclonal antibody

INDICATIONS

- For the rapid reversal of the anticoagulant activity of dabigatran when required for an emergency surgery/procedure or for life-threatening or uncontrolled bleeding.
- NOT indicated for the reversal of other anticoagulants.

ADMINISTRATION

- IV direct: physician or RN; administer each dose of 2.5 g (50 mL) one after the other, as a bolus drawn up in a syringe, for a total of 5 g (100 mL).
- IV intermittent infusion: infuse undiluted each 2.5 g dose (50 mL) over 5-10 minutes for a total of 5 g (100 mL). May infuse directly from the vial. Infuse the second vial within 15 minutes of the end of the infusion of the first vial.
- Flush IV line with NS before and after administration; do NOT mix with other drugs or solutions or run any infusions via the same IV line.
- Once solution is withdrawn from the vial, administration should begin within 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, hyperventilation, bronchospasm, anaphylaxis.
- GI: constipation.
- CNS: delirium, headache.
- Respiratory: pneumonia.
- Hypokalemia.
- Fever.

DOSAGE

- 5 g (2 X 2.5 g vials).
- Repeat dose only in exceptional circumstances such as 1) recurrence of clinically significant bleeding with a prolonged clotting time; 2) if patient requires a second emergency procedure/surgery and has a prolonged clotting time.
- Dosage in renal impairment: no dosage adjustment necessary.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge; do not freeze or expose vials to direct heat. Do not shake. Protect from light.
- Unopened vials are stable for 48 hours at room temp if protected from light and for 6 hours if exposed to light.
- Opened vials can be kept for up to 1 hour between 15-25°C if away from direct heat and light and temperature does not exceed 25°C.

MISCELLANEOUS

- If patient is clinically stable and adequate hemostasis has been achieved, dabigatran may be restarted 24 hours after idarucizumab administration, and other antithrombotic drugs (e.g. low-molecular weight heparins) can be started any time after idarucizumab.
- Caution in patients with hereditary fructose intolerance as idarucizumab contains sorbitol.
- The affinity of idarucizumab for dabigatran is approximately 350 times that of dabigatran for thrombin.

REFERENCES

5, 135, 379.

New monograph 2016; limited revision 2018

NAME OF MEDICATION
IFOSFAMIDE

OTHER NAMES
Ifex ®, Isophosphamide

CLASSIFICATION
Antineoplastic

INDICATIONS
- Treatment of soft tissue sarcoma, pancreatic carcinoma, cervical carcinoma, and other neoplasms.

ADMINISTRATION
- Reconstitute 1 and 3 g vials with 20 and 60 mL of SWFI, respectively, to obtain a solution containing 50 mg/mL. Shake well until dissolved.
- Intermittent IV infusion: dilute to 0.6 to 20 mg/mL with D5W, NS or RL; infuse over 30 minutes to 4 hours.
- Continuous IV infusion: dilute to 0.6 to 20 mg/mL with D5W, NS or RL; infuse continuously over 24 hours or longer.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rare; anaphylactoid reactions.
- GI: nausea and vomiting.
- CNS: encephalopathy (confusion, hallucinations, somnolence, dizziness, disorientation, agitation, seizures, coma).
- Renal: hemorrhagic cystitis, hematuria, dysuria, urinary frequency.
- Syndrome of inappropriate antidiuretic hormone (SIADH).
- Local reactions: chemical phlebitis.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- Usual: 50-60 mg/kg or 2-2.4 g/m^2 IV once daily for 5 consecutive doses OR 5-8 g/m^2 by continuous IV infusion over 24 hours, repeated every 3-4 weeks.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 60	60-40	40-20	less than 20%
Dose	100%	75%	50%	discontinue

- Dosage in hepatic impairment:

Bilirubin:	1-2 x ULN	2-4 x ULN	greater than 4 x ULN
AST/ALT:	less than 2 x ULN	2-5 x ULN	greater than 5 x ULN
Dose:	100%	75%	discontinue

- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials at room temp. Protect from heat.
- Reconstituted solution is stable for 7 days at room temp and 3-6 weeks under refrigeration.
- Stable for 7 days at room temp or 6 weeks in the fridge in D5W, NS or RL at concentrations of 0.6-20 mg/mL in glass, PVC and polyolefin containers.
- Stable for 24 hours at room temp in D5W, NS or RL at concentrations of 0.6 and 20 mg/mL in polypropylene syringes.
- Compatible with mesna.

MISCELLANEOUS
- Patient should receive adequate hydration prior to and during ifosfamide infusion (2 litres of oral or IV fluid per day).
- Concomitant mesna therapy is **mandatory** to prevent ifosfamide-induced hemorrhagic cystitis.
- Morning urinalysis for red blood cells is recommended before each dose of ifosfamide.

MISCELLANEOUS
1, 4, 5, 6, 40, 129, 165.

Full revision 2015; limited revision 2016, 2017, 2018, 2020

NAME OF MEDICATION
ILOPROST *

OTHER NAMES
Ilomedin, Ilomedin 20, Ilomedine, Iloprost trometamol

CLASSIFICATION
Vasodilator

INDICATIONS

- Treatment of advanced thromboangiitis obliterans (Buerger's disease) with critical limb ischemia.
- Treatment of severe peripheral arterial occlusive disease.
- Treatment of severe disabling Raynaud's phenomenon.

ADMINISTRATION

- **There are 2 concentrations: 20 mcg/mL (Ilomedin 20) and 100 mcg/mL (Ilomedin). Ensure using the correct one.**
- Intermittent IV infusion (mandatory):
  - Infusion pump: *using the 20 mcg/mL concentration*, dilute the 0.5 mL, 1 mL, and 2.5 mL ampoule in 50 mL, 100 mL, and 250 mL of NS or D5W, respectively; *using the 100 mcg/mL concentration*, dilute the 0.5 mL and 1 mL ampoule in 250 mL and 500 mL of NS or D5W, respectively. Final concentration is 0.2 mcg/mL.
  - Syringe driver: *using the 20 mcg/mL concentration*, dilute the 0.5 mL, 1 mL, and 2.5 mL ampoule to a final volume of 5 mL, 10 mL, and 25 mL with NS or D5W, respectively; *using the 100 mcg/mL concentration*, dilute the 0.5 mL and 1 mL ampoule to a final volume of 25 mL and 50 mL with NS or D5W, respectively. Final concentration of 2 mcg/mL.
  - Initiate with an infusion rate of 0.5 ng/kg/min for 30 minutes; if tolerated, increase the rate in increments of 0.5 ng/kg/min every 30 minutes to a maximum rate of 2 ng/kg/min and a total duration of 6 hours. Must be administered by an infusion pump or syringe driver.
- Vital signs should be obtained pre-infusion and after each dose increase.

POTENTIAL ADMINISTRATION HAZARDS

- **Non-cytotoxic hazardous drug**: consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity.
- Cardiovascular: hypotension/hypertension, tachycardia/bradycardia, flushing, angina, myocardial infarction, cardiac failure, stroke, pulmonary embolism.
- GI: nausea, vomiting, diarrhea, abdominal discomfort/pain, anorexia.
- CNS: headache, dizziness, vertigo, paresthesias, hyperesthesia, restlessness, agitation, sedation, confusion, apathy, convulsions.
- Dermatologic: hyperhidrosis.
- Hematologic: bleeding.
- Musculoskeletal: pain in jaw, trismus, myalgia, arthralgia.
- Respiratory: dyspnea, asthma, pulmonary edema.
- Fever, chills, asthenia, malaise, fatigue.
- Local reactions: erythema, pain, phlebitis at infusion site.

DOSAGE

- 0.5 to 2 ng/kg/min IV for 6 hours once daily.
- The individually tolerated dose is determined during the first 2 to 3 days of treatment; refer to Administration section. Treatment is then continued with the individually tolerated dose for up to 4 weeks.
- Dosage in renal impairment: reduce dose by 50% in patients requiring dialysis.
- Dosage in hepatic impairment: reduce dose by 50% in patients with liver cirrhosis.

…/Cont.

NAME OF MEDICATION
ILOPROST *

OTHER NAMES
Ilomedin, Ilomedin 20, Ilomedine, Iloprost trometamol

CLASSIFICATION
Vasodilator

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp.
- Product should be used immediately after its preparation as manufacturer has no data on stability once diluted.

MISCELLANEOUS

- As iloprost inhibits platelet function, use with caution in combination with anticoagulants or inhibitors of platelet aggregation due to an increased risk of bleeding.

REFERENCES

*Available via Health Canada's Special Access Programme

577, 578.

NAME OF MEDICATION
IMIGLUCERASE

OTHER NAMES
Cerezyme ®

CLASSIFICATION
Enzyme, Replacement therapy

INDICATIONS

- Long term enzyme replacement therapy for patients with non-neuronopathic (Type I) or chronic neuronopathic (Type 3) Gaucher disease who exhibit non-neurological manifestations of the disease.

ADMINISTRATION

- For IV use: allow vials to reach room temp. Reconstitute 200 and 400 unit vials with 5.1 and 10.2 mL of SWFI, respectively, to give a concentration of 40 units/mL. Avoid excessive agitation during reconstitution. Allow several minutes for all bubbles to dissipate and drug to completely dissolve.
- Intermittent IV infusion: dilute dose with NS to a final volume of 100-200 mL and administer through a low-protein-binding 0.2 micron in-line filter over 1-2 hours; maximum infusion rate of 1 unit/kg/min.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, flushing, angioedema and anaphylactoid reactions. Using caution, reducing the rate of administration and pretreating with antihistamines and/or corticosteroids have allowed some hypersensitive patients to continue treatment.
- Cardiovascular: hypotension, tachycardia.
- GI: abdominal discomfort, nausea, vomiting, diarrhea.
- CNS: dizziness, headache, fever.
- Respiratory: dyspnea, cough.
- Local reactions: burning, swelling or sterile abcess at injection site.

DOSAGE

- Initial adult range: 2.5 units/kg IV three times per week up to 60 units/kg IV once every two weeks.
- Dose is individualized based on disease severity and patient's response; dosages up to 240 units/kg IV once every 2 weeks have been administered.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge, although they are stable at room temp for 48 hours.
- Reconstituted vial is stable for 12 hours at room temp or in the fridge.
- Diluted solution in NS is stable for 24 hours in the fridge or at room temp.

MISCELLANEOUS

REFERENCES

1, 5, 40, 95, 135.

Full revision 2015

NAME OF MEDICATION
IMIPENEM

OTHER NAMES
Imipenem-cilastatin, Primaxin ®

CLASSIFICATION
Antibiotic - carbapenem

INDICATIONS

- Moderate to severe infections due to susceptible organisms. Imipenem is effective against the majority of gram-positive and gram-negative aerobes and anaerobes.

ADMINISTRATION

- For IV use: reconstitute vial contents with 10 mL of selected solution for dilution (refer to Compatibility, Stablility section), shake well and add to 90 mL of solution for infusion. Repeat this procedure using 10 mL of the diluted solution to ensure complete transfer of vial contents. Shake solution for infusion and let stand until clear.
- Intermittent IV infusion: dilute 250-500 mg doses with at least 100 mL and infuse over 20-30 minutes. Dilute doses of 750 mg and 1 g with at least 200 mL and infuse over 40-60 minutes. Do not exceed a maximum concentration of 5 mg/mL.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, fever, pruritus and urticaria.
- GI: nausea, vomiting, diarrhea. Decrease the infusion rate if nausea and/or vomiting occur during administration.
- CNS: seizures have been reported in predisposed patients (e.g., renal failure, prior history of seizures). It is **VERY** important to adjust dose in patients with renal impairment.
- Local reactions: phlebitis/thrombophlebitis, infused vein pain, erythema, vein induration.

DOSAGE

- 250-1000 mg IV q6-8h depending on severity of infection. Maximum daily dose: 4 g or 50 mg/kg (whichever is less).
- Dosage in renal impairment:

CrCl (mL/min)	70-31	30-21	20-0
Dose (mg)	500	500	250-500
Interval (hr)	6-8	8-12	12 *

* dose is to be given after hemodialysis

- Reduce dose for patients less than 70 kg.
- Pediatrics: 3 months of age and older: 60-100 mg/kg/day IV divided into four equal doses (e.g., q6h). Maximum daily dose of 4 g.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Reconstituted solutions are stable for 4 hours at room temp and 24 hours at 5°C in the following diluents: D5W, D10W, dextrose-saline solutions, NS.
- Stable for 24 hours at room temp or in the fridge in NS or D5W at a concentration of 2.5 mg/mL in PVC, polyethylene or glass containers.
- Solutions of imipenem may range from clear to yellow. The potency is not affected by a yellow discolouration. Solutions should be discarded if they darken to brown.

MISCELLANEOUS

- Each 500 mg vial of Primaxin ® contains 500 mg of imipenem and 500 mg of cilastatin; the latter prevents metabolism of imipenem in the kidney.
- Potential for cross-allergenicity with other beta-lactams antibiotics.
- Sodium content: 3.2 mmol/g.

REFERENCES

1, 2, 4, 40, 41, 82.

Full revision 2015

NAME OF MEDICATION
IMMUNE GLOBULIN, HUMAN (INTRAVENOUS)

OTHER NAMES
Gammagard liquid, Gammagard S/D ®, Gamunex ®, IGIVnex ®, Octagam ®, Panzyga ®, Privigen ®

CLASSIFICATION
Passive immunizing agent

Do NOT confuse immune globulins for IV administration with those for SC administration. This monograph is specific to immune globulins for IV administration.

INDICATIONS

- Treatment of primary immune deficiency syndromes.
- May also be used as replacement therapy in patients with secondary immune deficiency syndromes (e.g., B-cell chronic lymphocytic leukemia, HIV) for prophylaxis of infections and in combination with antibiotics in severe infections.
- Treatment of immune thrombocytopenic purpura (ITP).
- Kawasaki syndrome; Guillain-Barré syndrome (GBS); chronic inflammatory demyelinating polyneuropathy (CIDP).
- To decrease the risk of infections and acute graft-versus-host disease following allogeneic bone marrow transplantation (BMT).

ADMINISTRATION

- For Gammagard S/D ®: reconstitute with SWFI (provided) as per manufacturer's recommendations. Other preparations are ready-to-use solutions.
- Intermittent IV infusion: for Gammagard S/D ®: use provided vented infusion set that contains a 15 micron filter.
- Must be administered by an infusion pump.

Product	Gammagard S/D 5%	Gammagard S/D 10%, Gammagard Liquid	Gamunex, IGIVnex	Panzyga	Privigen	Octagam
Initial rate for the first 30 minutes: (mL/kg/hr)	0.5	0.5	0.6-1.2	0.6	0.3	0.6
Maximum rate: (mL/kg/hr)	4	8	8.4 (4.8 for 1st dose)	4.8* or 8.4	7.2	7.2
Maximum rate if risk of renal failure: (mL/kg/hr)	4	2	4.8	-	-	-

* for 1st dose, if last dose was greater than 8 weeks ago, or for all ITP patients.

- The following simplified infusion schedule can be used for all preparations, including first infusion of intravenous immune globulin (IVIG) or when switching from one IVIG brand to another for subsequent doses. Slower rates may be ordered due to previous reactions; however, infusions rates should not exceed those listed below:
  - 30 mL/hr for 30 minutes
  - 60 mL/hr for 15 minutes
  - 100 mL/hr for 15 minutes (remain at this rate until the end if patient is at risk of renal failure)
  - 200 mL/hr for 15 minutes (remain at this rate until the end if at least one of the following criteria is met: 1) first dose of IVIG, 2) switching brand of IVIG, 3) any dose of Gammagard S/D 5%, or 4) all ITP patients receiving Panzyga)
  - 400 mL/hr for the remainder of the infusion when using IVIG preparations for other cases **if all of the following criteria are met**:
    - patient weighs at least 60 kg
    - first dose of this specific brand was well tolerated at a rate of 200 mL/hr and the same brand is used for the current dose (i.e., do not use this rate when switching to another brand of IVIG)
    - patient is not at risk of renal failure
- Obtain vital signs pre-infusion, at 15 minutes after the start of infusion, prior to each infusion rate increase, then hourly until completion once the maximum rate is reached and after completion of the infusion.
- Higher infusion rates are possible based on patient's weight; refer to table to calculate patient specific rate.
- Lower infusion rate may be necessary based on patient tolerance.
- At TOH, there is no need to restart infusion at initial rate or initial vital signs monitoring when multiple bags/vials are used during the same administration session, assuming all vials/bags have the same lot number; however, in case of different lot numbers, the vital signs monitoring protocol needs to be restarted but the infusion can be continued at the same rate.

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ADMINISTRATION (Cont.)
- Above infusion rate protocol to be followed each time with subsequent doses on different days.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0045 (Blood products and blood derivatives: Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, pruritus, urticaria, anaphylactoid reactions.
- Infusion-related reactions: fever, chills, flushing of the face, chest tightness, dizziness, nausea, vomiting, diaphoresis, hypotension/hypertension and acute lung injury (rare). These reactions generally occur 30 to 60 minutes after the start of the infusion. They are usually related to the rate of the infusion; therefore slowing or temporarily stopping the infusion may alleviate these reactions.
- CNS: headache, aseptic meningitis (rare).
- Hematologic: hemolysis, hemolytic anemia. Thrombotic effects: chest pain, myocardial infarction, CHF, cerebral infarct, ischemic encephalopathy, pulmonary embolism, peripheral deep vein thrombosis, retinal vein thrombosis.
- Renal: acute renal failure, osmotic nephrosis.
- Risk of transmission of infectious agents, including viruses, and theoretically, the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.

DOSAGE
- Dosing may vary based on specific product, therefore, it is best to consult manufacturer's information for dosing recommendations.
- Primary and secondary immune deficiency syndromes: range of 100-800 mg/kg IV every 3-4 weeks. B-cell chronic lymphocytic leukemia: 400 mg/kg IV every 3-4 weeks. Pediatric HIV infection: 400 mg/kg IV every 4 weeks.
- ITP (induction): 1000 mg/kg/day IV for 1-2 days OR
  400 mg/kg/day IV for 5 days.
  ITP (maintenance): 400-1000 mg/kg IV as a single infusion as needed.
- Kawasaki syndrome: 2 g/kg IV as a single dose; if no response may administer second dose.
- GBS: total dose of 2 g/kg IV given in divided doses over 2-5 days.
- CIDP: initial: total dose of 2 g/kg IV given in divided doses over 2-4 days. Maintenance: 1 g/kg IV every 3 weeks or 0.5 g/kg IV on 2 consecutive days every 3 weeks.
- BMT: 500 mg/kg IV on days -7 and -2 pre-transplant, then weekly through day 90 post-transplant.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Gamunex ®: store in the fridge; however, it is stable for 6 months at room temp not exceeding 25°C. Can be diluted with D5W but not with NS. May use D5W or NS for flushing IV line.
- Gammagard S/D ®: store unopened bottles at room temp not exceeding 25°C. Solution is stable for 24 hours in the fridge, or for 12 hours at room temp, or for 12 hours at room temp followed by 12 hours in the fridge when reconstituted with SWFI and stored in original glass bottles or transferred in a PVC bag.
- Privigen ®: store at room temp or in the fridge. Keep in its original carton to protect from light. Can be diluted with D5W. Line can be flushed with D5W or NS. Discard partially used product after 24 hours.
- Octagam ® and Panzyga ®: store in the fridge; however, it is stable for 9 months at temperatures not exceeding 25°C.
- IGIVnex ®: store in the fridge; however, it is stable for 6 months at temperature not exceeding 25°C. Can be diluted with D5W but not NS. May use D5W or NS for flushing the line.

MISCELLANEOUS
- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- Avoid use in patients with IgA deficiency as they are more prone to develop anti-IgA antibodies which can lead to severe hypersensitivity reactions.

REFERENCES
1, 2, 4, 5, 40, 82, 95, 171, 192, 258, 259, 544, 572.

Full revision 2015; limited revision 2017, 2019, 2020, 2021

NAME OF MEDICATION
IMMUNE GLOBULIN, HUMAN (SUBCUTANEOUS)

OTHER NAMES
Cutaquig ®, Cuvitru ®, Hizentra ®, SCIG, SC IgG

CLASSIFICATION
Passive immunizing agent

Do NOT confuse immune globulin for SC administration with those for IV administration. This monograph is specific to immune globulin for SC administration.

INDICATIONS

- Treatment of primary or secondary immune deficiency (Hizentra ®, Cutaquig ®, Cuvitru ®).
- Measles prophylaxis in patients with primary humoral immunodeficiency at risk of exposure or exposed to measles (Hizentra ®, Cutaquig ®).
- Treatment of chronic inflammatory demyelinating polyneuropathy (CIDP) as maintenance therapy to prevent relapse of neuromuscular disability and impairment (Hizentra ®).

ADMINISTRATION

- SC infusion: allow product to reach room temperature (at least 90 minutes for Cutaquig ® and Cuvitru ®) before infusion. See chart below for more details. Must be administered by an infusion pump.

	Hizentra ®	**Cutaquig ®**	**Cuvitru ®**
Infusion sites	- Abdomen, thigh, upper arm, upper leg/hip. - Rotate site with each infusion. - Dose may be infused simultaneously in different sites, each being 5 cm apart.	- Abdomen, thigh, upper arm, upper leg/hip. - Rotate site with each infusion. - Dose may be infused simultaneously in different sites, each being 5 cm apart.	- Abdomen, thigh, upper arm, lateral hip. - Rotate site with each infusion. - Dose may be infused simultaneously in a maximum of 4 different sites, each being 9 cm (4 inches) apart.
Rate	*For the first infusion:* maximum of 20 mL/hr per site. *For subsequent infusions:* maximum of 50 mL/hr per site as tolerated.	*For the first 6 infusions:* maximum of 15-20 mL/hr per site with a maximum of 30 mL/hr for all sites combined. *For subsequent infusions:* maximum of 25 mL/hr per site with a maximum of 50 mL/hr for all sites combined, gradually increased to 80 mL/hr then, if tolerated, up to 100 mL/hr.	*For the first 2 infusions*: maximum of 10-20 mL/hr per site. *For subsequent infusions*: maximum of 60 mL/hr per site as tolerated, with a maximum of 240 mL/hr for all sites combined.
Volume	*Initial volume:* maximum of 20 mL/site. *For subsequent infusions:* may increase as tolerated to a maximum volume of 50 mL/site.	*Initial volume:* maximum of 25 mL/site. *For subsequent infusions:* may increase as tolerated to a maximum volume of 40 mL/site.	*Initial volume for the first 2 infusions*: - Patient weight less than 40 kg: maximum of 20 mL/site. - Patient weight of 40 kg and over: maximum of 60 mL/site. *For subsequent infusions*: For all weights: maximum of 60 mL/site.

- Do NOT shake.
- Refer to manufacturer's instructions for a more detailed procedure.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, anaphylaxis (rare).
- Cardiovascular: myocardial infarction, hypotension.
- GI: nausea, diarrhea.

…/Cont.

…/Cont.

POTENTIAL ADMINISTRATION HAZARDS (Cont.)

- CNS: headache, aseptic meningitis.
- Hematologic: rare; thrombotic events: deep vein thrombosis, pulmonary embolism, stroke.
- Respiratory: acute respiratory distress, pulmonary edema, respiratory failure, bronchospasm, cough.
- Back pain, pain at extremities.
- Fatigue, fever, chills.
- Risk of transmission of infectious agents, including viruses, and theoretically, the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.
- Local reactions: erythema, induration, swelling, local heat, pruritus, pain.

DOSAGE

- Adults and pediatrics (2 years of age and older) (Cuvitru ®):
  - Treatment of immune deficiency: loading dose of 0.2-0.5 g/kg SC to be divided over several days as needed with a maximum daily dose of 0.1-0.15 g/kg SC. After steady state IgG levels are attained, maintenance dose to be administered at repeated intervals to reach 0.3-1 g/kg as a cumulative monthly dose. Doses can be administered at regular intervals from daily up to every 2 weeks. Monitor serum IgG trough levels to guide subsequent doses and dosing intervals; dose regimen should achieve a trough level of at least 5-6 g/L and aim to be within the reference interval of serum IgG for age. Refer to manufacturer's monograph for more detailed dosage information.
  - Switching from another SC immune globulin (SCIG): same weekly dose as the weekly dose of the previous SCIG.
  - Switching from monthly dose of IV immune globulin (IVIG): beginning 1 week after the last IVIG dose, give the same dose via the SC route, though divide the dose in equivalent weekly doses.
- Adults and pediatrics (2 years of age and older) (Hizentra ®, Cutaquig ®):
  - Treatment of immune deficiency: 0.1-0.2 g/kg SC once weekly. For dosing frequencies greater than once a week (e.g., 2-3 times a week), divide the calculated weekly dose by the desired number of administration per week. If a loading dose is required, may give 0.2-0.5 g/kg SC divided over several days. Switching from every 3-4 weeks IVIG therapy: beginning 1 week after the last IVIG dose, give the same dose via the SC route, though divide the dose in equivalent weekly doses.
  - Measles prophylaxis: at least 0.2 g/kg SC weekly for 2 consecutive weeks.
- Adults with CIDP (Hizentra ®): 0.2-0.4 g/kg SC once weekly, beginning one week after the last IVIG dose. The calculated weekly dose can be doubled for administration every 2 weeks or can be administered more frequently (2-7 times a week) by dividing the weekly dose by the desired number of times per week.
- Dose conversion from g to mL: Hizentra ® and Cuvitru ®: multiply the calculated dose in g by 5.
  Cutaquig ®: multiply the calculated dose in g by 6.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Hizentra ®:
- Store vials and prefilled syringes at room temp or in the fridge (range of 2 to 25°C). Protect from light. Do not freeze. Do not shake.
- The solution is clear and pale yellow to light brown.

Cutaquig ®:
- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Stable for 6 months at room temp (maximum of 25°C) but do not return to the fridge once it has been stored at room temp.
- The solution is clear and colourless.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY (Cont.)

Cuvitru ®:
- Store vials between 2-8°C for up to 36 months. Protect from light. Do not freeze. Do not shake.
- Stable for 24 months (from the date of manufacture) at room temp not exceeding 25°C. Do not return to the fridge once it has been stored at room temp.
- The solution is clear, colourless or pale yellow or light brown.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- Avoid use in patients with IgA deficiency as they are more prone to develop anti-IgA antibodies which can lead to severe hypersensitivity reactions.
- Hizentra ® contains L-proline as a stabilizer and is therefore contra-indicated in patients with hyperprolinemia.

REFERENCES

1, 5, 82, 135.

NAME OF MEDICATION
INDIGOTINDISULFONATE SODIUM

OTHER NAMES
Ceruleinum, FD&C Blue No. 2, Indigo Carmine ®

CLASSIFICATION
Diagnostic agent

INDICATIONS

- To localize ureteral orifices during cystoscopy and ureteral catheterization, and as a marker dye to identify severed ureters and fistulous communications.
- Originally used to determine renal function, but has largely been replaced by other agents that yield more precise results.
- Has been used a a marker dye in amniocentesis and for chromoendoscopy of the gastrointestinal tract.

ADMINISTRATION

- IV direct (preferred): physician only. Administer undiluted (dilution may lead to precipitation).
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, bronchoconstriction.
- Cardiovascular: hypertension, bradycardia.
- GI: nausea, vomiting.
- Skin discolouration following large doses.

DOSAGE

- 5 mL of a 0.8% solution (40 mg).
- Larger doses may be required when used IM.
- Smaller doses may be required in infants, children, and underweight patients to prevent skin discolouration.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Protect from light.

MISCELLANEOUS

REFERENCES

1, 3, 5, 95.

Full revision 2015

NAME OF MEDICATION
INDOCYANINE GREEN

OTHER NAMES
IC-Green ®

CLASSIFICATION
Diagnostic agent

INDICATIONS

- For determining cardiac output, hepatic function and blood flow, and for ophthalmic fluorescence imaging.

ADMINISTRATION

- For IV use: reconstitute only with provided diluent to obtain a final concentration of 5 mg/mL or as required for procedure. The syringe to be used for administration should be rinsed with the provided diluent.
- IV direct: physician or RN.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria and anaphylactoid reactions; hypotension, bronchospasm, shock – may be related to dose and/or rate of administration.
- Transient elevation in unconjugated bilirubin.

DOSAGE

- Cardiac output test: adult: 5 mg; child: 2.5 mg; infant: 1.25 mg. Several doses are required but total dose should not exceed 2 mg/kg.
- Hepatic function test: 0.5 mg/kg (range 0.1-5 mg/kg).
- Ophthalmic fluorescence imaging: up to 40 mg in 2 mL of diluent.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Reconstituted solution is stable for 6 hours at room temp as per manufacturer's package insert.

MISCELLANEOUS

- Solutions contain a small amount of sodium iodide and should be used with caution in patients sensitive to iodide.

REFERENCES

1, 3, 5, 95.

Full revision 2015

NAME OF MEDICATION
INDOMETHACIN *

OTHER NAMES

CLASSIFICATION
Nonsteroidal anti-inflammatory

INDICATIONS

- To close patent ductus arteriosus in premature infants when usual medical management is ineffective.

ADMINISTRATION

- Reconstitute 1 mg vial with 1 or 2 mL of SWFI or NS without bacteriostatic agent (preserved diluents should not be used), to obtain a solution of 1 mg/mL or 0.5 mg/mL, respectively.
- Intermittent IV infusion: may be further diluted in NS to 0.1 mg/mL; infuse over 20-30 minutes.
- Do NOT administer via an umbilical catheter into vessels near the superior mesenteric artery.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypertension, edema.
- GI: vomiting, melena, transient ileus, necrotizing enterocolitis, gastric perforation.
- Hematologic: bleeding problems, decreased platelet aggregation.
- Renal: transient oliguria, anuria, hypercreatinemia, renal dysfunction.
- Acidosis, alkalosis, hypoglycemia, electrolyte disturbances.
- Local reactions: irritating to extravascular tissues; avoid extravasation.

DOSAGE

- Course of therapy usually consists of 3 doses administered at 12-24 hour intervals, with careful attention to urinary output. In general, may use 12-hour dosing interval if urine output is greater than 1 mL/kg/hr; use 24-hour dosing interval if urine output is less than 1 mL/kg/hr.
- According to age:

Age at 1st dose	dosage mg/kg		
	1st	2nd	3rd dose
younger than 48 hrs of age	0.2	0.1	0.1
2-7 days	0.2	0.2	0.2
older than 7 days of age	0.2	0.25	0.25

- If ductus arteriosus closes or is significantly reduced in size 48 hours or more after completion of the first course, no further doses are necessary.
- If ductus arteriosus re-opens, a second course of 1 to 3 doses may be administered, each dose given at a 12-24 hour interval; use age at 1st dose of the first course.
- If anuria or marked oliguria (urine output less than 0.6 mL/kg/hr) is evident at the time of the second or third dose, do not give additional doses until the renal function has returned to normal.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Manufacturer suggests to prepare immediately prior to each administration as solutions are preservative-free.
- Reconstituted solution with SWFI (0.5 mg/mL) is stable for 14 days at room temp or in the fridge in polypropylene syringes.
- Reconstituted solution with SWFI (0.5 mg/mL) is stable for 14 days in the fridge or 12 days at room temp in its original glass vial.
- Reconstituted solution with NS (1 mg/mL) is stable for 16 days at room temp.
- Stable for 10 days at room temp in NS at a concentration of 0.1 mg/mL in glass vials.

MISCELLANEOUS

REFERENCES

1, 5, 40, 82, 89, 208, 319, 513.

* Available via Health Canada's Special Access Programme.

Full revision 2015; limited revision 2018

NAME OF MEDICATION
InFLIXimab

OTHER NAMES
Avsola ™, Inflectra ®, Remicade ™, Remsima ™, Renflexis ™

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS

- Moderate to severe Crohn's disease in adults and children 9 years of age and older, and fistulizing Crohn's disease in adults.
- Moderately to severely active ulcerative colitis in adults and children 6 years of age and older.
- Moderately to severely active rheumatoid arthritis (in combination with methotrexate).
- Active ankylosing spondylitis.
- Active psoriatic arthritis.
- Moderate to severe plaque psoriasis.

ADMINISTRATION

- **Consider administration of premedication prior to each dose; refer to Dosage section.**
- For IV use: reconstitute each 100 mg vial with 10 mL of SWFI (can also use NS or 1/2NS) and allow to stand for 5 minutes. Do NOT shake to avoid excessive foaming. Inspect the reconstituted solution; it should be colourless to light yellow and opalescent; do NOT use if discoloured or opaque.
- Intermittent IV infusion: further dilute the reconstituted solution to 250 mL with NS. The infusion concentration should range between 0.4-4 mg/mL. Administer over at least 2 hours. **Exception**: for patients with rheumatoid arthritis and adults patients with Crohn's disease or ulcerative colitis receiving doses of 5 mg/kg: if they have tolerated 3 initial 2-hour infusions, subsequent infusions may be given over at least 60 minutes. Safety of shortened infusions at doses greater than 6 mg/kg has not been studied.
- Use a low-protein-binding in-line filter of 1.2 microns or less.
- Monitor patient closely (including vital signs) during and for at least for 30 minutes (at TOH, monitor for 60 minutes) after the end of the infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity (during or within 2 hours of infusion): anaphylaxis, rash, hypotension, severe bronchospasm.
- Infusion-related reactions: urticaria, dyspnea, hypotension, hypertension, fever, chills, headache, pruritus, chest pain. Most of the reactions are mild and transient and may resolve with slowing of the infusion rate, discontinuing the infusion or providing treatment (e.g., diphenhydramine). If patient has experienced a previous infusion reaction to infliximab, consult manufacturer's guidelines for more information.
- Delayed infusion reaction: serum sickness-like reaction, arthralgia, myalgia, fever, rash. Occurs in patients who have developed infliximab-specific antibodies, usually when restarting therapy after an extended period without treatment.
- GI: nausea, diarrhea, abdominal pain, dyspepsia.
- CNS: headache, dizziness.
- Flu-like symptoms.
- Infections: upper respiratory tract infections, urinary tract infections.
- Increase in ALT.

DOSAGE

- **Premedication with oral diphenhydramine (25-50 mg) and acetaminophen (650 mg) may be administered prior to infusion to minimize risk of infusion-related reactions. Add oral prednisone 40 mg or IV hydrocortisone 100 mg or IV methylprednisolone 20-40 mg if patient has a history of acute infusion reactions and is not currently receiving steroids.**
- Crohn's disease: in adults and pediatrics: 5 mg/kg IV at weeks 0, 2 and 6, followed by 5 mg/kg every 8 weeks thereafter; dose may be increased in adults only up to 10 mg/kg in patients with an initial response but a subsequent loss of response or with an incomplete response.
- Fistulizing Crohn's disease: 5 mg/kg IV at weeks 0, 2 and 6, followed by 5 mg/kg every 8 weeks thereafter; dose may be increased up to 10 mg/kg in patients with an initial response but a subsequent loss of response or with an incomplete response.

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DOSAGE (Cont.)

- Ulcerative colitis: in adults and pediatrics: 5 mg/kg IV at weeks 0, 2 and 6, followed by 5 mg/kg every 8 weeks; dose may be increased in adults only to 10 mg/kg to sustain clinical response and remission.
- Rheumatoid arthritis: 3 mg/kg IV at weeks 0, 2 and 6, then repeated every 8 weeks. Dose can increase up to 10 mg/kg as often as every 4 weeks.
- Ankylosing spondylitis: 5 mg/kg IV at weeks 0, 2 and 6, followed by 5 mg/kg every 6-8 weeks.
- Psoriatic arthritis, plaque psoriasis: 5 mg/kg IV at weeks 0, 2 and 6, followed by 5 mg/kg every 8 weeks.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge.
- Unopened vials are stable for 6 months at 30°C (Avsola ™, Inflectra ®, Remicade ®). Do not return vials to the fridge once stored at room temp.
- Reconstituted solution (with SWFI, NS or 1/2NS) is stable for 24 hours at room temp.
- Diluted solution is stable for 24 hours at room temp or in the fridge in NS at concentrations of 0.4 and 4 mg/mL in PVC, ethyl vinyl acetate, polyethylene or polyolefin containers.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity or infusion-related reactions.
- Patients should be evaluated for active or latent tuberculosis and for risk factors before starting therapy and periodically during therapy.
- Do not administer live vaccines concurrently with infliximab.
- Avsola ™ , Inflectra ®, Remsima ™ and Renflexis ™ are classified as being biosimilar to Remicade ®; therefore, these products are not interchangeable.

REFERENCES

1, 2, 5, 40, 135, 143, 351.

Full revision 2015; limited revision 2016, 2017, 2019, 2020, 2021

NAME OF MEDICATION
INFLUENZA VIRUS VACCINE (INACTIVATED)

OTHER NAMES
Afluria ® Tetra, Agriflu ®, Fluad ®, Flucelvax ® Quad, Flulaval ® Tetra, Fluviral ®, Fluzone ®, Influvac ® Tetra

CLASSIFICATION
Vaccine

INDICATIONS

- Vaccine to provide active immunization against the influenza virus. Recommended in adults and children 6 months of age and older, with focus on people at high risk of influenza-related complications or hospitalization, people capable of transmitting influenza to those at high risk, people who provide essential community services and people who are in direct contact with poultry infected with avian influenza during culling operations.

ADMINISTRATION

- Shake well before use (vial and prefilled syringe).
- IM: adults and older children, administer into the deltoid muscle; young children, administer into the anterolateral aspect of thigh.
- Influvac ® Tetra: IM or deep SC.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare.
- Fever, malaise, myalgias.
- Neurological disorders (rare).
- Oculorespiratory syndrome (rare).
- Local reactions: erythema, soreness, ecchymosis and/or induration at injection site.

DOSAGE

- Administered annually.
- Adult dose: 0.5 mL.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 2-8ºC. Protect from light. Do not freeze.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- Influvac ® Tetra is contraindicated in patients allergic to gentamicin.
- Fluad ® and Agriflu ® are contraindicated in patients allergic to kanamycin or neomycin.
- Afluria ® Tetra is contraindicated in patients allergic to neomycin or polymyxin B.
- Patients with a history of allergy to egg or egg products may be vaccinated using an inactivated vaccine in a setting where vaccines are routinely administered; Flucelvax ® Quad does not contain egg protein.
- Multi-dose vials from all brands contain thimerosal as a preservative.
- Administration should be delayed, if possible, during an acute febrile illness or active infection of moderate to severe intensity.
- Vaccine can inhibit the clearance of warfarin and theophylline.

REFERENCES

1, 2, 5, 31.

Full revision 2015; limited revision 2020, 2021

NAME OF MEDICATION
INOTUZUMAB OZOGAMICIN

OTHER NAMES
Besponsa ™

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS
- Treatment of relapsed or refractory CD22-positive B-cell precursor acute lymphocytic leukemia (ALL).

ADMINISTRATION
- **Ensure premedication has been administered prior to each infusion; refer to Dosage section.**
- This product is light sensitive and should be protected from light at all times by using an ultraviolet protector (amber bag or foil) to cover the IV bag during its preparation and administration. A foil should also cover the syringe barrel if the reconstituted solution is not transferred immediately into the IV bag. The IV line can be exposed to light for approximately 1 hour duration i.e., the time required to administer the drug and flush the line.
- Allow vials to reach room temp before reconstitution. Reconstitute each 0.9 mg vial with 4 mL of SWFI to obtain a solution with a final concentration of 0.25 mg/mL. Gently mix; do NOT shake. Protect the reconstituted solution from light if not used immediately.
- Intermittent IV infusion: method 1: withdraw from a 50 mL PVC, polyolefin or ethinyl vinyl acetate bag containing NS a volume equal to the volume of the reconstituted solution required for the patient's dose, then add the dose of inotuzumab ozogamicin to the NS bag. Method 2: add the dose of inotuzumab ozogamicin to an empty PVC, polyolefin or ethinyl vinyl acetate bag and then complete with NS to obtain a final volume of 50 mL. With both methods, the final concentration should be between 0.01 and 0.1 mg/mL. Gently mix; do NOT shake. Infuse over 60 minutes; protect the bag from light during the infusion. PVC, polyolefin or polybutadiene IV lines are recommended for infusion; infusion line does not need to be protected from light.
- Observe the patient during and for at least 60 minutes post-infusion for infusion-related reactions.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Infusion-related reactions: fever, chills, hypotension. If an infusion-related reaction occurs, stop the infusion and contact physician for appropriate treatment (e.g., corticosteroids, epinephrine, bronchodilators, oxygen). Depending on the reaction and its severity, could consider starting again the infusion but at a reduced rate.
- Cardiovascular: QTc interval prolongation.
- GI: nausea, vomiting, abdominal pain, constipation, diarrhea, anorexia.
- CNS: headache
- Hematologic: anemia, thrombocytopenia, neutropenia, lymphopenia.
- Hepatic: elevation of liver enzymes and bilirubin, veno-occlusive liver disease, ascites.
- Hyperuricemia, tumour lysis syndrome.
- Respiratory: dyspnea, epistaxis, upper respiratory tract infection, pneumonia.
- Fatigue, asthenia.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- **Premedication: methylprednisolone (or another corticosteroid), acetaminophen and diphenhydramine (or another antihistamine), 30 minutes to 2 hours before starting the infusion, especially with first dose.**
- For cycle 1 (21 day duration; can extend to 28 days if patient experiences toxicity or achieves a complete response): day 1: 0.8 mg/m^2 IV; days 8 and 15: 0.5 mg/m^2 IV.
- For cycle 2 and beyond (28 day duration): day 1: 0.8 mg/m^2 IV (or 0.5 mg/m^2 IV if a dose reduction is needed or if patient achieves a complete response); days 8 and 15: 0.5 mg/m^2 IV.
- Based on toxicity, delay treatment or decrease dose by 25%; do not re-escalate after a dose reduction.
- Dosage in renal impairment: no adjustment required for CrCl of 15 mL/min or greater; no data for CrCl less than 15 mL/min.
- Dosage in hepatic impairment: no adjustment required for bilirubin of 1.5 x ULN or less and AST/ALT of 2.5 x ULN or less; no data for bilirubin greater than 1.5 x ULN and AST/ALT greater than 2.5 x ULN.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light.
- The reconstituted solution is stable for 4 hours in the fridge. Protect from light. Do not freeze.
- If not used immediately, the diluted solution may be stored in the fridge or at room temp. Protect from light. Do not freeze. The maximum time from reconstitution of vials to the end of the infusion should not exceed 8 hours. If the diluted solution is refrigerated, it should be brought to room temp for at least 1 hour before its administration.
- Discoloured solution or solution with particles should not be used.
- Compatible with NS only.

MISCELLANEOUS

REFERENCES

5, 129, 165, 381.

New monograph 2016; limited revision 2017, 2018, 2019, 2020

NAME OF MEDICATION
INSULIN ASPART

OTHER NAMES
Fiasp ®, NovoRapid ®

CLASSIFICATION
Antidiabetic

INDICATIONS
- Treatment of diabetes mellitus for adults who require a very short rapid-acting insulin for the maintenance of normal glucose homeostasis.
- Treatment of type 1 diabetes mellitus in children aged 2 years and older who require a very short rapid-acting insulin for the maintenance of normal glucose homeostasis.

ADMINISTRATION
- Continuous IV infusion: for NovoRapid ®: dilute in a polypropylene bag of NS, D5W or D10W to obtain a final concentration between 0.05-1 unit/mL; for Fiasp ®: dilute in a polypropylene bag of NS or D5W to obtain a final concentration between 0.5-1 unit/mL. Note: insulin aspart has been administered IV in a limited number of patients. However, as it does not offer any clinical advantage over IV regular insulin and experience is limited, prefer regular insulin when this route is required. Insulin is known to bind to IV tubing. Priming IV tubing prior to initiation of the infusion will improve the accuracy of the insulin dose received by the patient during the first hour after initiation of therapy, or after a tubing change. Prior to connecting IV line to the patient, fill the IV line with the prescribed insulin infusion and let it stand 15 minutes. After 15 minutes, open the IV and rapidly flush 20 mL through the line to prime it. If there is no time to let insulin stand in the line, flush line with 20 mL of the insulin infusion over a 1 minute period immediately prior to connecting the tubing to the patient. Must be administered by an infusion pump.
- SC: in the upper arms, thighs, buttocks, or abdomen. Rotate injection site.
- SC infusion: undiluted as a continuous infusion. Rotate infusion site. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; urticaria, rash, anaphylaxis.
- Hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Follow blood glucose carefully.
- Hypokalemia (especially with IV administration) as insulin shifts potassium into the cells.
- Local reactions: itching, swelling, redness; lipodystrophy (if no rotation of injection sites).

DOSAGE
- Dosage based on glycemia.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store in the fridge. Do not freeze.
- When in use, vials or cartridges may be kept at room temp for up to 28 days. Protect from excessive heat and sunlight.
- NovoRapid ®: stable for 24 hours at room temp in D5W, D10W or NS at concentrations of 0.05-1 unit/mL in polypropylene containers.
- Fiasp ®: stable for 24 hours at room temp in D5W and NS at concentrations of 0.5-1 unit/mL in polypropylene containers.
- Once diluted, an initial loss of insulin aspart occurs due to sorption to the bag; this sorptive loss is variable.

MISCELLANEOUS
- The Fiasp ® formulation contains niacinamide, which accelerates absorption.
- Kinetic profile of insulin aspart administered SC: onset of action: 10-15 minutes for NovoRapid ® and an average of 4 minutes for Fiasp ®; peak of action: 1-1.5 hours for NovoRapid ® and 1-3 hours for Fiasp ®; duration of action: 3-5 hours for both products.

REFERENCES
1, 2, 5, 40, 95, 117, 135, 150.

Full revision 2015; limited revision 2016, 2017, 2020, 2021

NAME OF MEDICATION
INSULIN DEGLUDEC

OTHER NAMES
Tresiba ®

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of type 1 or type 2 diabetes mellitus in patients aged 1 year and older who require a long-acting (basal) insulin for the maintenance of normal glucose homeostasis.

ADMINISTRATION

- SC: upper arms, thighs or abdomen. Do NOT use if solution is viscous or cloudy. Rotate injection sites.
- Note: although insulin degludec is a clear insulin, it should NOT be administered IV.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus, urticaria, angioedema, anaphylaxis.
- Endocrine and metabolic: hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Monitor blood glucose carefully.
- Local reactions: rare; pain, hematoma, hemorrhage, erythema, nodules, swelling, rash, discolouration, pruritus, warmth sensation; lipodystrophy (if no rotation of injection sites).

DOSAGE

- Type 1 diabetes mellitus:
  - Insulin-naïve patients (adults and pediatrics): the initial dose of insulin degludec should correspond to 1/3-1/2 of total daily insulin requirement (the remainder of the total daily dose should be administered as a short/rapid acting insulin in divided doses). Administer dose SC once daily; monitor blood glucose closely. Adjust dose based on glycemia.
  - Switching from another basal (long- or intermediate-acting) insulin to insulin degludec (adults and pediatrics): the dose of insulin degludec should be 20% lower than that of the total daily dose of the previous basal insulin. Administer insulin degludec SC once daily; monitor blood glucose closely. Adjust dose based on glycemia.
- Type 2 diabetes mellitus:
  - Insulin-naïve patients (adults and pediatrics): initial dose of 10 units SC once daily. Adjust dose based on glycemia.
  - Switching from another basal (long- or intermediate-acting) insulin to insulin degludec:
    - From once-daily basal insulin, except insulin glargine 300 units/mL (adults): the dose of insulin degludec should be the same as that of the previous basal insulin. Administer insulin degludec SC once daily; monitor blood glucose closely. Adjust dose based on glycemia.
    - From once-daily basal insulin, except insulin glargine 300 units/mL (pediatrics): the dose of insulin degludec should be 20% lower than that of the total daily dose of the previous basal insulin. Administer insulin degludec SC once daily; monitor blood glucose closely. Adjust dose based on glycemia.
    - From twice-daily basal insulin or insulin glargine 300 units/mL (adults and pediatrics): the dose of insulin degludec should be 20% lower than that of the total daily dose of the previous basal insulin. Administer insulin degludec SC once daily; monitor blood glucose closely. Adjust dose based on glycemia.
- Dosage in renal impairment: no dosage adjustment necessary. Monitor blood glucose closely.
- Dosage in hepatic impairment: no dosage adjustment necessary. Monitor blood glucose closely.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled pens and cartridges between 2-8°C. Protect from light. Do not freeze.
- **Unopened** prefilled pens and cartridges are stable for 56 days at room temp (up to 30°C).
- **When in use**:
  - Prefilled pens: stable for 56 days at room temp (up to 30°C) or in the fridge, protected from heat and light (cap kept on the pen).
  - Cartridges: stable for 56 days at room temp (up to 30°C) protected from light. Do not refrigerate.
- Do not mix in the same syringe with any other types of insulin.

MISCELLANEOUS

- Kinetic profile of insulin degludec administered SC: onset of action: 1 hour; peak of action: 9-12 hours, although relatively flat curve; duration of action: longer than 42 hours.

REFERENCES

1, 5, 82, 95, 135.

New monograph 2019; limited revision 2020

NAME OF MEDICATION
INSULIN DETEMIR

OTHER NAMES
Levemir ®

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of adult patients with type 1 or type 2 diabetes mellitus who require a long-acting (basal) insulin for the maintenance of normal glucose homeostasis.

ADMINISTRATION

- SC: in the upper arms, thighs, buttocks or abdomen. Rotate injection sites.
- Note: although insulin detemir is a clear insulin, it should NOT be administered IV.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; urticaria, rash, anaphylaxis.
- Hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Follow blood glucose carefully.
- Local reactions: redness, swelling, inflammation, itching, pain; lipodystrophy (if no rotation of injection sites).

DOSAGE

- Dosage based on glycemia.
- In insulin-naïve type 2 diabetic patients: initial dose of 0.1-0.2 units/kg SC once daily in the evening or 10 units SC once to twice daily and subsequently adjusted according to patient's need.
- In type 2 diabetic patients transferred from NPH insulin: initial dose at transfer can be identical (on a unit-for-unit basis) but dose may need to be retitrated according to glycemia; higher doses of insulin detemir may be required for similar hypoglycemic activity.
- To be administered once or twice daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 2-8°C. Do not freeze. Protect from light.
- When in use, vials or cartridges may be kept at room temp for up to 42 days, providing temperatures do not exceed 30°C.
- Do not mix in the same syringe with any other types of insulin.

MISCELLANEOUS

- Kinetic profile of insulin detemir administered SC: onset of action: 1.5 hours; peak of action: relatively flat curve; duration of action: 12-24 hours.

REFERENCES

1, 2, 95, 117, 150.

Full revision 2015; limited revision 2017, 2020

NAME OF MEDICATION
INSULIN GLARGINE

OTHER NAMES
Basaglar ™, Lantus ®, Toujeo ™

CLASSIFICATION
Antidiabetic

INDICATIONS
- Treatment of adult patients with type 1 or type 2 diabetes mellitus who require a long-acting (basal) insulin for the maintenance of normal glucose homeostasis.
- Treatment of children over 6 years of age with type 1 diabetes mellitus who require a long-acting (basal) insulin for the maintenance of normal glucose homeostasis.

ADMINISTRATION
- SC: in the upper arms, thighs, buttocks or abdomen. Rotate injection sites.
- Note: although insulin glargine is a clear insulin, it should NOT be administered IV.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; urticaria, rash, anaphylaxis.
- Hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Follow blood glucose carefully.
- Local reactions: redness, swelling, itching, pain; lipodystrophy (if no rotation of injection sites).

DOSAGE
- Dosage based on glycemia.
- In insulin-naïve type 2 diabetic patients: initial dose of 10 units for Lantus ® and Basaglar ™ and 0.2 units/kg for Toujeo ™ SC once daily, to be adjusted according to patient's needs.
- In patients transferred from NPH insulin to insulin glargine (Lantus ®, Basaglar ™ or Toujeo ™): if patient on a once daily NPH regimen, transfer to same dose of insulin glargine; if patient on a twice daily NPH insulin regimen, transfer to a 20% total lower dose of insulin glargine given once daily.
- In patients switched from Lantus ® to Toujeo ™: initial dose of Toujeo ™ should be the same as the Lantus ®, but a higher dose may be needed to achieve target glucose levels.
- In patients switched from Toujeo ™ to Lantus ® or Basaglar ™: initial dose of Lantus ® or Basaglar ™ should be 20% lower than that of Toujeo ™.
- In patients switched from Lantus ® to Basaglar ™: initial dose of Basaglar ™ should be the same as the Lantus ®.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 2-8°C. Do not freeze.
- Lantus ® and Basaglar ™: when in use, vials or cartridges may be kept at room temp for 28 days, providing temperatures do not exceed 30°C and they are kept away from direct light and heat.
- Toujeo ™: when in use, prefilled pens may be kept at room temp for 42 days, providing temperatures do not exceed 30°C and they are kept away from direct light and heat.
- Do not mix in the same syringe with any other types of insulin.

MISCELLANEOUS
- Kinetic profile of insulin glargine administered SC: onset of action: 1.5 hours for Lantus ® and over 6 hours for Toujeo ™; peak of action: relatively flat curve, but even flatter for Toujeo ™; duration of action: 24 hours for Lantus ® and more than 24 hours for Toujeo ™.
- Lantus ® (100 units/mL) and Basaglar ™ (100 units/mL) compared to Toujeo ™ (300 units/mL) are not bioequivalent as Toujeo ™ is less potent on an unit per unit basis; these products are therefore not interchangeable without dose adjustment.
- Basaglar ™ is classified as being biosimilar to Lantus ®; therefore, these products are not interchangeable.

REFERENCES
1, 5, 95, 117, 127, 135, 150.

Full revision 2015; limited revision 2016, 2017, 2020, 2021

NAME OF MEDICATION
INSULIN GLULISINE

OTHER NAMES
Apidra ™

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of diabetes mellitus for patients who require a very short rapid-acting insulin for the maintenance of normal glucose homeostasis.

ADMINISTRATION

- Continuous IV infusion: dilute in NS to obtain a final concentration of 0.05 to 1 unit/mL. Note: insulin glulisine has been administered IV in a limited number of patients. However, as it does not offer any clinical advantage over IV regular insulin and experience is limited, prefer regular insulin when this route is required. Insulin is known to bind to IV tubing. Priming IV tubing prior to initiation of the infusion will improve the accuracy of the insulin dose received by the patient during the first hour after initiation of therapy, or after a tubing change. Prior to connecting IV line to the patient, fill the IV line with the prescribed insulin infusion and let it stand 15 minutes. After 15 minutes, open the IV and rapidly flush 20 mL through the line to prime it. If there is no time to let insulin stand in the line, flush line with 20 mL of the insulin infusion over a 1 minute period immediately prior to connecting the tubing to the patient. Must be administered by an infusion pump.
- SC: in the upper arms, thighs, buttocks or abdomen. Rotate injection site.
- SC infusion: undiluted as a continuous infusion in the abdomen. Rotate infusion site. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; urticaria, rash, anaphylaxis.
- Hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Follow blood glucose carefully.
- Hypokalemia (especially with IV administration) as insulin shifts potassium into the cells.
- Local reactions: itching, swelling, redness; lipodystrophy (if no rotation of injection sites).

DOSAGE

- Dosage based on glycemia.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store in the fridge. Do not freeze. Protect from sunlight.
- When in use, vials or cartridges may be kept at room temp (between 15-25°C) for up to 28 days.
- Infusion sets (reservoir, tubing, catheter) and insulin glulisine in the reservoir should be discarded after 2 days of use or after exposure to temperatures above 37°C.
- Stable for 48 hours at room temp in NS in PVC containers.
- Incompatible with dextrose-containing solutions and with Ringer's injection.

MISCELLANEOUS

- Kinetic profile of insulin glulisine administered SC: onset of action: 10-15 minutes; peak of action: 1-1.5 hours; duration of action: 3-5 hours.

REFERENCES

1, 2, 40, 95, 117, 135, 150, 208.

Full revision 2015; limited revision 2016, 2017, 2020

NAME OF MEDICATION
INSULIN LISPRO

OTHER NAMES
Admelog ®, Humalog ®

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of diabetes mellitus for patients who require a very short rapid-acting insulin for the maintenance of normal glucose homeostasis.

ADMINISTRATION

- Continuous IV infusion: dilute in NS to obtain a final concentration of 0.1 to 1 unit/mL. Note: insulin lispro has been administered IV in a limited number of patients. However, as it does not offer any clinical advantage over IV regular insulin and experience is limited, prefer regular insulin when this route is required. Insulin is known to bind to IV tubing. Priming IV tubing prior to initiation of the infusion will improve the accuracy of the insulin dose received by the patient during the first hour after initiation of therapy, or after a tubing change. Prior to connecting IV line to the patient, fill the IV line with the prescribed insulin infusion and let it stand 15 minutes. After 15 minutes, open the IV and rapidly flush 20 mL through the line to prime it. If there is no time to let insulin stand in the line, flush line with 20 mL of the insulin infusion over a 1 minute period immediately prior to connecting the tubing to the patient. Must be administered by an infusion pump.
- SC: in the upper arms, thighs, buttocks or abdomen. Rotate injection site.
- SC infusion: undiluted as a continuous infusion. Rotate infusion site. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; urticaria, rash, anaphylaxis.
- Hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Follow blood glucose carefully.
- Hypokalemia (especially with IV administration) as insulin shifts potassium into the cells.
- Local reactions: itching, swelling, redness; lipodystrophy (if no rotation of injection sites).

DOSAGE

- Dosage based on glycemia.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store in the fridge. Do not freeze. Protect from sunlight.
- When in use, vials, cartridges or prefilled pens may be kept at room temp (below 30°C) for up to 28 days.
- Stable for 48 hours in the fridge followed by an additional 48 hours at room temp in NS at concentrations of 0.1-1 unit/mL.
- Compatible with D5W, D10W and NS.

MISCELLANEOUS

- Kinetic profile of insulin lispro administered SC: onset of action: 10-15 minutes; peak of action: 1-2 hours; duration of action: 3.5-4.75 hours.
- Admelog ® is classified as being biosimilar to Humalog ®; therefore, these products are not interchangeable.

REFERENCES

1, 5, 40, 95, 117, 135, 150.

Full revision 2015; limited revision 2016, 2017, 2020

NAME OF MEDICATION
INSULIN NPH

OTHER NAMES
Humulin N ®, Hypurin NPH ®, Novolin ® ge NPH

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of diabetes mellitus patients (Type 1 or Type 2) who require an intermediate-acting insulin for the maintenance of normal glucose homeostasis.

ADMINISTRATION

- SC: in the upper arms, thighs, buttocks or abdomen. Rotate injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; urticaria, rash, anaphylaxis.
- Hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Follow blood glucose carefully.
- Local reactions: itching, swelling, redness; lipodystrophy (if no rotation of injection sites).

DOSAGE

- Dosage based on glycemia.
- To be administered once or twice daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 2-8°C. Do not freeze. Protect from sunlight.
- When in use, vials or cartridges may be kept at room temp for up to 28 days.
- Stable for 5-7 days in the fridge in either glass or plastic syringes.
- Mixtures of the same manufacturer of human NPH insulin with human regular insulin are stable for 28 days in the fridge when stored in polypropylene and polypropylene-ethylene copolymer syringes.

MISCELLANEOUS

- Kinetic profile of NPH insulin administered SC: onset of action: 1-3 hours; peak of action: 5-8 hours; duration of action: up to 18 hours.

REFERENCES

1, 2, 4, 6, 117, 135.

Full revision 2015; limited revision 2017, 2020

NAME OF MEDICATION
INSULIN REGULAR

OTHER NAMES
Entuzity ™, Humulin R ®, Hypurin Regular ®, Novolin ® ge Toronto

CLASSIFICATION
Antidiabetic

INDICATIONS

For the 100 units/mL solution:
- Treatment of diabetes mellitus for patients who require a short-acting insulin for the maintenance of normal glucose homeostasis.
- Treatment of emergency situations such as diabetic acidosis or coma in diabetics.
- Treatment of hyperglycemia in patients on Total Parenteral Nutrition.
- Treatment of hyperkalemia with dextrose.

For the 500 units/mL solution (Entuzity ™):
- Treatment of diabetes mellitus for patients requiring more than 200 units of insulin per day.

ADMINISTRATION

- IV direct (using 100 units/mL solution only): physician or RN; administer over 1 minute.
- Continuous IV infusion (using 100 units/mL solution only): may dilute with commonly used IV fluids to give by infusion at concentrations of 0.05-1 unit/mL. At TOH, recommended infusion dilution is 2 units/10 mL (0.2 units/mL as 50 units in 250 mL) except for critical care areas and acute monitoring areas, where the recommended dilution is 1 unit/mL (100 units in 100 mL or 50 units in 50 mL). Insulin is known to bind to IV tubing. Priming IV tubing prior to initiation of the infusion will improve the accuracy of the insulin dose received by the patient during the first hour after initiation of therapy, or after a tubing change. Prior to connecting IV line to the patient, fill the IV line with the prescribed insulin infusion and let it stand 15 minutes. After 15 minutes, open the IV and rapidly flush 20 mL through the line to prime it. If there is no time to let insulin stand in the line, flush line with 20 mL of the insulin infusion over a 1 minute period immediately prior to connecting the tubing to the patient. Must be administered by an infusion pump.
- SC (using 100 units/mL or 500 units/mL solution): in the upper arms, thighs, buttocks or abdomen. Rotate injection site.
- IM (using 100 units/mL solution only) (uncommon).

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; urticaria, rash, anaphylaxis.
- Hypoglycemia (**Antidote**: oral carbohydrates or 25-50% dextrose IV or glucagon IV/IM/SC). Follow blood glucose carefully.
- Hypokalemia (especially with IV administration) as insulin shifts potassium into the cells.
- Local reactions: itching, pain, swelling, redness; lipodystrophy (if no rotation of injection sites).

DOSAGE

- Dosage based on glycemia.
- In diabetic ketoacidosis (using 100 units/ mL solution only): 0.15 units/kg OR 10 units initially IV direct, followed by an IV infusion of 0.1 units/kg/hr or 5-10 units/hr. If plasma glucose unchanged after 60 minutes, double the infusion rate.
- Hyperkalemia (using 100 units/mL solution only): 5-10 units IV direct **followed immediately** by 25-50 g of dextrose (50-100 mL 50% solution) IV administered over 5 minutes.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

For the 100 units/mL and 500 units/mL solutions:
- Store unopened vials, cartridges and pens in the fridge. Do not freeze. Protect from sunlight.
- Do not use if solution is not clear and colourless.
- When in use, vials, cartridges and pens should be kept at room temp and discarded after 28 days.

For the 100 units/mL solution:
- Human insulin is compatible with saline, dextrose and RL.
- Stable for 14 days in the fridge undiluted in polypropylene syringes.
- Mixtures of the same manufacturer of human NPH insulin with human regular insulin are stable for 28 days in the fridge when stored in polypropylene and polypropylene-ethylene copolymer syringes.

MISCELLANEOUS

- Kinetic profile of human regular insulin (100 units/mL) administered SC: onset of action: 30 minutes; peak of action: 2-3 hours; duration of action: 6.5 hours.
- After SC administration, mean peak serum insulin concentrations are lower for Entuzity ™ compared to Humulin R ®, and half-life of Entuzity ™ is 4.5 hours compared to 3.6 hours for Humulin R ®.

REFERENCES

1, 2, 4, 5, 6, 40, 95, 117, 328.

Full revision 2015; limited revision 2017, 2018, 2020, 2021

NAME OF MEDICATION
INTERFERON ALPHA-2B

OTHER NAMES
Intron A ®

CLASSIFICATION
Antineoplastic, Immunomodulator

Do NOT confuse interferon alpha-2B with interferon beta.
This monograph is specific to interferon ALPHA-2B.

INDICATIONS

- For the treatment of patients with chronic hepatitis C, chronic active hepatitis B, chronic myelogenous leukemia (CML), thrombocytosis associated with CML, multiple myeloma, non-Hodgkin's lymphoma (NHL), malignant melanoma, AIDS-related Kaposi's sarcoma, and hairy cell leukemia.

ADMINISTRATION

- **Consider administration of premedication prior to each dose to decrease flu-like symptoms; refer to Dosage section.**
- Intermittent IV infusion:
  Intron A ® powder: reconstitute by adding 1 mL of supplied SWFI or bacteriostatic water. Do NOT shake. Dilute dose in 100 mL of NS for a final concentration of no less than 0.1 million units/mL. Infuse over 20 minutes.
  Ready-to-use solution: dilute dose in 50 mL of NS for a final concentration of no less than 0.3 million units/mL. Infuse over 20 minutes.
- Dilution and rate of administration may be dependent on the specific dosing protocol used.
- SC: reconstitute powder with 1 mL of SWFI or bacteriostatic water or use ready-to-use solution. Do NOT shake. Inject in anterior thighs, outer upper arms or abdomen (avoid navel or below waistline). Rotate injection sites.
- IM: reconstitute powder with 1 mL of SWFI or bacteriostatic water or use ready-to-use solution. Do NOT shake. Inject in anterolateral thighs, upper arms or outer area of buttocks. Rotate injection sites. Avoid IM use if thrombocytopenia.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: urticaria, angioedema, bronchoconstriction, anaphylaxis (rare).
- Cardiovascular: hypotension (during or up to 2 days post therapy), cardiac arrhythmias.
- GI: anorexia, nausea, vomiting.
- CNS: depression, headache, confusion, altered mental status.
- Flu-like symptoms: fever, fatigue, chills, rigors, myalgia, anorexia, headache (reversible within 72 hours). Can be alleviated with **premedication**; refer to Dosage section.
- Local reactions: redness, pain, burning, bleeding, bruising, pruritus.

DOSAGE

- **Premedication with acetaminophen 500 mg-1 g may be administered 30 minutes prior to interferon (and q4h after for up to 4 g per day) to alleviate flu-like symptoms. Meperidine 50 mg IV prior to interferon may decrease chills and rigors.**
- Chronic hepatitis C: 3 million units SC/IM 3 times per week. Dose may increase up to 10 million units 3 times per week.
- Chronic active hepatitis B: 30-35 million units per week SC/IM, administer as 5 million units daily or 10 million units 3 times per week.
- CML and thrombocytosis associated with CML: 4-5 million units/m^2 SC daily. Dose range: 0.5-10 million units/m^2 daily. Can be decreased to 3 times a week when white blood cells are controlled.
- Multiple myeloma: 3 million units/m^2 SC 3 times per week.
- NHL: 5 million units SC 3 times per week.
- Malignant melanoma: induction with 20 million units/m^2 IV for 5 consecutive days per week for 4 weeks followed by maintenance treatment with 10 million units/m^2 SC 3 times per week.
- Kaposi's sarcoma: 30 million units/m^2 SC/IM 3 times per week.
- Hairy cell leukemia: 2 million units/m^2 SC 3 times per week.
- Specialized references should be consulted for specific dosing protocols.
- Dose can be adjusted according to efficacy and toxicity.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials (powder and ready-to-use solution) in the fridge. However, the powder may also be stored at room temp for up to 4 weeks and the ready-to-use solution for up to 7 days.
- Stable for two weeks between 15-30°C or 30 days between 2-8°C after reconstitution with bacteriostatic water. Stable for up to 24 hours in the fridge after reconstitution with SWFI.
- Opened vials of the ready-to-use solution: the10 million unit vials are stable for up to 7 days in the fridge and the 18 and 25 million unit vials are stable for up to 4 weeks in the fridge.
- Stable for 24 hours at room temp or in the fridge when diluted in NS.
- Stable for 24 hours at room temp or in the fridge in Ringer's injection, RL, amino acids and sodium bicarbonate 5% at concentrations of 0.05-1 million unit/mL in glass bottles.
- Incompatible with D5W.

MISCELLANEOUS

- Adequate hydration should be maintained.
- Electrocardiograms prior to and during the course of treatment should be done in patients with pre-existing cardiac abnormalities and/or advanced stages of cancer.

REFERENCES

1, 2, 129, 135, 165.

Full revision 2015; limited revision 2016, 2017, 2019, 2020

NAME OF MEDICATION
INTERFERON BETA

OTHER NAMES
Avonex ®, Betaseron ®, Extavia ®, Rebif ®

CLASSIFICATION
Immunomodulator

Do NOT confuse interferon beta with interferon alpha-2B.
This monograph is specific to interferon BETA.

INDICATIONS
- Treatment of the relapsing forms of multiple sclerosis (MS).
- Treatment of secondary progressive MS.
- Treatment of patients who have experienced a single demyelinating event with lesions typical of MS.

ADMINISTRATION

- Avonex ® and Rebif ® are provided as ready-to-use solutions.
- For Betaseron ® and Extavia ®, reconstitute 0.3 mg vial with the prefilled diluent syringe of sodium chloride 0.54% (1.2 mL) for a final concentration of 0.25 mg/mL. Do NOT shake.
- RN in presence of physician for first dose only.
- IM (Avonex ®): thigh or upper arm.
- SC (Betaseron ®, Extavia ®, Rebif ®): abdomen (except waistline), buttock, thigh or arm.
- Rotate site of injection.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: skin rash, urticaria, dyspnea, orolingual edema, anaphylaxis (rare).
- GI: abdominal pain, constipation, diarrhea, dry mouth, dyspepsia, nausea.
- CNS: mood disorders, depression.
- Flu-like symptoms: headache, fever, fatigue, chills, malaise, myalgia, rigors, arthralgia. Occur within hours or days after injection. Subside within a few months of therapy. Can be alleviated by acetaminophen or NSAID 4 hours before, at the time and 4 hours after injection.
- Myleosuppression.
- Local reactions: pain, inflammation, redness, induration. Tissue necrosis reported with SC administration.

DOSAGE
- Avonex ®: 30 mcg (6 million international units) IM once weekly. May start at 1/4 dose (7.5 mcg) and increase dose weekly by 7.5 mcg until reach full dose of 30 mcg.
- Betaseron ® and Extavia ®: 0.25 mg (8 million international units) SC every other day. May start at 1/4 dose and increase gradually (25% every 1-2 weeks) over 6 weeks to the full dose.
- Rebif ®: 44 mcg (12 million international units) SC three times weekly. May be reduced to 22 mcg (6 million international units) three times weekly if poorly tolerated. May start at 20% of the dose during the first 2 weeks, increase at 50% during weeks 3 and 4, and to the full dose thereafter.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Do not freeze.
- Avonex ®: store prefilled syringes and autoinjectors in the fridge; however, they are stable 7 days at room temp. Protect from light.
- Betaseron ® and Extavia ®: store vials at room temp or fridge. After reconstitution, use product immediately or within 3 hours if stored in the fridge.
- Rebif ®: store prefilled syringes, cartridges and pens in the fridge; however, they are stable for up to 1 month at room temp. Protect from direct light.

MISCELLANEOUS
- Avonex ® and Rebif ® are interferon beta-1a while Betaseron ® and Extavia ® are interferon beta-1b. These products are not interchangeable.

REFERENCES
1, 2, 95.

Full revision 2015; limted revision 2016, 2019

NAME OF MEDICATION
IPILIMUMAB

OTHER NAMES
Yervoy ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of unresectable or metastatic melanoma, as monotherapy or in combination with nivolumab.
- Treatment of intermediate/poor-risk advanced or metastatic renal cell carcinoma in combination with nivolumab.

ADMINISTRATION

- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards section.**
- Let the vials at room temp for approximately 5 minutes.
- Intermittent IV infusion (mandatory): withdraw the required dose and put into an empty glass bottle or IV bag (PVC or non-PVC). May be administered undiluted (5 mg /mL) or further diluted in NS or D5W to a final concentration of 1 to 4 mg/mL. Gently mix the solution by gentle inversion. Do NOT shake. The solution is clear to pale yellow and may contain translucent-to-white amorphous particles; discard if cloudy. Administer in a separate infusion line over 90 minutes for melanoma and over 30 minutes for renal cell carcinoma, through a low-protein-binding in-line filter (0.2 or 1.2 micron polyethersulfone filter or 0.2 micron nylon filter). Flush the line with NS or D5W at the end of infusion.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: immune-mediated reactions, can be fatal; can involve any organ. Most common reactions include enterocolitis, intestinal perforation, hepatitis, dermatitis (e.g., Stevens-Johnson syndrome, toxic epidermal necrolysis, severe rash), neuropathy (e.g., Guillain-Barré syndrome, severe sensory or motor neuropathy, myasthenia gravis), endocrinopathy (hypopituitarism, adrenal insufficiency or crisis, hypogonadism, hypothyroidism, hyperthyroidism, Cushing's syndrome). Discontinue ipilimumab if reaction is severe and administer high-dose corticosteroids (i.e., 1 to 2 mg/kg/day of prednisone or equivalent) with or without additional immunosuppressive therapy; ocular manifestations should be treated with ophthalmic corticosteroids.
- Infusion-related reactions: pruritus, urticaria, flushing, rash, dyspnea, bronchospasm, hypotension, angioedema. If severe: discontinue ipilimumab permanently and treat reactions as needed (bronchodilators, epinephrine, diphenhydramine, corticosteroids). If mild to moderate: give diphenhydramine and decrease infusion rate; corticosteroids may be given though would decrease any beneficial immunologic effect of ipilimumab. May **consider premedication** for subsequent doses (diphenhydramine, acetaminophen).
- GI: diarrhea, nausea, vomiting, abdominal pain, anorexia, weight loss.
- CNS: headache.
- Dermatologic: rash, pruritus.
- Fatigue, fever, chills, arthralgia, myalgia.
- Risk of hemophagocytic lymphohistiocytosis (a condition where a large number of immune cells attack and destroy other blood cells) when ipilimumab is used alone or in combination with nivolumab.
- Local reactions at injection site.

DOSAGE

- For melanoma (with or without nivolumab): 3 mg/kg IV every 3 weeks for a total of 4 doses.
- For renal cell carcinoma: 1 mg/kg IV every 3 weeks for a total of 4 doses.
- Dosage in renal impairment: no dosage adjustment required in mild to moderate renal impairment.
- Dosage in hepatic impairment: no dosage adjustment necessary for mild hepatic insifficiency; has not been studied in patients with moderate or severe hepatic impairment.
- Consult specific protocol.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Protect from light.
- Stable for 24 hours at room temp or in the fridge undiluted (5 mg/mL) or in NS or D5W at concentrations of 1-4 mg/mL.

MISCELLANEOUS

- Monitor LFTs, thyroid function and electrolytes before each dose.
- Contraindicated in patients with active, life-threatening autoimmune disease and in organ recipients.

REFERENCES

1, 5, 40, 95, 129, 165, 208, 438.

NAME OF MEDICATION
IRINOTECAN

OTHER NAMES
Camptosar ®

CLASSIFICATION
Antineoplastic

Do NOT confuse irinotecan with liposomal irinotecan.
This monograph is specific to IRINOTECAN.

INDICATIONS

- First-line therapy for patients with metastatic carcinoma of the colon or rectum, in combination with other antineoplastics.
- As a single agent for recurrent metastatic carcinoma of the colon or rectum in patients who received previously 5-fluorouracil.
- Also used for other types of cancer.

ADMINISTRATION

- Intermittent IV Infusion (mandatory): dilute in 250-500 mL D5W (preferred) or NS for a final concentration of 0.12-3 mg/mL; infuse over 90 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity (rare).
- Cardiovascular: bradycardia, vasodilation (flushing).
- GI: nausea, vomiting, abdominal pain, anorexia, weight loss, flatulence, salivation, diarrhea. Early onset diarrhea (occurring up to 24 hours of administration) is cholinergic in nature. It may be severe but is usually transient; may be managed by use of atropine (see cholinergic syndrome below). For late onset diarrhea (occurring more than 24 hours after administration), prompt use of loperamide is recommended, as diarrhea may be prolonged, leading to electrolyte disturbances; may be life-threatening.
- CNS: dizziness, somnolence, insomnia, headache, speech disorder.
- Respiratory: dyspnea, non-productive cough, rhinitis, pneumonitis (infrequent).
- Fatigue, fever, chills, sweating, pain, tumour lysis, infection.
- Cholinergic syndrome: diarrhea, rhinitis, hypersalivation, miosis, lacrimation, diaphoresis, flushing and cramping. To prevent or treat: atropine 0.25-1 mg IV or SC.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Combination therapy: 125 mg/m^2 IV once weekly on days 1, 8, 15, 22, followed by a rest period until next course begins on day 43 OR 180 mg/m^2 IV once every 2 weeks on days 1, 15, 29, followed by a rest period until next course begins on day 43.
- Monotherapy: 125 mg/m^2 IV once weekly on days 1, 8, 15, 22, followed by a rest period until next course begins on day 43 OR 350 mg/m^2 IV once every 3 weeks.
- Consider dose reduction in patients 70 years of age and older, patients with prior pelvic/abdominal radiotherapy, with performance status of 2 (as per WHO/ECOG scale), with moderately increased bilirubin (17-35 mcmol/L) and with Gilbert's syndrome.
- Consult the manufacturer's monograph for dosage modification guidelines.
- Consult specific protocol.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp; protect from light.
- Stable for 28 days at room temp or in the fridge, protected from light in both cases, in NS and D5W at concentrations of 0.4-2.8 mg/mL in PVC and low-density polyethylene containers.
- Stable for 72 hours at room temp and exposed to light in NS and D5W at concentrations of 0.4-2.8 mg/mL in PVC and low-density polyethylene containers.
- Some solutions diluted in NS resulted in precipitation when refrigerated.

MISCELLANEOUS

- Contraindicated in patients on azole antifungals (e.g., ketoconazole, fluconazole, itraconazole) or receiving concurrent irradiation.

REFERENCES

2, 40, 129, 143, 165.

Full revision 2015; limited revision 2016, 2017, 2018, 2019, 2020

NAME OF MEDICATION
IRINOTECAN (LIPOSOMAL)

OTHER NAMES
Irinotecan liposome, Nanoliposomal irinotecan, Onivyde ®, Pegylated liposomal irinotecan

CLASSIFICATION
Antineoplastic

Do NOT confuse liposomal irinotecan with irinotecan.
This monograph is specific to LIPOSOMAL irinotecan.

INDICATIONS

- Treatment of metastatic adenocarcinoma of the pancreas, in combination with 5-fluorouracil and leucovorin, in adult patients who have disease progression following gemcitabine-based therapy.

ADMINISTRATION

- **Ensure premedication has been administered; refer to dosage section.**
- Intermittent IV infusion: withdraw the required amount from the vials and dilute dose in D5W or NS to a final volume of 500 mL. Gently invert the bag to mix. Infuse over 90 minutes. Do NOT use in-line filters.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: includes anaphylaxis.
- Infusion-related reactions: rash, urticaria, pruritus, periorbital edema.
- GI: nausea, vomiting, stomatitis, anorexia. Early onset diarrhea: occurs during or within 24 hours of administration, transient. May be accompanied by cholinergic symptoms (see Cholinergic syndrome below). Late-onset diarrhea: can be life-threatening; occurs more than 24 hours after IV administration (median onset of 8 days after infusion) and may lead to electrolyte disturbances and dehydration (see Electrolyte disturbances); initiate loperamide at the first sign and administer until the patient is without diarrhea for at least 12 hours, maximum duration of loperamide therapy of 48 hours. If diarrhea persists after 24 hours of loperamide therapy, add an antibiotic (e.g., a fluoroquinolone); if diarrhea is still present after 48 hours of loperamide therapy, discontinue loperamide and continue the antibiotic.
- Cholinergic syndrome: diarrhea, rhinitis, hypersalivation, miosis, lacrimation, bradycardia, diaphoresis, flushing, abdominal cramping. To prevent or treat: atropine 0.25 mg to 1 mg IV or SC.
- Electrolyte disturbances: due to severe diarrhea: hypokalemia, hyponatremia, hypocalcemia, hypophosphatemia, hypomagnesemia.
- Endocrine and metabolic: weight loss, hypoalbuminemia.
- Hematologic: neutropenia, lymphopenia, anemia, thrombocytopenia.
- Hepatic: increase in LFTs.
- Respiratory: interstitial lung disease (has been observed with conventional irinotecan; unknown risk with liposomal irinotecan; closely monitor).
- Renal: increased serum creatinine.
- Fever, fatigue.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- **Premedicate at least 30 minutes prior to each dose with a corticosteroid (dexamethasone or equivalent) and an antiemetic.**
- 70 mg/m^2 (free base) IV every 2 weeks; administer liposomal irinotecan first, then leucovorin, followed by 5-fluorouracil.

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DOSAGE (Cont.)

- If patient is known to be homozygous for the UGT1A1*28 allele: initial dose of 50 mg/m^2 (free base) IV to be increased to 70 mg/m^2 if no drug toxicities within the first 2 weeks.
- Dosage in renal impairment: usual dose if CrCl is 30 mL/min or greater; do not use if CrCl is lower than 30 mL/min (lack of data).
- Dosage in hepatic impairment: do not use in patients if bilirubin is greater than 34.2 mcmol/L, or AST/ALT are greater than 2.5 x ULN or 5 x ULN in presence of liver metastasis.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze.
- The diluted suspension may be stored for 4 hours at room temp (approximately 25°C), protected from light prior to infusion.
- The diluted suspension may be stored for 24 hours in the fridge protected from light prior to infusion. Do not freeze.

MISCELLANEOUS

- Do not use live vaccines; inactivated vaccines may be administered, but may lead to a decreased response.

REFERENCES

5, 95, 129, 135, 165.

New monograph 2018; limited revision 2019, 2020

NAME OF MEDICATION
IRON ISOMALTOSIDE 1000

OTHER NAMES
Ferric derisomaltose, Monoferric ™

CLASSIFICATION
Hematinic

Do NOT confuse iron isomaltoside 1000 with iron sucrose.
This monograph is specific to iron ISOMALTOSIDE 1000.

INDICATIONS

- Treatment of iron deficiency anemia in adults who are intolerant or unresponsive to oral therapy.

ADMINISTRATION

- IV direct: physician or RN; undiluted or diluted with 20 mL of NS; administer at a rate of 250 mg/min; maximum single dose of 500 mg.
- Intermittent IV infusion: dilute dose in NS (using a maximum volume of 500 mL) to obtain a concentration of at least 1 mg/mL (not including the volume of iron). Administer doses of up to 1000 mg over at least 20 minutes and doses exceding 1000 mg over at least 30 minutes; maximum single dose for infusion of 1500 mg. At TOH: dilute the 500 mg dose in 100 mL of NS and infuse over 30 minutes; dilute the 1000 mg and 1500 mg doses in 100 mL of NS and infuse over 60 minutes. For each dose administered, always run at 25 mL/hr for the first 5 minutes to assess patient's tolerance.
- Observe patient for signs and symptoms of hypersensitivity and hypotension during and for at least 30 minutes after administration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, rash, bronchospasm, cardiorespiratory arrest, syncope, angioedema, anaphylactoid/anaphylactic reactions.
- Cardiovascular: hypotension, edema.
- GI: nausea, vomiting, constipation.
- CNS: headache.
- Electrolyte disturbances: hypophosphatemia.
- Nasopharyngitis.
- Local reactions: erythema; if extravasated: irritation, pain, long standing brown discolouration.

DOSAGE

- The total cumulative dose should first be determined, using either the Ganzoni equation or the Simplified table:
  - Ganzoni formula (usually used for patients with chronic kidney disease):
    Iron need [mg] = body weight [kg] x (Target Hb - Actual Hb) [g/dl] x 2.4 + iron stores [mg]
    body weight: use ideal body weight for obese patients; *Target Hb:* default is 15 g/dL but a lower value may be used based on clinical judgment; *iron stores* may vary from 500 to 1000 mg or use 10-15 mg/kg.

  - Simplified table (usually used for patients with iron deficiency from causes other than renal):

Hemoglobin [g/dL]	Dose for weight less than 70 kg	Dose for weight of 70 kg and over
10 and greater	1000 mg	1500 mg
Less than 10	1500 mg	2000 mg

- The total cumulative dose can be given as a single IV direct injection not exceeding 500 mg OR as a single IV infusion not exceeding 20 mg/kg or 1500 mg. If the total cumulative dose exceeds these limits, divide in 2 doses by giving the maximum allowable dose in the first administration, if feasible; administer the 2 doses at least one week apart.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Do not freeze.
- Opened vials should be used immediately.
- Once diluted in NS, iron solution should be administered immediately, as recommended by the manufacturer.
- Stable for 48 hours at 25°C and 30°C in NS at concentrations of 1 mg/mL and 4 mg/mL in non-PVC non-DEHP containers.
- Compatible only with NS.

MISCELLANEOUS

- Emergency drugs and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- Large doses may give a brown colour to serum when blood sample is drawn 4 hours after administration.
- Injectable solution contains 100 mg/mL of elemental iron.

REFERENCES

5, 415.

New monograph 2019; limited revision 2020, 2021

NAME OF MEDICATION
IRON SUCROSE

OTHER NAMES
Iron Saccharate, Venofer ®

CLASSIFICATION
Hematinic

Do NOT confuse iron sucrose with iron isomaltoside 1000.
This monograph is specific to iron SUCROSE.

INDICATIONS
- Treatment of iron deficiency anemia associated with chronic kidney disease (CKD).

ADMINISTRATION
- IV direct: physician or RN; undiluted solutions may be administered direct by slow IV injection over 2 to 5 minutes, not to exceed 200 mg (10 mL) per injection.
- Intermittent IV Infusion: 100 mg (5 mL) and 200 mg (10 mL) to be diluted in a maximum of 100 mL of NS only. Infuse over a minimum of 15 minutes. Doses of 300 to 500 mg should be diluted in a maximum of 250 mL of NS and infused over 1.5 hours for a 300 mg dose, 2.5 hours for a 400 mg dose and 3.5-4 hours for a 500 mg dose (TOH note: diluting to a final concentration of 1 to 2 mg/mL is adequate for IV infusion).
- Observe patient for signs and symptoms of hypersensitivity during and for at least 30 minutes after administration.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, pruritus, wheezing, dyspnea, hypotension, anaphylaxis.
- Cardiovascular: hypotension (occurs with rapid administration; infusions tend to lessen the risk), peripheral edema, chest pain.
- GI: nausea, vomiting, diarrhea, metallic taste, taste perversion.
- CNS: headache, dizziness.
- Muscle cramps, arthralgia, back pain, pain in extremity.
- Local reactions: phlebitis, burning, pain.

DOSAGE
For iron deficiency anemia in CKD:
- Hemodialysis patients: 100 mg IV during consecutive dialysis sessions to a total cumulative dose of 1000 mg (10 doses); may repeat treatment if needed.
- Peritoneal dialysis patients: two IV infusions of 300 mg 14 days apart followed by a single 400 mg IV infusion 14 days later (total cumulative dose of 1000 mg in 3 divided doses); may repeat treatment if needed.
- Non-dialysis patients: 200 mg IV on 5 different occasions within a 14-day period (total cumulative dose of 1000 mg in a 14-day period); may repeat treatment if needed. Limited experience with two 500 mg doses infused on day 1 and 14.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C. Do not freeze.
- Mix only with NS; mixing with other additives is not advised.
- Stable for 7 days at room temp or in the fridge undiluted (20 mg/mL) or diluted in NS at concentrations of 2-10 mg/mL and stored in plastic syringes.
- Stable for 7 days at room temp diluted in NS at concentrations 1-2 mg/mL in a bag.

MISCELLANEOUS
- Emergency drugs and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- As transferrin saturation values increase rapidly after IV administration of iron sucrose, serum iron values may only be reliably obtained 48 hours after the last IV dose.
- Vial contains 100 mg/5 mL of elemental iron.

REFERENCES
1, 5, 40, 135.

Full revision 2015; limited revision 2019, 2020, 2021

NAME OF MEDICATION
ISATUXIMAB

OTHER NAMES
Sarclisa ™

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of relapsed and refractory multiple myeloma (in combination with pomalidomide and dexamethasone).

ADMINISTRATION

- **Ensure premedication has been administered; refer to Dosage section.**
- Intermittent IV infusion (mandatory): withdraw dose to be administered from the vial and dilute in 250 mL of NS or D5W bag made of PVC with DEHP, polyolefin, polyethylene, polypropylene, or ethyl vinyl acetate. Gently mix; do NOT shake. Administer with an IV infusion set made of polyethylene, PVC with or without DEHP, polybudadiene, or polyurethane and using a 0.2 to 0.22 micron in-line filter made of polyethersulfone, polysulfone, or nylon, as per the following rates:
  - First infusion: initial rate of 25 mL/hr for 60 minutes. If no infusion reaction, increase rate in increments of 25 mL/hr every 30 minutes to a maximum rate of 150 mL/hr.
  - Second infusion: initial rate of 50 mL/hr for 30 minutes. If no infusion reaction, increase rate to 100 mL/hr for 30 minutes and then to 200 mL/hr for remainder of infusion.
  - Subsequent infusions: infuse at 200 mL/hr.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication**: consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, angioedema, anaphylaxis.
- Infusion-related reactions: dyspnea, cough, nasal congestion, chills, nausea, hypotension/hypertension, hypoxia, pulmonary edema, tachycardia, syncope, bronchospasm, cytokine release syndrome, face edema, and hyperglycemia; usually occur within 24 hours of the start of the first infusion. For mild reactions, continue infusion at physician's discretion with close monitoring. For moderate reactions, interrupt infusion and administer additional doses of diphenhydramine 25 mg IV (or equivalent) and/or methylprednisolone 100 mg IV (or equivalent) as needed; after patient recovery, resume infusion at reduced rate with close monitoring. For severe reactions, stop infusion and administer additional doses of diphenhydramine 25 mg IV (or equivalent) and/or methylprednisolone 100 mg IV (or equivalent) and/or epinephrine as needed; discontinue treatment.
- Cardiovascular: arrhythmias, peripheral edema.
- GI: nausea, vomiting, diarrhea, anorexia, stomatitis.
- CNS: headache, tremor, dizziness.
- Hematologic: neutropenia, febrile neutropenia, anemia, lymphopenia, thrombocytopenia.
- Musculoskeletal: chest pain (musculoskeletal in nature), bone pain, myalgia, muscle weakness.
- Infection.

DOSAGE

- **Premedicate with dexamethasone 40 mg PO/IV (reduce to 20 mg PO/IV in patients aged 75 years and older), acetaminophen 650 mg to 1000 mg PO (or equivalent), H_2 antagonist or oral proton pump inhibitor, and diphenhydramine 25-50 mg PO/IV (or equivalent; the IV route for diphenhydramine is preferred for at least the first 4 infusions) at least 15 to 60 minutes prior to each infusion to prevent infusion-related reactions. For patients who do not experience an infusion-related reaction with the first 4 infusions, premedication for subsequent infusions may be reevaluated. The dexamethasone is used as both premedication and anti-myeloma treatment on the day of isatuximab infusion; do not administer a second dose.**
- Cycle 1: 10 mg/kg IV on days 1, 8, 15, and 22 (28-day cycle).
- Cycles 2 and beyond: 10 mg/kg IV on days 1 and 15 (28-day cycles).
- Dosage in renal impairment: no dosage adjustment required.

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…/Cont.

DOSAGE (Cont.)

- Dosage in hepatic impairment: no dosage adjustment required for mild impairment; limited data for moderate impairment; no data for severe impairment.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Diluted solution is stable for 48 hours between 2-8°C followed by 8 hours at room temp (includes infusion time) in NS or D5W in PVC with DEHP, polyolefin, polyethylene, polypropylene, or ethyl vinyl acetate containers.

MISCELLANEOUS

- Emergency drugs (e.g., epinephrine, corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity and infusion-related reactions.
- Isatuximab interferes with cross-matching and red blood cell antibody screening; if possible, type and screen patient's blood before initiating therapy.

REFERENCES

5, 129, 135, 165, 208, 573.

New monograph 2021

NAME OF MEDICATION
ISAVUCONAZOLE

OTHER NAMES
Cresemba ™, Isavuconazonium

CLASSIFICATION
Antifungal

INDICATIONS

- Treatment of invasive aspergillosis.
- Treatment of invasive mucormycosis.

ADMINISTRATION

- Reconstitute each 200 mg vial with 5 mL of SWFI to obtain a concentration of 40 mg/mL; shake vial gently to completely dissolve powder.
- Intermittent IV infusion (mandatory): dilute entire contents of vial in 250 mL of NS or D5W to obtain a final concentration of approximately 0.8 mg/mL. Fine white-to-translucent particles that do not sediment may appear after dilution; these particles will be removed by in-line filtration. Mix gently or roll infusion bag to minimize particle formation; do NOT vigorously shake and avoid unnecessary vibration. Infuse over at least 60 minutes using a 0.2-1.2 micron in-line filter of polyethersulfone. Flush line with NS or D5W before and after each infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, rash, pruritus, Stevens-Johnson syndrome, hypotension, respiratory failure, dyspnea.
- Infusion-related reactions: chills, hypotension, dizziness, paresthesias, hypoesthesia, headache, nausea, dyspnea.
- Cardiovascular: peripheral edema, chest pain, hypotension, arrhythmias.
- GI: nausea, vomiting, diarrhea, constipation, abdominal pain.
- CNS: headache, anxiety, delirium, insomnia.
- Electrolyte disturbances: hypokalemia, hypomagnesemia.
- Hepatic: elevated LFTs, increased bilirubin.
- Musculoskeletal: back pain.
- Renal: renal failure.
- Respiratory: cough, dyspnea, respiratory failure.
- Fatigue.
- Local reactions: injection site reactions.

DOSAGE

- Loading dose: 200 mg IV q8h <u>for 6 doses</u>.
- Maintenance dose: 200 mg IV once daily, to start 12-24 hours after administration of the last loading dose.
- Dosage in renal impairment: no dosage adjustment required.
- Dosage in hepatic impairment: no dosage adjustment required for mild to moderate impairment; use with caution in severe impairment as it has not been studied.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Reconstituted solution is stable at temperatures below 25°C for up to 60 minutes prior to dilution.
- Diluted solutions are stable for 6 hours at room temp and 24 hours in the fridge in NS and D5W.

…/Cont.

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MISCELLANEOUS

- 200 mg of isavuconazole is equivalent to 372.6 mg of isavuconazonium sulfate.
- Isavuconazonium sulfate is the prodrug of isavuconazole.
- When switching to the oral route, use the same dose as the IV dose.
- Monitor liver function tests prior to starting therapy and during therapy as required.
- Contraindicated with concomitant use of strong CYP3A4 inhibitors (e.g., ketoconazole, high-dose ritonavir), strong CYP3A4 inducers (e.g., rifampin, carbamazepine, phenobarbital), moderate CYP3A4/5 inducers (e.g., efavirenz, etravirine) or in patients with familial short QT syndrome.

REFERENCES

4, 5, 40, 135.

New monograph 2020

NAME OF MEDICATION
ISONIAZID *

OTHER NAMES
Nydrazid ®

CLASSIFICATION
Antitubercular

INDICATIONS
- Treatment of active tuberculosis when oral administration is not possible; to use in conjunction with other antitubercular drugs.
- Treatment of latent tuberculosis infection in patients at risk when oral administration is not possible; to use as the sole antitubercular drug.

ADMINISTRATION
- IM.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, fever, lymphadenopathy, vasculitis.
- GI: nausea, vomiting, epigastric distress, dry mouth.
- CNS: paresthesia of the feet and hands which may progress to peripheral neuropathy (dose-related).
- Hepatic: elevated serum transaminases, elevated bilirubin, jaundice, hepatitis; common prodromal symptoms of hepatitis: anorexia, nausea, vomiting, fatigue, malaise, weakness, fever lasting over 3 days.
- Hematologic: agranulocytosis, anemia (hemolytic, sideroblastic, or aplastic), thrombocytopenia, eosinophilia, methemoglobinemia.
- Endocrine and metabolic: gynecomastia, pyridoxine deficiency, pellagra, hyperglycemia, metabolic acidosis.
- Rheumatic syndrome and lupus-like syndrome.
- Local reactions: irritation at site of injection.

DOSAGE
- Adults and children 15 years of age and older:
  - For active tuberculosis: 5 mg/kg up to 300 mg IM once daily; if intermittent therapy: 15 mg/kg up to 900 mg IM 2 to 3 times a week.
  - For latent tuberculosis infection: 5 mg/kg up to 300 mg IM once daily; if intermittent therapy: 15 mg/kg up to 900 mg IM twice a week.

- Pediatrics (children 14 years of age and younger):
  - For active tuberculosis: 10-15 mg/kg up to 300 mg IM once daily; if intermittent therapy: 20-30 mg/kg up to 900 mg IM twice a week.
  - For latent tuberculosis infection: 10-20 mg/kg up to 300 mg IM once daily; if intermittent therapy: 20-40 mg/kg up to 900 mg IM twice a week.

- Dosage in renal impairment: no dosage adjustment required. Use with caution in severe renal impairment.
- Dosage in hepatic impairment: no dosage adjustment but caution is required; consider discontinuation if liver transaminases are over 3 to 5 x ULN.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp and protect from light. Do not freeze.
- Crystals may form when isoniazid is exposed to low temperatures; warming the vial to room temp will redissolve the crystals.

MISCELLANEOUS
- Pyridoxine oral administration (10-50 mg daily for adults, 1-2 mg/kg daily for children) is recommended in patients who are malnourished or prone to neuropathy.
- Monitor liver function periodically.

REFERENCES
1, 5, 95, 135, 216.

* Available via Health Canada's Special Access Programme

Full revision 2015; limited revision 2016

NAME OF MEDICATION
ISOPROTERENOL

OTHER NAMES
Isoprenaline, Isopropylarterenol, Isopropylnoradrenaline, Isuprel ®

CLASSIFICATION
Beta agonist, Antiarrhythmic

INDICATIONS
- Atrioventricular (AV) heart block, cardiac arrest, atropine-resistant bradycardia, Adams-Stokes syndrome.
- As an adjunct in the treatment of shock.
- Treatment of bronchospasm occurring during anesthesia.
- Aid in diagnosing the etiology of mitral regurgitation and in the diagnosis of coronary artery disease.

ADMINISTRATION
- IV direct (emergency situations only): physician or RN; ***cardiac monitoring***; dilute 0.2 mg (1 mL) with NS or D5W to obtain a total volume of 10 mL at a concentration of 0.02 mg/mL; inject at a rate of 1 mL/min. Follow with a 20 mL flush with NS to ensure distribution to circulation.
- Continuous IV Infusion (preferred): ***cardiac monitoring***. Dilute 1 mg in 250 or 500 mL of NS or D5W to obtain a final concentration of 0.004 or 0.002 mg/mL (4 or 2 mcg/mL), respectively. At TOH, dilute 1 mg in 250 mL of NS or D5W to obtain a final concentration of 4 mcg/mL. Infusion rate varies according to indication. Physician should be present to establish rate. Must be administered by an infusion pump.
- SC, IM.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: tachycardia, palpitation, arrhythmias, coronary insufficiency, facial flushing.
- GI: nausea and vomiting.
- CNS: nervousness, restlessness, dizziness, headache.
- Weakness, tremor.
- Sweating.

DOSAGE
- Emergency treatment of cardiac arrhythmias: 20-60 mcg IV direct i.e., 1-3 mL of a 0.02 mg/mL diluted solution. May administer subsequent IV doses of 10 to 200 mcg (0.5-10 mL of a 0.02 mg/mL diluted solution) or use an IV infusion given at an initial rate of 5 mcg/min.
- Treatment of cardiac arrhythmias in less urgent situations: 200 mcg (1 mL of the undiluted 0.2 mg/mL solution) SC or IM initially, followed by 150-200 mcg SC or 20-1000 mcg IM depending on clinical response.
- Shock: use an IV infusion given at a rate of 0.5-5 mcg/min. Rates greater than 30 mcg/min have been used in advanced stages of shock.
- Bronchospasm during anesthesia: 10-20 mcg IV (0.5-1 mL of a 0.02 mg/mL diluted solution), repeated when necessary.
- Diagnostic aid: mitral regurgitation: 4 mcg/min IV infusion.
  coronary artery disease: 1-3 mcg/min IV infusion.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Solution should be colourless.
- Exposure to air or light may turn the solution to a pink to brownish-pink discolouration. Discard if discoloured or contains a precipitate.
- Stable for 24 hours in the fridge in D5W, NS, D5-NS, RL and D5-RL at a concentration of 2 mcg/mL in glass and PVC containers.
- Incompatible with sodium bicarbonate.

MISCELLANEOUS

REFERENCES
1, 4, 5, 40, 95, 135.

Full revision 2015; limited revision 2016, 2017, 2019, 2020

NAME OF MEDICATION
ITRACONAZOLE *

OTHER NAMES
Sporanox ®

CLASSIFICATION
Antifungal

INDICATIONS

- Treatment of blastomycosis, histoplasmosis, and aspergillosis in seriously ill patients.
- Empiric therapy in febrile neutropenic patients with suspected fungal infections.

ADMINISTRATION

- Intermittent IV Infusion: dilute 250 mg (25 mL) in 50 mL of NS (special bag provided by the manufacturer) for a total of 75 mL. Final concentration of 3.33 mg/mL. Either discard 15 mL or administer only 60 mL (200 mg) as a 60 minute infusion (1 mL/min). Use the filter set provided by the manufacturer. After administration, flush the infusion set with 15-20 mL of NS over 30 seconds to 15 minutes using the two-way stopcock.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, anaphylaxis (rare), angioedema, Stevens-Johnson Syndrome.
- Cardiovascular: transient asymptomatic decreased ejection fraction, CHF, pulmonary edema, peripheral edema.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: headache, dizziness.
- Fever, fatigue.
- Hepatotoxicity and abnormal LFTs.
- Local reactions: infusion site reactions.

DOSAGE

- Loading dose: 200 mg IV twice daily on days 1 and 2 (4 doses).
- Maintenance dose: 200 mg IV once daily for the next 5 to 12 days.
- May be followed by oral treatment.
- Dosage in renal impairment: no dosage adjustment if CrCl is 30 mL/min or greater; do not use if CrCl is below 30 mL/min as one of the excipients, hydroxypropyl-B-cyclodextrin, is eliminated through glomerular filtration.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility).

- Store at room temp; protect from light and freezing.
- Stable for 48 hours at room temp when protected from light in NS at a concentration of 3.33 mg/mL in the special bag provided by the manufacturer.
- Compatible only with NS and stable only at a concentration of 3.33 mg/mL; other concentrations will lead to precipitation.
- Should not be co-administered with other fluids or drugs.

MISCELLANEOUS

- Quinidine, pimozide, CYP3A4-metabolized HMG-CoA reductase inhibitors (e.g., simvastatin, lovastatin, atorvastatin), oral midazolam, triazolam and ergot alkaloids (e.g., dihydroergotamine) are contraindicated with itraconazole IV.
- High potential for drug interactions through the cytochrome P4503A4.
- No information regarding cross-hypersensitivity with other azole antifungal agents; caution should be used.

REFERENCES

1, 5, 40, 95, 208.

* Available via Health Canada's Special Access Programme

Full revision 2015

NAME OF MEDICATION
KADCYLA ® TRASTUZUMAB EMTANSINE

OTHER NAMES
Ado-trastuzumab emtansine, Kadcyla ®, Trastuzumab emtansine, Trastuzumab-MCC-DM1

CLASSIFICATION
Antineoplastic, Monoclonal antibody

Do NOT confuse Kadcyla ® trastuzumab emtansine with Enhertu ™ trastuzumab deruxtecan or trastuzumab (Herceptin ®, Herzuma ®, Kanjinti ™, Ogivri ™ and Trazimera ®). Kadcyla ® trastuzumab emtansine and Enhertu ™ trastuzumab deruxtecan both contain a cytotoxic component, unlike trastuzumab. These products have different uses, toxicities, administration and dosages. This monograph is specific to KADCYLA ® TRASTUZUMAB EMTANSINE.

INDICATIONS

- Monotherapy for patients with HER2-positive, metastatic breast cancer who received both prior treatment with trastuzumab and a taxane, separately or in combination.
- Monotherapy for the adjuvant treatment of HER2-positive early breast cancer in patients who have residual invasive disease following neoadjuvant taxane and trastuzumab-based treatment.

ADMINISTRATION

- Reconstitute each 100 mg or 160 mg vial by slowly injecting 5 mL or 8 mL, respectively, of SWFI to obtain a concentration of 20 mg/mL. Gently swirl until complete dissolution. Do NOT shake.
- Intermittent IV infusion (mandatory): dilute required amount into 250 mL of 1/2NS or NS. Gently invert bag to mix; do NOT shake. When mixed in NS, must use an in-line filter (0.2 micron non-protein adsorptive or 0.22 micron polyethersulfone (PES) filter) during the infusion; when mixed in 1/2NS, filtering is not necessary. Administer initial dose over 90 minutes. If well tolerated, may administer subsequent doses over 30 minutes. Observe patient closely during infusion, for at least 90 minutes after the first infusion and at least 30 minutes after subsequent infusions.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactoid reaction, rash, pruritus.
- Infusion-related reactions: flushing, fever, chills, dyspnea, wheezing, bronchospasm, hypotension and tachycardia. Patient should be observed for infusion-related reactions, especially during the first infusion. Symptoms are usually mild to moderate and resolve within 24 hours of infusion. If needed, decrease infusion rate or interrupt infusion depending on severity of the reactions. Supportive therapy such as beta-agonists, antihistamines, antipyretics and corticosteroids may be required.
- Cardiovascular: reduced left ventricular ejection fraction, hypertension.
- GI: nausea, vomiting, constipation, diarrhea, abdominal pain, dry mouth, stomatitis, dyspepsia, dysgeusia.
- CNS: peripheral neuropathy (mostly sensory), paresthesia, headache, dizziness, insomnia.
- Hematologic: thrombocytopenia (more frequent in the Asian population), hemorrhage, epistaxis, anemia, neutropenia, leukopenia.
- Hepatic: increased LFTs, liver failure, nodular regenerative hyperplasia, portal hypertension, hyperbilirubinemia.
- Ophthalmic: blurred vision, conjunctivitis.
- Respiratory: cough, dyspnea, interstitial lung disease, pneumonitis, acute respiratory distress syndrome.
- Fatigue, peripheral edema, urinary tract infection, hypokalemia, musculoskeletal pain, arthralgia, myalgia.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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NAME OF MEDICATION
KADCYLA ® TRASTUZUMAB EMTANSINE

OTHER NAMES
Ado-trastuzumab emtansine, Kadcyla ®, Trastuzumab emtansine, Trastuzumab-MCC-DM1

CLASSIFICATION
Antineoplastic, Monoclonal antibody

…/Cont.

DOSAGE

- 3.6 mg/kg IV every 3 weeks. Dose reductions or treatment interruption may be necessary if not well tolerated. Doses should not be re-escalated after a dose reduction. Consult manufacturer's dosing instructions for further details.
- Dosage in renal impairment: no dosage adjustment required in case of mild or moderate renal impairment (CrCl of 30 mL/min or greater).
- Dosage in hepatic impairment: consult manufacturer's monograph.
- Dosage adjustment due to toxicity: in case of cardiotoxicity, refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Do not freeze.
- Reconstituted solution is stable for 24 hours in the fridge. It should be clear to slightly opalescent, colourless to pale brown.
- Diluted solutions are stable for 24 hours in the fridge in NS or 1/2NS in PVC or polyolefin bags.
- Incompatible with D5W.

MISCELLANEOUS

- Baseline cardiac assessment should be done for all patients and repeated every 3 months during treatment or more often if clinically indicated.
- LFTs should be performed prior to each dose.

REFERENCES

1, 5, 40, 129, 135, 165, 333, 334.

Full revision 2016; limited revision 2017, 2018, 2019, 2020, 2021

NAME OF MEDICATION
KETAMINE

OTHER NAMES
Ketalar ®

CLASSIFICATION
General Anesthetic, Analgesic

INDICATIONS

- As monotherapy for induction and maintenance of anesthesia during diagnostic and surgical procedures.
- For the induction of anesthesia prior to the administration of other general anesthetic agents or as a supplement to low potency anesthetic agents such as nitrous oxide.
- For pain management in patients refractory to standard treatments.
- For procedural analgesia/sedation.

ADMINISTRATION

- IV direct (for doses of 10 mg or less): physician or RN; ***respiratory support***; inject first dose over 1 minute or at a rate of 0.5 mg/kg/min (35 mg/min for a 70 kg patient).
- IV direct (for doses greater than 10 mg): physicians trained in anesthesiology only for first dose. RN may give subsequent doses for patients on ventilator support; ***respiratory support***; undiluted; inject over 1 minute or at a rate of 0.5 mg/kg/min (35 mg/min for a 70 kg patient).
- Continuous IV infusion: ***respiratory support***; undiluted (10 mg/mL formulation) or diluted to 1-2 mg/mL in NS or D5W. At TOH: can also dilute to 4-5 mg/mL. Must be administered by an infusion pump.
- Continuous SC infusion: ***respiratory support.*** Dilute with NS to reduce discomfort. Must be administered by an infusion pump.
- IM, SC.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis, erythema, morbilliform rash.
- Cardiovascular: hypertension, tachycardia; infrequently arrhythmias, hypotension, bradycardia.
- GI: increased salivation.
- CNS: postanesthesia: hallucinations, vivid dreams, confusion, excitement and delirium; may last up to 24 hours; more frequent in patients 15-65 years old and with IV administration. Prophylaxis and treatment with lorazepam, midazolam or haloperidol can help alleviate such symptoms. Sedation, increased intracranial pressure; may exacerbate psychotic symptoms.
- Hepatic: increase in liver enzymes, biliary ductal dilatations, hepatic fibrosis.
- Respiratory: respiratory depression with rapid administration of high doses.
- Hypertonia, tonic and clonic movements.
- Renal: may cause severe and irreversible urinary tract toxicity with chronic administration.
- Ophthalmic: blurred vision, diplopia, nystagmus.
- Local reactions: pain and exanthema at injection site.

DOSAGE

Anesthesia:
- IV: induction 1-2 mg/kg (range 0.5-4.5 mg/kg).
  Maintenance: 50-100% induction dose prn or IV infusion of 0.1-0.5 mg/min. A continuous maintenance infusion can also be used with a range from 0.6 to 1.8 mg/kg/hr (42-126 mg/hr for a 70 kg patient).
- IM: induction 5-10 mg/kg (range 4-13 mg/kg).
  Maintenance: 50-100% induction dose prn.

Analgesia/Sedation:
- IV: usually 2.5-5 mg prn for cancer pain; 5-50 mg for procedural pain/sedation.
  Can be given as an initial dose of 1 mg/kg (70 mg for a 70 kg patient) IV with incremental doses of 0.5 mg/kg (35 mg for a 70 kg patient) every 5 to 15 minutes prn or followed by a continuous infusion of 0.05-1.2 mg/kg/hr (3.5-84 mg/hr for a 70 kg patient); higher initial infusion rates (0.24-0.5 mg/kg/hr or 17-35 mg/hr for a 70 kg patient) have also been used. Titrate infusion rate as needed.

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DOSAGE (Cont.)

- IV: as an adjunct to opioid analgesia: 0.1-0.8 mg/kg (7-56 mg for a 70 kg patient) IV as an initial dose; may follow with a continuous infusion of 0.05 mg/kg/hr (3.5 mg/hr for a 70 kg patient), up to 0.4 mg/kg/hr (28 mg/hr for a 70 kg patient).
- SC: usual dose 10-25 mg prn (range 2.5-25 mg) OR 0.5-1 mg/kg prn (range 0.125-2 mg/kg); increase in steps of 25-33%.
- Continuous SC infusion: 0.04-0.1 mg/kg/hr (2.8-7 mg/hr for a 70 kg patient). Daily dose can be increased by 0.1 mg/kg/hr or by 2-4 mg/hr q24h up to an usual maximum dose of 500 mg/24 hours; up to 3600 mg/24 hours have been used.
- IM: 0.5-5 mg/kg.
- Dosage in renal impairment: no dosage adjustment appears to be necessary.
- Dosage in hepatic impairment: prolonged duration of effect may warrant dose reduction.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and ampoules at room temp. Protect from light.
- Undiluted solution is clear, colourless to slightly yellow. Exposure to light darkens the solution though does not cause product degradation. Do not use in case of precipitation.
- Incompatible with barbiturates and diazepam.
- Compatible with NS and D5W.
- Stable for 182 days at room temp exposed to light in SWFI at a concentration of 10 mg/mL stored in glass vials.
- Stable for 12 months between 23-27°C, 38-42°C and refrigerated in NS at a concentration of 1 mg/mL in polypropylene syringes.

MISCELLANEOUS

REFERENCES

5, 85, 95, 135, 208, 256, 310, 311, 457.

Full revision 2016; limited revision 2017, 2020, 2021

NAME OF MEDICATION
KETOROLAC

OTHER NAMES
Ketorolac trometamol, Ketorolac tromethamine, Toradol ®

CLASSIFICATION
Nonsteroidal anti-inflammatory

INDICATIONS

- Short-term treatment of moderate to severe acute pain.
- Treatment of cancer pain in palliative care patients.

ADMINISTRATION

- IV direct: physician or RN; administer undiluted over at least 15 seconds, but preferably over 1-2 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of compatible solution and infuse over 15-30 minutes.
- Although ketorolac is not approved by Health Canada for IV administration, the IM product may be given IV as there is strong documentation in the literature supporting its efficacy and safety by this route.
- SC (palliative care): intermittent injection, continuous infusion (must be administered by an infusion pump).
- IM (deep).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactic/anaphylactoid reactions, angioedema, Stevens-Johnson syndrome, rash, toxic epidermal necrolysis, exfoliative dermatitis, erythema multiforme, bronchospasm.
- Cardiovascular: fluid retention/edema leading to hypertension and CHF, vasodilation.
- GI: nausea, vomiting, abdominal pain, diarrhea, dyspepsia, GI ulceration and bleeding.
- CNS: somnolence, headache, dizziness.
- Hepatic: increased LFTs.
- Sweating.
- Local reactions: pain at IM and intermittent SC injection site.

DOSAGE

- 10-30 mg IV/IM q4-6h prn, maximum 120 mg/day OR single dose of 60 mg IM or 30 mg IV.
- Geriatric patients, patients weighing less than 50 kg, or with renal impairment: 15 mg IV/IM q6h, maximum of 60 mg/day IV/IM OR single dose of 30 mg IM or 15 mg IV.
- Various studies indicate that single doses of 10 mg are equally effective as higher doses for acute pain.
- Parenteral therapy should not continue beyond 2 days, except in palliative care patients.
- Palliative care: 15-30 mg SC TID OR 30-60 mg/24 hours as a continuous SC infusion, up to 120 mg/24 hours.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and vials at room temp. Protected from light.
- Compatible with D5W, NS, D5-NS, Ringer's, RL.
- Stable for 21 days at room temp or under refrigeration in NS or D5W at concentrations of 0.3 mg/mL and 0.6 mg/mL in PVC plastic bags.
- Stable for 3 months when frozen in D5W at a concentration of 0.2 mg/mL in polyolefin bags; after subsequent microwave thawing, solution is still stable in the fridge for another 60 days, protected from light.

MISCELLANEOUS

- Do not administer to patients with ASA intolerance (i.e., nasal polyps/rhinosinusitis, urticaria/angioedema or asthma after taking ASA or other NSAIDs) as cross-hypersensitivity has been reported.
- Do not administer for perioperative pain in patients undergoing coronary artery bypass graft surgery.

REFERENCES

1, 4, 5, 6, 40, 95, 143, 256, 313, 353, 413.

Full revision 2016; limited revision 2017, 2019, 2020

NAME OF MEDICATION
LABETALOL

OTHER NAMES
Trandate ®

CLASSIFICATION
Antihypertensive

INDICATIONS

- For the emergency treatment of severe hypertension when prompt and urgent reduction of blood pressure is essential.

ADMINISTRATION

- IV direct: physician or RN; ***cardiac monitoring; continuous BP monitoring***. Administer undiluted at a rate not exceeding 20 mg over 2 minutes.
- Continuous IV infusion: ***cardiac monitoring; blood pressure monitoring***. Dilute 200 mg (40 mL) in 250 mL of a compatible solution for an approximate concentration of 2 mg/3 mL. At TOH: withdraw 10 mL from a 250 mL bag of D5W or NS and discard, then add 300 mg (60 mL from labetalol 5 mg/mL vials) in the bag for an approximate concentration of 1 mg/mL OR, for fluid-restricted patients, use undiluted labetalol (transfer contents of vials to an empty bag to obtain 1500 mg/300 mL or 600 mg/120 mL). Infuse as per Dosage section. Must be administered by an infusion pump.
- Patient should be in supine position.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; anaphylactoid reactions, rash, pruritus, urticaria, dyspnea, angioedema.
- Cardiovascular: ventricular arrhythmias, bradycardia; orthostatic hypotension, likely to occur if supine patients are tilted upward or allowed to assume an upright position within 3 hours of IV administration.
- GI: nausea, vomiting, dyspepsia.
- CNS: somnolence, dizziness, headache.
- Renal: elevated urea and creatinine.
- Tingling of skin/scalp.
- Fatigue, malaise.
- Blurred vision.

DOSAGE

- IV direct: Initial dose 20 mg, may repeat 20-80 mg at 10 minute intervals until desired supine BP or total of 300 mg has been given.
- Continuous IV infusion: initiate rate at 0.5-2 mg/min, adjust rate according to blood pressure; up to 8 mg/min has been used. The usual effective dose is 50-200 mg. Maximum cumulative dose of 300 mg; however, occasional clinical situation may require prolonged infusions and cumulative doses greater than 300 mg.
- Dosage in hepatic impairment: patients have required, on average, dose reductions of 50%.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Stable for 24 hours at room temp or in the fridge in NS at concentrations of 1.25 and 3.75 mg/mL.
- Stable for 72 hours at room temp or in the fridge in D5W, RL, Ringer's, D5-Ringer's, D5-RL, dextrose-saline solutions at concentrations of 1.25, 2.5 and 3.75 mg/mL.
- Incompatible with 5% sodium bicarbonate.

MISCELLANEOUS

- Maximum hypotensive effect occurs in 5-15 minutes.
- Concomitant use of IV labetalol with IV diltiazem or IV verapamil may cause significant cardiac depression. Therefore, co-administration (within a few hours) is contraindicated.

REFERENCES

1, 4, 5, 6, 40, 95, 135, 208.

Full revision 2016; limited revision 2017, 2019, 2020

NAME OF MEDICATION
LACOSAMIDE

OTHER NAMES
Vimpat ®

CLASSIFICATION
Anticonvulsant

INDICATIONS

- Management of partial-onset seizures in adult patients.

ADMINISTRATION

- Intermittent IV infusion: administer undiluted or diluted in NS, D5W or RL over 30-60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: multiorgan hypersensitivity: triad of rash, fever and other organ involvement (e.g., eosinophilia, hepatitis, nephritis, lymphadenopathy and/or myocarditis).
- Cardiovascular: PR interval prolongation, AV block, atrial flutter or fibrillation in patients with diabetic neuropathy and/or cardiovascular disease, ventricular tachyarrhythmia.
- GI: nausea, vomiting, diarrhea.
- CNS: dizziness, ataxia, vertigo, headache, somnolence, tremor, memory impairment, balance disorder.
- Ophthalmic: diplopia, nystagmus, blurred vision.
- Fatigue.
- Local reactions: pain or discomfort, irritation, erythema.

DOSAGE

- Monotherapy: initial dose of 100 mg IV BID; can be increased if needed at weekly intervals by 50 mg BID to a maximum of 300 mg BID (600 mg/day).
- In combination with other anticonvulsants: initial dose of 50 mg IV BID, to be increased to 100 mg BID one week later; can be further increased if needed by 50 mg BID every week to a maximum of 200 mg BID (400 mg/day).
- Dosage in renal impairment: titrate with caution; no dosage adjustment needed if CrCl is greater than 30 mL/min; maximum dose of 300 mg/day if CrCl is 30 mL/min or less, including patients with end-stage renal disease. Hemodialysis will remove approximately 50% of the dose after a 4-hour session.
- Dosage in hepatic impairment: titrate with caution to a maximum dose of 300 mg/day in patients with mild to moderate impairment; not evaluated and therefore not recommended in patients with severe liver impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Stable for 24 hours at room temp when diluted in D5W, NS and RL in glass or PVC containers.

MISCELLANEOUS

- When transferring from IV to oral (and *vice versa*), administer at equivalent dose and frequency.
- IV administration has been used for up to 5 days; switch patients to oral route as soon as practical.
- An ECG before initiating therapy and when therapy is at steady-state is recommended in patients with known conduction problems and/or a history of severe cardiac disease.

REFERENCES

1, 5, 40, 135.

Full revision 2016; limited revision 2017, 2019

NAME OF MEDICATION
LEUCOVORIN CALCIUM

OTHER NAMES
Calcium folinate, Citrovorum factor, Folinic acid

CLASSIFICATION
Folic acid derivative, Antidote

INDICATIONS

- Treatment or prevention of hematologic toxicity associated with folic acid antagonists (such as methotrexate).
- Treatment of megaloblastic anemia due to folate deficiency or congenital dihydrofolate reductase deficiency, when oral administration is not possible.
- As adjuvant therapy to 5-fluorouracil (5-FU) in the palliative treatment of patients with advanced colorectal cancer.

ADMINISTRATION

- IV direct: physician or RN; doses less than 100 mg may be given undiluted; due to the calcium content, inject at a rate no greater than 160 mg/min into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion: dilute in 50 mL (for doses up to 500 mg) or 100-500 mL (for higher doses) of a compatible IV solution; administer over 15-60 minutes, not exceeding 160 mg per minute due to the calcium content. Large doses may be infused over 1-6 hours.
- IM. Absorption is better from deltoid than gluteus. Rotate injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; anaphylactic/anaphylactoid reactions, rash, hives, pruritus, wheezing.
- GI: mucositis, stomatitis, diarrhea, enterocolitis (can be fatal) when combined with 5-FU.
- Leukopenia when combined with 5-FU.

DOSAGE

Treatment of folic acid antagonist/methotrexate overdose:
- Administer leucovorin IM/IV in amounts equal or greater to the suspected dose of the folic acid antagonist, preferably within the first hour, or within 24-36 hours when there is delayed excretion (e.g., presence of ascites, pleural effusion, renal insufficiency or dehydration); OR
- For large overdoses, up to 75 mg as an intermittent IV infusion within 12 hours of overdose, followed by 12 mg IM q6h for 4 doses; for less severe overdoses, 6-12 mg IM q6h for 4 doses; OR
  10 mg/m^2 IV/IM q6h until methotrexate levels are under 0.1 mcmol/L.
- If creatinine has increased by more than 50% from pre-methotrexate administration or if methotrexate levels are still elevated 24 hours after leucovorin administration, increase the leucovorin dose to 100 mg/m^2 IV q3h until methotrexate level is under 0.1 mcmol/L.

Prevention of hematologic toxicity from methotrexate chemotherapy:
- 15 mg or 10-25 mg/m^2 IV/IM, starting 24 hours after methotrexate administration, q6h for 8-10 doses or until serum methotrexate level is under 0.05 mcmol/L.
- Increase the dose to 100 mg/m^2 IV q3h if serum creatinine has increased by more than 50% from pre-methotrexate administration or if methotrexate levels are above 1 mcmol/L, then decrease the dose to 10 mg/m^2 or 15 mg IV q3h until methotrexate level is 0.1 mcmol/L. Consult specific protocol.

Megaloblastic anemia:
- Due to folate deficiency: up to 1 mg/day IM/IV.
- Due to congenital deficiency of dihydrofolate reductase: 3-6 mg IM.

Colorectal cancer:
- 20-200 mg/m^2 IV daily X 5 days every 4-5 weeks, given before 5-FU OR 400 mg/m^2 IV for one dose on day 1 every 2 weeks. Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light.
- Stable for 7 days at room temp or in the fridge, protected from light, in NS at a concentration of 1 mg/mL in glass and polyolefin containers.
- Stable for 4 days at room temp or in the fridge, protected from light, in NS at concentrations of 1-1.5 mg/mL and in D5W at concentrations of 0.1-1.5 mg/mL in glass and PVC containers.
- Stable for 4 days at room temp or in the fridge, protected from light, in NS at concentrations of 0.1-0.5 mg/mL in glass containers.
- Stable for 24 hours at room temp, protected from light, in D10W or D10-NS at a concentration of 0.05 mg/mL.
- Stable for 24 hours at room temp in NS, D5W, RL and Ringer's at a concentration of 0.05 mg/mL.
- Stable for 24 hours at room temp in RL and Ringer's at concentrations of 0.06-1 mg/mL.
- Stable for 12 hours at room temp in D10W at concentrations of 0.06-1 mg/mL.
- Stable for 8 hours at room temp in D5-NS at a concentration of 0.05 mg/mL.
- Stable for 6 hours at room temp in D10-NS at concentrations of 0.06-1 mg/mL.
- Incompatible with 5-FU.

MISCELLANEOUS

- Timing of administration is critical, thus administer exactly as prescribed by physician.
- Oral absorption is saturable, thus doses greater than 25 mg should be given parenterally.
- Hydration (at least 3 litres/day) and alkalinization of the urine (pH greater or equal to 7) are recommended.

REFERENCES

1, 4, 5, 40, 95, 129, 135, 143, 165.

NAME OF MEDICATION
LEUPROLIDE

OTHER NAMES
Eligard ®, Lupron ®, Lupron Depot ®, Zeulide Depot ™

CLASSIFICATION
Gonadotropin - releasing hormone analog

INDICATIONS

- Palliative treatment of advanced hormone-dependent prostate cancer.
- Treatment of breast cancer.
- Treatment of endometriosis, alone or in combination with norethindrone.
- In combination with iron therapy to treat anemia associated with uterine leiomyomata prior to surgery.

ADMINISTRATION

- Lupron Depot ® Zeulide Depot ™ and Eligard ® injections are to be reconstituted with supplied diluent and prepared according to manufacturer's directions.
- SC:
  - Lupron ® 5 mg/mL solution: allow product to reach room temp. Inject in abdomen or anterior thigh region. Rotate injection site.
  - Eligard ® all strengths: allow product to reach room temp. Inject in abdomen, upper buttock or any region with an adequate amount of SC tissue.
- IM:
  - Lupron Depot ® : inject in gluteus, anterior thigh or deltoid. Rotate injection site.
  - Zeulide Depot ™ : allow product to reach room temp. Inject in gluteus. Rotate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis (rare), rash.
- Cardiovascular: hot flashes (from mild to severe; usually decrease with continued therapy), QT interval prolongation, edema.
- GI: nausea, vomiting.
- CNS: headache, seizures, depression, dizziness, paresthesia, insomnia.
- Fatigue, fever, malaise, musculoskeletal pain, asthenia.
- Disease flare (e.g., for prostate cancer: increase in bone pain, urinary tract obstruction, neuropathy, hematuria, spinal cord compression) during the first few weeks of therapy. May need an antiandrogen (cyproterone 100 mg BID, flutamide 250 mg TID, bicalutamide 50 mg daily or nilutamide 150 mg daily) given concurrently for the initial 1-2 weeks of therapy.
- Local reactions: bruising, redness, burning, pain at injection site.

DOSAGE

- Prostate cancer: 1 mg daily; 3.75 mg or 7.5 mg monthly; 22.5 mg every 3 months; 30 mg every 4 months; or 45 mg every 6 months. Refer to each specific product for route of administration.
- Endometriosis: 3.75 mg as a monthly IM depot injection or 11.25 mg every 3 months as an IM depot injection for 6 months. Retreatment can be performed for another 6 months only in combination with norethindrone.
- Uterine leiomyomata: 3.75 mg as a monthly IM depot injection for 3 months, or a single injection IM of 11.25 mg depot formulation.
- Breast cancer: 3.75 mg as a monthly IM depot injection or 11.25 mg every 3 months as an IM depot injection.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Lupron ®: store 5 mg/mL vials in the fridge. Protect from light.
- Lupron Depot ®: store at room temp. Reconstituted solution is stable for 24 hours; however, immediate use is recommended.
- Eligard ®: store in the fridge; the unreconstituted product is stable for 8 weeks at room temp; the reconstituted product is stable for only 30 minutes.
- Zeulide Depot ™: store below 25°C. Protect from light. Do not freeze.

MISCELLANEOUS

- Due to different release characteristics, the different formulations or fractional doses of the same brand of product are not interchangeable.
- Do not use a combination of syringes to achieve a particular dose.

REFERENCES

1, 5, 95, 129, 135, 165.

Full revision 2016; limited revision 2018, 2019, 2020

NAME OF MEDICATION
LevETIRAcetam

OTHER NAMES
Keppra ®

CLASSIFICATION
Anticonvulsant

INDICATIONS

- For the management of status epilepticus.
- As an adjunct therapy in the management of patients with epilepsy when oral administration is not possible.
- In palliative care patients to control seizures.

ADMINISTRATION

- Intermittent IV infusion: dilute dose in 100 mL of D5W, NS or RL; if a smaller volume of diluent is required (e.g., pediatrics), do NOT exceed a concentration of 15 mg/mL. Infuse over 15 minutes.
- SC infusion, Continuous SC infusion: palliative care patients. For SC infusion: dilute dose in 100 mL of NS and infuse over 30 minutes. Must be administered by an infusion pump. For continuous SC infusion: dilute in as much volume of SWFI or NS that is pratical to preserve infusion site/avoid infusion site irritation. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, Stevens-Johnson syndrome, toxic epidermal necrolysis, angioedema, anaphylaxis.
- CNS: behavioral problems (aggression, apathy, anger, agitation, anxiety, depersonalization, irritability, neurosis), headache, somnolence, dizziness, coordination difficulties, depression, increased risk of suicidality.
- Respiratory: nasopharyngitis, pharyngitis.
- Asthenia, fatigue.
- Infection.
- Local reactions: with SC administration: irritation at the infusion site; rotating infusion site and using a more diluted solution decrease irritation.

DOSAGE

Adults and adolescents (12 years of age and older) weighing at least 50 kg:
- Status epilepticus: 1000-3000 mg OR 40-60 mg/kg as a single IV dose; maximum dose: 4500 mg.
- Maintenance dose: 500 mg IV q12h; dose may be increased by 1000 mg/day (500 mg q12h) every 2 weeks to a maximum dose of 3000 mg/day (1500 mg IV q12h).
- Seizures in palliative care patients: initial dose: 500 mg SC per 24 hours. Usual range of 500-3000 mg SC per 24 hours; up to 4000 mg/day SC have been used in some cases.
- Dosage in renal impairment (for maintenance dose):

CrCl (mL/min)	80 and above	79-50	49-30	less than 30	dialysis*
Dose (mg)	500-1500	500-1000	250-750	250-500	500-1000
Interval (hr)	12	12	12	12	24

  * A 750 mg loading dose is recommended on the first day of treatment with levetiracetam. As hemodialysis removes a substantial amount, give an additional dose of 250-500 mg after dialysis.
- Dosage in hepatic impairment: no dosage adjustment is necessary, except in severe impairment where a 50% reduction of the daily maintenance dose is recommended if patient also has a CrCl below 60 mL/min.

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DOSAGE (Cont.)

Pediatrics (children weighing less than 50 kg):
- Maintenance dose:
  - 1 month to less than 6 months of age: 7-21 mg/kg IV twice daily.
  - 6 months to less than 12 years of age: 10-30 mg/kg IV twice daily.
- Dosage in renal impairment (weight under 50 kg):

CrCl (mL/min)	80 and above	79-50	49-30	less than 30	dialysis
Dose (mg/kg)					
• 1 to less than 6 months	7-21	7-14	3.5-10.5	3.5-7	7-14
• 6 months to less than 12 years	10-30	10-20	5-15	5-10	10-20**
Interval (hr)	12	12	12	12	24

** A 15 mg/kg loading dose is recommended on the first day of treatment. A 5-10 mg/kg supplemental dose is recommended after dialysis.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light. Do not freeze.
- Compatible with D5W, NS and RL.
- Compatible with lorazepam, diazepam and valproate sodium.
- Stable for 24 hours at room temp in NS, D5W and RL at concentrations up to 15 mg/mL in PVC and polypropylene containers.

MISCELLANEOUS

- An initial ratio of 1:1 for dosage is used when converting from oral to IV or oral to SC and *vice versa*.
- Switch to oral therapy as soon as possible.

REFERENCES

5, 40, 95, 135, 256, 410, 411, 491, 561, 562, 563, 564.

New monograph 2018; limited revision 2019, 2020, 2021

NAME OF MEDICATION
LevOCARNitine

OTHER NAMES
Carnitor ®, L-carnitine

CLASSIFICATION
Nutritional supplement

INDICATIONS

- Treatment of primary systemic carnitine deficiency.
- Acute and chronic treatment of patients with an inborn error of metabolism that results in a secondary carnitine deficiency.
- Prevention and treatment of carnitine deficiency in patients undergoing hemodialysis.

ADMINISTRATION

- IV direct: physician or RN; undiluted; inject slowly over 2-3 minutes.
- Intermittent IV infusion: dilute in NS or RL to a concentration of 0.5-8 mg/mL.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity.
- Cardiovascular: hypertension, hypotension, chest pain.
- GI: transient nausea and vomiting, diarrhea, abdominal pain, gastritis.
- CNS: seizures, headache, dizziness.
- Hypercalcemia.
- Fever, flu syndrome, asthenia, body odour.
- Local reactions: redness, swelling, tenderness.

DOSAGE

- Metabolic disorder:
  - Initial dose: in case of metabolic crisis: 50 mg/kg IV followed by an equivalent dose over the following 24 hours (see Daily dose).
  - Daily dose: 50 mg/kg IV in divided doses q3-4h (and not less frequent than every 6 hours due to its short half-life).
  - Maximum daily dose: 300 mg/kg IV.
- Hemodialysis patients: 5-20 mg/kg IV in the venous return line after each dialysis session.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp and protect from light.
- Stable for 24 hours at room temp in NS and RL at concentrations of 0.5 to 8 mg/mL in PVC bags.

MISCELLANEOUS

- Plasma carnitine levels should be monitored prior to initiating and during therapy.

REFERENCES

5, 40, 95, 135.

Full revision 2016; limited revision 2017

NAME OF MEDICATION
LevoFLOXacin

OTHER NAMES
Levaquin ®

CLASSIFICATION
Antibiotic - quinolone

INDICATIONS

- Broad spectrum antibiotic indicated for treatment of infections due to susceptible organisms when oral administration is not possible.

ADMINISTRATION

- Available in bags of premixed solution of 5 mg/mL in D5W.
- Intermittent IV infusion: infuse over a minimum of 60 minutes for doses of 250-500 mg and 90 minutes for a dose of 750 mg.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: mild to severe dermatologic reactions and anaphylaxis; can occur after the first dose. Discontinue therapy if rash or any symptoms develop.
- Cardiovascular: hypotension with rapid or bolus injection, QT interval prolongation and arrhythmias. Aortic aneurism/aortic dissection (rare but serious); avoid using in patients with known history or at risk of aortic aneurism.
- GI: diarrhea, nausea, constipation.
- CNS: headache, dizziness, insomnia.
- Hypoglycemia, hyperglycemia.
- Myasthenia gravis exacerbation (can be fatal); avoid using in patients with this condition.
- Tendinitis and tendon rupture, especially in the elderly, patients receiving corticosteroids or transplant recipients.
- Local reactions: injection site reactions.

DOSAGE

- 250-750 mg IV once daily depending on indication.
- Dosage in renal impairment: initial and subsequent doses vary according to indication and degree of renal impairment. Consult manufacturer's monograph for details.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store premixed solutions in the fridge or at room temp. Do not freeze. Keep infusion bags in overwrap until ready to use.
- Compatible IV solutions include NS, D5W, D5-NS and sodium lactate 1/6 M.
- Incompatible with sodium bicarbonate, mannitol and multivalent cations (e.g., magnesium, calcium).

MISCELLANEOUS

- Patient should be well hydrated to prevent crystalluria or cylindruria.
- Use with caution in patients predisposed to seizures.
- Use in pediatrics is not routinely recommended but may be justified in special circumstances.

REFERENCES

1, 4, 5, 6, 40, 95, 135, 208.

Full revision 2016; limited revision 2017, 2020

NAME OF MEDICATION
LEVOTHYROXINE

OTHER NAMES
Synthroid ®, T4, Thyroxine sodium

CLASSIFICATION
Thyroid hormone

INDICATIONS
- Replacement therapy for diminished or absent thyroid function of any etiology where a rapid repletion is required or when oral administration is not possible.
- Myxedematous stupor or coma.
- As an aid for organ preservation before harvesting for organ transplantation in brain-dead organ donor.

ADMINISTRATION
- Reconstitute lyophilized powder with 5 mL of NS without preservative to obtain a concentration of approximately 40 mcg/mL and 100 mcg/mL for the 200 mcg and 500 mcg vials, respectively; shake until a clear solution is obtained.
- IV direct (preferred): physician or RN; undiluted; administer at rate of 100 mcg/min.
- Continuous IV infusion: in brain-dead organ donor; dilute 250 mcg in 250 mL D5W for a final concentration of 1 mcg/mL. As a substantial amount of drug binds to PVC, use glass bottle or polyolefin bag for dilution. For administration, use non-PVC tubing or if a PVC tubing is used, flush tubing with levothyroxine solution before administration.
- IM (NOT preferred as absorption may be erratic).

POTENTIAL ADMINISTRATION HAZARDS
- Cardiovascular: tachycardia, chest pain; caution in patients with cardiovascular disease.
- GI: diarrhea, abdominal cramps, vomiting.
- CNS: nervousness, tremors, insomnia.
- Sweating.
- May potentiate effects of anticoagulants. Frequent prothrombin determinations are recommended.

DOSAGE
- Hypothyroidism: dosage is adjusted to clinical response and lab values; typical adult parenteral maintenance dose is 50-100 mcg/day. The usual initial parenteral dosage for maintenance therapy in adults and infants/children who cannot take oral treatment should be approximately 75-80% of the previously established oral dose.
- Myxedema coma: initial dose of 100-500 mcg IV. Depending on response to first dose, 100-300 mcg IV or even greater on the second day; follow by daily doses of 50-100 mcg IV or 1.2 mcg/kg/day until oral replacement is feasible. Lower doses should be considered in geriatric patients and in those with cardiovascular disease.
- Organ preservation in brain-dead donor: 20 mcg IV bolus followed by 10 mcg/hr IV infusion OR 100 mcg IV bolus followed by 50 mcg IV bolus q12h.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- The manufacturers recommend to use immediately following reconstitution and to discard the unused portion; however, the product monograph for the Riva brand states that the 40 mcg/mL and 100 mcg/mL reconstituted solutions are stable for 4 hours at 25°C.
- The reconstituted solution of 100 mcg/mL in NS is stable for 7 days in the fridge when repackaged in polypropylene syringes.
- Stable for 24 hours at room temp unprotected from light in D5W at a concentration of 1 mcg/mL in glass bottles and polyolefin bags.

MISCELLANEOUS
- Onset of effect in myxedema coma is 6-12 hours; maximum effect in up to 24 hours.
- Contraindicated in thyrotoxicosis, acute myocardial infarction or uncorrected adrenal insufficiency.

REFERENCES
1, 4, 5, 6, 40, 82, 95, 102, 108, 135.

Full revision 2016; limited revision 2020, 2021

NAME OF MEDICATION
LIDOCAINE

OTHER NAMES
Lignocaine, Xylocaine ®, Xylocard ®

CLASSIFICATION
Antiarrhythmic, Anesthetic

INDICATIONS

- Treatment of acute ventricular tachyarrhythmias.
- Treatment of acute and chronic neuropathic pain [refer to Corporate nursing standard operating procedure C-SOP NUR 0063 (Pain: Intravenous and subcutaneous lidocaine administration)].
- As a diluent to reconstitute some specific medications to decrease the pain associated with IM administration (lidocaine 0.5% or 1%, without epinephrine).

ADMINISTRATION

- IV direct: physician or RN; for cardiac or pain indications: ***cardiac monitoring, continuous BP monitoring*****;** use 100 mg/5 mL prefilled syringes or 100 mg/5 mL solution. If administered for pain: physician to remain in attendance with the patient for at least 15 minutes after bolus dose. Rate as per Dosing section for both indications.
- Intermittent IV infusion, Continuous IV infusion: for cardiac indication: ***cardiac monitoring, continuous BP monitoring*** **(not needed for pain indication)**. Dilute 1-2 g in 250-500 mL of D5W or NS or dilution as per Dosage section; may also use premixed bag containing D5W. Rate as per Dosage section. Must be administered by an infusion pump.
- SC infusion: for pain management only. Must be administered by an infusion pump.
- IM: only when used as a diluent to reconstitute some specific medications to be administered IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactoid/anaphylactic reactions, urticaria, edema.
- Cardiovascular: hypotension, bradycardia, cardiovascular collapse.
- GI: vomiting.
- CNS: drowsiness, apprehension, paresthesias, dizziness, euphoria, seizures.
- Respiratory depression.
- Muscle tremors.
- Local reactions: thrombophlebitis with IV administration; pain with IM administration.

DOSAGE

Adults (for cardiac indications):
- Loading: 0.5-1.5 mg/kg IV (e.g., 50-100 mg) given at a rate of 0.35-0.7 mg/kg/min (approximately 25-50 mg/min); if refractory after 5-10 minutes, a second bolus of 0.5-0.75 mg/kg IV (e.g., 25-50 mg) may be given. Maximum single dose should not exceed 100 mg. Maximum total loading dose: 3 mg/kg.
- Maintenance: a continuous infusion may be administered at a rate of 1-4 mg/min IV (20-50 mcg/kg/min). Reappearance of arrhythmias during an infusion should be treated with a small IV bolus dose (0.5 mg/kg) and an incremental infusion rate increase (maximum rate 4 mg/min). Use slower infusion rates (e.g., 1/2 of above infusion rates) in patients with decreased cardiac output (e.g., CHF), bradycardia, liver disease, or patients older than 65 years of age.
- No more than 200-300 mg should be administered during a 60 minute period.

Adults [for pain indications as per Corporate nursing standard operating procedure C-SOP NUR 0063 (Pain: Intravenous and subcutaneous lidocaine administration)]:
- Palliative/chronic pain: 1.5 mg/kg slow IV direct over 2-4 minutes followed by 3.5-5 mg/kg in 500 mL of NS IV over 30-60 minutes.
- Acute pain: 1.5 mg/kg slow IV direct over 2-4 minutes followed by an infusion of 0.5-2 mg/kg/hr.
- Ideal body weight is to be used if body mass index is greater than 30.

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DOSAGE

Pediatrics (for cardiac indications):
- Loading dose of 0.5-1 mg/kg IV (maximum of 100 mg), repeated according to response with the total dose not exceeding 3-5 mg/kg.
- IV Maintenance: 10-50 mcg/kg/min.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Do not freeze.
- Compatible with D5W, NS, RL, 1/2NS, 1/4NS, D5-NS, D5-1/2NS, D5-RL.
- Stable for 90 days at room temp and in the fridge undiluted at 20 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Duration of IV bolus dose is 10-20 minutes.
- Desired range of plasma concentration is approximately 6-21 mcmol/L (1.5-5 mcg/mL) for the treatment of ventricular arrhythmia, and 6-12 mcmol/L for pain control.

REFERENCES

1, 4, 5, 40, 95, 135.

NAME OF MEDICATION
LINCOMYCIN

OTHER NAMES
Lincocin ®

CLASSIFICATION
Antibiotic

INDICATIONS

- Treatment of serious infections due to susceptible streptococci or staphylococci when treatment with a penicillin is considered inappropriate.

ADMINISTRATION

- Intermittent IV infusion (mandatory): dilute each gram in at least 100 mL of a compatible fluid to obtain a maximum concentration of 10 mg/mL; infuse at a rate not exceeding 1 g per hour.
- Deep IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, erythema multiforme, angioedema, serum sickness, anaphylactic/anaphylactoid reactions.
- Cardiovascular: hypotension with too rapid administration.
- GI: nausea, vomiting, diarrhea (including *C. difficile*-associated diarrhea), colitis, abdominal pain, esophagitis.
- CNS: headache, dizziness.
- Hepatic: jaundice, abnormal LFTs, increased bilirubin.
- Otic: tinnitus.
- Myalgia, vaginitis.
- Local reactions: irritation, pain, induration, sterile abscess with IM administration; pain, erythema, swelling, thrombophlebitis with IV administration.

DOSAGE

- Adults: 600-1000 mg IV q8-12h.
  600 mg IM q12-24h.
  Dose may be increased in more severe infections. Maximum of 8 g/day in life-threatening infections.
- Pediatrics, over 1 month of age: 10-20 mg/kg/day IV in 2 or 3 divided doses.
  10 mg/kg IM q12-24h.
  Maximum pediatric parenteral dose is 40 mg/kg/day or 8 g/day.
- Dosage in renal impairment: patients with severe renal impairment should receive 25-30% of the normal dose.
- Dosage in hepatic impairment: use with caution as the serum half-life may be 2-fold longer.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Stable for 24 hours at room temp in D5W, D10W, NS, D5-NS at a concentration of 1.2 mg/mL.
- Compatible with D10-NS, Ringer's and sodium lactate 1/6 M.

MISCELLANEOUS

- Contraindicated in patients allergic to clindamycin.
- Antagonizes the activity of erythromycin. Avoid such combination.
- Contains benzyl alcohol. Avoid use in pregnancy and in neonates.

REFERENCES

1, 4, 5, 40, 95, 135, 208.

Full revision 2016

NAME OF MEDICATION
LINEZOLID

OTHER NAMES
Zyvoxam ®

CLASSIFICATION
Antibiotic

INDICATIONS
- Vancomycin-resistant *Enterococcus faecium* (VREF) infections.
- Nosocomial and community acquired pneumonias caused by *S. pneumoniae* (penicillin-susceptible strains only) and *S. aureus* (methicillin-susceptible and -resistant strains).
- Methicillin-resistant staphylococcal (MRSA) infections if failure or intolerance to vancomycin.
- Complicated skin and skin structure infections (SSSI), including non-limb threatening diabetic foot infections, without osteomyelitis, caused by *S. aureus* (methicillin-susceptible and -resistant strains), *S. pyogenes*, or *S. agalactiae*.
- Uncomplicated SSSI caused by *S.aureus* (methicillin-susceptible strains only) and *S. pyogenes*.

ADMINISTRATION
- Intermittent IV infusion: premixed bags of 600 mg/300 mL in D5W; infuse over 30-120 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, urticaria, angioedema, anaphylaxis, Stevens-Johnson syndrome.
- Cardiovascular: hypertension.
- GI: nausea, vomiting, diarrhea.
- CNS: headache, dizziness, seizures.
- Hematologic: myelosuppresssion (e.g., anemia, leukopenia and thrombocytopenia).
- Hepatic: elevated ALT and AST.
- Peripheral and optic neuropathy (can lead to vision loss). May be irreversible. Occurs mostly with treatments longer than 28 days.
- Lactic acidosis, hypoglycemia.

DOSAGE
- VREF: 600 mg IV q12h for 14 to 28 days.
- Pneumonia: 600 mg IV q12h for 7 to 21 days.
- SSSI:
  - non-diabetic foot infections: 600 mg IV q12h for 10 to 14 days.
  - diabetic foot infections: 600 mg IV q12h for 14 to 28 days.
- Dosage in renal impairment: no dosage adjustment is necessary; however, use with caution as metabolites may accumulate. Administer dose after hemodialysis session.
- Dosage in hepatic impairment: no adjustment necessary in mild to moderate impairment. Has not been evaluated in severe hepatic impairment (Child-Pugh C).

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store bags at room temp. Protect from light. Do not freeze. Keep infusion bags in overwrap until ready to use.
- Compatible with D5W, NS, RL.
- Solutions may exhibit a yellow colour that can intensify over time without affecting potency.

MISCELLANEOUS
- Monitor blood counts weekly.
- Linezolid is a reversible non-selective inhibitor of monoamine oxidase. Has the potential to interact with adrenergic and serotonergic drugs.
- Avoid consuming foods/beverages with high tyramine content. Tyramine consumption should be less than 100 mg per meal.
- Linezolid should not be used in patients taking SSRIs or SNRIs because of risk of development of serotonin syndrome.
- Recommended to reduce doses of sympathomimetics, vasopressors, and dopaminergic agents prior to initiating linezolid.
- Switch to oral treatment at the same dosage when clinically appropriate.

REFERENCES
1, 5, 40, 68, 95, 135.

Full revision 2016

NAME OF MEDICATION
LIRAGLUTIDE

OTHER NAMES
Saxenda ®, Victoza ®

CLASSIFICATION
Antidiabetic, Weight management

INDICATIONS

- Treatment of type 2 diabetes mellitus in adults as monotherapy when metformin is inappropriate or as add-on therapy to metformin, with or without a sulfonylurea or basal insulin or a sodium-glucose cotransporter-2 (SGLT2) inhibitor.
- Treatment of type 2 diabetes mellitus in children and adolescents aged 10 years and older as an adjunct to metformin with or without basal insulin.
- Add-on combination to reduce the incidence of cardiovascular death in patients with type 2 diabetes mellitus and established cardiovascular disease.
- Chronic weight management in adults with a body mass index (BMI) of at least 30 kg/m^2, or at least 27 kg/m^2 if there is also one weight-related comorbidity (e.g., hypertension, type 2 diabetes mellitus, dyslipidemia).
- Chronic weight management in children and adolescents aged 12 years and older with an inadequate response to diet and exercise, with a body weight above 60 kg and an initial BMI corresponding to 30 kg/m^2 and above for adults by international cut-offs; refer to manufacturer's instructions for more details.

ADMINISTRATION

- SC: in the upper arm, thigh, or abdomen. If basal insulin is part of the treatment regimen, both products can be administered in the same region, although NOT adjacent to each other.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, angioedema, antibody formation, anaphylactic reaction.
- Cardiovascular: tachyarrhythmia, tachycardia, PR interval prolongation.
- GI: nausea, vomiting, dyspepsia, diarrhea, constipation; minimized by increasing dose at weekly intervals. Pancreatitis.
- CNS: headache, dizziness.
- Hypoglycemia, fatigue.
- Renal dysfunction secondary to dehydration.
- Local reactions: bruising, pain, rash, erythema.

DOSAGE

- Adults:
  - Diabetes: initial dose of 0.6 mg SC once daily, without regard to meals, for at least 1 week then increase to the therapeutic dose of 1.2 mg SC daily. Based on clinical response, dose can be increased after at least 1 week to the maximum dose of 1.8 mg SC once daily.
  - Weight management: initial dose of 0.6 mg SC daily; daily dose increased by 0.6 mg at weekly intervals to the maintenance and maximum dose of 3 mg SC daily. Discontinue treatment if a reduction of at least 5% of initial body weight has not been achieved after 12 weeks at the 3 mg daily dose.
- Pediatrics:
  - Diabetes: initial dose of 0.6 mg SC once daily, without regard to meals, for at least 1 week then increase to the therapeutic dose of 1.2 mg SC daily. Based on clinical response, dose can be increased after at least 1 week to the maximum dose of 1.8 mg SC once daily.
  - Weight management: initial dose of 0.6 mg SC once daily; daily dose increased by 0.6 mg at weekly intervals to the maintenance and maximum dose of 3 mg SC daily. If the patient cannot tolerate an increased dose during dose escalation, consider going back to the previous level. Discontinue treatment if patient does not tolerate the 2.4 mg daily dose or if a reduction in BMI of at least 1% has not been achieved after 12 weeks on the maintenance dose.

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DOSAGE (Cont.)

- Dosage in renal impairment: no dosage adjustment required in patients with mild, moderate or severe renal impairment; not recommended in patients with end-stage renal disease.
- Dosage in hepatic impairment: use with caution as there is limited data.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled multidose pens between 2-8°C. Do not freeze. Protect from light.
- When in use, the pen may be kept between 2-30°C for up to 30 days. Protect from excessive heat and sunlight; protect from light by using the pen cap when not in use.

MISCELLANEOUS

- Liraglutide is a glucagon-like peptide-1 (GLP-1) receptor agonist, an incretin mimetic agent.
- Liraglutide should not be used in the treatment of type 1 diabetes mellitus or diabetic ketoacidosis, in patients with multiple endocrine neoplasia syndrome type 2, or in patients with a personal or family history of medullary thyroid carcinoma.

REFERENCES

1, 5, 95, 135.

Full revision 2016; limited revision 2018, 2021

NAME OF MEDICATION
LIXISENATIDE

OTHER NAMES
Adlyxine ™

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of type 2 diabetes mellitus in adults in combination with metformin and/or a sulfonylurea, pioglitazone or basal insulin.

ADMINISTRATION

- SC: thigh, abdomen or upper arm. Rotate injection site. After administering the dose, remove the needle and replace the cap on the pen to protect from light.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis, angioedema.
- Cardiovascular: PR interval prolongation.
- GI: nausea, vomiting, diarrhea, gastroparesis (avoid in patients with severe gastroparesis), pancreatitis.
- CNS: headache, dizziness.
- Endocrine and metabolic: hypoglycemia (risk increased if used in combination with basal insulin or a sulfonylurea; refer to Miscellaneous section).
- Renal: renal failure.
- Local reactions: pain, pruritus, rash, erythema, hematoma, inflammation, irritation, nodule.

DOSAGE

- Initial dose of 10 mcg SC once daily for 14 days, then increase to maintenance dose of 20 mcg SC once daily. Administer dose within 1 hour before a meal (preferably the same meal of the day).
- If the 20 mcg daily maintenance dose is not tolerated, reduce to 10 mcg SC once daily and consider increasing the dose back to 20 mcg once daily maintenance dose within 4 weeks.
- Dosage in renal impairment: no dosage adjustment necessary when CrCl is 15 mL/min or greater, however closely monitor renal function and for occurrence of adverse events in patients with CrCl between 15-89 mL/min; not recommended when CrCl is less than15 ml/min as has not been studied.
- Dosage in hepatic impairment: no dosage adjustment necessary.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled pen between 2-8°C. Protect from light and excessive heat. Do not freeze.
- When in use, the pen may be kept at up to 30°C for 14 days.
- Solution in the pen should be clear and colourless.

MISCELLANEOUS

- Lixisenatide is a glucagon-like peptide-1 (GLP-1) receptor agonist, an incretin mimetic agent.
- Lixisenatide should not be used in the treatment of type 1 diabetes mellitus or diabetic ketoacidosis.
- When lixisenatide is added to basal insulin or a sulfonylurea, consider decreasing the dose of these latters to reduce the risk of hypoglycemia.

REFERENCES

1, 5, 95, 135.

New monograph 2019

NAME OF MEDICATION	
LORazepam	
OTHER NAMES	CLASSIFICATION
Ativan ®	Benzodiazepine

INDICATIONS

- Preoperatively to produce sedation, relieve anxiety, and provide anterograde amnesia.
- Sedation in the critical care setting.
- Initial treatment of status epilepticus.
- Antiemetic adjunct for chemotherapy.
- To alleviate symptoms of acute alcohol withdrawal.
- Palliative care (e.g., agitation, sedation).

ADMINISTRATION

- IV direct: physician or RN; ***respiratory support***; dilute dose in a syringe with an equal volume of a compatible solution. Gently invert several times to mix; do NOT shake vigorously to avoid air entrapment. Rate should not exceed 2 mg/min.
- Continuous IV infusion: dilute in D5W for a concentration of 2 mg/mL. At TOH, mix 40 mg (10 mL from lorazepam 4 mg/mL vials) with 10 mL D5W to obtain a final concentration of 2 mg/mL. Use a glass or a polyolefin container for dilution. Rate to be titrated according to response (refer to Dosage section).
- IM (deep): undiluted.
- SC: in palliative care, as an injection and infusion. For SC infusion, dilute to 2 mg/mL to reduce the risk of precipitation; must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis, anaphylactoid reactions, angioedema, rash.
- CNS: sedation, dizziness.
- Respiratory depression.
- Risk of propylene glycol toxicity when high dose continuous infusion (greater than 0.1 mg/kg/hr or 6 mg/hr) used for longer than 48 hours; monitor acid-base status, renal function, serum osmolality, electrolyte levels and osmol gap.
- Risk of polyethylene glycol toxicity with high doses (acute tubular necrosis).
- Local reactions: pain and erythema at injection site.
- Extravasation hazard: vesicant. Can cause arteriospasm in perivascular tissue, which can lead to gangrene. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH electronic formulary.

DOSAGE

- Preoperative, IV direct:
  - for sedation and relief of anxiety: 0.044 mg/kg IV (up to 2 mg) 15-20 minutes preoperatively.
  - for anterograde amnesia: 0.05 mg/kg IV (up to 4 mg, or 2 mg in patients over 50 years old).
- Preoperative, IM: 0.05 mg/kg (up to 4 mg, or 2 mg in patients over 50 years old) 2 hours preoperatively.
- Sedation in critical care: 0.02-0.06 mg/kg IV q2-6h OR 2 mg IV, followed by an IV infusion at 0.5-1 mg/hr; titrate to desired response and lowest effective dose. Dose range 0.25-10 mg/hr or 0.01-0.1 mg/kg/hr.
- Status epilepticus: 0.05-0.1 mg/kg IV (up to 4 mg); an additional dose of 0.05 mg/kg IV (up to 4 mg) may be required (after 10-15 minutes) if seizures recur or continue. Maximum: 8 mg per 12 hours.
- Antiemetic: 0.025-0.05 mg/kg (up to 4 mg) or 1.5 mg/m^2 (up to 3 mg) IM or IV 30-45 minutes pre-chemo then q4h prn.
- Alcohol withdrawal: 1-4 mg IV or IM; to be repeated as needed (IV: every 5-15 minutes; IM: every 30-60 minutes) to keep patient comfortable then every hour to maintain light somnolence.
- Palliative care: 0.5-2 mg SC q4-8h OR 4-20 mg SC infusion/24 hours.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge, protect from light. Do not freeze.
- Solution of 4 mg/mL in unopened vials from Sandoz is stable for 2 months at room temp; product should be discarded after this period. If exposure is up to a maximum of 7 days, the vial can be put back in the fridge and the product will be stable until the printed expiry date.
- Solution of 2 mg/mL in unopened vials from Sandoz is stable for 30 days at room temp; product should be discarded after this period.
- Compatible in SWFI, NS, and D5W, but solubility is highest in D5W.
- Lorazepam adsorbs to a significant amount to PVC bags.
- Stable for up to 24 hours at room temp at a concentration of 2 mg/mL and at concentrations of less than 0.08 mg/mL. Concentrations in the middle range of 0.08 to 1 mg/mL may be problematic as occasional precipitation may occur. Note that solutions in D5W at a concentration of 1 mg/mL have been stable for 24 hours at room temp only when prepared with the 2 mg/mL strength; precipitation occurred when prepared with the 4 mg/mL strength. This difference is explained by the fixed concentrations of solubilizing agents.

MISCELLANEOUS

- Lorazepam injection 4 mg/mL contains 80% v/v of propylene glycol and 18% v/v of polyethylene glycol.
- Lorazepam injection 2 mg/mL contains 77.5% v/v of propylene glycol and 20.3% v/v of polyethylene glycol.

REFERENCES

1, 4, 5, 40, 95, 135, 245, 254, 256, 335, 354, 457.

Full revision 2016; limited revision 2017, 2019, 2020, 2021

NAME OF MEDICATION
LOXAPINE

OTHER NAMES
Loxapac ®

CLASSIFICATION
Antipsychotic

INDICATIONS

- Symptomatic management of psychotic episodes associated with schizophrenia in adults.

ADMINISTRATION

- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: orthostatic hypotension, hypotension (**Antidote** if severe: norepinephrine or phenylephrine; do not use epinephrine), tachycardia, QTc interval prolongation (rare).
- GI: nausea, vomiting, constipation (can be severe), dry mouth.
- CNS: drowsiness, confusion, extrapyramidal symptoms (e.g., pseudoparkinsonism, akathisia and dystonia), seizures.
- Hematologic: venous thromboembolism.
- Neuroleptic malignant syndrome: hyperpyrexia, muscle rigidity, altered mental status, autonomic instability.
- Blurred vision, urinary retention, hyperglycemia.

DOSAGE

- Usual dose: 12.5-50 mg IM q4-6h or longer, adjusted to patient's response.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 15-30°C.

MISCELLANEOUS

- Switch to oral therapy when feasible.

REFERENCES

5, 135.

Full revision 2016

NAME OF MEDICATION
MAGNESIUM SULFATE, AQUEOUS

OTHER NAMES
$MgSO_4$

CLASSIFICATION
Electrolyte

INDICATIONS

- Prevention and treatment of seizures in toxemia of pregnancy (pre-eclampsia and/or eclampsia).
- For fetal neuroprotection of the preterm in women with imminent preterm birth (less than 34 weeks' gestation).
- Treatment of hypomagnesemia when oral administration is not possible.
- For ventricular fibrillation/pulseless ventricular tachycardia cardiac arrest associated with torsades de pointes.
- For polymorphic ventricular tachycardia with QT interval prolongation (torsades de pointes).
- For severe exacerbations of asthma in patients unresponsive to 1 hour of intensive conventional therapy.
- For severe migraine unresponsive to standard treatment.

ADMINISTRATION

- IV direct (in cardiac arrest situation only): physician or RN in presence of physician; ***respiratory support; cardiac monitoring***. Dilute 1-2 g in 10 mL of D5W or NS (1 g = 5 mL of magnesium sulfate 20% solution [200 mg/mL]) and administer over 1-2 minutes.
- Intermittent IV infusion, Continuous IV infusion:
  - dilute with a compatible solution: for intermittent IV infusion: MAXIMUM CONCENTRATION of 200 mg/mL (20%). For continuous IV infusion, 20 g should be mixed in 500 mL; it is suggested to avoid using the 1 litre bags to distinguish from other large volume infusions.
  - infuse at a maximum rate of 1 g/hr when treating asymptomatic hypomagnesemia as higher infusion rate may exceed the renal threshold and result in disproportionate excretion of magnesium in the urine.
  - can infuse at a MAXIMUM RATE of 150 mg/min if situation is warranted; EXCEPT in severe eclampsia with seizures where a higher rate may be used.
  - in severe eclampsia with seizures: 4 g (i.e., 20 mL of magnesium sulfate 20% [200 mg/mL]) undiluted or diluted in 20 mL of solution (to obtain a 10% solution) administered over 3-4 minutes OR 4-6 g in 100-250 mL of solution over 20-30 minutes.
- Must be administered by an infusion pump when used for eclampsia, preeclampsia and fetal neuroprotection.
- IM: concentrations of 25 or 50% are generally used. As IM injections are very painful, this route should only be used when IV access is not possible. May be mixed with lidocaine 2% solution to reduce pain at IM injection.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Side effects associated with parenteral magnesium are caused by magnesium intoxication. See below for signs of hypermagnesemia.
- Cardiovascular: flushing, hypotension, ECG abnormalities, heartblock, circulatory collapse, hypothermia, cardiac arrest.
- GI: nausea, vomiting.
- CNS: drowsiness, confusion, ataxia.
- Muscle weakness, depression of reflexes, respiratory paralysis and depression.

Antidote: 5-10 mEq of calcium IV (e.g., approximately 10-20 mL of calcium gluconate 10%) to reverse respiratory depression and heart block.

DOSAGE

- Toxemia of pregnancy (pre-eclampsia and/or eclampsia): loading dose of 4-6 g IV followed by a continuous IV infusion of 1-2 g/hr. If IV administration is not possible, use a loading dose of 10 g IM followed by 5 g IM q4h. Continue for at least 24 hours after delivery.
- For fetal neuroprotection: loading dose of 4 g IV over 30 minutes, with or without being followed by a continuous IV infusion of 1 g/hr until birth or until delivery is no longer imminent, not exceeding an infusion duration of 24 hours.
- Treatment of hypomagnesemia: dose depending on severity of hypomagnesemia. Mild to moderate (0.4-0.7 mmol/L): 1-4 g IV/IM. Severe (less than 0.4 mmol/L): 4-8 g IV/IM.

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DOSAGE (Cont.)

- For ventricular fibrillation/pulseless ventricular tachycardia cardiac arrest associated with torsades de pointes: 1-2 g IV over 1-2 minutes.
- For polymorphic ventricular tachycardia associated with QT interval prolongation (torsades de pointes): 1-2 g IV over 15 minutes.
- For severe exacerbations of asthma: a single dose of 2 g IV over 20-60 minutes.
- For severe migraine: a single dose of 1-2 g IV at a rate of 1 g/15 min.
- Dosage in renal impairment: dosage to be decreased; use with caution.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Stable for 3 months at room temp diluted in NS or RL at a concentration of 37 mg/mL in PVC bags and glass bottles.
- Stable for 30 days at room temp and 60 days in the fridge at a concentration of 200 mg/mL (undiluted 20% solution) in PVC bags.
- Stable for 30 days at room temp, fridge and -20°C diluted in RL at a concentration of 83.3 mg/mL in polyolefin bags.
- Stable for 24 hours at room temp diluted in D5W, D5-RL, D5-1/4NS, D5-1/2NS, D5-NS, RL, NS at a concentration of 1 mg/mL in PVC bags.

MISCELLANEOUS

- Contraindicated in patients with heart block.
- Observe patients carefully for signs and symptoms of hypermagnesemia and monitor serum magnesium concentrations at least once daily during repletion.
- 1 g of magnesium sulfate contains 4 mmol (8 mEq) magnesium.
- When hypomagnesemia and hypokalemia coexist, magnesium deficiency should be corrected to facilitate the correction of hypokalemia.

REFERENCES

1, 4, 6, 13, 16, 35, 36, 40, 95, 110, 135, 235, 272, 356, 357, 474, 498, 500.

Full revision 2016; limited revision 2017, 2019, 2020, 2021

NAME OF MEDICATION
MANNITOL

OTHER NAMES
Osmitrol ®

CLASSIFICATION
Diuretic, Diagnostic agent

INDICATIONS

- Measurement of glomerular filtration rate (GFR). However, inulin is a better choice.
- Prevention and/or treatment of the oliguric phase of acute renal failure before irreversible renal failure occurs.
- To reduce intracranial pressure and to treat cerebral edema in neurologic emergencies.
- To reduce intraocular pressure.
- Treatment of edema and ascites of nephrotic, cirrhotic, or cardiac origin.
- Promotion of urinary excretion of toxins following chemotherapy, radiation, or drug intoxications.

ADMINISTRATION

- Intermittent IV infusion, Continuous IV infusion: the rate, dosage and concentration will vary with the indication and patient's clinical state.
- May need a central vein for administration of the more concentrated solutions due to high osmolarity and depending on duration of the infusion.
- Use an administration set with a filter (5 microns or less) when infusing solutions containing 15% or more since mannitol crystals may be present (refer to Compatibility, Stability section if presence of crystals).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis and anaphylactoid reactions, rash, urticaria, chills.
- Cardiovascular: hypervolemia (may result in pulmonary edema), CHF, edema, angina-like chest pain, hypertension, hypotension.
- GI: dry mouth, nausea, vomiting.
- CNS: headache, dizziness, confusion, convulsions, lethargy, coma.
- Electrolyte disburbances: hyponatremia, hypernatremia, hyperkalemia, hypokalemia, thirst.
- Osmotic nephrosis (high doses), urinary retention.
- Blurred vision.
- Local reactions: thrombophlebitis.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Measurement of GFR: 20 g (100 mL of 20% solution) diluted in 180 mL of NS (total: 280 mL) and infused IV at a rate of 20 mL/min.
- Prevention of oliguria of acute renal failure: 50-100 g IV as a 5-25% solution. Generally, a more concentrated solution is used initially, followed by a 5 or 10% solution.
- Treatment of oliguria: initial test dose of approximately 0.2 g/kg as a 15 or 20% solution IV over 3-5 minutes; if urine output is at least 30-50 mL/hr over the next 2-3 hours, may proceed with full dose. A second test dose may be given if no response is seen. If still no response, re-evaluate need for mannitol. Full dose: 100 g IV as a 15-20% solution over 90 minutes to several hours.
- Intracranial/Intraocular pressure: 0.25-2 g/kg IV of a 15-25% solution given over 30-60 minutes has been used; may repeat q6-8h prn when used to decrease intracranial pressure.
- Edema/ascites: 100 g of a 10-20% solution given IV over 2-6 hours.
- Promotion of urinary excretion of toxins: initial IV loading dose of 25 g followed by an infusion at a rate that will maintain urinary output of at least 100 mL/hr but preferably 500 mL/hr with a positive balance of 1-2 litres. Total dose should not exceed 200 g.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Solutions of mannitol are stable at room temp but in concentrations of 15% or more, mannitol may crystallize when exposed to low temperatures.
- Crystals indicate a supersaturated solution that should be warmed to dissolve crystals then cooled to body temperature before administration. Do not use if crystals have not been completely dissolved.
- Compatible with D5W, D5-NS, LR, NS and Ringer's solution.

MISCELLANEOUS

- Contraindicated in pulmonary edema, severe pulmonary congestion, severe CHF, severe dehydration, active intracranial bleeding (except during craniotomy) or in patients with well established anuria.

REFERENCES

1, 4, 5, 40, 95, 135, 208, 366, 367, 462.

NAME OF MEDICATION
MEASLES, MUMPS AND RUBELLA VACCINE

OTHER NAMES
MMR, MMR-II ®, Priorix ®

CLASSIFICATION
Vaccine

INDICATIONS

- For immunization against measles, mumps and rubella in persons 12 months of age or older.

ADMINISTRATION

- Reconstitute the vial with the entire volume of diluent provided; mix thoroughly until complete dissolution. The reconstituted solution should be clear yellow (MMR-II ®) or clear peach to fuchsia pink (Priorix ®).
- SC: preferably into the outer aspect of the upper arm.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, angioedema, anaphylaxis.
- Fever, sore throat, malaise, irritability, arthralgia, myalgia.
- Skin rash usually minimal but may be generalized; usually appears between the 5th and the 12th day.
- Local reactions: redness, pain, induration and tenderness at the injection site.

DOSAGE

- 0.5 mL SC.
- Refer to specific manufacturer's monograph for complete dosage schedule.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vaccine between 2-8°C; diluent may be stored between 2-8°C or at room temp. Protect from light.
- Use the vaccine as soon as possible after reconstitution; however, it can be stored in the dark between 2-8°C for up to 8 hours. Discard if not used within this time.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The MMR vaccine is a live attenuated vaccine.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity or in immunocompromised individuals.
- Contains trace amounts of neomycin (MMR-II ®, Priorix ®) and gelatin (MMR-II ® only).
- Although the MMR vaccine contains trace amount of egg protein, it has been administered safely in most egg-allergic patients; have adequate facilities available to manage anaphylaxis should it occur.
- Do not give human immune globulin concurrently with the vaccine as it could interfere with the immune response.
- The MMR vaccine is generally contraindicated in pregnant women because there is a theoretical risk to the fetus. Although one manufacturer (MMR-II ®) recommends to avoid pregnancy for 3 months following vaccination, health authorities consider that a one-month waiting period is sufficiently safe.

REFERENCES

1, 5, 31.

Full revision 2016; limited revision 2020

NAME OF MEDICATION
MedroxyPROGESTERone

OTHER NAMES
Depo-Provera ®

CLASSIFICATION
Progestin

INDICATIONS

- Treatment of endometriosis.
- Adjunctive and/or palliative treatment of endometrial and renal cell carcinoma.
- Hormonally dependent metastatic breast cancer.
- Contraception.

ADMINISTRATION

- Shake vigorously immediately before use to obtain a uniform suspension.
- IM: deep injection, undiluted.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, angioedema, anaphylactoid reactions, anaphylaxis.
- GI: nausea, vomiting.
- Menstrual abnormalities, including amenorrhea.
- Local reactions: pain, residual lump, skin discolouration, sterile abscess, local atrophy, necrosis, edema.

DOSAGE

- Endometriosis: 50 mg IM weekly or 100 mg IM every 2 weeks for at least 6 months.
- Endometrial and renal cell carcinoma: 400-1000 mg IM every 1-4 weeks.
- Breast cancer: 500 mg IM daily for 28 days, followed by 500 mg IM twice weekly.
- Contraception: 150 mg IM every 3 months. First injection to be given within first five days of onset of normal menstrual period, within first five days post-partum if not breastfeeding or during the sixth post-partum week in women who breastfeed.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Protect from freezing.

MISCELLANEOUS

REFERENCES

1, 5, 165.

Full revision 2016

NAME OF MEDICATION
MELPHALAN

OTHER NAMES
Alkeran ®, L-phenylalanine mustard (PAM), L-sarcolysin

CLASSIFICATION
Antineoplastic

INDICATIONS

- Multiple myeloma, malignant melanoma, ovarian carcinoma, breast cancer, Ewing's sarcoma, neuroblastoma, Waldenstrom's macroglobulinemia when oral therapy is not feasible or desirable.
- Conditioning regimen for bone marrow transplantation (BMT).

ADMINISTRATION

- Reconstitute the 50 mg vial with 10 mL of provided diluent for a final concentration of 5 mg/mL.
- IV direct: physician or RN; undiluted; inject slowly into an injection port or a central line of a freely running IV solution of NS.
- Intermittent IV infusion: dilute in 100 mL of NS to a concentration of 0.1-0.45 mg/mL; infuse over 15-30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: urticaria, pruritus, edema, tachycardia, bronchospasm, dyspnea, hypotension, chest pain, anaphylactoid reactions, anaphylaxis.
- Cardiovascular: flushing with high doses.
- GI: nausea, vomiting.
- Increase in creatine phosphokinase, rhabdomyolysis.
- Local reactions: pain, irritation, warmth, tingling.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Multiple myeloma: 16 mg/m^2 IV every 2-4 weeks.
- BMT: 200 mg/m^2 IV for one dose on day-1 of BMT (these doses are fatal without BMT).
- Dosage in renal impairment (not for BMT dosing):

CrCl (mL/min)	50-10	less than 10
Dose	75%	50%

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Use as soon as possible after preparation; ideally, administration should be completed within 60 minutes of reconstitution.
- Stable for 2 hours after reconstitution at room temp. Do not store reconstituted vials in the fridge due to risk of precipitation.
- Stable for 50 minutes at 30°C and 3 hours at 20°C in NS at a concentration between 0.1 to 0.45 mg/mL.
- Stable for 6 hours in the fridge in NS at a concentration of 0.2 mg/mL in PVC containers.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 129, 135, 165.

Full revision 2016; limited revision 2017, 2018, 2020

NAME OF MEDICATION
MENINGOCOCCAL VACCINE, GROUP B

OTHER NAMES
Bexsero ®, Trumenba ™

CLASSIFICATION
Vaccine

INDICATIONS

- For active immunization for the prevention of invasive disease caused by *N. meningitidis* serogroup B in individuals from 2 months to 25 years old (Bexsero ®) and in patients 10 to 25 years old (Trumenba ™).

ADMINISTRATION

- Shake well prefilled syringe before use to obtain a homogenous white suspension.
- IM: in the anterolateral thigh for infants; in the deltoid region for older children, adolescents and adults.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis.
- GI: nausea, vomiting, diarrhea, change in appetite.
- CNS: headache, irritability, drowsiness.
- Respiratory: apnea may occur in premature infants, particularly if history of respiratory immaturity.
- Malaise, myalgia, arthralgia, fatigue, chills.
- Fever: prophylactic use of acetaminophen reduces the incidence and severity of fever without affecting the immunogenicity.
- Local reactions: pain, redness, induration, swelling at injection site.

DOSAGE

- 0.5 mL IM.
- Refer to specific manufacturer's monograph for complete dosage schedule.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled syringes flat and between 2-8°C. Protect from light. Do not freeze.
- Trumenba ™ is stable for a cumulative time of 96 hours at 25°C.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The two group B meningococcal vaccines (Bexsero ®, Trumenba ™) are not interchangeable as they contain different antigens. Therefore, the same vaccine product should be used for all doses in a vaccination series.
- The tip cap of the Bexsero ® prefilled syringe may contain latex.
- The meningococcal vaccine is an inactivated vaccine.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.

REFERENCES

5, 31, 135, 442.

Full revision 2016; limited revision 2018, 2019, 2020

NAME OF MEDICATION
MENINGOCOCCAL VACCINE, GROUP C

OTHER NAMES
Meningitec ®, Menjugate ®, Menjugate ® Liquid, NeisVac-C ®

CLASSIFICATION
Vaccine

INDICATIONS

- For active immunization of children from 2 months of age, adolescents and adults for the prevention of invasive disease caused by *N. meningitidis* serogroup C.

ADMINISTRATION

- For Menjugate ®: reconstitute vial with 0.6 mL of the diluent provided. Gently shake the vial until dissolution.
- No reconstitution necessary for Menjugate ® Liquid, Meningitec ® and NeisVac-C ®. Shake immediately prior to use to obtain a uniform suspension.
- IM: in the deltoid region for older children, adolescents and adults; in the anterolateral thigh for infants.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis, angioedema.
- GI: nausea, vomiting, diarrhea, change in appetite.
- CNS: headache, irritability, drowsiness, impaired sleeping.
- Fever, malaise, arthralgia, myalgia.
- Local reactions: pain, redness, swelling at injection site.

DOSAGE

- 0.5 mL IM.
- Refer to specific manufacturer's monograph for complete dosage schedule.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vaccine between 2-8°C. Do not freeze. Protect from light (for Menjugate ® and Menjugate ® Liquid).
- Menjugate ®: stable at room temp (up to 25°C) for 6 months before its reconstitution. Use reconstituted vaccine immediately.
- NeisVac-C ®: stable at room temp (up to 25°C) for 9 months.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The meningococcal vaccine is an inactivated vaccine.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.
- Menjugate ®: tip cap of the syringe contains dry natural rubber (latex).

REFERENCES

5, 31, 135.

Full revision 2016; limited revision 2020

NAME OF MEDICATION **MENINGOCOCCAL VACCINE, GROUPS A, C, Y, W-135**

OTHER NAMES
Menactra ®, Menomune ®, Menveo ®, Nimenrix ®

CLASSIFICATION
Vaccine

INDICATIONS

- For active immunization of children, adolescents and adults for the prevention of invasive disease caused by *N. meningitidis* serogroups A, C, Y and W-135.
- Note: Menveo ® can be given in children from 2 months of age, Menactra ® from 9 months of age, Nimenrix ® from 6 weeks of age and Menomume ® from 2 years of age.

ADMINISTRATION

- Menomune ®, Menveo ®, Nimenrix ®: reconstitute vial with the diluent provided (in the case of Menveo ®, the diluent is the MenCWY component and the powder vial is the MenA component). Shake the vial until dissolution.
- Menactra ®: no reconstitution necessary. Shake the vial to obtain a uniform suspension before administration.
- Menactra ®, Menveo ®, Nimenrix ®: IM, preferably in the deltoid region for older children, adolescents and adults; in the anterolateral thigh for infants.
- Menomune ®: SC.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis.
- GI: nausea, vomiting, diarrhea, change in appetite.
- CNS: headache, irritability, drowsiness.
- Fever, malaise, arthralgia, myalgia.
- Local reactions: pain, redness, swelling at injection site.

DOSAGE

- 0.5 mL IM or SC, depending on brand of meningococcal vaccine (refer to Administration section).
- Refer to specific manufacturer's monograph for complete dosage schedule.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vaccine between 2-8°C. Do not freeze. Protect from light (Menveo ®, Nimenrix ®).
- Menomune ® single dose vial should be used within 24 hours of reconstitution and the multi-dose vial 35 days after reconstitution, provided the solution is kept between 2-8°C.
- Menveo ® is stable for 2 hours at 25°C following its reconstitution.
- Nimenrix ® is stable for 8 hours at 30°C following its reconstitution.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The meningococcal vaccine is an inactivated vaccine.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.
- Menomune ® contains thimerosal and its vial stopper contains latex.

REFERENCES

5, 31, 135, 541.

Full revision 2016; limited revision 2018, 2020

NAME OF MEDICATION
MEPERIDINE

OTHER NAMES
Demerol ®, Pethidine

CLASSIFICATION
Opioid analgesic

INDICATIONS
- For preoperative sedation, as a supplement to anesthesia.
- For severe pain when oral administration is not possible.
- For prevention or treatment of shaking chills caused by some medications (e.g., amphotericin B).

ADMINISTRATION
- IV direct: physician or RN; ***respiratory support***. Administer undiluted or preferably, dilute dose to 10 mL with NS or D5W; inject over 4-5 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W; administer over 15-30 minutes. Must be administered by an infusion pump.
- Continuous IV infusion: use a dilute solution of 1 mg/mL. Must be administered by an infusion pump.
- IM (preferred over SC if repeat doses).
- SC (NOT preferred over IM if repeat doses): may cause local tissue irritation and induration.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: pruritus, flushing, urticaria (histamine-release reaction); anaphylaxis.
- Cardiovascular: facial flushing, bradycardia, tachycardia, palpitation, hypotension.
- GI: nausea, vomiting, constipation.
- CNS: dizziness, sedation, seizures.
- Respiratory depression.
- Local reactions: pain; irritation and induration with SC injection.

Antidote: naloxone (for respiratory depression).

DOSAGE
- Adults:
  - For preoperative sedation: 50-100 mg IM/SC 30-90 minutes before anesthesia.
  - For pain: 50-150 mg IM/SC q3-4h prn; use a lower dose when administered intermittent IV. For continuous IV infusion: 15-35 mg/hr.
  - For shaking chills: 0.5 mg/kg IV 20 minutes before expected time of onset of shaking chills OR 25-50 mg IV after onset of chills.
  - Patient is at increased risk of CNS toxicity from the normeperidine metabolite, when meperidine is used for longer than 48 hours or in total dosage exceeding 600 mg/24 hours.

- Pediatrics:
  - For pain: over 6 months of age: 0.8-1 mg/kg/dose IV/IM/SC q2-3h prn. Maximum dose: 75 mg.

- Dosage in renal impairment:

CrCl (mL/min)	50-10	less than 10
Dose	75%	do not use due to accumulation of toxic metabolite (normeperidine)

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Stable for 28 days at room temp or in the fridge, protected from light, in D5W or NS at concentrations from 0.25 to 30 mg/mL when stored in plastic syringes.
- Stable for 24 hours at room temp in NS at a concentration of 2.5 mg/mL in PVC bags.
- Stable for 36 hours at room temp in D5W at a concentration of 1.2 mg/mL in PVC bags.

MISCELLANEOUS
- 75 mg meperidine IM = 10 mg morphine IM.
- Contraindicated within 14 days of monoamine oxidase inhibitors administration.

REFERENCES
1, 2, 4, 5, 40, 82, 95, 135, 216, 293.

Full revision 2016; limited revision 2017, 2020

NAME OF MEDICATION
MEROPENEM

OTHER NAMES
Merrem ®

CLASSIFICATION
Antibiotic - carbapenem

INDICATIONS

- Treatment of the following infections when caused by susceptible organisms: skin and skin structure infections, intra-abdominal infections, meningitis, respiratory tract infections, septicemia, urinary tract infections and febrile neutropenia.

ADMINISTRATION

- Reconstitute vial with SWFI (10 mL per 500 mg vial, 20 mL per 1 g vial) for a final concentration of 50 mg/mL. Shake vial until dissolution and then allow it to stand until solution is clear.
- IV direct: physician or RN. Administer over 5 minutes.
- Intermittent IV infusion: dilute in NS or D5W and infuse over 15-30 minutes.
- Continuous IV infusion (NOT preferred): dilute in NS; change solution regularly due to limited stability; refer to Compatibility, Stability section. (At TOH, infuse each 500 mg minibag over 6 hours for continuous infusion).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, Stevens-Johnson syndrome, toxic epidermal necrolysis, drug reaction with eosinophilia and systemic symptoms (DRESS), erythema multiforme, acute generalized exanthematous pustulosis, anaphylaxis.
- GI: diarrhea, nausea, vomiting.
- CNS: headache, seizures (rare).
- Anemia.
- Local reactions: inflammation, phlebitis, thrombophlebitis, pain, edema at injection site.

DOSAGE

- Adults:
  - Usual dose 500-1000 mg IV q8h, up to 2000 mg IV q8h for meningitis.
  - Dosage of 500 mg IV q6h has been found as effective clinically and microbiologically as a regimen of 1 g IV q8h and is significantly less expensive.

- Pediatrics:
  - Over 3 months of age and weighing up to 50 kg: 10-40 mg/kg IV q8h.
  - Weighing more than 50 kg: use adult doses.

- Dosage in renal impairment, when dosed according to product monograph:

CrCl (mL/min)	greater than 50	50-26	25-10	less than 10
Dose (mg)	500-2000	500-2000	250-1000	250-1000
Interval (hr)	8	12	12	24 *

- Dosage in renal impairment for 500 mg q6h dosing:

CrCl (mL/min)	50 and greater	49-25	24-10	less than 10
Dose (mg)	500 mg	500 mg	500 mg	500 mg
Interval (hr)	6	8	12	24 *

* removed by dialysis, therefore give after dialysis

…/Cont.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Vials reconstituted with SWFI are stable for 3 hours at room temp and up to 16 hours in the fridge.
- Stable for 22 hours in NS and 4 hours in D5W at room temp at a concentration of 1 mg/mL in PVC bags.
- Stable for 7 days in NS and 24 hours in D5W in the fridge at a concentration of 1 mg/mL in PVC bags.
- Stable for 12 hours at 25°C in NS at concentrations of 6 mg/mL, 8 mg/mL and 12 mg/mL in polyolefin containers.
- Stable for 12 hours at room temp in NS at a concentration of 10 mg/mL in PVC bags.
- Stable for 5 days in the fridge in NS at concentrations of 10 mg/mL and 20 mg/mL in PVC bags.
- Stable for 7 hours at 22°C in NS at a concentration of 14.3 mg/mL in PVC bags.
- Stable for 17 hours in NS at room temp at a concentration of 22 mg/mL in PVC bags.
- Stable for 3 days in NS, and 24 hours in D5W in the fridge at a concentration of 22 mg/mL.

MISCELLANEOUS

- May exhibit cross-allergy with penicillins or cephalosporins.
- Each gram of meropenem provides 3.92 mmol or 90.2 mg of sodium.

REFERENCES

1, 4, 5, 57, 82, 95, 123, 135, 143, 208, 261, 443, 501, 502, 523, 524.

Full revision 2016; limited revision 2017, 2018, 2020

NAME OF MEDICATION
MESNA

OTHER NAMES
Sodium mercaptoethane sulfonate, Uromitexan ™

CLASSIFICATION
Uroprotector

INDICATIONS

- Prevention of hemorrhagic cystitis associated with oxazaphosphorines (ifosfamide, cyclophosphamide).

ADMINISTRATION

- IV direct: physician or RN; dilute dose with NS, D5W, D5-1/2NS, or RL to a concentration of 20 mg/mL; inject over 3-5 minutes.
- Intermittent IV infusion: dilute with NS, D5W, D5-1/2NS, or RL to a concentration of 1 to 20 mg/mL; infuse over 15-30 minutes.
- Continuous IV infusion: dilute in an appropriate volume of NS, D5W, D5-1/2NS, or RL to a concentration of 1 to 20 mg/mL; infuse as per protocol.

POTENTIAL ADMINISTRATION HAZARDS

Note: at recommended doses, there are usually no side effects.
- Hypersensitivity: urticaria, rash.
- Cardiovascular: hypotension, tachycardia, flushing.
- GI: nausea, vomiting, diarrhea, abdominal pain, unpleasant taste in the mouth.
- CNS: headache, dizziness.
- Flu-like symptoms.
- Local reactions: thrombophlebitis and irritation (when undiluted).
- Extravasation hazard: irritant (when undiluted).

DOSAGE

- Different dosage regimens have been tried.
- Intermittent dosing regimen: mesna total dose equivalent to 60% of the oxazaphosphorine daily dose, given in 3 divided doses, i.e., mesna dose equivalent to 20% of the oxazaphosphorine dose given at 0, 4 and 8 hours.
- Continuous infusion: mesna dose equivalent to 20% of the oxazaphosphorine dose given IV direct as a loading dose, followed by a continuous infusion of the drug at a dose equivalent to 40% of the oxazaphosphorine dose, administered concomitantly with the oxazaphosphorine. The infusion must be continued for 8-24 hours after completion of the chemotherapy infusion.
- Mesna total daily dose is usually equivalent to 60-160% of the total daily dose of the oxazaphosphorines.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Stable for 48 hours at room temp in D5-1/2NS, RL and NS at concentrations of 1-20 mg/mL.
- Stable for 24 hours at room temp in D5W at a concentration of 1 mg/mL.
- Stable for 48 hours at room temp in D5W at a concentration of 20 mg/mL.
- Stable for 24 hours at room temp and 48 hours in the fridge in D5W and NS once diluted (no concentration specified; manufacturer's information)

MISCELLANEOUS

- May cause false positive reactions in tests for ketone bodies in urine.
- Oral mesna dose = double IV dose.

REFERENCES

1, 5, 95, 129, 165, 208.

Full revision 2016; limited revision 2018

NAME OF MEDICATION
METHOTREXATE

OTHER NAMES
Amethopterin sodium, Metoject ®

CLASSIFICATION
Antineoplastic, Immunomodulator

INDICATIONS

- Acute lymphocytic leukemia, choriocarcinoma, carcinoma of head and neck area, mycosis fungoides, breast cancer, bladder cancer, gastric cancer, lymphoma and sarcoma.
- Treatment of psoriasis, rheumatoid arthritis and Crohn's disease.
- Management of ectopic pregnancy.

ADMINISTRATION

- IV direct: physician or RN; undiluted; inject at a rate of 10 mg/min into the tubing of a freely running IV solution of D5W or NS.
- Intermittent IV infusion, Continuous IV infusion: dilute with D5W or NS; infuse as per protocol, range from 20 minutes to 42 hours.
- IM.
- SC (usually for patients with psoriasis, rheumatoid arthritis or Crohn's disease).

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis (rare).
- GI: nausea, vomiting, stomatitis.
- CNS: seizures with high doses, dizziness, headache.
- Dermatologic: radiation recall reaction, vasculitis, rash.
- Hematologic: leukopenia, thrombocytopenia, anemia.
- Hepatic: elevated liver function tests (usually transient and asymptomatic).
- Ophthalmic: blurred vision.
- Respiratory: cough, pleuritic chest pain, pulmonary alveolar hemorrhage.
- Hyperuricemia, tumour lysis syndrome.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

Antidote for overdosage: leucovorin. Refer to leucovorin calcium in this manual for more details.

DOSAGE

- Oncology indications: usual doses: 25-40 mg/m^2 IV. High-dose therapy with leucovorin "rescue": 100 mg-15 g/m^2 IV have been given by infusion.
- Psoriasis and rheumatoid arthritis: 10-25 mg IM/SC weekly, with a maximum of 50 mg per week.
- Crohn's disease: 15-25 mg IM/SC weekly.
- Ectopic pregnancy: 50 mg/m^2 IM; may be repeated once or twice depending on the serum beta-HCG level decline in the successive weeks.
- Dosage in renal impairment:

Method 1:	CrCl (mL/min)	80-61	60-51	50-10	less than 10
	Dose	75%	70%	30-50%	avoid

Method 2:	CrCl (mL/min)	greater than 50	50-10	less than 10
	Dose	100%	50%	avoid

- Dosage in hepatic impairment: if bilirubin is 2.5-4 x ULN or transaminases are greater than 3 x ULN, give 75% of usual dose. Discontinue dose if bilirubin greater than 4 x ULN.
- Consult specific protocol.

.../Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Diluted solutions should be protected from light.
- Stable for 24 hours at room temp and 30 days in the fridge, protected from light, in D5W or NS at concentrations of 0.225 mg/mL and 24 mg/mL in PVC containers.

MISCELLANEOUS

- Formulation preserved with benzyl alcohol should not be used for high-dose therapy or intrathecal injection; in these cases, preservative-free formulations should be used.
- High-dose methotrexate therapy requires leucovorin rescue, hydration and alkalinization of urine with sodium bicarbonate to reduce toxicity.

REFERENCES

1, 4, 5, 37, 40, 95, 129, 143, 165, 216, 482.

Full revision 2016; limited revision 2017, 2018, 2020, 2021

NAME OF MEDICATION
METHOTRIMEPRAZINE

OTHER NAMES
Levomepromazine, Nozinan ®

CLASSIFICATION
Antipsychotic

INDICATIONS

- Management of moderate to severe pain.
- Treatment of nausea and vomiting.
- In palliative care patients to control agitation or for sedation.

ADMINISTRATION

- Intermittent IV infusion: ***blood pressure monitoring***. Dilute dose in 50 mL of NS or D5W; infuse over 30 minutes.
- Continuous IV infusion: ***blood pressure monitoring***. Dilute dose in 250-500 mL of NS or D5W; infuse as per Dosage section.
- IM (preferred).
- SC, SC infusion: for palliative care patients. Must be administered by an infusion pump.
- Only administer to non-ambulatory patients; patient must be kept supine for at least 6 hours but preferably 12 hours after administration.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: orthostatic hypotension presenting as dizziness, fainting, or weakness. Usually occurs within 10-20 minutes following injection and may last up to 12 hours. (**Antidote**: norepinephrine or phenylephrine; not epinephrine). Tachycardia; may prolong QT interval.
- CNS: sedation, confusion, seizures, extra-pyramidal symptoms.
- Anticholinergic effects: dry mouth, urinary retention, constipation.
- Local reactions: irritation at SC injection site.

DOSAGE

Adults:
- For pain: 10-25 mg IV/IM q8h.
- For nausea and vomiting: 25 mg IM q6-8h OR 5-25 mg/day as a SC infusion.
- For agitation or sedation in palliative care patients: 12.5-50 mg SC/IV q4-6h, up to 300 mg/24 hours for agitation.

Pediatrics:
- For pain, nausea or vomiting: 0.0625-0.125 mg/kg/day IM in one or divided doses.
- In the context of palliative care: 0.0625 mg/kg/day in 250 mL D5W as a continuous IV infusion.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp, protect from light.
- Discard solutions if markedly discoloured.
- Compatible with D5W and NS.

MISCELLANEOUS

- May potentiate the action of other CNS depressants (barbiturates, analgesics, opioids, antihistamines); dose reductions by half are suggested during concomitant use.

REFERENCES

3, 4, 5, 95, 135, 260, 457.

Full revision 2016; limited revision 2019, 2020

NAME OF MEDICATION
METHYLENE BLUE

OTHER NAMES
Tetramethylthionine chloride trihydrate

CLASSIFICATION
Antidote, Diagnostic agent

INDICATIONS
- For the treatment of idiopathic and drug-induced methemoglobinemia.
- To visualize parathyroid gland before surgery.
- For refractory hypotension associated with septic shock and/or cardiac surgery.
- Treatment and prevention of ifosfamide-induced encephalopathy.

ADMINISTRATION
- IV direct: physician only; undiluted; inject over at least 5 minutes.
- Intermittent IV infusion: dilute in an appropriate amount of D5W (50-500 mL) and infuse over 10-60 minutes; refer to Dosage section.
- Continuous IV infusion: dilute in an appropriate amount of D5W (at TOH, suggest 250 to 500 mL); rate as per Dosage section.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: urticaria, hypotension, tachycardia, bronchospasm, anaphylaxis.
- Cardiovascular: precordial pain, hypertension.
- GI: nausea, vomiting, abdominal pain.
- CNS: dizziness, headache, confusion.
- Serotonin syndrome if administered in a patient already receiving drugs with serotonin reuptake inhibition properties (e.g., SSRIs [fluoxetine, fluvoxamine, sertraline, paroxetine, citalopram, escitalopram], SNRIs [venlafaxine, desvenlafaxine, duloxetine], clomipramine, etc): agitation or diaphoresis or hypertonia accompanied with pyrexia (above 38˚C), and tremor, hyperreflexia or clonus.
- Diaphoresis.
- Additional methemoglobinemia may occur if methylene blue is injected too fast or if high doses are used.
- Transient blue colouring of the skin, feces and urine.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE
- Methemoglobinemia: 1-2 mg/kg IV or 25-50 mg/m^2 IV; repeat dose after 60 minutes if needed.
- To visualize parathyroid gland: 7.5 mg/kg in 500 mL of solution infused IV over 45-60 minutes prior to surgery.
- Refractory hypotension: 1-2 mg/kg IV; if needed, after a 2-hour interval, start an IV infusion of 0.25 mg-2 mg/kg/hr titrated to arterial blood pressure, for up to 6 hours.
- Ifosfamide-induced encephalopathy: <u>prevention</u>: 50 mg (at TOH, suggest to dilute in 50 mL of D5W) IV q6-8h during ifosfamide therapy; <u>treatment</u>: 50 mg (at TOH, suggest to dilute in 50 mL of D5W) IV once OR q4-8h until symptoms resolve.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Compatible with D5W.
- Avoid dilution is NS as may cause precipitation.

MISCELLANEOUS
- If patient is taking drugs with serotonin reuptake inhibition properties, consider stopping them before administration of methylene blue to avoid a serotonin syndrome reaction. A washout period equivalent to at least 4 to 5 half-lives is recommended. Treatment may be restarted 24 hours after the last dose of methylene blue.
- Provided as a 1% solution (10 mg/mL).
- Use with caution in patients with glucose-6-phosphate dehydrogenase deficiency.
- Caution in severe renal dysfunction.

REFERENCES
1, 50, 58, 63, 79, 80, 95, 135, 165, 208, 367, 462, 492, 496.

Full revision 2016; limited revision 2018, 2019, 2020, 2021

NAME OF MEDICATION
METHYLNALTREXONE

OTHER NAMES
Relistor ®

CLASSIFICATION
Opioid antagonist

INDICATIONS

- Treatment of opioid-induced constipation in patients with advanced illness, receiving palliative care and had an insufficient response to laxative therapy.
- Treatment of opioid-induced constipation in patients with chronic non-cancer pain.

ADMINISTRATION

- SC: in the upper arm, abdomen or thigh. Patient should be seated or recumbent during administration. Rotate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- GI: nausea, diarrhea, flatulence, abdominal pain. Increased risk of GI perforation (stomach, duodenum, colon) if patient has localized or diffused reduction of structural integrity of GI mucosa, or has GI lesions; severe persistent abdominal symptoms could be signs of GI perforation.
- CNS: dizziness.

DOSAGE

- Opioid-induced constipation with advanced illness: dosage based on patient's weight, administered as a SC injection every other day as needed with a maximum of 1 dose/24 hours:
  - 33 to less than 38 kg: 6 mg (0.3 mL)
  - 38 to less than 62 kg: 8 mg (0.4 mL)
  - 62 to 114 kg: 12 mg (0.6 mL)
  - 115 to 126 kg: 18 mg (0.9 mL)
  - For weight less than 33 kg or greater than 126 kg: 0.15 mg/kg with **volume** to be administered as per following equation:
    Patient's weight (kg) x 0.0075 = volume (mL) to be injected; round to the nearest 0.1 mL.
- Opioid-induced constipation with chronic non-cancer pain: 12 mg SC once daily.
- Dosage in renal impairment: 50% of the dose if CrCl is less than 30 mL/min.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp; do not freeze; protect from light.

MISCELLANEOUS

- Caution in patients with cancer, GI malignancy or ulcer, Ogilvie's syndrome or receiving concomitant therapy with NSAIDs, corticosteroids or bevacizumab as these conditions increase the risk of GI perforation with methylnaltrexone.

REFERENCES

1, 5, 95, 135, 243.

Full revision 2016

NAME OF MEDICATION
MethylPREDNISolone (acetate)

OTHER NAMES
Depo-Medrol ®

CLASSIFICATION
Corticosteroid

Do NOT confuse methylprednisolone acetate with methylprednisolone sodium succinate.
This monograph is specific to methylprednisolone ACETATE.

INDICATIONS

- For local or systemic anti-inflammatory, antirheumatic or antiallergic effects for the treatment of corticosteroid-responsive disorders.

ADMINISTRATION

- IM: do NOT inject in the deltoid muscle due to a high incidence of SC atrophy. Do NOT inject into areas that have evidence of acute local infection.
- Intraarticular, intrasynovial, intralesional and at soft tissue sites: physician only; use **only** preservative-free product (i.e., without benzyl alcohol).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, angioedema, anaphylactic/anaphylactoid reaction (rare).
- Cardiovascular: fluid retention, hypertension, CHF.
- GI: peptic ulcer, GI hemorrhage.
- CNS: delirium, mood changes, headache, insomnia, depression, seizure.
- Electrolyte disturbances: sodium retention, hypokalemia.
- Endocrine and metabolic: hyperglycemia.
- Increased susceptibility to infections.
- Local reactions: injection site reaction.

DOSAGE

- For systemic effect: dosage variable depending upon clinical condition; usual range is 40-120 mg IM at weekly to every 2-week intervals, as needed.
- For local effect: 4-80 mg depending on size of tissue/joint injected. Dose may be repeated if needed at intervals of 1 to 5 or more weeks. Refer to manufacturer's recommendations.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30ºC. Do not freeze.

MISCELLANEOUS

- Methylprednisolone acetate IM and methylprednisolone oral daily doses are equivalent on a mg per mg basis; for weekly IM administration, multiply methylprednisolone oral daily dose by 7.
- Each mL contains 28-29.5 mg of polyethylene glycol.

REFERENCES

1, 5, 135.

New monograph 2021

NAME OF MEDICATION
MethylPREDNISolone (sodium succinate)

OTHER NAMES
Solu-Medrol ®

CLASSIFICATION
Corticosteroid

Do NOT confuse methylprednisolone sodium succinate with methylprednisolone acetate.
This monograph is specific to methylprednisolone SODIUM SUCCINATE.

INDICATIONS

- For anti-inflammatory or immunosuppressant effect.
- Situations when a rapid and intense hormonal effect is desired, e.g., hypersensitivity reactions, status asthmaticus, organ transplants, cerebral edema, hypercalcemia, spinal cord injury, shock, multiple sclerosis, lupus nephritis.

ADMINISTRATION

- Reconstitute the 40 mg, 125 mg, 500 mg and 1 g vial with 1 mL, 2 mL, 7.8 mL and 15.6 mL respectively of bacteriostatic water for injection or SWFI.
- IV direct: physician or RN; undiluted; for doses of 125 mg or less; inject over at least 5 minutes.
- Intermittent IV infusion: for doses greater than 125 mg; dilute dose in 50-100 mL of D5W or NS; infuse over 15-60 minutes or according to protocol.
- Continuous IV infusion.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis (rare).
- Cardiovascular: cardiac arrhythmias and/or circulatory collapse and/or cardiac arrest following rapid administration of high doses, bradycardia with large doses, fluid retention, hypertension, CHF.
- GI: peptic ulcer, GI hemorrhage.
- CNS: delirium, mood changes, headache.
- Electrolyte disturbances: sodium retention, hypokalemia.
- Endocrine and metabolic: hyperglycemia.
- Increased susceptibility to infections.

DOSAGE

Adults:
- Dosage variable depending upon clinical condition; usual range is 10 to 250 mg IM/IV q4-24h.
- Adjunctive therapy in life-threatening conditions (e.g., shock): 30 mg/kg IV over a period of at least 30 minutes. The large doses may be repeated q4-6h for up to 48 hours. Also 100-250 mg q2-6h have been recommended.
- Status asthmaticus: 40-80 mg/day IV in 1 or 2 divided doses until peak expiratory flow is 70% of predicted or personal best.
- Multiple sclerosis exacerbations: 160 mg IV once daily for 7 days, followed by 64 mg every other day for 1 month. Dose regimens of 500 mg or 1 g IV daily for 3 to 5 days have also been used.
- Lupus nephritis: as pulse therapy; 500 mg to 1 g IV over 60 minutes daily for 3 days.

Pediatrics:
- For anti-inflammatory or immunosuppressive effect: 0.5-1.7 mg/kg/day IM or IV in divided doses q6-12h; dosage to be determined by severity of condition and response to therapy.
- Status asthmaticus: 0.5-1 mg/kg IV q12h (maximum 60 mg/day) until peak expiratory flow is 70% of predicted or personal best OR 1 mg/kg IV q6h on day 1.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Reconstituted solutions are stable for 48 hours at room temp when reconstituted with bacteriostatic water and for 24 hours when reconstituted with SWFI (due to risk of microbial contamination).
- Compatible with D5W, NS, D5-NS and D5-1/2NS.
- Stable for 48 hours at room temp in D5W, NS, D5-1/2NS at a concentration of 0.25 mg/mL.
- Stable for 4 days at room temp and for 3 weeks in the fridge in NS at concentration of 10 mg/mL stored in polypropylene syringes.

MISCELLANEOUS

- 4 mg methylprednisolone is equivalent to 20 mg hydrocortisone.
- Solu-Medrol ® 40 mg Act-O-Vial ® contains lactose produced from cow's milk; it is contraindicated in patients with a known hypersensitivity to cow's milk or its components or other dairy products.

REFERENCES

1, 4, 5, 6, 40, 82, 83, 95, 135, 208, 356, 357.

NAME OF MEDICATION
METOCLOPRAMIDE

OTHER NAMES
Maxeran ®, Reglan ®

CLASSIFICATION
Antiemetic

INDICATIONS

- An adjunct in the management of delayed gastric emptying associated with subacute and chronic gastritis or following vagotomy and pyloroplasty and other surgical procedures if oral administration is not possible.
- An adjunct to facilitate small bowel intubation.
- During radiographic examination of the upper GI tract.
- For the prevention of nausea and vomiting associated with cancer chemotherapy or surgery.
- For diabetic or post-surgical gastric stasis.

ADMINISTRATION

- IV direct: physician or RN; for doses of 10 mg or less; administer undiluted over 1-2 minutes.
- Intermittent IV infusion (preferred over IV direct): further dilute in 50-100 mL of D5W or NS; infuse over at least 15 minutes.
- IM.
- SC.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, bronchospasm, angioedema.
- Cardiovascular: atrioventricular block, CHF, hypotension, hypertension, arrhythmias, flushing (with large IV doses).
- GI: nausea, diarrhea.
- CNS: drowsiness, restlessness, headache, dizziness.
- Transient akathisia manifested clinically as an urge to move the extremities or torso and anxiety with agitation (more frequent and more severe with IV direct injection).
- Extrapyramidal reactions including acute dystonic reactions (**Antidote**: diphenhydramine 50 mg IM or benztropine 1-2 mg IV/IM).
- Fatigue.

DOSAGE

- Adults:
  - Usual dose: 10-20 mg IV/IM, may be repeated up to 4 times daily.
  - To facilitate small bowel intubation: 10 mg IV as a single dose.
  - Radiographic examination of the upper GI tract: 10 mg IV as a single dose.
  - Cancer chemotherapy nausea and vomiting: 1-2 mg/kg IV; may repeat twice q2h then three times q3h. Some clinicians use doses as high as 2.75 mg/kg (by IV infusion). Administer first dose 30 minutes prior to chemotherapy.
- Pediatrics:
  - Radiographic examination of small bowel or for intubation: 6-14 years: 2.5-5 mg IV as a single dose; younger than 6 years of age: 0.1 mg/kg IV as a single dose.
  - Gastroesophageal reflux: 0.1-0.2 mg/kg IV q6h as needed.
- Dosage in renal impairment: when initiating therapy, 50% of usual recommended doses if CrCl is less than 40 mL/min.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Protect ampoules from light and store at room temp. Solution should be colourless.
- Incompatible with sodium bicarbonate due to modification of pH.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 48 hours at room temp in 50 mL of NS, D5W, D5-1/2NS, Ringer's and RL when protected from light or for up to 24 hours when stored at these temperatures and exposed to normal light.
- Stable for 90 days at 4°C, 60 days at 23°C and 7 days at 32°C at a concentration of 5 mg/mL in polypropylene syringes.
- Stable for 21 days at room temp in NS at a concentration of 0.5 mg/mL in polypropylene syringes.

MISCELLANEOUS

- Anticholinergic drugs will antagonize the effects of metoclopramide.
- Can modify absorption of some medications administered orally.
- Not recommended for use in patients with epilepsy or extrapyramidal symptoms.

REFERENCES

1, 4, 5, 6, 40, 95, 135, 208, 336, 420, 457, 546.

NAME OF MEDICATION
METOPROLOL

OTHER NAMES
Betaloc ®, Lopresor ®

CLASSIFICATION
Beta-blocker

INDICATIONS

- Management of acute myocardial infarction.
- Treatment of cardiac arrhythmias.
- For ventricular rate control or treatment of hypertension when oral administration is not possible.
- Prophylaxis of atrial fibrillation after cardiac surgery.

ADMINISTRATION

- IV direct: physician for first dose only; RN may administer subsequent doses; ***cardiac monitoring; continuous BP monitoring***. Undiluted; inject over at least 1 minute.
- Intermittent IV infusion (when oral administration is not possible): ***cardiac monitoring***; ***continuous BP monitoring***. Dilute dose in 50 mL of D5W or NS and administer over 30-60 minutes. Must be administered by an infusion pump.
- Continuous IV infusion: ***cardiac monitoring; continuous BP monitoring***. Infuse undiluted or further dilute in a compatible solution as per Dosage section. At TOH: transfer 15 mg (15 mL from metoprolol 1 mg/mL vials/ampoules) to an empty sterile minibag for a concentration of 1 mg/mL OR withdraw 15 mL from a 50 mL minibag of D5W or NS and discard, then add 15 mg (15 mL from metoprolol 1 mg/mL vials/ampoules) to the minibag to obtain a final concentration of 0.3 mg/mL. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, urticaria.
- Cardiovascular: bradycardia (**Antidote**: atropine 0.25-0.5 mg; use isoproterenol if atropine not effective), congestive heart failure, hypotension, heart block.
- CNS: vertigo, dizziness, sedation.
- Respiratory: bronchospasm, dyspnea.
- Fatigue.

DOSAGE

- Acute myocardial infarction: 5 mg IV direct followed by two additional 5 mg doses at approximately 2-5 minute intervals. Thereafter, initiate oral metoprolol therapy 15 minutes after last IV dose.
- Cardiac arrhythmias: 2.5-5 mg IV direct every 2-5 minutes with a maximum of 15 mg over 10-15 minutes.
- Ventricular rate control/hypertension: 1.25-5 mg IV q6-12h, with a maximum of 15 mg IV q3-6h.
- Prophylaxis of atrial fibrillation after cardiac surgery: 1-3 mg/hr IV; to titrate as per patient response.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Stable for 36 hours at room temp in NS and D5W at a concentration of 0.3 mg/mL in PVC containers.

MISCELLANEOUS

- Concomitant use of IV metoprolol with IV diltiazem or IV verapamil may cause significant cardiac depression. Therefore, co-administration (within a few hours) is contraindicated.
- Equivalent beta-blocking effect when switching between oral and IV route of administration: can use 2.5:1 (oral: IV) ratio (example: 25 mg po q12h to 5 mg IV q6h). However, this ratio varies from one patient to the other (range 2:1 to 5:1).

REFERENCES

1, 4, 5, 40, 95, 107, 135.

Full revision 2016; limited revision 2017, 2019; 2020

NAME OF MEDICATION
MetroNIDAZOLE

OTHER NAMES
Flagyl ®

CLASSIFICATION
Antibiotic

INDICATIONS

- Treatment of infections due to susceptible anaerobic bacteria.
- Treatment of amebic liver abscess.
- Treatment of *Clostridium difficile*-associated diarrhea when oral administration is not possible.
- Prevention of postoperative anaerobic sepsis in patients undergoing colorectal surgery.

ADMINISTRATION

- Intermittent IV infusion: use premixed bags of 500 mg/100 mL (5 mg/mL); infuse each 500 mg over 20 minutes.
- Consult Medical staff rules and regulations MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis) for specific dosing regimens and infusion time.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus.
- GI: diarrhea, nausea, vomiting, abdominal discomfort, metallic taste.
- CNS: seizures, ataxia, dizziness, drowsiness, headache, peripheral neuropathy (most commonly reported as numbness and paresthesias in extremities).
- Local reactions: thrombophlebitis.

DOSAGE

- Adults:
  - Treatment of infections: 500 mg IV q8-12h OR 7.5 mg/kg IV q6-8h. For anaerobic infections, may initiate treatment with a loading dose of 15 mg/kg IV. Maximum 4 g/day.
  - Prophylaxis in colorectal surgery: 15 mg/kg IV 60 minutes before skin incision, followed if necessary by 7.5 mg/kg IV at 6 and 12 hours after the initial dose OR 500 mg IV within 60 minutes prior to surgical incision.
- Pediatrics (anaerobic infections): 30-40 mg/kg/day IV divided q6-8h. Maximum dose: 4 g/day.
- Dosage in renal impairment: no need to adjust dosage.
- Dosage in hepatic impairment: decrease dose or frequency if severe liver impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Compatible with D5W, NS, RL.

MISCELLANEOUS

- Consumption of alcohol may produce a disulfiram-like reaction.

REFERENCES

1, 2, 5, 40, 82, 95, 135, 216, 422, 444, 497.

Full revision 2016; limited revision 2020

NAME OF MEDICATION
MICAFUNGIN

OTHER NAMES
Mycamine ®

CLASSIFICATION
Antifungal

INDICATIONS

- Treatment of patients with candidemia, acute disseminated candidiasis, candidal peritonitis and abscess infections, or esophageal candidiasis.
- Prophylaxis of *Candida* infections in patient undergoing hematopoietic stem cell transplantation (HSCT).

ADMINISTRATION

- Reconstitute each 50 mg or 100 mg vial with 5 mL of NS or D5W, without a bacteriostatic agent to get a concentration of 10 mg/mL or 20 mg/mL, respectively. Gently dissolve the powder by swirling vial to minimize excessive foaming. Do NOT shake.
- Intermittent IV infusion: dilute dose in 100 mL NS or D5W. For pediatrics, dilute dose in an appropriate volume to obtain a final concentration between 0.5 to 4 mg/mL. For concentrations above 1.5 mg/mL, administer through a central catheter. Flush IV line with NS before and after infusion. Administer over 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactoid and anaphylaxis, including shock.
- Histamine-mediated reactions: rash, pruritus, facial swelling, vasodilation. Occur most often with a rapid infusion over less than 60 minutes.
- GI: nausea, vomiting, abdominal pain, diarrhea.
- Hematologic: leukopenia, neutropenia, thrombocytopenia, hemolysis, hemolytic anemia.
- Hepatic: increases in ALT, AST, alkaline phosphatase and bilirubin.
- Rigors, fever, headache, delirium.
- Local reactions: inflammation, phlebitis, thrombophlebitis; occur more often if micafungin is infused peripherally.

DOSAGE

- Adults:
  - Treatment of candidemia and other *Candida* infections (other than esophageal candidiasis): 100 mg IV once daily. Mean duration of treatment is 15 days.
  - Treatment of esophageal candidiasis: 150 mg IV once daily. Mean duration of treatment is 15 days.
  - Prophylaxis of *Candida* infections in HSCT recipients: 50 mg IV once daily.

- Pediatrics (4 months and older):
  - Treatment of candidemia and other *Candida* infections (other than esophageal candidiasis): 2 mg/kg IV once daily with a maximum daily dose of 100 mg.
  - Treatment of esophageal candidiasis: weight of 30 kg or less: 3 mg/kg IV once daily; weight greater than 30 kg: 2.5 mg/kg IV once daily with a maximum daily dose of 150 mg.
  - Prophylaxis of *Candida* infections in HSCT recipients: 1 mg/kg IV once daily with a maximum daily dose of 50 mg.

- Dosage in renal impairment: no need to adjust dosage.
- Dosage in hepatic impairment: no need to adjust dosage in mild to moderate hepatic impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp; protect from light.
- Reconstituted solution is stable for 24 hours at room temp.

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COMPATIBILITY, STABILITY (Cont.)

- Diluted solution is stable for 24 hours at room temp, protected from light. However, it is not necessary to cover the infusion drip chamber or the tubing.
- Stable for 4 days at room temp, protected from light, in NS or D5W at concentrations of 0.25 mg/mL and 4 mg/mL in plastic bags.
- Stable for 7 days in the fridge, protected from light, in NS at concentrations of 0.125 mg/mL and 1 mg/mL in plastic bags.

MISCELLANEOUS

- Efficacy of micafungin against infections caused by fungi other than *Candida* has not been established.
- Micafungin has not been adequately studied for endocarditis, osteomyelitis and meningitis due to *Candida*.

REFERENCES

1, 5, 40, 143, 158.

NAME OF MEDICATION
MIDAZOLAM

OTHER NAMES
Versed ®

CLASSIFICATION
Benzodiazepine

INDICATIONS

- For induction of anesthesia.
- As a sedative agent.
- For agitation, seizures and sedation in palliative care patients.

ADMINISTRATION

- **Prerequisites for IV injection (direct, continuous IV infusion) or IM administration are either**
  i) ventilator support and ECG monitor or
  ii) pulse oximeter; supplemental oxygen should be available if needed.
- Rapid IV direct: physician only for induction of anesthesia; ***respiratory support***; use 1 mg/mL solution; inject over 20-30 seconds.
- IV direct: physician or RN for sedation; ***respiratory support***; use 1 mg/mL solution; inject over 2-3 minutes.
- Continuous IV infusion: ***respiratory support***; dilute with NS or D5W for a final concentration of 0.5 mg/mL or infuse undiluted (1 mg/mL). At TOH: withdraw 20 mL from a 100 mL minibag of D5W or NS and discard, then add 100 mg (20 mL from midazolam 5 mg/mL vials/ampoules) to the minibag to obtain a final concentration of 1 mg/mL. Titrate rate to patient response. Must be administered by an infusion pump.
- IM.
- SC, SC infusion: for palliative care patients. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylactoid, rash, pruritus, hives.
- Cardiovascular: hypotension, increase or decrease in heart rate.
- GI: hiccups, nausea, vomiting.
- CNS: headache, drowsiness, excessive sedation, rarely paradoxical agitation and involuntary movements.
- Respiratory depression and/or arrest, apnea, coughing; rarely tachypnea.
- Local reactions: pain with IM injection; tenderness, erythema, induration at IV administration site and pain during IV injection.

DOSAGE

- Induction of anesthesia (administer over 20-30 seconds):
  - Premedicated patients: Initial dose: - Adults: range 0.15-0.35 mg/kg IV.
    - 55 years or older or patients with severe systemic disease: 0.15-0.2 mg/kg IV.
  - Unpremedicated patients: Initial dose: - Adults: 0.3-0.35 mg/kg IV.
    - 55 years of older: 0.3 mg/kg IV.
    - With severe systemic disease/debilitated: 0.15-0.25 mg/kg IV.
  - After approximately 2 minutes, increments of approximately 25% of initial dose may be given to complete induction.
  - Maintenance: administer in incremental doses of approximately 25% of initial induction dose when lightening of anesthesia is evident; repeat as necessary.
- Conscious sedation (administer over 2-3 minutes):
  - Initial dose: - Adult: 1-2.5 mg IV (approximately 0.015-0.03 mg/kg).
    - Elderly, chronically ill or debilitated: 1-1.5 mg IV.
  - After 2 minutes, dosage may be further titrated incrementally to a total dose of 5 mg. Elderly, chronically ill or debilitated patients or patients with limited pulmonary reserve may require 30% lower dose (total dose of 3.5 mg).
  - Maintenance: administer in increments of 25% of initial total dose.

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DOSAGE (Cont.)

- Preoperative sedation: 0.07-0.08 mg/kg IM 30-60 minutes preoperatively; decrease dose to 0.02-0.05 mg/kg IM for the elderly, chronically ill or debilitated.
- Infusion for sedation: loading dose of 0.01-0.05 mg/kg IV followed by 1-7 mg/hr IV OR 0.02-0.1 mg/kg/hr IV.
- Palliative care:
  - Intermittent injection: 0.5-5 mg SC/IV every 30-60 minutes prn OR 1-10 mg SC/IV q2-4h prn.
  - Continuous infusion: 0.2-10 mg/hr SC/IV; some patients may require up to 20 mg/hr SC/IV; may administer an initial loading dose of 1-5 mg SC/IV.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Compatible with D5W, NS and RL.
- Stable for 10 days at room temp, protected from or exposed to light, in NS at a concentration of 1 mg/mL in PVC bags.
- Stable for 27 days at room temp or in the fridge, protected from or exposed to light, in D5W at a concentration of 1 mg/mL in PVC or polyolefin bags.
- Ready-to-use solution of 1 mg/mL is stable for 1 year in the freezer (-20°C) and in the fridge (5°C) and for 90 days at room temp (25°C) in polypropylene syringes.
- Stable for 36 days at room temp, protected from light, at a concentration of 5 mg/mL (undiluted) in polypropylene syringes.

MISCELLANEOUS

REFERENCES

1, 4, 5, 6, 40, 95, 111, 135, 256, 457, 485, 486, 488, 489, 565.

Full revision 2016; limited revision 2019, 2020, 2021

NAME OF MEDICATION
MILRINONE

OTHER NAMES
Primacor ®

CLASSIFICATION
Inotropic agent

INDICATIONS

- Short-term management of severe congestive heart failure, including low output states following cardiac surgery.

ADMINISTRATION

- Intermittent IV infusion: ***cardiac monitoring, continuous BP monitoring***; for loading dose; undiluted or dilute in NS, 1/2NS or D5W to a total volume of 10 or 20 mL; administer over 10 minutes.
- Continuous IV infusion: ***cardiac monitoring***; dilute the 20 mg/20 mL vial with 180, 113 or 80 mL of NS, 1/2NS or D5W to get a final concentration of 100 mcg/mL, 150 mcg/mL or 200 mcg/mL respectively. At TOH: withdraw 20 mL from a 100 mL minibag of D5W or NS and discard, then add 20 mg (20 mL from milrinone 1 mg/mL vials) to the bag to obtain a final concentration of 200 mcg/mL.
- Note: loading dose may also be administered from the bag prepared for the continuous infusion if rate set up accordingly.
- Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash.
- Cardiovascular: arrhythmias (ventricular and supraventricular), hypotension, chest pain/angina.
- CNS: headache.

DOSAGE

- Loading dose (optional) of 50 mcg/kg IV over 10 minutes.
- Infusion dose of 0.125-0.75 mcg/kg/min IV; maximum dose of 1.13 mg/kg/day; titrate to clinical status.
- Dosage in renal impairment:

CrCl (mL/min/1.73 m^2)	50	40	30	20	10	5
Infusion dose (mcg/kg/min)	0.43	0.38	0.33	0.28	0.23	0.2

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Milrinone is a clear colourless to pale yellow solution.
- Stable for 14 days at room temp or refrigerated in D5W or NS at a concentration of 200 mcg/mL in PVC containers.
- Stable for 7 days at room temp in D5W, NS, 1/2NS, and RL at a concentration of 400 mcg/mL in PVC containers.
- Do not mix with furosemide solutions.

MISCELLANEOUS

REFERENCES

1, 2, 4, 6, 40, 135, 283.

Full revision 2016; limited revision 2017, 2018, 2019, 2020, 2021

NAME OF MEDICATION
MitoMYcin

OTHER NAMES
Mitomycin C, Mutamycin ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Adenocarcinoma of stomach and colon.
- Has also been used for breast, cervical, head and neck, pancreatic, lung (non-small cell) and anal cancers.

ADMINISTRATION

- Reconstitute the 5 mg and 20 mg vials with 10 mL and 40 mL SWFI, respectively, to obtain an approximate concentration of 0.5 mg/mL. Shake well until dissolved.
- IV direct: physician or RN; undiluted over 5-10 minutes into the tubing of a freely flowing IV solution of NS or D5W.
- Intermittent IV infusion: dilute with 50 mL of NS (preferred) or D5W; infuse over 15-30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- GI: nausea, vomiting.
- Hematologic: leukopenia, thrombocytopenia.
- Dermatologic: radiation recall reaction (rare); dermatitis, often manifested as palmar rash with desquamation.
- Local reactions: pain at injection site, induration, thrombophlebitis.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Various dosage regimens and protocols are used:
  - 10-15 mg/m^2 IV on day 1 or 3 every 4-8 weeks.
  - 10-20 mg/m^2 IV on day 1 every 6-8 weeks.
  - 2 mg/m^2/day IV for 5 days, stop for 2 days, repeat X 1 (i.e., to give on days 1-5 and days 8-12), every 6-8 weeks.
- Dosage in renal impairment: if CrCl is less than 10 mL/min, give 75% of usual dose.
- Dosage adjustement due to toxicity:
  - if leucocytes: 2-2.9 x 10^9/L and platelets: 25-74.9 X 10^9/L, give 70% of previous dose.
  - If leucocytes are less than 2 X 10^9/L and platelets are less than 25 X 10^9/L, give 50% of previous dose.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp; protect from light.
- Reconstituted solutions (0.5 mg/mL) prepared with SWFI are stable for 1 week at room temp and 2 weeks under refrigeration, protected from light.
- Stable for 3 hours in D5W and for 12 hours in NS at room temp at a concentration of 20-40 mcg/mL.

MISCELLANEOUS

REFERENCES

1, 4, 5, 6, 40, 129, 165, 216.

Full revision 2016; limited revision 2017, 2018, 2020

NAME OF MEDICATION
MitoXANTRONE

OTHER NAMES
Mitozantrone, Novantrone ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Metastatic carcinoma of the breast.
- Non-Hodgkin's lymphoma.
- Acute leukemia, including non-lymphocytic.
- Hepatocellular cancer.
- Hormone refractory prostate cancer.
- Ovarian cancer.

ADMINISTRATION

- IV direct: physician or RN; dilute to at least 50 mL with NS or D5W; inject over at least 3-5 minutes into the tubing of a freely running IV solution of NS or D5W.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W; infuse over 15-30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylactic reactions (rare).
- Cardiovascular: ECG changes and arrhythmias (transient), cardiomyopathy.
- GI: nausea, vomiting, diarrhea, stomatitis.
- Renal: blue-green discolouration of urine for 1-2 days, proteinuria, nephrotoxicity.
- Ophthalmic: conjunctivitis, blurred vision.
- Tumour lysis syndrome (rare).
- Local reactions: phlebitis, blue discolouration of vein.
- Extravasation hazard: vesicant; blue discolouration of the skin that slowly fades; rare cases of skin necrosis. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Solid tumours: initial dose of 12-14 mg/m^2 IV q3weeks.
- Induction dose for acute leukemia: 12 mg/m^2/day IV for 5 consecutive days, as a single agent; 10-12 mg/m^2/day for 3 consecutive days when combined with cytarabine.
- Consolidation dose for acute leukemia: 12 mg/m^2/day IV for 2 consecutive days.
- Dosage adjustment based on clinical status, degree and duration of myelosuppression.
- Dosage in hepatic impairment: reduce dose by 50%.
- Maximum lifetime dose: 140 mg/m^2.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Partial vials may be stored for 7 days at room temp and for 14 days in the fridge.
- Cooling of the intact vials may cause precipitation that will redissolve upon warming.
- Stable for 7 days at room temp or refrigerated in D5W or NS at a concentration of 20 to 500 mcg/mL in PVC containers.
- Stable for 42 days at room temp or in the fridge at a concentration of 2 mg/mL in plastic syringes and glass vials.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 129, 165, 547, 548, 549.

Full revision 2016; limited revision 2017, 2018, 2020

NAME OF MEDICATION	
MORPHINE	
OTHER NAMES	CLASSIFICATION
	Opioid analgesic

Do NOT confuse morphine with apomorphine.
This monograph is specific to MORPHINE.

INDICATIONS
- Supplement to anesthesia.
- Treatment of acute pulmonary edema.
- Relief of moderate to severe acute and chronic pain.
- For pain associated with myocardial infarction (analgesic of choice).
- Treatment of dyspnea in palliative care patients.

ADMINISTRATION
- IV direct: physician or RN; ***respiratory support***. Administer undiluted or dilute dose to 10 mL with NS or D5W and inject very slowly, at a rate of 3 mg/min.
- Intermittent IV infusion: dilute dose in 50 to 100 mL NS or D5W. Must be administered by an infusion pump.
- Continuous IV infusion: Standard concentration is 1 or 2 mg/mL in D5W or NS. Patients on high doses may require more concentrated solutions. Must be administered by an infusion pump.
- SC, IM.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis (rare); pseudoallergic reactions due to histamine release: pruritus, urticaria.
- Cardiovascular: bradycardia, circulatory depression, hypotension (increased incidence with rapid IV administration).
- GI: nausea, vomiting, constipation, dry mouth.
- CNS: dizziness, confusion, drowsiness.
- Respiratory depression.
- Local reactions: pain at injection site; wheals with IV injection.

Antidote: naloxone (for respiratory depression).

DOSAGE
Adults:
- Opioid-naïve patients: SC/IM: 2.5-5 mg q4h prn; IV: 1-2 mg q1h prn. Titrate as required. These initial doses are for the average healthy patient. Elderly patients and those with reduced renal or hepatic function require lower doses while opioid-experienced patients may need higher doses.
- Continuous IV Infusion: initiate at 0.8-10 mg/hr; increase stepwise until analgesia is achieved; in severe chronic pain, maintenance doses have ranged from 0.8-80 mg/hr, and occasionally higher doses (e.g., 150 mg/hr) have been required.
- Analgesia in palliative care and cancer patients: usual dosage range can be exceeded and is titrated to effect.
- Dyspnea in palliative care patients: 2-5 mg IV every 5-10 minutes until relief.

Pediatrics:
- 0.05 mg/kg/dose IV/IM/SC q2-4h prn as an initial dose; usual range 0.1-0.2 mg/kg/dose. Usual maximum dose: 2 mg for an infant, 4 mg for a child 1-6 years old, 8 mg for a child 7-12 years old and 10 mg for an adolescent.
- Continuous IV Infusion: for postoperative pain: 0.01-0.04 mg/kg/hr; for cancer pain: 0.025-2.6 mg/kg/hr.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store at room temp; protect from light.
- Stable for 7 days at room temp or refrigerated in D5W or NS at concentrations of 0.04-0.4 mg/mL in glass and PVC containers.
- Stable for 30 days at room temp in NS or D5W at a concentration of 5 mg/mL in PVC containers.

MISCELLANEOUS
REFERENCES
1, 4, 5, 40, 82, 135, 253.

Full revision 2016; limited revision 2017, 2019, 2020

NAME OF MEDICATION
MOXIFLOXACIN

OTHER NAMES
Avelox ®

CLASSIFICATION
Antibiotic - quinolone

INDICATIONS

- Treatment of respiratory tract infections (sinusitis, bronchitis, pneumonia) caused by susceptible micro-organisms when oral administration is not possible.
- Treatment of complicated intra-abdominal infections caused by susceptible microorganisms.
- Treatment of complicated skin and skin structure infections caused by susceptible organisms.

ADMINISTRATION

- Intermittent IV infusion: 400 mg in ready-to-use 250 mL bag of 0.8% saline to be administered IV over 60 minutes. A slow infusion into a large vein minimizes patient discomfort and reduces risk of venous irritation.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, rash, pruritus, facial or pharyngeal edema, dyspnea, anaphylaxis.
- Cardiovascular: QT interval prolongation. Aortic aneurism/aortic dissection (rare, but serious); avoid using in patients with known history or at risk of aortic aneurism.
- GI: nausea, vomiting, diarrhea, abdominal pain.
- CNS: dizziness, anxiety, agitation, confusion, tremor and seizures.
- Blood glucose disturbances (hyper- and hypoglycemia), usually in diabetic patients.
- Myasthenia gravis exacerbation (rare, but serious); avoid using in patients with this condition.

DOSAGE

- 400 mg IV once daily; duration of therapy (from 5 to 21 days) is dependent upon the type and severity of infection.
- Dosage in renal impairment: no dosage adjustment necessary.
- Dosage in hepatic impairment: no dosage adjustment necessary.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store bags at room temp; do not refrigerate; protect from light.
- The intravenous solution usually has a normal yellow appearance.
- Compatible with NS, D5W, D10W and RL.

MISCELLANEOUS

- When switching from IV to oral administration, no dosage adjustment is necessary.
- Use in pediatrics is not routinely recommended, but may be justified in special circumstances.

REFERENCES

1, 5, 28, 40, 44, 135.

Full revision 2016; limited revision 2020

NAME OF MEDICATION
MULTIVITAMINS

OTHER NAMES
MVI-12 ®, MVI-1000 ®

CLASSIFICATION
Vitamins

INDICATIONS

- Vitamin supplement for patients receiving parenteral nutrition or for those who cannot take oral therapy.

ADMINISTRATION

- Intermittent IV infusion, Continuous IV infusion: dilute with not less than 500 mL but preferably 1000 mL of any commonly-used IV solution. Administer at prescribed rate of infusion fluids.

POTENTIAL ADMINISTRATION HAZARDS

- Rare when administered as recommended.
- Hypersensitivity: anaphylaxis, angioedema, urticaria, shortness of breath, pruritus, rash.
- Local reactions: irritation at injection site.

DOSAGE

- 10 mL IV daily; may be increased to 20 mL daily if required.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Refrigerate ampoules or vials and protect from light.
- Compatible with dextrose, saline, RL, amino acid solutions.
- Incompatible with alkaline solutions.
- Dilute immediately after opening ampoule and use within 24 hours after adding to IV solution. Do not use if any crystals have formed.

MISCELLANEOUS

REFERENCES

4, 5, 40.

Full revision 2016

NAME OF MEDICATION
MYCOPHENOLATE MOFETIL

OTHER NAMES
CellCept ®

CLASSIFICATION
Immunosuppressant

INDICATIONS

- Prophylaxis of organ rejection in patients receiving allogeneic renal, cardiac or hepatic transplants.

ADMINISTRATION

- Reconstitute each 500 mg vial with 14 mL of D5W (to obtain 500 mg/15 mL); shake gently to dissolve.
- Intermittent IV infusion (mandatory): to obtain a final concentration of 6 mg/mL, dilute 1 g dose in 140 mL D5W (for a total of 1 g in approximately 170 mL) or 1.5 g dose in 210 mL D5W (for a total of 1.5 g in approximately 255 mL). (At TOH, dilute 250 mg doses in 50 mL of D5W; dilute 500 mg, 750 mg and 1 g doses in 100 mL of D5W and dilute 1.5 g doses in 250 mL of D5W). Infuse dose over at least 2 hours.

POTENTIAL ADMINISTRATION HAZARDS

- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- GI: diarrhea, nausea, vomiting.
- Hematologic: leukopenia, phlebitis, thrombosis, pure red cell aplasia (very rare).
- Sepsis.
- Numerous adverse effects have been reported in clinical trials. While not specific to the IV preparation, they may occur during IV therapy. Please refer to manufacturer's monograph for a complete listing.

DOSAGE

Mycophenolate mofetil IV administration should start within 24 hours following transplantation.
- Renal transplant: 1 g IV twice daily.
- Cardiac transplant: 1.5 g IV twice daily.
- Hepatic transplant: 1 g IV twice daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Manufacturer recommends to begin the infusion within 4 hours of reconstitution and dilution of the product.
- Incompatible with solutions other than D5W.
- Stable for 7 days at 4° and 25° C in D5W at concentrations from 1 to 10 mg/mL in PVC bags.

MISCELLANEOUS

- Product can be used IV for up to 14 days, but patients should be switched to the oral route as soon as they can tolerate oral medication.
- Mycophenolate mofetil should be used with standard cyclosporine and corticosteroid therapy.

REFERENCES

1, 4, 5, 40, 490.

Full revision 2016; limited revision 2018, 2019, 2020

NAME OF MEDICATION
NADROPARIN

OTHER NAMES
Fraxiparine ®, Fraxiparine Forte ®

CLASSIFICATION
Anticoagulant

INDICATIONS

- Prophylaxis of thrombosis related to surgery.
- Treatment of deep vein thrombosis (DVT).
- Treatment of unstable angina and non-Q-wave myocardial infarction (NQWMI).
- Prevention of clotting during hemodialysis.

ADMINISTRATION

- IV direct: physician or RN. For first dose only when used in unstable angina and NQWMI. Administer undiluted over 30 seconds to 1 minute. Flush with NS after administration.
- SC: in the fat tissue of the lower abdomen; as an alternative, can also be injected into the side of the thigh (but NOT into the muscle tissue). With the thickness of skin held between the operator's thumb and finger, introduce the entire length of the needle vertically into the skin. To minimize bruising, injection sites should NOT be massaged after injection. Rotate site of injection daily.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, angioedema, allergic reactions, anaphylactoid reactions.
- Hematologic: bleeding (less than 5% incidence with prophylactic treatment, up to 10% with active treatment), thrombocytopenia.
- Risk of epidural or spinal hematomas that can result in permanent paralysis when epidural or spinal anesthesia or spinal puncture is used in conjunction with nadroparin. No spinal invasion should be performed for at least 12 hours after a phophylactic dose and at least 24 hours after a therapeutic dose.
- Hepatic: transient increase of liver transaminases (AST, ALT); reversible when drug is discontinued.
- Local reactions: hematomas at the injection site; skin necrosis (rare).

Antidote: The anticoagulant effect of nadroparin can be partially neutralized by protamine. Refer to protamine in this manual for more details.

DOSAGE

- Prophylaxis of thrombosis related to surgery:
  - General surgery: 2850 anti-Xa international units SC once daily for at least 7 days, starting 2-4 hours before surgery.
  - Hip surgery: 38 anti-Xa international units/kg SC once daily for the first 3 postoperative days, starting 12 hours before surgery, if potential benefits outweigh the potential risks; starting postoperative day 4, dose should be increased to 57 anti-Xa international units/kg/day. Treatment to be continued for a total of at least 10 days.
  - Dosage in obesity: dose should be calculated using actual body weight.

- Treatment of DVT:
  - 171 anti-Xa international units/kg SC once daily until adequately anticoagulated with an oral anticoagulant.
  - In patients at increased risk of bleeding, a dose of 86 anti-Xa international units/kg SC twice daily is recommended.
  - Dosage in obesity: dose should be calculated using actual body weight.

- Treatment of unstable angina and NQWMI:
  - Initial IV bolus dose of 86 anti-Xa international units/kg followed by SC injections of 86 anti-Xa international units/kg twice daily. Usual treatment duration is 6 days.

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DOSAGE (Cont.)

- Prophylaxis of clotting during hemodialysis:
  - 65 anti-Xa international units/kg into the arterial line at the start of each session; an additional dose may be given if session lasts longer than 4 hours; adjust dose for subsequent sessions to achieve plasma anti-Xa levels between 0.5-1 anti-Xa international units/mL.
  - In patients at risk of bleeding, use halved doses (i.e., 32.5 anti-Xa international units/kg); an additional smaller dose may be given if session lasts longer than 4 hours; adjust dose for subsequent sessions to achieve plasma anti-Xa levels between 0.2-0.4 anti-Xa international units/mL.

- Dosage in renal impairment: consider dosage adjustment and monitoring of anti-Xa levels.

CrCl (mL/min)	50-30	less than 30
Dose	decrease by 25-33%	decrease by 25-33% if prophylaxis do not use if treatment

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.
- Compatible by Y-site with NS.

MISCELLANEOUS

- Nadroparin should not be given to patients with a history of heparin-induced thrombocytopenia (risk of cross-reactivity), unless an *in vitro* platelet aggregation test is negative.
- For laboratory monitoring of effect, anti-Xa methods are recommended (although not routinely done, monitoring is recommended in special cases such as obesity, pregnancy and renal failure). Peak activity occurs 4 hours after SC administration.

REFERENCES

5, 95, 119, 135, 296.

Full revision 2016; limited revision 2017, 2020

NAME OF MEDICATION
NALBUPHINE

OTHER NAMES
Nubain ®

CLASSIFICATION
Opioid analgesic

INDICATIONS

- Supplement to surgical anesthesia.
- Relief of moderate to severe pain.
- Preoperative and postoperative analgesia.
- Obstetrical analgesia during labour.
- As an adjunct to epidural opioids to reduce the severity of nausea, vomiting and pruritus.

ADMINISTRATION

- IV direct: physician or RN; ***respiratory support***. Administer undiluted or dilute dose to 10 mL with NS or D5W; inject each 10 mg over 3-5 minutes.
- Intermittent IV infusion: dilute dose in 50 mL of D5W or NS; infuse over at least 15 minutes. Must be administered by an infusion pump.
- SC, IM.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylactic/anaphylactoid reactions, rash, pruritus.
- GI: nausea, vomiting, dry mouth.
- CNS: sedation, dizziness, vertigo, headache.
- Respiratory depression (less than with morphine).
- Sweaty or clammy sensation.

Antidote: naloxone (for respiratory depression).

DOSAGE

- Analgesia: 10 mg (or 0.14 mg/kg) IV/IM/SC q3-6h. Single doses may be increased to 20 mg. Possesses analgesic ceiling effect, with no increase in pain relief with doses in excess of 0.4 mg/kg IV. Maximum total daily dose is 160 mg.
- Anesthesia induction: 0.3-3 mg/kg IV over 10-15 minutes, followed by maintenance doses of 0.25-0.5 mg/kg in single IV injections as needed.
- Pruritus: 5 mg IV; may repeat dose if pruritus not resolved.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Stable for 48 hours at room temp in NS, D5-NS, or RL at concentrations of 3.3 mg/mL, 5 mg/mL, 6.7 mg/mL and 10 mg/mL.

MISCELLANEOUS

- Similar onset and duration of action as morphine.
- Nalbuphine is a mixed agonist-antagonist and may precipitate withdrawal in opioid-dependent individuals.

REFERENCES

1, 5, 40, 95, 208, 285.

Full revision 2016; limited revision 2017, 2020

NAME OF MEDICATION
NALOXONE

OTHER NAMES
Narcan ®

CLASSIFICATION
Opioid antagonist, Antidote

INDICATIONS

- To reverse postoperative opioid depression.
- For diagnosis and treatment of respiratory depression in overdose situations induced by natural or synthetic opioids (including pentazocine, propoxyphene, butorphanol, nalbuphine, and methadone).
- For opioid-induced pruritus.

ADMINISTRATION

- IV direct: physician or RN; administer undiluted (preferred) at a rate of 0.4 mg/15 seconds. To facilitate administration of smaller doses, may further dilute with NS as follows: 0.4 mg (1 mL from the 0.4 mg/mL vial or ampoule) in 9 mL of NS for a final concentration of 0.04 mg/mL (10 mL volume) OR 1 mg (1 mL from the 1 mg/mL vial) in 9 mL of NS for a final concentration of 0.1 mg/mL (10 mL volume).
- Continuous IV infusion: dilute 2 mg in 500 mL D5W or NS to obtain a final concentration of 4 mcg/mL or 0.004 mg/mL; refer to Dosage section for administration rate. In fluid-restricted patients, may prepare a more concentrated solution such as 0.01 mg/mL (10 mcg/mL) or 0.1 mg/mL (100 mcg/mL). At TOH, add 2 mg (2 mL from a naloxone 1 mg/mL vial) to 500 mL of D5W or NS to obtain a final concentration of 0.004 mg/mL (4 mcg/mL). If need a high concentration for opioid overdose, withdraw 10 mL from a 100 mL bag of D5W or NS and discard, then add 12 mg (30 mL from naloxone 0.4 mg/mL, 10 mL vials) to the 90 mL bag for a total volume of 120 mL and a final concentration of 0.1 mg/mL (100 mcg/mL).
- IM (NOT preferred).
- SC (NOT preferred).

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension, hypertension, ventricular tachycardia and fibrillation, pulmonary edema, cardiac arrest, especially in postoperative patients with pre-existing cardiovascular disease or taking concurrent cardiotoxic drugs.
- Reversal of analgesia in postoperative patients.
- May precipitate withdrawal symptoms in opioid-dependent patients (pain, hypertension, sweating, agitation, irritability).

DOSAGE

Adults:
- Postoperative opioid depression: initial dose of 0.1-0.2 mg IV, repeat at 2-3 minute intervals prn until desired response is obtained.
  For TOH patients from Acute Pain Service, refer to Corporate nursing standard operating procedures C-SOP NUR 0059, C-SOP NUR 0060, C-SOP NUR 0061, appendices for the naloxone administration clinical guideline.
- Known or suspected opioid overdose: 0.4-2 mg IV/IM/SC; repeat at 2-3 minute intervals prn; if no response is observed after a total of 10 mg, diagnosis of opioid overdose should be questioned. Although an infusion is rarely indicated, it can be used initially at 0.4 mg/hr; titrate according to patient's response. Others recommend to administer on an hourly basis the equivalent of 2/3 of the bolus dose that resulted in reversal (e.g., if a patient responds to a 3 mg bolus, initiate at 2 mg/hr); titrate to effect. A second bolus equal to 1/2 the initial bolus should be given 15 minutes after the continuous infusion has started to prevent a decrease in naloxone blood levels.
- For opioid-induced pruritus: IV infusion of 0.25 mcg/kg/hr; may increase to 2.4 mcg/kg/hr depending on clinical response (for a 70 kg patient, range from 0.018 mg/hr to 0.17 mg/hr); monitor pain and check if naloxone is not reversing analgesia.

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DOSAGE (Cont.)

Pediatrics:
- Postoperative opioid depression: 0.005-0.01 mg/kg IV, repeat at 2-3 minute intervals prn until desired response is obtained.
- Known or suspected opioid overdose: up to 5 years old or weight up to 20 kg: 0.1 mg/kg/dose IV/IM/SC, repeat at 2-3 minute intervals prn until desired response is obtained. Over 5 years old or weight greater than 20 kg: 2 mg/dose IV/IM/SC, repeat at 2-3 minute intervals prn until desired response is obtained.

Note: The duration of action of opioids is often greater than that of naloxone; therefore, repeat doses (e.g., every 20-60 minutes) or an IV infusion may be necessary.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp and protect from light.
- Do not mix with parenteral preparations containing bisulfite, sulfite or any solution or drug with an alkaline pH.
- Stable for 24 hours at room temp in D5W or NS at a concentration of 0.004 mg/mL (4 mcg/mL).

MISCELLANEOUS

- Naloxone is a pure antagonist with little or no agonist activity. In the absence of an opioid agent, the drug has no effect.
- Following IV administration: onset of action within 1-2 minutes; duration of action of 45 minutes.

REFERENCES

1, 2, 5, 40, 82, 95, 135, 208, 324, 446.

Full revision 2016; limited revision 2017, 2019, 2020

NAME OF MEDICATION
NATALIZUMAB

OTHER NAMES
Tysabri ®

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS

- Monotherapy for relapsing-remitting multiple sclerosis.
- Management of moderate to severely active Crohn's disease in patients with inadequate response to or unable to tolerate conventional therapy and tumour-necrosis factor inhibitors.

ADMINISTRATION

- Intermittent IV infusion: dilute 300 mg in 100 mL of NS. Invert diluted solution gently to mix; do NOT shake. Infuse over 60 minutes. Flush line with NS after infusion is complete.
- Patient should be observed during the infusion and for 60 minutes after the end of the infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; anaphylaxis, urticaria, dizziness, fever, rash, rigors, pruritus, nausea, angioedema, flushing, hypotension, dyspnea, chest pain. Usually occurs within 2 hours of the start of the infusion. If a hypersensitivity reaction occurs, discontinue drug immediately and initiate appropriate therapy.
- Infusion-related reactions (defined as any adverse event occurring within 2 hours of the start of the infusion): headache, dizziness, fatigue, urticaria, pruritus, rigors. Treat symptomatically, as appropriate.
- GI: gastroenteritis, nausea, abdominal discomfort, diarrhea, cholelithiasis.
- CNS: depression, suicidal ideation.
- Hepatic: increase in hepatic enzymes and total bilirubin.
- Progressive multifocal leukoencephalopathy or PML (rare opportunistic viral infection of the brain that can lead to severe disability or death).
- Arthralgia or pain in extremities.
- Infections: increased risk of infections as natalizumab is an immunosuppressant; respiratory tract infections, urinary tract infections, influenza, vaginal infections, tooth infections, tonsillitis/pharyngitis, herpes infections.
- Dermatitis.

DOSAGE

- 300 mg IV once every 4 weeks.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge and protect from light. Do not shake and do not freeze.
- Store the diluted solution in the fridge (do not freeze) if immediate infusion is not possible; administration of the solution should be completed within 8 hours of its preparation.

MISCELLANEOUS

- Monitor for any new sign or symptom of PML.

REFERENCES

1, 5, 40, 95.

Full revision 2016

NAME OF MEDICATION
NELARABINE

OTHER NAMES
Arranon ®, Atriance ™

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of T-cell acute lymphoblastic leukemia (T-ALL) or T-cell lymphoblastic lymphoma (T-LBL) in patients who have not responded to or whose disease has relapsed following treatment with at least two chemotherapy regimens.

ADMINISTRATION

- Ready-to-use solution. Transfer dose into an empty, sterile glass or PVC container.
- Intermittent IV infusion: do NOT dilute prior to administration. Infuse over 2 hours.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- GI: diarrhea, nausea, vomiting, anorexia, constipation.
- CNS: somnolence (can progress to coma and death), confusion, dizziness, headache, ataxia, paresthesia, hypoesthesia, Guillain-Barré (GB) syndrome, peripheral neuropathy.
- Hematologic: anemia, leukopenia, neutropenia, thrombocytopenia.
- Respiratory: cough, dyspnea, pleural effusion.
- Asthenia, fatigue, myalgia, pyrexia.
- Edema.
- Tumour lysis syndrome, hyperuricemia: can be prevented by administration of fluids, allopurinol and alkalinization of urine.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Adult: 1500 mg/m^2/day IV administered on days 1, 3 and 5 and repeated every 21 days.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Stable for 8 hours at room temp in PVC infusion bags and glass containers.

MISCELLANEOUS

REFERENCES

1, 5, 25, 165, 301.

Full revision 2016; limited revision 2017, 2018, 2020

NAME OF MEDICATION
NEOSTIGMINE

OTHER NAMES

CLASSIFICATION
Cholinesterase inhibitor

INDICATIONS

- Reversal of postoperative neuromuscular blockade.
- Treatment of acute exacerbations of myasthenia gravis.
- Prophylaxis and treatment of postoperative urinary retention and intestinal atony.
- Treatment of acute intestinal pseudo-obstruction.

ADMINISTRATION

- IV direct: physician only; undiluted; rate should not exceed 0.5 mg/min.
- SC, IM.

POTENTIAL ADMINISTRATION HAZARDS

- Cholinergic symptoms: nausea, vomiting, diarrhea, abdominal cramps, miosis, salivation, sweating, bradycardia, hypotension, increased respiratory secretions, bronchoconstriction, etc.

Antidote: atropine 0.6-1.2 mg IV or glycopyrrolate 0.2-0.6 mg IV.

DOSAGE

- Reversal of neuromuscular blockade: 0.5-2.5 mg OR 0.03-0.07 mg/kg IV. Repeat as needed; total dose does not usually exceed 5 mg.
- Myasthenia gravis: 0.5-2.5 mg IV/SC/IM; repeat q1-3h as needed to a maximum 10 mg/24 hours.
- Postoperative urinary retention or intestinal atony: Prophylaxis: 0.25 mg SC/IM immediately before or after surgery, repeat q4-6h for 2-3 days. Treatment: 0.5 mg SC/IM/IV q4-5h. For urinary retention: if no response within 60 minutes, the patient should be catheterized; after the bladder has been emptied, continue with 0.5 mg SC/IM q3h for at least 5 doses.
- Treatment of intestinal pseudo-obstruction: 2-2.5 mg IV; repeat once if needed.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp and protect from light.
- Compatible with commonly-used IV solutions.
- Compatible with glycopyrrolate when mixed in the same syringe.

MISCELLANEOUS

- Not effective in neutralizing neuromuscular blockade produced by depolarizing agents such as succinylcholine.

REFERENCES

1, 4, 5, 6, 40, 95, 135.

Full revision 2016

NAME OF MEDICATION
NESIRITIDE *

OTHER NAMES
Natrecor ®

CLASSIFICATION
Vasodilator

INDICATIONS

- Treatment of acute decompensated heart failure presenting with moderate to severe dyspnea.

ADMINISTRATION

- Reconstitute the 1.5 mg vial with 5 mL of diluent removed from a prefilled 250 mL plastic bag (D5W, D5-1/4NS, D5-1/2NS, NS) to obtain a 0.32 mg/mL solution. Roll vial gently to ensure complete reconstitution. Do NOT shake. Further dilute entire content of vial by transferring it in the original 250 mL plastic bag to obtain a final concentration of 6 mcg/mL. Invert bag several times for complete mixing.
- Prime the tubing with 5 mL of the prepared infusion solution before connecting to the IV access port and before withdrawing the loading dose.
- IV direct: physician only; ***blood pressure monitoring***. Withdraw the appropriate loading dose from the prepared infusion bag and administer over 60 seconds.
- Continuous IV infusion: ***blood pressure monitoring***; administer solution from the prepared infusion bag, as per dosage section.
- Nesiritide must NOT be administered through a heparin coated catheter.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: pruritus, rash, anaphylaxis (rare).
- Cardiovascular: arrhythmias, angina and hypotension; if marked hypotension, reduce dosage or discontinue infusion. May restart infusion without a loading dose and with a 30% reduction in the infusion rate following an appropriate period of observation and only after the patient has been stabilized.
- GI: nausea, vomiting.
- CNS: headache, insomnia, dizziness, anxiety.
- Back pain.
- Nephrotoxicity.
- Local reactions: catheter pain.

DOSAGE

- Loading dose: 2 mcg/kg IV. Volume of loading dose to be taken from the prepared infusion bag can be calculated using the following formula: Loading dose (mL) = patient weight (kg)/3.
- Infusion dose: to follow immediately loading dose, at a rate of 0.01 mcg/kg/min (0.1 mL/kg/hr). Infusion rate in mL/hr can be calculated using the following formula: Infusion rate (mL/hr) = patient weight (kg) x 0.1.
  Dose may be increased in increments of 0.005 mcg/kg/min (after a bolus of 1 mcg/kg) no more frequently than every 3 hours, up to a maximum dosage of 0.03 mcg/kg/min.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Reconstituted vial is stable for 24 hours in the fridge or at room temp; protect from light.
- Diluted solution is stable for 24 hours at room temp.
- Compatible with D5W, NS, D5-1/2NS and D5-1/4NS.

MISCELLANEOUS

- Limited experience with administration of nesiritide for longer than 96 hours.

REFERENCES

1, 40, 95, 135.

* Available via Health Canada's Special Access Programme

Full revision 2016

NAME OF MEDICATION
NIACINAMIDE

OTHER NAMES
Nicotinamide

CLASSIFICATION
Vitamin

INDICATIONS

- Acute treatment of pellagra.

ADMINISTRATION

- Intermittent IV infusion (preferred): dilute dose in 500 mL of NS; concentration should not exceed 10 mg/mL; infuse at a rate not exceeding 2 mg/min.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- GI: nausea, vomiting, diarrhea.
- CNS: headache, dizziness.
- Dermatologic (rare): flushing, rash, hives, facial erythema.

DOSAGE

- 25-100 mg IV q2-3h (maximum dose: 1 g daily) OR 50-100 mg/day IM, administered in 5 or more divided doses.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Compatible with NS.

MISCELLANEOUS

REFERENCES

95.

Full revision 2016

NAME OF MEDICATION
NitroGLYCERIN

OTHER NAMES
Glyceryl trinitrate, Nitroject, Trinitroglycerin

CLASSIFICATION
Vasodilator

INDICATIONS
- Control of blood pressure in perioperative hypertension and hypertensive emergencies.
- Congestive heart failure associated with acute myocardial infarction.
- Treatment of unstable angina.
- As a tocolytic agent for the treatment of uterine tachysystole during pregnancy/labour.

ADMINISTRATION
- IV direct (for the treatment of uterine tachysystole): physician or RN in presence of physician; ***BP monitoring, fetal monitoring***. From a nitroglycerin premixed bottle of 50 mg/250 mL, withdraw 1 mL (200 mcg) and dilute with 9 mL of NS to get a final concentration of 20 mcg/mL (200 mcg/10 mL). Administer dose (usually 50 mcg=2.5 mL) over 2-3 minutes.
- Continuous IV infusion: ***cardiac monitoring; continuous BP monitoring***. Use a premixed bottle OR dilute 25-50 mg from vials in 250-500 mL of D5W or NS in a glass or non-PVC plastic container and invert container several times to ensure uniform dilution. Concentrations up to 400 mcg/mL (0.4 mg/mL) may be used. At TOH, use premixed bottle OR add 50 mg (10 mL from a nitroglycerin 5 mg/mL vial) to 250 mL of D5W or NS in a glass or a non-PVC plastic container to obtain a final concentration of 200 mcg/mL (0.2 mg/mL). Administer via an infusion pump at a rate titrated to patient response (refer to Dosage section). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- Cardiovascular: excessive hypotension, reflex tachycardia or bradycardia.
- GI: nausea, vomiting, abdominal pain.
- CNS: headache, dizziness, weakness, restlessness
- Tachyphylaxis with continued use.

DOSAGE
- When special non-PVC administration sets are used, initiate with 5-10 mcg/min, and increase every 3 to 5 minutes in 5 mcg/min increments. If no response at 20 mcg/min, increments of 10 mcg/min and later 20 mcg/min can be used. Maximum rate for unstable angina is 400 mcg/min, and that for hypertension is 100 mcg/min.
- When PVC administration sets are used, higher dosages are generally required; initiate with 25 mcg/min. Dosage is then titrated according to response and tolerance of the patient.
- Each patient must be individually titrated.
- For uterine tachysystole: 50 mcg IV every 3-5 minutes to a maximum of 200 mcg.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials and premixed bottles at room temp. Protect from light; do not freeze.
- Baxter premixed nitroglycerin solution in glass bottle can be exposed to ambient light for up to 72 hours including time for administration.
- Only infusion containers made from glass or a plastic known to be compatible with nitroglycerin (i.e., polyolefin, polyethylene) are to be used.
- Nitroglycerin readily undergoes sorption to PVC containers and tubing. Special administration sets are marketed to avoid these potential problems.
- Stable for 28 days at room temp or in the fridge in D5W or NS at concentrations of 200 and 400 mcg/mL in glass and polyolefin containers.
- Compatible with RL, D5-RL, D5-1/2NS, D5-NS, 1/2NS, sodium lactate 1/6 M.
- Baxter premixed nitroglycerin solution in glass bottle is incompatible with furosemide administered by Y-site.

MISCELLANEOUS
- Contraindicated in patients with hypotension or uncorrected hypovolemia, increased intracranial pressure, constrictive pericarditis, pericardial tamponade, and when taking riociguat, sildenafil, tadalafil or vardenafil.

REFERENCES
1, 4, 5, 6, 40, 70, 95, 135, 314, 315, 382.

Full revision 2016; limited revision 2017, 2018, 2019

NAME OF MEDICATION
NitroPRUSSIDE

OTHER NAMES
Nipride ®

CLASSIFICATION
Antihypertensive

INDICATIONS
- To treat hypertensive emergencies.
- To control blood pressure during surgery.
- To improve cardiac performance in refractory heart failure or acute myocardial infarction.

ADMINISTRATION
- Continuous IV infusion (mandatory): ***cardiac monitoring; continuous BP monitoring***. Usually, dilute 50 mg in D5W or NS to obtain a final concentration of 50 mcg/mL (50 mg/1000 mL), 100 mcg/mL (50 mg/500 mL) or 200 mcg/mL (50 mg/250 mL). At TOH, dilute 50 mg (2 mL from a nitroprusside 25 mg/mL vial) in 100 mL D5W or NS to obtain a final concentration of 500 mcg/mL (0.5 mg/mL). Promptly wrap the infusion container in aluminium foil or other opaque material to protect from light (it is not necessary to cover the tubing). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- Cardiovascular: retrosternal discomfort, palpitations, hypotension.
- GI: nausea, retching, vomiting, abdominal pain.
- CNS: apprehension, headache, agitation, somnolence, dizziness.
- Sweating, muscle twitching.
- Thiocyanate accumulation may occur at infusion rates faster than 2 mcg/kg/min; symptoms can include blurred vision and miosis, hyperreflexia, skin rash, tinnitus, bradycardia, metabolic acidosis, seizures and confusion.
- Local reactions: irritation at injection site.

Antidote: hydroxocobalamin or sodium thiosulfate.

DOSAGE
- Initiate with 0.25-0.3 mcg/kg/min IV and titrate every few minutes to effect. Usual rate is 3 mcg/kg/min. Maximum rate is 10 mcg/kg/min; the absence of a response to a dose of 10 mcg/kg/min after 10 min of therapy requires cessation of therapy as may indicate cyanide accumulation.
- Smaller doses are required in patients receiving concomitant antihypertensive medications and elderly patients.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials at room temp. Protect from light; do not freeze.
- Freshly prepared solutions range in colour from faint brown, brownish-pink, light orange to straw. Upon exposure to light, the colour of solutions changes from brown to blue. Highly discoloured solutions (blue, green or dark red) should be discarded.
- Stable for 48 hours at room temp when protected from light (with an opaque covering such as aluminium foil, and not amber plastic covering) in D5W, NS or RL at concentrations of 50 mcg/mL and 100 mcg/mL.

MISCELLANEOUS
- Onset: 30-60 seconds.
- Nitroprusside is metabolized to cyanide and then to thiocyanate which is then excreted by the kidney. Monitor thiocyanate levels if treatment is prolonged or high doses are used; levels should be less than 1 mmol/L (or 60 mg/L).
- Contraindicated in the treatment of compensatory hypertension, in patients with uncorrected hypovolemia or anemia and in patients with inadequate cerebral circulation; when used during surgery, it is also contraindicated in patients with hepatic or severe renal impairment, congenital (Leber's) optic atrophy, tobacco amblyopia or disease states associated low vitamin B_{12} plasma levels.
- Caution in hypothyroidism and in patients with increased intracranial pressure.

REFERENCES
1, 5, 40, 95, 135, 208, 263.

Full revision 2016; limited revision 2017, 2019

NAME OF MEDICATION	
NIVOLUMAB	
OTHER NAMES	CLASSIFICATION
Opdivo ®	Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of unresectable or metastatic melanoma or for the adjuvant treatment of melanoma.
- Treatment of locally advanced or metastatic non-small cell lung cancer (NSCLC).
- Treatment of advanced or metastatic renal cell carcinoma (RCC).
- Treatment of recurrent or metastatic squamous cell carcinoma (SCC) of the head and neck.
- Treatment of patients with classical Hodgkin lymphoma (cHL).
- Treatment of advanced or metastatic hepatocellular carcinoma (HCC).

ADMINISTRATION

- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazards section.**
- Intermittent IV infusion: withdraw the required amount from the 10 mg/mL vials and transfer to a compatible container. The dose may also be diluted in NS or D5W. The 240 or 480 mg dosage may be diluted to a maximum total infusion volume of 120 mL. The 1 mg/kg and 3 mg/kg dosages may be diluted to a final concentration ranging from 1-10 mg/mL (if 10 mg/mL: no dilution is necessary). Gently invert container to mix; do NOT shake. Administer over 30 minutes through an in-line low-protein-binding 0.2-1.2 micron filter. Flush line with NS or D5W after administration. When in combination with ipilimumab, infuse nivolumab first, followed by ipilimumab on the same day.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus, anaphylactic reaction; immune-mediated reactions, can be fatal; can involve any organ. Most common reactions include endocrinopathies (e.g., hypothyroidism, hyperthyroidism, adrenal insufficiency, hypophysitis, hypopituitarism, diabetes mellitus and diabetic ketoacidosis), colitis, hepatitis, myocarditis, pneumonitis or interstitial lung disease, nephritis and renal failure, Stevens-Johnson syndrome, toxic epidermal necrolysis, encephalitis. Consult manufacturer's recommendations for management of these reactions and when to withhold or discontinue treatment.
- Infusion-related reactions: chills, tremor, pruritus, rash, flushing, dyspnea, dizziness, fever, faintness. Patients with mild or moderate infusion reaction may continue to receive nivolumab (infusion rate may be slowed) with close monitoring; **consider premedication**. If reaction is severe, discontinue treatment.
- Cardiovascular: can be severe with combination therapy; edema, peripheral edema.
- GI: nausea, vomiting, diarrhea, constipation, decreased appetite, abdominal pain.
- CNS: headache, peripheral neuropathy.
- Electrolyte disturbances: hyponatremia, hypocalcemia, hypercalcemia, hyperkalemia, hypokalemia, hypomagnesemia.
- Hematologic: anemia, leukopenia, lymphopenia, thrombocytopenia.
- Respiratory: respiratory tract infection, cough, dyspnea.
- Elevated markers: LFTs, total bilirubin, amylase, lipase, creatinine, triglycerides, cholesterol.
- Fever, fatigue, vitiligo, musculoskeletal pain, weakness, urinary tract infection.
- Risk of hemophagocytic lymphohistiocytosis (a condition where a large number of immune cells attack and destroy other blood cells) when nivolumab is used alone or in combination with ipilimumab.

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DOSAGE

- Monotherapy for unresectable or metastatic melanoma, NSCLC, RCC, SCC of the head and neck, cHL: 3 mg/kg IV every 2 weeks OR 240 mg IV every 2 weeks OR 480 mg IV every 4 weeks.
- Monotherapy for adjuvant treatment of melanoma and HCC: 3 mg/kg IV every 2 weeks.
- Combination therapy for unresectable or metastatic melanoma: 1 mg/kg IV every 3 weeks for the first 4 doses in combination with ipilimumab, followed by nivolumab as single-agent at 3 mg/kg IV every 2 weeks OR 240 mg IV every 2 weeks OR 480 mg IV every 4 weeks.
- Combination therapy for NSCLC:
    - Previously untreated metastatic, PD-L1 expressing: 3 mg/kg IV every 2 weeks in combination with ipilimumab every 6 weeks;
    - Metastatic in combination with ipilimumab and chemotherapy: 360 mg IV every 3 weeks in combination with ipilimumab every 6 weeks and platinum-doublet chemotherapy every 3 weeks for the first 2 cycles. Thereafter, nivolumab 360 mg IV every 3 weeks in combination with ipiliumumab every 6 weeks.
- Combination therapy for RCC: 3 mg/kg IV every 3 weeks for the first 4 doses in combination with ipilimumab, followed by nivolumab as single agent at 3 mg/kg IV every 2 weeks OR 240 mg IV every 2 weeks OR 480 mg IV every 4 weeks.
- Dosage in renal impairment: no adjustment in patients with CrCl 30 mL/min and above; insufficient data when CrCl is below 30 mL/min.
- Dosage in hepatic impairment: no adjustment in patients with mild hepatic impairment (abnormal AST or total bilirubin up to 1.5 x ULN); no data available for moderate to severe impairment.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze or shake.
- Solution in vial is clear to opalescent, colourless to pale-yellow and may contain a few translucent-to-white amorphous particles.
- Infusion must be completed within 24 hours of preparation.
- If not given immediately, the infusion solution is stable for 24 hours between 2-8°C, protected from light, undiluted or in NS or D5W in PVC, non-PVC or glass containers. Note that a maximum of 8 hours of the total 24 hours can be between 20-25°C and exposed to light.

MISCELLANEOUS

- Thyroid function, LFTs, blood glucose and electrolytes should be monitored at the start and during treatment.
- Contains 0.1 mmol (2.30 mg) of sodium per 10 mg of nivolumab.

REFERENCES

5, 129, 135, 165, 438.

New monograph 2017; limited revision 2018, 2019, 2020, 2021

NAME OF MEDICATION
NOREPINEPHRINE BITARTRATE

OTHER NAMES
Levarterenol, Levophed ®, Noradrenaline bitartrate

CLASSIFICATION
Sympathomimetic

INDICATIONS

- Temporary maintenance of BP in acute hypotensive states produced by trauma, surgery, myocardial infarction, poliomyelitis, pheochromocytomectomy, sympathectomy, spinal anesthesia, septicemia, drug reactions, hemorrhage and blood transfusion reactions.
- Limited use in shock states while attempting to replace circulating blood volume.
- Adjunct in the treatment of cardiac arrest and profound hypotension.

ADMINISTRATION

- Continuous IV infusion (mandatory): ***cardiac monitoring; continuous BP monitoring***. Must dilute before use. Concentration of solution will depend on dosage and fluid requirement of each patient. Dilute 4 mg or 8 mg (norepinephrine base) in D5W, D5-NS or NS to obtain a final concentration of 4 mcg/mL (4 mg/1000 mL), 8 mcg/mL (4 mg/500 mL), 16 mcg/mL (4 mg/250 mL or 8 mg/500 mL) or 32 mcg/mL (8 mg/250 mL). At TOH, add 8 mg or 16 mg (8 mL or 16 mL from norepinephrine 1 mg/mL vials) to 250 mL of NS to obtain a concentration of 32 mcg/mL (0.032 mg/mL) or 64 mcg/mL (0.064 mg/mL) respectively. Rate as per Dosage section. When possible, infuse into a large vein.
- Check blood pressure every 2 minutes until desired level is reached, then every 5 minutes. Patient should never be left unattended while receiving the drug.
- Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: contains metabisulfite that may cause allergic-type reaction such as anaphylaxis, asthma.
- Cardiovascular: bradycardia (**Antidote**: atropine), arrhythmia.
- CNS: headache, weakness, dizziness, tremor, anxiety.
- Respiratory difficulty or apnea.
- Local reactions: blanching along the infused vein, may progress to superficial sloughing. If occurs, change infusion site.
- Extravasation hazard: vesicant. (**Antidote**: phentolamine 5-10 mg diluted in 10-15 mL of NS). Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Doses are expressed as the base form of norepinephrine.
- Average dose: initially 8-12 mcg/min IV or 0.1-0.5 mcg/kg/min IV; titrate to desired blood pressure.
- Usual maintenance dose: 2-4 mcg/min IV.
- Continue therapy until patient can maintain own BP then decrease gradually to prevent sudden drop in BP.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp. Protect from light; do not freeze.
- Do not use if pink or darker than slightly yellow or contains a precipitate.
- Compatible with NS, D5W, D10W, D5-NS, D5-1/2NS, D5-RL, RL.
- Stable for 7 days at room temp exposed to light in D5W or NS at concentrations of 4 and 16 mcg/mL in PVC containers.

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COMPATIBILITY, STABILITY (Cont.)

- Stable for 61 days at room temp (protected from light) or in the fridge (protected from light) in NS or D5W at a concentration of 64.5 mcg/mL in PVC containers.
- Stable for 26 days at room temp (exposed to fluorescent light) or 39 days in the fridge (exposed to fluorescent light) in NS or D5W at a concentration of 64.5 mcg/mL in PVC containers.
- Stability is affected when mixed with solutions that have a final pH of greater than 5.5-6.

MISCELLANEOUS

- Pressor therapy is not a substitute for replacement of blood/fluids.
- 1 mL of solution contains 1.88 mg of norepinephrine bitartrate which is equivalent to 1 mg of the base.

REFERENCES

1, 4, 5, 6, 40, 95, 101, 135, 208, 366, 367, 462.

Full revision 2016; limited revision 2017, 2019, 2020, 2021

NAME OF MEDICATION
OBINUTUZUMAB

OTHER NAMES
Gazyva ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS
- Treatment of previously untreated chronic lymphocytic leukemia (CLL) in combination with chlorambucil.
- Treatment of Non-Hodgkin follicular lymphoma (FL) (in combination with bendamustine, followed by obinutuzumab monotherapy) that has relapsed after or is refractory to a rituximab-containing regimen.
- Treatment of previously untreated stage II, III or IV FL (in combination with chemotherapy, followed by obinutuzumab monotherapy).

ADMINISTRATION
- **Ensure premedication has been administered as recommended; refer to Dosage section.**
- For IV use: withdraw from vials the required amount of concentrated solution and dilute as follows: 4 mL (100 mg), 36 mL (900 mg) and 40 mL (1000 mg) added to 100 mL, 250 mL and 250 mL, respectively, of NS only. Gently invert bag to mix. Do NOT shake.
- Intermittent IV infusion (mandatory):
  - CLL: for the first dose of the first cycle: infuse the first 100 mg over 4 hours at 25 mg/hr; if no infusion-related reaction, administer the remaining 900 mg starting at 50 mg/hr, which can be increased by 50 mg/hr every 30 minutes (maximum: 400 mg/hr). For subsequent infusions, if no infusion-related reaction in previous infusion where the final infusion rate was 100 mg/hr or faster, initial infusion rate of 100 mg/hr which can be increased by 100 mg/hr every 30 minutes (maximum: 400 mg/hr).
  - FL: for day 1 of cycle 1: initial infusion rate of 50 mg/hr, which can be increased by 50 mg/hr every 30 minutes (maximum: 400 mg/hr). For subsequent infusions, if no infusion-related reaction or a mild reaction occurred in previous infusion where the final infusion rate was 100 mg/hr or faster, initial infusion rate of 100 mg/hr, which can be increased by 100 mg/hr every 30 minutes (maximum: 400 mg/hr).
  - Infusion rate may be slowed in the event of infusion-related reactions. Consult manufacturer's recommendations for subsequent rate adjustment.
- Patients with pre-existing cardiac or pulmonary conditions should be monitored carefully during and after the infusion.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis, rash, pruritus.
- Infusion-related reactions: nausea, vomiting, diarrhea, hypotension, hypertension, flushing, tachycardia, atrial fibrillation, pyrexia, fatigue, chills, headache, dizziness, dyspnea, bronchospasm, laryngeal edema, larynx and throat irritation, wheezing, anaphylactic reactions; occur mainly during infusion of the first 1000 mg. If reactions occur, slow and/or stop the infusion. If reaction is severe, treatment should be interrupted or discontinued.
- Cardiovascular: arrhythmia, tachycardia, myocardial infarction, heart failure, acute coronary syndrome, angina; hypotension: withholding antihypertensive medications should be considered for 12 hours prior, during and for the first hour after the infusion.
- GI: nausea, vomiting, diarrhea, GI perforation (rare).
- Hematologic: myelosuppression, neutropenia, lymphopenia, leukopenia. Thrombocytopenia: can occur as soon as within 24 hours after first infusion and can lead to life-threatening hemorrhage; consider withholding concomitant medications which may increase bleeding risk, especially during the first cycle.
- Hepatic: elevated LFT, may be severe; possible reactivation of hepatitis B virus.
- Infections: increased risk of bacterial, fungal and viral infections from myelosuppression; may need antimicrobial prophylaxis.
- Respiratory: cough, dyspnea, upper respiratory tract infection, sinusitis.
- Progressive multifocal leukoencephalopathy (PML): rare, can be fatal. Report new onset of or changes in pre-existing neurological signs and symptoms. Discontinue obinutuzumab if PML is suspected.
- Tumour lysis syndrome: hydrate patient and administer allopurinol or rasburicase in patients with high tumour load, high lymphocyte count (greater than 25 x 10^9/L) and/or renal impairment (CrCl less than 70 mL/min).
- Fever, arthralgia, asthenia, hyperkalemia, hypocalcemia, hypokalemia, hyponatremia, increased creatinine, hypoalbuminemia.
- Local reactions: pain, erythema, hematoma, swelling, induration at injection site.

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DOSAGE

- **Premedication for CLL cycle 1 days 1 and 2 and FL cycle 1 day 1: IV corticosteroid (prednisolone 100 mg, dexamethasone 20 mg or methylprednisolone 80 mg) completed at least 1 hour prior infusion, in combination with oral or injectable diphenhydramine 50 mg and oral acetaminophen 1000 mg given at least 30 minutes prior to the infusion. Premedication requirements for further cycles include oral acetaminophen 1000 mg; additional requirements vary according to infusion-related reaction in previous cycle or lymphocyte count. Consult manufacturer's recommendations.**
- CLL: for the first 28-day cycle: 100 mg IV on day 1; if this dose is completed without changing the infusion rate, give the remaining 900 mg on the same day without delay; otherwise, administer the remaining 900 mg IV dose on day 2, then 1000 mg IV on days 8 and 15. For cycles 2 to 6: 1000 mg IV on day 1.
- Relapsed/refractory FL (28-day cycles): 1000 mg IV on days 1, 8 and 15 of the first cycle, then on day 1 for cycles 2 to 6 followed by every 2 months for up to 2 years.
- Previously untreated FL:
  - In combination with bendamustine (28-day cycles): as for relapsed/refractory FL;
  - In combination with CHOP (cyclophosphamide, doxorubicin, vincristine, prednisone) or CVP (cyclophosphamide, vincristine, prednisone) (21-day cycles): 1000 mg IV on days 1, 8 and 15 of the first cycle, then on day 1 for cycles 2-8, followed by every 2 months for up to 2 years.
- Dosage in renal impairment: no dose adjustment required in patients with CrCl 30 mL/min and above; use with caution when CrCl is between 30 and 50 mL/min; no data if CrCl is below 30 mL/min.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake. Solution in vials should be clear, colourless to slight brown.
- Compatible with NS; incompatible with D5W. Do not mix with other IV solutions.
- Stable for 24 hours in the fridge then 48 hours at room temp (up to 30 °C) in NS at concentrations 0.4-4 mg/mL in PVC, polyethylene, polypropylene or polyolefin bags.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity and infusion-related reactions.
- Patient should be tested for hepatitis B virus infection before initiating therapy.
- Do not administer live vaccines concurrently with obinutuzumab therapy and until B-cell recovery.

REFERENCES

1, 5, 129, 135, 165.

New monograph 2017; limited revision 2019, 2020

NAME OF MEDICATION
OCTREOTIDE

OTHER NAMES
Sandostatin ®, Sandostatin LAR ®

CLASSIFICATION
Somatostatin analog

INDICATIONS

- Control of diarrhea associated with vasoactive intestinal peptide (VIP) secreting tumours.
- Control of symptoms associated with metastatic carcinoid tumours (diarrhea, flushing) and carcinoid crisis (hypotension).
- Treatment of acromegaly.
- Treatment of GI and pancreatic fistulas.
- Treatment of bleeding gastro-esophageal varices.
- To prevent complications following pancreatic surgery in patients undergoing high risk procedures.

ADMINISTRATION

For octreotide solution:
- IV direct (for emergency situations such as carcinoid crisis, bleeding, gastro-esophageal varices): physician only; undiluted; inject over 3 minutes.
- Intermittent IV infusion: dilute dose in 50-100 mL of NS or D5W; infuse over 15-30 minutes.
- Continuous IV infusion: dilute 500 mcg in 250 mL of NS or D5W and administer as a continuous infusion at a rate of 25-200 mcg/hr. More concentrated solutions can be prepared depending on patient's fluid requirements.
- SC: at the thigh, hip or abdomen; rotate injection site. Usual route of administration for VIP secreting tumours, metastatic carcinoid tumours and acromegaly because of delayed absorption, somewhat prolonged activity and patient convenience.
- Administer octreotide between meals and at bedtime to decrease GI side effects.

For octreotide suspension (LAR):
- Reconstitute as per instructions in the package insert.
- IM: deep intragluteal injection; alternate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: sinus bradycardia, conduction abnormalities and arrhythmias observed in patients with acromegaly and carcinoïd syndrome.
- GI: nausea, vomiting, diarrhea, abdominal pain/cramps; incidence decreases with time; refer to Administration section to decrease GI effects. Also cholelithiasis, biliary sludge without stones, biliary duct dilation.
- Endocrine and metabolic: mild and transient hyperglycemia and hypoglycemia.
- Local reactions: irritation at the injection site: pain, burning, redness and swelling; usually last no more than 15 minutes.

DOSAGE

For octreotide solution:
- For most indications: usually start with 50-100 mcg SC 1-3 times daily. Increase dose based on patient response; octreotide is usually given 2-4 times daily. Maximum recommended dose has not been established, but doses up to 1500-3000 mcg daily in divided doses have been given.
- Carcinoid crisis:
  - Prevention of carcinoid crisis in patients undergoing invasive procedures: for patients well controlled with octreotide LAR IM: supplemental dose of octreotide solution 250-500 mcg SC within 1-2 hours prior to procedure; for somatostatin analog-naïve patients with functional neuroendocrine tumours and going for emergency surgery: 500-1000 mcg IV bolus OR 500 mcg SC, 1-2 hours prior to procedure.
  - Carcinoid crisis with hypotension, intraoperative: 500-1000 mcg IV bolus, this may be repeated every 5 minutes until symptoms are controlled OR 500-1000 mcg IV bolus followed by a continuous IV infusion of 50-200 mcg/hr during the procedure.
  - Carcinoid crisis with hypotension, postoperative: if supplemental doses were required during the procedure, continuous IV infusion of 50-200 mcg/hr for 24 hours followed by resumption of preoperative dosing regimen.

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DOSAGE (Cont.)

For octreotide solution (cont.):
- Bleeding gastro-esophageal varices: 50-100 mcg IV bolus (optional) followed by an IV infusion of 25-50 mcg/hr for 48 hours (up to 5 days in high risk patients).
- Prevention of complications following pancreatic surgery: 100 mcg SC TID for 7 days, starting on the day of the operation, at least 60 minutes before laparotomy.

For octreotide suspension (LAR):
- IM: initial dose of 20 mg IM every 4 weeks; may increase to 30 mg if symptoms persist after 2-3 months or may decrease to 10 mg if patient is well controlled. Before starting IM therapy, it is recommended to initiate SC therapy to control symptoms. Consult product monograph for how to switch patients from SC to IM therapy.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

For octreotide solution:
- Store multidose vials and ampoules in the fridge; protect from light.
- Vials and ampoules are stable for 2 weeks at room temp, protected from light.
- Stable for 30 days in the fridge undiluted (100 mcg and 500 mcg/mL) in polypropylene syringes.
- Stable for 96 hours at room temp exposed to light in NS at concentrations of 5, 50 and 250 mcg/mL in PVC containers.
- Stable for 24 hours in D5W.

For octreotide suspension (LAR):
- Store octreotide suspension in the fridge; protect from light.
- Octreotide suspension in vial can remain at room temp on the day of the injection; however, once reconstituted, the suspension should be administered immediately.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 95, 135, 143, 412.

NAME OF MEDICATION	
OLANZapine	
OTHER NAMES	CLASSIFICATION
Zyprexa ®	Antipsychotic

INDICATIONS

- Rapid control of agitation in patients with schizophrenia and related psychotic disorders, and bipolar mania.

ADMINISTRATION

- Reconstitute 10 mg vial with 2.1 mL of SWFI for a final concentration of 5 mg/mL.
- IM (deep).
- SC (palliative care).
- Monitor patient for hypotension (including postural hypotension), bradyarrhythmia and/or hypoventilation for the first 2 to 4 hours following injection.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension and/or syncope associated with bradycardia (infrequent).
- CNS: somnolence, dizziness, extrapyramidal reactions (e.g., acute dystonias).
- Neuroleptic malignant syndrome (rare): hyperpyrexia, muscle rigidity, altered mental status, autonomic instability.

DOSAGE

- Initial dose: usually 10 mg IM. Lower doses of 2.5, 5 or 7.5 mg may also be appropriate.
- Repeat doses: 5-10 mg IM may be repeated 2 hours after the first injection and a third dose no sooner than 4 hours after the second dose, if required.
- Maximum total dose per day is 30 mg, with no more than 3 injections in a 24 hour period.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C.
- Reconstituted solution is stable for 60 minutes at room temp.

MISCELLANEOUS

- If ongoing therapy with olanzapine is required, the patient should be switched to oral therapy as soon as possible.

REFERENCES

1, 5, 135, 453, 457.

Full revision 2017; limited revision 2020

NAME OF MEDICATION
OMALIZUMAB

OTHER NAMES
Xolair ®

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS

- Treatment of moderate to severe persistent asthma in patients 6 years of age and older with a positive reaction to a perennial aeroallergen and who are inadequately controlled with inhaled steroids.
- Treatment of chronic idiopathic urticaria in patients 12 years of age and older who remain symptomatic despite H_1 antihistamine treatment.

ADMINISTRATION

- Reconstitute the 150 mg vial with 1.4 mL of SWFI to obtain a solution of 150 mg/1.2 mL (125 mg/mL). Swirl the vial gently for 1 minute to evenly wet the powder. Do NOT shake. Gently swirl the vial for 5-10 seconds approximately every 5 minutes to dissolve any remaining solids. The drug may take longer than 20 minutes to dissolve completely. Do NOT use the solution if content of the vial is not completely dissolved after 40 minutes.
- Also available as prefilled syringes of 75 and 150 mg. Take the box containing the syringe out of the fridge and leave it for about 20 minutes to reach room temp. Refer to manufacturer's instructions for how to use the syringe.
- SC: in the deltoid region of the arm or in the thigh. Solution is slightly viscous therefore it may take 5 to 10 seconds to inject the drug; no more than 150 mg is to be injected at one site.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis (rare), urticaria, dermatitis, pruritus, throat and/or tongue edema.
- CNS: dizziness, headache.
- Infections: nasopharyngitis, sinusitus, upper respiratory tract infections, viral infections.
- Local reactions: bruising, burning, itching, pain, induration, swelling, erythema, bleeding.

DOSAGE

- Adults and children 12 years of age and older:
  - Asthma: 150-300 mg SC every 4 weeks OR 225-375 mg SC every 2 weeks. Dosing is based on the serum total IgE level before therapy and body weight; refer to manufacturer's monograph for further details.
  - Chronic idiopathic urticaria: 150 mg or 300 mg SC every 4 weeks.

- Pediatrics (6 to 11 years of age):
  - Asthma: 75-300 mg SC every 4 weeks OR 225-375 mg SC every 2 weeks. Dosing is based on the serum total IgE level before therapy and body weight; refer to manufacturer's monograph for further details.

- Dosage in renal impairment: no dosage adjustment necessary.
- Dosage in hepatic impairment: no dosage adjustment necessary.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes in the fridge; do not freeze.
- The reconstituted solution should be clear to slightly opalescent, colourless to pale brownish-yellow.
- The reconstituted solution should be used within 4 hours when stored at room temp or within 8 hours when stored in the fridge; protect from direct sunlight.
- Prefilled syringes can be kept at room temp for a cumulative time of 4 hours.

MISCELLANEOUS

- Each vial of 150 mg of omalizumab (1.2 mL) contains 145 mg of sucrose (0.5 calories or 2.3 Joules).
- The needle cap of the prefilled syringe may contain latex.

REFERENCES

1, 5.

Full revision 2017; limited revision 2018

NAME OF MEDICATION
ONABOTULINUMTOXINA

OTHER NAMES
Botox ®, Botulinum toxin type A

CLASSIFICATION
Neuromuscular paralytic agent

INDICATIONS

- Treatment of strabismus and blepharospasm associated with dystonia.
- To reduce subjective symptoms and objective signs of cervical dystonia (spasmodic torticollis).
- Management of upper and lower limb spasticity related to stroke, spinal injury or cerebral palsy.
- Treatment of hemifacial spasm.
- Prophylaxis of headaches in adults with chronic migraine.
- Use in achalasia, chronic anal fissure and dysphonia (laryngeal dystonia).

ADMINISTRATION

- Reconstitute 100 unit vial with 1, 2, 4 or 8 ml of NS without preservative to get a final concentration of 10 units, 5 units, 2.5 units or 1.25 units per 0.1 mL, respectively. Reconstitute 200 unit vial with 1, 2, 4 or 8 ml of NS without preservative to get a final concentration of 20 units, 10 units, 5 units or 2.5 units per 0.1 mL, respectively. Inject diluent slowly during reconstitution; gently swirl; do NOT shake.
- IM (into affected muscles): physician only, with the appropriate qualifications and experience in the treatment and the use of required equipment; localization of the exact affected muscles with electromyographic guidance or nerve stimulation techniques may be useful.

POTENTIAL ADMINISTRATION HAZARDS

- Side effects may vary depending on sites of administration.
- GI: nausea, dry mouth.
- CNS: headache, dizziness.
- Ocular: diplopia, dry eyes, blurred vision, eyelid ptosis.
- Generalized fatigue, flu-like syndrome, fever.
- Potential for possible distant spread of the toxin from the site of injection with the following resulting symptoms: muscle weakness remote to the site of injection, dysphagia, aspiration pneumonia, speech disorders, respiratory depression.
- Development of antibodies with repeated injections (rare); will result in decreased efficacy.
- Local reactions: pain, swelling, hematoma, ecchymosis.

DOSAGE

- Dose depends on indication, number and location of the muscles that need to be injected, severity of the condition, presence of local weakness and patient's prior response to treatment.
- Consult product monograph and specialized literature for exact dose to be injected.
- Adult maximum cumulative dose is 400 units or 7 units/kg, whichever is lower, in a 3-month interval.
- Pediatric maximum cumulative dose is 200 units or 6 units/kg, whichever is lower, in a 3-month interval.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store unopened vials in the fridge or freezer.
- Reconstituted solution is stable for 24 hours in the fridge; do not freeze.

MISCELLANEOUS

- Caution when administered with an aminoglycoside as the latter can also affect neuromuscular transmission.

REFERENCES

1, 5, 95, 237.

Full revision 2012; limited revision 2013, 2019, 2021

NAME OF MEDICATION
ONDANSETRON

OTHER NAMES
Zofran ®

CLASSIFICATION
Antiemetic

INDICATIONS

- Management of nausea and vomiting associated with chemotherapy.
- For the prevention and treatment of postoperative nausea and vomiting.
- Treatment of epidural opioid-induced pruritus.

ADMINISTRATION

- IV direct **(restricted to patients younger than 65 years of age)**: physician or RN; **only for 4 mg dose or less**; administer undiluted or diluted to 10 mL with NS (dilution recommended to facilitate administration), over 2-5 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W; infuse over 15 minutes.
- IM, SC (alternative routes if IV route not possible).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylactic/anaphylactoid reactions, bronchospasm, hypotension, angioedema, urticaria, rash.
- Cardiovascular: dose-dependent QTc interval prolongation which can lead to torsades de pointes.
- GI: constipation, diarrhea.
- CNS: headache, dizziness.
- Transient increases in liver enzymes.

DOSAGE

Adults:
- Chemotherapy-induced nausea and vomiting:
  - highly emetogenic chemotherapy: 8 mg IV up to a maximum of 16 mg IV (in patients 75 years of age and older, do not exceed 8 mg as the initial dose) given at least 30 minutes prior to chemotherapy. Two additional doses of 8 mg IV may be given 4 and 8 hours after the initial dose. After 24 hours, follow with oral therapy as needed.
  - less emetogenic chemotherapy: 8 mg IV given at least 30 minutes prior to chemotherapy, followed by oral therapy post chemotherapy.
- Postoperative nausea and vomiting: 4 mg IV/IM given immediately before induction of anesthesia or postoperatively.
- Treatment of epidural opioid-induced pruritus: 4 mg IV.

Pediatrics:
- Chemotherapy-induced nausea and vomiting (6 months of age and older): 3-5 mg/m^2 IV or 0.15 mg/kg/dose IV (maximum: 16 mg/dose) infused over 15 minutes, at least 30 minutes before chemotherapy; to repeat twice at 4-hour intervals for 3 doses total; may also switch to oral therapy post chemotherapy.
- Postoperative nausea and vomiting (1 month of age and older): 0.1 mg/kg IV, to a maximum of 4 mg; to be given immediately before induction of anesthesia or postoperatively.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules below 30°C; protect from light. Do not freeze.
- Compatible with NS, D5W, dextrose-saline solutions, RL.
- Stable for 48 hours at room temp, 14 days at 5°C and 3 months at -20°C in D5W or NS at concentrations of 0.03 mg/mL and 0.3 mg/mL in PVC containers.
- Stable for 48 hours at room temp, 14 days at 4°C and 90 days at -20°C undiluted (2 mg/mL) or diluted with NS or D5W at concentrations of 0.25 mg/mL, 0.5 mg/mL and 1 mg/mL in plastic syringes.

…/Cont.

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MISCELLANEOUS

- Avoid use of ondansetron in patients with congenital long QT syndrome. Use with caution in patients with other risk factors for QT interval prolongation such as electrolyte disturbances (hypokalemia, hypomagnesemia, hypocalcemia), congestive heart failure, bradyarrhythmias or taking other drugs that can lead to either QT interval prolongation or electrolyte disturbances.

REFERENCES

1, 4, 5, 6, 40, 82, 95, 143, 211, 226, 227, 239, 330.

NAME OF MEDICATION
OXALIPLATIN

OTHER NAMES
Eloxatin ®, Oxalatoplatin, Oxaliplatinum

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of metastatic colorectal cancer in combination with 5-fluorouracil and leucovorin.
- As an adjuvant treatment in stage III (Duke's C) colon cancer after complete resection of primary tumour, in combination with 5-fluorouracil and leucovorin.
- Also effective for a number of other cancers, including breast, stomach, head and neck, lung, pancreas, ovary, testicular, prostate and cancer of unknown primary origin.

ADMINISTRATION

- Intermittent IV infusion: dilute in 250 to 500 mL of D5W only (NOT in NS or any chloride solution) to obtain a concentration not less than 0.2 mg/mL and not more than 0.7 mg/mL; infuse over 2 to 6 hours. The infusion line should be flushed with D5W prior to administration of oxaliplatin.
- To be given before 5-fluorouracil.
- Aluminium-containing IV needles, syringes or sets should not be used to prepare or administer oxaliplatin.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis, skin rash.
- Cardiovascular: hypotension, hypertension.
- GI: nausea, vomiting, diarrhea, mucositis.
- CNS: sensory neuropathy (cumulative, dose-related, usually reversible); pharyngolaryngeal dysesthesia (minimized by extending infusion time to 6 hours; **antidote**: antihistamines, bronchodilators); dysphasia; ataxia; cranial neuropathy.
- Hematologic: myelosuppression; immune hemolytic anemia (rare).
- Ophthalmic: conjunctivitis, lacrimation.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- 85 mg/m^2 IV repeated every 2 weeks when used in combination with 5-fluorouracil and leucovorin.
- Dosage adjustment required in patients who develop sensory neuropathies, severe diarrhea or severe neutropenia or thrombocytopenia. Refer to product monograph for further details.
- Dosage in renal impairment: CrCl of 30 mL/min and greater: 100% of dose; CrCl less than 30 mL/min: contraindicated.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light. Do not freeze.
- Stable for 24 hours at room temp and 48 hours in the fridge in D5W in the concentration range of 0.2 mg/mL to 2 mg/mL.
- Do not mix with sodium chloride or chloride containing solutions as it causes degradation of oxaliplatin.
- Compatible with leucovorin through a Y-site connection if leucovorin is diluted in D5W; do not combine in the same infusion bag.

MISCELLANEOUS

- Contraindicated in patients with a hypersensitivity to other platinum agents (cisplatin, carboplatin).

REFERENCES

1, 5, 40, 129, 165.

Full revision 2017; limited revision 2018, 2020

NAME OF MEDICATION
OXYTOCIN, SYNTHETIC

OTHER NAMES
Pitocin ®, Syntocinon ®

CLASSIFICATION
Oxytocic

INDICATIONS

- Induction of labour when indicated in term or near-term pregnancies.
- Augmentation of labour in the first or second stage of labour when there is evidence of uterine hypocontractility or labour dystocia.
- To produce uterine contractions during the third stage of labour and to control postpartum bleeding and hemorrhage.
- Adjunctive therapy in the management of inevitable or incomplete abortion, or midtrimester elective abortion.

ADMINISTRATION

For induction/augmentation of labour:
- Oxytocin must NOT be administered IV direct in the antepartum patient.
- Continuous IV infusion: ***blood pressure monitoring***; ***fetal monitoring.*** Dilute 10 units in 1000 mL of NS, D5W or RL. (At TOH birthing unit: dilute 20 units in 500 mL of RL, as per Obstetrics policy 00659 BU-Induction/Augmentation of labour: Oxytocin administration). Mix solution thoroughly. Must be piggybacked to mainline IV solution.
- Must be administered by an infusion pump.

Third stage of labour/postpartum:
- IV direct: physician or RN under physician supervision, with or after the delivery of the anterior shoulder of the infant; ***blood pressure monitoring***. Dilute 5-10 units with 3-5 mL of NS and inject IV over 1-2 minutes.
- Continuous IV infusion: dilute 20-40 units in 1000 mL of NS, D5W or RL. (At TOH suggest to dilute 20 units in 500 mL or 1000 mL of RL). Mix thoroughly.
- IM.

Abortion:
- Continuous IV infusion: dilute 10 units in 500 mL of NS or D5W. Mix thoroughly.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis.
- Cardiovascular: hypotension, tachycardia, arrhythmia. Severe hypertension may occur following prophylactic administration of a vasoconstrictor in conjunction with caudal block anesthesia. Also fetal bradycardia.
- GI: nausea, vomiting.
- Uterine hyperstimulation.
- Water intoxication and seizures with prolonged IV infusion of oxytocin with excessive fluid load.

DOSAGE

- Induction/augmentation of labour: administer at initial rate of 0.5-1 milliunits/min IV, and increase slowly in increments of 1-2 milliunits/min at 30-60 minute intervals depending on response up to suggested maximum of 20 milliunits/min; rates over 9-10 milliunits/min are rarely needed.
- Third stage of labour and postpartum hemorrhage: give 10 units IM OR 5 to 10 units IV direct OR administer a continuous IV infusion at a rate of 50-100 milliunits/min (3-6 units/hr), rate to be adjusted to maintain uterine contraction and control uterine atony.
- Abortion: 10 units as an IV infusion at 10-20 millliunits/min after suction, curettage or abortifacients; total dose should not exceed 30 units in a 12-hour period to avoid water intoxication.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Do not freeze.
- Stable for 28 days at room temp protected from light in RL at a concentration of 0.08 units/mL in PVC bags.
- Stable for 90 days at room temp protected from light in D5W or NS at a concentration of 0.08 units/mL in PVC bags.

MISCELLANEOUS

- 1 unit of oxytocin = 2-2.2 mcg of pure oxytocin.
- 1 milliunit = 0.001 unit.

REFERENCES

1, 4, 5, 40, 64, 95, 135, 143, 208.

NAME OF MEDICATION	
PACLitaxel	
OTHER NAMES	CLASSIFICATION
Taxol ®	Antineoplastic

Do NOT confuse paclitaxel with nanoparticle, albumin-bound paclitaxel.
This monograph is specific to PACLITAXEL.

INDICATIONS
- Alone or in combination for the treatment of carcinoma of the ovary, breast or lung, or AIDS-related Kaposi's sarcoma.
- Also used in other tumours (e.g., head and neck, gastrointestinal, genitourinary, gynecological, thyroid, thymoma, Ewing's sarcoma, skin cancer, cancer of unknown primary origin).

ADMINISTRATION
- **Ensure premedication has been administered; refer to Dosage section.**
- Intermittent IV infusion, Continuous IV infusion: dilute with NS, D5W, D5-NS or D5-RL to a final concentration of 0.3-1.2 mg/mL; infuse over 1-3 hours. May also be administered as a 24-hour infusion, depending on specific protocol.
- Use non-PVC IV containers and administration sets for dilution and administration (provided by Pharmacy). Infuse through a 0.22 micron in-line filter.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: reactions commonly occur during the first hour of infusion of the first two cycles and are occasionally severe. Frequent monitoring of vital signs is recommended, particularly during the first hour of infusion. (At TOH, vital signs are monitored pre-infusion and then once every hour up to the end of the infusion; patients should be closely monitored throughout the infusion for other signs and symptoms of paclitaxel hypersensitivity).

  Mild hypersensitivity (i.e., mild flushing, rash, pruritus):
  - Continue paclitaxel infusion and observe closely for worsening symptoms.

  Moderate hypersensitivity (i.e., moderate rash, moderate flushing, moderate dyspnea, chest discomfort, mild hypotension):
  - Stop paclitaxel infusion, give diphenhydramine 50 mg IV and/or hydrocortisone 100 mg IV.
  - Once symptoms resolved, re-start paclitaxel infusion at 25% of the rate at the time of the reaction for 5 minutes, then at 50% of the rate at the time of the reaction for 5 minutes then at 75% of the rate at time of the reaction for 5 minutes and, if no further symptoms, resume to the rate at the time of the reaction.

  Severe hypersensitivity (i.e., respiratory distress requiring treatment, generalized urticaria, angioedema, hypotension requiring treatment):
  - Stop paclitaxel infusion.
  - Cardiac monitoring (if available) plus NS IV bolus if hypotension.
  - Give 0.5 mL of epinephrine (1 mg/mL) IM plus diphenhydramine 50 mg IV and hydrocortisone 100 mg IV.
  - If no or inadequate response, repeat 0.5 mL of epinephrine (1 mg/mL) IM plus oxygen and bronchodilators as needed.
  - May refer to The Ottawa Hospital Anaphylaxis Algorithm.
  - Further treatment with paclitaxel is possible, but is at the discretion of the treating physician.
- Cardiovascular: bradycardia, severe conduction abnormality (rare), hypotension during infusion. If patient experiences significant cardiac conduction abnormalities during administration, continuous ECG should be performed during subsequent therapy.
- GI: nausea, vomiting, diarrhea, inflammation of the cecum (rare).
- Dermatologic: radiation recall reactions (rare).
- Local reactions: phlebitis, erythema, discomfort, tenderness at site of injection.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

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DOSAGE

- **To minimize severe hypersensitivity reactions, premedicate with dexamethasone 20 mg PO 12 and 6 hours before paclitaxel infusion or 20 mg IV 30 minutes before paclitaxel infusion, diphenhydramine 50 mg IV (at TOH, give PO) and famotidine 20 mg IV 30-60 minutes before paclitaxel infusion**.
- Initial dose is 135-175 mg/m^2 IV every three weeks.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp and protect from light. Once punctured, the 5, 16.7 and 25 mL vials are stable for 28 days at room temp; the 50 mL and 100 mL pharmacy bulk vials should be used within 24 hours after puncture.
- Stable for 3 days at room temp when diluted in NS or D5W at a concentration of 0.3-1.2 mg/mL in polyolefin or polyethylene containers.
- Stable for 3 days at room temp or in the fridge in NS or D5W at a concentration of 0.1-1 mg/mL in polyolefin containers.

MISCELLANEOUS

REFERENCES

1, 4, 5, 129, 135, 143, 165, 184, 294.

Full revision 2017; limited revision 2018, 2019, 2020, 2021

NAME OF MEDICATION	**PACLitaxel (nanoparticle, albumin-bound)**
OTHER NAMES Abraxane ®, Nab-paclitaxel	CLASSIFICATION Antineoplastic

Do NOT confuse nanoparticle, albumin-bound paclitaxel with paclitaxel.
This monograph is specific to NANOPARTICLE, ALBUMIN-BOUND paclitaxel.

INDICATIONS

- Treatment of metastatic breast cancer.
- First-line treatment of metastatic adenocarcinoma of the pancreas, in combination with gemcitabine.
- Other uses: bladder/urothelial cancer and melanoma.

ADMINISTRATION

- Reconstitute the 100 mg vial by injecting 20 mL of NS slowly over 1 minute and directing the solution flow onto the inside wall of the vial. Then allow vial to sit for a minimum of 5 minutes to ensure proper wetting of the lyophilized cake/powder. Gently swirl vial and/or invert vial slowly for at least 2 minutes until complete dissolution. If foaming or clumping occurs, stand solution for at least 15 minutes until foam subsides. Reconstituted solution should be milky and homogenous without visible particulates. Final concentration is 5 mg/mL.
- Intermittent IV infusion: inject the appropriate amount into an empty sterile PVC or non-PVC bag and infuse over 30 minutes (30-40 minutes in the case of pancreatic cancer).
- Use of specialized DEHP-free containers or administration sets is NOT necessary.
- Use of syringes and IV bags containing silicone oil as a lubricant during reconstitution and administration may cause proteinaceous strands. If strands are observed in IV bag, use a 15 micron filter (do not use a smaller pore size) to administer the solution; if the strands are observed and a 15 micron filter is not available, discard the product.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rare (less than 4%) and minor. Premedication to prevent hypersensitivity reactions is not required before administration.
- Cardiovascular: hypotension and bradycardia during the 30-minute infusion (usually asymptomatic), ECG abnormalities (usually asymptomatic), AV block.
- GI: nausea, vomiting, diarrhea, mucositis.
- CNS: sensory neuropathy.
- Respiratory: dyspnea, cough.
- Hematologic: neutropenia, leukopenia.
- Arthralgia, myalgia.
- Local reactions: at injection site.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Metastatic breast cancer: 260 mg/m^2 IV every 3 weeks. Dose can be reduced to 220 mg/m^2 then to 180 mg/m^2 or held/discontinued depending on severity of toxicity (neutropenia, sensory neuropathy), as per recommendations from the manufacturer.
- Metastatic cancer of the pancreas: 125 mg/m^2 IV on days 1, 8, 15 of each 28-day cycle. Follow with the infusion of gemcitabine. Nab-paclitaxel dose can be reduced to 100 mg/m^2 then to 75 mg/m^2 or discontinued depending on severity of toxicity (neutropenia, thrombocytopenia, sensory neuropathy, cutaneous toxicity, gastrointestinal toxicity), as per recommendations of the manufacturer.

…/Cont.

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DOSAGE (Cont.)

- Dosage in renal impairment:

CrCl (mL/min)	Dose
30-90	100%
Less than 30	discontinue

- Dosage in hepatic impairment: refer to manufacturer's instructions for dosage modification based on AST and serum bilirubin.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the original containers at room temp, protected from light.
- Reconstituted solution should be used immediately but is stable for 8 hours in the fridge and protected from light when left in the vial.
- Stable for 8 hours at room temp at ambient light at a concentration of 5 mg/mL (undiluted) in a PVC container.

MISCELLANEOUS

- Each 100 mg vial of paclitaxel contains approximately 900 mg of albumin.
- The use of nab-paclitaxel in patients who had hypersensitivity reactions with "conventional" paclitaxel has not been studied.

REFERENCES

1, 5, 40, 129, 165.

NAME OF MEDICATION
PALIPERIDONE

OTHER NAMES
Invega Sustenna ®, Invega Trinza ®

CLASSIFICATION
Antipsychotic

INDICATIONS

- Long-acting depot medication for the treatment of schizophrenia (1-month injection Invega Sustenna ® and 3-month injection Invega Trinza ®) and schizoaffective disorder (1-month injection Invega Sustenna ®).
- Not indicated in the elderly with dementia due to increased risk of mortality.

ADMINISTRATION

1-month injection (Invega Sustenna ®):
- IM only: shake the prefilled syringe vigorously for at least 10 seconds. Attach appropriate needle (see Note at the bottom of this section for selecting the proper size). Remove air. Inject slowly deep into the muscle. The initial 2 doses must be administered in the deltoid muscle. The subsequent maintenance doses can be administered in either the deltoid or the gluteal muscle. Each dose should be administered as a single injection. Alternate sides (right, left) at each dose.
- Note: use needles included in the kit: use the 23 gauge needle for deltoid administration in patients weighing less than 90 kg; use the 22 gauge needle in all other situations.

3-month injection (Invega Trinza ®):
- IM only: shake the prefilled syringe vigorously with the tip up and a loose wrist for at least 15 seconds to ensure a uniform suspension (milky white colour). Attach appropriate needle (see Note at the bottom of this section for selecting the proper size). Within 5 minutes of shaking vigorously, inject slowly deep into the deltoid or gluteal muscle.
- In the event of an incomplete administration: do NOT re-inject the dose remaining in the syringe and do NOT administer another dose. Monitor and treat patient appropriately until the next scheduled 3-month injection.
- Note: use the thin wall needles included in the kit; use the 22 gauge 1-inch for deltoid administration in patients weighing less than 90 kg and the 22 gauge 1.5 inch in all other situations.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity (rare): rash, hives, angioedema, anaphylaxis, dyspnea, hypotension, Stevens-Johnson syndrome, toxic epidermal necrolysis, and drug reaction with eosinophilia and systemic symptoms (DRESS).
- Cardiovascular: QT interval prolongation, orthostatic hypotension, syncope, bradycardia, tachycardia.
- GI: dysphagia, constipation (can be severe), nausea, vomiting.
- CNS: extrapyramidal symptoms (tardive dyskinesia, parkinsonism, akathisia), sedation, dizziness, seizures, agitation, suicidal ideation, anxiety, headache.
- Hematologic: leukopenia, neutropenia, granulocytopenia; deep vein thrombosis, pulmonary embolism (very rare).
- Disruption of body temperature regulation.
- Hyperglycemia.
- Hyperprolactinemia.
- Neuroleptic malignant syndrome (rare): hyperpyrexia, muscle rigidity, altered mental status, autonomic instability.
- Local reactions: mild; pain, induration, redness, swelling.

DOSAGE

1-month injection (Invega Sustenna ®)
- For patients who have never taken long-acting injectable antipsychotics:
  - Initiation: 150 mg IM on day 1 and 100 mg IM on day 8.
  - Maintenance for schizophrenia: the usual dose is 75 mg (range 25-150 mg) IM every month, starting one month after the end of the initiation regimen.
  - Maintenance for schizoaffective disorder: range of 50 to 150 mg IM every month, starting one month after the end of initiating regimen.

…/Cont.

…/Cont.

DOSAGE (Cont.)

- For patients who are switching from another long-acting injectable antipsychotic: start directly with the maintenance dose at the next scheduled injection, usually 75 mg (range 25-150 mg) IM every month. Refer to manufacturer's monograph for specific dose conversion from long-acting risperidone to long-acting paliperidone.
- Dosage in renal impairment: in patients with mild renal impairment (CrCl from 50 to less than 80 mL/min), the initiation regimen is reduced to 100 mg IM on day 1 and 75 mg IM on day 8, and the maintenance dose is reduced to 50 mg IM once monthly. Not recommended in patients with CrCl under 50 mL/min.
- Dosage in hepatic impairment: no dose adjustment required in patients with mild to moderate impairment.

3-month injection (Invega Trinza ®)
- To be used only after patient has been stabilized for at least 4 months on the 1-month paliperidone injectable formulation (Invega Sustenna ®), with the last 2 doses being the same dosage strength.
- First dose: 3.5 x dose of 1-month injection (Invega Sustenna ®) IM, to give when the next dose of Invega Sustenna ® would have been due or up to 7 days before or after the next monthly dose date.
- Subsequent doses: 175-525 mg IM every 3 months; on exceptional occasions, may give the injection up to 2 weeks before or after the 3-month due date.
- Missed doses (i.e., more than 2 weeks past due date): refer to manufacturer's instructions.
- Dosage in renal impairment: for mild renal impairment (CrCl from 50 to less than 80 mL/min), adjust dose using the 1-month injection (Invega Sustenna ®) then transition to 3-month injection (Invega Trinza ®) in a 3.5 to 1 ratio with a maximum dose of 350 mg. Not recommended in patients with a CrCl less than 50 mL/min.
- Dosage in hepatic impairment: no dosage adjustment required in patients with mild to moderate impairment.
- Refer to manufacturer's instructions for switching from the 3-month injection (Invega Trinza ®) to the 1-month injection (Invega Sustenna ®) or to oral paliperidone sustained-release tablets.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled syringes at room temp.

MISCELLANEOUS

- Paliperidone is the active metabolite of risperidone.

REFERENCES

5, 95.

NAME OF MEDICATION
PAMIDRONATE

OTHER NAMES
Aredia ®

CLASSIFICATION
Bisphosphonate

INDICATIONS

- For the treatment of hypercalcemia of malignancy following saline rehydration.
- Symptomatic Paget's disease of bone.
- Conditions associated with increased osteoclastic activity: predominantly lytic bone metastases and multiple myeloma.

ADMINISTRATION

- Intermittent IV infusion (mandatory): dilute in 250-500 mL of D5W or NS; infuse over 2-4 hours depending on the indication (refer to Dosage section).
- Note: dilution and slow IV administration prevent thrombophlebitis, severe local reactions and renal failure.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; angioedema, dyspnea, hypotension and anaphylaxis.
- Cardiovascular: arrhythmia, hypertension, hypotension (rare).
- GI: nausea, vomiting.
- Hypocalcemia, hypomagnesemia, hypophosphatemia.
- Fever (may last up to 48 hours), influenza-like symptoms (usually at first infusion only), malaise.
- Transient bone pain, arthralgia, myalgia.
- Local reactions: pain, redness, swelling, induration, thrombophlebitis.

DOSAGE

Hypercalcemia:

- Dose based on initial corrected serum calcium concentration (refer to Miscellaneous section for calculation of corrected serum calcium):

Serum calcium (mmol/L)	less than 3	3-3.5	3.5-4	greater than 4
Dose of pamidronate (mg) IV	30	30-60	60-90	90

- Recommended maximum infusion rate is 22.5 mg/hr (i.e., 90 mg in 500 mL over 4 hours).

Bone metastases:

- 90 mg diluted in 250 mL and given IV over 2 hours every 3-4 weeks.
- Recommended infusion rate is 45 mg/hr.
- In a small trial, 90 mg of pamidronate in 250 mL NS was given over 60 minutes to patients with bone metastases without untoward effects.

Multiple myeloma:

- 90 mg diluted in 500 mL and given IV over 4 hours every 4 weeks.
- Recommended infusion rate is 22.5 mg/hr.

Paget's disease:

- Usual recommended total dose for a treatment course: 180-210 mg IV administered as 30 mg weekly for 6 weeks OR 30 mg on week 1, then 60 mg on weeks 3, 5 and 7 OR 60 mg every 2 weeks X 3 doses if retreatment.
- A 90 mg total dose for a treatment course (30 mg IV daily for 3 consecutive days) has also been used.
- Dilute each dose of 30 mg or 60 mg in at least 250 mL or 500 mL, respectively.
- Recommended infusion rate is 15 mg/hr.

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…/Cont.

DOSAGE (Cont.)

- Dosage in renal impairment: maximum rate of 22.5 mg/hr. Not to administer if CrCl is below 30 mL/min unless in cases of life-threatening tumor-induced hypercalcemia where the benefit exceeds risk.
- In hemodialysis at TOH: usual dose of 60 mg in 250 mL NS given IV over 60 minutes after dialysis. Range dose of 30-90 mg; 90 mg dose to be administered over 90 minutes.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from heat.
- Stable for 24 hours in the fridge (protected from light) in D5W or NS at concentrations of 0.06-0.36 mg/mL in PVC bags followed by 24-hour exposure at room temp exposed to light, for a total of 48 hours.
- Stable for 7 days at room temp and 30 days in the fridge in D5W at a concentration of 0.36 mg/mL in PVC bags.
- Calcium containing solutions should not be used (e.g., RL).

MISCELLANEOUS

- In tumour-induced hypercalcemia, corrected calcium levels should be monitored; use the following equation to correct serum calcium values:
  Corrected Ca (mmol/L) = measured Ca (mmol/L) + (0.02 X [40-measured albumin in g/L]).

REFERENCES

1, 5, 40, 129, 165, 187, 208, 267.

NAME OF MEDICATION
PANCURONIUM

OTHER NAMES

CLASSIFICATION
Neuromuscular blocker

INDICATIONS
- As an adjunct to anesthesia to induce muscle relaxation during surgery.
- To facilitate the management of patients undergoing endotracheal intubation or mechanical ventilation.

ADMINISTRATION
- IV direct: physician trained in anesthesiology, RN may administer subsequent doses; ***ventilator support; cardiac monitoring***. Inject undiluted over 60-90 seconds.
- Continuous IV infusion: ***ventilator support; cardiac monitoring***. Dilute in a compatible solution; titrate rate to patient response (refer to Dosage section). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; from histamine release: cutaneous flushing, bronchospasm, hypotension.
- Cardiovascular: increased heart rate, transient rise in BP.
- GI: excessive salivation if no anticholinergic premedication given.

Antidote: Anticholinesterase agents such as neostigmine in conjunction with an anticholinergic agent such as atropine or glycopyrrolate.

DOSAGE
- Adults:
  - As an adjunct to anesthesia during surgery: initial doses of 0.04-0.1 mg/kg (40-100 mcg/kg) IV with additional doses of 0.01 mg/kg (10 mcg/kg) IV at 25-60 minute intervals if needed to sustain neuromuscular blockade for longer periods.
  - For endotracheal intubation: 0.06-0.1 mg/kg (60-100 mcg/kg) IV; doses up to 0.16 mg/kg (160 mcg/kg) have been used.
  - For mechanical ventilation: IV direct injection of 0.1-0.2 mg/kg q1-3h OR continuous IV infusion beginning with a loading dose of 0.03-0.1 mg/kg followed by a maintenance dose of 0.06-0.1 mg/kg/hr (60-100 mcg/kg/hr).

- Pediatrics:
  - As an adjunct to anesthesia during surgery: initial doses of 0.04-0.1 mg/kg (40-100 mcg/kg) IV with additional doses of 0.01 mg/kg (10 mcg/kg) IV at 25-60 minute intervals if needed to sustain neuromuscular blockade for longer periods. For neonates, due to extreme sensitivity to pancuronium, begin with a test dose of 0.02 mg/kg (20 mcg/kg) IV.

- Dosage in renal impairment: decrease dose to 50% if CrCl is 10-50 mL/min and avoid in patients with a CrCl below 10 mL/min.
- Dosage in hepatic impairment: use with caution; adjust dosage as necessary.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and vials in the fridge; however, pancuronium is stable for 6 months at room temp.
- Stable for 48 hours at room temp in D5W, D5-NS, D5-1/2NS, RL and NS in glass or plastic containers.

MISCELLANEOUS
- Onset: 2-3 minutes. Duration: 30-45 minutes.
- Caution in patients with myasthenia gravis and debilitated states.
- Effect is potentiated by inhalation anesthetics, aminoglycosides, opioids, corticosteroids, acid-base and/or electrolyte disturbances.

REFERENCES
1, 4, 5, 40, 95, 216, 263.

Full revision 2017; limited revision 2018, 2020

NAME OF MEDICATION
PANITUMUMAB

OTHER NAMES
Vectibix ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Monotherapy treatment of patients with epidermal growth factor receptors (EGFR) expressing metastatic colorectal carcinoma with non-mutated (wild-type) RAS after failure of fluoropyrimidine-, oxaliplatin-, and irinotecan-containing chemotherapy regimens.
- Treatment of previously untreated patients with non-mutated (wild-type) RAS metastatic colorectal carcinoma in combination with 5-fluorouracil, leucovorin and oxilaplatin (FOLFOX).

ADMINISTRATION

- Intermittent IV infusion (mandatory): doses up to 1000 mg: dilute in 100 mL NS and infuse over 60 minutes. Can be administered over 30-60 minutes for subsequent infusions if first one was well tolerated. Doses over 1000 mg: dilute in 150 mL NS and infuse over 90 minutes. Dilute with NS only. Mix solution by gentle inversion. Do NOT shake. Final concentration not to exceed 10 mg/mL. Solution should be clear and colourless and may contain a small amount of visible, translucent-to-white amorphous, panitumumab particles. Administer via a low-protein-binding 0.2 micron or 0.22 micron in-line filter. Flush line with NS before and after administration.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: early and late onset reactions.
- Infusion-related reactions: anaphylaxis, angioedema, bronchospasm, fever, chills, and hypotension. For severe infusion reactions, stop the infusion immediately and treatment with panitumumab should be permanently discontinued. For mild to moderate reactions, slow the infusion rate by 50%.
- GI: nausea, diarrhea, abdominal pain, anorexia, vomiting, constipation.
- Dermatologic: dermatitis, paronychia, pruritus, erythema, acneiform rash, skin fissures. May need topical hydrocortisone 1% and oral antibiotics.
- Respiratory: dyspnea, cough.
- Fatigue.
- Hypomagnesemia, hypocalcemia, hypokalemia.

DOSAGE

- RAS testing must be performed prior to administration.
- 6 mg/kg (actual body weight) IV every 14 days, until disease progression or undesirable toxicity.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Protect from light. Do not freeze and do not shake.
- Diluted solution is stable 6 hours at room temp and 24 hours in the fridge (according to manufacturer). Do not freeze.
- Do not mix with other drugs or dilute with infusion solutions other than NS.

MISCELLANEOUS

- Patients should limit sun exposure; use of skin moisturizers and sunscreen is encouraged to avoid exacerbation of dermatologic effects.

REFERENCES

1, 5, 40, 95, 129, 165.

Full revision 2017; limited revision 2018, 2020

NAME OF MEDICATION
PANTOPRAZOLE

OTHER NAMES
Panto IV

CLASSIFICATION
Proton pump inhibitor

INDICATIONS

- Treatment of conditions where a rapid reduction of gastric acid secretion is required, such as reflux esophagitis or Zollinger-Ellison syndrome, in patients who cannot tolerate oral medications.
- Prophylaxis of recurrent upper GI bleeding.

ADMINISTRATION

- Reconstitute each 40 mg vial with 10 mL of NS to obtain a concentration of 4 mg/mL.
- IV direct: physician or RN; undiluted; inject over 2-5 minutes.
- Intermittent IV infusion: dilute dose in 50-100 mL of NS or D5W; infuse over 15 minutes.
- Continuous IV infusion: withdraw 20 mL from a 100 mL NS or D5W minibag and add 80 mg of pantoprazole (20 mL) for a final concentration of 0.8 mg/mL; infuse at 8 mg/hr (10 mL/hr).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: pruritus, exanthema, anaphylaxis.
- GI: diarrhea, abdominal pain.
- CNS: headache.
- Local reactions: inflammation, bruises.

DOSAGE

- Gastro-duodenal ulcer or reflux esophagitis: 40 mg IV once daily.
- Zollinger-Ellison syndrome: 80 mg IV q12h; can be increased to q8h if needed.
- Prophylaxis of recurrent upper GI bleeding: loading dose of 80 mg IV, followed by a continuous IV infusion at a rate of 8 mg/hr for 72 hours OR loading dose of 80 mg IV followed by an intermittent IV infusion of 40 mg q6-12h for 72 hours.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Reconstituted solutions must be used within 24 hours of initial puncture of the stopper.
- According to manufacturers: the reconstituted solution must be diluted within 3 hours of the initial puncture of the stopper; when further diluted, stable for 21 hours at room temp in NS at a concentration of 0.4 mg/mL or 0.8 mg/mL and for 12 hours at room temp when diluted in D5W at the same concentrations. However, longer stability data are available from other sources; see below.

For brands <u>with</u> EDTA (Teva, Frenesius Kabi):
- Stable for 20 days at 4°C in NS at concentrations between 0.16 mg/mL and 0.8 mg/mL in PVC bags; these solutions are stable for an additional 6 hours at room temp after their fridge storage.
- Stable for 11 days at 4°C in D5W at concentrations between 0.16 mg/mL and 0.8 mg/mL in PVC bags; these solutions are stable for an additional 8 hours at room temp after their fridge storage.
- Stable for 48 hours at 23°C in NS at concentrations between 0.16 mg/mL and 0.8 mg/mL in PVC bags.
- Stable for 24 hours at 23°C in D5W at concentrations between 0.16 mg/mL and 0.8 mg/mL in PVC bags.
- Stable for 96 hours in the fridge and at room temp in NS at a concentration of 4 mg/mL in polypropylene syringes.
- Compatible with RL.

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COMPATIBILITY, STABILITY (Cont.)

For brands without EDTA (Sandoz, Pharmascience):
- Reconstituted solution (4 mg/mL) is stable for 3 days at room temp, exposed to light, in glass vials.
- Reconstituted solution (4 mg/mL) is stable for 28 days in the fridge, protected from light, in polypropylene syringes.
- Solution stable for 3 days at room temp (normal lighting) and 28 days in the fridge (protected from light) in NS at concentrations of 0.4 mg/mL and 0.8 mg/mL in PVC bags.
- Solution stable for 2 days at room temp (normal lighting) and 14 days in the fridge (protected from light) in D5W at a concentration of 0.4 mg/mL in PVC bags.
- Solution stable for 3 days at room temp (normal lighting) and 28 days in the fridge (protected from light) in D5W at a concentration of 0.8 mg/mL in PVC bags.

MISCELLANEOUS

REFERENCES

1, 2, 4, 5, 6, 40, 77, 125, 223, 359, 517.

NAME OF MEDICATION	
PAPAVERINE	
OTHER NAMES	CLASSIFICATION
	Vasodilator

INDICATIONS

- To decrease vascular spasms such as those associated with myocardial infarction, angina, peripheral and pulmonary embolism, peripheral vascular disease, and cerebral angioplastic states.
- To decrease visceral spasms, such as ureteral, biliary and GI colic.

ADMINISTRATION

- IV direct: physician only, undiluted or diluted in an equal volume of SWFI; inject slowly over a 1-2 minute period since arrhythmias and fatal apnea may result from rapid injection.
- IM, SC.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: arrhythmias with rapid IV injection; hypertension or hypotension.
- GI: anorexia, nausea, abdominal distress.
- CNS: drowsiness, headache, vertigo.
- Respiratory: apnea with rapid IV injection.
- Sweating, facial flushing.
- Local reactions: thrombophlebitis at injection site.

DOSAGE

- Adults: 30-120 mg IV/IM/SC. May repeat q3h as necessary. In treating cardiac extrasystoles: 2 doses may be administered 10 minutes apart.
- Pediatrics: 6 mg/kg/day IV/IM/SC in 4 divided doses.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp, protect from light. Crystallization may occur at low temperatures; in such cases, warm the vial and shake well until complete dissolution.
- Yellow discolouration of papaverine does not appear to be related to drug decomposition.
- Compatible with NS, dextrose, Ringer's injection and sodium lactate 1/6 M solutions.
- Incompatible with RL.

MISCELLANEOUS

- Administer with caution to patients with glaucoma or liver disease.

REFERENCES

1, 4, 5, 40, 95, 135.

Full revision 2017

NAME OF MEDICATION
PARICALCITOL *

OTHER NAMES
19-nor-1 alpha-25-dihydroxyvitamin D_2, Zemplar ®

CLASSIFICATION
Vitamin D analog

INDICATIONS

- Prevention and treatment of secondary hyperparathyroidism associated with chronic renal failure stage 5 patients on hemodialysis.

ADMINISTRATION

- IV direct: (via a hemodialysis line), physician or RN. Administer undiluted as a bolus at any time during hemodialysis.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: pruritus, rash, urticaria, oral and facial edema, angioedema.
- GI: nausea, vomiting, taste changes (metallic taste), dry mouth, GI bleed.
- CNS: headache, dizziness.
- Hypercalcemia related to vitamin D intoxication: patient may be asymptomatic or presents with early signs of intoxication such as weakness, headache, somnolence, nausea, vomiting, dry mouth, constipation, muscle pain, bone pain and metallic taste.
- Hyperphosphatemia.
- Fever, chills.

DOSAGE

- Dose depends on parathyroid hormone (PTH) level; target range for intact PTH level in chronic renal failure stage 5 patients is no more than 1.5 to 3 x non-uremic ULN.
- Initial dose: 0.04 to 0.1 mcg/kg IV at dialysis, no more frequently than every other day.
- If a satisfactory response is not obtained, dose may be increased by 2-4 mcg at every 2 to 4 week intervals; single doses as high as 0.24 mcg/kg have been safely administered.
- Refer to product monograph for suggested dosing guidelines according to PTH level.
- Relative dosing of paricalcitol to calcitriol is 4:1. When converting a patient from calcitriol to paricalcitol, the initial dose of paricalcitol should be 4 times greater than the patient's dose of calcitriol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 15-25°C; protect from light and freezing.
- Do not mix with heparin as its effect will be neutralized by propylene glycol, an excipient of the paricalcitol formulation.

MISCELLANEOUS

- Monitor serum calcium and phosphorus levels frequently, at least twice a week during dosage adjustment then at least monthly once dosage is established.
- Monitor serum PTH level at least every 3 months or more often during titration.
- Caution if patient is on digoxin as digoxin toxicity is potentiated by hypercalcemia.
- Caution if patient is starting or discontinuing an inhibitor of cytochrome P450 3A4 isoenzyme as paricalcitol is partly metabolized by these enzymes.
- Contains ethanol 20% v/v and propylene glycol 30% v/v.

REFERENCES

1, 5, 40, 95.

* Available via Health Canada's Special Access Programme

Full revision 2017

NAME OF MEDICATION
PEGFILGRASTIM

OTHER NAMES
Fulphila ™, Lapelga ®, Neulasta ®, Nyvepria ™, PEG filgrastim, Pegylated filgrastim, Ziextenzo ®

CLASSIFICATION
Hematopoietic

Do NOT confuse pegfilgrastim with filgrastim.
This monograph is specific to PEGFILGRASTIM.

INDICATIONS
- To decrease the incidence of febrile neutropenia in patients with non-myeloid malignancies receiving myelosuppressive chemotherapy.

ADMINISTRATION
- Do NOT shake; allow prefilled syringe to reach room temp for at least 30 minutes before injection.
- SC: back of the upper arms, abdomen (2 inches away from navel), front upper thighs, upper outer buttocks.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylaxis, rash, urticaria.
- Cardiovascular: capillary leak syndrome, manifested as hypotension, edema, hypoalbuminemia, hemoconcentration and circulatory shock.
- Respiratory: adult respiratory distress syndrome.
- Sickle cell crisis in susceptible patients (rare).
- Splenic rupture (rare).
- Medullary bone pain.
- Local reactions: pain, induration, local erythema.

DOSAGE
- A dose of 6 mg SC given once per cycle of chemotherapy.
- Should not be administered less than 14 days before next chemotherapy and no sooner than 24 hours after chemotherapy.
- Dosage in renal impairment: no dosage adjustment required, including in end-stage renal disease.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Brand Names	Store in the fridge	Protect from light	Stability at room temp	Freezing
Fulphila ™ Neulasta ®	Yes	Yes	72 hours	Avoid freezing; if frozen, allow to thaw in the fridge; if frozen for a second time, discard.
Lapelga ®	Yes	Yes	Single excursion of 15 days (not above 25°C); do not put back in the fridge.	Avoid freezing; if frozen, allow to thaw in the fridge; if frozen for a second time or frozen for more than 24 hours, discard.
Nyvepria ™	Yes	Yes	Single excursion of 15 days (not above 25°C).	Avoid freezing; if frozen, allow to thaw in the fridge; if frozen for a second time, discard.
Ziextenzo ®	Yes	Yes	Single excursion of 120 hours (not above 35°C).	Avoid freezing; if frozen, allow to thaw in the fridge; if frozen for a second time, discard.

MISCELLANEOUS
- The needle cover on the prefilled syringes of Neulasta ® and Lapelga ® contains latex; the needle cover of the prefilled syringes of Fulphila ™, Nyvepria ™ and Ziextenso ® is free of latex.
- Lapelga ®, Fulphila ™, Nyvepria ™ and Ziextenzo ® are classified as being biosimilar to Neulasta ®; therefore, these products are not interchangeable.

REFERENCES
1, 5, 135, 165, 451, 566.

Full revision 2017; limited revision 2018, 2019, 2020, 2021

NAME OF MEDICATION
PEMBROLIZUMAB

OTHER NAMES
Keytruda ®, Lambrolizumab

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of unresectable or metastatic melanoma and adjuvant treatment for stage III melanoma.
- Treatment of metastatic or stage III non-small cell lung carcinoma (NSCLC).
- Treatment of locally advanced or metastatic urothelial carcinoma.
- Treatment of advanced or metastic renal cell carcinoma.
- Treatment of adults and pediatric patients with refractory or relapsed classical Hodgkin lymphoma (cHL).
- Treatment of adult and pediatric patients with relapsed or refractory primary mediastinal B-cell lymphoma (PMBCL).
- Treatment of advanced endometrial cancer.
- Treatment of advanced unresectable or metastatic colorectal cancer.
- Treatment of metastatic or unresectable recurrent head and neck squamous cell carninoma (HNSCC).
- Treatment of Bacillus Calmette-Guerin (BCG)-unresponsive non-muscle invasive bladder cancer (NMIBC).
- Treatment of locally advanced unresectable or metastatic esophageal carcinoma.

ADMINISTRATION

- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential Administration Hazard section.**
- Use the ready-to-use vials of 100 mg/4 mL OR reconstitute the 50 mg vial of lyophilized powder with 2.3 mL of SWFI to obtain a concentration of 25 mg/mL. Direct the stream toward wall of the vial. Slowly swirl the vial until dissolved. Do NOT shake. Allow up to 5 minutes for the bubbles to clear.
- Intermittent IV infusion: dilute required dose in NS or D5W to obtain a final concentration of 1-10 mg/mL. Gently invert bag to mix. Administer over 30 minutes using a sterile, non-pyrogenic, low-protein-binding 0.2 to 5 micron in-line or add-on filter.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: immune-mediated reactions, can be fatal; can involve any organ. Most common reactions include pruritus, rash, Stevens-Johnson syndrome, toxic epidermal necrolysis, pneumonitis, colitis, hepatitis, nephritis and endocrinopathies (adrenal insufficiency, hypophysitis, hypopituitarism, hyperthyroidism, hypothyroidism, diabetes mellitus, diabetic ketoacidosis). Consult manufacturer's recommendations for the management of these reactions and when to withhold or discontinue treatment.
- Infusion-related reactions: rigors, chills, wheezing, pruritus, flushing, rash, hypotension, hypoxemia, fever. Patients with mild or moderate infusion reaction may continue to receive pembrolizumab with close monitoring. **Consider premedication** with antipyretic and antihistamine. If reaction is severe, discontinue treatment.
- Cardiovascular: facial and peripheral edema.
- GI: nausea, vomiting, diarrhea, constipation, abdominal pain, decreased appetite.
- CNS: headache.
- Electrolyte disturbances: hyponatremia, hypocalcemia, hyperkalemia.
- Hematologic: anemia, lymphopenia, leukopenia, thrombocytopenia.
- Hepatic: increased LFTs.
- Renal: increase creatinine.
- Hyperglycemia, hypercholesterolemia, hypertriglyceridemia, decreased serum bicarbonate, hypoalbuminemia.
- Fever, fatigue, arthralgia, myalgia, cough, weight loss, transient tumour size increase.

DOSAGE

- Adults: all indications: 200 mg IV as a fixed dose every 3 weeks OR 400 mg IV 6 weeks.
- Pediatrics: cHL and PMBCL: 2 mg/kg IV (maximum of 200 mg) every 3 weeks.
- Dosage in renal impairment: no dose adjustment needed for patients with CrCl 30 mL/min or greater; no data available if CrCl is less than 30 mL/min.
- Dosage in hepatic impairment: no dose adjustment needed for patients with mild hepatic impairment (abnormal AST with normal total bilirubin OR total bilirubin 1-1.5 x ULN); no data available in case of moderate to severe impairment.
- Consult specific protocol.

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store the ready-to-use vials of 100 mg/4 mL between 2-8°C. Protect from light. Do not freeze. Do not shake.
- Store 50 mg vials of lyophilized powder between 2-8°C. Stable for 24 hours at room temp.
- Reconstituted solution should be clear to slightly opalescent and colorless to slightly yellow; can contain translucent to white proteinaceous particles.
- Reconstituted and diluted solutions are stable for 6 hours at room temp, or for no more than 96 hours in the fridge from the time of reconstitution. Do not freeze.

MISCELLANEOUS

- Thyroid function, renal function, blood glucose and LFTs should be monitored at the start and during treatment.

REFERENCES

1, 5, 129, 135, 165, 208.

New monograph 2017; limited revision 2018, 2019, 2020, 2021

NAME OF MEDICATION
PEMEtrexed

OTHER NAMES
Alimta ®

CLASSIFICATION
Antineoplastic

INDICATIONS
- Treatment of malignant pleural mesothelioma in combination with cisplatin in patients whose disease is unresectable or who are not candidates for curative surgery.
- Treatment of nonsquamous non-small cell lung cancer (NSCLC) as monotherapy or with cisplatin.

ADMINISTRATION
- **Ensure premedication has been administered as recommended; refer to Dosage section.**
- Reconstitute each 100 mg and 500 mg vial with 4.2 mL and 20 mL of preservative-free NS, respectively, for a concentration of 25 mg/mL.
- Intermittent IV infusion (mandatory): dilute dose with preservative-free NS to a final total volume of 100 mL; infuse over 10 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: includes anaphylaxis.
- GI: nausea, vomiting, anorexia, constipation, diarrhea (refer to **premedication** in the Dosage section to decrease GI toxicity).
- Dermatologic: rash (refer to **premedication** in the Dosage section to decrease dermatologic toxicity).
- Hematologic: anemia, leukopenia, neutropenia, thrombocytopenia (refer to **premedication** in the Dosage section to decrease hematologic toxicity).
- Fatigue.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- **Premedication:**
  1) **to decrease hematologic & GI toxicity: folic acid 0.4-1 mg PO daily and vitamin B_{12} 1000 mcg IM every 9 weeks (i.e., every 3 cycles) beginning 1 week before starting pemetrexed and continued for 3 weeks after the last dose of pemetrexed.**
  2) **to decrease dermatologic toxicity: dexamethasone 4 mg PO BID for 3 days beginning the day before pemetrexed administration.**
- For mesothelioma and NSCLC: 500 mg/m^2 IV every 21 days.
- Dosage in renal impairment: do not administer to patients with CrCl below 45 mL/min.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution may range in colour from colourless to yellow or green-yellow.
- Reconstituted solution (25 mg/mL) is stable for 2 days at room temp and 31 days in the fridge in polypropylene syringes.
- Stable for 24 hours at room temp or in the fridge in NS or D5W at concentrations of 2, 10 and 20 mg/mL in PVC containers. Longer storage period may lead to formation of microparticulates.
- Incompatible with solutions containing calcium (e.g., RL, Ringer's).

MISCELLANEOUS

REFERENCES
1, 5, 40, 84, 129, 165, 380.

Full revision 2017; limited revision 2018, 2020

NAME OF MEDICATION
PENICILLIN G

OTHER NAMES
Benzyl penicillin, Crystalline penicillin G, Crystapen ®

CLASSIFICATION
Antibiotic - penicillin

INDICATIONS

- Treatment of infections caused by pneumococci, *N. meningitidis*, beta-hemolytic and non-hemolytic streptococci and gonococci, treatment of syphilis, actinomycosis and anthrax.

ADMINISTRATION

- **CHECK ALLERGY STATUS OF PATIENT**.
- Reconstitute vials as per manufacturer's directions.
- Intermittent IV infusion (preferred): dilute in 100 mL of a compatible solution; for adults, infuse over 30 minutes-2 hours and for children, over 15-30 minutes.
- Continuous IV infusion: dilute in 1-2 L of a compatible solution (volume of solution to be determined based on daily fluid requirements).
- IM: solutions containing up to 100 000 units/mL may be used with minimum discomfort; higher concentrations are physically possible and may be used when needed.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylaxis.
- GI: diarrhea.
- CNS: with very high dosage, particularly in patients with renal insufficiency: hallucinations, confusion, lethargy, hyperreflexia, seizures.
- Hypernatremia.
- Renal: acute interstitial nephritis (rare).
- Local reactions: phlebitis and thrombophlebitis with IV administration; pain and sterile abscess with IM administration.

DOSAGE

- Adults and children 12 years and older: usual daily dose: 1 000 000 to 24 000 000 units/day IV/IM in 4-12 divided doses.
- Pediatrics (1 month to 12 years): 25 000 to 400 000 units/kg/day IV/IM in 4-6 divided doses.
- Dosage in renal impairment:

Method 1	CrCl (mL/min)	50-10	less than 10
	Dose	75%	20-50%
Method 2	CrCl (mL/min)	50-10	less than 10
	Dosing interval (hr)	q8-12h	q12-18h

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solutions stable for 7 days in the fridge or 24 hours at room temp.
- Infusions usually stable for 24 hours at room temp or fridge in D5W and NS.
- Most stable at pH of 6-7.2. Rapid inactivation of penicillin occurs at a pH of below 5 or above 8. Examples of additives with a low pH include ascorbic acid, tetracycline; examples of additives with a high pH include aminophylline, sodium bicarbonate and sodium salts of barbiturates.

MISCELLANEOUS

- 1 million units of penicillin G Na yield 2 mmol Na^+ (46 mg).
- Approximately 300 mg are equivalent to 500 000 units of penicillin G.

REFERENCES

1, 4, 5, 40, 82, 95, 135, 216.

Full revision 2017

NAME OF MEDICATION
PENICILLIN G BENZATHINE

OTHER NAMES
Benzathine Benzyl penicillin, Bicillin LA ®

CLASSIFICATION
Antibiotic - penicillin

INDICATIONS

- Prophylaxis of recurrent rheumatic fever.
- Treatment of treponematosis: syphilis (caused by *Treponema pallidum*), bejel (caused by *Treponema pallidum endemicum*), yaws (caused by *Treponema pertenue*) and pinta (caused by *Treponema carateum*).
- Treatment of streptococcal infections (Group A without bacteremia): mild to moderate infections of the upper respiratory tract (e.g., pharyngitis).

ADMINISTRATION

- **CHECK ALLERGY STATUS OF PATIENT**.
- IM (deep): inject slowly at a steady rate into the gluteus muscle (for infants and small children, midlateral thigh); may need more than one injection site for administration of one dose; rotate injection site if dose is repeated.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, angioedema, anaphylaxis.
- GI: nausea, vomiting, diarrhea.
- Hoigne's syndrome: immediate toxic reaction consisting of bizarre behaviour and neurologic reactions (auditory/visual disturbances, unusual taste, anxiety, confusion, agitation, weakness, dizziness, palpitation, seizures, fear of impending death); generally transient, lasting 5 to 30 minutes.
- Jarisch-Herxheimer reaction: headache, fever, chills, sweating, sore throat, myalgia, arthralgia, malaise, tachycardia, hypertension followed by hypotension. Occurs generally 2-12 hours post-injection and subsides within 12-24 hours. This reaction is due to release of toxins from bacteria when killed; it is not considered a drug allergy; can be treated with antipyretics.
- Neurological damage if injected into or near a nerve.
- Local reactions: pain, sterile abcess (rare).

DOSAGE

Adults:
- Prophylaxis of recurrent rheumatic fever: 1.2 million units IM every 4 weeks (every 3 weeks in high-risk patients) OR 600 000 units IM every 2 weeks.
- Syphilis:
  - Primary, secondary and early latent (less than 1 year duration): 2.4 million units IM as a single dose.
  - Late latent syphilis, latent syphilis of unknown duration or tertiary syphilis not involving the CNS: 2.4 million units IM once weekly for 3 weeks.
  - Sexual contacts in the preceding 90 days to primary, secondary and early latent syphilis: 2.4 million units IM as a single dose.
- Yaws, bejel and pinta: 1.2 million units IM as a single dose.
- Upper respiratory streptococcal infections: 1.2 million units IM as a single dose.

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DOSAGE (Cont.)

Pediatrics:
- Prophylaxis of recurrent rheumatic fever:
  - 27 kg or less: 600 000 units IM every 4 weeks (every 3 weeks in high-risk patients).
  - Greater than 27 kg: 1.2 million units IM every 4 weeks (every 3 weeks in high-risk patients); maximum dose: 1.2 million units/dose.
- Syphilis:
  - Congenital (less than 2 years of age): 50 000 units/kg IM as a single dose.
  - Primary, secondary, early latent (less than 1 year duration): 50 000 units/kg IM as a single dose; maximum dose: 2.4 million units/dose.
  - Late latent syphilis or latent syphilis of unknown duration: 50 000 units/kg IM every week for 3 weeks; maximum dose: 2.4 million units/dose.
- Upper respiratory streptococcal infections:
  - Less than 27 kg: 300 000-600 000 units IM as a single dose.
  - 27 kg and over: 900 000-1.2 million units IM as a single dose.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store syringes in the fridge; protect from freezing.
- Stable for 7 days at room temp not exceeding 30°C.

MISCELLANEOUS

- Time to peak concentration: within 12-24 hours; serum levels usually detectable for 1-4 weeks, depending on the dose.
- Available as a pre-packed syringe of 1.2 million units/2 mL.

REFERENCES

1, 5, 59, 82, 95.

NAME OF MEDICATION
PENTAMIDINE ISETHIONATE

OTHER NAMES
Pentacarinat ®, Pentamidine isetionate

CLASSIFICATION
Antiparasitic

INDICATIONS

- Treatment and prophylaxis of *Pneumocystis jirovecii* (formerly *Pneumocystis carinii)* pneumonia (PCP).

ADMINISTRATION

- For IV and IM use: reconstitute the 300 mg vial with 3 mL of SWFI to yield final concentration of 100 mg/mL.
- Intermittent IV infusion: ***blood pressure monitoring***. Dilute with 50-500 mL of NS or D5W to a concentration of 1-2 mg/mL; administer over at least 60 minutes, preferably over 2 to 3 hours.
- IM: normally not recommended due to poor local tolerance; reserve for patients in whom IV therapy is not possible.
- Keep patient in a supine position; monitor blood pressure immediately before, during administration and several times thereafter until blood pressure is stable.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, anaphylactoid reactions, anaphylaxis.
- Cardiovascular: arrhythmias, hypotension, especially with rapid IV administration.
- GI: nausea, vomiting, abdominal pain, dysgeusia, pancreatitis.
- CNS: dizziness, confusion, hallucinations, neuralgia.
- Renal: acute renal failure, increased serum creatinine.
- Hematologic: leukopenia, thrombocytopenia.
- Dermatologic: facial flushing, dry skin.
- Hypoglycemia (can be severe and/or prolonged) and hyperglycemia.
- Local reactions: phlebitis with IV use; pain, erythema, and tenderness with IM administration.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- PCP treatment:
  - 4 mg/kg IV/IM once daily for 14-21 days.
  - Dosage in renal impairment (CrCl less than 35 mL/min): for life-threatening infections: 4 mg/kg once daily for 7-10 days then 4 mg/kg on alternate days to complete the course of 14 doses. For less severe infections, mg/kg on alternate days for 14 doses.
- PCP prophylaxis: 4 mg/kg IV once monthly.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp; protect from light.
- Reconstituted solution stable for 48 hours at room temp, protected from light.
- Stable for 48 hours at room temp, exposed to normal fluorescent light in NS or D5W at a concentration of 1-2 mg/mL in PVC containers.

MISCELLANEOUS

- 1.74 mg of pentamidine isethionate = 1 mg of pentamidine base.

REFERENCES

1, 4, 5, 40, 95, 143, 367.

Full revision 2017

NAME OF MEDICATION
PENTAZOCINE

OTHER NAMES
Talwin ®

CLASSIFICATION
Opioid analgesic

INDICATIONS

- Analgesic for moderate to severe pain when oral administration is not possible.
- Adjunct to surgical anesthesia.
- As a preoperative or preanesthetic sedative and analgesic.
- As an obstetric analgesic during labour.
- Treatment of epidural opioid-induced pruritus.

ADMINISTRATION

- IV direct: physician or RN; ***respiratory support***. Undiluted or dilute each 5 mg with at least 1 mL of SWFI; inject at a rate of 5 mg/min.
- IM (preferred), SC; rotate site of injection.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus, urticaria, edema of the face.
- Cardiovascular: hypertension, hypotension, tachycardia, shock.
- GI: nausea, vomiting, constipation.
- CNS: confusion, visual hallucinations, euphoria, dizziness, sedation, lightheadedness.
- Respiratory: respiratory depression, dyspnea, laryngospasm.
- May precipitate withdrawal symptoms in opioid-dependent patients.
- Local reactions: stinging, induration, nodules, cutaneous depression at injection site; possible severe tissue damage (sclerosis) with SC injection.

Antidote: naloxone (for respiratory depression).

DOSAGE

- Analgesic (other than labour pain):
  - IM/SC: 30-60 mg q3-4h prn. A single dose should normally not exceed 1 mg/kg. Maximum of 360 mg/day.
  - IV: 30 mg q3-4h prn OR 5-15 mg q2h prn. A single dose should normally not exceed 0.5 mg/kg. Maximum of 360 mg/day.
- Labour pain:
  - IM: 30 mg as a single dose.
  - IV: 20 mg; may be repeated q2-3h up to 3 times (total dose not to exceed 60 mg).
- Pruritus induced by epidural opioid administration: 15 mg IV.
- Dosage requirements may be greater in smokers.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 20-25°C. Protect from light; do not freeze.

MISCELLANEOUS

- Behaves qualitatively as an opioid, but also has weak opioid antagonist properties.
- Onset of action: 2-3 minutes after IV injection or 15-20 minutes after IM/SC injections.

REFERENCES

1, 4, 5, 40, 95, 135, 215.

Full revision 2017; limited revision 2020

NAME OF MEDICATION
PENTOSTATIN *

OTHER NAMES
2-Deoxycoformycin, Nipent ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of hairy cell leukemia, chronic lymphocytic leukemia (CLL) and cutaneous T-cell lymphoma in adults.

ADMINISTRATION

- Reconstitute 10 mg vial with 5 mL of SWFI (concentration 2 mg/mL). Shake the vial until dissolution.
- IV direct: physician or RN; undiluted; inject over 5 minutes.
- Intermittent IV infusion: dilute with 25-50 mL of NS or D5W; infuse over 20-30 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, pruritus, anaphylactoid reaction.
- GI: nausea, vomiting, diarrhea, abdominal pain, anorexia, stomatitis.
- CNS: headache, seizures.
- Hematologic: leukopenia, anemia, thrombocytopenia.
- Respiratory: cough, respiratory tract infection, dyspnea, rhinitis.
- Renal: increased serum creatinine. Can be avoided with proper hydration (refer to Miscellaneous section).
- Fatigue, lethargy, malaise, asthenia, myalgia, chills.
- Local reactions: inflammation and bleeding at injection site.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Hairy cell leukemia: 4 mg/m^2 IV every other week.
- CLL: 2-4 mg/m^2 IV every 3 weeks.
- Cutaneous T-cell lymphoma: 4 mg/m^2 IV once weekly for 3 weeks, then every other week for 6 weeks, then once monthly for 6 months.
- Dosage in renal impairment: there is insufficient data on appropriate dosage reductions in patients with CrCl less than 60 mL/min.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Reconstituted solution is stable for 8 hours at room temp.
- Compatible with D5W, NS and RL.
- Stable for 48 hours at room temp or 96 hours in the fridge in NS at a concentration of 0.02 mg/mL.
- Stable for 24 hours at room temp or 96 hours in the fridge in D5W at a concentration of 0.02 mg/mL.
- Stable for 48 hours at room temp in RL at a concentration of 0.02 mg/mL.

MISCELLANEOUS

- Patients should receive hydration with 500-1000 mL of D5-1/2NS before administration and 500 mL of D5W after administration of pentostatin to minimize risk of adverse renal effects.

REFERENCES

1, 4, 95, 135.

* Available via Health Canada's Special Access Programme

Full revision 2017; limited revision 2018, 2020

NAME OF MEDICATION
PERTUZUMAB

OTHER NAMES
Monoclonal antibody 2C4, Perjeta ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of HER2-positive metastatic breast cancer in patients who have not received prior anti-HER2 therapy or chemotherapy for metastatic disease (in combination with trastuzumab and docetaxel).
- Treatment of HER2-positive early breast cancer in patients with lymph node positive and/or hormone receptor negative disease (in combination with trastuzumab and chemotherapy).

ADMINISTRATION

- **Ensure premedication has been administered if patient experienced an infusion reaction with a prior pertuzumab infusion; refer to Potential Administration Hazards section**.
- Intermittent IV infusion (mandatory): dilute dose in a 250 mL PVC or non-PVC polyolefin bag of NS. Do NOT use D5W (refer to Compatibility, Stability section). Invert the bag gently to ensure mixing without foaming. Do NOT shake. Infuse over 60 minutes for an 840 mg dose and over 30 to 60 minutes for a 420 mg dose.
- If trastuzumab and docetaxel are administered on the same day, ensure they are administered 30 to 60 minutes after the end of the pertuzumab infusion.
- Close observation of the patient during the infusion, as well as for 60 minutes after first infusion and for 30 minutes after subsequent infusions.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, anaphylaxis, acute respiratory distress syndrome.
- Infusion-related reactions: fever, chills, fatigue, headache, asthenia, dysgeusia, myalgia, hypersensitivity and vomiting. If a reaction occurs, infusion rate should be decreased or stopped depending on reaction severity. Monitor patient and administer beta-agonists, antipyretics, antihistamines and corticosteroids if needed. **Consider premedication** (with antipyretics, antihistamines, corticosteroids) for subsequent infusions.
- Cardiovascular: edema, left ventricular dysfunction (patients who have received prior anthracyclines or prior radiotherapy to the chest area may be at higher risk).
- GI: nausea, vomiting, diarrhea (may be severe), anorexia, mucositis, dyspepsia, dysguesia.
- CNS: headache, peripheral neuropathy, insomnia, dizziness.
- Hematologic: neutropenia, febrile neutropenia (more frequent in Asian patients), leukopenia, bleeding.
- Respiratory: cough, dyspnea.

DOSAGE

- 840 mg IV loading dose on day 1 of the first cycle, followed by a maintenance dose of 420 mg IV on day 1 of each subsequent 3-week cycle.
- If treatment has been interrupted for 6 weeks or more, repeat the loading dose of 840 mg followed by the 420 mg maintenance dose every 3 weeks.
- Dosage in renal impairment: no dosage adjustment for mild to moderate renal impairment. No data available for CrCl of less than 30 mL/min.
- Dosage in hepatic impairment: no data available.
- Dosage adjustment due to toxicity: withhold or discontinue drug; dose reductions are not recommended.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C; protect from light; do not shake; do not freeze.
- The diluted solution is stable for 24 hours in the fridge in NS in a PVC or non-PVC polyolefin infusion bag.
- Compatible with polyethylene bags.
- Not stable in D5W.

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MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable for corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity and infusion-related reactions.
- Patients should have a thorough cardiac assessment before starting therapy, every 3 months during therapy and every 6 months during the 24-month period following treatment discontinuation.

REFERENCES

5, 40, 129, 135, 165.

NAME OF MEDICATION
PHENobarbital

OTHER NAMES
Phenobarbitone sodium

CLASSIFICATION
Barbiturate, Anticonvulsant

INDICATIONS

- Treatment of status epilepticus and other acute seizure states.
- Maintenance anticonvulsant therapy.
- To alleviate preoperative anxiety and produce sedation.
- For terminal agitation and sedation in the palliative setting.
- To control intracranial hypertension.

ADMINISTRATION

- IV direct: physician or RN; ***blood pressure monitoring; respiratory support***. Inject slowly over at least 1 minute not to exceed 60 mg/min in adults and 1 mg/kg/min up to 30 mg/min in children and infants, unless for status epilepticus (50-100 mg/min).
- Intermittent IV infusion: ***respiratory support.*** Dilute in 50-100 mL of compatible solution; infuse over 20-30 minutes.
- Continuous IV infusion: for intracranial hypertension; at TOH, dilute in NS to obtain a final concentration of 10 mg/mL.
- IM, deep in a large muscle.
- SC, continuous SC infusion (palliative care); must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, exfoliative dermatitis, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Cardiovascular: hypotension with rapid IV administration, circulatory collapse.
- CNS: drowsiness, lethargy, headache, dizziness; paradoxical agitation (especially in children).
- Respiratory: respiratory depression especially with rapid IV administration, laryngospasm.
- Local reactions: thrombophlebitis, pain at injection site; irritation, redness, tenderness or necrosis with SC administration.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Adults:
  - Status epilepticus: loading dose of 20 mg/kg IV; may give an additional 5-10 mg/kg IV, if necessary, 10-15 minutes later up to a total of 30 mg/kg.
  - Maintenance anticonvulsant therapy: 100-600 mg/day IV/IM/SC in 1-2 divided doses OR 1-3 mg/kg/day IV in divided doses.
  - Preoperative: 100-200 mg IV/IM given 60-90 minutes prior to surgery.
  - Hypnotic: 100-320 mg IV/IM.
  - Terminal agitation: 100-200 mg IV/IM. Can repeat dose every 30-60 minutes prn (maximum 1600 mg/24 hours) OR start a continuous SC infusion at 800 mg/24 hours which can gradually be increased up to 3800 mg/24 hours.
  - Sedation in palliative care: 15-60 mg IV/SC q4-6h (up to 1600 mg/day).
  - Intracranial hypertension: (TOH use) loading dose (optional) of 10 mg/kg IV followed by an IV infusion of 0.5-3 mg/kg/hr; adjust dose according to intracranial pressure and hemodynamics.

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DOSAGE (Cont.)

- Pediatrics:
  - Status epilepticus: loading dose of 15-20 mg/kg IV (maximum 1000 mg/dose); may give an additional 5-20 mg/kg, if necessary, every 15-20 minutes up to a total of 40 mg/kg.
  - Maintenance anticonvulsant therapy: 1-8 mg/kg/day IV/IM in 1-2 divided doses.
  - Preoperative: 1-3 mg/kg IV/IM, given 60-90 minutes prior to surgery.
  - Hypnotic: 2-5 mg/kg IV/IM.

- Dosage in renal impairment: if CrCl is less than 10 mL/min, adjust dosing interval to every 12 to 16 hours OR reduce dosage by 25-50%.
- Dosage in hepatic impairment: contraindicated in patients with marked liver impairment.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 15-25°C; protect from light. Do not freeze.
- Stable for 28 days in the fridge in NS at a concentration of 10 mg/mL in glass vials.
- Compatible in NS, D5W, Ringer's, RL, sodium lactate 1/6 M solutions.
- Incompatible with acidic solutions.

MISCELLANEOUS

- Contraindicated in patients with porphyria.
- When used as an anticonvulsant, it may take 2-4 weeks to reach adequate serum concentrations; desired serum concentration: 43-172 mcmol/L (10-40 mg/L).

REFERENCES

1, 4, 5, 40, 82, 95, 102, 135, 156, 256, 312, 457, 462.

Full revision 2017; limited revision 2020, 2021

NAME OF MEDICATION
PHENTOLAMINE

OTHER NAMES
Rogitine ®

CLASSIFICATION
Alpha-blocker

INDICATIONS

- Diagnostic agent for pheochromocytoma.
- Prevention and treatment of hypertensive episodes prior to or during surgery for removal of pheochromocytoma.
- As an antagonist to catecholamines in other acute forms of hypertension.
- Treatment of alpha-adrenergic drug extravasation (e.g., dobutamine, dopamine, epinephrine, norepinephrine, phenylephrine).
- Treatment of left ventricular failure secondary to acute myocardial infarction (AMI).

ADMINISTRATION

- IV direct: physician only; ***blood pressure monitoring; norepinephrine (Levophed ®) must be available;*** inject over 1 minute.
- Continuous IV infusion: ***blood pressure monitoring, cardiac monitoring***. Dilute in NS; concentrations up to 0.8 mg/mL have been used. Infusion rate as per Dosage section. Must be administered by an infusion pump.
- IM: ***blood pressure monitoring***.
- SC (for infiltration of area where alpha-adrenergic drug extravasated): dilute 5-10 mg in 10-15 mL of NS; use a fine hypodermic needle.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: tachycardia, flushing, cardiac arrhythmias, anginal pain; hypotension, can be severe or prolonged and can lead to myocardial infarction and cerebrovascular spasm or occlusion.
- GI: abdominal pain, nausea, vomiting, diarrhea.
- CNS: dizziness, headache.
- Weakness, nasal congestion.

Antidote: norepinephrine for treatment of hypotension. Refer also to the Miscellaneous section.

DOSAGE

Adults:
- Diagnostic: 5 mg IV/IM.
- Prevention or control of hypertensive episodes prior to or during surgical excision for pheochromocytoma: 2-5 mg IV/IM given 1-2 hours prior to surgery. Repeat as needed via IV route during surgery.
- For management of hypertensive crisis due to catecholamine release: 1-5 mg IV direct, maximum dose of 20 mg IV. May continue with 1 mg/hr as a continuous IV infusion; titrate to response (maximum 40 mg/hr).
- Treatment of alpha-adrenergic drug extravasation: 5-10 mg SC to be infiltrated in the area as soon as possible and within 12 hours of the extravasation, otherwise treatment is ineffective.
- Left ventricular failure secondary to AMI: continuous IV infusion of 0.17-0.4 mg/min OR 10-24 mg/hr.

Pediatrics:
- Diagnostic: 1 mg IV or 0.05-0.1 mg/kg (maximum: 5 mg/dose) IV or 3 mg/m^2 IV or 3 mg IM.
- Prevention or control of hypertensive episodes prior to or during surgical excision for pheochromocytoma: 1 mg or 0.05-0.1 mg/kg (maximum: 5 mg/dose) or 3 mg/m^2 IV/IM given 1-2 hours prior to surgery. Repeat as needed via IV route during surgery.
- Treatment of alpha-adrenergic drug extravasation: local SC infiltration of small amounts in 0.2 mL aliquots of a 0.5-1 mg/mL solution as soon as possible and within 12 hours of the extravasation. Total volume depends on extravasation area size.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and vials between 2-8°C; protect from light and heat. Do not freeze.
- Compatible with NS.

MISCELLANEOUS

- Do not use epinephrine if hypotension occurs as it may cause a paradoxical fall in BP.
- Maximum effect on blood pressure is 2 minutes following IV administration and 20 minutes with IM dosing.
- Contains sulfites.

REFERENCES

1, 5, 40, 82, 95, 135, 504.

NAME OF MEDICATION
PHENYLEPHRINE

OTHER NAMES
Neo-Synephrine ®

CLASSIFICATION
Sympathomimetic

INDICATIONS

- Prevention and treatment of acute hypotension during spinal anesthesia (including in obstetric patients).
- Treatment of acute hypotension due to peripheral circulatory collapse.
- Treatment of paroxysmal supraventricular tachycardia.

ADMINISTRATION

- IV direct: physician only; ***blood pressure monitoring***. Use prefilled syringes (50 mcg/mL) OR use vials (10 mg/mL) by mixing 10 mg (1 mL) in 9 mL of NS, D5W or SWFI, discard 9 mL, and dilute the remaining 1 mL with 9 mL NS, for a final concentration of 100 mcg/mL; inject over 20-30 seconds.
- Continuous IV infusion: ***continuous BP monitoring***. Dilute in NS or D5W to obtain a final concentration of 20 mcg/mL (10 mg/500 mL) or 40 mcg/mL (10 mg/250 mL). More concentrated solutions may be prepared depending on dosage and fluid requirement of each patient (e.g., 100 mcg/mL, 200 mcg/mL and 400 mcg/mL). At TOH, add 10 mg (1 mL from phenylephrine 10 mg/mL vial) in 100 mL of D5W or NS to obtain a final concentration of 100 mcg/mL (0.1 mg/mL) OR add 100 mg (10 mL from phenylephrine 10 mg/mL vials) in 250 mL of NS to obtain a final concentration of 400 mcg/mL (0.4 mg/mL) OR add 200 mg (20 mL from phenylephrine 10 mg/mL vials) in 250 mL of NS to obtain a final concentration of 800 mcg/mL (0.8 mg/mL). A central line is preferred. Must be administered by an infusion pump.
- IM, SC.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: ventricular extrasystoles, ventricular tachycardia, reflex bradycardia (**Antidote**: atropine for bradycardia), hypertension.
- GI: nausea, vomiting.
- CNS: restlessness, nervousness, anxiety, excitability, dizziness, headache, paresthesia, sensation of fullness in head.
- Dermatologic: peripheral vasoconstriction, pallor.
- Local reactions: blanching; sloughing or necrosis with SC administration.
- Extravasation hazard: vesicant. **Antidote**: phentolamine 5-10 mg in 10-15 mL of NS. Consult the document "Extravasation of noncytotoxic intravenous agents associated with tissue necrosis" in TOH Electronic Drug Formulary.

DOSAGE

- Treatment of hypotension during anesthesia:
  - IV direct: 40-200 mcg every 1 to 2 minutes prn; titrate to effect. Do not exceed 500 mcg for subsequent dose.
  - Continuous IV infusion: if initial IV direct dose was not sufficient, initiate infusion at an initial rate of 10-35 mcg/min OR 0.5-1.4 mcg/kg/min; titrate to effect. Do not exceed 200 mcg/min.
- Prevention of hypotension during anesthesia: 2-3 **mg** IM/SC given 3-4 minutes before injection of spinal anesthetic.
- Treatment of hypotension in other acute states:
  - Mild to moderate hypotension: usual initial dose of 200 mcg (range from 100 mcg to a maximum of 500 mcg) IV direct can be given prn. Do not give more frequently than every 10-15 minutes OR 2-5 **mg** SC/IM (range 1-10 **mg**; initial dose not to exceed 5 **mg**).
  - Severe hypotension or shock: continuous IV infusion at starting rate of 100-180 mcg/min OR 0.5-6 mcg/kg/min. Maintenance rate of 40-60 mcg/min. Titrate to maintain BP.
- Paroxysmal supraventricular tachycardia: up to 500 mcg rapid IV direct initially. Subsequent doses may be increased in increments of 100-200 mcg (not exceeding 1 **mg** per dose).
- Dosage in hepatic impairment: patients with liver cirrhosis may require doses in the upper end of the dosing range.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes between 15-30°C. Protect from light.
- Do not use if the solution darkens or becomes cloudy.
- Stable for 14 days at room temp (25°C) in NS at a concentration of 100 mcg/mL in PVC bags.
- Stable for 60 days at room temp (between 23-25°C) exposed to light in NS at concentrations of 200 mcg/mL and 400 mcg/mL in PVC bags.

MISCELLANEOUS

- Use with extreme caution in patients on monoamine oxidase inhibitors or tricyclic antidepressants.
- Vials contain sulfites.

REFERENCES

1, 4, 5, 40, 95, 135, 208, 366, 367, 462.

Full revision 2017; limited revision 2019, 2020, 2021

NAME OF MEDICATION
PHENYTOIN

OTHER NAMES
Dilantin ®, Diphenylhydantoin, Tremytoine ™

CLASSIFICATION
Anticonvulsant

Do NOT confuse phenytoin with fosphenytoin.
This monograph is specific to PHENYTOIN.

INDICATIONS

- Treatment of status epilepticus.
- Prophylaxis of seizures during neurosurgery.
- Treatment of digitalis-induced arrhythmias and of ventricular tachycardia or paroxysmal atrial tachycardia particularly in persons refractory to (or unable to tolerate) conventional antiarrhythmic agents or cardioversion.

ADMINISTRATION

- IV direct: physician or RN; ***cardiac monitoring***, ***blood pressure monitoring***, ***respiratory support***; undiluted; rate should not exceed 50 mg/min in adults, 25 mg/min in the elderly or in patients with cardiovascular disease, 1-3 mg/kg/min (maximum of 50 mg/min) in infants and children, and 0.5-3 mg/kg/min in neonates.
- Intermittent IV infusion (preferred for loading dose): ***blood pressure monitoring***, ***respiratory support***; doses may be diluted in NS at a concentration of 5 mg/mL or greater. (At TOH, loading doses less than 500 mg are diluted in 50 ml of NS and loading doses of 500 mg to 2000 mg in 100 mL of NS). Rate should not exceed 50 mg/min in adults, 25 mg/min in the elderly or in patients with cardiovascular disease, 1-3 mg/kg/min (maximum of 50 mg/min) in infants and children, and 0.5-3 mg/kg/min in neonates. **For all doses diluted in a bag, use of an in-line filter of 0.22 micron is suggested to remove any crystalline phenytoin which forms when phenytoin is diluted. Administration of the drug should be completed within 1 to 4 hours after preparation of the diluted solution.**
- Administer IV doses via a large peripheral vein through a large gauge needle or via a large gauge central line. **At TOH, phenytoin should NOT be administered through a PICC line (re: risk of precipitation with catheter blockage).**
- **Always flush IV line before and after with NS to decrease risk of incompatibility and to reduce venous irritation; flush line SLOWLY after administration as some residual phenytoin may still sit in the line.**
- The preferred method of administration for the loading dose is with an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash; can be resumed if rash is not severe and has completely resolved. Toxic epidermal necrolysis, Stevens-Johnson syndrome. Drug reactions with eosinophilia and systemic symptoms (DRESS). Anaphylaxis.
- Cardiovascular: hypotension, cardiovascular collapse, bradycardia, heart block, ventricular tachycardia, ventricular fibrillation. Minimize by administering at a rate not exceeding the rates above.
- CNS: confusion, ataxia, slurred speech, somnolence; may be signs of toxicity. Suicidal ideation.
- Ophthalmic: nystagmus (may be a sign of toxicity).
- Respiratory: respiratory depression, especially if exceeding the rates above.
- Hyperglycemia, lymphadenopathy.
- Local reactions: tenderness, irritation, inflammation, necrosis, sloughing, purple glove syndrome (distal pain, edema and discolouration).
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

Adults:
- Status epilepticus: loading dose: 15-20 mg/kg IV (may give an additional dose of 5-10 mg/kg IV 10 minutes later prn).
  maintenance: 100 mg IV q6-8h OR 4-7 mg/kg/day IV in 2-4 divided doses. Refer to Miscellaneous section for therapeutic serum levels.

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DOSAGE (Cont.)

- Dosage in obesity for status epilepticus:
  Method 1: loading dose (mg): [14 mg/kg x IBW] + [19 mg/kg x (TBW-IBW)]
  maintenance dose: use IBW
  Method 2: loading dose (mg): 15 mg/kg x TBW for males
  20 mg/kg x TBW for females
  maintenance dose: use IBW
  IBW: ideal body weight; TBW: total body weight

- Prophylaxis during neurosurgery: 100-200 mg IV q4h during surgery and the immediate postoperative period.
- Antiarrhythmic: 100 mg IV direct, repeated at 5 minute intervals prn until arrhythmias stop or undesirable side effects occur or up to a total dose of 1 g.

Pediatrics:
- Status epilepticus: loading dose: 15-20 mg/kg IV (maximum 1000 mg); may give an additional dose of 5-10 mg/kg IV 10 minutes later prn.
  maintenance: 4-10 mg/kg/day IV (maximum 300 mg/day) in 2-3 divided doses.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and ampoules between 15-30°C. Protect from light. Do not freeze. Solution should be colourless to light yellow and clear.
- Incompatible with dextrose-containing solutions.
- Stable for a maximum of 4 hours at room temp in NS; do not refrigerate the diluted solution.

MISCELLANEOUS

- When switching from IV to oral phenytoin or *vice versa*, use the same total daily dose. The IV formulation should be administered every 6-8 hours as opposed to the oral extended-release capsules that can be administered once or twice daily. Monitor serum levels after the switch for the following reasons: 1) bioavailability of oral preparations vs. the IV formulation is 90%; 2) there is an 8% increase in the drug content with the free acid form (50 mg tablets, oral suspension) over that of the sodium salt (IV formulation, 100 mg extended-release capsules).
- Therapeutic serum levels: 40-80 mcmol/L (10-20 mcg/mL). Correct levels in patients with low albumin levels or renal failure.
- Rule out hypoglycemia before giving phenytoin for convulsions.
- Undiluted product contains 10% ethyl alcohol.

REFERENCES

1, 4, 5, 40, 82, 95, 135, 208, 257, 319, 366, 367, 435, 436, 513.

Full revision 2017; limited revision 2018, 2019, 2020

NAME OF MEDICATION
PHOSPHATE (as sodium or potassium salt)

OTHER NAMES

CLASSIFICATION
Electrolyte

INDICATIONS

- Prevention or treatment of hypophosphatemia, when oral administration is not possible.

ADMINISTRATION

- Intermittent IV infusion: dilute dose in NS or D5W and infuse as per chart below*:

Access	Phosphate dose	To dilute in at least	To infuse over at least**	Maximum phosphate infusion rate**
Peripheral	7.5 mmol	100 mL	60 minutes	7.5 mmol/hr
	15 mmol	250 mL	2 hours	7.5 mmol/hr
	30 mmol	500 mL	4 hours	7.5 mmol/hr
Central	Less than 30 mmol	100 mL	-	7.5 mmol/hr
	30 to 60 mmol	250 mL	-	7.5 mmol/hr

* More diluted solutions, especially for peripheral administration, and slower infusion rates than above would be better tolerated.
** This is of particular importance for potassium phosphate.

- Continuous IV infusion (in parenteral nutrition solutions).

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypocalcemia: occurs when phosphate is infused too rapidly or at doses that lead to hyperphosphatemia; may lead to convulsions, tetany and cardiac arrhythmias.
- Hyperkalemia (with potassium phosphate): characterised by paresthesia of extremities, flaccid paralysis, listlessness, confusion; may precipitate cardiac standstill especially from rapid IV infusion (**Antidote**: calcium gluconate 1000 mg IV given as 10 mL of 10% solution over 5 minutes, repeat in 5 minutes as required. NB: calcium is given only in the setting of ECG changes. Insulin/glucose: 5-10 units of regular insulin IV direct **followed immediately** by 25-50 g of dextrose (50-100 mL of 50% solution) administered IV over 5 minutes).
- Hypernatremia, edema (with sodium phosphate).
- Local reactions (with potassium phosphate): phlebitis and pain with injection of concentrated solutions.
- Extravasation hazard (with potassium phosphate): vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Adult maintenance dose in parenteral nutrition: 10-15 mmol/L of solution or 20-40 mmol/day.
- Mild hypophosphatemia (0.6-0.89 mmol/L): 0.08-0.16 mmol/kg phosphate, up to 15-30 mmol/dose.
- Moderate hypophosphatemia (0.4-0.59 mmol/L): 0.16-0.32 mmol/kg phosphate, up to 15-45 mmol/dose.
- Severe symptomatic hypophosphatemia (less than 0.4 mmol/L): 0.24-0.64 mmol/kg phosphate, up to 30-60 mmol/dose.
- Weight-based dosing: use actual body weight (ABW). If ABW is greater than 130% of ideal body weight (IBW), the adjusted body weight is calculated as follows: IBW + 0.25 (ABW-IBW).
- Maximum dose: 80 mmol/24 hours.
- Dosage in renal impairment: use lower end of the dosage range or decrease dose by at least 50% and administer over a longer period (e.g., 4 to 6 hours).

…/Cont.

NAME OF MEDICATION
PHOSPHATE (as sodium or potassium salt)

OTHER NAMES

CLASSIFICATION
Electrolyte

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light.
- Stable for 24 hours at room temp in D5W, NS, D5-NS.
- Variable compatibility with calcium-containing solutions, including parenteral nutrition.

MISCELLANEOUS

- Use potassium phosphate if serum K less than 4 mmol/L and sodium phosphate if serum K greater than 4 mmol/L.
- 1 mmol phosphate or phosphorous contains 31 mg elemental phosphorus.
- Each mL of potassium phosphate solution contains 4.4 mmol of potassium and 3 mmol of phosphate.
- Each mL of sodium phosphate solution contains 4 mmol of sodium and 3 mmol of phosphate.
- Obtain a phosphorous level 2 to 4 hours after administering a dose.

REFERENCES

4, 5, 40, 90, 95, 135, 208, 328, 366, 367, 388, 474.

Full revision 2017; limited revision 2020

NAME OF MEDICATION
PHYSOSTIGMINE *

OTHER NAMES
Antilirium ®, Eserine salicylate

CLASSIFICATION
Cholinesterase inhibitor

INDICATIONS

- To reverse severe CNS effects caused by atropine or other anticholinergic agents.

ADMINISTRATION

- IV direct: physician only; undiluted; inject slowly, not to exceed 1 mg/min in adults and 0.5 mg/min in pediatrics.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, bronchoconstriction (mostly due to presence of sulfites).
- Cardiovascular: bradycardia (minimized by slow IV injection), hypotension or hypertension.
- GI: excessive salivation (can lead to respiratory distress, minimized by slow IV injection), nausea, vomiting, diarrhea, epigastric pain.
- CNS: confusion, seizures (minimized by slow IV injection), coma.
- Sweating, miosis, severe muscle weakness, paralysis, lacrimation.

Antidote: for cholinergic crisis, atropine 2-4 mg IV every 3-10 minutes in adults and 1 mg IV doses in children; IV pralidoxime may be useful to reverse respiratory paralysis.

DOSAGE

Adults:
- 0.5-2 mg IV or IM. May be repeated every 10-30 minutes. May be necessary to use additional doses of 1-4 mg given every 30-60 minutes as life-threatening signs (arrhythmias, seizures, coma) recur.

Pediatrics
- 0.02 mg/kg IV or IM repeated at 5-10 minute intervals as needed up to a total dose of 2 mg.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 20-25°C.

MISCELLANEOUS

- In contrast to neostigmine and pyridostigmine, physostigmine readily penetrates the blood-brain barrier.
- Onset: 3-8 minutes with IV administration, 20-30 minutes with IM administration. Duration: 30-60 minutes.
- Contraindicated in patients with asthma, gangrene, diabetes, cardiovascular disease, mechanical obstruction of the intestinal or urogenital tracts, or any vagotonic state and in patients receiving choline esters or depolarizing neuromuscular blocking agents.
- Caution in patients with epilepsy, parkinsonism or bradycardia.
- Contain sulfites.

REFERENCES

1, 5, 40, 82, 95, 135.

* Available via Health Canada's Special Access Programme

Full revision 2017

NAME OF MEDICATION
PIPERACILLIN

OTHER NAMES
Pipracil ®

CLASSIFICATION
Antibiotic - penicillin

INDICATIONS

- Treatment of infections due to non-penicillinase producing susceptible strains of bacteria including *Pseudomonas aeruginosa*, *E. coli*, *Enterobacter*, *Proteus*, *Klebsiella*, streptococci, enterococci in the following conditions: intra-abdominal infections, skin and skin structure infections, gynecological infections, lower respiratory tract infections, septicemia, urinary tract infections, uncomplicated urethritis (caused by *N. gonorrhoeae*) and bone and joint infections.

ADMINISTRATION

- **CHECK ALLERGY STATUS OF PATIENT**.
- For IV use: reconstitute 2, 3, and 4 g vials with 10, 15, and 20 mL of SWFI, respectively, for an approximate concentration of 0.18 g/mL. Shake well until dissolved.
- IV direct: physician or RN; undiluted; inject over 3-5 minutes.
- Intermittent IV infusion (preferred): dilute in at least 50 mL of a compatible solution; infuse over 20-30 minutes.
- For IM use: reconstitute 2, 3, and 4 g vials with 4, 6, or 8 mL of SWFI or 0.5-1% lidocaine (without epinephrine), respectively, for an approximate concentration of 0.4 g/mL. Shake well until dissolved.
- IM: reserved for uncomplicated UTIs and gonorrhea or follow-up to IV therapy; give no more than 2 g per injection site. Inject into the gluteus muscle (preferred); use deltoid area only if well developed.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria, anaphylactic/anaphylactoid reaction, erythema multiforme, Stevens-Johnson syndrome.
- GI: diarrhea, nausea, vomiting.
- CNS: headache, dizziness; convulsions with very high doses, particularly in patients with renal insufficiency.
- Hematologic: leukopenia, thrombocytopenia, eosinophilia, neutropenia.
- Local reactions: phlebitis, pain, erythema, induration.

DOSAGE

- Urinary tract infections:
  - uncomplicated: 6-8 g (100-125 mg/kg) IM/IV daily in 2 to 4 divided doses
  - complicated: 8-16 g (125-200 mg/kg) IV daily in 3 to 4 divided doses
- Serious infections (e.g., septicemia, pneumonia): 12-18 g (200-300 mg/kg) IV daily in 4 to 6 divided doses.
- Uncomplicated gonorrhea: 2 g IM as a single dose (probenecid [not marketed in Canada] 1 g is given orally 30 minutes prior to injection)
- Maximum daily dose is 24 g; however higher doses have been used.
- Dosage in renal impairment:

CrCl (mL/min)	greater than 40	40-20	less than 20	hemodialysis *
Dose (g)	usual	3 to 4	3 to 4	2
Interval (hr)	usual	8	12	8

* Give an additional 1 g IV after each dialysis session or give scheduled dose right after dialysis, as 30 to 50% is removed after a 4-hour hemodialysis session.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light.
- Reconstituted solutions are stable for 24 hours at room temp, 7 days when refrigerated and 1 month when frozen at -20°C.

…/Cont.

…/Cont.

COMPATIBILITY, STABILITY (Cont.)

- Stable for 24 hours at room temp and for 72 hours in the fridge in NS, D5W, D5-NS, RL in plastic containers (manufacturer data).
- Stable for 24 hours at room temp, one week in the fridge and one month when frozen at -20°C in D5W, D5-NS, NS or RL at a concentration of 120 mg/mL in glass or PVC containers.

MISCELLANEOUS

- Cross-allergenicity with other beta-lactam antibiotics.
- Sodium content: 1.85 mmol/g.
- Since piperacillin can inactivate the activity of aminoglycosides, they should be administered at least 60 minutes apart.

REFERENCES

5, 40, 95, 135, 208.

NAME OF MEDICATION
PIPERACILLIN/TAZOBACTAM

OTHER NAMES
Tazocin ®

CLASSIFICATION
Antibiotic - penicillin with beta-lactamase inhibitor

INDICATIONS

- Treatment of infections due to susceptible strains, including beta-lactamase producing bacteria, in the following conditions: intra-abdominal infections, skin and skin structure infections, gynecological infections, respiratory tract infections, septicemia and urinary tract infections.

ADMINISTRATION

- **CHECK ALLERGY STATUS OF PATIENT.**
- For IV use: reconstitute with NS, D5W or SWFI as follows: the 2 g/0.25 g vial with 10 mL, the 3 g/0.375 g vial with 15 mL and the 4 g/0.5 g vial with 20 mL; shake vial until dissolved.
- Intermittent IV infusion: further dilute with 50-150 mL of compatible solution and administer over at least 30 minutes.
- Continuous IV infusion (NOT preferred): at TOH, for critically ill patients on sustained low-efficiency dialysis (SLED); infuse each minibag of 3 g/0.375 g over the dosing interval (i.e., 8 hours).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, anaphylactic/anaphylactoid reaction, Stevens-Johnson syndrome, toxic epidermal necrolysis, drug reactions with eosinophilia and systemic symptoms (DRESS).
- GI: diarrhea, nausea, vomiting, constipation.
- CNS: headache, insomnia; convulsions with very high doses particularly in patients with renal insufficiency.
- Hematologic: leukopenia, thrombocytopenia, eosinophilia, neutropenia.
- Local reactions: phlebitis, pain, inflammation, edema.

DOSAGE

- Usual dosage: 3 g/0.375 g piperacillin/tazobactam IV q6h.
- Higher dosage: for nosocomial pneumonia and febrile neutropenia: 4 g/0.5 g IV q6h. At TOH, the most commonly used dose for nosocomial pneumonia is 3 g/0.375 g IV q6h although 4 g/0.5 g IV q6h may be used in select cases.
- Dosage in renal impairment:

	Usual dose	Higher dose
CrCl greater than 40 mL/min	3 g/0.375 g IV q6h	4 g/0.5 g IV q6h
CrCl 20-40 mL/min	2 g/0.25 g IV q6h	3 g/0.375 g IV q6h
CrCl less than 20 mL/min	2 g/0.25 g IV q8h	2 g/0.25 g IV q6h
Hemodialysis *	2 g/0.25 g IV q12h	2 g/0.25 g IV q8h
SLED		3 g/0.375 g IV q8h as a continuous infusion

* As hemodialysis removes 30-40% of the dose in 4 hours, give an additional 0.67 g/0.08 g after each hemodialysis or give scheduled dose right after dialysis.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Once reconstituted, solutions are stable for 24 hours at room temp and 48 hours in the fridge (72 hours in the fridge for Alveda brand).
- All brands: diluted solution is stable for 24 hours at room temp and 72 hours in the fridge in NS or D5W.
- Alveda, Sandoz and Sterimax brands only: diluted solution is stable for 24 hours at room temp and 72 hours in the fridge in RL.
- Sandoz brand only: compatible with amikacin (1.75-7.5 mg/mL in D5W or NS) or gentamicin (0.7-3.32 mg/mL in NS) through Y-site connection; incompatible with tobramycin.

…/Cont.

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COMPATIBILITY, STABILITY (Cont.)

- Sandoz brand only: diluted solution is stable for 30 days in the fridge in D5W or NS at concentrations ranging from 18 mg piperacillin/2.3 mg tazobactam/mL to 67 mg piperacillin/8.3 mg tazobactam/mL in PVC bags.
- Sterimax brand only: diluted solution is stable for 32 days in the fridge in D5W or NS at a concentration ranging from 20 mg piperacillin/2.5 mg tazobactam/mL to 60 mg piperacillin/7.5 mg tazobactam/mL in PVC bags.

MISCELLANEOUS

- Vials contain:
  Piperacillin 2 g and tazobactam 0.25 g: usually ordered as 2.25 g. Also contains 108-207 mg of sodium.
  Piperacillin 3 g and tazobactam 0.375 g: usually ordered as 3.375 g. Also contains 162-312 mg of sodium.
  Piperacillin 4 g and tazobactam 0.5 g: usually ordered as 4.5 g. Also contains 216-415 mg of sodium.
- Tazobactam is included as a beta-lactamase inhibitor, which extends piperacillin's spectrum of coverage to include certain beta lactamase producing bacteria (e.g., *H. influenzae, S. aureus*).
- Cross-allergenicity with other beta-lactam antibiotics.

REFERENCES

1, 5, 135, 177, 179, 193, 194, 422, 455, 478.

NAME OF MEDICATION
PLERIXAFOR

OTHER NAMES
Mozobil ®

CLASSIFICATION
Hematopoietic

INDICATIONS

- Peripheral stem cell collection and transplantation: to be used in combination with filgrastim (granulocyte-colony stimulating factor [G-CSF]) for harvesting of peripheral blood stem cells in: 1) adults with non-Hodgkin's lymphoma and multiple myeloma; 2) children (1 year of age and older), with lymphoma or other solid malignant tumors.

ADMINISTRATION

- SC: in the abdomen. Observe the patient for 60 minutes after each dose.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis/anaphylactic shock, urticaria, periorbital swelling, dyspnea, hypoxia.
- Cardiovascular: vasovagal reaction: orthostatic hypotension and/or syncope within 60 minutes of administration (uncommon).
- GI: diarrhea, nausea, vomiting, flatulence.
- CNS: headache, dizziness, insomnia, anxiety.
- Hematologic: leukocytosis, thrombocytopenia.
- Fatigue, arthralgia.
- Local reactions: erythema, hematoma, hemorrhage, induration, inflammation, irritation, pain, paresthesia, pruritus, rash, urticaria.

DOSAGE

- Adults: after receiving daily filgrastim for 4 days, administer plerixafor in the evening 10-11 hours prior to initiation of each apharesis:
  - Patients weighing 83 kg or less: 20 mg as a fixed dose OR 0.24 mg/kg SC daily.
  - Patients weighing more than 83 kg: 0.24 mg/kg (maximum dose: 40 mg) SC daily.
  - Continue treatment until achieved target collection, up to a maximum of 4 consecutive days.
  - Dosage in renal impairment:

CrCl (mL/min)	Dose
greater than 50	no dosage adjustment necessary
50-20	0.16 mg/kg, not to exceed 27 mg
less than 20 or dialysis	insufficient data to recommend a dose

- Pediatrics: after receiving daily filgrastim for 4 days, administer plerixafor in the evening 8-12 hours prior to initiation of each apharesis:
  - 0.24 mg/kg SC daily.
  - Continue treatment until achieved target collection, up to a maximum of 3 consecutive days.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials of ready-to-use solution at room temp.
- Stable for 7 days at room temp (25°C) and 40°C at a concentration of 20 mg/mL in plastic syringes.

MISCELLANEOUS

- There is a potential for tumour cell mobilization from the bone marrow.

REFERENCES

1, 5, 95, 135, 219.

Full revision 2017; limited revision 2020

NAME OF MEDICATION
PNEUMOCOCCAL VACCINE

OTHER NAMES
Pneumovax ® 23 (23-valent)
Prevnar ® 13 (13-valent), Synflorix ® (10-valent)

CLASSIFICATION
Vaccine

INDICATIONS

- Active immunization against pneumonia and invasive diseases, including sepsis, meningitis, pleural empyema and bacteriemia, caused by covered serotypes of *Streptococcus pneumoniae* in specific age groups (Pneumovax ® 23: 2 years of age and older; Prevnar ® 13: 6 weeks of age and older; Synflorix ®: 6 weeks up to 5 years of age).
- Active immunization against active otitis media caused by covered serotypes of *Streptococcus pneumoniae* in patients 6 weeks up to 5 years of age.

ADMINISTRATION

- Pneumovax ® 23: SC, IM (deltoid muscle or lateral midthigh).
- Prevnar ® 13 and Synflorix ®: IM only in the anterolateral aspect of the thigh in infants or the deltoid muscle in older children and adults. Shake well to obtain a uniform white suspension.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylactoid reaction, serum sickness, angioneurotic edema, erythema multiforme.
- GI: diarrhea, vomiting, decreased appetite.
- CNS: headache, irritability, drowsiness.
- Fever, arthralgia, myalgia, malaise, fatigue.
- Local reactions: erythema, warmth, swelling, soreness, induration at injection site.

DOSAGE

- Pneumovax ® 23: 0.5 mL SC/IM.
- Prevnar ® 13 and Synflorix ®: 0.5 mL IM. Refer to manufacturer's monograph for full schedule, depending on age of patient.
- Revaccination or booster doses may be indicated depending on patient age and comorbidities. Refer to manufacturer's monograph and the Canadian Immunization Guide for schedule recommendations.
- Administration of Pneumovax ® 23, and Prevnar ® 13 or Synflorix ® should be spaced by at least 8 weeks if Prevnar ® 13 or Synflorix ® is used for the first dose, and by at least one year if Pneumovax ® 23 is used for the first dose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes between 2-8°C. Protect Synflorix ® from light. Do not freeze Prevnar ® 13 and Synflorix ®.
- Prevnar ® 13: potency not affected if exposed to temperatures up to 25°C for cumulative period of 4 days.
- Synflorix ®: potency not affected if exposed to temperatures between 8-25°C for 3 days.
- Pneumovax ® 23: potency not affected if exposed at temperatures between -15° and 0°C on 2 occasions for a total maximum period of 14 days.

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MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The pneumococcal vaccine is an inactivated vaccine; Pneumovax ® 23 is a polysaccharide vaccine while Prevnar ® 13 and Synflorix ® are conjugate vaccines.
- Administration of pneumococcal vaccine should be delayed, if possible, during an acute febrile illness or active infection of moderate to severe intensity.
- Pneumovax ® 23: may be administered concomitantly with influenza, shingles, *Haemophilus influenza* type b conjugate and meningococcal vaccines, providing separate syringes and different sites are used.
- Prevnar ® 13 and Synflorix ®: may be administered concomitantly with diphteria, tetanus, acellular pertussis, *Haemophilus influenza* type B conjugate, inactivated polyomyelitis, hepatitis B, measles, mumps, rubella, meningococcal serogroup C conjugate, rotavirus and varicella virus vaccines, providing separate syringes and different sites are used.

REFERENCES

1, 5, 31, 135, 201.

NAME OF MEDICATION
PORFIMER

OTHER NAMES
Dihematoporphyrin ether, Photofrin ®

CLASSIFICATION
Antineoplastic, Photosensitizing agent

INDICATIONS
- Treatment of recurring superficial papillary bladder cancer.
- Reduction of obstruction and palliation of dysphagia in patients with completely or partially obstructing esophageal cancer.
- Reduction of obstruction and palliation of symptoms in patients with completely or partially obstructing endobronchial non-small cell lung cancer and treatment of superficial endobronchial non-small cell lung cancer in patients for whom surgery and radiotherapy are not indicated.
- Ablation of high-grade dysplasia associated with Barrett's Esophagus in patients who refuse esophagectomy and are in overall good health.

ADMINISTRATION
- Reconstitute the 75 mg and 15 mg vials with 31.8 mL and 6.6 mL of D5W, respectively, to obtain a final concentration of 2.5 mg/mL. Shake well until dissolved.
- IV direct: physician or RN; undiluted, inject over 3-5 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rare.
- Cardiovascular: atrial fibrillation, peripheral edema, cerebrovascular accident (rare), thromboembolism (rare).
- GI (mainly when used for GI lesions): nausea, vomiting, constipation, diarrhea, esophageal stenosis, abdominal pain, dehydration, dysphagia, hiccups.
- CNS: anxiety, insomnia, headache.
- Hematologic: anemia.
- Respiratory (mainly when used to treat lung cancer): dyspnea, hemoptysis, chest pain (also when used to treat esophageal lesions), cough, pleural effusion, respiratory failure, pneumonia.
- Dermatologic: photosensitivity lasting for at least 4-6 weeks, skin discolouration and fragility, hair growth, skin nodules, wrinkles.
- Urogenital (mainly when used to treat bladder cancer): dysuria, hematuria, urinary frequency, urgency, strangury, genital edema, suprapubic pain, urinary incontinence, nocturia, urinary tract infection.
- Fever, pain, photophobia.
- Extravasation hazard: irritant after exposure to light. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy). If extravasation occurs, protect the area from light for a minimum of 30 days.

DOSAGE
- 2 mg/kg IV administered 40-50 hours prior to laser light. Light therapy may be repeated 96-120 hours after porfimer administration for patients with lung or esophageal cancer only. Further courses are permitted, except for bladder cancer.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials at room temp.
- Reconstituted solutions are opaque and are stable for 24 hours in the fridge, when protected from light.
- Porfimer sodium is incompatible with NS.

MISCELLANEOUS
- Patients should protect eyes and skin from exposure to direct or indirect sunlight or bright focused indoor light for a period of at least 30 days; use full-length clothing, hat, gloves, dark sunglasses; use of sunscreens does not provide adequate protection from sunlight; sunblock (e.g., zinc oxide or titanium dioxide) should be applied to exposed skin when outdoors.
- Patients should be advised to avoid total darkness as exposure of skin to normal ambient lighting is important to aid in the clearance of porfimer from the skin through the process of photobleaching.

REFERENCES
5, 40, 129, 165.

Full revision 2017; limited revision 2020

NAME OF MEDICATION
POSACONAZOLE

OTHER NAMES
Posanol ®

CLASSIFICATION
Antifungal

INDICATIONS
- Prophylaxis of *Aspergillus* and *Candida* infections in adult patients at high risk of developing these infections, such as patients with prolonged neutropenia secondary to chemotherapy or receiving hematopoietic stem cell transplant.
- Treatment of invasive aspergillosis in patients resistant or intolerant to amphotericin B or itraconazole.
- Treatment of mucormycosis.

ADMINISTRATION
- Intermittent IV infusion: allow vial to reach room temp. Dilute the 300 mg dose (16.7 mL) in NS, 1/2NS, D5W, D5-1/2NS or D5-NS to obtain a final concentration of 1-2 mg/mL. Administer through a central IV line over 90 minutes; a single dose can be administered only once through a peripheral line over 30 minutes (as vein irritation will occur).

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, pruritus, anaphylaxis.
- Cardiovascular: peripheral edema, hypertension, arrhythmias, QT interval prolongation.
- GI: diarrhea, nausea, vomiting, abdominal pain, decreased appetite, mucositis.
- CNS: headache.
- Hepatic: elevated LFT.
- Hypokalemia, hypomagnesemia, hypocalcemia.
- Fever, chills.
- Epistaxis.
- Local reactions: thrombophlebitis with multiple doses administered peripherally.

DOSAGE
- Prophylaxis and treatment of fungal infections: 300 mg IV twice daily for the first day followed by 300 mg IV once daily. When instituting prophylaxis in patients with acute myelogenous leukemia or myelodysplastic syndromes, start a few days before estimated onset of neutropenia and continue for 7 days after neutrophil count rises above 500 cells/mm.[3]
- Patients weighing over 120 kg may have lower posaconazole levels; closely monitor for breakthrough fungal infections.
- Dosage in renal impairment: no dosage adjustment required if CrCl is 50 mL/min or greater. Should be avoided in patients with a CrCl below 50 mL/min as the vehicle (Betadex Sulfobutyl Ether Sodium) may accumulate; if administration in such cases, monitor serum creatinine level and consider changing to oral posaconazole if creatinine increases.
- Dosage in hepatic impairment: no dosage adjustment required for mild to severe disease but use with caution in severe liver disease.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Once diluted, solution should be colourless to pale yellow.
- Stable for 24 hours in the fridge once product is diluted.

MISCELLANEOUS
- CYP3A4 substrates that prolong QTc interval (e.g., quinidine, pimozide), statins that are primarily metabolized by CYP3A4 (e.g., simvastatin, lovastatin, atorvastatin), ergot alkaloids (e.g., dihydroergotamine) and sirolimus are contraindicated with posaconazole IV.
- No information regarding cross-hypersensitivity with other azole antifungal agents; caution should be used.

REFERENCES
1, 4, 5, 6, 40, 135, 208.

Full revision 2017; limited revision 2020, 2021

NAME OF MEDICATION
POTASSIUM CHLORIDE

OTHER NAMES

CLASSIFICATION
Electrolyte

INDICATIONS

- Prevention or treatment of hypokalemia.
- For treatment of severe hypokalemic-hypochloremic alkalosis.

ADMINISTRATION

- Intermittent IV infusion, Continuous IV infusion: use premixed bag; see table below for rate and concentration.

Location	Maximum Concentration For Peripheral Access		Maximum Concentration for Central Access		Maximum Rate of Infusion	Comments
	Infusion	Small volume Infusion	Infusion	Small volume infusion		
Regular Wards	60 mmol/L	10 mmol/ 100 mL	60 mmol/L	20 mmol/ 50 mL	20 mmol/hr	
Critical Care Areas	60 mmol/L	10 mmol/ 100 mL	60 mmol/L	20 mmol/ 50 mL	40 mmol/hr	ECG monitoring required if rate greater than 20 mmol/hr

- The preferred method of infusion is with an infusion pump; if rate is greater than 10 mmol/hr or concentration greater than 40 mmol/L, then it must be administered by an infusion pump.
- Consult Pharmacy policy 00627 (Intravenous administration of potassium chloride (KCL)) if more information is needed.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus, angioedema, anaphylaxis.
- GI: nausea, vomiting.
- Hyperkalemia: especially from rapid IV infusion. Signs and symptoms include numbness and tingling in the extremities, confusion, flaccid paralysis, cold skin, ECG changes. (**Antidote**: calcium gluconate 1000 mg IV given as 10 mL of 10% solution over 5 minutes, repeat in 5 minutes as required. NB: calcium is given only in the setting of ECG changes. Insulin/glucose: 5-10 units of regular insulin IV direct **followed immediately** by 25-50 g of dextrose (50-100 mL 50% solution) administered IV over 5 minutes.
- Because of slow shifts between extracellular and intracellular compartments, it is impossible to replace a substantial potassium deficit quickly; attempts to quickly replace large potassium deficits will result in derangement of transcellular potassium balance and potentially dangerous neuromuscular and cardiac complications.
- Local reactions: phlebitis, vesicles, swelling, erythema and pain at site of injection with concentrated solutions.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Usual dose: 20-60 mmol (at a slow rate as above) per 24 hours. Dose depends on requirements.
- Usual maximum daily dosage: 200 mmol. However, up to 400 mmol/24 hours have been given in selected situations with extreme caution.
- Dosage in renal impairment: reduce dose in renally impaired patients as potassium is mainly eliminated by kidney; monitor frequently.

…/Cont.

NAME OF MEDICATION
POTASSIUM CHLORIDE

OTHER NAMES

CLASSIFICATION
Electrolyte

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and bags at room temp. Avoid excessive heat. Do not freeze.
- Stable 24 hours at room temp when mixed with NS or D5W.
- Compatible with all commonly used IV solutions.

MISCELLANEOUS

- Refer to the phosphate monograph for information on potassium phosphate.
- Oral route is usually preferable.
- Monitor serum potassium and chloride concentrations.
- 1 mmol KCl =1 mEq KCl.
- When hypokalemia and hypomagnesemia coexist, magnesium deficiency should be corrected to facilitate the correction of hypokalemia.

REFERENCES

1, 2, 4, 5, 6, 40, 95, 135, 272, 328, 366, 367, 431, 462.

Full revision 2017; limited revision 2020, 2021

NAME OF MEDICATION	
PRALAtrexate	
OTHER NAMES Folotyn ®	CLASSIFICATION Antineoplastic

INDICATIONS

- Treatment of relapsed or refractory peripheral T-cell lymphoma (PTCL).

ADMINISTRATION

- **Ensure folic acid and vitamin B_{12} have been administered; refer to Dosage section.**
- IV direct: physician or RN; administer undiluted over 3-5 minutes into the tubing of a freely running IV solution of NS.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication**: consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug**: consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Infusion-related reactions.
- Cardiovascular: edema, tachycardia, hypertension.
- GI: mucosal inflammation (may be minimized by taking vitamin B_{12} and folic acid; refer to Dosage section); nausea, vomiting, diarrhea, constipation, abdominal pain, anorexia, dyspepsia, dehydration.
- CNS: headache.
- Dermatologic: urticaria, rash, pruritus, skin exfoliation, ulceration, toxic epidermal necrolysis; can be severe and life-threatening; can occur after the first dose; may increase in severity with continued treatment.
- Electrolyte disturbances: hypokalemia
- Hematologic: leukopenia, neutropenia, thrombocytopenia, anemia; may be minimized by taking vitamin B_{12} and folic acid (refer to Dosage section).
- Hepatic: elevated LFTs.
- Musculoskeletal: pain in extremities, back pain, weakness.
- Renal: acute kidney injury, renal failure.
- Respiratory: cough, dyspnea, epistaxis, pharyngolaryngeal pain.
- Tumour lysis syndrome.
- Mucosal inflammation of GI (including stomatitis), respiratory and genitourinary tracts; may be minimized by taking vitamin B_{12} and folic acid (refer to Dosage section).
- Infections (e.g., upper respiratory tract infection, pneumonia, febrile neutropenia, sepsis, septic shock, herpes zoster); may be fatal.
- Fever, chills, fatigue, night sweats.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation policy of the Nursing Policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- **Folic acid 1-1.25 mg PO daily should be started 10 days before the first pralatrexate dose and continued during therapy and for 30 days after last dose. Vitamin B_{12} 1000 mcg IM should be given within 10 weeks before the first dose and every 8-10 weeks thereafter; subsequent injections may be given on the same day as pralatrexate.**
- 7-week cycles: 30 mg/m^2 IV once weekly for 6 weeks.
- 4-week cycles (Columbia protocol; minimizes risk of mucositis):
  - Cycle 1: 10 mg/m^2 IV on day 1, 20 mg/m^2 IV on day 8, 30 mg/m^2 IV on day 15.
  - Cycle 2 onwards: 30 mg/m^2 IV on days 1, 8, and 15.
  - For all cycles: leucovorin 15 mg PO twice daily on days 3-6, 10-13, and 17-20. Leucovorin is given in addition to folic acid to further reduce the risk of mucositis.

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DOSAGE (Cont.)

- Dosage in renal impairment: reduce dose to 15 mg/m^2 in patients with CrCl of 15 to less than 30 mL/min; avoid in end-stage renal disease.
- Dosage in hepatic impairment: no data available in patients with total bilirubin greater than ULN; AST or ALT greater than 2.5 x ULN; or AST or ALT greater than 5 x ULN if documented hepatic lymphoma.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light.
- Unopened vials are stable for 72 hours at room temp protected from light.
- Stable for 24 hours at room temp protected from light undiluted in plastic syringes.

MISCELLANEOUS

- Monitor liver function, renal function and complete blood cell count prior to starting therapy and during therapy.
- Live vaccines should not be administered concurrently with pralatrexate and for at least 3 months after the last dose of pralatrexate.

REFERENCES

5, 129, 135, 165, 470.

New monograph 2020

NAME OF MEDICATION
PRALIDOXIME *

OTHER NAMES
2-PAM, Protopam ® chloride,
2-Pyridine aldoxime methochloride

CLASSIFICATION
Antidote

INDICATIONS

- Treatment of poisoning due to organophosphate anticholinesterase pesticides and chemicals (including nerve agents).
- Control of overdose of carbamate anticholinesterase agents used in the treatment of myasthenia gravis (e.g., neostigmine, pyridostigmine).

ADMINISTRATION

- For IV use: reconstitute each 1 g vial with 20 mL of SWFI for a concentration of 50 mg/mL.
- IV direct: physician only; undiluted; inject over at least 5 minutes, not to exceed a rate of 200 mg/min.
- Intermittent IV infusion: (preferred): dilute in 100 mL of NS; infuse over 15-30 minutes, not to exceed a rate of 200 mg/min.
- Continuous IV infusion: dilute 2 g in 250 mL of NS.
- IM, SC (when IV administration is not feasible): reconstitute each 1 g vial with 3.3 mL of SWFI for a final concentration of approximately 300 mg/mL. For children, administer in the anterolateral aspect of the thigh.
- Except in emergency situations, should only be used in hospitalized patients when respiratory and other supportive measures are available.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: tachycardia with rapid IV administration; hypertension.
- GI: nausea, vomiting.
- CNS: dizziness, drowsiness, headache.
- Hepatic: transient ALT/AST elevation.
- Ophthalmic: blurred vision, diplopia, impaired accommodation.
- Respiratory: laryngospasm and apnea with rapid IV administration.
- Muscle rigidity and transient neuromuscular blockade with rapid IV administration. May precipitate myasthenic crisis when use in patients treated for myasthenia gravis.
- Local reactions: mild pain at injection site with IM administration.

DOSAGE

- Adults:
  - Organophosphate poisoning:
    - IV: 1-2 g or 30 mg/kg IV. May repeat dose 60 minutes later prn then every 10-12 hours prn OR follow initial dose with a continuous IV infusion of 8 mg/kg/hr (maximum of 12 g in 24 hours).
    - IM/SC: for mild symptoms, 600 mg IM/SC and repeat every 15 minutes prn for a maximum total dose of 1800 mg; for severe symptoms, 1800 mg IM/SC administered as 3 divided doses of 600 mg given in rapid succession. May repeat 60 minutes after the last dose prn.
  - In anticholinesterase overdose: 1-2 g IV followed by 250 mg IV every 5 minutes as needed.

- Pediatrics:
  - Organophosphate poisoning:
    - IV: 20-50 mg/kg IV (maximum 2 g/dose). May repeat 60 minutes later prn then every 10-12 hours prn OR follow initial dose with a continuous IV infusion of 10-20 mg/kg/hr.
    - IM/SC: for children weighing 40 kg or more, refer to adult dosing. For children weighing less than 40 kg: for mild symptoms, 15 mg/kg IM/SC and repeat every 15 minutes prn for a maximum total dose of 45 mg/kg; for severe symptoms, 45 mg/kg IM/SC administered as 3 divided doses given in rapid succession. May repeat 60 minutes after the last dose prn.

  - Reduce dose in patients with renal dysfunction. Use with caution.

…/Cont.

NAME OF MEDICATION	
PRALIDOXIME *	
OTHER NAMES 2-PAM, Protopam ® chloride, 2-Pyridine aldoxime methochloride	CLASSIFICATION Antidote

…/Cont.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Stable for 28 days between -20 and 50°C in NS at concentrations of 8 and 10 mg/mL in PVC plastic bags.

MISCELLANEOUS

- Treatment will be most effective if given within a few hours after poisoning.
- In most cases of organophosphate anticholinesterase poisoning, administer atropine before pralidoxime.
- Pralidoxime has only limited value as an antagonist to carbamate anticholinesterases (neostigmine, pyridostigmine, physostigmine).

REFERENCES

1, 3, 4, 5, 6, 40, 82, 95, 208.

* Available via Health Canada's Special Access Programme

Full revision 2017; limited revision 2019

NAME OF MEDICATION	
PROCAINAMIDE	
OTHER NAMES	CLASSIFICATION
Pronestyl ®	Antiarrhythmic

INDICATIONS

- Treatment of supraventricular arrhythmias: atrial fibrillation and paroxysmal atrial tachycardia uncontrolled by other antiarrhythmic agents.
- Treatment of ventricular arrhythmias.

ADMINISTRATION

- IV direct: physician or RN; ***cardiac monitoring, blood pressure monitoring***. Dilute each mL (100 mg) with 5-10 mL of D5W; usual rate is 20 mg/min; up to a maximum of 50 mg/min.
- Intermittent IV infusion: ***cardiac monitoring, blood pressure monitoring***. For loading dose only. Dilute dose with NS or D5W to a maximal concentration of 20 mg/mL (e.g., 1 g in 50 mL) and administer at a rate of 20-50 mg/min. Must be administered by an infusion pump.
- Continuous IV infusion: ***cardiac monitoring, blood pressure monitoring***. Dilute 0.5-1 g in 250 mL of NS or D5W or 1-2 g in 500 mL of NS or D5W to obtain a final concentration of 2 or 4 mg/mL. At TOH, add 1 g (10 mL from a procainamide 100 mg/mL vial) to 250 mL of NS to obtain a final concentration of 4 mg/mL. Must be administered by an infusion pump.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: angioedema, urticaria, pruritus, rash.
- Cardiovascular: conduction defects (widening of QRS complex, QT interval prolongation, PR interval prolongation), ventricular tachycardia, bradycardia, ventricular asystole or fibrillation; hypotension especially with rapid IV administration (**Antidote**: IV phenylephrine or norepinephrine).
- GI: nausea, vomiting, diarrhea.
- CNS: dizziness, depression, confusion, hallucinations, convulsions.
- Hematologic: blood dyscrasias (e.g., thrombocytopenia, agranulocytosis, leukopenia, hypoplastic anemia, hemolytic anemia).

DOSAGE

- A number of loading dosing regimens have been suggested:
  - 100 mg IV direct every 5 minutes until arrhythmia is suppressed or a total of 1 g has been given.
    or
  - loading dose IV infusion at a rate of 20-50 mg/min until arrhythmia is suppressed, side effects develop, or a total of 17 mg/kg has been given.
    or
  - loading dose of 500-600 mg IV over 25-30 minutes.
- Maintenance continuous IV infusion of 1-6 mg/min OR 0.02-0.08 mg/kg/min.
- IM: 50 mg/kg/day in divided doses q3-6h. For cardiac arrhythmias associated with anesthesia or surgery: 100-500 mg IM.
- Dosage in renal impairment: if CrCl is between 10-50 mL/min, reduce continuous infusion rate by 25-50%; if CrCl is less than 10 mL/min, reduce continuous infusion rate by 50-75%.
- Dosage in hepatic impairment: if Child-Pugh score is between 8-10, reduce continuous infusion rate by 25%; if Child-Pugh score is greater than 10, reduce continuous infusion rate by 50%.

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COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Do not freeze.
- Solutions darker than light amber should not be used.
- Stable for 24 hours at room temp and 7 days if refrigerated when diluted in NS at a concentration of 2-4 mg/mL.
- Stable for 6 hours at room temp in D5W at a concentration of 2-4 mg/mL.
- Stable for 193 days at room temp (23°C, exposed to light) and in the fridge (5°C, protected from light) at a concentration of 100 mg/mL (original solution with no dilution) in clear glass vials.
- Stable for 193 days at room temp (23°C, exposed to light) and in the fridge (5°C, protected from light) in NS at a concentration of 3 mg/mL in PVC bags.

MISCELLANEOUS

- Do not administer in patients with: 1) torsades de pointes; 2) complete AV heart block or with second or third degree AV nodal block unless an electrical pacemaker is operative; 3) myasthenia gravis; 4) systemic lupus erythematosus; 5) hypersensitivity to ester-type local anesthetics (e.g., tetracaine, benzocaine, cocaine).
- Potentiates neuromuscular blocking agents.

REFERENCES

1, 4, 5, 6, 40, 135, 383.

Full revision 2017; limited revision 2018, 2019, 2020

NAME OF MEDICATION
PROMETHAZINE

OTHER NAMES
Phenergan ®

CLASSIFICATION
Antihistamine, Antiemetic, Analgesic

INDICATIONS
- For the control of nausea and vomiting.
- Adjunctive use when immediate sedation is required (preoperative, postoperative, labour).
- Adjunctive analgesic for the control of preoperative and postoperative pain.
- Treatment of allergic disorders (low efficacy).

ADMINISTRATION
- IV administration is NOT preferred; use only large-bore vein, preferably via a free-flowing IV line at a port farthest from the vein or a central venous access site, but absolutely no hand or wrist vein; check patency of the access site before administration. Before IV administration of the drug, tell patients to report immediately any burning or pain that occurs during or after the injection.
- IV direct: physician or RN; ***blood pressure monitoring***. Dilute with 10-20 mL NS; inject into the tubing of a freely running IV solution over 10-15 minutes.
- Intermittent IV infusion: dilute in 50-100 mL of NS and infuse over 10-15 minutes.
- Deep IM (preferred).

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, angioedema, urticaria.
- Cardiovascular: hypotension, hypertension, tachycardia, bradycardia.
- CNS: drowsiness, seizures, extrapyramidal symptoms, confusion.
- Dry mouth, blurred vision.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE
- Consider limiting initial dose to 6.25-12.5 mg to prevent tissue damage.

Adults:
- Nausea and vomiting: 12.5-25 mg IV/IM q4-6h as needed.
- Pre/postoperative sedation or analgesia: 25-50 mg IV/IM.
- Sedation during labour: 50 mg IV/IM in early labour or 25-75 mg IV/IM in established labour q4h prn. Maximum of 100 mg/24 hours during labour.
- Allergic disorders: 25 mg IV/IM, dose may be repeated in 2 hours prn.

Pediatrics (only if 2 years old or older):
- For sedation, analgesia and treatment of nausea/vomiting: 0.25-1.1 mg/kg/dose (not to exceed 25 mg) IV/IM q4-6h as needed. Use the lowest effective dose.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 15-30°C. Protect from light.
- Do not use solutions if discolouration (darker than slightly yellow) or if a precipitate is present.
- Compatible with NS, D5W, Ringer's, RL and sodium lactate 1/6 M solutions.

MISCELLANEOUS

- Use in children younger than 2 years of age has been associated with potentially fatal risk of respiratory depression; do not use in this patient population.
- In children 2 years of age and older: avoid combining with other drugs that may also slow breathing.
- Contains sulfites.

REFERENCES
1, 4, 5, 40, 82, 95, 135, 366, 367.

Full revision 2017

NAME OF MEDICATION
PROPOFOL

OTHER NAMES
Diprivan ®

CLASSIFICATION
General anesthetic, Sedative

INDICATIONS

- For sedation in critically ill, mechanically ventilated patients.
- For conscious sedation for surgical and diagnostic procedures.
- For induction and maintenance of general anesthesia.

ADMINISTRATION

- Shake gently before use.
- IV direct: physician only. RN may administer in patients on ventilator support.
- Continuous IV infusion: physician or RN. ***Ventilator support***. Infuse undiluted or dilute only with D5W to a final concentration greater than 2 mg/mL. At TOH, use the ready-to-use vials of 10 mg/mL with no further dilution. Must be administered by an infusion pump.
- Do NOT use filters with a pore size less than 5 microns as it may restrict the administration and/or cause the breakdown of the emulsion.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; anaphylactic/anaphylactoid reaction, rash, pruritus, erythema, angiodema.
- Cardiovascular: bradycardia; hypotension (10-30%) more frequent in elderly patients and those with pre-existing cardiovascular disease; flushing (especially in children).
- GI: nausea, vomiting.
- CNS: headache.
- Respiratory: apnea with large induction doses (2.5 mg/kg or greater), respiratory depression, cough, hypoxia.
- Renal: urine may turn green in some patients.
- Excitatory effects (e.g., spontaneous movements, twitching).
- Propofol infusion syndrome: may be fatal, includes metabolic acidosis, hyperkalemia, lipemia, rhabdomyolysis, hepatomegaly, cardiac failure, renal failure, especially with doses above 5 mg/kg/hr for longer than 48 hours.
- Priapism.
- Local reactions: burning/stinging and pain occur in 30% of cases when hand veins are used for injection and 5% when larger veins are used.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

Adults:

- <u>ICU Sedation</u>: Initiate at 0.3 mg/kg/hr IV (0.005 mg/kg/min IV) and increase by increments of 0.3-0.6 mg/kg/hr IV (0.005-0.01 mg/kg/min IV) every 5-10 minutes. Most patients require IV infusions of 0.3-3 mg/kg/hr (0.005-0.05 mg/kg/min). A maximum rate of 3.6-4.8 mg/kg/hr (0.06-0.08 mg/kg/min) has been suggested in the literature; however, maximum infusion rate is not well defined and may vary by institution and treatment plan. Intermittent bolus doses of 10-20 mg IV prn if hypotension is not likely.

- <u>Conscious sedation</u>:
  - In adults younger than 55 years of age, administer initial dose of 0.5-1 mg/kg IV over 3-5 minutes OR 6-9 mg/kg/hr IV (0.1-0.15 mg/kg/min IV) for 3-5 minutes; follow by maintenance IV infusion at 1.5-4.5 mg/kg/hr (0.025-0.075 mg/kg/min) titrated to desired effect OR give intermittent bolus doses of 10-15 mg IV prn.
  - In adults 55 years of age and older, debilitated, hypovolemic or ASA class III or IV patients, reduce initial dose by 20-30% then follow by maintenance IV infusion at 4-6 mg/kg/hr (0.066-0.1 mg/kg/min) titrated to desired effect.

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DOSAGE (Cont.)

- Induction dose for general anesthesia:
  - In adults younger than 55 years of age, approximately 40 mg IV every 10 seconds until induction onset (usual adult dose 2-2.5 mg/kg IV).
  - In adults 55 years of age and older, debilitated or ASA class III or IV patients (American Society of Anesthesiology physical status classification of surgical patients), the total dose required is likely to be 1-1.5 mg/kg IV and should be given at a rate of approximately 20 mg IV every 10 seconds.
  - For cardiac anesthesia: approximately 20 mg IV every 10 seconds until induction onset (0.5-1.5 mg/kg IV).
  - For neurosurgical anesthesia: approximately 20 mg IV every 10 seconds until induction onset (1-2 mg/kg IV).

- Maintenance dose for general anesthesia:
  - In adults younger than 55 years of age: infusion of 6-12 mg/kg/hr IV (0.1-0.2 mg/kg/min IV) titrated to desired effect AND/OR intermittent bolus in increments of 25-50 mg IV as required.
  - In adults 55 years of age and older, debilitated or ASA class III or IV patients: infusion of 3-6 mg/kg/hr IV (0.05-0.1 mg/kg/min IV).

Pediatrics (3-18 years of age):
- General anesthesia: Induction dose of 2.5 mg-3.5 mg/kg IV over 20-30 seconds (reduced dosage for patients of ASA class III or IV), followed by a maintenance IV infusion at 6-15 mg/kg/hr (0.1-0.25 mg/kg/min) titrated to desired effect.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Available as a white, oil in water emulsion. Store between 2-25°C; do not freeze.
- Use strict aseptic technique when handling propofol as the solution is susceptible to microbial contamination.
- Compatible with the following IV solutions when administered into a running line (Y-site compatibility): D5W, NS, RL, D5-RL, D5-1/2NS, D5-1/4NS.
- Infusion of propofol straight from the vial (with no further dilution) is stable for 12 hours after spiking; discard infusion and tubing after this time.
- Infusion of diluted propofol transferred from the original container is stable for 6 hours after dilution; discard infusion and tubing after this time.
- When diluted in D5W, more stable in a glass container than plastic (95% potency after 2 hours of running infusion in plastic).

MISCELLANEOUS

- Recovery time is usually 10-30 minutes when discontinued, even after prolonged infusion.
- Contraindicated in patients with known hypersensitivity to egg or egg components.

REFERENCES

1, 4, 5, 40, 82, 95, 135, 198, 366, 462.

Full revision 2017; limited revision 2019, 2020, 2021

NAME OF MEDICATION
PROPRANOLOL

OTHER NAMES
Inderal ®

CLASSIFICATION
Beta-blocker

INDICATIONS

- In cardiac arrhythmias: supraventricular tachyarrhythmias, ventricular tachyarrhythmias, tachyarrhythmias associated with digitalis intoxication.
- Prevention of beta blockade withdrawal symptoms.
- Used in combination with alpha-blocking agents in the management of tachycardia induced by pheochromocytoma.
- In cardiac disturbances associated with thyrotoxicosis.
- In idiopathic hypertrophic subaortic stenosis.
- In hypertension and angina when oral administration is not possible.

ADMINISTRATION

- IV direct: physician only for first dose, RN may administer subsequent doses; ***cardiac monitoring; continuous BP monitoring***. Undiluted but preferably dilute each 1 mg in 10 mL of D5W or NS; inject at a maximum rate of 1 mg/min.
- Intermittent IV infusion: ***cardiac monitoring; continuous BP monitoring***. Dilute in 50 mL of a compatible solution and administer over 10-30 minutes. Must be administered by an infusion pump.
- Continuous IV infusion: ***cardiac monitoring; continuous BP monitoring.*** Dilute 15 mg in 250-500 mL of D5W; initial infusion rate of 2-3 mg/hr. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; rash, anaphylactic/anaphylactoid reactions, Stevens-Johnson syndrome, toxic epidermal necrolysis, exfoliative dermatitis, erythema multiforme, urticaria. Response to epinephrine may be greatly diminished.
- Cardiovascular: hypotension, syncope, shock, bradycardia, partial heart block, heart failure. (**Antidote** for severe bradycardia: atropine 0.25-1 mg IV).
- CNS: lightheadedness, dizziness, giddiness, confusion, drowsiness.
- Respiratory: bronchospasm, dyspnea, wheezing.
- Hyperglycemia and hypoglycemia; may also mask the signs and symptoms of hypoglycemia and of hypothyroidism.

DOSAGE

Adults:

- Arrhythmias - IV Direct: 1-3 mg IV, may repeat in 2-5 minutes, if necessary (total of 5 mg) OR 0.5-1 mg IV, may repeat prn, maximum of 0.1 mg/kg. Once response or maximum dose is achieved, additional doses may be given at intervals of no less than 4 hours.
  - IV Infusion: 2-5 mg/hr IV have been used.
- For beta-blockade therapy when oral administration is not possible: IV infusion: 3 mg/hr (range 2-5 mg/hr).
- Thyrotoxicosis: 0.5-1 mg IV over 10 minutes; repeat q3h prn OR give subsequent doses of up to 3 mg IV over 15 minutes every several hours prn.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light and excessive heat. Do not freeze.
- Stable 24 hours at room temp in D5W, NS, dextrose-saline solutions and RL at concentrations of 0.5 and 20 mg/L in PVC and polyolefin containers.

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MISCELLANEOUS

- No standard conversion from PO to IV because of wide variance of bioavailability with oral route (16-60%) and first-pass metabolism by the liver.
- Contraindicated in patients with sinus bradycardia, uncompensated heart failure, cardiogenic shock, bronchial asthma and bronchospasm.
- Concomitant use of IV propranolol with IV diltiazem or with IV verapamil may cause significant cardiac depression; therefore, co-administration (within a few hours) is contra-indicated.

REFERENCES

1, 4, 5, 6, 40, 95, 135, 234, 235.

NAME OF MEDICATION
PROTAMINE

OTHER NAMES

CLASSIFICATION
Antidote

INDICATIONS

- To neutralize the anticoagulant activity of heparin in severe heparin overdosage, during extracorporeal circulation or dialysis procedures.
- To partially neutralize the anticoagulant activity of LMWHs (dalteparin, enoxaparin, nadroparin, tinzaparin) in overdose situations.

ADMINISTRATION

- IV direct: physician or RN may give direct to a maximum dose of 50 mg; undiluted; inject at a maximum rate of 5 mg/min.
- Intermittent IV infusion: dilute with NS or D5W (at TOH, suggest 50 mL) and infuse at a maximum rate of 5 mg/min.
- Continuous IV infusion: dilute with NS or D5W (at TOH, suggest 250 mL) and infuse over 8-16 hours.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, angioedema, anaphylaxis and anaphylactoid reactions.
- If too rapid injection or use of high doses of protamine: severe hypotension, bradycardia, dyspnea, pulmonary edema, pulmonary hypertension, transitory flushing and a feeling of warmth (may minimize these by administering drug slowly).
- Heparin "rebound" usually within 8-9 hours after protamine administration, may occur as heparin is released from the protamine-heparin complex or from extravascular compartments; additional protamine may be required.

DOSAGE

Heparin neutralization:
- Dose of protamine to administer depends on heparin dose, route of administration, time elapsed since heparin was given and blood coagulation results.
- No more than 50 mg should be administered in any 10 minute period.
- **For heparin continuous IV infusion neutralization**: after stopping the heparin IV infusion, administer 25-50 mg IV of protamine immediately.
- **For heparin IV injection neutralization**: because heparin disappears rapidly from circulation, the dose of protamine required also decreases rapidly with the time elapsed following the IV injection of heparin.

Time elapsed since heparin injection	A few minutes	30-60 minutes	60-120 minutes	Greater than 120 minutes
Dose of protamine to neutralize 100 units of heparin	1 mg IV	0.5-0.75 mg IV	0.375 mg-0.5 mg IV	0.25-0.375 mg IV

- **For heparin SC injection neutralization**: based on the ratio of 1-1.5 mg of protamine IV to neutralize 100 units of SC heparin, administer protamine 25-50 mg IV slowly over 10 minutes followed by the remainder of the calculated dose as an IV infusion over 8-16 hours (the expected absorption time of the SC heparin dose).
- **For heparin neutralization during extracorporeal circulation**: administer 1.5 mg IV of protamine for each 100 units of heparin administered; may also determine dose of protamine according to coagulation studies.

LMWH neutralization:
- Protamine partially neutralizes LMWH anti-Xa activity (to a maximum of 60-75%). Excess protamine administered may contribute to bleeding.
- Dose of protamine to administer depends on LMWH dose, time elapsed since the LMWH was given and blood coagulation results.
- No more than 50 mg as a single dose.
- Refer to the following tables for more specific recommendations (on next page).

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DOSAGE (Cont.)

- **For enoxaparin**:

Time elapsed since enoxaparin injection	8 hours or less	8-12 hours	Greater than 12 hours
Dose of protamine to neutralize 1 mg of enoxaparin	1 mg IV	0.5 mg IV	May not be required

Note: a second dose of protamine at an equivalent of 0.5 mg per 1 mg of enoxaparin may be administered IV if aPTT is still prolonged 2-4 hours after the first protamine dose or if bleeding continues.

- **For dalteparin, nadroparin and tinzaparin**:

Time elapsed since LMWH injection	8 hours or less	Over 8 hours
Dose of protamine to neutralize 100 anti-Xa units of LMWH	1 mg IV	0.5 mg IV

Note: a second dose of protamine at an equivalent of 0.5 mg per 100 anti-Xa units of LMWH may be administered IV if aPTT is still prolonged 2-4 hours after the first protamine dose or if bleeding continues.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Do not freeze.
- Compatible with NS and D5W.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- Onset of action: within 5 minutes.
- Use with caution in patients allergic to fish, if previous exposure to protamine (including exposure to protamine-containing insulin) and in infertile or vasectomized men (may have antiprotamine antibodies); pretreat with corticosteroids and antihistamines if there is a concern about a potential protamine allergy.

REFERENCES

1, 4, 5, 40, 135.

NAME OF MEDICATION
PYRIDOXINE

OTHER NAMES
Vitamin B_6

CLASSIFICATION
Vitamin

INDICATIONS

- To treat pyridoxine deficiency when oral administration is not possible (generally preferable to give all B components instead of a single one).
- In the treatment of isoniazid, cycloserine, or hydrazine intoxication.
- For the treatment of poisoning from mushrooms of the genus *Gyromitra*.
- For treatment of pyridoxine-dependent seizures in neonates or infants.

ADMINISTRATION

- IV direct: physician only; undiluted; inject at a rate not exceeding 50 mg/min.
- Intermittent IV infusion: dilute in a compatible solution; rate as per Dosage section.
- IM.
- SC.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity.
- CNS: seizures (if IV administration of very large doses), headache, ataxia, somnolence, sensory neuropathy (after chronic administration of large doses).
- Folate deficiency.
- Local reactions: burning and stinging at injection site with IM and SC administration.

DOSAGE

- Pyridoxine deficiency: 10-20 mg IV/IM/SC daily for 3 weeks. Doses up to 600 mg per day may be needed for pyridoxine dependency syndrome.
- For isoniazid overdose (greater than 10 g): pyridoxine dose equal to amount of isoniazid ingested. Generally, 1-4 g is given by IV at a rate of 0.5-1 g/min; may repeat every 5-10 minutes prn OR follow by 1 g IM every 30 minutes until entire dose is given.
- For cycloserine overdose: 300 mg/day IV. Higher doses may be required.
- For hydrazine overdose: 25 mg/kg; give 1/3 of the dose IM and rest by IV infusion over 3 hours.
- For *Gyromitra* mushrooms poisoning: 25 mg/kg IV infused over 15-30 minutes, repeated as needed to a maximal dose of 15-20 g daily.
- Treatment of pyridoxine-dependent seizures in neonates or infants: 10-100 mg IV/IM.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light. Do not freeze.
- Compatible with D5W, NS, RL, sodium lactate 1/6 M.
- Incompatible with alkaline solutions, iron salts and oxidizing agents.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 82, 95, 135.

Full revision 2017

NAME OF MEDICATION
QuiNINE base *

OTHER NAMES
Quinimax ®

CLASSIFICATION
Antimalarial

Do NOT confuse quinine base with quinine dihydrochloride.
This monograph is specific to quinine BASE.

INDICATIONS
- Treatment of severe and complicated malaria when IV artesunate is contraindicated or unavailable.
- Treatment of non-complicated malaria in patients unable to tolerate oral therapy.

ADMINISTRATION
- Intermittent IV infusion (preferred): for loading dose: dilute dose in D5W or NS (at TOH, dilute in 100 mL of D5W) and infuse over 30 minutes. For maintenance dose: dilute dose in D5W or NS (at TOH, dilute in 500 mL of D5W) and infuse over 4 hours. Must be administered by an infusion pump.
- IM: only if IV administration is not possible; inject by deep and slow injection.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylaxis, urticaria, rash, pruritus, angioedema.
- Cardiovascular: cardiac conduction disturbances, hypotension.
- GI: nausea, vomiting, abdominal pain, diarrhea.
- Cinchonism: tinnitus, nausea, vomiting, vertigo, headache, disturbed vision.
- Endocrine and metabolic: hypoglycemia.
- Hematologic: hemolysis, thrombocytopenia, hypoprothrombinemia.
- Local reactions (IM only): pain, irritation, fibrosis, necrosis, abscess at injection site.

DOSAGE
Note: Loading dose not recommended if IV quinine is used to treat non-severe malaria in a patient who cannot tolerate oral therapy, or if patient received quinine or quinidine within the preceding 24 hours, or mefloquine within the preceding 2 weeks.

- Loading dose: 5.8 mg/kg (quinine base) IV, immediately followed by maintenance dose.
- Maintenance dose: 8.3 mg/kg (quinine base) IV, repeated q8h for a minimum of 24 hours if treating severe malaria. After 24 hours, switch to oral therapy if patient can take oral therapy and parasitemia is 1% or less. When IV quinine is used for non-severe malaria when oral administration is not possible, switch to oral therapy as soon as feasible. In patients requiring more than 48 hours of IV therapy or continuing to have acute renal injury, extend the dosing interval to q12h after 48 hours. For patients receiving hemodialysis, the full dose can be given; give after dialysis as quinine is removed by dialysis.
- IM (only if IV administration is not possible): 8.3 mg/kg (quinine base) IM q8h.
- Dosage in obesity: dose should be calculated using ideal body weight.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at a temperature not exceeding 25°C.
- Compatible with D5W, D10W and NS.

MISCELLANEOUS
- It is essential that appropriate follow-on oral therapy be started after IV quinine discontinuation.
- Solution is a mixture of quinine gluconate, quinidine gluconate, cinchonine hydrochloride, and cinchonidine hydrochloride. Each mL contains 125 mg of quinine base.
- Monitor vital signs and blood glucose. Monitor ECG if history of underlying cardiac disease.

REFERENCES
1, 5, 95, 97, 135, 346

* Available from the Canadian Malaria Network via Health Canada's Special Access Programme

New monograph 2020; limited revision 2021

NAME OF MEDICATION
QuiNINE dihydrochloride *

OTHER NAMES
Quininject ®

CLASSIFICATION
Antimalarial

Do NOT confuse quinine dihydrochloride with quinine base.
This monograph is specic to quinine DIHYDROCHLORIDE.

INDICATIONS
- Treatment of severe and complicated malaria when IV astesunate is contraindicated or unavailable.
- Treatment of non-complicated malaria in patients unable to take oral therapy.

ADMINISTRATION
- Intermittent IV infusion (preferred): for loading dose: dilute dose in 100 mL of D5W or NS (at TOH, dilute in D5W) and infuse over 30 minutes. For maintenance dose: dilute dose in 10 mL/kg of D5W or NS (at TOH, dilute in 500 mL of D5W) and infuse over 4 hours. Must be administered by an infusion pump.
- IM (only if IV administration is not possible): may dilute to 60-100 mg/mL to reduce pain; inject into anterior thigh (avoid gluteal area as may cause sciatic nerve injury).

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash.
- Cardiovascular: cardiac conduction disturbances, hypotension.
- GI: nausea, vomiting, abdominal pain, diarrhea.
- Cinchonism: tinnitus, nausea, vertigo, headache, disturbed vision.
- Hemolysis (rare), hypoprothrombinemia, hypoglycemia.
- Local reactions: with IM use: pain, local irritation at injection site.

DOSAGE
Note: Loading dose not recommended if IV quinine is used to treat non-severe malaria in a patient who cannot tolerate oral therapy, or if patient received quinine or quinidine within the preceding 24 hours, or mefloquine within the preceding 2 weeks.

- Loading dose: quinine dihydrochloride 7 mg/kg (quinine base 5.8 mg/kg) IV, immediately followed by maintenance dose.
- Maintenance dose: quinine dihydrochloride 10 mg/kg (quinine base 8.3 mg/kg) IV, repeated q8h for a minimum of 24 hours if treating severe malaria. After 24 hours, switch to oral therapy if patient can take oral therapy and parasitemia is 1% or less. When IV quinine is used for non-severe malaria when oral administration is not possible, switch to oral therapy as soon as feasible. In patients requiring more than 48 hours of IV therapy or continuing to have acute renal injury, extend the dosing interval to q12h after 48 hours. For patients receiving hemodialysis, the full dose can be given; give after dialysis as quinine is removed by dialysis.
- IM: (only if IV route is not possible) quinine dihydrochloride 10 mg/kg (quinine base 8.3 mg/kg) IM q8h.
- Dosage in obesity: dose should be calculated using ideal body weight.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp. Protect from light.
- Compatible with D5W and NS.

MISCELLANEOUS
- It is essential that appropriate follow-on oral therapy be started after IV quinine discontinuation.
- Quinine dihydrochloride 20 mg = quinine base 16.7 mg.
- Monitor vital signs and blood glucose. Monitor ECG if history of underlying cardiac disease.

REFERENCES
1, 5, 97, 134, 135, 208, 346, 456.

* Available from the Canadian Malaria Network via Health Canada's Special Access Programme

Full revision 2018; limited revision 2020, 2021

NAME OF MEDICATION
QUINUPRISTIN/DALFOPRISTIN *

OTHER NAMES
Synercid ®

CLASSIFICATION
Antibiotic

INDICATIONS

- Treatment of vancomycin-resistant *Enterococus faecium* (VREF) infections refractory to other antiinfectives.
- Treatment of complicated skin and skin structure infections in adults caused by *Staphylococcus aureus* (methicillin sensitive) and *Streptococcus pyogenes*.

ADMINISTRATION

- Reconstitute immediately before use. Reconstitute by **slowly** injecting 5 mL of D5W or SWFI to prepare a solution of 100 mg/mL; gently swirl to mix; do NOT shake. Allow to sit for a few minutes until all foam disappears. Resulting solution should be clear.
- Intermittent IV infusion: within 30 minutes of reconstitution, further dilute dose in 250 mL of D5W for peripheral administration or in 100 mL of D5W for central line administration. If peripheral vein irritation occurs, can increase infusion volume to 500 or 750 mL of D5W, change the infusion site or use a central/PICC line. In all cases, infuse over 60 minutes. Flush infusion line with D5W before and after administration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, angioedema, anaphylaxis.
- GI: diarrhea, vomiting, nausea.
- CNS: headache.
- Endocrine and metabolic: elevated lactate dehydrogenase (LDH).
- Hepatic: elevated alkaline phosphatase, total and conjugated bilirubin, AST, ALT, gamma glutamyl transferase.
- Musculoskeletal: arthralgia, myalgia (may be alleviated by reducing dose frequency to q12h).
- Local reactions: venous irritation, pain, burning, inflammation, edema, thrombophlebitis, thrombosis

DOSAGE

- VREF infections: 7.5 mg/kg IV q8h.
- Skin/skin structure infections: 7.5 mg/kg IV q12h.
- Dosage in renal impairment: no dosage adjustment is necessary.
- Dosage in hepatic impairment: dosage reduction may be required for patients with hepatic cirrhosis, but exact recommendations are not available.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store unreconstituted vial in the fridge; stable for 7 days at room temp.
- Reconstituted solution is stable for only 30 minutes.
- Diluted infusion solution stable for 5 hours at room temp or 54 hours in the fridge. Do not freeze.
- Compatible with D5W.
- Incompatible with NS and heparin.

MISCELLANEOUS

- Vial contains 500 mg of freeze-dried powder composed of quinupristin 150 mg/dalfopristin 350 mg.
- Quinupristin/dalfopristin is a strong inhibitor of cytochrome P450 3A4; monitor for any drug interaction with concomitant medications.

REFERENCES

1, 4, 40, 95, 135.

* Available via Health Canada's Special Access Programme

Full revision 2018

NAME OF MEDICATION
RABIES IMMUNE GLOBULIN, HUMAN

OTHER NAMES
HyperRAB ®, Imogam ®, KamRAB ™

CLASSIFICATION
Passive immunizing agent

INDICATIONS

- Post-exposure prophylaxis in patients of all ages suspected of exposure to rabies, who have not previously been fully immunized with a rabies vaccine.

ADMINISTRATION

- Infiltration (preferred): the full dose should be thoroughly infiltrated in the wound(s) and surrounding area, when possible. In case of insufficient immunoglobulin volume to infiltrate the entire wound area, the dose may be diluted as follows: 1) HyperRAB ®: with an equal volume of D5W (do NOT use NS); 2) Imogam ®: 2- to 3- fold in NS; 3) KamRAB ™: do NOT dilute as no data is available. Administer any remaining volume IM using a separate needle.
- IM: preferred sites are the deltoid or anterolateral aspect of the thigh. Upper outer buttocks may be used if wounds are located in that area or very large volumes/multiple doses are required, although absorption is unpredictable; do NOT administer in the central gluteal region (risk of sciatic nerve injury).
- SC: for KamRAB ™: may be administered SC when IM administration is contraindicated, however no data is available.
- Do NOT administer the rabies immune globulin in the same syringe or near the site of the rabies vaccine.
- Patients at increased risk of thrombosis should be monitored for at least 24 hours after administration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, angioedema, anaphylactic shock.
- Cardiovascular: hypotension, tachycardia.
- GI: nausea, vomiting, diarrhea, abdominal pain, flatulence.
- CNS: headache, dizziness.
- Hematology: hemolysis, thrombosis.
- Musculoskeletal: myalgia, arthralgia.
- Renal: leukocyturia, hematuria.
- Respiratory: nasal congestion, oropharyngeal pain, upper respiratory tract infection.
- Fever, chills, fatigue.
- Risk of transmission of infectious agents, including viruses, and theoretically the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.
- Local reactions: nodule and pain at injection site.

DOSAGE

- 20 international units/kg as an infiltration (with/without IM administration) promptly after exposure. Do not exceed recommended dose. Administer at the time of the first rabies vaccine dose (preferred) or any time up to and including the 7th day after the first dose of rabies vaccine. Do not repeat the dose of the immune globulin once the rabies vaccine treatment has been initiated. These instructions need to be followed, otherwise the efficacy of the rabies vaccine will be reduced.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

HyperRAB ®:
- Store vials between 2-8°C. Do not freeze.
- Vials are stable at room temp up to 25°C for 6 months; once stored at room temp, vials should not be returned to the refrigerator.

Imogam ®:
- Store vials between 2-8°C. Do not freeze.
- During storage, the clear to opalescent solution may become slight turbid or may have a small amount of visible particulate matter.

KamRAB ™:
- Store vials between 2-8°C. Do not freeze.
- Vials are stable at room temp up to 25°C for 1 month; once stored at room temp, vials should not be returned to the refrigerator.

MISCELLANEOUS

- Emergency drugs (e.g., epinephrine) should be immediately available for the treatment of hypersensitivity reactions.
- The vial containing 300 international units is sufficient for a patient weighing 15 kg, that of 900 international units for a patient weighing 45 kg, and that of 1500 international units for a patient weighing 75 kg.
- May contain immunoglobulin A (IgA); use with caution in patients with IgA deficiency due to increased risk of anaphylaxis.
- Live vaccines should not be administered during the following period after the administration of:
  - HyperRAB ®: 3 months;
  - Imogam ®: 4 months;
  - KamRAB ™: 4 months for the measles vaccine, and 3 months for all other live vaccines.

REFERENCES

1, 5, 31, 135.

New monograph 2021

NAME OF MEDICATION
RABIES VACCINE

OTHER NAMES
Imovax ® Rabies, RabAvert ®

CLASSIFICATION
Vaccine

INDICATIONS

- Pre-exposure and post-exposure vaccinations against rabies in all age groups.

ADMINISTRATION

- Reconstitute vaccine with 1 mL of the diluent provided (SWFI). Mix gently to avoid foaming. RabAvert ® vaccine is prepared under negative pressure; after reconstitution, disconnect syringe from the needle to allow vacuum to exhaust. Do NOT create positive pressure (e.g., by injecting air into the vial) as this may interfere with withdrawal of the reconstituted solution.
- IM: into the deltoid muscle (never the gluteal area); for small children and infants, administer into the anterolateral upper thigh.
- Do NOT administer the rabies vaccine in the same syringe or at the same site as the rabies immune globulin.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, angioedema, anaphylaxis.
- GI: nausea, abdominal pain.
- CNS: headache, dizziness.
- Myalgia, malaise.
- Lymphadenopathy.
- Local reactions: erythema, swelling, induration, pain, itching.

DOSAGE

- Pre-exposure vaccination: a) primary immunization: a total of 3 doses given as 1 mL IM on days 0, 7 and 21 (or anytime between days 21 to 28). b) booster immunization: if high risk of exposure continues, 1 mL IM every 2-5 years depending on serological test.
- Post-exposure vaccination: a) previously unvaccinated individuals: a total of 5 doses given as 1 mL IM on days 0, 3, 7, 14 and 28; to begin as soon as possible after exposure; to administer after and at a distant site from the rabies immune globulin. b) previously immunized individuals: a total of 2 doses given as 1 mL IM on days 0 and 3; rabies immune globulin should not be given in this case.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 2-8°C. Protect from light. Imovax ® Rabies: do not use if frozen.
- The reconstituted vaccine should be used immediately.
- Reconstituted Imovax ® Rabies is clear or slightly opalescent, red to purplish red suspension.
- Reconstituted RabAvert ® is a clear to slightly opalescent, colourless to slightly pink solution.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The rabies vaccine is an inactivated vaccine.
- Imovax ® Rabies: contains trace amounts of neomycin.
- RabAvert ®: contains trace amounts of neomycin, chlortetracycline, amphotericin B, chicken proteins and bovine gelatin.

REFERENCES

1, 5, 31, 135.

Full revision 2018; limited revision 2020

NAME OF MEDICATION
RALTITREXED

OTHER NAMES
Tomudex ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of advanced colorectal cancer.

ADMINISTRATION

- Reconstitute each 2 mg vial with 4 mL SWFI to obtain a 0.5 mg/mL solution.
- Intermittent IV infusion: dilute in 50-250 mL of NS or D5W; infuse over 15 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- GI: nausea, vomiting, diarrhea.
- Hematologic: leukopenia.
- Hepatic: increased LFTs.
- Flu-like symptoms: fever, chills, cramps, arthralgia, myalgia.
- Dermatologic: rash.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

Antidote: leucovorin 25 mg/m^2 IV q6h until resolution of symptoms.

DOSAGE

- 3 mg/m^2 IV repeated every 3 weeks.
- Dosage in renal impairment:

CrCl (mL/min):	Greater than 65	65-55	54-25	less than 25
Dose:	100%	75%	% equivalent to CrCl in mL/min (e.g., if 30 mL/min, give 30% of the dose)	discontinue
Interval:	3 weeks	4 weeks	4 weeks	---

- Dosage in hepatic impairment: 100% of the dose if WHO grade 1 or 2 liver impairment; use with extreme caution if WHO grade 3 (no data) and do not use if grade 4 liver impairment.
- Dosage adjustment due to toxicity: reductions are recommended for subsequent treatment based on toxicity observed on previous treatment:
  - WHO Grade 3 hematological toxicity or WHO Grade 2 GI toxicity: withhold until complete recovery then continue at a dose reduced by 25%.
  - WHO Grade 4 hematological toxicity or WHO Grade 3 GI toxicity: withhold until complete recovery then continue at a dose reduced by 50%.
  - WHO Grade 4 GI toxicity: discontinue treatment.
  - WHO Grade 4 hematological toxicity with WHO Grade 3 GI toxicity: discontinue treatment.
  - Once a dose reduction has been made, all subsequent doses should be given at the reduced dose level.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-25ºC. Protect from light.
- Reconstituted solution is stable for 24 hours at room temp but it is recommended to store it in the fridge to avoid bacterial contamination.
- Diluted solution is stable for 24 hours at room temp.

MISCELLANEOUS

- Concomitant folic acid, leucovorin (folinic acid) or vitamin preparations containing these agents may reduce the efficacy of raltitrexed; avoid immediately before or during raltitrexed administration.

REFERENCES

5, 129, 165.

NAME OF MEDICATION	
RAMUCIRUMAB	
OTHER NAMES Cyramza ®	CLASSIFICATION Antineoplastic, Monoclonal antibody

INDICATIONS

- As monotherapy or in combination with paclitaxel for the treatment of advanced or metastatic gastric cancer or gastro-esophageal junction adenocarcinoma with disease progression on or after prior platinum and fluoropyrimidine chemotherapy.
- In combination with irinotecan, leucovorin and fluorouracil for the treatment of metastatic colorectal cancer with disease progression on or after prior therapy with bevacizumab, oxaliplatin and a fluoropyrimidine.
- In combination with docetaxel for the treatment of metastatic non-small cell lung cancer (NSCLC) with disease progression on or after platinum-based chemotherapy.

ADMINISTRATION

- **Ensure premedication has been administered as recommended; refer to Dosage section.**
- Intermittent IV infusion (mandatory): dilute total dose in NS to obtain a final volume of 250 mL using one of the following 2 methods:
  1) withdraw from a 250 mL NS bag a volume equal to the drug volume required for the patient's dose, then add the patient's dose (from the 10 mg/mL vials) to the NS bag; OR
  2) transfer the required amount of drug volume to an empty IV container and add NS to obtain a final volume of 250 mL.

  Gently invert container to mix; do NOT shake. Administer over approximatively 60 minutes at a maximum rate of 25 mg/min with an in-line protein-sparing 0.2-0.22 micron filter. Flush line with NS after administration. When given on the same day as other agents for chemotherapy, infuse ramucirumab first.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus.
- Infusion-related reactions: chills, rigors, tremor, back pain/spasms, chest pain/tightness, dyspnea, wheezing, hypoxia, bronchospasm, paresthesia, flushing, supraventricular tachycardia, hypotension; usually with first or second infusion. If mild to moderate reaction, treat with dexamethasone and reduce ramucirumab infusion rate by 50% (as well as for subsequent infusions). If reaction is severe, discontinue treatment.
- Cardiovascular: hypertension: can be severe, in such case temporarily withhold product until BP is controlled, otherwise discontinue treatment; peripheral edema, CHF, arterial thromboembolism (including myocardial infarction, cardiac arrest, cerebrovascular accident/ischemia).
- GI: diarrhea, stomatitis, abdominal pain, GI hemorrhage, GI perforation (rare), fistula formation, intestinal obstruction.
- CNS: headache, reversible posterior leukoencephalopathy syndrome (rare).
- Hematologic: neutropenia, leukopenia, thrombocytopenia.
- Hepatic: deterioration of patients with cirrhosis (e.g., encephalopathy, ascites, hepatorenal syndrome).
- Renal: proteinuria, nephrotic syndrome. Consult manufacturer's recommendations as may require dose adjustment or treatment discontinuation.
- Impairment of wound healing, hypothyroidism, fatigue, asthenia, epistaxis, hypoalbuminemia, hyponatremia.

DOSAGE

- **Premedication with IV diphenhydramine prior to each ramucirumab administration. If previous mild to moderate infusion-related reaction, also premedicate with dexamethasone (or equivalent) and acetaminophen.**
- Gastric cancer: monotherapy: 8 mg/kg IV every 2 weeks; in combination with paclitaxel: 8 mg/kg IV on days 1 and 15 of each 28-day cycle, to administer before paclitaxel infusion.
- Colorectal cancer: 8 mg/kg IV every 2 weeks.
- NSCLC: 10 mg/kg IV on day 1 of each 21-day cycle.

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DOSAGE (Cont.)

- Dosage in renal impairment: no adjustment required when CrCl is 15 mL/min or greater; no data available below 15 mL/min.
- Dosage in hepatic impairment: no adjustment in patients with mild or moderate impairment (abnormal AST or total bilirubin up to 3 x ULN); no data in case of severe impairment.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze. Do not shake. Solution in vials should be clear to slightly opalescent and colourless to slightly yellow.
- Diluted solution is stable for 24 hours in the fridge and 4 hours at room temp in NS. Do not freeze. Do not shake.
- Compatible with NS; do not mix with other IV solutions.
- Incompatible with D5W.

MISCELLANEOUS

- Medications for the treatment of infusion-related reactions (e.g., acetaminophen, antihistamines, corticosteroids) should be available for immediate use in the event of a reaction.
- Monitor for proteinuria at baseline and before each cycle, and thyroid function at baseline and every 2-3 cycles.
- As the product interferes with the wound healing process, withhold ramucirumab prior to scheduled surgery until the wound is fully healed. Discontinue if wound healing complications occur.

REFERENCES

1, 5, 129, 135, 165, 208.

NAME OF MEDICATION
RASBURICASE

OTHER NAMES
Fasturtec ®, Recombinant urate-oxidase enzyme

CLASSIFICATION
Enzyme, Antihyperuricemic

INDICATIONS

- Prophylaxis and treatment of hyperuricemia in cancer patients.

ADMINISTRATION

- Reconstitute vials of 1.5 mg and 7.5 mg with 1 mL and 5 mL, respectively, of the provided diluent to obtain a concentration of 1.5 mg/mL. Gently swirl to mix. Do NOT shake.
- Intermittent IV infusion (mandatory): dilute further in 50 mL of NS; infuse over 30 minutes.
- Do NOT use a filter for infusion.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, severe allergic reactions (bronchospasm, hypotension, urticaria), including anaphylaxis.
- GI: nausea, vomiting, diarrhea, constipation, mucositis, abdominal pain.
- CNS: headache.
- Hematologic: hemolysis, methemoglobinemia (rare).
- Fever.

DOSAGE

- Weight-based dosing (adults and pediatrics over the age of 1 month): 0.2 mg/kg IV once daily for up to 7 days.
- Fixed-dosing (adults only): 3 mg, 4.5 mg, 6 mg or 7.5 mg IV once; to repeat q24h once or twice if needed.
- Chemotherapy may be initiated as soon as 4 hours after the first dose.
- Dosing exceeding 7 days or greater than one course of treatment is not recommended.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Do not freeze. Protect from light.
- Reconstituted solution and solution further diluted in NS are stable for 24 hours in the fridge.
- Do not mix with any other solutions.

MISCELLANEOUS

- When collecting blood for uric acid analysis, use a pre-chilled tube containing heparin anticoagulant. Immerse immediately in an ice water bath. A pre-cooled centrifuge (4ºC) should be used to prepare the plasma samples; the samples should be maintained in the ice water bath and analysed for uric acid within 4 hours. This is to avoid rapid breakdown of uric acid in blood sample and falsely lowered level results.
- Contraindicated in patients with Glucose-6-phosphate dehydrogenase (G6PD) deficiency.
- Standard medical practice for the management of plasma uric acid in patients at risk for tumour lysis syndrome should be followed.

REFERENCES

1, 4, 5, 40, 95, 377, 378.

Full revision 2018

NAME OF MEDICATION
REMDESIVIR

OTHER NAMES
Veklury ®

CLASSIFICATION
Antiviral

INDICATIONS

- Treatment of coronavirus disease 2019 (COVID-19; SARS-CoV-2 infection) in moderately ill adults and children (12 years of age and older weighing at least 40 kg) with pneumonia requiring supplemental oxygen (not recommended in critically ill patients receiving mechanical ventilation). Its use is confined to healthcare facilities in which patients can be monitored closely.

ADMINISTRATION

- Vial with lyophilized powder: reconstitute each 100 mg vial with 19 mL of SWFI to obtain a 5 mg/mL solution. Shake the vial for 30 seconds; allow the contents to settle for 2-3 minutes. Repeat process (shaking and settling) as necessary until vial contents are completely dissolved. From a 250 mL bag of NS (or a 100 mL bag for patients with severe fluid restriction), withdraw a volume of solution equal to the volume of remdesivir solution that will be added to the bag (20 mL for a 100 mg dose; 40 mL for a 200 mg dose); discard the withdrawn NS solution. Add the remdesivir dose to the bag; discard any unused portion remaining in the remdesivir vial. Invert bag gently 20 times to mix the solution; do NOT shake.
- Vial with solution concentrate (5 mg/mL): allow the vials to reach room temp (20-25°C). From a 250 mL bag of NS, withdraw a volume of solution equal to the volume of remdesivir solution that will be added to the bag (20 mL for a 100 mg dose; 40 mL for a 200 mg dose); discard the withdrawn NS solution. Add the remdesivir dose to the bag; discard any unused portion remaining in the remdesivir vial. Invert bag gently 20 times to mix the solution; do NOT shake.
- Intermittent IV infusion: infuse over 30-120 minutes. Flush the line with at least 30 mL of NS after administration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: includes infusion-related reactions and anaphylaxis: hypotension, hypertension, tachycardia, bradycardia, hypoxia, fever, dyspnea, wheezing, angioedema, rash, nausea, vomiting, diaphoresis, shivering. Slower infusion times (up to a maximum of 120 min) may prevent the occurrence of these reactions; in case of reaction, discontinue administration and give appropriate treatment.
- GI: nausea.
- CNS: headache.
- Dermatology: rash.
- Endocrine and metabolic: hyperglycemia.
- Hepatic: increased transaminases.
- Renal: increased creatinine, acute renal failure.

DOSAGE

- Loading dose: 200 mg IV on day 1; maintenance dose: 100 mg IV once daily, starting on day 2. Treatment duration of at least 5 days, maximum of 10 days.
- Dosage in renal impairment: if CrCl is 30 mL/min or greater, no dosage adjustment is necessary; if CrCl is less than 30 mL/min, do not administer as the excipient betadex sulfobutyl ether sodium (also called cyclodextrin) may accumulate.
- Dosage in hepatic impairment: not studied in patients with liver impairment. Do not administer if ALT is 5 x ULN or greater (may resume when ALT is less than 5 x ULN), or if ALT is increased and is accompanied with signs or symptoms of liver inflammation or increased conjugated bilirubin, alkaline phosphatase or INR.

…/Cont.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Vial with lyophilized powder:
- Store vials below 30°C.
- Reconstituted solution should be clear.
- Reconstituted and diluted solutions are stable for 4 hours at up to 25°C and 24 hours between 2-8°C.

Vial with solution concentrate:
- Store vials between 2-8°C; can be stored up to 12 hours at room temp prior to dilution.
- Diluted solution is stable for 4 hours at up to 25°C and 24 hours between 2-8°C.

MISCELLANEOUS

- Monitor renal and hepatic functions prior to and during treatment.
- Do not administer with chloroquine or hydroxychloroquine as these drugs may antagonize the antiviral activity of remdesivir.

REFERENCES

5, 95, 135, 506.

New monograph 2020; limited revision 2021

NAME OF MEDICATION
REMIFENTANIL

OTHER NAMES
Ultiva ®

CLASSIFICATION
Opioid analgesic, Anesthesia adjunct

INDICATIONS

- As an analgesic during the induction and maintenance of general anesthesia.

ADMINISTRATION

- Reconstitute vial with 1 mL of a compatible diluent per 1 mg of remifentanil to obtain a concentration of 1 mg/mL. Shake well.
- IV direct: physician only; dilute to a final concentration of 20, 25, 50 or 250 mcg/mL; inject over 30-60 seconds.
- Continuous IV infusion: dilute to a final concentration of 20, 25, 50 or 250 mcg/mL; infusion rate as per Dosage section. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Cardiovascular: hypotension, bradycardia.
- GI: nausea, vomiting.
- CNS: headache.
- Dermatologic: pruritus.
- Respiratory: respiratory depression, apnea.
- Muscle rigidity.

Antidote: naloxone (for respiratory depression)

DOSAGE

- Induction of anesthesia: 0.5-1 mcg/kg/min IV. If patient is to be intubated within 8 minutes of the start of remifentanil infusion, an initial dose of 1 mcg/kg IV may be given over 30-60 seconds.
- Maintenance of anesthesia when combined with other agents: 0.05-2 mcg/kg/min IV. Infusion rate may be titrated upward in 25% to 100% increments or downward in 25% to 50% decrements every 2-5 minutes. Supplemental IV boluses of 0.5-1 mcg/kg may be given every 2-5 minutes if needed. For infusion rates greater than 1 mcg/kg/min, consider increasing doses of concomitant anesthetics to increase the depth of anesthesia.
- Dosage in renal impairment: no dosage adjustment is necessary.
- Dosage in hepatic impairment: no dosage adjustment is necessary.
- Dosage in obesity: dose should be calculated using ideal body weight.
- Dosage in geriatric patients: reduce starting dose by 50% then titrate to effect.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp or in the fridge.
- Reconstituted solution is stable for 24 hours at room temp. Compatible with SWFI.
- Stable for 24 hours at room temp in D5W, NS, 1/2NS, D5-NS and D5-RL at a concentration of 20-250 mcg/mL.
- Stable for 4 hours at room temp in RL at a concentration of 20-250 mcg/mL.

MISCELLANEOUS

- Analgesic effect stops within 5 to 10 minutes of discontinuation of remifentanil but respiratory depression can continue for up to 30 minutes due to residual effects of concomitant anesthetics.
- Risk of intraoperative awareness when used with propofol at infusion rates of less than 75 mcg/kg/min in patients under 55 years of age.

REFERENCES

4, 5, 40, 95.

Full revision 2018; limited revision 2020

NAME OF MEDICATION
RH_o(D) IMMUNE GLOBULIN (HUMAN)

OTHER NAMES
WinRho ® SDF

CLASSIFICATION
Passive immunizing agent

INDICATIONS

- To prevent the formation of antibodies to Rh_o(D) factor in unsensitized Rh_o(D)-negative women with an Rh-incompatible pregnancy: 1) fetus/baby is Rh_o(D) positive or unknown or 2) father is RH_o(D) positive or unknown.
- To prevent alloimmunization in Rh_o(D)-negative female child or female of childbearing age transfused with Rh_o(D)-positive red blood cells or blood components with Rh_o(D) red blood cells.
- To increase the platelet count and to control bleeding in Rh_o(D)-positive patients with immune thrombocytopenic purpura (ITP). DO NOT administer the product in ITP patients: 1) with ITP secondary to other conditions including leukemia, lymphoma, or active viral infections with EBV (Epstein-Barr virus) or HCV (hepatitis C); 2) who are elderly with co-morbidities predisposing them to complications of acute hemolytic reaction (AHR); 3) with evidence of autoimmune hemolytic anemia (Evan's Syndrome), systemic lupus erythematosus (SLE) or antiphospholipid antibody syndrome (APS); 4) who are IgA deficient; 5) who have antibodies to IgA or a history of IgA hypersensitivity; 6) who are Rh_o(D) negative; 7) who are splenectomised.

ADMINISTRATION

- Allow vials to reach body or room temp prior to use.
- IV direct: physician or RN; undiluted; inject at a rate of 300 mcg (1500 international units) per 5-15 seconds. Can also dilute dose to 10 mL with NS and administer at the slower rate of 300 mcg (1500 international units) per minute.
- IM: into the deltoid, anterolateral aspect of the upper thigh; gluteral region NOT preferred.
- When used for the prevention of Rh immunization, observe patient for at least 20 minutes for the development of any adverse events.
- When used for ITP treatment, observe patient in a healthcare setting for a period of 8 hours for signs and symptoms of intravascular hemolysis. Urine dipstick testing for blood should be conducted before dosing and at 2, 4 and 8 hours after receiving the dose.
- Consult Corporate nursing standard operating procedure C-SOP NUR 0045 (Blood products and blood derivatives: Administration of) for more information.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, anaphylactoid reaction.
- CNS: headache.
- Fever, chills.
- Risk of transmission of infectious agents, including viruses, and theoretically the Creutzfeldt-Jakob disease agent, cannot be completely eliminated as product is made from pooled human plasma.
- Local reactions: discomfort, swelling, induration, pruritus.

DOSAGE

Pregnancy and obstetric conditions:
- 300 mcg (1500 international units) IV/IM at 28-30 weeks gestation predelivery.
- 120 mcg (600 international units) IV/IM given within 72 hours of delivery (only if baby is Rh_o(D) positive or unknown). If more than 72 hours has elapsed, dose should not be held; give as soon as possible, up to 28 days after delivery.
- 300 mcg (1500 international units) IV/IM within 72 hours of invasive procedures during pregnancy (e.g., amniocentesis or chorionic villus sampling); repeat every 12 weeks during pregnancy.
- 300 mcg (1500 international units) IV/IM within 72 hours of obstetric complications (e.g., miscarriage, abortion, threatened abortion, ectopic pregnancy or hydatidiform mole, transplacental hemorrhage).

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DOSAGE (Cont.)

Transfusion: to administer within 72 hours after exposure of incompatible blood transfusion.
- IV administration: 9 mcg (45 international units)/mL of whole blood or 18 mcg (90 international units)/mL of red blood cells as per following schedule: administer 600 mcg (3000 international units) IV q8h until total dose is given.
- IM administration: 12 mcg (60 international units)/mL of blood or 24 mcg (120 international units)/mL of red blood cells as per following schedule: administer 1200 mcg (6000 international units) IM q12h until total dose is given.

ITP (IV treatment only):
- Initial dose: 50 mcg/kg (250 international units/kg) IV. If patient has a hemoglobin level between 80-100 g/L, use a dose of 25-40 mcg/kg (125-200 international units/kg). Initial dose may be divided in two and administered on separate days.
- Subsequent dose: 25-60 mcg/kg (125-300 international units/kg) IV. Frequency of dosing based on clinical response.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Do not freeze.
- Do not mix with other drugs.
- Compatible with NS.
- Incompatible with D5W.

MISCELLANEOUS

- Conversion: 1 mcg = 5 international units.
- For prophylaxis of Rh immunization, do not administer to: 1) the infant; 2) Rh_o(D)-positive individual; 3) Rh_o(D)-negative females who are Rh-immunized; 4) IgA-deficient patients; 5) patient with antibodies to IgA or history of IgA hypersensititivity.
- ITP: platelet count usually increases within 1-2 days with peak effect at 7 to 14 days. Response usually lasts for 30 days.
- Contains maltose; may give false elevated glucose readings with glucometers using glucose dehydrogenase pyrroloquinoline-quinone (GDH-PQQ) or glucose-dye-oxidoreductase methods for testing.

REFERENCES

1, 2, 5, 40, 95, 135, 136.

NAME OF MEDICATION
RifAMPin *

OTHER NAMES
Rifadin ®, Rifampicin, Rimactane ®

CLASSIFICATION
Antibiotic - antitubercular

INDICATIONS

- Treatment of tuberculosis when oral administration is not possible.
- Treatment of leprosy, brucellosis, *Mycobacterium avium* complex and staphylococcal infections.
- Treatment of asymptomatic carriers of *Neisseria meningitidis* when oral administration is not possible.

ADMINISTRATION

- Reconstitute each 600 mg vial with 10 mL of SWFI. Swirl gently until dissolved.
- Intermittent IV infusion (mandatory): dilute in 250 mL of D5W or in 500 mL of D5W or NS and infuse over 1.5-3 hours (preferred). In patients with volume restriction, may dilute in 100 mL of D5W or NS; infuse over 30 minutes immediately after mixing as no stability data is available at this concentration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: flu-like syndrome, pruritus, urticaria, rash, Stevens-Johnson syndrome, toxic epidermal necrolysis, anaphylaxis (rare).
- GI: nausea, vomiting, diarrhea, abdominal cramps.
- Endocrine and metabolic: elevations of serum urea and uric acid.
- Hematologic: thrombocytopenia, purpura, hemolytic anemia, leukopenia.
- Hepatic: increased LFTs.
- Body fluids and excrements may be coloured red-orange. May permanently stain soft contact lenses.
- Local reactions: irritation and inflammation if extravasation.

DOSAGE

Dose depends on type of infection. Refer to specialized infectious diseases references. Oral and intravenous doses are the same. Switch to oral therapy as soon as possible.

Adults:
- Usual dose: 600 mg IV daily given in one or divided doses.
- Severe acute infections: 900-1200 mg IV daily. Higher doses are usually divided in 2-3 doses.

Pediatrics:
- 10-20 mg/kg/day IV in 1-2 divided doses, up to a maximum of 600 mg daily.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Reconstituted solution is stable for 24 hours at room temp.
- Stable for 24 hours at room temp in 250 or 500 mL of D5W at concentrations of 2.4 mg/mL or 1.2 mg/mL respectively, and in 500 mL of NS at a concentration of 1.2 mg/mL. Note that this stability is specific to the product from France; recommendations for products from other countries may be different.

MISCELLANEOUS

- Resistance to rifampin monotherapy develops rapidly; therefore, it should be used in combination with other agents for most infections.

REFERENCES

1, 4, 5, 6, 40, 95, 135, 458.

* Available via Health Canada's Special Access Programme

Full revision 2018; limited revision 2019, 2020, 2021

NAME OF MEDICATION
RisperiDONE

OTHER NAMES
Risperdal Consta ®

CLASSIFICATION
Antipsychotic

INDICATIONS

- Treatment of schizophrenia and related psychotic disorders.

ADMINISTRATION

- Prepare the dose immediately prior to administration as described in the manufacturer's directions. Allow kit to reach room temp for at least 30 minutes prior to reconstitution. Use only the diluent, syringe and needles provided in the kit for reconstitution and administration.
- Immediately prior to injection, resuspend by vigorous shaking of the syringe.
- IM: in the gluteal muscle using the 2-inch needle, alternating between the 2 buttocks OR in the deltoid muscle using the 1-inch needle, alternating between the 2 arms. The two sites are bioequivalent and therefore interchangeable.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rare; rash, hives, anaphylaxis, angioedema, Stevens-Johnson syndrome, toxic epidermal necrolysis, and drug reaction with eosinophilia and systemic symptoms (DRESS).
- Cardiovascular: hypotension.
- GI: dyspepsia, constipation, dry mouth.
- CNS: somnolence, headache, dizziness, extrapyramidal symptoms.
- Fatigue, weight gain, pain in the extremities.
- Local reactions: redness, swelling, induration.

DOSAGE

- 25 to 50 mg IM every 2 weeks.
- Initial dose may be reduced to 12.5 mg IM in patients with hepatic or renal impairment, or to minimize drug interactions.
- Before starting IM risperidone, it is recommended to initiate therapy with oral risperidone to assess tolerability.
- Oral risperidone should be continued for the first 3 weeks following the first injection of IM risperidone which will then take effect.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store kit in the fridge. Protect from light.
- Kit can be stored for no more than 7 days at room temp not exceeding 25°C.
- Reconstituted dose is stable for 6 hours at temperatures below 25°C.

MISCELLANEOUS

- Extended-release suspension.

REFERENCES

1, 5, 95.

Full revision 2018; limited revision 2021

NAME OF MEDICATION	
RiTUXimab	
OTHER NAMES Rituxan ®, Riximyo ®, Ruxience ™, Truxima ™	CLASSIFICATION Antineoplastic, Immunomodulator, Monoclonal antibody

INDICATIONS

- Treatment of low-grade or follicular, CD20 positive, B-cell non-Hodgkin's lymphoma (NHL).
- Treatment of CD20 positive, diffuse large B-cell NHL.
- Treatment of patients with previously untreated or previously treated B-cell chronic lymphocytic leukemia (CLL), Binet Stage B or C, in combination with fludarabine and cyclophosphamide.
- In combination with methotrexate to reduce signs and symptoms in adult patients with moderately to severely active rheumatoid arthritis who have had an inadequate response or intolerance to one or more tumour necrosis factor inhibitor therapies.
- In combination with corticosteroids for the induction of remission in adult patients with severely active anti-neutrophil cytoplasmic antibody (ANCA)-associated vasculitis, including, granulomatosis with polyangiitis (GPA, also known as Wegener's granulomatosis) and microscopic polyangiitis (MPA).

ADMINISTRATION

- **Ensure premedication has been administered prior to each IV infusion or SC injection; refer to Potential Administration Hazards and Dosage sections.**
- **Ensure that the appropriate formulation (IV or SC for Rituxan ® and IV only for Riximyo ®, Ruxience ™ and Truxima ™) will be administered to the patient as prescribed**.
- **Consider withholding antihypertensive agents for 12 hours prior to and during rituximab administration as rituximab may cause transient hypotension.**
- Intermittent IV infusion (mandatory): dilute dose in NS or D5W to a final concentration of 1 to 4 mg/mL. To avoid foaming, invert the bag gently to mix solution.
- At TOH, follow specific infusion rates from physician's medication orders.
- First infusion: start IV infusion at 50 mg/hr. If no hypersensitivity or infusion-related events occur then increase rate in 50 mg/hr increments every 30 minutes, to a maximum of 400 mg/hr. If reactions occur, stop or slow down the infusion temporarily and notify physician. Once patient improves, restart at half the previous infusion rate.
- Subsequent Infusions: If patient tolerated the first infusion without adverse effects, start IV infusion at 100 mg/hr. Increase rate in 100 mg/hr increments every 30 minutes, to a maximum of 400 mg/hr as tolerated.
- Subsequent infusions – Alternative for rheumatoid patients only: if patient did not experience a serious reaction from the previous infusion and has no significant cardiovascular disease, 250 mL of a 4 mg/mL solution can be administered as follows: 125 mg IV in the first 30 minutes (62.5 mL/hr) and 875 mg IV over the next 90 minutes (150 mL/hr), for a total infusion time of 120 minutes. If well tolerated, this method can be used for the subsequent infusions.
- Subsequent infusions – Alternative for NHL only: In patients who had no reactions to the first dose, with steroid-containing chemotherapy and have no significant cardiovascular disease, more rapid infusions have been used safely as follows: mix dose in 250 mL NS and administer over a total infusion time of 90 minutes, giving 20% of the dose in the first 30 minutes then the remaining 80% over 60 minutes.
- For patients with NHL or CLL: infusion rate may be decreased or dose split over 2 days for any cycle in patients with bulky disease or over 25 x 10^9 circulating malignant cells.
- SC injection (using the SC formulation; Rituxan ® brand only): for non-Hodgkin's lymphoma and CLL, for second or subsequent cycles if patient was able to receive the full rituximab IV infusion dose during first cycle. Inject SC into the abdominal wall over approximately 5 minutes for a 1400 mg dose (NHL) and over approximately 7 minutes for a 1600 mg dose (CLL). Rotate injection sites. Observe patient for at least 15 minutes (or longer in patients with an increased risk of hypersensitivity reactions) following SC injection.
- Consult specific protocol.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis; usually with the second or subsequent doses.

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POTENTIAL ADMINISTRATION HAZARDS (Cont.)

- Infusion-related reactions: occur in most patients during the first dose of rituximab and tend to decrease with subsequent infusions. Caution, as some of these reactions have been fatal. Reactions generally occur within 30 minutes to 2 hours but could be delayed until 24 hours after the start of infusion and include: fever, rigors, chills, urticaria, rash, pruritus, angioedema, hypotension, flushing. Severe respiratory reactions can occur within 1-2 hours of starting the first infusion and include: dyspnea, bronchospasm, hypoxia, pulmonary infiltrates or edema, acute respiratory failure. If any severe reaction occurs, the infusion should be stopped and a physician notified immediately. **Premedication** (acetaminophen, diphenhydramine, corticosteroids) may reduce frequency and severity of infusion-related reactions; refer to Dosage section.
- Cardiovascular: hypertension, hypotension, arrhythmia.
- GI: nausea, vomiting, constipation, diarrhea, abdominal pain. Complaints of abdominal pain require investigation.
- Respiratory: rhinitis, cough, wheezing, pneumonitis. Refer also to infusion-related reactions.
- Tumour lysis syndrome (hyperuricemia, hyperkalemia, hypocalcemia, hyperphosphatemia, acute renal failure, elevated lactate dehydrogenase, fever) occurs rarely and usually within 1-2 hours after the first dose, but may be delayed until 12-24 hours after the infusion.
- Headache, myalgia, arthralgia, pain at disease site, fatigue, flu-like symptoms.
- Increase in serum IgM which can lead to hyperviscosity of blood in patients with Waldenstrom's macroglobulinemia.
- Progressive multifocal leukoencephalopathy (PML) (rare).
- Possible reactivation of tuberculosis or hepatitis B virus in seropositive patients.
- Dermatologic: rash, rare but severe mucocutaneous reactions including Stevens-Johnson syndrome, lichenoid dermatitis, vesiculobullous dermatitis, toxic epidermal necrolysis and paraneoplastic pemphigus.
- Local reactions (with SC administration): pain, swelling, induration, hemorrhage, erythema, pruritus, rash at injection site. Can occur more than 24 hours after SC administration. Usually resolve within 1-2 days without treatment.

DOSAGE

- **Premedication with acetaminophen and diphenhydramine is mandatory. In rheumatoid arthritis patients, methylprednisolone 100 mg IV 30 minutes before the administration of rituximab is also recommended to decrease rate and severity of acute infusion reactions. In lymphoma and leukemic patients, premedication with corticosteroids should also be considered, particularly if rituximab is not given in combination with steroid-containing chemotherapy.**

<u>Non-Hodgkin's lymphoma</u>:
- Low grade or follicular NHL:
  - Induction treatment: as a single agent: 375 mg/m^2 IV infusion once weekly for 4 doses. In combination with CVP (cyclosphosphamide, vincristine, prednisolone) or CHOP (cyclophosphamide, doxorubicin, vincristine, prednisone) chemotherapy: 375 mg/m^2 IV infusion every 3 weeks for 6-8 cycles administered as an IV infusion on day 1 of each chemotherapy cycle after IV administration of the corticosteroid component of CVP or CHOP. If patient was able to receive the first full IV infusion dose, may use for subsequent cycles 1400 mg SC injection on day 1 of each cycle after administration of the corticosteroid component of the chemotherapy for a total of 8 cycles, including the IV infusion cycle.
  - Maintenance treatment: in previously untreated patients with advanced high-tumour burden follicular lymphoma after complete or partial response to induction treatment: 375 mg/m^2 IV infusion every 8 weeks for a maximum of 12 doses or 1400 mg SC injection once every 2 months until disease progression or for a maximum period of 2 years. For relapsed or refractory patients after response to induction treatment: 375 mg/m^2 IV infusion every 3 months or 1400 mg SC injection once every 3 months until disease progression or for a maximum period of 2 years.

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DOSAGE (Cont.)

- Diffuse large B-cell NHL: in combination with CHOP (cyclophosphamide, doxorubicin, vincristine, prednisone): 375 mg/m^2 IV infusion every 3 weeks for 6-8 cycles, administered on day 1 of each chemotherapy cycle after IV administration of the corticosteroid component of CHOP. The other components of CHOP should be given after rituximab. If patient tolerated the IV infusion on day 1 of first cycle, may give 1400 mg SC injection on day 1 of each chemotherapy cycle, after administration of the corticosteroid component of the chemotherapy if applicable. Total of 8 cycles.

CLL in combination with fludarabine and cyclophosphamide:

- Cycle 1: 375 mg/m^2 IV infusion on day 1.
- Cycles 2-6: 500 mg/m^2 IV infusion on day 1 OR if patient tolerated the IV infusion on day 1 of first cycle, may give 1600 mg SC on day 1 of each chemotherapy cycle after administration of the corticosteroid component of the chemotherapy, if applicable.

Rheumatoid arthritis:

- Usual dose: 1000 mg IV infusion on days 1 and 15, for a total of 2 doses.
- Retreatment: not well established; patients may receive other courses of treatment (2 infusions per course), with an interval of 16-24 weeks between courses.

ANCA-associated vasculitis (GPA and MPA):

- 375 mg/m^2 IV infusion once a week for 4 weeks in combination with methylprednisolone 1000 mg IV for 1-3 days followed by daily oral prednisone 1 mg/kg/day (maximum of 80 mg/day).

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Rituxan ® IV formulation:

- Store vials in the fridge. Protect from light.
- Stable for 24 hours in the fridge and for an additional 24 hours at room temp in D5W at concentrations of 1-4 mg/mL in PVC and polyethylene bags.
- Stable for 30 days in the fridge and for an additional 24 hours at room temp (30°C or lower) in NS at concentrations of 1-4 mg/mL in PVC and polyethylene bags.

Rituxan ® SC formulation:

- Store vials in the fridge. Protect from light. Do not freeze.
- Stable for 48 hours in the fridge and subsequently 8 hours at 30°C in diffused daylight once SC solution is transferred from the vial to the syringe.

Riximyo ® IV formulation:

- Store vials between 2-8°C. Protect from light. Do not shake. Do not freeze.
- Stable for 30 days in the fridge and for an additional 24 hours at room temp (up to 25°C) in NS at concentrations of 1-4 mg/mL in polyethylene and polypropylene bags.
- Stable for 24 hours in the fridge and for an additional 12 hours at room temp (up to 25°C) in D5W at concentrations of 1-4 mg/mL in polyethylene and polypropylene bags.
- Stable for 24 hours in the fridge and for an additional 24 hours at room temp (up to 25°C) in NS at concentrations of 1-4 mg/mL in PVC bags.

Ruxience ™ and Truxima ™ IV formulations:

- Store vials in the fridge. Protect from light.
- Stable for 24 hours in the fridge and for an additional 12 hours for Truxima ™ and 24 hours for Ruxience ™ at room temp in D5W or NS at concentrations of 1-4 mg/mL in PVC and polyethylene bags.

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MISCELLANEOUS

- Emergency drugs (epinephrine, antihistamines, corticosteroids and salbutamol nebules) should be immediately available for the treatment of hypersensitivity reactions with both IV and SC administration.
- Do not administer live viral vaccines while on rituximab; may administer inactivated vaccines but response rate to vaccination may be reduced.
- Riximyo ®, Ruxience ™ and Truxima ™ are classified as being biosimilar to Rituxan ®; therefore these products are not interchangeable.

REFERENCES

1, 4, 5, 40, 95, 129, 135, 143, 165, 567.

NAME OF MEDICATION
ROCURONIUM

OTHER NAMES
Zemuron ®

CLASSIFICATION
Neuromuscular blocker

INDICATIONS

- Adjunct to general anesthesia to facilitate both rapid sequence and routine endotracheal intubation, and to provide skeletal muscle relaxation during surgery or mechanical ventilation.

ADMINISTRATION

- IV direct: physician trained in anesthesiology. RN may administer subsequent doses; ***ventilator support, cardiac monitoring***. Undiluted or diluted in SWFI; given as a rapid injection over 5-15 seconds.
- Continuous IV infusion: ***ventilator support, cardiac monitoring***. Dilute in a compatible solution at concentrations up to 5 mg/mL; refer to Dosage section for administration rate. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: angioedema, urticaria, rash, anaphylactic/anaphylactoid reactions.
- Cardiovascular: arrhythmia, abnormal ECG, tachycardia, hypertension, hypotension.
- Respiratory: prolonged respiratory insufficiency and apnea lasting beyond the required time period.
- Histamine release not clinically significant following usual doses.
- Local reactions: pain, edema.

Antidote: anticholinesterase agents such as neostigmine or edrophonium, in conjunction with an anticholinergic agent such as atropine.

DOSAGE

- Rapid sequence intubation: 0.6-1.2 mg/kg IV.
- Routine endotracheal intubation:
  - Initial dose: 0.6 mg/kg IV (range 0.45-2 mg/kg).
  - Maintenance dose: 0.1-0.2 mg/kg/dose IV (usually required 15-85 minutes after a 0.6 mg/kg dose). Repeat doses based on neuromuscular activity or use a continuous IV infusion of 0.01-0.012 mg/kg/min and titrate dose to desired effect (usual maintenance rate: 0.004-0.016 mg/kg/min).
- Dosage in obesity: dose should be calculated using actual body weight if body mass index is 40 kg/m^2 or less; if body mass index is greater than 40 kg/m^2, may use ideal body weight.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Do not freeze. Unopened vials may be stored for 18 months at room temp (15-30°C).
- Punctured vials must be used within 30 days.
- Stable 24 hours at room temp in D5W, NS, D5-NS, RL and SWFI at concentrations up to 5 mg/mL in plastic bags and glass bottles.
- Incompatible with alkaline solutions.

MISCELLANEOUS

- Effect is dependent on dose.
- After a 0.6 mg/kg dose:
  - Onset: 1 minute (range: 0.4-6 minutes).
  - Duration: 31 minutes (range 15-85 minutes).

REFERENCES

1, 5, 6, 40, 95, 135, 208, 511.

Full revision 2018; limited revision 2020

NAME OF MEDICATION
RomiDEPsin

OTHER NAMES
Istodax ®

CLASSIFICATION
Antineoplastic

INDICATIONS
- Treatment of relapsed/refractory peripheral T-cell lymphoma in patients who are not eligible for transplant and have received at least one prior systemic therapy.

ADMINISTRATION
- Reconstitute each 10 mg vial by injecting slowly 2.2 mL of the supplied diluent (note that each diluent vial contains 2.4 mL i.e., 0.2 mL overfill) into the vial to obtain a solution of 5 mg/mL. Swirl the vial until there are no more visible particles. The reconstituted solution will be slightly viscous.
- Intermittent IV infusion: withdraw from the vial the exact dose needed and inject into a 500 mL bag of NS. Infuse over 4 hours.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rare; rash.
- Cardiovascular: hypotension, tachycardia, edema, QT interval prolongation, venous thromboembolism.
- GI: nausea, vomiting, abdominal pain, diarrhea, constipation, anorexia, dysgeusia, dyspepsia, mucositis.
- CNS: headache, dizziness.
- Electrolyte disturbances: especially hypokalemia and hypomagnesemia.
- Hematologic: anemia, neutropenia (leading to infections), thrombocytopenia, bleeding.
- Hepatic: elevation of liver enzymes, hyperbilirubinemia.
- Respiratory: cough, dyspnea.
- Asthenia, fatigue, pyrexia, musculoskeletal pain.
- Hyperuricemia, tumour lysis syndrome.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- 14 mg/m^2 IV on days 1, 8, 15 of a 28-day cycle.
- Dosage may be reduced to 10 mg/m^2 IV if patient experiences some toxicity; refer to manufacturer's instructions for more details.
- Dosage in renal impairment: no dosage adjustment; caution in patients with end-stage renal disease as no data.
- Dosage in hepatic impairment:

Bilirubin	AST	Starting dose
1.5 x ULN or less	any value	no dosage adjustment
above 1.5 x ULN, up to 3 x ULN	any value	7 mg/m^2 (50% reduction)
above 3 x ULN	any value	not recommended

- Missed dose: administer as soon as possible, except when within 5 days of next scheduled dose.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Stores vials at room temp.
- The reconstituted solution (5 mg/mL) is stable for 8 hours at room temp.
- The diluted solution is stable for 24 hours at room temp in NS when stored in PVC, polyethylene, ethylene vinyl acetate or glass containers.

MISCELLANEOUS
- Monitor potassium and magnesium levels at baseline and before each cycle.
- Avoid concurrent use with strong CYP3A4 inhibitors or inducers; use with caution with other drugs that can prolong QT interval.

REFERENCES
5, 129, 165.

Full revision 2018; limited revision 2019, 2020

NAME OF MEDICATION	
RomiPLOStim	
OTHER NAMES	CLASSIFICATION
Nplate ®	Hematopoietic

INDICATIONS

- To increase platelet counts in adults with chronic immune thrombocytopenic purpura who have not responded adequately to corticosteroids, immunoglobulins, or surgical splenectomy and are at risk of bleeding.

ADMINISTRATION

- Reconstitute vials of 250 mcg and 500 mcg with 0.72 mL and 1.2 mL, respectively, of SWFI to obtain a final concentration of 500 mcg/mL. Gently swirl to mix. Do NOT shake.
- SC.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: angioedema, rash.
- GI: dyspepsia, diarrhea, nausea, abdominal pain.
- CNS: headache, dizziness, insomnia, paresthesia, confusion.
- Hematologic: bruising, epistaxis, bleeding, increased bone marrow reticulin (risk for bone marrow fibrosis), thrombosis (with excessive increase in platelet count). Thrombocytopenia and serious bleeding have occurred upon termination of treatment.
- Respiratory: nasopharyngitis, upper respiratory infection.
- Fatigue, arthralgia, myalgia, pain in shoulder and extremities.
- Antibody production to romiplostim.

DOSAGE

- 1 mcg/kg (actual body weight) SC once weekly. Adjust dose using actual body weight at start of treatment to maintain a platelet count greater or equal to 50 X 10^9/L:
  - Platelet count less than 50 X 10^9/L: increase dose by 1 mcg/kg every 1-2 weeks.
  - Platelet count greater than 200 X 10^9/L for 2 consecutive weeks: decrease dose by 1 mcg/kg every 2 weeks.
  - Platelet count greater than 400 X 10^9/L: withhold therapy. Once platelet count is less than 200 X 10^9/L, resume romiplostim with a dose reduction of 1 mcg/kg.
- Maximum dose: 10 mcg/kg/week.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Protect from light. Do not freeze.
- Reconstituted solution is stable for 24 hours at room temp or in the fridge, protected from light.

MISCELLANEOUS

- Complete blood counts including platelet counts should be monitored: prior to start of therapy, weekly during dose adjustment, monthly after stable platelet count (greater or equal to 50 X 10^9/L) for at least 4 weeks without dose adjustment, and weekly for at least 2 weeks upon discontinuation.
- Peripheral blood smears should be evaluated prior to start therapy then monthly for morphological changes once a stable dose has been established.

REFERENCES

1, 5, 95.

Full revision 2018; limited revision 2019

NAME OF MEDICATION
ROPIVACAINE

OTHER NAMES
Naropin ®

CLASSIFICATION
Local anesthetic

INDICATIONS

- Local or regional anesthesia and analgesia.

ADMINISTRATION

- Local infiltration, major nerve block (e.g., brachial plexus block), epidural block.
- Epidural (for anagesia): continuous infusion and intermittent bolus administration; nurses may not administer epidural boluses unless using an infusion pump and following Unit policy.

POTENTIAL ADMINISTRATION HAZARDS

- Cardiovascular: hypotension, bradycardia, fetal bradycardia.
- GI: nausea, vomiting.
- CNS: headache, paresthesia.
- Backache.
- Pruritus.

DOSAGE

- Surgical anesthesia: major nerve block: 35-50 mL of the 5 mg/mL solution. Local infiltration: 1-40 mL of the 5 mg/mL solution.
- Analgesia: epidural using the 2 mg/mL solution: 6-14 mL/hr as a continuous infusion or 10-15 mL as intermittent boluses. Up to 770 mg ropivacaine administered over 24 hours and up to 28 mg/hr for 72 hours have been well tolerated in adults when used for postoperative pain.
- Refer to manufacturer's monograph for more specific dosing information.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Do not freeze.
- Stable for 30 days at 30°C in the dark in NS at a concentration of 1-2 mg/mL when mixed with fentanyl (1-10 mcg/mL), sufentanil (0.4-4 mcg/mL), morphine (20-100 mcg/mL) or clonidine (5-50 mcg/mL) in polypropylene infusion bags.
- Compatible with NS.
- Incompatible with alkaline solutions.

MISCELLANEOUS

- Amide-type anesthetic.
- Contraindicated for IV regional anesthesia (Bier block) and in obstetric paracervical block anesthesia.
- Do not use for the production of retrobulbar block or spinal anesthesia (subarachnoid block) due to insufficient data to support such use.

REFERENCES

4, 5, 95, 135, 143.

Full revision 2018

NAME OF MEDICATION
SALBUTAMOL

OTHER NAMES
Albuterol, Ventolin ®

CLASSIFICATION
Bronchodilator

INDICATIONS

- Treatment of severe bronchospasm associated with chronic bronchitis and bronchial asthma.
- Treatment of status asthmaticus.

ADMINISTRATION

Adults:
- Continuous IV infusion (mandatory): ***cardiac monitoring***; withdraw 5 mL (5 mg) from the ampoule and dilute in 500 mL of D5W, NS or dextrose-saline to obtain a final concentration of 10 mcg/mL (0.01 mg/mL). Must be administered by an infusion pump.

Pediatrics:
- Intermittent IV infusion (loading dose): ***cardiac monitoring;*** dilute in a convenient volume and infuse over 5 to 10 minutes. Must be administered by an infusion pump.
- Continuous IV infusion (maintenance dose): ***cardiac monitoring;*** dilute to get a final concentration of 200 mcg/mL (0.2 mg/mL). Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, angioedema, oropharyngeal edema, bronchospasm, hypotension, collapse, anaphylaxis.
- Cardiovascular: palpitations, tachycardia.
- CNS: headache.
- Fine muscle tremor, muscle cramps.
- Hypokalemia (can be severe).
- Hyperglycemia in patients with diabetes mellitus; NS admixture may be preferable.

DOSAGE

Adults:
- Continuous IV infusion: 5 mcg/min initially, increase prn to 10 mcg/min and 20 mcg/min at intervals of 15-30 minutes. Usual range: 3-20 mcg/min, though higher doses have been used.

Pediatrics:
- Loading dose: 1 month-1 year of age: 5 mcg/kg IV, followed by maintenance dose.
  2-17 years of age: 15 mcg/kg (maximum 250 mcg), followed by maintenance dose.
- Maintenance dose: 1 month-17 years of age: 1-2 mcg/kg/min, adjust dose according to response and heart rate; can increase rate up to 5 mcg/kg/min.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules at room temp. Protect from light.
- The solution in ampoules is clear and the colour can vary from colourless to pale straw.
- Stable for 24 hours at room temp and in the fridge in NS, D5W and RL at a concentration of 10 mcg/mL.
- Stable for 24 hours at room temp in dextrose-saline solutions at a concentration of 10 mgc/mL.

MISCELLANEOUS

- Monitor serum potassium and glucose during therapy.

REFERENCES

4, 5, 95, 109, 113, 170, 208.

Full revision 2018; limited revision 2020

NAME OF MEDICATION
SCOPOLAMINE HYDROBROMIDE

OTHER NAMES
Hyoscine hydrobromide

CLASSIFICATION
Anticholinergic

Do NOT confuse scopolamine hydrobromide with scopolamine butylbromide (hyoscine butylbromide). This monograph is specific to SCOPOLAMINE HYDROBROMIDE.

INDICATIONS

- Adjunct to anesthesia to produce sedation and amnesia, to reduce salivary and respiratory secretions, and to prevent other cholinergic effects during surgery.
- For relief of nausea and vomiting.
- To reduce upper airway secretions, including “death rattle”, or for sedation in palliative care patients.

ADMINISTRATION

- IV direct: physician or RN. Dilute with an equal volume or 10 mL of SWFI. Give over 2-3 minutes.
- IM, SC.
- SC infusion (palliative care). Dilute with SWFI, NS or D5W. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, angioedema, anaphylaxis.
- Cardiovascular: tachyarrhythmia, tachycardia, palpitations. Paradoxical bradycardia with low doses (less than 0.5 mg).
- GI: dry mouth, constipation.
- CNS: sedation, amnesia. Agitation, confusion and delirium with higher doses and in the elderly.
- Ophthalmic: blurred vision, mydriasis, increased intraocular pressure.
- Renal: urinary retention.

DOSAGE

Adults:
- Perioperative use: 0.3-0.65 mg IV/IM/SC. May repeat 3-4 times daily.
- Antiemetic: 0.6-1 mg SC.
- In palliative care patients:
  - To reduce upper airways secretions: 0.2-0.8 mg SC q2-4h prn.
  - Death rattle caused by excessive secretions: 0.3-0.8 mg IV/SC q2-8h prn OR 0.4 mg SC, followed with 1.2 mg/24 hours continuous SC infusion with repeat 0.4 mg SC prn.
  - Sedation: 0.3-0.8 mg IV/SC q2-4h.

Pediatrics:
- 0.006 mg/kg (6 mcg/kg) or 0.2 mg/m^2 IV/IM/SC repeated q6-8h prn. Maximum of 0.3 mg/dose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and vials at room temp. Protect from light. Do not freeze.
- Incompatible with alkaline solutions.

MISCELLANEOUS

REFERENCES

1, 2, 4, 5, 82, 95, 135, 256, 318, 457.

Full revision 2018; limited revision 2019, 2020

NAME OF MEDICATION
SELENIUM

OTHER NAMES
Selenious acid

CLASSIFICATION
Trace element

INDICATIONS

- Prevention or treatment of selenium deficiency in patients receiving total parenteral nutrition (TPN).

ADMINISTRATION

- Continuous IV infusion: must be added to IV TPN solution.

POTENTIAL ADMINISTRATION HAZARDS

- Unlikely to occur at recommended dosages and dilution.
- Local reactions: irritation and phlebitis when administered undiluted.

DOSAGE

- Prevention of deficiency:
  - Adults: 60-100 mcg/day IV.
  - Pediatrics: 2 mcg/kg/day IV.
- Treatment of deficiency: higher doses may be required.
- Dosage in renal impairment: use with caution as selenium retention may occur.
- Dosage in GI malfunction: use with caution as selenium retention may occur.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Do not freeze.
- Stable 24 hours in TPN.

MISCELLANEOUS

- Various components of TPN may be contaminated with multiple trace elements, including selenium, which can significantly add to total intake.
- Routine monitoring of selenium blood levels is suggested as a guideline for administration. Range for plasma levels: 78-157 ng/mL, whole blood levels: 70-229 ng/mL.

REFERENCES

1, 5, 82, 135, 512.

Full revision 2018

NAME OF MEDICATION
SEMAGLUTIDE

OTHER NAMES
Ozempic ®

CLASSIFICATION
Antidiabetic

INDICATIONS

- Treatment of type 2 diabetes mellitus as monotherapy when metformin is inappropriate, or as add-on therapy to metformin with or without a sulfonylurea or basal insulin, or as add-on therapy to a sodium-glucose cotransporter-2 (SGLT2) inhibitor with metformin or a sulfonylurea.

ADMINISTRATION

- SC: in the abdomen, thigh or upper arm. Rotate injection site.
- May be injected in the same body region as other SC injections (e.g., insulin), but injections should NOT be adjacent to each other.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, pruritus, anaphylactic reaction, angioedema.
- Cardiovascular: tachycardia, PR interval prolongation.
- GI: nausea, vomiting, diarrhea, abdominal pain, dyspepsia, bloating, pancreatitis.
- CNS: dizziness.
- Endocrine and metabolic: hypoglycemia.
- Renal: acute renal failure or worsening of chronic renal failure due to dehydration.
- Fatigue.
- Local reactions: injection site discomfort, erythema, bruising, pain, irritation, pruritus and rash.

DOSAGE

- Start initially with 0.25 mg SC once weekly for 4 weeks, then increase to a therapeutic dose of 0.5 mg SC once weekly. After another 4 weeks, dose may be further increased if needed to a maximum dose of 1 mg SC once weekly.
- Dosage in renal impairment: no dosage adjustment necessary; do not use in end-stage renal disease as only limited data available.
- Dosage in hepatic impairment: use with caution as it has not been studied.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store prefilled multidose pens between 2-8°C. Protect from excessive heat and light. Do not freeze.
- After first use, keep at room temp below 30°C or in the fridge between 2-8°C for 8 weeks; protect from light by keeping pen cap on when pen is not in use.

MISCELLANEOUS

- Semaglutide is a glucagon-like peptide-1 (GLP-1) receptor agonist, an incretin mimetic agent.
- Should not be used in treatment of type 1 diabetes mellitus or diabetic ketoacidosis, in patients with multiple endocrine neoplasia syndrome type 2, or in patients with a personal or family history of medullary thyroid carcinoma.
- An oral formulation of semaglutide is also available (Rybelsus ®). Although manufacturer does not provide dosage equivalency between the oral and SC routes, some experts suggest the following: 1) oral semaglutide 14 mg once daily may be converted to SC semaglutide 0.5 mg once weekly, beginning the day after the last oral dose; 2) SC semaglutide 0.5 mg once weekly may be converted to oral semaglutide 7 mg or 14 mg once daily, beginning within 7 days of the last injection. Monitor blood glucose during route transition.

REFERENCES

1, 5, 95, 135.

New monograph 2021

NAME OF MEDICATION
SILTUXIMAB

OTHER NAMES
Sylvant ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

INDICATIONS

- Treatment of patients with multicentric Castleman's disease who are HIV-negative and human herpes virus-8 (HHV-8)-negative.

ADMINISTRATION

- **Consider premedication if patient previously experienced an infusion-related reaction; refer to Potential administration hazards section**.
- Bring vials to room temp for approximately 30 minutes. Using a syringe with a 21-gauge 1-1/2 inch needle, reconstitute 100 mg and 400 mg vials with 5.2 mL and 20 mL of SWFI, respectively, to obtain a concentration of 20 mg/mL. Gently swirl (do NOT shake) until complete dissolution (should take less than 60 minutes). Do NOT use if foreign particles and/or solution is discoloured or visibly opaque.
- Intermittent IV infusion: withdraw from a 250 mL D5W bag (PVC, polyethylene, polypropylene or polyolefin) a volume equal to the volume of reconstituted drug required for the patient's dose, then slowly add the required volume of reconstituted drug into this bag. Gently mix. Administer over 60 minutes using infusion sets lined with PVC, polyurethane or polyethylene containing a 0.2-micron in-line polyethersulfone (PES) filter.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, pruritus, anaphylaxis.
- Infusion-related reactions: pruritus, erythema, chest pain, nausea, vomiting, flushing, palpitations, back pain. Upon resolution of mild to moderate reactions, restart at a slower infusion rate and **consider premedication** with antihistamines, acetaminophen and/or corticosteroids; if infusion is still not tolerated, discontinue.
- Cardiovascular: hypertension, peripheral edema, hot flashes.
- GI: nausea, vomiting, abdominal pain, diarrhea, constipation, GI perforation (rare).
- CNS: headache, peripheral neuropathy.
- Endocrine and metabolic: elevation of triglycerides, cholesterol, uric acid; decrease of eletrolytes: K, Na, Ca, Mg.
- Hematologic: myelosuppression, elevated hemoglobin.
- Hepatic: elevation of LFTs.
- Renal: elevation of creatinine.
- Infections: upper respiratory infections, nasopharyngitis, pneumonia, sepsis.
- Fever, musculoskeletal pain, fatigue.

DOSAGE

- 11 mg/kg IV every 3 weeks until treatment failure.
- Dose administration may be delayed based on results from hematology lab tests done on a regular basis (absolute neutrophil count, platelet count and hemoglobin); refer to manufacturer's monograph for specific retreatment criteria. Consider delaying treatment until retreatment criteria are met; dose reduction is not recommended.
- Dosage in renal impairment: CrCl 15 mL/min and greater: no dosage adjustment required; if CrCl less than 15 mL/min: not studied.
- Dosage in hepatic impairment: mild to moderate impairment (Child Pugh class A or B): no dosage adjustment required; severe impairment (Child Pugh class C): not studied.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge; protect form light. Do not freeze.
- Reconstituted solution with SWFI is stable for 2 hours at room temp. Do not refrigerate.
- Diluted solution is stable for 6 hours at room temp in D5W in PVC, polyethylene, polypropylene or polyolefin bags. Do not refrigerate.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity and infusion-related reactions.
- Live attenuated vaccines should not be given concurrently with siltuximab or within the 4 weeks prior to initiating siltuximab as clinical safety has not been established.

REFERENCES

1, 5, 40, 95, 129, 135, 165.

NAME OF MEDICATION
SINCALIDE

OTHER NAMES
Cholecystokinin C-terminal octapeptide, Kinevac ™

CLASSIFICATION
Diagnostic agent

INDICATIONS

- To stimulate gallbladder contraction for diagnostic imaging or for bile sampling.
- To stimulate pancreatic secretion for duodenal aspirate sampling.
- To stimulate intestinal motility for the relief of postoperative ileus or to accelerate the transit of a barium meal during examination of the intestinal tract.

ADMINISTRATION

- Reconstitute 5 mcg vial with 5 mL of SWFI to obtain a 1 mcg/mL solution.
- IV direct: physician or RN; undiluted; inject over 30-60 seconds.
- Intermittent IV infusion:
  - for 0.12 mcg/kg dose: dilute with 100 mL NS.
  - for 0.02 mcg/kg/dose: dilute with 30 mL NS.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity.
- GI: nausea, abdominal pain or discomfort, urge to defecate. More common with rapid IV administration.
- Decreasing the infusion rate can lessen the incidence of adverse reactions.
- Local reactions: pain at injection site (rare).

DOSAGE

Gallbladder stimulation:
- 0.02 mcg/kg IV direct; if satisfactory contractions do not occur within 15 minutes, a dose of 0.04 mcg/kg IV direct may be administered OR 0.12 mcg/kg IV infusion over 50 minutes OR 0.1 mcg/kg IM.

Pancreas stimulation:
- 0.02 mcg/kg IV infusion over 30 minutes; when used in combination with secretin, start sincalide administration 30 minutes after the initiation of the secretin infusion.

Intestinal stimulation:
- For postoperative ileus: 0.04 mcg/kg IV direct; additional dose of 0.04 or 0.08 mcg/kg q4h may be administered prn (maximum of 5 doses).
- For the acceleration of a barium meal: when the barium meal is beyond the proximal jejunum, administer sincalide 0.04 mcg/kg IV direct; if satisfactory contractions do not occur within 30 minutes, administer another 0.04 mcg/kg dose IV direct, OR 0.12 mcg/kg IV infusion over 30 minutes.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution is stable for 8 hours at room temp or 24 hours in the fridge.
- Diluted solutions should be used within 60 minutes of mixing.

MISCELLANEOUS

- Caution in pregnancy; may induce premature labour.

REFERENCES

1, 5, 95, 135, 208.

Full revision 2018

NAME OF MEDICATION
SODIUM BICARBONATE

OTHER NAMES
Baking soda, $NaHCO_3$, Sodium acid carbonate, Sodium hydrogen carbonate

CLASSIFICATION
Alkalinizing agent

INDICATIONS

- Treatment of metabolic acidosis.
- Advanced cardiac life support during cardiac arrest in certain circumstances (e.g., pre-existing metabolic acidosis, hyperkalemia, tricyclic antidepressant overdosage, prolonged cardiac arrest).
- Adjunct in the treatment of hyperkalemia-induced cardiotoxicity.
- Treatment of cardiotoxic effects from drug overdoses involving sodium channel blockade.
- Alkalinization of urine in cases of drug overdoses (e.g., salicylates) or to prevent nephrotoxicity caused by contrast media, chemotherapy (e.g., methotrexate) or by heme pigments.

ADMINISTRATION

- IV direct: physician or RN in emergency situations only; use the 8.4% solution in adults and children 2 years of age and older, and the 4.2% solution in children less than 2 years old; inject over 1-5 minutes. Do not use if solution is unclear or contains a precipitate.
- Intermittent IV infusion (preferred): dilute in NS or D5W to obtain a concentration of 0.5 mmol/mL or less. Administer over 4-8 hours. Flush IV tubing before and after administration. Do not use if solution is unclear or contains a precipitate.
- SC: if IV administration is not possible. Must be diluted with a compatible solution to obtain an isotonic 1.5% solution (e.g., dilute each mL of the 8.4% solution with 4.6 mL of NS).

POTENTIAL ADMINISTRATION HAZARDS

- Alkalosis (hyperirritability, tetany) if injected too quickly, with high doses or in patients with renal impairment (**Antidote**: calcium gluconate or ammonium chloride).
- Electrolyte disturbances: hypokalemia, hypocalcemia, hyperosmolality and hypernatremia if injected too quickly or with high doses.
- Cardiovascular: peripheral edema, CHF exacerbation due to sodium and water retention.
- CNS: decreased cerebrospinal fluid pressure and intracranial haemorrhage with the rapid administration (above 0.33 mmol/kg/hr) of the 8.4% solution in children less than 2 years of age.
- Respiratory: pulmonary edema.
- Local reactions: pain, irritation, phlebitis.
- Extravasation hazard: vesicant (undiluted 8.4% solution). Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

Adult:
- Metabolic acidosis: usual initial dose of 2-5 mmol/kg IV infused over 4 to 8 hours. Do not exceed 50 mmol/hr. Administer subsequent doses prn. Full correction should not be attempted during the first 24 hours because of the risk of delayed compensation and alkalosis.
- Cardiac arrest: 1 mmol/kg initially (1 mL of 8.4% solution per kg of body weight) IV direct over 5 minutes, then 0.5 mmol/kg IV direct at 10 minute intervals prn.
- Hyperkalemia-induced cardiotoxicity: 50-100 mmol IV direct over 5 minutes, repeat as needed.
- Overdosage cardiotoxic effects: 1-2 mmol/kg IV direct over 1-2 minutes, repeat dose prn depending on patient’s acid-base status.

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DOSAGE (Cont.)

- Alkalinization of urine:
  - Usual: 100-150 mmol mixed in 1000 mL of D5W administered at an IV infusion rate of 150-200 mL/hr (urinary pH target of 7-8).
  - Prevention of renal dysfunction associated with contrast media: remove 150 mL from a 1000 mL bag of D5W then add 150 mL from vials of 8.4% solution (150 mmol) to the bag to obtain a concentration of 150 mmol/1000 mL; administer IV at a rate of 3 mL/kg/hr for 60 minutes prior to the contrast injection, at 1 mL/kg/hr during the contrast exposure and for 6 hours post procedure.

Pediatrics:
- Metabolic acidosis: Full correction should not be attempted during the first 24 hours because of the risk of delayed compensation and alkalosis.
  - Children less than 2 years old: initial dose of 1-2 mmol/kg IV direct over at least 2 minutes; maximum of 8 mmol/kg/day and maximum rate 10 mmol/min.
  - Children 2 years of age and older: initial dose of 2-5 mmol/kg IV infused over 4 to 8 hours, not exceeding 1 mmol/kg/hr. Administer subsequent doses prn.
- Cardiac arrest: initial dose of 1 mmol/kg IV direct over at least 2 minutes (maximum rate of 10 mmol/min); may administer additional doses of 0.5 mmol/kg IV every 10 minutes prn.
- Dosage in renal impairment: use with caution as sodium content may induce sodium and water retention.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp. Do not freeze. Protect from excessive heat.
- Compatible with NS and dextrose solutions.
- Incompatible with acids, acidic salts, many alkaloidal salts and RL.
- Sodium bicarbonate inactivates catecholamines (e.g., epinephrine, norepinephrine, dobutamine) and precipitates when mixed with calcium.
- Stable for 7 days at room temp (exposed to light) or 21 days in the fridge (unexposed to light) or frozen (-20°C, unexposed to light) in D5W at concentrations of 0.1 mmol/mL or 0.15 mmol/mL in polyolefin bags.

MISCELLANEOUS

- 8.4% sodium bicarbonate = 84 mg/mL of sodium bicarbonate = 1 mmol (1 mEq) of sodium bicarbonate per mL = 1 mmol (1 mEq) each of sodium and bicarbonate ions per mL.
- Sodium bicarbonate infusion has not been found to be more effective than NS infusion in the prevention of renal dysfunction associated with contrast media.

REFERENCES

1, 2, 4, 5, 40, 95, 102, 135, 159, 220, 321, 328, 366, 367, 441, 460, 462.

Full revision 2018; limited revision 2020, 2021

NAME OF MEDICATION
SODIUM CHLORIDE, HYPERTONIC 3%

OTHER NAMES
Hypertonic saline

CLASSIFICATION
Electrolyte

INDICATIONS

- Treatment of acute or symptomatic hypotonic hyponatremia or when high sodium and/or chloride content without large amounts of fluid is/are required (e.g., electrolyte and fluid loss replaced with sodium-free fluids, excessive water intake resulting in drastic dilution of body water, emergency treatment of severe salt depletion, Addisonian crisis, diabetic coma).
- Treatment of intracranial hypertension (intracranial pressure greater than 20 mmHg) after traumatic brain injury.

ADMINISTRATION

- Intermittent IV infusion: refer to Dosage section for details.
- Continuous IV infusion: administer half of calculated dose over 8 hours. Infuse at a rate that corrects the serum sodium concentration by approximately 1 mmol/L/hr, with a maximum correction of serum sodium concentration of 8-10 mmol/L/24 hours and to less than 18 mmol/L in 48 hours (refer to Dosage section for calculations). Do not exceed a maximum of 100 mL/hr. Must be administered by an infusion pump.
- Infuse in a large vein.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication**: consult Corporate policy ADM VIII 860 (High alert medications).
- Due to sodium excess: edema, pulmonary edema, hypertension, hyperchloremic acidosis, deep respiration, disorientation, nausea, weakness, potassium loss.
- A too rapid correction of sodium deficit can result in osmotic demyelination syndrome with resultant severe brain injury and potentially death.
- Local reactions: phletitis. Local pain and venous irritation with too-rapid infusion; decrease rate for tolerance.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Hyponatremia with severe symptoms:
  - Intermittent IV infusion using one of the following 2 methods: 1) give 150 mL or 2 mL/kg IV over 20 minutes, then check serum sodium concentration (SSC) after 20 minutes and repeat dose 2-3 times prn until SSC increase of 5 mmol/L; OR 2) give 100 mL IV over 10 minutes; repeat twice prn at 30 minute intervals until SSC increases by 4-6 mmol/L. Note: 100 mL should increase SSC by about 2 mmol/L.
  - If symptoms have not improved despite an increased SSC of 5 mmol/L in the first hour, start a continuous IV infusion aiming for a SSC increase of 1 mmol/L/hr; check SSC every 4 hours and stop infusion when the first of the following occurs: 1) symptoms improve, 2) the SSC increases by 10 mmol/L, or 3) SSC reaches 130 mmol/L.

- Hyponatremia with moderately severe symptoms: single intermittent IV infusion of 150 mL over 20 minutes. Target of a SSC increase of 5 mmol/L/24 hr. Check SSC after 1, 6 and 12 hours.

- Hyponatremia with mild to moderate symptoms and a low risk of brain herniation:
  - If acute decrease in SSC greater than 10 mmol/L: give 150 mL intermittent IV infusion over 20 minutes and check SSC after 4 hours.
  - Otherwise, give a continuous IV infusion at a rate of 0.5-2 mL/kg/hr.

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DOSAGE (Cont.)

- May also use the following equations in hyponatremia:
  - To determine the total amount of sodium required (mmol) to be administered:
    [desired serum sodium concentration (mmol/L) – actual serum sodium concentration (mmol/L)] x weight (kg) x BWF*.

  - To determine the rate of continuous IV infusion, the following is to estimate the effect of 1 litre of NaCl 3% administered to a patient:

$$\text{change in serum sodium concentration (mmol/L)} = \frac{513 \text{ mmol/L} - \text{actual serum sodium concentration (mmol/L)}}{[(\text{weight in kg} \times \text{BWF*}) + 1]}$$

  * BWF = bodyweight fraction = 0.6 for nonelderly males; 0.5 for elderly males; 0.5 for nonelderly females; 0.45 for elderly females.

- For intracranial hypertension after brain injury:
  - Intermittent IV infusion: 250-300 mL IV over 20 minutes; to repeat dose until intracranial pressure is controlled or a serum sodium level of 155 mmol/L is achieved.
  - Continuous IV infusion: initial rate of 1-2 mL/kg/hr.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store at room temp.

MISCELLANEOUS

- A litre of NaCl 3% contains 513 mmol of sodium and of chloride.

REFERENCES

1, 5, 40, 95, 135, 325, 332, 341, 343, 349, 367.

NAME OF MEDICATION
SODIUM CITRATE 4%

OTHER NAMES
Citrosodine, Trisodium citrate

CLASSIFICATION
Anticoagulant

INDICATIONS

- For maintenance of catheter patency of vascular access devices (VAD), as an alternative to heparin.

ADMINISTRATION

- IV direct: fill up the VAD catheter with sodium citrate 4% slowly over 5-10 seconds.

POTENTIAL ADMINISTRATION HAZARDS

- No adverse reactions have been reported when used as indicated.

DOSAGE

- Dose according to size of catheter lumen; range from 1 to 5.5 mL.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 15 and 30°C. Protect from direct light.
- The solution should be colourless and clear.

MISCELLANEOUS

- Monitor serum levels of ionized calcium, magnesium and arterial pH.
- Do not use the hemodialysis catheter as a conduit for sampling blood as sodium citrate can contaminate blood samples.

REFERENCES

5, 43, 361, 372.

Full revision 2018

NAME OF MEDICATION
SODIUM STIBOGLUCONATE *

OTHER NAMES
Pentavalent antimony compound, Pentostam ®, Sodium antimony gluconate

CLASSIFICATION
Antiparasitic

INDICATIONS

- Treatment for visceral, cutaneous, and mucocutaneous leishmaniasis.

ADMINISTRATION

- IV direct: physician or RN. Undiluted, slowly at a rate of 2-3 mL/min, over at least 5 minutes. Discontinue immediately if cough, vomiting or substernal pain occurs.
- Intermittent IV infusion: dilute in 50-100 mL of D5W and infuse over 10-30 minutes.
- Continuous IV infusion: dilute in 1000 mL of D5W; infuse over 24 hours.
- Solution must be filtered with a sterile filter of 5 microns or less immediately before use, such as using an in-line filter.
- IM: when IV administration is not possible.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, anaphylactic shock.
- Cardiovascular: QT interval prolongation, T-wave flattening or inversion, arrhythmia (dose and duration dependent effect; can be fatal).
- GI: nausea, vomiting, anorexia, diarrhea, abdominal pain, pancreatitis.
- CNS: headache, lethargy.
- Endocrine and metabolic: increase in amylase, lipase.
- Hematologic: reduction in platelets, white blood cells and hemoglobin.
- Hepatic: increase in liver enzymes; discontinue if hepatotoxicity.
- Respiratory: cough.
- Severe inflammation around mucocutaneous lesions; corticosteroids may be used in life-threatening pharyngeal or tracheal involvement.
- Arthralgia, myalgia, malaise.
- Local reactions: pain with or without abcess-like swelling at site of IM injection; pain and thrombophlebitis with IV administration; minimize by administration through a fine needle.

Antidote: In case of overdose, dimercaprol 200 mg IM q6h until recovery, can be used.

DOSAGE

- Cutaneous leishmaniasis: 20 mg/kg/day IV/IM for 20 days (varies from 10 to 28 days).
- Visceral or mucocutaneous leishmaniasis: 20 mg/kg/day IV/IM for at least 28 days.
- Dosage in renal impairment: avoid use if CrCl is less than 50 mL/min.
- Dosage in hepatic impairment: use with caution.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp up to 25°C. Do not freeze. Protect from light.
- Stable 1 month after first puncture.

MISCELLANEOUS

- Sodium stibogluconate injection contains the equivalent of 100 mg/mL pentavalent antimony.
- Cardiac monitoring recommended before and during therapy.

REFERENCES

3, 5, 95, 135, 200, 373.

* Available via Health Canada's Special Access Programme

Full revision 2018

NAME OF MEDICATION
SODIUM THIOSULFATE

OTHER NAMES
Disodium thiosulfate pentahydrate, Seacalphyx ®, Sodium hyposulfite

CLASSIFICATION
Antidote

INDICATIONS

- Treatment of acute and severe cyanide poisoning.
- Prevention of nitroprusside-induced cyanide toxicity.
- Treatment of calciphylaxis also called "calcific uremic arteriolopathy" (systemic medial calcification of the arterioles).
- Treatment of mechlorethamine extravasation.

ADMINISTRATION

- IV direct (preferred) (for cyanide poisoning): physician only; ***blood pressure monitoring***; administer undiluted slowly over at least 10 minutes. Administer immediately after sodium nitrate injection in the same IV line.
- Intermittent IV infusion (for cyanide poisoning): ***blood pressure monitoring***; dose may be diluted in D5W and administered over 10-30 minutes. Administer immediately after sodium nitrate injection in the same IV line.
- IV direct/SC (for mechlorethamine extravasation): prepare a 4% (1/6 M) solution by mixing 1.6 mL of a 25% solution with 8.4 mL SWFI; either 1) inject dose of 2.5 mL into existing IV line immediately after extravasation and preferably through the same needle to ensure injection is in the extravasated area OR 2) inject 2 mL SC into the extravasated area for each mg of mechlorethamine suspected to have extravasated.
- Intermittent IV infusion (for calciphylaxis): administer undiluted over 30-60 minutes; may dilute in 100 mL of NS and administer over 30-60 minutes (note that the sodium content needs to be taken into consideration, e.g., a 25 g dose diluted in 100 mL of NS will yield 4.8 g of sodium). (At TOH: for patients undergoing hemodialysis in the dialysis unit: administer undiluted through the hemodialysis catheter; infuse initial doses over 3-4 hours to minimize nausea and vomiting; may infuse subsequent doses at a higher rate based on patient's tolerance).

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity.
- Cardiovascular: hypotension; reduce administration rate if significant.
- GI: nausea, vomiting; reduce administration rate if significant. Salty taste.
- CNS: headache, disorientation.
- Electolyte disturbances: hypernatremia with large doses.
- Hematologic: prolonged bleeding time.
- Renal: diuresis, hypovolemia.
- Metabolic acidosis.
- Muscle cramps.

DOSAGE

- Adults:
  - Cyanide poisoning: 12.5 g IV (50 mL of a 25% solution): sodium thiosulfate may be repeated prn at <u>half</u> the original dose.
  - Prevention of nitroprusside-induced cyanide toxicity: administer IV concurrently with nitroprusside at a dose 5-10 times that of nitroprusside.
  - Calciphylaxis: 25 g (100 mL of a 25% solution) IV 3 times a week; if patient on hemodialysis, administer during the last hour or after the dialysis session.
  - Mechlorethamine extravasation: 2.5 mL of a 4% solution (1/6 M) through the existing IV line OR 2 mL of a 4% solution SC into the extravasated area per mg of extravasated mechlorethamine.

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DOSAGE (Cont.)

- Pediatrics:
  - Cyanide poisoning: 250-500 mg/kg IV or 7.5-10 g/m^2 IV. Maximum dose of 12.5 g; sodium thiosulfate may be repeated prn at half the original dose.

- Dosage in renal impairment: caution as patients may be at increased risk of adverse effects.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from direct light. Do not freeze.
- Stable for 24 hours in NS, D5W and D5-1/2NS at concentrations of 1.5% and 9.76%.
- Incompatible with hydroxocobalamine.
- Compatible with sodium nitrate.

MISCELLANEOUS

- Observe patient for 24-48 hours for reoccurrence of symptoms of poisoning.

REFERENCES

4, 5, 40, 82, 95, 129, 135, 339, 374.

NAME OF MEDICATION
STREPTOMYCIN

OTHER NAMES

CLASSIFICATION
Antibiotic – aminoglycoside – antitubercular

INDICATIONS

- Treatment of tuberculosis in conjunction with other antituberculosis agents, when the infecting organisms are susceptible.
- Treatment and post-exposure prophylaxis of tularemia and plague (including in the context of bioterrorism).
- Treatment of brucellosis and severe *M. avium* complex, when the infecting organisms are susceptible.

ADMINISTRATION

- Reconstitute the 1 g vial with 1.8, 3.2 or 4.2 mL of SWFI to obtain a solution of 400, 250 or 200 mg/mL, respectively.
- Intermittent IV infusion: not an officially approved route of administration but documented in the literature for patients who cannot tolerate IM injections. Dilute to a concentration of 5-10 mg/mL (usually with 100 mL of NS or D5W); infuse over 30-60 minutes.
- IM (preferred): deep into large muscle mass; for adults, into the buttocks or the mid-lateral thigh (deltoid only if well developed); in children, preferably in the mid-lateral thigh. Rotate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, rash, urticaria, angioneurotic edema, exfoliative dermatitis.
- CNS: facial and peripheral paresthesia; tingling or dizziness during the infusion (increase infusion time to 60 minutes to alleviate this problem).
- Hematologic: eosinophilia.
- Otic: ototoxicity (may be irreversible): vestibular (nausea, vomiting, vertigo) and cochlear (tinnitus, hearing loss) toxicity (related to dose and duration of therapy).
- Renal: nephrotoxicity; alkalinization of the urine may minimize renal irritation with long-term treatment.
- Potential for neuromuscular blockade (rare).
- Fever.

DOSAGE

Adults:

- Tuberculosis: 15 mg/kg (up to 1 g) IM/IV once daily OR 25-30 mg/kg (up to 1.5 g) IM/IV 2-3 times weekly in patients younger than 60 years old; 10 mg/kg (up to 750 mg) IM/IV once daily in patients of 60 years of age and older. Daily dosing can be reduced to 2-3 times a week after 2 to 4 months or culture conversion.

- Tularemia:
  - naturally occurring: 1-2 g IM daily in 2 divided doses for 7-14 days, until the patient is afebrile for 5-7 days.
  - in the context of bioterrorism: 1 g IM q12h for at least 10-14 days.

- Plague: in the context of bioterrorism: 15 mg/kg (up to 1 g) IM q12h for at least 10-14 days and patient afebrile for at least 2-3 days.

- Brucellosis: in patients up to 60 years of age: 15 mg/kg (up to 1 g) IM daily in 2 to 4 divided doses for 14-21 days in combination with doxycycline regimen; for patients older than 60 years of age, streptomycin dose is reduced to 1 g IM every other day.

- Severe *M. avium* complex: 15 mg/kg IM (range from 8-25 mg/kg; up to 1 g in patients up to 50 years old and 500 mg in patients older than 50 years of age) three times weekly for the first 2-3 months in combination with a macrolide, rifampin and ethambutol.

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DOSAGE (Cont.)

- Dosage in renal impairment: daily dosage intervals should be adjusted as follows and as per serum drug levels:

CrCl (mL/min)	greater than 50	50-10	less than 10
Interval (hr)	24	24-72	72-96

Pediatrics:
- Tuberculosis:
  - Children younger than 15 years old and weighing 40 kg or less: 15-20 mg/kg/day (may go up to 40 mg/kg/day; maximum of 1 g/day) IM/IV given once daily or in 2 divided doses OR 25-30 mg/kg (up to 1 g) IM/IV twice weekly.
  - Children younger then 15 years old and weighing more than 40 kg, or children 15 years old and greater: 15 mg/kg (up to 1 g) IM/IV once daily OR 25 mg/kg (up to 1 g) IM/IV 3 times per week.

- Tularemia and plague: in the context of bioterrorism: 15 mg/kg IM twice daily (maximum 2 g daily) for at least 10-14 days.

- Brucellosis: 15 mg/kg (up to 1 g) IM once daily OR 20-40 mg/kg/day IM in 2 to 4 divided doses, in combination with other anti-infectives, such as doxycycline.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light.
- Reconstituted solution is stable for 7 days at room temp, when protected from light.
- Stable for 24 hours at room temp in NS or D5W.
- Exposure to light causes darkening of solution without apparent loss of potency.

MISCELLANEOUS

- Monitoring of serum concentrations is suggested in patients with renal disease; peak (60 minutes after IM injection) should be 20-30 mcg/mL for most indications (tuberculosis: 35-45 mcg/mL with daily administration and 65-80 mcg/mL with twice weekly administration); trough (just before administration) should be below 5 mcg/mL.
- Audiography tests are recommended before treatment and at least monthly during treatment.

REFERENCES

1, 2, 4, 5, 40, 82, 95, 135, 208, 233, 537.

NAME OF MEDICATION
SUCCINYLCHOLINE

OTHER NAMES
Quelicin ®, Suxamethonium chloride

CLASSIFICATION
Neuromuscular blocker

INDICATIONS

- As an adjunct to general anesthesia, to facilitate endotracheal intubation, endoscopy and short manipulative procedures, and to produce muscle relaxation during surgery or mechanical ventilation.

ADMINISTRATION

- IV direct: physician trained in anesthesiology; RN may administer subsequent doses; ***ventilator support, cardiac monitoring***. Undiluted; inject over 30 seconds.
- Intermittent IV infusion, Continuous IV infusion: ***ventilator support, cardiac monitoring***. Dilute to 1-2 mg/mL in D5W or NS (500 mg in 250-500 mL or 1 g in 500-1000 mL); do not exceed 10 mg/min or 500 mg/hr. Must be administered by an infusion pump.
- Deep IM (NOT preferred): ***ventilator support, cardiac monitoring***. Use this route only when IV administration is not possible.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: circulatory collapse, flushing, skin rash, urticaria, bronchospasm, anaphylactoid/anaphylactic reaction.
- Cardiovascular: transient arrhythmias, bradycardia with repeated doses in adults and single doses in children. Pre-treatment with atropine is recommended in children. Cardiac arrest from acute rhabdomyolysis with hyperkalemia in children (rare).
- GI: excessive salivation.
- Ophthalmic: increase in intraocular pressure.
- Respiratory: respiratory depression, apnea.
- Transient muscle fasciculation, especially if infused rapidly; myalgia.
- Malignant hyperthermia.
- Hyperkalemia.

DOSAGE

Adults:
- A test dose of 5-10 mg (or 0.1 mg/kg) IV or a slow infusion of a 1 mg/mL solution may be given to patients suspected of having an atypical plasma pseudocholinesterase activity or to assess individual patient response.
- For short procedures/intubation: 0.6 mg/kg (range 0.3-1.1 mg/kg) IV direct, repeated as required.
- For rapid sequence intubation: 1-1.5 mg/kg IV.
- For maintenance of neuromuscular blockade during prolonged procedures:
  - Continuous IV infusion: usually 2.5-4.3 mg/min (range of 0.5-10 mg/min).
  - IV direct or intermittent IV infusion: after initial dose of 0.3-1.1 mg/kg, followed by 0.04-0.07 mg/kg as required.
- IM: up to 3-4 mg/kg. Total dose should not exceed 150 mg.

Pediatrics:
- IV direct or intermittent IV infusion: 1-2 mg/kg IV; maintenance: 0.3-0.6 mg/kg every 5-10 minutes if required.
- Continuous IV infusion is not recommended in children.
- IM: up to 3-4 mg/kg. Total dose should not exceed 150 mg.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Unopened vials are stable for 6 months at room temp.
- Stable for 24 hours at 5°C in D5W or NS at concentrations of 1 and 2 mg/mL.
- Compatible with NS, D5W, dextrose-saline solutions and RL.
- Incompatible with alkaline solutions (e.g., barbiturates); also decomposes in c with pH of greater than 4.5.

MISCELLANEOUS

- Onset of muscle relaxation is observed within 1 minute after IV administration and 2-3 minutes after IM administration in healthy adults.
- Use is contraindicated in patients with skeletal muscle myopathy, severe burns, multiple trauma, extensive denervation of skeletal muscle, angle-closure glaucoma, upper motor neuron lesions (recent) and other unstable neurological diseases, and malignant hyperthermic susceptibility.
- Use with caution in patients with glaucoma or with reduced plasma cholinesterase activity (e.g., severe renal or liver disease, anemia, organophosphate insecticide poisoning).
- Succinycholine effects are not necessarily reversed by anticholinesterase agents.

REFERENCES

1, 3, 4, 5, 40, 82, 95, 135, 204, 208, 375.

NAME OF MEDICATION
SUFentanil

OTHER NAMES
Fentathienil, Sufenta ®, Sufentanyl, Sulfentanil

CLASSIFICATION
Opioid analgesic, anesthesia adjunct

INDICATIONS
- For the induction and maintenance of anesthesia (with 100% oxygen).
- As an analgesic adjunct in the maintenance of balanced general anethesia.

ADMINISTRATION
- IV direct (preferred): physician only; ***respiratory support***. Undiluted; inject over 2-10 minutes.
- Continuous IV infusion: dilute dose in D5W or NS. Administer at a rate of 250-300 mcg/min. Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: pruritus, anaphylaxis.
- Cardiovascular: hypotension/hypertension, tachycardia, arrhythmias, bradycardia.
- GI: nausea, vomiting.
- Respiratory: apnea, respiratory depression, bronchospasm (**Antidote**: epinephrine).
- Skeletal muscle and chest wall rigidity (**Antidote**: neuromuscular blocker).

Antidote: naloxone (for respiratory depression).

DOSAGE
- Must be individualized to each patient, and titrated to clinical response.
- Reduce dose in elderly and debilitated patients.
- Dosage should be based on lean body weight in those patients weighing more than 20% above ideal body weight.

Adults:
- Primary anesthetic agent (with oxygen only): initial dose is individualized; for maintenance increments: 25-50 mcg; usual total dose 8-30 mcg/kg.
- Adjunct to surgery (balanced anesthesia, i.e., with nitrous oxide and oxygen):

Expected duration of surgery	Total Dose*	Initial Dose	Maintenance Dose		
			INCREMENTAL DOSE	OR	CONTINUOUS INFUSION
Up to 2 hours	0.5 -2 mcg/kg	75% or more of total dose	10-25 mcg OR 0.1-0.25 mcg/kg		0.3-1.5 mcg/kg/hr
2-8 hours	2-8 mcg/kg	75% or less of total dose	10-50 mcg		0.3-1.5 mcg/kg/hr

*Dosages above 1 mcg/kg per hour of surgery are frequently associated with respiratory depression.

Pediatrics (less than 12 years of age):
- Anesthetic initial dose of 10-25 mcg/kg. Supplemental doses up to 25-50 mcg OR 1-2 mcg/kg may be given as required.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store ampoules and vials at room temp. Protect from light.
- Compatible with D5W and NS.
- Undergoes hydrolysis in acidic solution.

MISCELLANEOUS
- Sufentanil is 5-7 times more potent than fentanyl.

REFERENCES
1, 4, 5, 82, 135.

Full revision 2018; limited revision 2020

NAME OF MEDICATION
SUGAMMADEX

OTHER NAMES
Bridion ™

CLASSIFICATION
Antidote, Selective relaxant binding agent

INDICATIONS

- For reversal of moderate to deep neuromuscular blockade induced by rocuronium or vecuronium in adults undergoing surgery.

ADMINISTRATION

- IV direct: physician trained in anesthesiology; ***ventilator support***; administer dose rapidly over 10 seconds with flushing the IV line with a compatible solution (e.g., NS) before and after administration.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, flushing, urticaria, anaphylactoid and anaphylactic reactions (hypotension, tachycardia, bronchospasm, swelling of the tongue and throat). Increased risk with doses greater than 4 mg/kg.
- Cardiovascular: bradycardia within a few minutes after the injection (**Antidote**: atropine), hypotension.
- GI: nausea, vomiting.
- CNS: headache.
- Hematologic: transient increase of aPTT and INR within an hour of administration.
- Local reactions: pain at injection site.

DOSAGE

- Moderate rocuronium or vecuronium blockade reversal: 2 mg/kg IV when spontaneous recovery has reached the reappearance of the second twitch (T_2) in response to train-of-four (TOF) stimulation.
- Deep rocuronium or vecuronium blockade reversal: 4 mg/kg IV when spontaneous recovery has reached 1-2 post-tetanic counts (PTC), and there are no twitch responses to TOF stimulation.
- Urgent reversal of rocuronium blockade: 16 mg/kg IV when urgent need to reverse neuromuscular blockade soon (approximately 3 minutes) after administration of a single dose of 1.2 mg/kg of rocuronium.
- Dosage in renal impairment: CrCl 30-80 mL/min: no dosage adjustment required; CrCl less than 30 mL/min, including dialysis: do not use.
- Dosage in hepatic impairment: no dosage adjustment required but use with caution if coagulopathy or severe edema.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light; stable for 5 days when left unprotected from light.
- Solution in the vial should be clear and colourless to slightly yellow-brown.
- Compatible with NS, D5W, D5-NS, D2.5-1/2NS, RL, Ringer's.

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitive reactions.
- Waiting time before readministration of rocuronium or vecuronium after using sugammadex varies between 5 minutes to 24 hours, depending on agent, dose and renal function. If neuromuscular blockade is required before the recommended waiting time has elapsed, use a nonsteroidal neuromuscular blocking agent (e.g., cisatracurium).
- Can decrease progestogen exposure by 1/3 in patients taking a hormonal contraceptive, considered equivalent to a missed dose of the hormonal contraceptive. In childbearing female patients taking hormonal contraception, counsel patient on taking a non-hormonal contraceptive method (back-up method of contraception such as condoms and spermicides) for the 7 days following sugammadex administration.

REFERENCES

1, 5, 40, 95, 135, 208.

Full revision 2018; limited revision 2020

NAME OF MEDICATION
TACROLIMUS

OTHER NAMES
Prograf ®

CLASSIFICATION
Immunosuppressant

INDICATIONS
- Prophylaxis of organ rejection and treatment of refractory rejection for patients receiving allogenic heart, liver or renal transplants in conjunction with other immunosupressants, including corticosteroids, when oral administration is not possible.

ADMINISTRATION
- Continuous IV infusion (mandatory): dilute with NS or D5W to a concentration of 0.004-0.02 mg/mL in glass or non-PVC containers; do NOT use PVC containers or administration sets; give as a continuous 24-hour infusion.
- Observe the patient for hypersensitivity reactions for the first 30 minutes following the start of the infusion and at frequent intervals thereafter.

POTENTIAL ADMINISTRATION HAZARDS
- **Non-cytotoxic hazardous drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis.
- Cardiovasvascular: hypertension, chest pain, edema.
- GI: nausea, vomiting, constipation, diarrhea, anorexia, abdominal pain.
- CNS: headache, insomnia, paresthesia, tremor, dizziness.
- Dermatologic: alopecia, erythema, pruritus.
- Endocrine and metabolic: hyperkalemia, hypomagnesemia, hypophosphatemia, hyperglycemia, hyperlipidemia.
- Hematologic: anemia, leukocytosis, thrombocytopenia.
- Renal: nephrotoxicity.

DOSAGE
- Adults & pediatrics: liver and kidney transplants: initial dose of 0.03-0.05 mg/kg/day IV; heart transplant: 0.01 mg/kg/day IV. The first dose should be given no sooner than 6 hours post liver or heart transplantation; administer within 24 hours of renal transplantation (may need to be delayed until renal function has improved).
- Dosage in renal or hepatic impairment: use the lowest recommended dose (further reductions may be required).
- If switching from cyclosporin to tacrolimus or *vice versa*, allow at least 24 hours prior to initiating the new medication to avoid excess nephrotoxicity.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store ampoules between 15-25°C. Protect from light.
- Stable for 24 hours at room temp in D5W or NS at a concentration of 0.004-0.02 mg/mL in glass or polyethylene containers.
- Stable for 48 hours at room temp in D5W or NS at a concentration of 0.01 mg/mL in polyolefin containers.
- Stable for 20 hours in the fridge followed by 28 hours at room temp in D5W or NS at a concentration of 0.01 mg/mL in polyolefin containers.
- PVC containers and administration sets should not be used due to decreased stability and the potential for extraction of phthalates.
- Incompatible with alkaline solutions with a pH of 9 or greater.

MISCELLANEOUS
- Emergency drug (epinephrine) and oxygen must be immediately available for the treatment of hypersensitivity reactions.
- Tacrolimus IV contains castor oil derivatives.
- Conversion from IV to PO immediate release: use a 1:4 ratio, i.e., total daily IV dose X 4 = total daily PO dose; administer total PO dose as 2 divided doses per day, 12 hours apart. The first oral dose shoud be given 8-12 hours post discontinuation of the IV infusion. Careful supervision and therapeutic monitoring are required after the conversion.

REFERENCES
1, 4, 5, 40, 95, 135, 208.

Full revision 2019; limited revision 2020

NAME OF MEDICATION
TELAVANCIN

OTHER NAMES
Vibativ ™

CLASSIFICATION
Antibiotic - glycopeptide

INDICATIONS

- Treatment of complicated skin and skin structure infections (cSSSI) caused by susceptible strains of the following gram-positive microorganisms: *Staphylococcus aureus* (methicillin-resistant and -susceptible strains), *Streptococcus pyogenes*, *Streptococcus agalactiae*, *Streptococcus anginosus* group (including *S. anginosus*, *S. intermedius* and *S. constellatus*), and *Enterococcus faecalis* (vancomycin-susceptible isolates only).
- Treatment of hospital-acquired pneumonia (HAP) and ventilator-associated pneumonia (VAP) caused by susceptible strains of *Staphylococcus aureus* (methicillin-resistant and -susceptible strains).

ADMINISTRATION

- Reconstitute 250 mg and 750 mg vials with 15 and 45 mL, respectively, of SWFI, D5W or NS to get a concentration of approximately 15 mg/mL. To minimize foaming, allow the vacuum of the vial to pull the diluent from the syringe into the vial. Mix well but do NOT shake forcefully. Reconstitution time is generally under 2 minutes but can take up to 20 minutes in rare occasions.
- Intermittent IV infusion: dilute in 100 to 250 mL of D5W, NS or RL to obtain a final concentration of 0.6 to 8 mg/mL. Do NOT shake the final infusion. Administer over a period of at least 60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, anaphylaxis.
- Infusion-related reactions: hypotension, flushing of the upper body, urticaria, pruritus or rash if administered in less than 60 minutes. If this occurs, stop or slow the infusion.
- Cardiovascular: QTc interval prolongation.
- GI: nausea, vomiting, diarrhea, altered taste (metallic or soapy taste).
- CNS: dizziness, headache, insomnia.
- Renal: increase in serum creatinine, foamy urine.
- Local reactions: erythema, pain, pruritus.

DOSAGE

- 10 mg/kg IV q24h for 7 to 14 days if cSSSI, and for 7 to 21 days if HAP/VAP.
- Dosage in renal impairment:

CrCl (mL/min)	50-30	29-10	less than 10
Dose (mg/kg)	7.5	10	no data *
Interval (hr)	24	48	

  * Not significantly removed by hemodialysis.
- Dosage in hepatic impairment: no dosage adjustment necessary for patients with mild to moderate hepatic impairment; not evaluated in severe hepatic impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C.
- Reconstituted or diluted solution is stable for 12 hours at room temp or 7 days in the fridge. However, the total time in the vial plus the time in the IV bag should not exceed 12 hours at room temp or 7 days in the fridge.

MISCELLANEOUS

- Contraindicated in patients with known hypersensitivity to other glycopeptides (e.g., vancomycin).
- Monitor renal function (e.g., serum creatinine) in all patients: at baseline, during treatment (every 2-3 days or more frequently if clinically indicated) and at the end of therapy.
- Use with caution in patients taking medications that could be implicated in QTc interval prolongation or torsades de pointes; avoid use in patients with congenital long QT syndrome.
- May interfere with the following coagulation tests when blood samples are drawn within 18 hours of telavancin administration: prothrombin time, INR, aPTT, activated clotting time and coagulation based factor Xa tests.

REFERENCES

5, 95, 103, 135.

Full revision 2019; limited revision 2021

NAME OF MEDICATION
TEMSIROLIMUS

OTHER NAMES
Torisel ™

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of metastatic renal cell carcinoma.
- Treatment of Mantle cell lymphoma.

ADMINISTRATION

- **Ensure premedication has been administered; refer to Dosage section.**
- Intermittent IV infusion: two-step dilution process: 1) dilute 30 mg vial (1.2 mL of drug concentrate) with 1.8 mL of the supplied diluent to obtain 3 mL of a 10 mg/mL solution. Allow sufficient time for air bubbles to subside; 2) withdraw exact dose required for the patient and further dilute in 250 mL NS. Use non-PVC bags (i.e., glass, polypropylene, polyolefin or polyethylene) for dilution. Mix by inverting the bag; avoid excessive shaking. Infuse over 30 to 60 minutes using an administration set consisting of non-DEHP, non PVC material with an in-line polyethersulfone filter of no greater than 5 microns.
- Must be administered by an infusion pump.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity/infusion-related reactions: dyspnea, apnea, flushing, chest pain, hypotension and/or loss of consciousness. If a hypersensitivity reaction occurs, stop the infusion and give appropriate medical care. Observe the patient for at least 30-60 minutes. A risk-benefit assessment should be performed before continuing therapy. Administer an IV H_1 antagonist such as diphenhydramine 25-50 mg (if not given already) and/or an IV H_2 antagonist such as famotidine 20 mg, 30 minutes before restarting the infusion at a slower rate (length of temsirolimus infusion of up to 60 minutes).
- Cardiovascular: edema, QTc interval prolongation.
- GI: mucositis, stomatitis, nausea, anorexia, vomiting, diarrhea, constipation, dysgeusia, abdominal pain.
- Dermatologic: rash, pruritus.
- Endocrine and metabolic: hyperglycemia, hyperlipidemia, hypophosphatemia, hypokalemia.
- Hematologic: bleeding, anemia, neutropenia, leukopenia, lymphopenia, thrombocytopenia.
- Hepatic: increased liver enzymes.
- Renal: increased serum creatinine, renal failure.
- Respiratory: cough, dyspnea, interstitial lung disease.
- Fatigue.
- Abnormal wound healing may occur with use in the perioperative period.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- **Premedicate with an antihistamine (e.g., IV diphenhydramine 25-50 mg) 30 minutes before the infusion**.
- Renal cell carcinoma: 25 mg IV once weekly.
- Mantle cell lymphoma: 175 mg IV once weekly for 3 weeks then 75 mg IV once weekly.
- Dosage adjustment due to toxicity: refer to manufaturer's recommendations.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials in the fridge. Protect from light. Do not freeze.
- Vial diluted with supplied diluent is stable for 24 hours at room temp protected from light.
- Diluted solution is stable for 6 hours at room temp. Admixtures should be protected from excessive room light and sunlight.

MISCELLANEOUS

- As the drug has immunosuppressant effect, avoid live vaccination and contact with those who have recently received live vaccines.
- Temsirolimus is metabolized primarily by CYP3A4; if possible, avoid co-administration with drugs which are CYP3A4 inducers or inhibitors, otherwise temsirolimus dose may have to be modified.

REFERENCES

1, 5, 40, 95, 129, 165.

NAME OF MEDICATION
TENECTEPLASE

OTHER NAMES
TNKase ®

CLASSIFICATION
Thrombolytic

INDICATIONS
- Lysis of coronary artery thrombi associated with acute myocardial infarction (AMI).

ADMINISTRATION
- Follow the manufacturer's instructions on the correct use of the cannula device during reconstitution and administration.
- Reconstitute each 50 mg vial with 10 mL of the provided SWFI diluent using the red hub cannula syringe filling device; inject diluent, directing stream into the powder. Slight foaming is not unusual; large bubbles will dissipate if allowed to stand for several minutes. Gently swirl contents until dissolved; do NOT shake.
- IV direct: physician or RN; undiluted. Administer dose over 5 seconds; flush dextrose-containing IV lines with a saline solution before and after tenecteplase administration (risk of precipitation when tenecteplase comes into contact with a dextrose solution).

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rare; anaphylaxis, angioedema, laryngeal edema, rash, urticaria.
- Cardiovascular: reperfusion arrhythmia, hypotension.
- GI: nausea, vomiting.
- Hematologic: superficial bleeding (at vascular sites, arterial punctures) and internal bleeding (intracranial, retroperitoneal, GI, genitourinary or respiratory tract).
- Fever.

DOSAGE
- Single IV dose according to patient weight. Dose not to exceed 50 mg.

Patient weight (kg)	Tenecteplase dose (mg)	* Volume of drug to be administered (mL)
less than 60	30	6
60 to less than 70	35	7
70 to less than 80	40	8
80 to less than 90	45	9
90 or more	50	10

* From one vial reconstituted with 10 mL of SWFI.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials at room temp or in the fridge.
- Reconstituted solution is stable for 8 hours in the fridge.
- Incompatible with dextrose.

MISCELLANEOUS
- Intramuscular injections and nonessential handling of the patient should be avoided for the first few hours following treatment with tenecteplase.
- If an arterial puncture is necessary during the first few hours following tenecteplase administration, it is preferable to use an upper extremity vessel that is accessible to manual compression; pressure should be applied for at least 30 minutes.

REFERENCES
1, 5, 40, 95.

Full revision 2019; limited revision 2020

NAME OF MEDICATION
TERIPARATIDE

OTHER NAMES
Forteo ®, Osnuvo ™

CLASSIFICATION
Parathyroid hormone

INDICATIONS

- Treatment of severe osteoporosis in postmenopausal women and men.
- Treatment of glucocorticoid-induced osteoporosis in women and men.

ADMINISTRATION

- To prime the pen and prepare a dose, consult the manufacturer's directions.
- SC: into abdomen or thigh.
- The patient should remain either seated or lying down for the first several doses due to the risk of orthostatic hypotension.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, angioedema, anaphylaxis.
- Cardiovascular: transient orthostatic hypotension (onset within 4 hours of administration of first several doses).
- GI: nausea.
- CNS: dizziness, headache.
- Leg cramps, joint pain.
- Local reactions: erythema.

DOSAGE

- 20 mcg SC once daily.
- Dosage in renal impairment: no dosage adjustment needed in mild to moderate renal impairment; not recommended in severe renal impairment.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store pens and cartridges between 2-8°C. Protect from light. Do not freeze.
- The pen may be used for up to 28 days after the first injection. The in-use pen should be stored in the fridge at all times. A warming period is not required prior to dose administration.

MISCELLANEOUS

- Osnuvo ™ is classified as being biosimilar to Forteo ®; therefore, these products are not interchangeable.
- Supplemental calcium and vitamin D may be necessary if dietary intake is not adequate.

REFERENCES

1, 5, 95, 135.

Full revision 2019; limited revision 2021

NAME OF MEDICATION
TETANUS & DIPHTHERIA TOXOIDS ADSORBED

OTHER NAMES
Td Adsorbed

CLASSIFICATION
Vaccine

INDICATIONS

- Prevention of tetanus and diphtheria in persons 7 years of age and older.
- Tetanus prophylaxis in wound management.

ADMINISTRATION

- Shake well to obtain a uniform suspension.
- IM: in the deltoid.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, angioedema, anaphylaxis.
- CNS: brachial neuritis.
- Influenza-like symptoms (myalgia, arthralgia, chills, malaise, fever).
- Local reactions: pain, induration, swelling, discomfort, redness at injection site; severe arthus-type reaction (rare).

DOSAGE

- Primary immunization: 3 injections of 0.5 mL, with the first 2 doses given 4-8 weeks apart and the third dose given 6 to 12 months after the second dose.
- Booster dose (for individuals previously immunized): 0.5 mL every 10 years.
- Tetanus wound prophylaxis: 0.5 mL with or without tetanus immune globulin, depending on immunization history and the likelihood of contamination with tetanus bacilli. Refer to manufacturer's monograph for further details.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Do not freeze.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- This vaccine is an inactivated vaccine.
- Administration should be delayed, if possible, during an acute febrile illness or active infection of moderate to severe intensity.
- Tetanus toxoid and tetanus immune globulin should be administered in separate syringes at different sites.
- A fixed-combination preparation containing tetanus toxoid, reduced diphtheria toxoid and acellular pertussis vaccine adsorbed (Tdap) is preferred in some situations.

REFERENCES

1, 5, 31.

Full revision 2019; limited revision 2020

NAME OF MEDICATION
THIAMINE

OTHER NAMES
Vitamin B_1

CLASSIFICATION
Vitamin

INDICATIONS

- Treatment and prevention of acute thiamine deficiencies including beriberi, Wernicke's encephalopathy, alcohol withdrawal syndrome, and peripheral neuritis associated with pregnancy or pellagra.

ADMINISTRATION

- IV direct: physician or RN; dilute 100 mg with 10 mL of NS; inject over at least 5 minutes.
- Intermittent IV infusion: dilute dose in 50-100 mL of NS, D5W, or RL; infuse over 30 minutes.
- IM: provides rapid and complete absorption.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, angioedema; usually after multiple large doses.
- GI: nausea.
- Dermatologic: sweating, pruritus, urticaria.
- Feeling of warmth, weakness.
- Local reactions: pain and burning with IV administration; tenderness, induration and skin sclerosis with IM administration.

DOSAGE

- Critically ill thiamine-deficient patient or with malabsorption syndrome: 5-100 mg IV/IM three times daily.
- Beriberi: 5-30 mg IV/IM three times daily for 2 weeks followed by oral therapy.
- Wernicke encephalopathy:
  - Standard dosing regimen: 100 mg IV initially followed by 50-100 mg IV/IM daily until patient is eating well.
  - High dosing regimen: 500 mg IV three times daily for 2-3 days, followed by 250 mg IV/IM once daily for 3-5 days followed by oral thiamine.
- Alcohol withdrawal syndrome: 100-250 mg IV/IM once daily for 3-5 days, followed by oral thiamine.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and vials between 15-30°C. Protect from light. Do not freeze.
- Compatible with D5W, NS, RL.
- Stable in acidic solutions; unstable in neutral or alkaline solutions.
- Rapidly inactivated in solutions containing sulfites or bisulfites.

MISCELLANEOUS

- In thiamine deficiency, administer thiamine prior to IV dextrose to prevent sudden onset of Wernicke's encephalopathy.

REFERENCES

1, 4, 5, 40, 95, 135, 328.

Full revision 2019

NAME OF MEDICATION
THIOTEPA

OTHER NAMES
Tepadina ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Myeloablation prior to allogeneic or autologous stem cell transplant.
- Treatment of ovarian and breast cancer in combination with other chemotherapy agents.

ADMINISTRATION

- Reconstitute each 15 mg and 100 mg vial with 1.5 mL and 10 mL, respectively, of SWFI to obtain a concentration of 10 mg/mL. Gently mix until dissolution.
- IV direct (only for low doses such as those used for the treatment of ovarian and breast cancer): physician or RN. Filter reconstituted solution with a 0.22 micron filter. Do NOT use if the filtered solution is not clear or contains particulate matter. Dilute dose in NS to obtain a final concentration between 0.5 and 1 mg/mL; inject over 1-2 minutes.
- Intermittent IV infusion (mandatory for high doses such as those used for myeloablation): dilute dose in NS to obtain a final concentration between 0.5 and 1 mg/mL (e.g., dilute doses less than 500 mg in 500 mL and doses between 500 mg and 1000 mg in 1000 mL). Infuse through a 0.2 micron polyethersulfone in-line filter over 2-4 hours via a central line. Flush line with NS before and after infusion.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis, wheezing, urticaria, pruritus, rash.
- Cardiovascular: hypertension, arrhythmia.
- GI: nausea, vomiting, diarrhea, abdominal pain, anorexia, stomatitis.
- CNS: confusion, somnolence, dizziness, headache, encephalopathy, convulsion, paresthesia.
- Dermatologic: skin toxicity at high doses since the product is excreted in sweat to an appreciable amount. Wash (bath or shower) the patient at least twice daily for 48 hours after thiotepa administration when used for myeloablation.
- Hematologic: myelosuppression, febrile neutropenia, anemia, thrombocytopenia, epistaxis.
- Hepatic: veno-occlusive disease, hepatomegaly, jaundice.
- Ophthalmic: conjunctivitis, blurred vision.
- Otic: tinnitus.
- Renal: urinary retention, hemorrhagic cystitis.
- Myalgia, back pain, fever, fatigue, asthenia.
- Local reactions: inflammation and pain at injection site.
- Extravasation hazard: non-vesicant/non-irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Myeloablation: can vary from 125 to 481 mg/m^2/day IV starting within the week prior to transplantation.
- Oncology: 0.2 mg/kg (6-8 mg/m^2) IV once daily from day 1 to day 4-5, every 2-4 weeks OR 0.3-0.4 mg/kg IV on day 1, every 1-4 weeks.
- Dosage in renal impairment: no dosage reduction necessary in mild to moderate impairment, though use with caution.
- Dosage in hepatic impairment: use with caution.
- Consult specific protocol.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze.
- The reconstituted solution is colourless or slightly opalescent.
- Reconstituted solution is stable for 8 hours in the fridge.
- Diluted solution is stable for 24 hours in the fridge and for 4 hours at 25°C in NS.

MISCELLANEOUS

- Do not administer live vaccines concurrently with thiotepa.

REFERENCES

1, 5, 40, 95, 129, 135, 363.

NAME OF MEDICATION
TIGECYCLINE

OTHER NAMES
Tygacil ®

CLASSIFICATION
Antibiotic - tetracycline

INDICATIONS

- Treatment of complicated intra-abdominal infections and complicated skin and skin structure infections caused by susceptible organisms (includes gram-positive, gram-negative and anaerobes).
- Treatment of mild to moderate community-acquired pneumonia caused by susceptible organisms.

ADMINISTRATION

- Reconstitute each 50 mg vial with 5.3 mL of NS, D5W or RL to obtain a concentration of 10 mg/mL (note: there is an overfill). Gently swirl to dissolve. The reconstituted solution should be yellow-orange in colour; if not, discard.
- Intermittent IV infusion: dilute in 100 mL of NS, D5W or RL to a maximum concentration of 1 mg/mL; flush the line with NS, D5W or RL before and after administration. Inspect prior to administration; do NOT administer in presence of particulate matter or discolouration (green or black). Administer over 30-60 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, urticaria, anaphylactic/anaphylactoid reactions, Stevens-Johnson syndrome.
- GI: nausea, vomiting, diarrhea, abdominal pain, pancreatitis.
- CNS: headache.
- Hematologic: decreased fibrinogen, bleeding; may be serious.
- Hepatic: increased transaminases and alkaline phosphatase; hyperbilirubinemia.
- Local reactions: edema, inflammation, pain, phlebitis (infrequent).

DOSAGE

- 100 mg IV as initial dose, then 50 mg IV q12h for 5 to 14 days.
- Dosage in renal impairment: no dosage adjustment required, including in hemodialysis.
- Dosage in hepatic impairment: no dosage adjustment required in mild to moderate impairment. In severe impairment: 100 mg IV as initial dose then 25 mg IV q12h.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C.
- Stable for 24 hours at room temp (for 6 hours as a reconstituted solution in the vial and the remaining time, diluted in an IV bag).
- Diluted solution is stable for 48 hours in the fridge in NS or D5W not exceeding the maximal concentration of 1 mg/mL if transferred immediately in a plastic minibag after reconstitution.
- Compatible with NS, D5W, and RL.

MISCELLANEOUS

- Monitor blood coagulation parameters, including blood fibrinogen, prior to and during treatment with tigecycline.

REFERENCES

1, 4, 5, 40, 95, 135.

Full revision 2019; limited revision 2020

NAME OF MEDICATION	
TINZAPARIN	
OTHER NAMES	CLASSIFICATION
Innohep ®	Anticoagulant

INDICATIONS

- Prophylaxis of thrombosis related to surgery.
- Treatment of acute deep vein thrombosis (DVT) and pulmonary embolism (PE).
- Prevention of clotting of indwelling IV lines for hemodialysis and extracorporeal circulation.

ADMINISTRATION

- IV: (or into the arterial line of the dialyser) at the beginning of the dialysis.
- SC: lower abdomen; as an alternative, can also inject into the side of the thigh, in the lower back or the upper arm; always inject in the fat tissue, NOT in the muscle. With the thickness of skin held between the operator's thumb and finger, introduce the entire length of the needle vertically into the skin. To minimize bruising, injection sites should NOT be massaged after injection. Rotate site of injection daily.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, allergic reactions.
- Hematologic: bleeding, thrombocytopenia.
- Risk of spinal or epidural hematomas that can result in permanent paralysis when epidural or spinal anesthesia or spinal puncture is used in conjunction with tinzaparin.
- Hepatic: transient increase of liver transaminases (AST, ALT); reversible within 2-4 weeks of the last dose of tinzaparin.
- Local reactions: ecchymosis, erythema, irritation, pain.

Antidote: The anticoagulant effect of tinzaparin can be partially neutralized by protamine. Refer to protamine in this manual for more details.

DOSAGE

- Prophylaxis of thrombosis related to surgery:
  - Hip surgery: 50 anti-Xa international units/kg SC once daily for 7-10 days, starting 2 hours before surgery OR 75 anti-Xa international units/kg SC once daily for 7-10 days, starting postoperatively.
  - Knee surgery: 75 anti-Xa international units/kg SC once daily for 7-10 days, starting postoperatively.
  - General surgery: 3500 anti-Xa international units SC once daily for 7-10 days, starting 2 hours before surgery.
  - Dosage in obesity: dose should be calculated using actual body weight.

- Treatment of DVT and PE:
  - 175 anti-Xa international units/kg SC once daily until adequately anticoagulated with an oral anticoagulant.
  - Dosage in obesity: dose should be calculated using actual body weight.

- Anticoagulation during extracorporeal circulation and hemodialysis:
  - No risk of hemorrhage: bolus dose of 4500 anti-Xa international units via arterial line of the dialyser or IV at the beginning of dialysis. If hemodialysis lasts longer than 4 hours, a larger dose may be given.
  - Patients at higher risk of hemorrhage: use half of the dose. If hemodialysis lasts longer than 4 hours, an additional smaller dose may be given.
  - At TOH: bolus dose of 2500 anti-Xa international units via arterial line of the dialyser within the first 5 minutes of dialysis session. If clotting in circuit occurs, increase dose to 3500 anti-Xa international units at the next session; if clotting persists, increase dose further at the next session.

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DOSAGE (Cont.)

- Dosage in renal impairment: no accumulation in patients with a CrCl of 20 mL/min and higher. For more severe renal impairment, consider dosage adjustment and/or monitoring of anti-Xa levels.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes between 15-25°C.
- Multi-dose vials are stable for up to 6 punctures over a 28-day period at room temp.
- Solution from multi-dose vials is stable for 10 days at room temp or in the fridge in tuberculin syringes. Upon storage, needle should be changed prior to administration.

MISCELLANEOUS

- Tinzaparin should not be given to patients with a history of heparin-induced thrombocytopenia (risk of cross-reactivity), unless an *in vitro* platelet aggregation test is negative.
- For laboratory monitoring of effect, anti-Xa methods are recommended (although not routinely done, monitoring is recommended in special cases such as extremes of weight, pregnancy and renal failure).
- Monitor closely the elderly patients with low body weight (less than 45 kg) or with decreased renal function.
- As the multi-dose vial contains benzyl alcohol, it is not recommended for use during pregnancy or in neonates.
- Derived from porcine tissue.

REFERENCES

5, 87, 95, 119, 135, 218, 275.

NAME OF MEDICATION
TIROFIBAN

OTHER NAMES
Aggrastat ®

CLASSIFICATION
Platelet aggregation inhibitor

INDICATIONS
- Management of patients with non-ST-elevation acute coronary syndrome, including those subsequently undergoing percutaneous coronary interventions (PCI). Use in conjunction with antiplatelet therapies (e.g., ASA with a P2Y12 platelet ADP-receptor antagonist such as clopidogrel, prasugrel and ticagrelor) and anticoagulants (e.g., heparin or a LMWH).

ADMINISTRATION
- Continuous IV infusion: use premixed bags of 250 mL (50 mcg/mL in NS) for all IV administration.
- For patients who undergo PCI after 4 hours of diagnosis: administer an initial loading dose over 30 minutes followed by a continuous maintenance IV infusion. Refer to Dosage section for rate.
- For patients who undergo PCI within 4 hours of diagnosis: administer an initial bolus dose over 3 minutes followed by a continuous maintenance IV infusion. Refer to Dosage section for rate.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: rash, urticaria, anaphylactic reactions.
- Cardiovascular: edema, bradycardia, dissection of the coronary artery.
- GI: nausea.
- CNS: headache, dizziness.
- Hematologic: bleeding, thrombocytopenia (reversible within 4-6 days after stopping therapy).
- Pelvic and leg pain.
- Fever.

DOSAGE
- Patients not planned to undergo angiography for at least 4 hours after diagnosis:
  - IV loading dose at an initial rate of 0.4 mcg/kg/min for 30 minutes, followed by a maintenance IV infusion at a rate of 0.1 mcg/kg/min.
  - In patients without signs of refractory ischemic symptoms and who do not proceed into angiography and angioplasty, continue infusion for at least 48 hours.
  - For patients proceeding into angiography and angioplasty, continue infusion throughout both procedures and for at least 12 hours, to a maximum of 24 hours after angioplasty; discontinue the infusion once patient is stable and no more coronary intervention is planned.
- High risk patients who undergo PCI within 4 hours of diagnosis: at the start of PCI, IV bolus dose of 25 mcg/kg over 3 minutes, followed by a maintenance IV infusion at a rate of 0.15 mcg/kg/min for 12-24 hours.
- Dosage in renal impairment: decrease dose by 50% if CrCl is less than 30 mL/min.
- Dosage in hepatic impairment: no dosage adjustment is necessary in mild to moderate disease; unknown in severe hepatic impairment.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store premixed bags between 15-30°C. Protect from light during storage. Do not freeze.
- Compatible with D5W and NS.
- May be co-administered with heparin through the same line.

MISCELLANEOUS
- Monitor for potential bleeding; caution when used with other drugs that affect hemostasis. When bleeding cannot be controlled with pressure, infusion of tirofiban and heparin should be discontinued. Transfusions may be given if required.
- Monitor platelet counts, hemoglobin and hematocrit prior to treatment, within 6 hours following the bolus or loading infusion and then daily during therapy.

REFERENCES
1, 4, 5, 40, 95.

Full revision 2019; limited revision 2020

NAME OF MEDICATION
TOBRAMYCIN

OTHER NAMES
Nebcin ®

CLASSIFICATION
Antibiotic - aminoglycoside

INDICATIONS

- Treatment of serious infections caused by gram-negative organisms such as *Pseudomonas aeruginosa*, *Proteus*, *Klebsiella*, *E. coli*, *Enterobacter*, *Acinetobacter*, *Citrobacter* and *Serratia* species.

ADMINISTRATION

- Reconstitute the 1.2 g vial with 30 mL of SWFI to obtain a concentration of 40 mg/mL. Also available in solutions of 10 and 40 mg/mL.
- Intermittent IV infusion: dilute in 50-100 mL of D5W or NS (for children, volume of diluent should be proportionally less than for adults). Administer over 20-60 minutes.
- IM.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, urticaria.
- GI: nausea, vomiting, diarrhea.
- CNS: headache, lethargy.
- Otic: vestibular and auditory toxicity (risk factor: peak serum levels greater than 12 mg/L with traditional dosing).
- Renal: nephrotoxicity (risk factor: trough serum levels greater than 2 mg/L).
- Neuromuscular blockade (rare).
- Fever.
- Local reactions: pain.

DOSAGE

- Traditional dosing: 1-2 mg/kg/dose IV or IM with dosing interval according to creatinine clearance:

CrCl (mL/min)	greater than 60	60-40	39-20	less than 20 or hemodialysis
Interval (hr)	8	12	24	frequency as per serum concentrations

  Refer to Miscellaneous section for monitoring serum levels.
- Extended interval dosing: 4-7 mg/kg/dose (10 mg/kg/dose for cystic fibrosis patients) IV or IM once daily in selected patients. Refer to Miscellaneous section for monitoring serum levels.
- Doses should be calculated using actual body weight, except in obese patients.
- Dosage in obesity (20-30% above ideal body weight [IBW]): dose should be calculated using dosing body weight (DBW) = IBW (kg) + 0.4 X (actual body weight - IBW).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store between 15-30°C. Protect from light.
- Compatible with D5W, NS and RL.
- Reconstituted solution is stable for 24 hours at room temp or 96 hours in the fridge.
- Stable for 48 hours at room temp in D5W, NS, D5-NS at concentrations of 0.2 mg/mL and 1 mg/mL in PVC containers.
- Incompatible with heparin.

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MISCELLANEOUS

- Monitoring serum levels:
  - Traditional dosing: serum levels pre-dose: taken 5 minutes before next dose; should be less than 2 mg/L.
    serum levels post-dose: taken 30 minutes after the end of IV infusion or 60 minutes after the IM injection; should be less than 12 mg/L.
  - Extended interval dosing: serum levels pre-dose: taken 5 minutes before next dose; should be undetectable (i.e., less then 0.3 mg/L).
    serum levels post-dose: no need to monitor.

REFERENCES

1, 5, 6, 40, 95, 135, 208, 303, 370, 371, 399.

NAME OF MEDICATION
TOCILIZUMAB

OTHER NAMES
Actemra ®

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS
- Treatment of adult patients with moderate to severe active rheumatoid arthritis (IV and SC formulations).
- Treatment of giant cell arteritis (GCA) in adults (SC formulation only).
- Treatment of active systemic juvenile idiopathic arthritis (SJIA) in children 2 years of age and older, who had an inadequate response to NSAIDs and systemic corticosteroids (IV and SC formulations).
- Treatment of polyarticular juvenile idiopathic arthritis (PJIA) in children 2 years of age and older who had an inadequate response to previous therapy with DMARDs (IV and SC formulations).
- Treatment of severe cytokine release syndrome (CRS) secondary to treatment with blinatumomab or with chimeric antigen receptor (CAR) T-cell therapy (e.g., tisagenlecleucel) in adults and children 3 years of age and older (IV formulation only).
- In addition to corticosteroids in the treatment of coronavirus disease (COVID-19; SARS-CoV-2 infection) in moderately- or critically-ill hospitalized patients requiring supplemental oxygen or respiratory support.

ADMINISTRATION
- **There are 2 formulations: one for IV use (vial) and one for SC use (prefilled syringe and autoinjector). Ensure using the correct one.**
- Intermittent IV infusion: from a 100 mL NS solution bag (50 mL NS if patient weighs less than 30 kg), withdraw and discard the exact same volume of solution as the volume of tocilizumab to be added to the bag. Using the IV formulation, add tocilizumab slowly into the bag to get a final volume of exactly 100 mL (50 mL if patient weighs less than 30 kg). Gently mix by inverting the bag slowly to avoid foaming. Allow bag to reach room temp prior to infusion. Administer over 60 minutes.
- SC: ensure using the SC formulation; allow the prefilled syringe or autoinjector to reach room temp. Inject SC the full amount in the syringe (162 mg/0.9 mL) into abdomen, thigh or upper arm; rotate injection sites. Ensure site of injection is free of moles, scars and open sores and is not tender, bruised, red or hard.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylactoid reactions, anaphylaxis.
- Infusion-related reactions: occur during or within 24 hours of infusion; hypertension, headache, dizziness, skin reactions (rash, pruritus, urticaria).
- GI: nausea, vomiting, diarrhea, dyspepsia, abdominal pain, mouth ulceration; gastrointestinal perforation, especially in patients with prior ulcers or diverticulitis.
- Endocrine and metabolic: elevation of lipids (total cholesterol, triglycerides, Low-Density-Lipoprotein cholesterol).
- Hematologic: reduction in neutrophil count and platelet count.
- Hepatic: elevation of liver enzymes (AST, ALT), acute liver failure.
- Immunosuppression with increased risk of infections (bacterial, viral, fungal); more frequent in the elderly.
- Local reactions: erythema, pruritus, pain and hematoma at SC injection sites.

DOSAGE
Adults:
- Rheumatoid arthritis using IV tocilizumab (to use IV formulation):
  - Initial dose: 4 mg/kg IV every 4 weeks; may increase to 8 mg/kg based on clinical response.
  - For patients weighing more than 100 kg, do not exceed 800 mg/dose.
- Rheumatoid arthritis using SC tocilizumab (to use SC formulation):
  - If patient weighs less than 100 kg: 162 mg SC every other week, followed by an increase to every week based on clinical response.
  - If patient weighs 100 kg or more: 162 mg SC every week.
- Rheumatoid arthritis switching from IV to SC administration: give the first SC dose at the time of the next scheduled IV dose.
- GCA using SC tocilizumab (to use SC formulation): 162 mg SC every week (with a tapering course of corticosteroids); may be decreased to 162 mg SC every other week (with a tapering course of corticosteroids) based on clinical considerations.
- CRS using IV tocilizumab (to use IV formulation): 8 mg/kg (maximum of 800 mg/dose) IV. May administer up to 3 additional doses; dosing intervals may differ between protocols. [At TOH: 8 mg/kg (maximum of 800 mg/dose) IV q6-8h prn, with a maximum of 3 doses in 24 hours. Total maximum of 4 doses]. .../Cont.

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DOSAGE (Cont.)
- In the treatment of COVID-19 disease: 8 mg/kg (maximum of 800 mg/dose) IV once, in addition to corticosteroids; may be repeated once 8-24 hours later if no improvement. (At TOH: single dose only of 400 mg IV).

Pediatrics:
- SJIA using IV tocilizumab (to use IV formulation):
  - If patient weighs less than 30 kg: 12 mg/kg IV every 2 weeks.
  - If patient weighs 30 kg or more: 8 mg/kg IV every 2 weeks.
- SJIA using tocilizumab (to use SC formulation):
  - If patient weighs less than 30 kg: 162 mg SC every 2 weeks.
  - If patient weighs 30 kg or more: 162 mg SC every week.
- PJIA using IV tocilizumab (to use IV formulation):
  - If patient weighs less than 30 kg: 10 mg/kg IV every 4 weeks.
  - If patient weighs 30 kg or more: 8 mg/kg IV every 4 weeks.
- PJIA using SC tocilizumab (to use SC formulation):
  - If patient weighs less than 30 kg: 162 mg SC every 3 weeks.
  - If patient weighs 30 kg or more: 162 mg SC every 2 weeks.
- CRS using IV tocilizumab (to use IV formulation):
  - If patient weighs less than 30 kg: 12 mg/kg IV.
  - If patient weighs 30 kg or more: 8 mg/kg IV (maximum of 800 mg/dose).
  - May administer up to 3 additional doses; dosing interval may differ between protocols.

Adults and pediatrics:
- Dosage in renal impairment: no dosage adjustment needed in mild renal impairment; has not been studied in patients with moderate to severe renal impairment.
- Dosage in hepatic impairment: not recommended.
- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials, prefilled syringes and autoinjectors between 2-8°C. Protect from light and freezing.
- Diluted IV solution is colourless to a pale yellow.
- The diluted IV solution is stable for 24 hours in the fridge and at room temp (up to 30°C) in NS in a PVC, polyethylene, polypropylene or glass container.
- Stable for 24 hours at 30°C followed by 14 days at 2-8°C in NS at concentrations of 0.1 mg/mL and 8 mg/mL in PVC and polypropylene bags.
- The prefilled syringe and the autoinjector for SC administration are stable for 8 hours at room temp (below 30°C).

MISCELLANEOUS
- Tocilizumab should not be used with other biologic DMARDs (e.g., abatacept, adalimumab, anakinra, etanercept, infliximab, rituximab) because of possible increased risk of immunosuppression and infections.
- It would be prudent not to use tocilizumab in patients on azathioprine or cyclophosphamide.
- Use with caution if ANC is less than 2 x 10^9/L, platelet count less than 100 x10^9/L, or AST or ALT exceeds 1.5 x ULN.
- Live and live attenuated vaccines should not be given concurrently with tocilizumab.
- Patients should be evaluated for tuberculosis risk factors and tested for latent and active infection before initiating therapy.

REFERENCES
1, 5, 12, 40, 95, 135, 401, 403, 404, 506, 568, 569, 570.

Full revision 2019; limited revision 2020, 2021

NAME OF MEDICATION
TOPOTECAN

OTHER NAMES
Hycamtin ®

CLASSIFICATION
Antineoplastic

INDICATIONS
- Treatment of metastatic carcinoma of the ovary after failure of initial or subsequent therapy.
- Treatment of sensitive small cell lung cancer after failure of first-line chemotherapy.
- Treatment of other cancers.

ADMINISTRATION
- Reconstitute the 4 mg vial with 4 mL of SWFI to obtain a 1 mg/mL solution.
- Intermittent IV infusion: dilute in 50 to 100 mL of NS or D5W to a concentration of 0.02-0.5 mg/mL; infuse over 30 minutes.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: angioedema, anaphylactic/anaphylactoid reactions.
- GI: nausea, vomiting, anorexia.
- Dermatologic: rash.
- Hematologic: anemia, neutropenia, thrombocytopenia.
- Respiratory: dyspnea, cough, pneumonia.
- Fever.
- Extravasation hazard: irritant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Usual: 1.5 mg/m^2 IV daily on days 1 to 5 of a 21 day cycle.
- Dosage in renal impairment:

CrCl (mL/min)	60-40	39-20	less than 20
Dose	1.5 mg/m^2	0.75 mg/m^2	contraindicated

- Dosage in hepatic impairment: no dosage adjustment if bilirubin is less than 171 mcmol/L.
- Dosage adjustment due to toxicity:
  - Grade 4 neutropenia for 7 days or longer, or febrile neutropenia, or cycle delay for hematologic toxicity: reduce subsequent dose by 0.25 mg/m^2 or add filgrastim (G-CSF) with next cycle.
  - Platelets of 25 X 10^9/L or less, or Grade 3 GI or organ toxicity: reduce subsequent dose by 0.25 mg/m^2.
  - Grade 4 GI or organ toxicity: discontinue treatment.
  - Do not retreat until toxicity of Grade 2 or less and absolute neutrophil count is at least 1 X 10^9/L, platelet count is at least 100 X 10^9/L, and hemoglobin is at least 90 g/L (after transfusion if necessary).
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light.
- Reconstituted solution is stable for 28 days in the fridge or at room temp at a concentration of 1 mg/mL.
- Stable for 24 hours at room temp or 7 days in the fridge in NS or D5W at a concentration of 0.025 mg/mL in PVC containers and at a concentration of 0.05 mg/mL in PVC, polyolefin and glass containers.
- Stable for 28 days in the fridge or at room temp, protected from light in NS or D5W at concentrations of 0.01-0.05 mg/mL in PVC containers.
- Stable for 17 days at room temp, exposed to light, in NS at a concentration of 0.01 mg/mL in PVC containers.

MISCELLANEOUS

REFERENCES
1, 4, 5, 6, 40, 95, 129, 135, 165.

Full revision 2019; limited revision 2020

NAME OF MEDICATION
TRACE METALS

OTHER NAMES
Micro+6 ®, Trace elements

CLASSIFICATION
Trace element

INDICATIONS

- Nutritional supplement to intravenous solutions given for total parenteral nutrition (TPN) to prevent depletion of endogenous stores of trace metals and development of subsequent deficiency symptoms.

ADMINISTRATION

- Intermittent IV infusion, Continuous IV infusion: dilute in TPN solutions and administer at prescribed rate.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: from iodine: urticaria, angioedema, anaphylaxis.
- Local reactions: phlebitis if not diluted.

DOSAGE

- Dose as required (in clinical practice, usually 1 mL daily).

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Protect from light. Do not freeze.
- Compatible in standard solutions used to manufacture TPN solutions.
- Compatible with electrolytes and vitamins that are commonly added to TPN solutions.

MISCELLANEOUS

- Each mL of Micro+6 ® concentrate from Sandoz contains: zinc 5 mg/77 mcmol, copper 1 mg/16 mcmol, manganese 0.5 mg/9 mcmol, chromium 10 mcg/0.2 mcmol, selenium 60 mcg/0.76 mcmol and iodine 75 mcg/0.6 mcmol.
- Each mL of Micro+6 ® Pediatric from Sandoz contains: zinc 3 mg/46 mcmol, copper 0.4 mg/6.4 mcmol, manganese 0.1 mg/1.8 mcmol, chromium 4 mcg/0.08 mcmol, selenium 20 mcg/0.25 mcmol and iodine 60 mcg/0.5 mcmol.
- Available as single elements and selected combined elements in various strengths.

REFERENCES

5, 40, 208.

Full revision 2019

NAME OF MEDICATION
TRANEXAMIC ACID

OTHER NAMES
Cyklokapron ®

CLASSIFICATION
Hemostatic

INDICATIONS
- Treatment or prophylaxis of systemic or local hyperfibrinolysis when oral administration is not possible.

ADMINISTRATION
- IV direct: physician only; undiluted; administration at a maximum rate of 100 mg/min.
- Intermittent IV infusion: dilute with at least 50 mL of a compatible solution and infuse at a maximum rate of 100 mg/min.
- Continuous IV infusion: dilute in a large volume of compatible solution (at TOH: dilute 1000 mg in 500 mL of D5W or NS to get a concentration of 2 mg/mL); refer to Dosage section for infusion rate.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylaxis, dyspnea, throat tightening, facial flushing, dermatitis.
- Cardiovascular: hypotension, especially when administered too rapidly.
- GI: nausea, vomiting, diarrhea.
- CNS: dizziness, seizures.
- Thrombosis.

DOSAGE
- Usual dose range is 500-1000 mg IV q8h OR 25-50 mg/kg IV daily (administered in divided doses or as a continuous IV infusion).
- Dental surgery in patients with coagulopathies: 10 mg/kg IV immediately or 2 hours before surgery and q6-8h postoperatively for 2-8 days.
  - Dosage in renal impairment:

Serum creatinine (mcmol/L)	Dosage in dental surgery
120-250	10 mg/kg twice daily
250-500	10 mg/kg q24h
greater than 500	10 mg/kg q48h or 5 mg/kg q24h

- Cardiac surgery (prophylaxis): Various protocols exist; for example: 100 mg/kg IV preoperatively with an additional 50 mg/kg IV postoperatively OR 15 mg/kg IV followed by 1 mg/kg/hr by continuous IV infusion for 5-6 hours OR 30 mg/kg IV followed by 16 mg/kg/hr by continuous IV infusion until sternal closure with an additional 2 mg/kg to cardiopulmonary bypass circuit OR 10-15 mg/kg IV followed by 1-1.5 mg/kg/hr by continuous IV infusion with an additional 2-2.5 mg/kg to cardiopulmonary bypass circuit.
  - Dosage in renal impairment:

Serum creatinine (mcmol/L)	Maintenance infusion during cardiac surgery (based on an infusion of 2 mg/kg/hr for normal renal function)
140-290	1.5 mg/kg/hr (25% reduction)
290-580	1 mg/kg/hr (50% reduction)
greater than 580	0.5 mg/kg/hr (75% reduction)

- Total knee replacement, blood loss reduction: various protocols exist; for example: 10 mg/kg IV given 10-30 minutes before tourniquet deflation followed by either 10 mg/kg IV given 3 hours later or 1 mg/kg/hr by continuous IV infusion beginning postoperatively and continued for 6 hours.
- Postpartum hemorrhage (treatment): 1000 mg IV or 10 mg/kg/dose given within 3 hours of birth. May give a second dose if bleeding continues after 30 minutes or if bleeding stops and restarts within 24 hours of the first dose.
- Traumatic hemorrhage: 1000 mg IV followed by a continuous IV infusion of 1000 mg over 8 hours.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules and vials at room temp. Protect from light.
- Compatible with D5W, NS and RL.
- Stable for 24 hours at room temp in D5W or NS at a concentration of 2 mg/mL in PVC bags.
- Stable for 24 hours at room temp or in the fridge in D5W at concentrations of 10 mg/mL and 20 mg/mL in PVC, polyethylene and polypropylene bags.
- Stable for 7 days at room temp or in fridge in NS at concentrations of 10 mg/mL and 20 mg/mL in PVC, polyethylene and polypropylene bags.
- Stable for 90 days at room temp or in the fridge, protected from light, in NS at a concentration of 15.4 mg/mL in ethylene/propylene copolymer plastic containers.
- Do not mix with solutions containing penicillin.

MISCELLANEOUS

- The preparation should not be given to patients with acquired disturbances of colour vision, active thromboembolic disease, a history or risk of thrombosis or subarachnoid hemorrhage.
- Tranexamic acid is 10 times more potent *in vitro* than aminocaproic acid.

REFERENCES

2, 3, 4, 5, 40, 95, 96, 135, 200, 208, 232, 395, 396.

NAME OF MEDICATION
TRASTUZUMAB

OTHER NAMES
Herceptin ®, Herceptin ® SC, HER2 monoclonal antibody, Herzuma ®, Kanjinti ™, Ogivri ™, Trazimera ®

CLASSIFICATION
Antineoplastic, Monoclonal antibody

Do NOT confuse trastuzumab (Herceptin ®, Herzuma ®, Kanjinti ™, Ogivri ™, Trazimera ®) with Enhertu ™ trastuzumab deruxtecan or Kadcyla ® trastuzumab emtansine. Trastuzumab does not contain a cytotoxic component, unlike Enhertu ™ trastuzumab deruxtecan and Kadcyla ® trastuzumab emtansine. These products have different uses, toxicities, administration and dosages. This monograph is specific to TRASTUZUMAB.

INDICATIONS
- Treatment of patients with early or metastatic breast cancer whose tumours overexpress the HER2 protein (IV and SC formulations).
- Treatment of patients with metastatic gastric cancer whose tumours overexpress the HER2 protein (IV formulation only).

ADMINISTRATION
- **Consider premedication if patient previously experienced a hypersensitivity or infusion-related reaction; refer to Potential Administration Hazards section**.
- **There are 2 formulations: one for IV use and one for SC use. Both are available as vials. Ensure using the correct vial.**
- For IV use: using the IV formulation, reconstitute each 440 mg vial by slowly injecting 20 mL of bacteriostatic water for injection; for patients with known hypersensitivity to benzyl alcohol, reconstitute with SWFI. Reconstitute each 150 mg vial by slowly injecting 7.2 mL of SWFI. Concentration for both vial sizes: 21 mg/mL. Do NOT shake the vial; swirl gently. Allow the vial to stand for 5 minutes before withdrawing dose.
- Intermittent IV Infusion (mandatory): further dilute dose in 250 mL NS. Gently invert the bag to mix solution. Do NOT shake to avoid foaming. Refer to Dosage section for administration rate. Observe patient closely during infusion, for 60 minutes after the first infusion and 30 minutes after subsequent infusions.
- SC (Herceptin ® SC brand only): using the SC formulation, withdraw the dose into a syringe. If the dose is not administered immediately, replace transfer needle by a syringe closing cap to avoid drying of the solution. Inject into thigh over 2-5 minutes. Alternate between right and left thigh, giving new injections at least 2.5 cm from previous site; do not administer into areas where skin is red, bruised, tender or hard.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: infrequent with IV use; may occur with SC administration; anaphylaxis, urticaria, erythema, rash, pruritus, angioedema, bronchospasm, dyspnea, wheezing, hypoxia, acute respiratory distress, hypotension, and tachycardia; reactions may be fatal; more common with initial IV infusion and generally occur during or immediately after IV infusion but may be delayed more than 6 hours post-infusion; rarely, symptoms may improve initially but then worsen up to one week post-infusion. Consider permamently discontinuing therapy in cases of severe reactions; however, some patients have been successfully re-treated with administration of **premedication** (antihistamines +/- corticosteroids); others have experienced a second severe reaction. Patients with dyspnea at rest due to advanced cancer and/or cormorbidities are at increased risk of a fatal infusion reaction; use with extreme caution.
- Infusion-related reactions (IV administration): serious reactions are less frequent and may manifest as hypersensitivity (see above). Mild to moderate reactions are more common and involve chills, fever, rigors, nausea, vomiting, pain, headache, cough, dizziness, dyspnea, rash and asthenia; more common with first infusion; management includes interrupting and/or reducing infusion rate and controlling symptoms with diphenhydramine, acetaminophen, meperidine, beta-agonists, and corticosteroids, prn; infusion may be restarted once symptoms have resolved. Patient should be observed for 24 hours for infusion-related reactions. **Consider premedication** with antihistamines +/- corticosteroids and reducing infusion rate for subsequent infusions.
- Cardiovascular: arrhythmia, tachycardia, hypertension, hypotension; may be serious.
- GI: nausea, vomiting, diarrhea.
- Respiratory: cough, dyspnea, pneumonitis, acute respiratory distress; may be serious.
- Local reactions: pain, erythema, edema, hemorrhage, inflammation.

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DOSAGE

- Early breast cancer:
  - Using the IV formulation: loading dose of 4 mg/kg IV over 90 minutes followed by weekly infusions of 2 mg/kg over 30 minutes if the loading dose was well tolerated, otherwise give over 90 minutes, OR
  - Using the IV formulation: loading dose of 8 mg/kg IV over 90 minutes followed by 6 mg/kg every 3 weeks over 30 minutes if the loading dose was well tolerated, otherwise give over 90 minutes, OR
  - Using the SC formulation: 600 mg SC every 3 weeks.
- Metastatic breast cancer:
  - Using the IV formulation: loading dose of 4 mg/kg IV over 90 minutes followed by weekly infusions of 2 mg/kg over 30 minutes if the loading dose was well tolerated; otherwise give over 90 minutes, OR
  - Using the SC formulation: 600 mg SC every 3 weeks.
- Metastatic gastric cancer: using the IV formulation: loading dose of 8 mg/kg IV over 90 minutes followed by 6 mg/kg every 3 weeks over 30 minutes if the loading dose was well tolerated; otherwise give over 90 minutes.
- If IV treatment is stopped for more than one week, the loading dose should be readministered.
- Dosage adjustment due to toxicity: in case of cardiotoxicity, refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

IV formulation:
- Store vials between 2-8ºC.
- Vials of Ogivri ™ and Trazimera ® can be stored at room temp for 3 months.
- Reconstituted vial with bacteriostatic water for injection is stable for 28 days in the fridge; if reconstituted with SWFI, use immediately and discard the vial.
- Diluted solution is stable for 24 hours in the fridge, at room temp, or at 30°C in NS in PVC, polyethylene or polypropylene bags.
- Herceptin ® brand only: stable for 28 days at room temp and in the fridge (both with or without protection from light) in NS at concentrations of 0.4 mg/mL and 4 mg/mL in polypropylene bags.
- Herceptin ® brand only: stable for 180 days at room temp and in the fridge protected from light in NS at concentrations of 0.8 mg/mL and 2.4 mg/mL in polyolefin bags.
- Incompatible with D5W.

*SC formulation (*Herceptin ® SC):
- Store vials between 2-8ºC. Protect from light. Do not freeze.
- The solution in the vial should be colourless to yellowish, clear to opalescent.
- Stable for 48 hours in the fridge in polypropylene syringes and then stable for a single excursion of 6 hours at room temp (up to 30ºC) in diffused daylight.

MISCELLANEOUS

- Baseline cardiac assessment and monitoring of cardiac function during treatment should be done for all patients.
- Herzuma ®, Kanjinti ™, Ogivri ™ and Trazimera ® are classified as being biosimilar to Herceptin ®; therefore these products are not interchangeable.

REFERENCES

1, 5, 40, 95, 129, 135, 165, 208.

Full revision 2019; limited revision 2020, 2021

NAME OF MEDICATION
TREPROSTINIL

OTHER NAMES
Remodulin ®, Treprostinil sodium

CLASSIFICATION
Vasodilator

INDICATIONS

- Long-term treatment of pulmonary arterial hypertension (PAH) in New York Heart Association Class II to IV patients who did not respond adequately to conventional therapy.

ADMINISTRATION

- Continuous IV infusion: ***continuous BP monitoring, cardiac monitoring, while patient is in critical care areas***. Dilute the appropriate amount of drug in a 50 or 100 mL size reservoir (in PVC, polypropylene or glass) with a sufficient volume of SWFI or NS to achieve the desired total volume of the reservoir and concentrations as low as 0.004 mg/mL. Refer to Dosage section and to manufacturer's product monograph to get the equations for calculation of the concentration to prepare and selection of appropriate vial. The IV infusion should be administered through a central venous catheter; a temporary peripheral IV cannula, placed in a large vein, may be used for short term administration. Using a peripheral access for more than a few hours may be associated with an increased risk of thrombophlebitis. Use an infusion set with a 0.2 or 0.22 micron in-line filter and an anti-siphon valve.
- SC infusion (preferred): ***continuous BP monitoring, cardiac monitoring, while patient is in critical care areas***. Transfer solution from vial into a syringe reservoir (in PVC, polypropylene or glass) without any dilution. Refer to Dosage section and to manufacturer's product monograph to get the equation for calculation of the SC infusion rate. Rotate site in case of continued local reactions.
- Must be administered by an infusion pump (SC and IV).
- **Avoid abrupt cessation or sudden large rate reductions as symptoms of PAH can recur promptly.**

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensititity: pruritus, rash, angioedema.
- Cardiovascular: hypotension, vasodilation, edema.
- GI: nausea, diarrhea.
- CNS: headache.
- Hematologic: increased risk of bleeding, due to antiplatelet effect.
- Jaw pain.
- Local reactions: very frequent, can be severe: pain, erythema, induration, rash; thrombophlebitis with peripheral IV access.

DOSAGE

- Initial IV or SC infusion rate: 1.25 ng/kg/min; if not tolerated initially, reduce rate to 0.625 ng/kg/min.
- Adjust rate slowly to achieve improvement of PAH symptoms with an acceptable side effect profile. Titrate in increments of no more than 1.25 ng/kg/min at weekly intervals for the first 4 weeks and then no more than 2.5 ng/kg/min at weekly intervals. Limited experience with rates greater than 40 ng/kg/min.
- If infusion is interrupted then restarted within a few hours, may resume infusion at the same rate; interruptions for longer periods may require retitration.
- Dosage in renal impairment: no dosage adjustment necessary; titrate rate slowly.
- Dosage in hepatic impairment: for patients with mild to moderate impairment, initial rate of 0.625 ng/kg/min (calculate rate by using **ideal** body weight) and titrate cautiously; not studied in patients with severe hepatic impairment.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C. Excursions are permitted between 2-30°C.
- Do not use a vial for more than 30 days after initial puncture.
- Undiluted solution is stable for 72 hours at 37°C in PVC, glass or polypropylene syringes.
- Undiluted solution is stable for 60 days at 4°C, 23°C, and 37°C in plastic syringe pump reservoirs.
- Prior to administration, diluted solution is stable for 4 hours at room temp or 24 hours in the fridge in NS or SWFI at concentrations as low as 0.004 mg/mL in PVC, polypropylene or glass containers. Administration time limit for the diluted solution is 48 hours at 40°C.

MISCELLANEOUS

REFERENCES

1, 4, 5, 40, 95, 135, 143, 208.

NAME OF MEDICATION
USTEKINUMAB

OTHER NAMES
Stelara ®

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS

- Treatment of adults with chronic moderate to severe plaque psoriasis who are candidates for phototherapy or systemic therapy.
- Treatment of adolescents (12 years of age and older) with chronic moderate to severe plaque psoriasis who are not controlled by or are intolerant to phototherapy or other systemic therapy.
- Treatment of adults with active psoriatic arthritis, as monotherapy or in combination with methotrexate.
- Treatment of adults with moderate to severe Crohn's disease or ulcerative colitis who had an inadequate response or are intolerant to other therapies.

ADMINISTRATION

- Do NOT shake. Inspect solution before administration; do NOT use if cloudy or discoloured.
- Intermittent IV infusion: from a 250 mL NS or 1/2NS bag, withdraw a volume equal to the drug volume for the patient's dose, then add the drug dose to the bag. Gently mix. Administer through a 0.2 micron in-line low-protein-binding filter over at least 60 minutes.
- SC: upper arm, buttock, top of the thigh or abdomen. Rotate injection site.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, angioedema, rash, urticaria.
- GI: nausea, vomiting, diarrhea.
- CNS: headache.
- Dermatologic: pruritus; exfoliative dermatitis and erythrodermic psoriasis within a few days of administration (rare but can be severe).
- Respiratory: nasopharyngitis, upper respiratory tract infection, oropharyngeal pain.
- Infection (may be severe), arthralgia, back pain, fatigue.
- Local reactions: pain, swelling, erythema, pruritus, induration, irritation, hemorrhage, hematoma.

DOSAGE

Adults:
- Plaque psoriasis:
  - Patients weighing 100 kg or less: 45 mg SC at weeks 0, 4, and then every 12 weeks.
  - Patients weighing over 100 kg: 45 mg or 90 mg SC at weeks 0, 4, and then every 12 weeks.
  - Patients who inadequately respond to dosing every 12 weeks may be treated as often as every 8 weeks.
  - In case of an interruption in therapy, retreatment may be initiated at the initial dosing interval.
- Psoriatic arthritis:
  - Patients weighing 100 kg or less: 45 mg SC at weeks 0, 4, and then every 12 weeks.
  - Patients weighing over 100 kg: 90 mg SC at weeks 0, 4, and then every 12 weeks.
- Crohn's disease and ulcerative colitis:
  - Induction dose:
    - Patients weighing 55 kg or less: 260 mg IV.
    - Patients weighing over 55 kg up to 85 kg: 390 mg IV.
    - Patients weighing over 85 kg: 520 mg IV.
  - Maintenance: 90 mg SC, starting 8 weeks after IV induction dose, then every 8-12 weeks.
  - In case of an interruption in therapy, resume SC treatment every 8 weeks.

Pediatrics (12-17 years of age):
- Plaque psoriasis: administer the weight-appropriate dose at weeks 0, 4, and then every 12 weeks.
  - Patients weighing less than 60 kg: 0.75 mg/kg SC.
  - Patients weighing 60-100 kg: 45 mg SC.
  - Patients weighing over 100 kg: 90 mg SC.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and prefilled syringes between 2-8°C; protect from light; do not freeze. Do not shake.
- Prefilled syringes are stable for 30 days at room temp (up to 30°C) in the original packaging, protected from light; once stored at room temp, do not return to the fridge.
- The ready-to-use solution for SC administration is colourless to light yellow and may contain a few small translucent or white particles of protein.
- The solution in the vial for IV administration is clear, colourless to light yellow.
- The diluted IV solution is stable for 8 hours at room temp in NS or 1/2NS.

MISCELLANEOUS

- The needle covers for the prefilled syringes contain latex.
- Patients should be tested for tuberculosis prior to initiating therapy.
- Do not administer in case of clinically important active infection.
- Do not administer live vaccines during the treatment with ustekinumab.

REFERENCES

1, 5, 95, 135.

Full revision 2019; limited revision 2020, 2021

NAME OF MEDICATION
VALPROIC ACID *

OTHER NAMES
Depacon ®, Epiject ® IV, Valproate sodium

CLASSIFICATION
Anticonvulsant

INDICATIONS

- Treatment of various seizure disorders in patients for whom oral administration is temporarily not feasible.

ADMINISTRATION

- Intermittent IV infusion: dilute dose in at least 50 mL of D5W, NS or RL; infuse over at least 60 minutes, at a rate not exceeding 20 mg/min.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, drug reaction with eosinophilia and systemic symptoms (DRESS).
- GI: nausea, pancreatitis (can be fatal).
- CNS: dizziness, headache, somnolence.
- Endocrine and metabolic: elevated amylase,
- Hematologic: thrombocytopenia (dose-related).
- Hepatic: hepatic failure.
- Weakness, tremors.
- Local reactions: pain, inflammation.

DOSAGE

- Conversion from oral to IV route: total IV daily dose should be equivalent to total oral daily dose of valproic acid or divalproex, given in divided doses at the same frequency or q6h.
- Initial dose of 10-15 mg/kg/day IV; may be increased by 5-10 mg/kg/day at weekly intervals until desired clinical response. Total daily dose exceeding 250 mg should be divided into 2-4 doses.
- Maximum recommended dosage is 60 mg/kg/day IV.
- Use of IV valproic acid for greater than 14 days has not been studied.
- Dosage in renal impairment: no dosage adjustment necessary; may need to monitor free valproate serum levels due to decreased protein binding.
- Dosage in hepatic impairment: not recommended in patients with hepatic disease or significant hepatic dysfunction.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Stable 6 days at room temp in D5W, NS, or RL at a concentration of 1.6 mg/mL in glass, polyethylene, polyolefin, and PVC containers.

MISCELLANEOUS

- Switch to oral therapy as soon as possible.
- Monitor serum levels when converting from oral to IV or from IV to oral therapy.
- Not recommended in patients with acute head trauma and for prophylaxis of post-traumatic seizures.

REFERENCES

1, 4, 5, 40, 95, 135.

* Available via Health Canada's Special Access Programme

Full revision 2019

NAME OF MEDICATION
VANCOMYCIN

OTHER NAMES
Vancocin ®

CLASSIFICATION
Antibiotic - glycopeptide

INDICATIONS

- Treatment of severe gram-positive infections that are resistant to other antibiotics.
- Perioperative prophylaxis in patients undergoing surgical procedures when first-line beta-lactam antibiotics (e.g., cefazolin, cefuroxime) cannot be used because of severe beta-lactam hypersensitivity.
- Prevention of neonatal Group B streptococcal (GBS) disease in penicillin-allergic patients at high risk of anaphylaxis, when bacteria is resistant to clindamycin or susceptibility is unknown.

ADMINISTRATION

- Reconstitute each vial of 500 mg, 1 g and 5 g with 10 mL, 20 mL and 100 mL of SWFI, respectively, to obtain a concentration of 50 mg/mL. The 10 g vial should be reconstituted with 95 mL of SWFI to obtain a 100 mg/mL solution.
- Intermittent IV infusion (preferred): dilute in 100-500 mL of NS or D5W to a maximum concentration of 5 mg/mL (10 mg/mL in fluid-restricted patients); maximum recommended infusion rates vary from 10 mg/min to 500 mg/30 min. At TOH, follow the following infusion guide, depending on the dose to be administered:

Dose	1000 mg or less	1001-1500 mg	1501-2000 mg	2001-2500 mg
Administration time	60 min.	90 min.	120 min.	150 min.

- Continuous IV infusion: may dilute daily dosage in sufficient quantity of NS, D5W or RL to run by continuous drip over 24 hours.
- Must be administered by an infusion pump.
- Consult Medical Staff Rules and Regulations MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis) for specific dosing regimens and infusion time.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: anaphylaxis, rash, urticaria, exfoliative dermatitis, angioedema, Stevens-Johnson syndrome, toxic epidermal necrolysis, vasculitis, drug reaction with eosinophilia and systemic symptoms (DRESS).
- Infusion-related reactions: vancomycin flushing syndrome (usually with rapid administration): symptoms include hypotension (can be severe), throbbing pain in the back and neck, flushing (face, neck, shoulders, trunk and/or extremities), wheezing, dyspnea, pruritus, urticaria and macular rashes. Slower administration may minimize or avoid these effects.
- To reduce risk of vancomycin flushing syndrome, vancomycin should be infused as per recommendations in the Administration section. Pre-treatment with antihistamines may attenuate (but not eliminate) this reaction.
- Renal and auditory toxicity with repeated dosage.
- Local reactions: pain, inflammation, induration, erythema. Thrombophlebitis (can be severe); can be minimized by increasing dilution and by alternating the injection site.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- Adults:
  - Treatment (intermittent IV infusion):
    - 15-20 mg/kg/dose IV q8-12h (usually 1000 mg q12h), based on actual body weight rounded to the nearest 250 mg.
    - Complicated infections in critically ill patients (e.g., septic shock): may consider a loading dose of 20-35 mg/kg IV (at TOH: 20-25 mg/kg) to achieve target level rapidly.
    - Dosage in obesity: use actual body weight for initial dosing and adjust according to serum levels; for serious infections, may consider a loading dose of 20-25 mg/kg based on actual body weight with a maximum dose of 3000 mg.
  - Treatment (continuous IV infusion):
    - A loading dose of 15-20 mg/kg IV should be administered to achieve target level rapidly. This should be followed by a continuous maintenance infusion of 30-40 mg/kg/24 hours (up to 60 mg/kg/24 hours) to achieve a target steady-state concentration of 20-25 mg/L.

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DOSAGE (Cont.)

- Perioperative prophylaxis: 15 mg/kg/dose IV, based on actual body weight, rounded to the nearest 250 mg; infusion to be started 1-2.5 hours (depending on dose administered) before skin incision. At TOH, follow this dosing guide for preoperative prophylaxis, depending on patient's weight, as per Medical Staff Rules and Regulations MED-RR 017 (Administration of preoperative antibiotics for surgical prophylaxis):

Weight	Less than 90 kg	90-129 kg	130-149 kg	150 kg and greater
Dose	1000 mg IV	1500 mg IV	2000 mg IV	2500 mg IV

- Prevention of neonatal GBS disease: 1000 mg IV q12h intrapartum OR 20 mg/kg (maximum of 2 g/dose) IV q8h intrapartum.

- Pediatrics (infants and children): 40-80 mg/kg/**day** IV in divided doses q6-8h. Adjust dose according to levels; maximum of 1000 mg/dose; usual maximum daily dose of 3600 mg.

- Dosage in renal impairment: at TOH, 15-20 mg/kg/dose IV (usually 1000 mg) with dosing interval according to creatinine clearance*:

CrCl (mL/min)	greater than 60	60-40	39-30	29-15	less than 15
Interval (hr)	8-12	12-24	24-36	48	72

*a loading dose of 20-25 mg/kg is recommended in critically ill patients with suspected/documented serious methicillin-resistant *Staphylococcus aureus* (MRSA) infections.

- Hemodialysis patients: following a loading dose, vancomycin is usually administered after or during the last 60-90 minutes of each hemodialysis session; consult specialized references for dosage adjustment and monitoring of serum levels.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp.
- Reconstituted solution is stable for 2 weeks at room temp and in the fridge.
- Stable for 24 days at room temp and 30 days in the fridge in NS at concentrations of 4 and 5 mg/mL in PVC containers.
- Stable for 17 days at room temp and 30 days in the fridge in D5W at a concentration of 4 mg/mL in PVC containers.
- Stable for 17 days at room temp and 58 days in the fridge in D5W at a concentration of 5 mg/mL in PVC containers.
- Stable for 57 days in the fridge in D5W at a concentration of 10 mg/mL in PVC containers.
- Stable for 24 hours at room temp and 96 hours in the fridge in RL at concentrations up to 5 mg/mL.
- Incompatible with alkaline solutions.

MISCELLANEOUS

- A target range of 10-20 mg/L for trough levels is recommended to decrease risk of resistance and to improve clinical outcomes; the upper end of the range (15-20 mg/L) is recommended for complicated infections due to MRSA.
- For serious infections due to MRSA, recent guidelines advocate monitoring individualized area-under-the-curve over 24 hours to minimum inhibitory concentration (AUC/MIC) ratio with a target of 400-600, assuming MIC of the MRSA is 1 mg/L.

REFERENCES

1, 2, 4, 5, 9, 40, 66, 82, 93, 95, 135, 138, 148, 208, 274, 338, 366, 367, 398, 422, 462, 463, 557.

Full revision 2019; limited revision 2020, 2021

NAME OF MEDICATION
VARICELLA VIRUS VACCINE

OTHER NAMES
Chickenpox vaccine, Varilrix ™, Varivax ® III

CLASSIFICATION
Vaccine

Do NOT confuse the varicella virus vaccine with the zoster vaccine for the prevention of shingles. The varicella virus vaccine is approximately 14 times less potent than the zoster vaccine. This monograph is specific to VARICELLA VIRUS vaccine.

INDICATIONS

- For the immunization against varicella in individuals 12 months of age and older.
- For preventing or reducing the severity of varicella if given within 3 to 5 days after exposure to varicella to healthy, previously unvaccinated individuals 12 months of age and older.

ADMINISTRATION

- Reconstitute with provided diluent; mix well until complete dissolution (by shaking Varilrix ™, by gently mixing Varivax ® III).
- SC: Varilrix ™: outer aspect of the upper arm (deltoid region); Varivax ® III: outer aspect of the upper arm (preferably) or the anterolateral aspect of the thigh.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, pruritus, Stevens-Johnson syndrome, erythema multiforme, urticaria, anaphylaxis.
- Fever.
- Varicella-like rash (generalized or at injection site), appearing within 5 to 30 days after immunization.
- Local reactions: pain, swelling, erythema, induration, pruritus, stiffness, hematoma, rash, numbness.

DOSAGE

- 0.5 mL SC per dose.
- For children from 12 months to 12 years of age, there should be a minimum interval of 3 months between doses if a second dose is administered; if rapid protection is required, a minimum interval of 4 weeks may be used.
- For adolescents and adults: administer two doses within a minimum interval of 4 weeks between doses.
- For patients with chronic conditions, intervals between the doses may be longer. Consult the Canadian Immunization Guide for specific recommendations.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Varilrix ™:
- Store vials containing the vaccine between 2-8°C; may also be kept in the freezer. Diluent may be stored between 2-8°C or at room temp.
- Reconstituted solution is stable for 90 minutes at room temp and 8 hours in the fridge; the solution may vary from clear peach to pink.

Varivax ® III:
- Store vials containing the vaccine between 2-8°C and protect from light; may also be kept in the freezer; multiple temperature excursions up to 25°C are allowed for a cumulative period not exceeding 6 hours. Diluent may be stored between 2-8°C or at roomp temp.
- Reconstituted solution is stable for 90 minutes at room temp; the solution should be clear, colourless to pale yellow.

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MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- The varicella virus vaccine is a live attenuated vaccine.
- Administration should be delayed during an acute febrile illness or active infection; immunization of immunocompromised individuals should be evaluated on a case-by-case basis, based on the type of immunodeficiency and degree of immunosuppression.
- Contains trace amounts of neomycin (Varilrix ™, Varivax ® III) and gelatin (Varivax ® III).
- Do not use salicylates for 6 weeks following vaccination.
- Varicella virus vaccine is contraindicated in pregnant women. Pregnancy should be avoided for one month after vaccination.
- Interchangeability between Varilrix ™ and Varivax ® III has not been studied.
- Inadvertent IM administration would provide similar seroconversion to that observed with SC administration.

REFERENCES

1, 5, 31, 95, 135.

Full revision 2019; limited revision 2020, 2021

NAME OF MEDICATION
VASOPRESSIN

OTHER NAMES
Pressyn ® AR

CLASSIFICATION
Antidiuretic hormone, Vasopressor

INDICATIONS

- For the control of GI bleeding, including from esophageal varices.
- For the treatment of vasodilatory shock.
- Symptomatic control of diabetes insipidus.
- Prevention and control of abdominal distension.
- To dispel interfering gas shadows in abdominal radiographic procedures.
- As an alternative to epinephrine in the setting of cardiac arrest.

ADMINISTRATION

- IV direct: physician only in setting of cardiac arrest; administer undiluted; at TOH, may dilute dose in 10 mL NS.
- Continuous IV infusion: physician to be present for initiation of therapy; dilute in D5W or NS to a concentration of 0.1 to 1 unit/mL; at TOH, withdraw 50 mL from a 250 mL bag of D5W or NS and discard, then add 20 units (1 mL from a vasopressin 20 units/mL vial) to the bag to obtain a 0.1 unit/mL concentration. Infuse as indicated in Dosage section.
- Continuous intra-arterial infusion: physician only.
- IM, SC.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- Hypersensitivity: anaphylaxis, urticaria, bronchoconstriction, angioedema, fever, rash.
- Cardiovascular: vasoconstriction, decrease in blood flow, cardiac arrest, cardiac dysrhythmia, reduced cardiac output and coronary vasoconstriction which could lead to anginal attacks and cardiac ischemia.
- GI: nausea, vomiting, flatulence, abdominal cramps.
- CNS: throbbing headache, tremor, vertigo.
- Dermatologic: sweating, pallor, gangrene.
- Hematologic: thrombosis.
- Water intoxication syndrome: somnolence, listlessness, headache, seizures, coma, hyponatremia, confusion, anuria, weight gain.
- Extravasation hazard: vesicant. Consult the document “Extravasation of noncytotoxic intravenous agents associated with tissue necrosis” in TOH Electronic Drug Formulary.

DOSAGE

- GI bleed, including from esophageal varices: initial continuous IV infusion of 0.2-0.4 units/min (12-24 units/hr); if bleeding is uncontrolled, may progressively increase to a suggested maximum of 1 unit/min (60 units/hr). A bolus dose of 20 units IV over 20-30 minutes may be given, but is probably not necessary. After the bleeding has been controlled for 12 hours, the vasopressin dose may be decreased by 50% and then the infusion discontinued within the next 12-24 hours. Vasopressin has been infused into the superior or inferior mesenteric artery at a rate of 0.1-0.5 units/min (6-30 units/hr) to control bleeding.
- Vasodilatory shock: continuous IV infusion of 0.01-0.1 units/min (0.6-6 units/hr) has been used.
- Diabetes insipidus: 5-10 units IM/SC 2-4 times daily as needed. Range of 5-60 units daily.
- Abdominal distension: 5 units IM/SC initially, repeated q3-4h. Dose may be increased to 10 units if needed.
- Radiographic procedures: 5-10 units IM/SC at 2 hours and at 30 minutes prior to exam.
- Cardiac arrest: 40 units IV as a single dose.

…/Cont.

NAME OF MEDICATION
VASOPRESSIN

OTHER NAMES
Pressyn ® AR

CLASSIFICATION
Antidiuretic hormone, Vasopressor

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials at room temp. Protect from light and heat. Do not freeze.
- Stable for 18 hours at room temp and 24 hours in the fridge in NS or D5W at concentrations between 0.1-1 unit/mL.
- Stable for 90 days at 23-25°C and 3-5°C in NS at a concentration of 0.4 units/mL in PVC bags or PVC syringes and at a concentration of 1 unit/mL in PVC syringes.

MISCELLANEOUS

REFERENCES

1, 5, 6, 40, 95, 135, 208, 331, 366, 367.

Full revision 2019; limited revision 2020, 2021

NAME OF MEDICATION
VEDOLIZUMAB

OTHER NAMES
Entyvio ®

CLASSIFICATION
Immunomodulator, Monoclonal antibody

INDICATIONS

- Treatment of moderately to severely active ulcerative colitis in patients who have had an inadequate/loss of response to or were intolerant to conventional therapy, infliximab, or a tumor necrosis factor (TNF) antagonist.
- Treatment of moderately to severely active Crohn's disease in patients who have had an inadequate response to, loss of response to, or intolerance to immunomodulators or a TNF antagonist; or have had an inadequate response to, intolerance to, or demonstrated dependence on corticosteroids.

ADMINISTRATION

- **Consider premedication if patient previously experienced a mild to moderate infusion-related/hypersensitivity reaction; refer to Potential Administration Hazards section.**
- **There are 2 formulations: one for IV use (vial) and one for SC use (prefilled syringe or pen). Ensure using the correct one.**
- Intermittent IV infusion: ensure using the IV formulation. Reconstitute each 300 mg vial with 4.8 mL of SWFI using a syringe with a 21-25 gauge needle to obtain a concentration of 60 mg/mL. Direct stream of SWFI to glass wall of vial to avoid excessive foaming; gently swirl the vial for at least 15 seconds; do NOT shake vigorously or invert; let vial stand for 20 minutes at room temp to allow for reconstitution and for any foam to settle; the vial can be swirled and inspected for dissolution during this time; if not fully dissolved after 20 minutes, wait another 10 minutes; do NOT use if not dissolved within 30 minutes; once dissolved, gently invert vial 3 times. Withdraw entire contents of vial using a syringe with a 21-25 gauge needle and dilute in 250 mL of NS or RL; gently mix bag; infuse over 30 minutes. Flush line with 30 mL of NS or RL after infusion.
- SC: ensure using the SC formulation; allow the prefilled syringe or pen to reach room temp (wait time: 30 minutes).
  Injection sites: front of the thighs, abdomen (at least 5 cm from the navel) or back of the upper arm. Ensure site of injection is free of moles, scars, bruises and skin is not tender, hard, red or damaged. Rotate injection site.
  When using the prefilled syringe: inject SC into pinched skin with a 45-degree angle.
  When using the pen: inject SC by placing the pen (yellow end toward the injection site) at 90 degrees to the injection site; push down as far as it will go and hold and count for 10 seconds (may hear 2 clicks, one at the start and one near the end of injection). Viewing window should be filled with purple before stopping pushing. Refer to the manufacturer's monograph for more detailed instructions.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: urticaria, rash, pruritus, flushing, anaphylaxis, dyspnea, bronchospasm, hypertension, tachycardia; can occur with first or subsequent infusions; can occur during infusion or up to several hours post infusion. For patients with mild to moderate reactions, **consider premedication** prior to next infusion (e.g., antihistamines, hydrocortisone and/or acetaminophen). For severe reactions, discontinue immediately and initiate appropriate treatment.
- Infusion-related reactions: nausea, vomiting, headache, dizziness, urticaria, pruritus, fever, fatigue; generally occur within 2 hours after infusion and are mild to moderate in severity. For mild to moderate reactions, interrupt infusion or slow infusion rate; resume infusion once reaction has subsided; **consider premedication** prior to next infusion (e.g., antihistamines, hydrocortisone and/or acetaminophen).
- Cardiovascular: hypertension.
- GI: nausea.
- CNS: headache.
- Musculoskeletal: arthralgia, back pain, pain in extremity.
- Hepatic: elevated LFTs, increased bilirubin, hepatitis.
- Respiratory: nasopharyngitis, cough, upper respiratory tract infection, bronchitis.
- Progressive multifocal leukoencephalopathy (PML).
- Fever, fatigue.
- Local reactions : rash, pruritus at SC injection sites.

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DOSAGE

- 300 mg IV at weeks 0, 2, and 6 followed by every 8 weeks starting at week 14.
- May switch to 108 mg SC every 2 weeks for maintenance therapy after at least 2 IV infusions. Administer the first SC dose in place of the next scheduled IV dose.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials, prefilled syringes and pens between 2-8°C. Protect from light. Do not freeze.
- The prefilled syringes and pens are stable at room temp (up to 25°C) in their original carton for 7 days.
- Reconstituted solution is stable for 8 hours between 2-8°C under normal light condition.
- Stable for 12 hours at room temp or 24 hours in fridge in NS at a concentration of 1.2 mg/mL (this stability includes the time since vial reconstitution). Stability in fridge includes the 12-hour stability at room temp.
- Stable for 8 hours in fridge in RL at a concentration of 1.2 mg/mL (this stability includes the time since vial reconstitution).

MISCELLANEOUS

- Emergency drugs (epinephrine, diphenhydramine and injectable corticosteroids) and resuscitation equipment must be immediately available for the treatment of hypersensitivity reactions.
- Patients should be evaluated for active or latent tuberculosis and for risk factors before starting therapy and periodically during therapy.
- Administer live vaccines concurrently with vedolizumab only if benefits outweigh risks.
- Each 300 mg vial contains 552 mg of sucrose.

REFERENCES

1, 5, 95, 135, 471, 507.

New monograph 2020; limited revision 2021

NAME OF MEDICATION
VERAPAMIL

OTHER NAMES
Isoptin ®

CLASSIFICATION
Antiarrhythmic

INDICATIONS
- For the rapid conversion of paroxysmal supraventricular tachycardia to sinus rhythm.
- For temporary control of rapid ventricular rate in atrial flutter or atrial fibrillation not associated with accessory bypass tracts.

ADMINISTRATION
- IV direct: physician for first dose only; RN may administer subsequent doses; ***cardiac monitoring, blood pressure monitoring***. Undiluted; inject over at least 2 minutes (over at least 3 minutes in the elderly).

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: broncholaryngeal spasm, urticaria, pruritus, anaphylaxis, angioedema, Stevens-Johnson syndrome, erythema multiforme.
- Cardiovascular: hypotension, bradycardia, asystole, tachycardia, worsening of heart failure and rhythm disturbances. Verapamil-induced hypotension, bradycardia and asystole may be controlled with IV calcium or vasoconstrictors.
- GI: nausea, abdominal discomfort.
- CNS: headache, dizziness, seizures (rare).
- Hepatic: increase in LFTs.

DOSAGE
- Adults:
  - Initial dose of 2.5-10 mg OR 0.075-0.15 mg/kg IV.
  - Repeat dose of 5-10 mg (0.075-0.15 mg/kg) IV 15-30 minutes after the first dose if the initial response is not adequate, up to a maximum total dosage of 20 to 30 mg.

- Pediatrics:
  - Less than 1 year of age: initial dose of 0.1-0.2 mg/kg (usually 0.75 to 2 mg) IV; maximum dose of 0.2 mg/kg and 2 mg. Repeat same dose 30 minutes after the first dose if response is not adequate.
  - 1 to 15 years of age: 0.1-0.3 mg/kg (usually 2 to 5 mg) IV; maximum 5 mg. Repeat dose (up to a maximum of 10 mg) 30 minutes after the first dose if response is not adequate.
  - Adolescents older than 15 years of age: 5-10 mg (range of 0.075-0.3 mg/kg) IV. Repeat dose (up to a maximum of 10 mg) 15-30 minutes after the first dose if the response is not adequate.

- Dosage in renal impairment: monitor carefully as the effect of single doses may be prolonged; if repeat doses are required, smaller doses should be given.
- Dosage in hepatic impairment: reduce dose by 50% in patients with cirrhosis and monitor closely. Otherwise, monitor carefully as the effect of single doses may be prolonged; if repeat doses are required, smaller doses should be given.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store ampoules and vials at room temp. Protect from light. Do not freeze.
- Compatible with NS, D5W, Ringer's, RL.
- Incompatible with solutions with a pH of greater than 6.

MISCELLANEOUS
- Concomitant use of IV verapamil with IV beta-blockers (propranolol, metoprolol, etc.) may cause significant cardiac depression. Therefore, co-administration (within a few hours) is contraindicated.

REFERENCES
1, 2, 4, 5, 40, 102, 135, 200.

Full revision 2019

NAME OF MEDICATION
VERTEPORFIN

OTHER NAMES
Visudyne ®

CLASSIFICATION
Photosensitizing agent

INDICATIONS
- Treatment of predominantly classic subfoveal choroidal neovascularization in patients with age-related macular degeneration, pathologic myopia or presumed ocular histoplasmosis.

ADMINISTRATION
- Reconstitute the 15 mg vial with 7 mL of SWFI. Final concentration is 2 mg/mL. Protect solution from light.
- Intermittent IV infusion: dilute dose with D5W to obtain a final volume of 30 mL. A free-flowing IV line should be established before starting verteporfin. It is recommended that the largest arm vein possible, preferably the antecubital vein, be used for the infusion; avoid small veins in the back of the hand. Infuse over 10 minutes (3 mL/min) using an appropriate syringe pump and in-line filter of 1.2 microns. Protect from light during administration. Monitor the infusion line carefully during administration.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: rash, urticaria, pruritus, dyspnea, flushing, sweating, dizziness, syncope, headache.
- Cardiovascular: chest pain, vasovagal reactions, blood pressure and heart rate changes.
- Dermatologic: photosensitivity.
- Ophthalmic: blurred vision, flashes of light, decreased visual acuity, visual field defects, eye pain.
- Back pain during administration.
- Local reactions: pain, edema, inflammation, rash, hemorrhage, discolouration. In case of extravasation, severe pain, inflammation, swelling, discolouration at the injection site or necrosis can occur, especially if affected area is exposed to light. In this case, stop the infusion immediately, protect the site from direct light until swelling and discolouration dissipate and apply cold compresses.

DOSAGE
- 6 mg/m^2 IV. If bilateral treatment is required, initial treatment to the second eye is performed 1 week after treatment of the first eye, if treatment to first eye was well tolerated. Subsequent unilateral or concurrent bilateral treatments can be repeated every 3 months prn, up to 2 years' duration.
- In order to activate the drug, light therapy from a nonthermal diode laser is to be initiated 15 minutes after the start of the verteporfin infusion.
- Dosage in hepatic impairment: use with caution in patients with moderate impairment or biliary obstruction as it has not been studied in these populations; contraindicated in patients with severe hepatic impairment.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 20-25°C.
- Reconstituted solution is stable for 4 hours between 20-25°C, protected from light; the solution is opaque dark green.
- Stable for 4 hours between 20-25°C, protected from light, in D5W.
- Incompatible with saline solutions; do not use parenteral solutions other than D5W.

MISCELLANEOUS
- Contraindicated in patients with porphyria.
- Care should be taken to avoid exposure of skin or eyes to direct sunlight, bright indoor light, or prolonged exposure to light from light-emitting medical devices for 2 days.
- If outdoor activity is essential during the first 2 days, protective clothing completely covering the whole body and dark sunglasses must be worn. Sunscreens are not effective protection.
- Patients should not stay in the dark and should be encouraged to expose their skin to ambient indoor light, as it will help inactivate the drug in the skin through a process called photobleaching.
- If emergency surgery is required with 48 hours of administration, internal tissues should be protected from intense light as much as possible.

REFERENCES
1, 5, 40, 95, 135.

Full revision 2019

NAME OF MEDICATION
VinBLAStine

OTHER NAMES
Velban, Vincaleukoblastine

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of breast cancer, choriocarcinoma, histiocytosis X, Hodgkin's disease, Kaposi's sarcoma, non-Hodgkin's lymphoma, mycosis fungoides, testicular cancer, and other cancers.

ADMINISTRATION

- Intermittent IV infusion: dilute in 25-50 mL of NS or D5W; infuse over 5-30 minutes. Avoid using larger volumes of diluent and longer infusion times as these can increase risk of vein irritation and extravasation.
- **FOR IV USE ONLY**. Fatal if given by intrathecal route.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: rash, toxic epidermal necrolysis.
- Cardiovascular: hypertension, arterial thromboembolism.
- GI: nausea, vomiting.
- CNS: parotid gland pain, jaw pain, tumour pain, malaise, paresthesia.
- Hematologic: leukopenia, thrombocytopenia.
- Respiratory: shortness of breath, bronchospasm.
- Tumour lysis syndrome, hyperuricemia; can be minimized with allopurinol, hydration and alkalinization of urine.
- Local reactions: phlebitis, irritation.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Range of 3-18.5 mg/m^2 IV every 1-4 weeks. Dosage adjustments may be required, depending on toxicity.
- Maximal weekly recommended dose: 18.5 mg/m^2 IV.
- Dosage in hepatic impairment:

Bilirubin	% of usual dose
25-50 mcmol/L or up to 2.5 x ULN	50%
greater than 50 mcmol/L or greater than 2.5 x ULN	25%

- Consult specific protocol.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze.
- Stable for 7 days in the fridge protected from light in D5W or NS at a concentration of 0.1 mg/mL in PVC containers.

MISCELLANEOUS

REFERENCES

1, 2, 4, 5, 95, 129, 135, 165, 208, 320.

Full revision 2019; limited revision 2020

NAME OF MEDICATION
VinCRIStine

OTHER NAMES
Leurocristine, Oncovin ®

CLASSIFICATION
Antineoplastic

INDICATIONS

- Treatment of acute leukemias, Hodgkin's disease, non-Hodgkin's lymphoma, sarcomas, breast cancer, small cell lung cancer, melanoma, cervical cancer, colorectal cancer, Wilms' Tumor and other cancers.

ADMINISTRATION

- Intermittent IV infusion: dilute with 25-50 mL of NS or D5W; infuse over 5-15 minutes. During administration, it is suggested to observe the IV site for extravasation.
- **FOR IV USE ONLY**. Fatal if given by intrathecal route.

POTENTIAL ADMINISTRATION HAZARDS

- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis, rash, edema.
- Cardiovascular: myocardial infarction (rare), hypertension, hypotension.
- GI: nausea, vomiting, constipation, abdominal pain, paralytic ileus.
- CNS: seizures (rare), hallucinations, peripheral neuropathy, jaw pain, joint pain, insomnia, agitation, headache, ataxia.
- Renal: urinary retention.
- Respiratory: shortness of breath, bronchospasm (acute and severe).
- Tumour lysis syndrome, hyperuricemia; can be minimized with allopurinol, hydration and alkalinization of urine.
- Fever, sweating.
- Local reactions: chemical phlebitis; pain, redness at injection site.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE

- Adults: 0.8-1.4 mg/**m**^{2} OR 2 mg IV every 1 to 7 weeks.
- Pediatrics: weighing over 10 kg: 1-2 mg/**m**^{2} IV every 1-3 weeks.
  weighing 10 kg or less: 0.03-0.05 mg/**kg** IV every 1-3 weeks.
- Some regimens may cap dose at 2 mg IV.
- Combination with l-asparaginase: administer vincristine 12-24 hours prior to l-asparaginase.
- Dosage in hepatic impairment:

Bilirubin	% of usual dose
25-50 mcmol/L or up to 2.5 x ULN	50%
greater than 50 mcmol/L or greater than 2.5 x ULN	25%

- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

…/Cont.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 2-8°C. Protect from light. Do not freeze.
- Stable for 7 days in the fridge followed by 2 days at room temp in NS at a concentration between 0.01-0.12 mg/mL in PVC containers.
- Stable 7 days in the fridge protected from light in D5W at a concentration of 0.02 mg/mL in PVC containers.
- Stable for 24 hours at room temp in D5W at a concentration of 0.0167 mg/mL in PVC and glass containers.
- Stable for 84 days at room temp and in the fridge protected from light in NS at concentrations of 0.01, 0.025, and 0.05 mg/mL in polyolefin containers.
- Stable for 24 hours at room temp exposed to light in D5W at a concentration of 0.02 mg/mL in polyolefin containers.
- Stable for 7 days at room temp in NS or D5W at concentrations of 0.001 and 0.02 mg/mL in polyethylene or glass containers.

MISCELLANEOUS

- Contains mannitol.

REFERENCES

1, 2, 4, 5, 129, 135, 165, 320.

NAME OF MEDICATION
VINORELBINE

OTHER NAMES
Navelbine ®

CLASSIFICATION
Antineoplastic

INDICATIONS
- Treatment of advanced non-small cell lung cancer and metastatic breast cancer.
- Treatment of other cancers such as cervical, small cell lung and ovarian cancers.

ADMINISTRATION
- Intermittent IV infusion: dilute with 50 mL of D5W, NS, ½NS, D5-1/2NS, Ringer's, RL to obtain a concentration of 0.5-2 mg/mL; infuse over 6-10 minutes into the tubing of a freely running IV solution of D5W or NS and then flush line with at least 75-125 mL of D5W or NS.
- Continuous IV infusion: dilution and infusion time as per protocol.
- **FOR IV USE ONLY**. Fatal if given by intrathecal route.

POTENTIAL ADMINISTRATION HAZARDS
- **High alert medication:** consult Corporate policy ADM VIII 860 (High alert medications).
- **Cytotoxic drug:** consult Corporate standard operating procedure C-SOP V 250 (Cytotoxic and hazardous drug waste and spills) AND Nursing policy 01651 (Administration of chemotherapy & biotherapy).
- Hypersensitivity: anaphylaxis, rash, pruritus, urticaria, angioedema.
- Cardiovascular: chest pain.
- GI: nausea, vomiting, constipation, diarrhea, abdominal pain, anorexia, mucositis.
- CNS: peripheral neuropathy.
- Hematologic: anemia, leukopenia, neutropenia, granulocytopenia.
- Hepatic: elevated AST and bilirubin.
- Renal: elevated serum creatinine.
- Respiratory: acute shortness of breath, severe bronchospasm.
- Myalgia, arthralgia, asthenia, fatigue, fever.
- Local reactions: pain, erythema, vein discolouration, phlebitis. May be minimized by administering over 6-10 minutes and by flushing the vein with 100 mL of fluid before administration followed with at least 75-125 mL (up to 400 mL) of fluid after its completion.
- Extravasation hazard: vesicant. Consult the extravasation section of the Nursing policy 01651 (Administration of chemotherapy & biotherapy).

DOSAGE
- 8 to 35 mg/m^2 IV (most common is 25 to 30 mg/m^2 IV) every 1-3 weeks.
- Dosage in hepatic impairment:

Bilirubin (mcmol/L)	% of usual dose
36-50	50%
greater than 50	25%

- Dosage adjustment due to toxicity: refer to manufacturer's recommendations.
- Consult specific protocol.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials between 2-8°C. Protect from light. Do not freeze. Colour change from colourless to dark yellow or light amber should not prevent its use.
- Unopened vials are stable for 72 hours at 25°C.
- Stable for 24 hours between 5-30°C in D5W, NS, ½NS, D5-1/2NS, Ringer's, RL at concentrations between 0.5-2 mg/mL in PVC bags.
- Stable for 7 days in the fridge protected from light in D5W at a concentration of 0.5 mg/mL in PVC containers.
- Stable for 3 days in the fridge protected from light in NS at a concentration of 0.5 mg/mL in PVC containers.
- Stable for 5 days at 24°C exposed to fluorescent light in D5W or NS at a concentration of 0.5 mg/mL or 2 mg/mL in PVC containers.

MISCELLANEOUS

REFERENCES
1, 4, 5, 6, 95, 129, 135, 165, 208, 320.

Full revision 2019; limited revision 2020

NAME OF MEDICATION
VITAMIN B_{12}

OTHER NAMES
Cyanocobalamin

CLASSIFICATION
Vitamin

Do NOT confuse vitamin B_{12} (cyanocobalamin) with hydroxocobalamin.
This monograph is specific to CYANOCOBALAMIN.

INDICATIONS
- Treatment of pernicious anemia and other macrocytic, megaloblastic anemias resulting from vitamin B_{12} deficiency when GI absorption is impaired or oral administration is not possible.
- Used in conjunction with an oral radiolabelled dose of vitamin B_{12} in the Schilling test to study vitamin B_{12} absorption and diagnose pernicious anemia.

ADMINISTRATION
- IM.
- SC (deep): avoid dermis or upper SC tissue.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: pruritus, urticaria, anaphylaxis.
- GI: mild transient diarrhea.
- Electrolyte disturbances: hypokalemia, occurs early in therapy in patients with megaloblastic anemia.
- Usually non toxic, even in large doses.

DOSAGE

Adults:
- Pernicious anemia: 100 mcg IM/SC once daily for 6-7 days followed by either 1) 100 mcg IM/SC every 2 days for 6-7 days, then 100 mcg IM/SC every 3-4 days for 2-3 weeks or 2) 200 mcg IM/SC once weekly. When blood components have normalized, the maintenance dose is 100 mcg IM/SC once monthly. Alternate dosing: 1000 mcg IM/SC once daily for 3-7 days followed by 1000 mcg IM/SC once weekly for 3-4 weeks followed by a maintenance dose of 1000 mcg IM/SC once monthly.
- Vitamin B_{12} deficiency: 30-100 mcg IM/SC once daily for 5-10 days. When clinical symptoms have improved and blood components have normalized, the maintenance dose is 100-200 mcg IM/SC once monthly.
- Schilling test: 1000 mcg IM as a single dose.

Pediatrics:
- Pernicious anemia: 30-50 mcg IM/SC once daily for at least 2 weeks, until a total dose of 1000-5000 mcg has been administered. Maintenance dose of 100 mcg IM/SC once monthly.
- Vitamin B_{12} deficiency: 0.2 mcg/kg/day IM/SC for 2 days, followed by 1000 mcg IM/SC once daily for 2-7 days, then 100 mcg IM/SC once weekly for 4 weeks. Maintenance dose of 100 mcg IM/SC once monthly. Alternate dosing: 1000 mcg IM/SC once daily for 10-15 days, followed by a maintenance dose of at least 60 mcg IM/SC once monthly.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store vials and ampoules between 15-30°C. Protect from light.

MISCELLANEOUS
- High oral doses (1000-2000 mcg/day) may be more cost effective than parenteral administration for maintenance therapy of pernicious anemia.

REFERENCES
1, 2, 5, 82, 95, 137, 328.

Full revision 2019

NAME OF MEDICATION
VITAMIN C

OTHER NAMES
Ascorbic acid, Ascor L 500 ®, Sodium ascorbate

CLASSIFICATION
Vitamin

INDICATIONS
- Prevention and treatment of ascorbic acid deficiency, including scurvy, and treatment of surgical wounds and severe burns, when oral administration is not possible or when malabsorption is suspected.

ADMINISTRATION
- Manipulate ampoules and vials carefully as increased pressure may develop during storage, especially if stored at room temp.
- Intermittent IV infusion: dilute in 50-100 mL of NS or D5W to obtain a final concentration between 1 and 25 mg/Ml; infuse slowly at a rate of 33 mg/min for patients 11 years of age and older, 3.3 mg/min for those between 1 and less than 11 years of age, and 1.3 mg/min for those between 5 and less than 12 months old.
- IM (preferred).
- SC.

POTENTIAL ADMINISTRATION HAZARDS
- GI: diarrhea with large doses.
- CNS: dizziness or faintness with too rapid IV administration.
- Renal: renal calculi with large doses or with doses greater than 100 mg administered in hemodialysis patients.
- May precipitate gouty arthritis in susceptible patients when large doses are used.
- Local reactions: pain and swelling at site of injection.

DOSAGE
- Adults:
  - Prevention of deficiency: 70-150 mg IM/IV/SC daily.
  - Prevention of deficiency in hemodialysis patients: 60-100 mg IV once daily.
  - Scurvy: 100-250 mg IM/IV/SC 1 to 2 times daily for several days up to 3 weeks OR 300-1000 mg IM/IV/SC daily.
  - Wound healing: usual dose of 300-500 mg IM/IV/SC daily for 7-10 days pre- and post-operatively.
  - Severe burns: 200-2000 mg IM/IV/SC daily; dose may be determined by extent of tissue injury.

- Pediatrics:
  - Scurvy: 100-300 mg IM/IV per day, in divided doses for several days OR initial dose of 100 mg IM/IV/SC 3 times daily for 7 days followed by 100 mg IM/IV/SC once daily, usually for 1-3 months.

- Dosage in renal impairment: use with caution due to increased risk of acute or chronic oxalate nephropathy.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)
- Store ampoules between 15-30°C and vials between 2-8°C. Protect from light. Do not freeze.
- Sensitive to air and alkaline pH.
- Stable for 24 hours at room temp protected from light in NS and D5W at concentrations of 1.25 and 10 mg/mL.
- Stable for 24 hours at room temp in D10W, D5-1/2NS, Ringer's and RL at a concentration of 1.25 mg/mL.

MISCELLANEOUS
- Ampoules contain sulphites.
- Each 1000 mg of injectable ascorbic acid contains approximately 5 mmol of sodium.
- May interfere with laboratory tests based on oxidation-reduction reaction (e.g., blood and urine glucose testing, nitrite and bilirubin levels, leukocyte count). Delay test until 24 hours after last dose.
- May interfere with stool occult blood tests. Delay test 48-72 hours after last dose.

REFERENCES
1, 4, 5, 82, 95, 135, 208.

Full revision 2019

NAME OF MEDICATION
VITAMIN K_1

OTHER NAMES
Phylloquinone, Phytomenadione, Phytonadione, Vitamin K

CLASSIFICATION
Antidote, Vitamin

INDICATIONS

- In hemorrhagic situations due to overdose of vitamin K antagonists (VKA) such as warfarin.
- For rapid reversal of warfarin or other VKA activity prior to surgery.
- Prevention and treatment of hemorrhagic disease of the newborn.
- Treatment and prevention of hypoprothrombinemia secondary to factors limiting absorption, synthesis or activity of vitamin K.

ADMINISTRATION

- IV direct (NOT preferred): physician or RN; in adults, dilute with at least 10 mL of NS, D5W or D5-NS; in neonates, administer undiluted; do not exceed rate of 1 mg/min (to avoid severe reactions).
- Intermittent IV infusion: in adults, dilute in at least 50 mL of NS, D5W or D5-NS and administer over at least 20 minutes; maximum rate of 1 mg/min. In the newborn, dilute in D5W to obtain a final concentration of 1 mg/mL and administer over at least 15 minutes.
- Continuous IV infusion: dilute in up to 1000 mL of a compatible solution (usually a total parenteral nutrition (TPN) solution); rate as per solution.
- IM: erratic absorption; increased risk of hematoma. Adult and older children: inject in the buttocks; infants and young children: inject in the thigh (vastus lateralis) or deltoid region; in neonates: inject in the thigh (vastus lateralis).
- SC: erratic and unpredictable absorption.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: with IV and IM routes. Rare; rash, urticaria, anaphylactoid reactions. It seems that both vitamin K_1 and its solubilizer/vehicle may be responsible for the anaphlylactoid reactions.
- Cardiovascular: flushing, chest pain, rapid and weak pulse, hypotension, cyanosis.
- CNS: dizziness.
- Respiratory: dyspnea.
- Hyperbilirubinemia and severe hemolytic anemia in infants following excessive doses.
- Local reactions: pain, swelling, tenderness at injection site (rare).

DOSAGE

Adults:
- Treatment of VKA-induced prothrombin deficiency: usual dose of 2.5-10 mg IV (preferred if active bleed)/ SC (preferred if no bleed)/IM; rarely, doses up to 50 mg may be given. Repeat q6-12h prn. The lowest dose should be used to minimize resistance to further VKA therapy.
- Rapid reversal of VKA activity prior to urgent surgery (time prior to operating room within the next 12 hours): 5-10 mg IV. May repeat the dose prn if major surgery or if surgery within the next 6 hours.
- Treatment of hypoprothrombinemia from other causes: 2-25 mg IV/IM/SC, up to 50 mg may be given. IV route preferred if bleed, SC route is preferred if no bleed.

Pediatrics:
- Prophylaxis of hemorrhagic disease of the newborn: 0.5-1 mg IM (preferred)/SC within 6 hours of birth.
- Treatment of hemorrhagic disease of the newborn: 1 mg/day IV/IM; higher doses may be required if mother took anticonvulsants or VKA during pregnancy.

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NAME OF MEDICATION
VITAMIN K_1

OTHER NAMES
Phylloquinone, Phytomenadione, Phytonadione, Vitamin K

CLASSIFICATION
Antidote, Vitamin

…/Cont.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 15-30°C. Protect from light.
- Compatible with D5W, NS, dextrose-saline solutions, Ringer's, RL and TPN solutions.
- Use as soon as possible after mixing since very sensitive to light.

MISCELLANEOUS

- The injectable solution can be administered orally, undiluted or diluted in a convenient volume of water or juice. May be repeated 12-48 hours after first dose.
- The oral route offers a more predictable and quicker absorption than the SC route when used for reversal of warfarin or other VKA.
- The IV route allows a predictable and rapid reduction of the INR in 4 to 6 hours.

REFERENCES

1, 2, 4, 5, 40, 82, 95, 102, 135, 251, 319, 379, 384, 392, 393.

NAME OF MEDICATION
VORICONAZOLE

OTHER NAMES
Vfend ®

CLASSIFICATION
Antifungal

INDICATIONS

- Treatment of acute invasive aspergillosis.
- Treatment of *Candida* infections.

ADMINISTRATION

- Reconstitute each 200 mg vial with 19 mL of SWFI to obtain a concentration of 10 mg/mL.
- Intermittent IV infusion: dilute in 100-250 mL of a compatible IV solution to obtain a concentration of 0.5-5 mg/mL; infuse over 1-2 hours at a maximum rate of 3 mg/kg/hr.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: serious cutaneous reactions (e.g., Stevens-Johnson syndrome, erythema multiforme, toxic epidermal necrolysis).
- Infusion-related reactions: flushing, fever, sweating, tachycardia, chest tightness, dyspnea, faintness, nausea, pruritus, rash. Consider stopping the infusion if these reactions occur.
- Cardiovascular: QT_c interval prolongation, arrhythmias.
- Hepatic: increase in LFTs, jaundice, hepatitis, cholestasis, fulminant hepatic failure.
- Ophthalmic: blurred vision, colour vision change, photophobia.

DOSAGE

- Adults and pediatrics (12 years of age and older):
  - Invasive aspergillosis: 6 mg/kg IV q12h for 2 doses followed by 4 mg/kg IV q12h; switch to oral therapy when feasible. The maintenance dose may be decreased to 3 mg/kg q12h if the drug is poorly tolerated.
  - *Candida* infections: 6 mg/kg IV q12h for 2 doses followed by 3-4 mg/kg IV q12h; switch to oral therapy when feasible. Treat for a minimum of 14 days following resolution of symptoms or following last positive culture, whichever is longer.
- Dosage in renal impairment: in patients with CrCl less than 50 mL/min, accumulation of the IV vehicle sulfobutyl ether beta-cyclodextrin (SBECD) occurs. In these patients, use of the oral formulation is preferred.
- Dosage in hepatic impairment: in patients with mild to moderate hepatitic cirrhosis, use standard loading dose and reduce maintenance dose by 50%. Not studied in patients with severe hepatitic cirrhosis.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-30°C.
- Reconstituted solution is stable for 24 hours in the fridge as per manufacturer's recommendation.
- Reconstituted solution is stable for 30 days at room temp.
- Stable for 15 days in the fridge in D5W at a concentration of 4 mg/mL in PVC bags.
- Stable for 11 days in the fridge, protected from light, in NS at a concentration of 0.5 mg/mL in PVC bags.
- Stable for 9 days in the fridge, protected from light, in D5W at a concentration of 0.5 mg/mL in PVC bags.
- Stable for 6 days in the fridge and 4 days at room temp, protected from light, in D5W at a concentration of 2 mg/mL in polyolefin bags.
- Stable for 8 days in the fridge or at room temp, protected from light, in NS at a concentration of 2 mg/mL in polyolefin bags.
- Compatible with NS, 1/2NS, D5W, dextrose-saline solutions, RL, D5-RL.

MISCELLANEOUS

REFERENCES

1, 2, 5, 6, 40, 95, 208, 550.

Full revision 2019

NAME OF MEDICATION
ZIDOVUDINE

OTHER NAMES
Azidothymidine, Retrovir ™

CLASSIFICATION
Antiretroviral

INDICATIONS
- Treatment of HIV infection when oral administration is not possible.
- Prevention of maternal-fetal HIV transmission.

ADMINISTRATION
- Intermittent IV infusion: must be diluted to a concentration no greater than 4 mg/mL in NS, D5-1/2NS, D5W, D5-RL or RL; infuse over 60 minutes in adults and children; infuse over 30 minutes in neonates and infants younger than 6 weeks of age for the prevention of maternal-fetal HIV transmission.
- Continuous IV infusion: dilute in compatible solution to no greater than 4 mg/mL; infuse as per Dosage section.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity: anaphylaxis, rash.
- GI: nausea, vomiting, anorexia.
- CNS: headache, malaise, dizziness.
- Hematologic: anemia, granulocytopenia, neutropenia, macrocytosis (dose reduction or interruption may be required).
- Hepatic: lactic acidosis, severe hepatomegaly with steatosis (rare).
- Respiratory: cough.
- Fatigue, fever, myalgia.
- Local reactions: phlebitis, irritation, pain at injection site.

DOSAGE
Adults and adolescents weighing 30 kg and above:
- Treatment of HIV infection:
  - Usual dose of 1-2 mg/kg IV q4h.
  - Dosage in renal impairment: if CrCl is below 15 mL/min or hemodialysis or peritoneal dialysis, administer 1 mg/kg IV q6-8h.
  - Dosage in hepatic impairment: data are lacking; use with caution in patients with severe impairment and monitor for hematologic toxicity.
- Prevention of maternal-fetal HIV transmission: during labour and delivery, loading dose of 2 mg/kg IV administered over 60 minutes followed by a continuous IV infusion of 1 mg/kg/hr until clamping of the umbilical cord. For elective cesarean delivery, begin IV treatment 3 hours before surgery. For urgent cesarean delivery, administer loading dose then proceed to delivery.

Pediatrics weighing less than 30 kg:
- Treatment of HIV infection:
  - Neonates to less than 4 weeks of age: 3 mg/kg IV q12h. Premature neonates require dosage adjustment in the first 6-10 weeks of life.
  - 4 weeks to less than 3 months of age: 9 mg/kg IV q12h.
  - 3 months of age and older: 120 mg/m^2 IV (maximum 160 mg/dose) IV q6h OR, for those who are at least 6 months old, a continuous IV infusion of 15-42 mg/m^2/hr OR 0.5-1.4 mg/kg/hr can be considered.
- Neonatal dosing for prevention of maternal-fetal HIV transmission: 1.5 mg/kg IV q6h OR 3 mg/kg IV q12h. Premature neonates may require a dosage adjustment. Begin infusion as soon as possible after birth, preferably within 2-12 hours of birth and continue until 4-6 weeks of age.
- Dosage in renal impairment: in severe impairment (CrCl below 10 mL/min), administer 50% of the normal dose q8h.
- Dosage in hepatic impairment: data are lacking; use with caution in patients with severe impairment and monitor for hematologic toxicity.

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COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials between 15-25°C. Protect from light. Do not freeze.
- Stable for 48 hours at room temp or in the fridge in D5W at a concentration of 2 mg/mL.
- Stable for 8 days at room temp or in the fridge in D5W or NS at a concentration of 4 mg/mL in PVC containers.

MISCELLANEOUS

- In adults, the IV dosing regimen of 1 mg/kg IV q4h is approximatively equivalent to the oral administration of 100 mg q4h. In children, the oral dose is one-third greater than the IV dose. In neonates, the IV dose is 75% of the oral dose.
- The rubber stopper of the vial contains latex.

REFERENCES

1, 5, 6, 40, 82, 102, 135, 208, 319.

NAME OF MEDICATION
ZOLEDRONIC ACID

OTHER NAMES
Aclasta ®, Zometa ®

CLASSIFICATION
Bisphosphonate

INDICATIONS

- Treatment of tumour-induced hypercalcemia following adequate saline rehydration to restore urine output (Zometa ®).
- Treatment of bone metastases of solid tumours (including prostate cancer, breast cancer, lung cancer and renal cell carcinoma) and osteolytic lesions of multiple myeloma (Zometa ®).
- Treatment of Paget's disease (Aclasta ®).
- Treatment of osteoporosis in men and postmenopausal women (Aclasta ®).
- Treatment and prevention of glucocorticoid-induced osteoporosis (Aclasta ®).
- Prevention of postmenopausal osteoporosis in women with osteopenia (Aclasta ®).

ADMINISTRATION

- **Consider premedication if patient previously experienced flu-like symptoms; refer to Potential Administration Hazards section**.
- Available as a concentrate of 4 mg/5 mL vial (Zometa ®) and as a ready-to-use solution of 5 mg/100 mL (Aclasta ®).
- Intermittent IV infusion: for Zometa ®, dilute dose in 100 mL of D5W or NS. For Aclasta ®, use the ready-to-use solution. Administer over at least 15 minutes.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: bronchoconstriction, urticaria, angioedema, anaphylaxis.
- GI: nausea, diarrhea, constipation.
- Electrolyte disturbances: hypocalcemia.
- Ophthalmic: eye redness, itching, pain.
- Renal: renal dysfunction, acute renal failure; risk factors include renal impairment, advanced age, concomitant nephrotoxic drugs, diuretics and dehydration.
- Flu-like symptoms (e.g., fever, chills, arthralgia, myalgia, bone pain, fatigue, headache); usually mild to moderate; occur within 3 days and normally subside within 3 days of onset. **Consider premedication** with acetaminophen or ibuprofen.
- Local reactions: infrequent; redness, pain, itching and swelling at infusion site.

DOSAGE

Zometa ®:
- Hypercalcemia of malignancy:
  - 4 mg IV as a single dose in conjunction with adequate hydration.
  - At least 1 week must elapse before retreatment to allow for a full response to the initial dose.
  - Dosage in renal impairment: no dosage adjustment for mild to moderate impairment; not recommended for severe impairment (CrCl less than 30 mL/min).

- Bone metastases and osteolytic lesions of multiple myeloma:
  - 4 mg IV every 3 to 4 weeks.
  - Dosage in renal impairment:

CrCl (mL/min)	Dose (mg)
60-50	3.5
49-40	3.3
39-30	3
less than 30	not recommended

  - Treatment interruption may be required for increases in baseline serum creatinine. Treatment may be restarted when creatinine returns to within 10% of baseline value.

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DOSAGE (Cont.)

Aclasta ®:
- Paget's disease: 5 mg IV as a single dose. Treatment may be repeated once after at least 1 year in patients who relapse.
- Treatment of osteoporosis in men and postmenopausal women: 5 mg IV once a year.
- Treatment and prevention of glucocorticoid-induced osteoporosis: 5 mg IV once a year.
- Prevention of postmenopausal osteoporosis: 5 mg IV as a single dose or every 2 years.
- Dosage in renal impairment: no dosage adjustment required for mild to moderate impairment; not recommended for severe impairment (CrCl less than 35 mL/min).

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store vials and ready-to-use solution between 15-30ºC.
- Stable for 14 days at room temp or in the fridge in D5W or NS at a concentration of 0.04 mg/mL in PVC bags.
- Refrigerated diluted solutions should be brought to room temp before administration.
- Incompatible with calcium-containing infusion solutions (e.g., RL).

MISCELLANEOUS

- **Do not confuse Zometa ® with Aclasta ®**. Both contain zoledronic acid; however, they have different concentrations, doses and uses and are therefore not interchangeable.
- Pre-existing hypocalcemia must be treated before initiating treatment with zoledronic acid.
- Serum creatinine, electrolytes, calcium, magnesium and phosphate should be monitored.
- Calcium and vitamin D supplementation are recommended for patients being treated for multiple myeloma, bone metastases, Paget's disease or osteoporosis.
- When administering Aclasta ®, it is recommended that patients (especially elderly patients and those receiving diuretics) drink 500 mL of fluid (such as water) before and after the administration of the drug to reduce the risk of renal toxicity.

REFERENCES

1, 5, 40, 95, 165, 515.

NAME OF MEDICATION
ZOSTER VACCINE

OTHER NAMES
Herpes zoster vaccine, Shingles vaccine, Shingrix ™, Zostavax ® II

CLASSIFICATION
Vaccine

Do NOT confuse the zoster vaccine with the varicella virus vaccine for the prevention of chickenpox. The zoster vaccine is approximately 14 times more potent than the varicella virus vaccine. This monograph is specific to the ZOSTER vaccine.

INDICATIONS
- Prevention of herpes zoster (shingles) in immunocompetent adults aged 50 years and older.

ADMINISTRATION
- Reconstitute immediately upon removal of the refrigerator. Withdraw the entire content of the supplied diluent vial (Zostavax ® II) or adjuvant vial (Shingrix ™). Slowly inject all the diluent or adjuvant into their respective vial of lyophilized vaccine to avoid excessive foaming; gently mix.
- Zostavax ® II: SC; preferred site of injection is the deltoid area.
- Shingrix ™: IM; preferred site of injection is the deltoid area.

POTENTIAL ADMINISTRATION HAZARDS
- Hypersensitivity, including anaphylaxis.
- GI (Shingrix ™): nausea, vomiting, diarrhea, abdominal pain.
- CNS: headache.
- Zostavax ® II: pain in the extremities.
- Shingrix ™: myalgia, fatigue, shivering, fever.
- Local reactions: pain or tenderness, erythema, swelling, pruritus.

DOSAGE
- Zostavax ® II: single dose of the entire contents of the reconstituted vial (approximately 0.65 mL) SC.
- Shingrix ™: 0.5 mL IM for 2 doses: initial dose at month 0 and second dose between 2-6 months later.

COMPATIBILITY, STABILITY
(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

Zostavax ® II:
- Store Zostavax ® II vaccine vial in the fridge (between 2-8°C) until reconstitution. Store its diluent at room temp or in the fridge. Protect from light until reconstitution.
- Reconstituted vaccine solution is semi-hazy to translucent, off-white to pale yellow.
- Do not freeze the reconstituted vaccine.
- Reconstituted vaccine is stable for 30 minutes at room temp; discard if not administered within this timeframe.

Shingrix ™:
- Store Shingrix ™ vaccine and its adjuvant in the fridge (between 2-8°C) until reconstitution. Protect from light until reconstitution. Do not freeze.
- Prior to reconstitution, Shingrix ™ vaccine and its adjuvant may be stored between 8-25°C for a maximum cumulative time of 72 hours.
- Reconstituted vaccine solution is opalescent, colourless to pale brownish.
- Reconstituted vaccine is stable for 6 hours in the fridge (between 2-8°C).

…/Cont.

…/Cont.

MISCELLANEOUS

- Emergency drug (epinephrine) must be immediately available for the treatment of hypersensitivity reactions.
- Administration should be delayed if possible during an acute febrile illness or active infection of moderate to severe intensity.
- Persons with a reported history of herpes zoster can be vaccinated, if at least one year has elapsed since the episode. However, this recommendation does not include those with prior herpes zoster ophthalmicus.

Zostavax ® II:
- Zostavax ® II is a live attenuated vaccine.
- Contains trace amounts of neomycin and porcine gelatin.
- Contraindicated in pregnant women. Although manufacturer recommends avoiding pregnancy for 3 months following vaccination, health authorities consider that a one-month waiting period is sufficient and safe.

Shingrix ™:
- Shingrix ™ is an inactivated recombinant vaccine.
- Best to delay vaccination during pregnancy because of the lack of data.

REFERENCES

1, 5, 30, 31, 112, 135, 213.

NAME OF MEDICATION
ZUCLOPENTHIXOL

OTHER NAMES
Clopixol-Acuphase ®, Clopixol ® Depot

CLASSIFICATION
Antipsychotic

INDICATIONS

- Initial treatment of acute psychosis or for exacerbation of psychosis associated with schizophrenia (acetate salt i.e., Acuphase).
- Long term management of schizophrenia in patients who have been stabilized on oral or other short-acting antipsychotic therapy (decanoate salt i.e., Depot).

ADMINISTRATION

- IM: deep into the gluteal muscle; if a volume of more than 2 mL is to be injected, divide dose between 2 injection sites.

POTENTIAL ADMINISTRATION HAZARDS

- Hypersensitivity: rash, anaphylaxis.
- Cardiovascular: tachycardia, QT interval prolongation.
- GI: dry mouth.
- CNS: drowsiness, dizziness, extrapyramidal symptoms (e.g., akathisia, dystonia, Parkinson-like reaction).
- Endocrine and metabolic: hyperprolactinemia, galactorrhea, hyperglycemia, diabetic ketoacidosis.
- Neuroleptic malignant syndrome.
- Fatigue.
- Local reactions: skin reactions, pain.

DOSAGE

- Acetate (Acuphase): 50-150 mg IM; may be repeated prn at intervals of 2-3 days (some patients may need a second dose 1-2 days following the first injection), for a maximum cumulative dose of 400 mg or 4 injections. Duration of treatment with the acetate salt should not exceed 2 weeks. If treatment needs to be continued, switch to the long acting decanoate salt (Depot) or to oral tablets; refer to manufacturer's instructions for dosage conversion tables.
- Decanoate (Depot): usual dose of 150-300 mg IM every 2-4 weeks. Some patients may require higher or lower doses and/or a more frequent administration. Refer to manufacturer's instructions to determine the decanoate (Depot) dose when converting from oral tablets or the acetate (Acuphase). Maximum dose of 400 mg every 2 weeks.
- Dosage in renal impairment: no dosage adjustment necessary; use with caution.
- Dosage in hepatic impairment: use with caution as the drug undergoes extensive hepatic metabolism.

COMPATIBILITY, STABILITY

(Consult NAPRA compounding standards; information provided below refers to physical/chemical stability and not to sterility)

- Store ampoules between 15-25ºC. Protect from light.
- The acetate and decanoate doses can be mixed in the same syringe and given as a single injection.

MISCELLANEOUS

- Peak levels occur 24-48 hours post-injection with the acetate formulation (Acuphase) and 3-7 days post-injection with the decanoate formulation (Depot).

REFERENCES

5, 51, 95, 135, 414.

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HyperRAB	See Rabies Immune Globuline, Human
Hypertonic Saline	See Sodium Chloride, Hypertonic 3%
Hypurin NPH	See Insulin NPH
Hypurin Regular	See Insulin Regular
Hyrimoz	See Adalimumab

I

Ibuprofen (with Arginine)	
Ibuprofen (with Tromethamine)	
Ibuprofen Lysine	
Ibutilide	
Icatibant	
IC-Green	See Indocyanine Green
Idamycin PFS	See IdaruBICIN
IdaruBICIN	
IdaruCIZUmab	
Ifex	See Ifosfamide
Ifosfamide	

IGIVnex	See Immune Globulin, Human (Intravenous)
Ilomedin	See Iloprost
Ilomedin 20	See Iloprost
Ilomedine	see Iloprost
Iloprost	
Iloprost Trometamol	See Iloprost
Imfinzi	See Durvalumab
Imiglucerase	
Imipenem	
Imipenem-Cilastatin	See Imipenem
Immune Globulin Human (Intravenous)	
Immune Globulin, Human (Subcutaneous)	
Immunine VH	See Factor IX Concentrate (Human & Recombinant)
Imogam	See Rabies Immune Globuline, Human
Imovax Rabies	See Rabies Vaccine
Imuran	See AzaTHIOprine
Inderal	See Propranolol
Indigo Carmine	See Indigotindisulfonate Sodium
Indigotindisulfonate Sodium	
Indocyanine Green	
Indomethacin	
Inflectra	See InFLIXimab
InFLIXimab	
Influenza Virus Vaccine (Inactivated)	
Influvac Tetra	See Influenza Virus Vaccine (Inactivated)
Innohep	See Tinzaparin
Inotuzumab Ozogamicin	
Insulin Aspart	
Insulin Degludec	
Insulin Detemir	
Insulin Glargine	
Insulin Glulisine	
Insulin Lispro	
Insulin NPH	
Insulin Regular	
Integrilin	See Eptifibatide
Interferon Alpha-2B	
Interferon Beta	
Interleukin-2	See Aldesleukin
Intralipid	See Fat Emulsions
Intron A	See Interferon Alpha-2B
Invanz	See Ertapenem

Invega Sustenna	See Paliperidone
Invega Trinza	See Paliperidone
Ipilimumab	
Irinotecan	
Irinotecan (Liposomal)	
Irinotecan liposome	See Irinotecan (Liposomal)
Iron Isomaltoside 1000	
Iron Saccharate	See Iron Sucrose
Iron Sucrose	
Isatuximab	
Isavuconazole	
Isavuconazonium	See Isavuconazole
Isoniazid	
Isophosphamide	See Ifosfamide
Isoprenaline	See Isoproterenol
Isopropylarterenol	See Isoproterenol
Isopropylnoradrenaline	See Isoproterenol
Isoproterenol	
Isoptin	See Verapamil
Istodax	See RomiDEPsin
Isuprel	See Isoproterenol
Itraconazole	

J

Jevtana	See Cabazitaxel
Jivi	See Factor VIII (Recombinant)

K

Kadcyla	See Kadcyla Trastuzumab Emtansine
Kadcyla Trastuzumab Emtansine	
KamRAB	See Rabies Immune Globuline, Human
Kanjinti	See Trastuzumab
Kefurox	See Cefuroxime
Kefzol	See CeFAZolin
Keppra	See LevETIRAcetam
Ketalar	See Ketamine
Ketamine	
Ketorolac	
Ketorolac Trometamol	See Ketorolac
Ketorolac Tromethamine	See Ketorolac
Keytruda	See Pembrolizumab

Kidrolase	See Asparaginase (E Coli source)
Kinevac	See Sincalide
Kovaltry	See Factor VIII (Recombinant)
Kyprolis	See Carfilzomib
Kytril	See Granisetron

L

Labetalol	
Lacosamide	
Ladakamycin	See AzaCITIDine
Lambrolizumab	See Pembrolizumab
Lanoxin	See Digoxin
Lantus	See Insulin Glargine
Lapelga	See Pegfilgrastim
L-Arginine	See Arginine
Lasix	See Furosemide
L-Asparaginase (E coli source)	See Asparaginase (E coli source)
L-Asparaginase (Erwinia)	See Asparaginase (Erwinia)
L-Carnitine	See LevOCARNitine
Lemtrada	See Alemtuzumab
Leucovorin Calcium	
Leuprolide	
Leurocristine	See VinCRIStine
Leustatin	See Cladribine
Levaquin	See LevoFLOXacin
Levarterenol	See Norepinephrine Bitartrate
Levemir	See Insulin Detemir
LevETIRAcetam	
LevOCARNitine	
LevoFLOXacin	
Levomepromazine	See Methotrimeprazine
Levophed	See Norepinephrine Bitartrate
Levothyroxine	
Libtayo	See Cemiplimab
Lidocaine	
Lignocaine	See Lidocaine
Lincocin	See Lincomycin
Lincomycin	
Linezolid	
Lipid Emulsions	See Fat Emulsions
Liposyn	See Fat Emulsions

Liraglutide	
Lixisenatide	
LMD-10%	See Dextran-40
Lopresor	See Metoprolol
LORazepam	
Lovenox	See Enoxaparin
Low Molecular Weight Dextran	See Dextran-40
Loxapac	See Loxapine
Loxapine	
Lupron	See Leuprolide
Lupron Depot	See Leuprolide
Lutrepulse	See Gonadorelin

M

MabCampath	See Alemtuzumab
Magnesium Sulfate, Aqueous	
Mannitol	
Marcaine	See Bupivacaine
Mar-Cidofovir	See Cidofovir
Maxeran	See Metoclopramide
Maxipime	See Cefepime
Measles, Mumps and Rubella Vaccine	
MedroxyPROGESTERone	
Mefoxin	See CefOXitin
Melphalan	
Menactra	See Meningococcal Vaccine, Groups A, C, Y W-135
Meningitec	See Meningococcal Vaccine, Group C
Meningococcal Vaccine, Group B	
Meningococcal Vaccine, Group C	
Meningococcal Vaccine, Groups A, C, Y, W-135	
Menjugate	See Meningococcal Vaccine, Group C
Menjugate Liquid	See Meningococcal Vaccine, Group C
Menomune	See Meningococcal Vaccine, Groups A, C, Y, W-135
Menveo	See Meningococcal Vaccine, Groups A, C, Y W-135
Meperidine	
Meropenem	
Merrem	See Meropenem
Mesna	

Methotrexate	
Methotrimeprazine	
Methylene Blue	
Methylnaltrexone	
MethylPREDNISolone (Acetate)	
MethylPREDNISolone (Sodium Succinate)	
Metoclopramide	
Metoject	See Methotrexate
Metoprolol	
MetroNIDAZOLE	
$MgSO_4$	See Magnesium Sulfate, Aqueous
Micafungin	
Micro+6	See Trace Metals
Micro Cu	See Copper
Midazolam	
Milrinone	
MitoMYcin	
MitoMYcin C	See MitoMYcin
MitoXANTRONE	
Mitozantrone	See MitoXANTRONE
MMR	See Measles, Mumps and Rubella Vaccine
MMR-II	See Measles, Mumps and Rubella Vaccine
Monoclonal Antibody 2C4	See Pertuzumab
Monoferric	See Iron Isomaltoside 1000
Monomethyl Auristatin E	See Brentuximab Vedotin
Morphine	
Movapo	See Apomorphine
Moxifloxacin	
Mozobil	See Plerixafor
Mucomyst	See N-Acetylcysteine
Multivitamins	
Mutamycin	See MitoMYcin
Mvasi	See Bevacizumab
MVI-12	See Multivitamins
MVI-1000	See Multivitamins
Mycamine	See Micafungin
Mycophenolate Mofetil	
Myleran	See Busulfan
Mylotarg	See Gemtuzumab ozogamicin
Myocet	See DOXOrubicin (Liposomal)
Myozyme	See Alglucosidase Alfa

N

Nab-paclitaxel	See PACLitaxel (nanoparticle, albumin-bound)
Nadroparin	
$NAHCO_3$	See Sodium Bicarbonate
Nalbuphine	
Naloxone	
Nanoliposomal irinotecan	See Irinotecan (Liposomal)
Narcan	See Naloxone
Naropin	See Ropivacaine
Natalizumab	
Natrecor	See Nesiritide
Navelbine	See Vinorelbine
Nebcin	See Tobramycin
NeisVAc-C	See Meningococcal Vaccine, Group C
Nelarabine	
NeoProfen	See Ibuprofen Lysine
Neoral	See CycloSPORINE
Neostigmine	
Neo-Synephrine	See Phenylephrine
Nesiritide	
Neulasta	See Pegfilgrastim
Neupogen	See Filgrastim
Niacinamide	
NiaStase RT	See Factor VIIA (Recombinant)
Nicotinamide	See Niacinamide
Nimbex	See Cisatracurium
Nimenrix	See Meningococcal Vaccine, Groups A, C, Y W-135
Nipent	See Pentostatin
Nipride	See NitroPRUSSIDE
NitroGLYCERIN	
Nitroject	See NitroGLYCERIN
NitroPRUSSIDE	
Nivestym	See Filgrastim
Nivolumab	
Nonacog Alfa	See Factor IX Concentrate (Human & Recombinant)
Noradrenaline Bitartrate	See Norepinephrine Bitartrate
19-nor-1 alpha-25-dihydroxyvitamin D_2	See Paricalcitol
Norepinephrine Bitartrate	
Novantrone	See MitoXANTRONE
Novolin ge NPH	See Insulin NPH
Novolin ge Toronto	See Insulin Regular

NovoRapid	See Insulin Aspart
Nozinan	See Methotrimeprazine
Nplate	See RomiPLOStim
Nubain	See Nalbuphine
Nydrazid	See Isoniazid
Nyvepria	See Pegfilgrastim

O

Obinutuzumab	
Obizur	See Factor VIII (Recombinant)
Octagam	See Immune Globulin, Human (Intravenous)
Octalbin	See Albumin, Normal Human Serum
Octaplex	See Human Prothrombin Complex
Octostim	See Desmopressin
Octreotide	
Ofirmev	See Acetaminophen
Ogivri	See Trastuzumab
OLANZapine	
Omalizumab	
OnabotulinumtoxinA	
Oncovin	See VinCRIStine
Ondansetron	
One-Alpha	See Alfacalcidol
Onivyde	See Irinotecan (Liposomal)
Opdivo	See Nivolumab
Orencia	See Abatacept
Orgalutran	See Ganirelix
Orgaran	See Danaparoid
Osmitrol	See Mannitol
Osnuvo	See Teriparatide
Oxalatoplatin	See Oxaliplatin
Oxaliplatin	
Oxaliplatinum	See Oxaliplatin
Oxytocin, Synthetic	
Ozempic	See Semaglutide

P

PACLitaxel	
PACLitaxel, Nanoparticle, Albumin-Bound	
Padcev	See Enfortumab Vedotin

Paliperidone	
PAM	See Melphalan
2-PAM	See Pralidoxime
Pamidronate	
Pancuronium	
Panhematin	See Hemin
Panitumumab	
Panto IV	See Pantoprazole
Pantoprazole	
Panzyga	See Immune Globulin, Human (Intravenous)
Papaverine	
Paracetamol	See Acetaminophen
Paraplatin	See CARBOplatin
Paricalcitol	
Parvolex	See N-Acetylcysteine
Pedea	See Ibuprofen (with Tromethamine)
PEG Filgrastim	See Pegfilgrastim
Pegfilgrastim	
Pegylated Filgrastim	See Pegfilgrastim
Pegylated Liposomal Irinotecan	See Irinotecan (Liposomal)
Pembrolizumab	
PEMEtrexed	
Penicillin G	
Penicillin G Benzathine	
Pentacarinat	See Pentamidine Isethionate
Pentamidine Isethionate	
Pentamidine Isetionate	See Pentamidine Isethionate
Pentavalent Antimony Compound	See Sodium Stibogluconate
Pentazocine	
Pentostam	See Sodium Stibogluconate
Pentostatin	
Pepcid	See Famotidine
Perjeta	See Pertuzumab
Persantine	See Dipyridamole
Pertuzumab	
Pethidine	See Meperidine
Peyona	See Caffeine Citrate
Pfizer-BioNTech COVID-19 Vaccine	See COVID-19 Vaccine (mRNA)
Pharmorubicin	See EpiRUBicin
Phenergan	See Promethazine
PHENobarbital	
Phenobarbitone Sodium	See PHENobarbital

Phentolamine	
L-Phenylalanine Mustard	See Melphalan
Phenylephrine	
Phenytoin	
Phosphate (as sodium or potassium salt)	
Photofrin	See Porfimer
Phylloquinone	See Vitamin K_1
Physostigmine	
Phytomenadione	See Vitamin K_1
Phytonadione	See Vitamin K_1
Piperacillin	
Piperacillin/Tazobactam	
Pipracil	See Piperacillin
Pitocin	See Oxytocin, Synthetic
Plasbumin	See Albumin, Normal Human Serum
Platinol	See CISplatin
Platinol-AQ	See CISplatin
Plerixafor	
Pneumococcal Vaccine	
Pneumovax 23 (23-valent)	See Pneumococcal Vaccine
Porfimer	
Posanol	See Posaconazole
Posaconazole	
Potassium Chloride	
Potassium Phosphate	See Phosphate
Pralatrexate	
Pralidoxime	
Praxbind	See IdaruCIZUmab
Precedex	See Dexmedetomidine
Premarin	See Estrogens, Conjugated
Pressyn AR	See Vasopressin
Prevnar 13 (13-valent)	See Pneumococcal Vaccine
Primacor	See Milrinone
Primaxin	See Imipenem
Priorix	See Measles, Mumps and Rubella Vaccine
Privigen	See Immune Globulin, Human (Intravenous)
Procainamide	
Procytox	See Cyclophosphamide
Prograf	See Tacrolimus
Prolastin-C	See Alpha 1-Proteinase Inhibitor
Proleukin	See Aldesleukin
Prolia	See Denosumab

Promethazine	
Pronestyl	See Procainamide
Propofol	
Propranolol	
Prostaglandin E_1	See Alprostadil
Prostaglandin F_2 analog	See Carboprost
Prostin VR	See Alprostadil
Protamine	
Protopam Chloride	See Pralidoxime
2-Pyridine Aldoxime Methochloride	See Pralidoxime
Pyridoxine	

Q

Quelicin	See Succinylcholine
Quinimax	See QuiNINE base
QuinNINE base	
QuiNINE dihydrochloride	
Quininject	See QuiNINE dihydrochloride
Quinupristin/Dalfopristin	

R

RabAvert	See Rabies Vaccine
Rabbit ATG	See Antithymocyte Globulin, Rabbit
Rabbit IgG	See Antithymocyte Globulin, Rabbit
Rabies Immune Globuline, Human	
Rabies Vaccine	
Radicava	See Edaravone
Raltitrexed	
Ramucirumab	
Rasburicase	
rATG	See Antithymocyte Globulin, Rabbit
Rebif	See Interferon Beta
Rebinyn	See Factor IX (Concentrate Human & Recombinant)
Recombinant Alpha-Galactosidase A Enzyme	See Agalsidase Beta
Recombinant Human Erythropoietin	See Erythropoietin
Recombinant Urate-Oxidase Enzyme	See Rasburicase
Recombivax HB	See Hepatitis B Vaccine
Recombivax HB Adult Dialysis	See Hepatitis B Vaccine
Reglan	See Metoclopramide
Relistor	See Methylnaltrexone

Remdesivir	
Remicade	See InFLIXimab
Remifentanil	
Remodulin	See Treprostinil
Remsima	See InFLIXimab
Renflexis	See InFLIXimab
Retrovir	See Zidovudine
Rho(D) Immune Globulin (Human)	
RiaSTAP	See Fibrinogen Concentrate (Human)
Rifadin	See RifAMPin
Rifampicine	See RifAMPin
RifAMPin	
Rimactane	See RifAMPin
Risperdal Consta	See RisperiDONE
RisperiDONE	
Rituxan	See RiTUXimab
RiTUXimab	
Riximyo	See RiTUXimab
Robinul	See Glycopyrrolate
Rocephin	See CefTRIAXone
Rocuronium	
Rogitine	See Phentolamine
RomiDEPsin	
RomiPLOStim	
Ropivacaine	
rt-PA	See Alteplase
r-TPA	See Alteplase
Ruxience	See RiTUXimab

S

Salbutamol	
Salcatonin	See Calcitonin Salmon
Sandimmune	See CycloSPORINE
Sandostatin	See Octreotide
Sandostatin LAR	See Octreotide
Sarclisa	See Isatuximab
L-Sarcolysin	See Melphalan
Saxenda	See Liraglutide
SCIG	See Immune Globulin, Human (Subcutaneous)
SC IgG	See Immune Globulin, Human (Subcutaneous)
Scopolamine Butylbromide	See Hyoscine Butylbromide

Scopolamine Hydrobromide	
Seacalphyx	See Sodium Thiosulfate
Selenious Acid	See Selenium
Selenium	
Semaglutide	
Sensorcaine	See Bupivacaine
Septra	See Co-Trimoxazole
Shingles Vaccine	See Zoster Vaccine
Shingrix	See Zoster Vaccine
Siltuximab	
Simponi	See Golimumab
Simponi IV	See Golimumab
Simulect	See Basiliximab
Sincalide	
SMOFlipid	See Fat Emulsions
Sodium Acid Carbonate	See Sodium Bicarbonate
Sodium Ascorbate	See Vitamin C
Sodium Bicarbonate	
Sodium Calcium Edetate	See Calcium Disodium Edetate
Sodium Chloride, Hypertonic 3%	
Sodium Citrate 4%	
Sodium Edecrin	See Ethacrynic Acid
Sodium Ethacrynate	See Ethacrynic Acid
Sodium Ferric Gluconate Complex	See Ferric Gluconate Complex
Sodium Hydrogen Carbonate	See Sodium Bicarbonate
Sodium Hyposulfate	See Sodium Thiosulfate
Sodium Mercaptoethane Sulfonate	See Mesna
Sodium Phosphate	See Phosphate
Sodium Stibogluconate	
Sodium Thiosulfate	
Soliris	See Eculizumab
Solu-Cortef	See Hydrocortisone
Solu-Medrol	See MethylPREDNISolone (Sodium Succinate)
Sporanox	See Itraconazole
Stelara	See Ustekinumab
Streptomycin	
Succinylcholine	
Sufenta	See SUFentanil
SUFentanil	
Sugammadex	
Suprefact	See Buserelin
Suprefact Depot	See Buserelin

Suxamethonium Chloride	See Succinylcholine
Sylvant	See Siltuximab
Synacthen Depot	See Cosyntropin
Synercid	See Quinupristin/Dalfopristin
Synflorix (10-valent)	See Pneumococcal Vaccine
Synthetic Gonadotropin Releasing Hormone	See Gonadorelin
Synthroid	See Levothyroxine
Syntocinon	See Oxytocin, Synthetic

T

T4	See Levothyroxine
Tacrolimus	
Talwin	See Pentazocine
Taxol	See PACLitaxel
Taxotere	See DOCEtaxel
Tazidime	See CefTAZidime
Tazocin	See Piperacillin/Tazobactam
Td Absorbed	See Tetanus & Diphtheria Toxoids Adsorbed
Tecentriq	See Atezolizumab
Telavancin	
Temsirolimus	
Tenecteplase	
Tensilon	See Edrophonium
Tepadina	See Thiotepa
Teriparatide	
Tetanus & Diphtheria Toxoids Adsorbed	
Tetramethylthionine chloride trihydrate	See Methylene Blue
Theophylline Ethylenediamine	See Aminophylline
Thiamine	
Thiotepa	
Thymoglobulin	See Antithymocyte Globulin, Rabbit
Thyroxine sodium	See Levothyroxine
Tigecycline	
Tinzaparin	
Tirofiban	
Tissue Plasminogen Activator	See Alteplase
TMP-SMX	See Co-trimoxazole
TNKase	See Tenecteplase
Tobramycin	

Tocilizumab	
Tomudex	See Raltitrexed
Tomvi	See Etomidate
Topotecan	
Toradol	See Ketorolac
Torisel	See Temsirolimus
Toujeo	See Insulin Glargine
Tozinameran	See COVID-19 Vaccine (mRNA)
tPA	See Alteplase
Trace Elements	See Trace Metals
Trace Metals	
Trandate	See Labetalol
Tranexamic Acid	
Trastuzumab	
Trastuzumab Deruxtecan	See Enhertu Trastuzumab Deruxtecan
Trastuzumab DXd	See Enhertu Trastuzumab Deruxtecan
Trastuzumab Emtansine	See Kadcyla Trastuzumab Emtansine
Trastuzumab-MCC-DM1	See Kadcyla Trastuzumab Emtansine
Trazimera	See Trastuzumab
Treanda	See Bendamustine
Tremytoine	See Phenytoin
Treprostinil	
Treprostinil Sodium	See Treprostinil
Tresiba	See Insulin Degludec
Trimethoprim/Sulfamethoxazole	See Co-Trimoxazole
Trinitroglycerin	See NitroGLYCERIN
Trisenox	See Arsenic Trioxide
Trisodium Citrate	See Sodium Citrate 4%
Trumenba	See Meningococcal Vaccine, Group B
Truxima	See Rituximab
Tygacil	See Tigecycline
Tysabri	See Natalizumab

U

Ultiva	See Remifentanil
Unfractionated heparin	See Heparin
Uromitexan	See Mesna
Ustekinumab	

V

Valium	See DiazePAM
Valproate sodium	See Valproic Acid
Valproic Acid	
Vancocin	See Vancomycin
Vancomycin	
Vaqta	See Hepatitis A Vaccine
Vaqta Pediatric/Adolescent	See Hepatitis A Vaccine
Varicella Virus Vaccine	
Varilrix	See Varicella Virus Vaccine
Varivax III	See Varicella Virus Vaccine
Vasopressin	
Vasotec IV	See Enalaprilat
Vectibix	See Panitumumab
Vedolizumab	
Veklury	See Remdesivir
Velban	See VinBLAStine
Velcade	See Bortezomib
Venofer	See Iron Sucrose
Ventolin	See Salbutamol
VePesid	See Etoposide
Verapamil	
Versed	See Midazolam
Versenate	See Calcium Disodium Edetate
Verteporfin	
Vfend	See Voriconazole
Vibativ	See Telavancin
Vibramycin	See Doxycycline
Victoza	See Liraglutide
Vidaza	See AzaCITIDine
Vimpat	See Lacosamide
VinBLAStine	
Vincaleukoblastine	See VinBLAStine
VinCRIStine	
Vinorelbine	
Vistide	See Cidofovir
Visudyne	See Verteporfin
Vitamin B_1	See Thiamine
Vitamin B_6	See Pyridoxine
Vitamin B_{12}	
Vitamin C	
Vitamin K	See Vitamin K_1
Vitamin K_1	

Vivacaine	See Bupivacaine
Vocarvi	See Foscarnet
Volulyte	See Hydroxyethyl Starch 130/0.4
Voluven	See Hydroxyethyl Starch 130/0.4
Voriconazole	
VP-16	See Etoposide

W

Wilate	See Factor VIII/Von Willebrand Factor Complex (Human)
WinRho SDF	See Rho(D) Immune Globulin (Human)

X

Xgeva	See Denosumab
Xolair	See Omalizumab
Xylocaine	See Lidocaine
Xylocard	See Lidocaine

Y

Yervoy	See Ipilimumab

Z

Zemplar	See Paricalcitol
Zemuron	See Rocuronium
Zerbaxa	See Ceftolozane/Tazobactam
Zeulide Depot	See Leuprolide
Zevtera	See Ceftobiprole
Zidovudine	
Ziextenzo	See Pegfilgrastim
Zinacef	See Cefuroxime
Zinecard	See Dexrazoxane
Zirabev	See Bevacizumab
Zithromax	See Azithromycin
Zofran	See Ondansetron
Zoladex	See Goserelin
Zoladex LA	See Goserelin
Zoledronic Acid	
Zometa	See Zoledronic Acid

Zostavax II	See Zoster Vaccine
Zoster Vaccine	
Zovirax	See Acyclovir
Zuclopenthixol	
Zyprexa	See OLANZapine
Zyvoxam	See Linezolid

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17.

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